

Received


WELL COMPLETION OR RECOMPLETION REPORT - FORM 6

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 2468 (04-2010)

ND Oil & Gas
Division

Well File No.
28658

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Designate Type of Completion			
☒ Oil Well	☐ EOR Well	☐ Recompletion	☐ Deepened Well
☐ Gas Well	☐ SWD Well	☐ Water Supply Well	☐ Other:
Well Name and Number Kline Federal 5300 41-18 12TX		Spacing Unit Description Sec. 17/18/19/20 T153N R100W	
Operator Oasis Petroleum North America		Telephone Number (281) 404-9591	
Address 1001 Fannin, Suite 1500		Pool Bakken	
City Houston	State TX	Zip Code 77002	Permit Type ☐ Wildcat ☒ Development ☐ Extension

LOCATION OF WELL

At Surface 434 F S L	237 F W L	Qtr-Qtr LOT4	Section 18	Township 153 N	Range 100 W	County McKenzie
Spud Date September 13, 2014	Date TD Reached January 26, 2015	Drilling Contractor and Rig Number Nabors 486		KB Elevation (Ft) 2082	Graded Elevation (Ft) 2057	
Type of Electric and Other Logs Run (See Instructions) MWD/GR from KOP to TD						

CASING & TUBULARS RECORD (Report all strings set in well)

Well Bore	String Type	Size (Inch)	Top Set (MD Ft)	Depth Set (MD Ft)	Hole Size (Inch)	Weight (Lbs/Ft)	Anchor Set (MD Ft)	Packer Set (MD Ft)	Sacks Cement	Top of Cement
Surface Hole	Surface	13 3/8	0	2158	17 1/2	54.5			950	0
Vertical Hole	Intermediate	9 5/8	0	6118	12 1/4	36			1210	0
Vertical Hole	Intermediate	7	0	11175	8 3/4	32			818	10279
Lateral1	Liner	4 1/2	10259	20579	6	13.5			509	

PERFORATION & OPEN HOLE INTERVALS

Well Bore	Well Bore TD Drillers Depth (MD Ft)	Completion Type	Open Hole/Perforated Interval (MD Ft) Top Bottom	Kick-off Point (MD Ft)	Top of Casing Window (MD Ft)	Date Perf'd or Drilled	Date Isolated	Isolation Method	Sacks Cement
Lateral1	20584	Perforations	11175 20579	10300		06/09/2015			

PRODUCTION

Current Producing Open Hole or Perforated Interval(s), This Completion, Top and Bottom, (MD Ft) Lateral 1- 11175' to 20579'						Name of Zone (If Different from Pool Name)				
Date Well Completed (SEE INSTRUCTIONS) September 1, 2015			Producing Method flowing		Pumping-Size & Type of Pump			Well Status (Producing or Shut-In) producing		
Date of Test 09/01/2015	Hours Tested 24	Choke Size 38 /64	Production for Test		Oil (Bbls) 1127	Gas (MCF) 877	Water (Bbls) 3644	Oil Gravity-API (Corr.) °	Disposition of Gas Sold	
Flowing Tubing Pressure (PSI)		Flowing Casing Pressure (PSI) 1200		Calculated 24-Hour Rate	Oil (Bbls) 1127	Gas (MCF) 877	Water (Bbls) 3644	Gas-Oil Ratio 778		

GEOLOGICAL MARKERS

[illegible]

PLUG BACK INFORMATION

[illegible]

CORES CUT

Top (Ft)	Bottom (Ft)	Formation	Top (Ft)	Bottom (Ft)	Formation

Drill Stem Test

Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								

Well Specific Stimulations

Date Stimulated 06/09/2015	Stimulated Formation Three Forks	Top (Ft) 11175	Bottom (Ft) 20579	Stimulation Stages 50	Volume 186083	Volume Units Barrels
Type Treatment Sand Frac	Acid %	Lbs Proppant 9450980	Maximum Treatment Pressure (PSI) 9405		Maximum Treatment Rate (BBLS/Min) 38.0	
Details 100 Mesh White: 227820 20/40 White: 6562940 40/70 White: 1237840 20/40 Resin Coated: 1422380						
Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						
Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						
Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						
Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						
Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

ADDITIONAL INFORMATION AND/OR LIST OF ATTACHMENTS

This report has been amended to correct the casing information to show both the 13 3/8" and the 9 5/8" casing.	
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I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.	Email Address dbusch@oasispetroleum.com	Date 08/15/2018
	Signature 	Printed Name Daniel Busch

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.

28658

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☐ Notice of Intent	Approximate Start Date
☒ Report of Work Done	Date Work Completed October 3, 2015
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☐ Other	Well is now on pump

Well Name and Number Kline Federal 5300 41-18 12TX					
Footages	Qtr-Qtr	Section	Township	Range	
434 F S L 237 F W L	LOT4	18	153 N	100 W	
Field Baker	Pool Bakken	County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Effective 10/03/2015 the above referenced well is on pump.

End of Tubing: 2-3/8" L-80 tubing @ 9857.40'

Pump: 2-1/2" x 2.0" x 24' insert pump @ 10054.54'

Company Oasis Petroleum North America LLC		Telephone Number 281 404-9436	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature 		Printed Name Jennifer Swenson	
Title Regulatory Specialist		Date November 19, 2015	
Email Address jswenson@oasispetroleum.com			

FOR STATE USE ONLY

☒ Received	☐ Approved
Date 11-30-2015	
By Original Signed By: TAYLOR ROTH	
Title Engineering Technician	

**AUTHORIZATION TO PURCHASE AND TRANSPORT OIL FROM LEASE - Form 8**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5898 (03-2000)

Well File No.

28658

NDIC CTB No.

115899

228654

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.

PLEASE SUBMIT THE ORIGINAL AND FOUR COPIES.

Well Name and Number KLINE FEDERAL 5300 41-18 12TX	Qtr-Qtr LOT4	Section 18	Township 153	Range 100	County McKenzie
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Operator Oasis Petroleum North America LLC	Telephone Number (281) 404-9573	Field BAKER
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Address 1001 Fannin, Suite 1500	City Houston	State TX	Zip Code 77002
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Name of First Purchaser Oasis Petroleum Marketing LLC	Telephone Number (281) 404-9627	% Purchased 100%	Date Effective September 1, 2015
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Principal Place of Business 1001 Fannin, Suite 1500	City Houston	State TX	Zip Code 77002
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Field Address	City	State	Zip Code
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Transporter Hiland Crude, LLC	Telephone Number (580) 616-2058	% Transported 75%	Date Effective September 1, 2015
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Address P.O. Box 3886	City Enid	State OK	Zip Code 73702
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The above named producer authorizes the above named purchaser to purchase the percentage of oil stated above which is produced from the lease designated above until further notice. The oil will be transported by the above named transporter.

Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
---	-------------	----------------

Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
---	-------------	----------------

Other Transporters Transporting From This Lease	% Transported	Date Effective
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Power Crude Transport	25%	September 1, 2015
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Other Transporters Transporting From This Lease	% Transported	Date Effective
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		September 1, 2015
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Comments

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.	Date October 22, 2015
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Signature 	Printed Name Brianna Salinas	Title Marketing Assistant
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Above Signature Witnessed By: Signature 	Printed Name Laura Whitten	Title Marketing Analyst II
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FOR STATE USE ONLY

Date Approved OCT 27 2015
By
Title Oil & Gas Production Analyst



AUTHORIZATION TO PURCHASE AND TRANSPORT OIL FROM LEASE - Form 8
INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5698 (03-2000)



Well File No.

28658

NDIC CTS No.

To be assigned

22-8654

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.

PLEASE SUBMIT THE ORIGINAL AND FOUR COPIES.

Well Name and Number KLINE FEDERAL 5300 41-18 12TX	Qtr-Qtr LOT4	Section 18	Township 153	Range 100	County Mckenzie
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Operator Oasis Petroleum North America LLC	Telephone Number (281) 404-9573	Field BAKER
--	---	-----------------------

Address 1001 Fannin, Suite 1500	City Houston	State TX	Zip Code 77002
---	------------------------	--------------------	--------------------------

Name of First Purchaser Oasis Petroleum Marketing LLC	Telephone Number (281) 404-9627	% Purchased 100%	Date Effective August 22, 2015
---	---	----------------------------	--

Principal Place of Business 1001 Fannin, Suite 1500	City Houston	State TX	Zip Code 77002
---	------------------------	--------------------	--------------------------

Field Address	City	State	Zip Code
---------------	------	-------	----------

Transporter Hiland Crude, LLC	Telephone Number (580) 616-2058	% Transported 75%	Date Effective August 22, 2015
---	---	-----------------------------	--

Address P.O. Box 3886	City Enid	State OK	Zip Code 73702
---------------------------------	---------------------	--------------------	--------------------------

The above named producer authorizes the above named purchaser to purchase the percentage of oil stated above which is produced from the lease designated above until further notice. The oil will be transported by the above named transporter.

Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
Other First Purchasers Purchasing From This Lease	% Purchased	Date Effective
Other Transporters Transporting From This Lease	% Transported	Date Effective
Power Crude Transport	28%	August 22, 2015
Other Transporters Transporting From This Lease	% Transported	Date Effective
		August 22, 2015
Comments		

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.		Date August 22, 2015
Signature <i>Dina Barron</i>	Printed Name Dina Barron	Title Mktg. Contracts Administrator

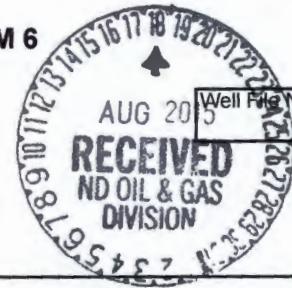
Above Signature Witnessed By:		
Signature <i>Jeremy Harris</i>	Printed Name Jeremy Harris	Title Marketing Scheduler

FOR STATE USE ONLY

Date Approved SEP 18 2015
By <i>Eric Peterson</i>
Title Oil & Gas Production Analyst

**WELL COMPLETION OR RECOMPLETION REPORT - FORM 6**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 2468 (04-2010)



Well File No. **28658**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
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Designate Type of Completion			
☒ Oil Well	☐ EOR Well	☐ Recompletion	☐ Deepened Well
☐ Gas Well	☐ SWD Well	☐ Water Supply Well	☐ Other:
Well Name and Number Kline Federal 5300 41-18 12TX		Spacing Unit Description Sec. 17/18/19/20 T153N R100W	
Operator Oasis Petroleum North America		Telephone Number (281) 404-9591	
Address 1001 Fannin, Suite 1500		Field Baker	
City Houston		Pool Bakken	
State TX		Zip Code 77002	
Permit Type		☐ Wildcat ☒ Development ☐ Extension	

LOCATION OF WELL

At Surface	Qtr-Qtr	Section	Township	Range	County
434 F S L 237 F WL	LOT4	18	153 N	100 W	McKenzie
Spud Date September 13, 2014	Date TD Reached January 26, 2015	Drilling Contractor and Rig Number Nabors 486	KB Elevation (Ft) 2082	Graded Elevation (Ft) 2057	
Type of Electric and Other Logs Run (See Instructions) MWD/GR from KOP to TD					

CASING & TUBULARS RECORD (Report all strings set in well)

Well Bore	String Type	Size (Inch)	Top Set (MD Ft)	Depth Set (MD Ft)	Hole Size (Inch)	Weight (Lbs/Ft)	Anchor Set (MD Ft)	Packer Set (MD Ft)	Sacks Cement	Top of Cement
Surface Hole	Surface	9 5/8	0	2160	13 1/2	36			950	0
Vertical Hole	Intermediate	7	0	11175	8 3/4	32			1210	
Lateral1	Liner	4 1/2	10259	20579	6	13.5			509	10279

PERFORATION & OPEN HOLE INTERVALS

Well Bore	Well Bore TD Drillers Depth (MD Ft)	Completion Type	Open Hole/Perforated Interval (MD,Ft)	Kick-off Point (MD Ft)	Top of Casing Window (MD Ft)	Date Perf'd or Drilled	Date Isolated	Isolation Method	Sacks Cement
Lateral1	20584	Perforations	Top Bottom	10300					

PRODUCTION

Current Producing Open Hole or Perforated Interval(s), This Completion, Top and Bottom, (MD Ft) Lateral 1-						Name of Zone (If Different from Pool Name)			
Date Well Completed (SEE INSTRUCTIONS)		Producing Method		Pumping-Size & Type of Pump		Well Status (Producing or Shut-In)			
Date of Test	Hours Tested	Choke Size 164	Production for Test	Oil (Bbbls)	Gas (MCF)	Water (Bbbls)	Oil Gravity-API (Corr.)	Disposition of Gas	
Flowing Tubing Pressure (PSI)		Flowing Casing Pressure (PSI)		Calculated 24-Hour Rate	Oil (Bbbls)	Gas (MCF)	Water (Bbbls)	Gas-Oil Ratio	

PLUG BACK INFORMATION

[illegible][illegible]

Top (Ft)	Bottom (Ft)	Formation	Top (Ft)	Bottom (Ft)	Formation

Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								
Test Date	Formation	Top (Ft)	Bottom (Ft)	BH Temp (°F)	CL ppm	H2S ppm	Shut-in 1 (PSIG)	Shut-in 2 (PSIG)
Drill Pipe Recovery								
Sample Chamber Recovery								

Well Specific Stimulations

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units Barrels
Type Treatment Sand Frac	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

Date Stimulated	Stimulated Formation	Top (Ft)	Bottom (Ft)	Stimulation Stages	Volume	Volume Units
Type Treatment	Acid %	Lbs Proppant	Maximum Treatment Pressure (PSI)		Maximum Treatment Rate (BBLS/Min)	
Details						

ADDITIONAL INFORMATION AND/OR LIST OF ATTACHMENTS

This is a preliminary completion report. A supplemental report will be filed upon first production of the well.

I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records.	Email Address	Date
	jswenson@oasispetroleum.com	08/17/2015
Signature	Printed Name	Title
	Jennifer Swenson	Regulatory Specialist

Industrial Commission of North Dakota
Oil and Gas Division

Well or Facility No
28658

Verbal Approval To Purchase and Transport Oil Tight Hole **Yes**

OPERATOR

Operator	Representative	Rep Phone
OASIS PETROLEUM NORTH AMERICA LL	Todd Hanson	(701) 577-1632

WELL INFORMATION

Well Name						Inspector
KLINE FEDERAL 5300 41-18 12TX						Richard Dunn
Well Location	QQ	Sec	Twp	Rng		County
	LOT4	18	153 N	100 W		MCKENZIE
Footages	434	Feet From the S Line				Field
	237	Feet From the W Line				BAKER
						Pool
						BAKKEN
Date of First Production Through Permanent Wellhead						This Is Not The First Sales

PURCHASER / TRANSPORTER

Purchaser	Transporter
Kinder Morgan	Kinder Morgan

TANK BATTERY

Central Tank Battery Number : 228654-01
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SALES INFORMATION This Is Not The First Sales

ESTIMATED BARRELS TO BE SOLD		ACTUAL BARRELS SOLD	DATE
15000	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	
	BBLS	BBLS	

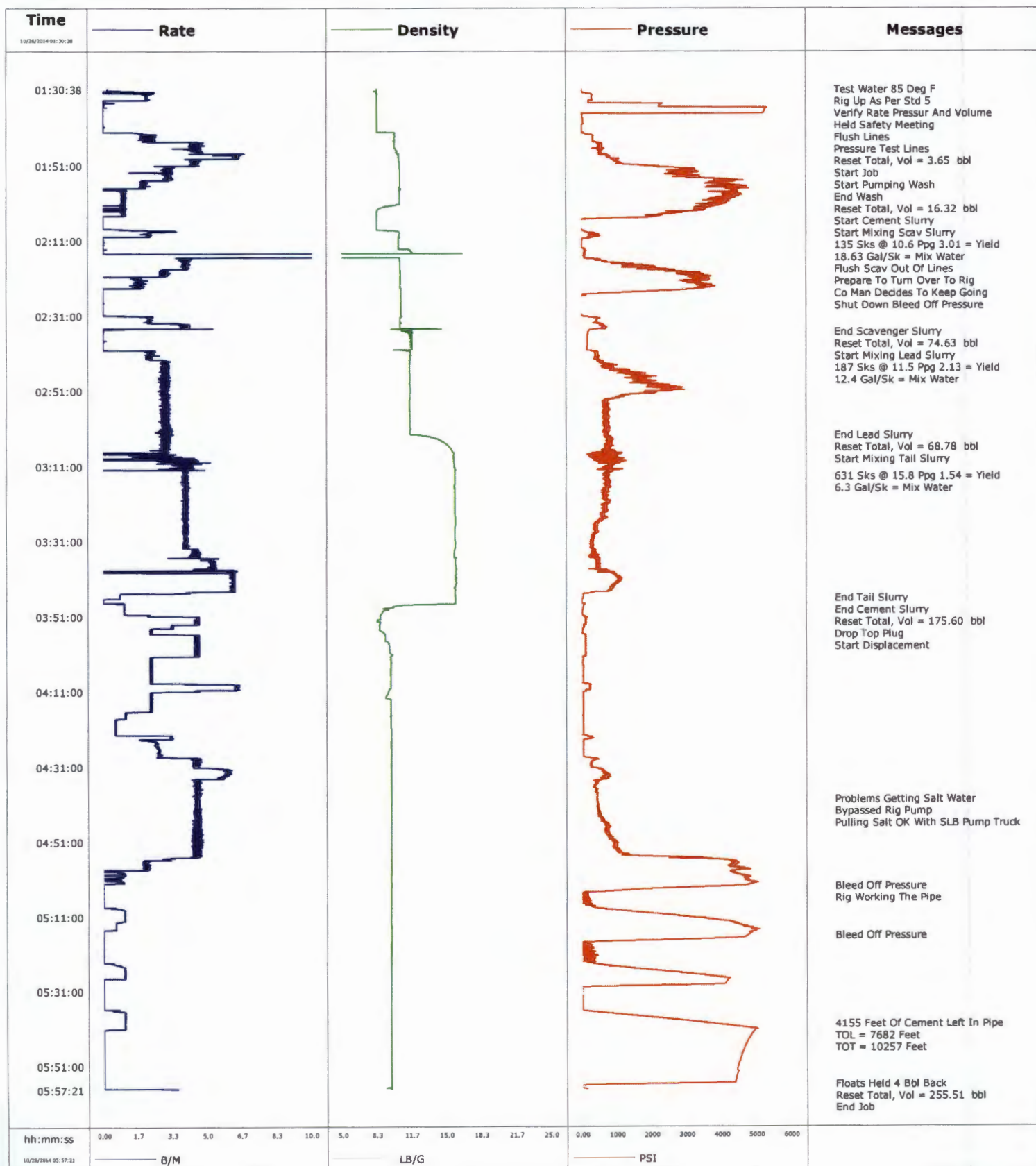
DETAILS

Must E-Mail or Call Inspector at 701-770-3554/rsdunn@nd.gov on first date of sales and report amount sold, date sold, and first date of production through the permanent wellhead. Must also forward Forms 6 & 8 to State prior to reaching 15000 Bbl estimate or no later than required time frame for submitting those forms.

Start Date **8/25/2015**
Date Approved **8/25/2015**
Approved By **Richard Dunn**

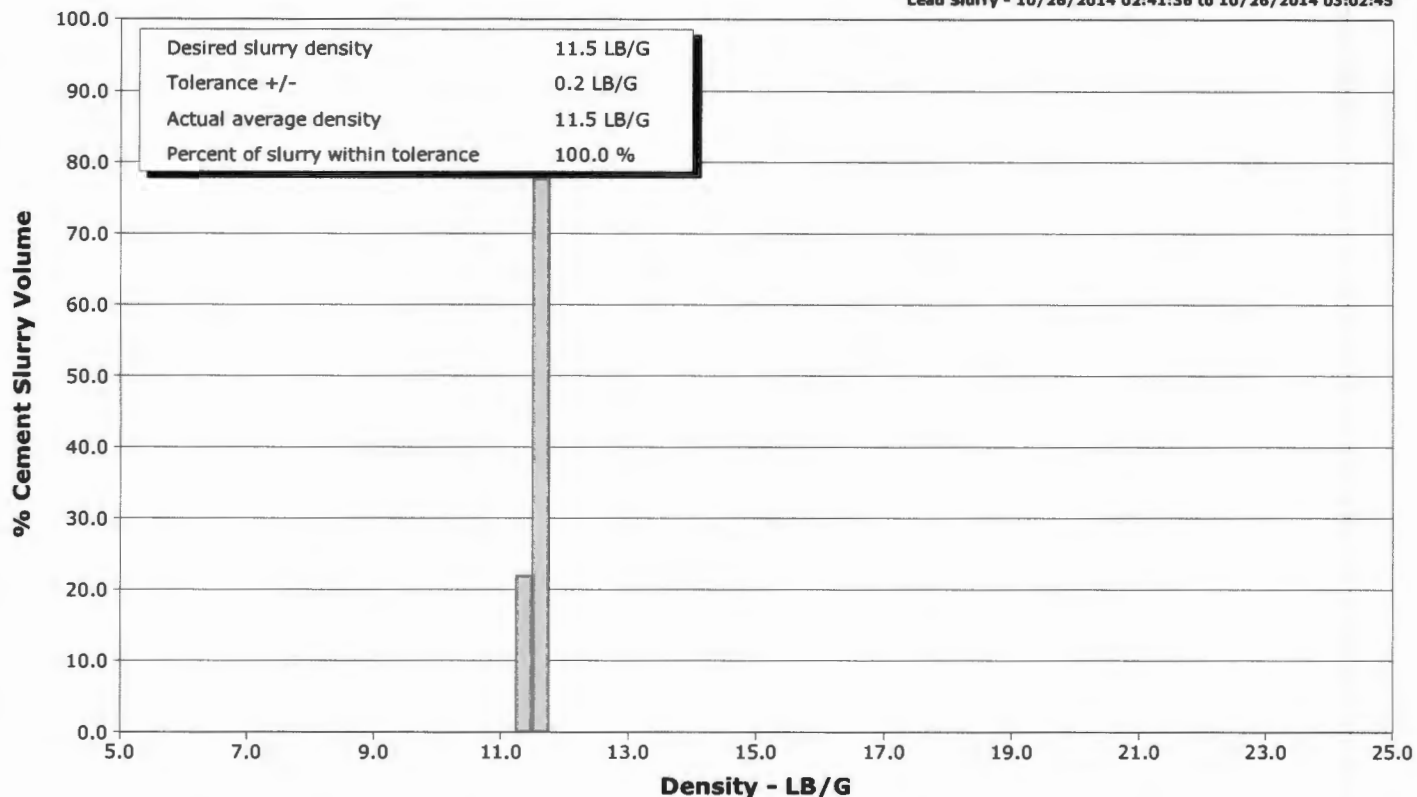
#28658

Well	Kline Federal 5300 41-18 12 TX	Client	Oasis
Field	Baker	SIR No.	2008051
Engineer	Charles Pitcher / Blaine Buxton	Job Type	Intermediate
Country	United States	Job Date	10-26-2014

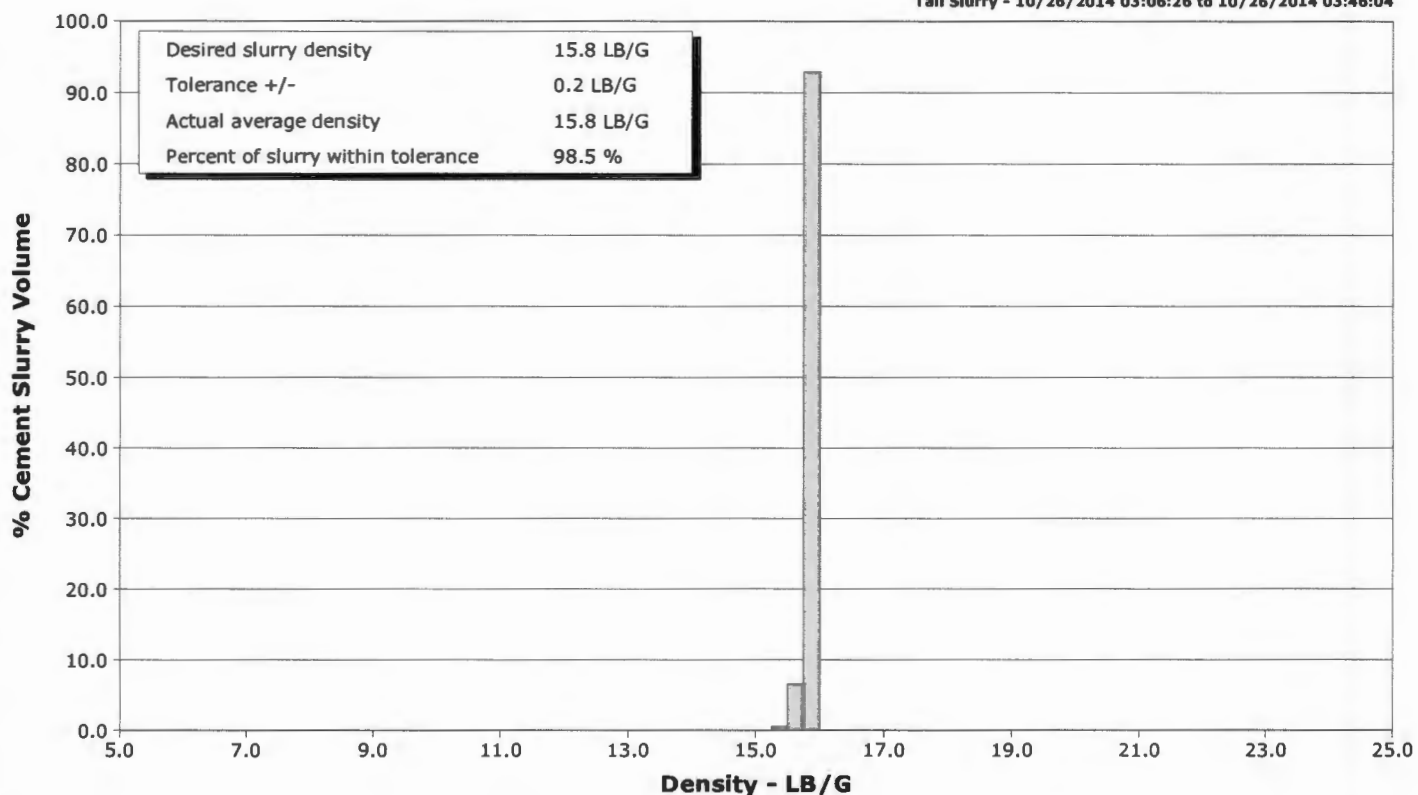


Well	Kline Federal 5300 41-18 12 TX	Client	Oasis
Field	Baker	SIR No.	2008051
Engineer	Charles Pitcher / Blaine Buxton	Job Type	Intermediate
Country	United States	Job Date	10-26-2014

Lead Slurry - 10/26/2014 02:41:36 to 10/26/2014 03:02:45



Tail Slurry - 10/26/2014 03:06:26 to 10/26/2014 03:46:04



Cementing Service Report

				Customer Oasis		Job Number 2008051	
Well Kline Federal 5300 41-18 12 TX Kline Federal 5300 41-18 12 TX			Location (legal) Nabors 684		Schlumberger Location Williston		Job Start Oct/26/2014
Field Baker		Formation Name/Type Shale		Deviation deg	Bit Size 8.8 in	Well MD 11182.0 ft	Well TVD 10832.0 ft
County McKenzie		State/Province North Dakota		BHP psi	BHST 253 degF	BHCT 205 degF	Pore Press. Gradient lb/gal
Well Master 0631576728		API/UWI 3305306030					
Rig Name Nabors 684		Drilled For Oil & Gas		Service Via Land		Casing/Liner	
				Depth, ft	Size, in	Weight, lb/ft	Grade
							Thread
Offshore Zone		Well Class New		Well Type Development			
				11182.0	7.0	32.0	P110
				0.0	0.0	0.0	8RD
Drilling Fluid Type Oil Mud		Max. Density lb/gal		Plastic Viscosity cP		Tubing/Drill Pipe	
Service Line Cementing		Job Type Intermediate					
Max. Allowed Tub. Press psi		Max. Allowed Ann. Press psi		WH Connection Single Cement head		Perforations/Open Hole	
				Top, ft	Bottom, ft	shot/ft	No. of Shots
				ft	ft		Total Interval ft
				ft	ft		Diameter in
				ft	ft		
				Treat Down Casing	Displacement 400.5 bbl	Packer Type	Packer Depth ft
				Tubing Vol. bbl	Casing Vol. 403.5 bbl	Annular Vol. 181.0 bbl	Openhole Vol. 136.0 bbl
Casing/Tubing Secured ☒		1 Hole Vol. Circulated prior to Cement ☒		Casing Tools		Squeeze Job	
Lift Pressure psi		Shoe Type Guide		Squeeze Type			
Pipe Rotated ☐		Pipe Reciprocated ☐		Shoe Depth 11182.0 ft		Tool Type	
No. Centralizers		Top Plugs 1		Bottom Plugs		Tool Depth ft	
Cement Head Type Single		Stage Tool Type		Tool Depth ft		Tail Pipe Size in	
Job Scheduled For Oct/26/2014 07:00		Arrived on Location Oct/25/2014 07:00		Leave Location Oct/26/2014 07:00		Collar Type Float	
						Tail Pipe Depth ft	
						Collar Depth 11095.0 ft	
						Sqz. Total Vol. bbl	
Date	Time 24-hr clock	Treating Pressure PSI	Flow Rate B/M	Density LB/G	Volume BBL	Message	
10/26/2014	01:30:38	4	0.2	8.41	0.0	Stopped Acquisition	
10/26/2014	01:30:41	4	0.2	8.41	0.0	Test Water 85 Deg F	
10/26/2014	01:30:42	4	0.2	8.41	0.0	Rig Up As Per Std 5	
10/26/2014	01:30:43	4	0.2	8.41	0.0	Verify Rate Pressur And Volume	
10/26/2014	01:30:44	4	0.2	8.41	0.0	Flush Lines	
10/26/2014	01:30:50	4	0.2	8.41	0.1	Pressure Test Lines	
10/26/2014	01:31:38	62	0.5	8.40	0.6		
10/26/2014	01:32:38	304	2.1	8.40	2.3		
10/26/2014	01:33:38	317	2.2	8.40	4.5		
10/26/2014	01:34:38	2318	0.0	8.40	4.6		
10/26/2014	01:35:38	5335	0.0	8.40	4.6		
10/26/2014	01:36:38	5258	0.0	8.40	4.6		
10/26/2014	01:37:38	31	0.0	8.40	4.6		
10/26/2014	01:38:38	45	0.0	8.40	4.6		
10/26/2014	01:39:34	55	0.0	8.40	4.6	Reset Total, Vol = 3.65 bbl	
10/26/2014	01:39:38	56	0.0	8.40	4.6		
10/26/2014	01:39:49	58	0.0	8.40	4.6	Start Job	
10/26/2014	01:40:38	0	0.0	8.40	4.6		
10/26/2014	01:41:38	1	0.0	8.39	4.7		
10/26/2014	01:42:38	136	1.9	10.06	5.3		
10/26/2014	01:43:38	332	2.4	10.09	7.4		

Well			Field	Job Start		Customer	Job Number
ie Federal 5300 41-18 12 TX Kline Federal 5300 41-18 12 TX			Baker	Oct/26/2014		Oasis	2008051
Date	Time 24-hr clock	Treating Pressure PSI	Flow Rate B/H	Density lb/g	Volume BBL	Message	
10/26/2014	01:45:38	507	4.5	10.08	13.7		
10/26/2014	01:46:38	521	4.3	10.18	18.1		
10/26/2014	01:47:14	526	4.8	10.30	20.8	End Wash	
10/26/2014	01:47:16	550	4.4	10.31	21.0	Reset Total, Vol = 16.32 bbl	
10/26/2014	01:47:18	529	4.7	10.30	21.1	Start Cement Slurry	
10/26/2014	01:47:25	561	4.6	10.29	21.7	135 Sks @ 10.6 Ppg 3.01 = Yield	
10/26/2014	01:47:38	578	4.7	10.23	22.7		
10/26/2014	01:48:38	773	6.5	10.47	28.6		
10/26/2014	01:49:38	950	4.4	10.50	34.4		
10/26/2014	01:50:38	1051	4.5	10.54	38.8		
10/26/2014	01:51:38	2720	3.1	10.56	42.6		
10/26/2014	01:52:38	3316	3.1	10.60	45.8		
10/26/2014	01:53:38	3169	3.2	10.60	48.9		
10/26/2014	01:54:38	3896	3.2	10.60	52.1		
10/26/2014	01:55:38	4165	2.0	10.59	54.5		
10/26/2014	01:56:38	4551	2.0	10.59	56.5		
10/26/2014	01:57:38	4390	0.9	10.52	57.5		
10/26/2014	01:58:38	4597	1.1	10.56	58.5		
10/26/2014	01:59:38	4381	0.9	10.55	59.5		
10/26/2014	02:00:38	4123	1.0	10.56	60.5		
10/26/2014	02:01:15	3313	1.0	10.55	61.1	Flush Scav Out Of Lines	
10/26/2014	02:01:38	3688	1.0	10.10	61.5		
10/26/2014	02:02:38	3478	1.0	8.51	62.4		
10/26/2014	02:03:38	2699	1.0	8.38	63.2		
10/26/2014	02:04:38	2125	0.0	8.37	64.0		
10/26/2014	02:05:38	-11	0.0	8.37	64.0		
10/26/2014	02:06:38	-8	0.0	8.37	64.0		
10/26/2014	02:07:38	-8	0.0	8.37	64.0		
10/26/2014	02:08:38	97	3.1	10.45	65.3		
10/26/2014	02:09:38	310	2.2	10.51	67.5		
10/26/2014	02:10:38	27	0.0	10.46	68.5		
10/26/2014	02:11:38	5	0.0	10.46	68.5		
10/26/2014	02:12:38	2	0.0	10.46	68.5		
10/26/2014	02:13:38	91	0.0	11.55	68.5		
10/26/2014	02:14:38	74	25.0	0.17	70.0		
10/26/2014	02:15:38	98	3.6	9.24	94.1		
10/26/2014	02:16:38	596	4.0	10.58	98.0		
10/26/2014	02:17:38	1656	4.0	10.60	102.0		
10/26/2014	02:18:38	2027	3.7	10.59	106.0		
10/26/2014	02:19:38	3352	2.7	10.66	109.0		
10/26/2014	02:20:38	3550	0.0	10.63	111.9		
10/26/2014	02:21:38	3401	1.9	10.62	113.6		
10/26/2014	02:22:38	3353	1.6	10.61	115.4		
10/26/2014	02:23:38	3446	1.8	10.62	117.1		
10/26/2014	02:23:51	3317	0.0	10.70	117.3	Shut Down Bleed Off Pressure	
10/26/2014	02:24:38	1746	0.0	10.70	117.3		
10/26/2014	02:25:38	57	0.0	10.70	117.3		
10/26/2014	02:26:38	-7	0.0	10.70	117.3		
10/26/2014	02:27:38	-5	0.0	10.69	117.3		
10/26/2014	02:28:38	-4	0.0	10.70	117.3		
10/26/2014	02:29:38	-3	0.0	10.70	117.3		
10/26/2014	02:30:38	0	0.0	10.70	117.3		
10/26/2014	02:31:38	478	2.3	10.59	118.4		
10/26/2014	02:32:38	442	2.3	10.60	120.7		

Well			Field	Job Start		Customer	Job Number	
ie Federal 5300 41-18 12 TX Kline Federal 5300 41-18 12 TX			Baker	Oct/26/2014		Oasis	2008b51	
Date	Time 24-hr clock	Treating Pressure PSI	Flow Rate B/M	Density LB/G	Volume BBL	Message		
10/26/2014	02:34:38	395	0.0	10.61	127.6			
10/26/2014	02:35:20	198	0.0	11.70	127.6	End Scavenger Slurry		
10/26/2014	02:35:22	197	0.0	11.65	127.6	Reset Total, Vol = 74.63 bbl		
10/26/2014	02:35:38	190	0.0	11.64	127.6			
10/26/2014	02:36:38	180	0.0	11.70	127.6			
10/26/2014	02:37:38	177	0.0	11.74	127.6			
10/26/2014	02:38:38	174	0.0	11.72	127.6			
10/26/2014	02:39:38	164	0.0	11.73	127.6			
10/26/2014	02:40:38	416	2.0	11.60	128.2			
10/26/2014	02:41:36	390	2.3	11.62	130.4	Start Mixing Lead Slurry		
10/26/2014	02:41:37	453	2.3	11.62	130.5	187 Sks @ 11.5 Ppg 2.13 = Yield		
10/26/2014	02:41:38	447	2.3	11.61	130.5			
10/26/2014	02:42:38	399	2.2	11.52	132.7			
10/26/2014	02:43:38	524	3.0	11.49	135.7			
10/26/2014	02:44:38	881	3.2	11.49	138.7			
10/26/2014	02:45:38	812	3.1	11.50	141.7			
10/26/2014	02:46:38	1464	3.2	11.53	144.7			
10/26/2014	02:47:38	1824	3.0	11.55	147.7			
10/26/2014	02:48:38	1715	3.1	11.53	150.7			
10/26/2014	02:49:38	1960	2.8	11.54	153.7			
10/26/2014	02:50:38	2708	3.1	11.58	156.7			
10/26/2014	02:51:38	2090	2.9	11.57	159.6			
10/26/2014	02:52:38	1061	3.1	11.54	162.7			
10/26/2014	02:53:38	772	3.1	11.52	165.7			
10/26/2014	02:54:38	626	3.2	11.50	168.7			
10/26/2014	02:55:38	709	3.1	11.49	171.7			
10/26/2014	02:56:38	684	3.1	11.52	174.8			
10/26/2014	02:57:38	721	3.0	11.55	177.8			
10/26/2014	02:58:38	736	3.0	11.56	180.8			
10/26/2014	02:59:38	679	2.8	11.57	183.9			
10/26/2014	03:00:38	647	2.8	11.57	186.9			
10/26/2014	03:01:38	635	2.9	11.57	189.9			
10/26/2014	03:02:38	732	3.0	11.56	192.9			
10/26/2014	03:02:45	700	3.2	11.62	193.3	End Lead Slurry		
10/26/2014	03:03:38	740	3.1	13.81	196.0			
10/26/2014	03:03:46	892	2.9	14.02	196.4	Reset Total, Vol = 68.78 bbl		
10/26/2014	03:04:38	796	3.0	14.65	199.0			
10/26/2014	03:05:38	765	3.1	15.18	202.0			
10/26/2014	03:06:26	717	3.0	15.42	204.4	Start Mixing Tail Slurry		
10/26/2014	03:06:38	688	3.1	15.47	205.0			
10/26/2014	03:07:38	1181	3.0	15.72	207.6			
10/26/2014	03:08:38	230	1.8	15.70	209.7			
10/26/2014	03:09:38	750	4.4	15.72	213.1			
10/26/2014	03:10:38	885	4.3	15.76	216.9			
10/26/2014	03:11:38	849	4.2	15.81	221.1			
10/26/2014	03:12:38	776	3.9	15.84	224.9			
10/26/2014	03:13:38	733	3.8	15.83	228.9			
10/26/2014	03:13:40	681	4.0	15.82	229.0	631 Sks @ 15.8 Ppg 1.54 = Yield		
10/26/2014	03:14:38	664	4.0	15.83	232.9			
10/26/2014	03:15:38	799	4.0	15.83	236.8			
10/26/2014	03:16:38	814	4.1	15.80	240.8			
10/26/2014	03:17:38	670	4.0	15.82	244.7			
10/26/2014	03:18:38	705	3.9	15.83	248.7			
10/26/2014	03:19:38	697	4.0	15.83	252.7			

Well			Field	Job Start	Customer	Job Number
ie Federal 5300 41-18 12 TX Kline Federal 5300 41-18 12 TX			Baker	Oct/26/2014	Oasis	2008051
Date	Time 24-hr clock	Treating Pressure PSI	Flow Rate B/M	Density LB/G	Volume BBL	Message
10/26/2014	03:21:38	615	4.0	15.82	260.6	
10/26/2014	03:22:38	688	4.0	15.82	264.6	
10/26/2014	03:23:38	649	3.9	15.83	268.6	
10/26/2014	03:24:38	661	3.9	15.86	272.5	
10/26/2014	03:25:38	529	4.0	15.87	276.5	
10/26/2014	03:26:38	412	4.0	15.86	280.5	
10/26/2014	03:27:38	398	4.1	15.85	284.5	
10/26/2014	03:28:38	449	4.1	15.84	288.5	
10/26/2014	03:29:38	359	3.9	15.84	292.4	
10/26/2014	03:30:38	340	4.0	15.85	296.4	
10/26/2014	03:31:38	304	4.0	15.87	300.4	
10/26/2014	03:32:38	277	3.9	15.88	304.4	
10/26/2014	03:33:38	369	4.4	15.80	308.7	
10/26/2014	03:34:38	341	4.5	15.80	313.1	
10/26/2014	03:35:38	493	5.0	15.88	317.5	
10/26/2014	03:36:38	479	5.3	15.88	322.8	
10/26/2014	03:37:38	504	5.4	15.91	328.1	
10/26/2014	03:38:38	752	5.1	15.92	333.3	
10/26/2014	03:39:38	935	6.2	15.95	338.0	
10/26/2014	03:40:38	1098	6.2	15.89	344.3	
10/26/2014	03:41:38	1079	6.2	15.93	350.5	
10/26/2014	03:42:38	917	6.2	15.92	356.8	
10/26/2014	03:43:38	888	6.2	15.83	363.1	
10/26/2014	03:44:38	378	4.2	15.85	369.2	
10/26/2014	03:45:38	55	0.8	15.85	371.3	
10/26/2014	03:46:04	55	0.8	15.86	371.7	End Tail Slurry
10/26/2014	03:46:09	55	0.8	15.86	371.8	End Cement Slurry
10/26/2014	03:46:38	1	0.0	15.86	372.0	
10/26/2014	03:47:32	11	0.0	15.89	372.0	Reset Total, Vol = 175.60 bbl
10/26/2014	03:47:36	20	1.0	15.88	372.0	Drop Top Plug
10/26/2014	03:47:38	35	0.7	15.88	372.0	
10/26/2014	03:47:39	48	0.6	15.88	372.1	Start Displacement
10/26/2014	03:48:38	49	1.0	9.99	373.0	
10/26/2014	03:49:38	22	1.0	9.00	374.1	
10/26/2014	03:50:38	18	1.6	8.74	375.1	
10/26/2014	03:51:38	121	4.5	8.68	378.6	
10/26/2014	03:52:38	129	4.4	8.67	383.1	
10/26/2014	03:53:38	64	3.3	8.62	387.2	
10/26/2014	03:54:38	42	2.4	8.55	390.2	
10/26/2014	03:55:38	47	2.2	9.03	392.5	
10/26/2014	03:56:38	116	4.5	9.15	396.5	
10/26/2014	03:57:38	108	4.6	9.17	401.0	
10/26/2014	03:58:38	117	4.5	9.52	405.5	
10/26/2014	03:59:38	121	4.5	9.56	410.1	
10/26/2014	04:00:38	115	4.5	9.59	414.6	
10/26/2014	04:01:38	127	4.6	9.70	419.1	
10/26/2014	04:02:38	53	2.2	9.66	421.7	
10/26/2014	04:03:38	33	2.2	9.71	424.1	
10/26/2014	04:04:38	48	2.4	9.67	426.4	
10/26/2014	04:05:38	48	2.4	9.67	428.7	
10/26/2014	04:06:38	47	2.4	9.67	431.0	
10/26/2014	04:07:38	48	2.4	9.71	433.3	
10/26/2014	04:08:38	48	2.2	9.70	435.6	
10/26/2014	04:09:38	238	6.4	9.69	440.5	

Well			Field	Job Start		Customer	Job Number
ie Federal 5300 41-18 12 TX Kline Federal 5300 41-18 12 TX			Baker	Oct/26/2014		Oasis	2008051
Date	Time 24-hr clock	Flowing Pressure PSI	Flow Rate B/M	Density LB/G	Volume BBL	Message	
10/26/2014	04:11:38	47	2.4	9.40	450.6		
10/26/2014	04:12:38	44	2.2	9.17	452.9		
10/26/2014	04:13:38	49	2.4	9.69	455.2		
10/26/2014	04:14:38	49	2.4	9.70	457.6		
10/26/2014	04:15:38	49	2.2	9.71	459.9		
10/26/2014	04:16:38	49	1.1	9.69	462.0		
10/26/2014	04:17:38	43	1.1	9.65	463.1		
10/26/2014	04:18:38	49	0.6	9.68	464.0		
10/26/2014	04:19:38	50	0.6	9.68	464.6		
10/26/2014	04:20:38	49	0.6	9.71	465.2		
10/26/2014	04:21:38	50	0.6	9.73	465.8		
10/26/2014	04:22:38	31	1.1	9.73	466.4		
10/26/2014	04:23:38	266	3.3	9.72	469.6		
10/26/2014	04:24:38	49	2.6	9.71	471.8		
10/26/2014	04:25:38	49	2.7	9.71	474.4		
10/26/2014	04:26:38	50	2.8	9.72	477.1		
10/26/2014	04:27:38	50	2.7	9.73	479.8		
10/26/2014	04:28:38	133	3.4	9.73	482.6		
10/26/2014	04:29:38	301	4.5	9.72	487.0		
10/26/2014	04:30:38	259	4.3	9.72	491.5		
10/26/2014	04:31:38	538	5.7	9.73	496.3		
10/26/2014	04:32:38	687	6.0	9.72	502.2		
10/26/2014	04:33:38	665	5.7	9.73	508.2		
10/26/2014	04:34:38	496	4.4	9.72	513.6		
10/26/2014	04:35:38	450	4.6	9.72	518.1		
10/26/2014	04:36:38	357	4.6	9.72	522.6		
10/26/2014	04:37:38	402	4.6	9.72	527.1		
10/26/2014	04:38:38	431	4.7	9.72	531.7		
10/26/2014	04:39:37	434	4.6	9.72	536.1	Problems Getting Salt Water	
10/26/2014	04:39:38	429	4.5	9.72	536.2		
10/26/2014	04:40:38	444	4.5	9.72	540.7		
10/26/2014	04:41:38	439	4.6	9.72	545.2		
10/26/2014	04:42:38	436	4.4	9.72	549.7		
10/26/2014	04:43:38	497	4.5	9.72	554.3		
10/26/2014	04:44:38	592	4.4	9.72	558.8		
10/26/2014	04:45:38	636	4.6	9.72	563.3		
10/26/2014	04:46:38	668	4.7	9.72	567.8		
10/26/2014	04:47:38	729	4.7	9.72	572.3		
10/26/2014	04:48:38	659	4.5	9.72	576.8		
10/26/2014	04:49:38	846	4.4	9.72	581.3		
10/26/2014	04:50:38	940	4.7	9.72	585.8		
10/26/2014	04:51:38	910	4.4	9.72	590.3		
10/26/2014	04:52:38	898	4.4	9.72	594.9		
10/26/2014	04:53:38	1024	4.5	9.72	599.3		
10/26/2014	04:54:38	1348	4.4	9.72	603.8		
10/26/2014	04:55:38	3202	3.1	9.72	607.8		
10/26/2014	04:56:38	4443	2.0	9.72	610.1		
10/26/2014	04:57:38	4237	2.0	9.72	612.1		
10/26/2014	04:58:38	4532	0.0	9.72	614.0		
10/26/2014	04:59:38	4384	0.8	9.71	614.6		
10/26/2014	05:00:38	4692	0.0	9.71	615.3		
10/26/2014	05:01:38	4786	0.9	9.71	615.8		
10/26/2014	05:02:38	4752	0.0	9.72	616.2		
10/26/2014	05:02:50	4427	0.0	9.72	616.2	Bleed Off Pressure	

Well			Field	Job Start	Customer	Job Number
Federal 5300 41-18 12 TX Kline Federal 5300 41-18 12 TX			Baker	Oct/26/2014	Oasis	2008051
Date	Time 24-hr clock	Flowing Pressure PSI	Flow Rate B/M	Density LB/G	Volume BBL	Message
10/26/2014	05:04:38	91	0.0	9.71	616.2	
10/26/2014	05:05:38	119	0.0	9.71	616.2	
10/26/2014	05:06:38	71	0.0	9.71	616.2	
10/26/2014	05:07:38	15	0.0	9.71	616.2	
10/26/2014	05:08:38	316	0.5	9.71	616.3	
10/26/2014	05:09:38	1286	1.0	9.71	617.1	
10/26/2014	05:10:38	2436	1.0	9.71	618.1	
10/26/2014	05:11:38	3544	1.0	9.71	619.1	
10/26/2014	05:12:38	4314	0.7	9.71	620.1	
10/26/2014	05:13:38	4651	0.6	9.71	620.7	
10/26/2014	05:14:38	4975	0.6	9.72	621.3	
10/26/2014	05:15:38	4853	0.0	9.72	621.3	
10/26/2014	05:16:08	4772	0.0	9.72	621.3	Bleed Off Pressure
10/26/2014	05:16:38	4694	0.0	9.72	621.3	
10/26/2014	05:17:38	1339	0.0	9.72	621.3	
10/26/2014	05:18:38	0	0.0	9.71	621.3	
10/26/2014	05:19:38	13	0.0	9.71	621.3	
10/26/2014	05:20:38	95	0.0	9.72	621.3	
10/26/2014	05:21:38	225	0.0	9.71	621.3	
10/26/2014	05:22:38	146	0.0	9.71	621.3	
10/26/2014	05:23:38	264	0.4	9.71	621.4	
10/26/2014	05:24:38	1033	1.0	9.71	622.1	
10/26/2014	05:25:38	2185	1.0	9.71	623.1	
10/26/2014	05:26:38	3273	1.0	9.71	624.1	
10/26/2014	05:27:38	4098	1.0	9.71	625.2	
10/26/2014	05:28:38	4147	0.0	9.71	625.3	
10/26/2014	05:29:38	2749	0.0	9.71	625.3	
10/26/2014	05:30:38	-5	0.0	9.71	625.3	
10/26/2014	05:31:38	-4	0.0	9.71	625.3	
10/26/2014	05:32:38	-1	0.0	9.71	625.3	
10/26/2014	05:33:38	-1	0.0	9.71	625.3	
10/26/2014	05:34:38	2	0.0	9.71	625.3	
10/26/2014	05:35:38	0	0.0	9.71	625.3	
10/26/2014	05:36:38	283	1.0	9.71	625.6	
10/26/2014	05:37:38	1388	1.0	9.71	626.7	
10/26/2014	05:38:38	2442	1.0	9.71	627.7	
10/26/2014	05:39:38	3486	1.0	9.70	628.7	
10/26/2014	05:39:56	3783	1.0	9.70	629.0	4155 Feet Of Cement Left In Pipe
10/26/2014	05:40:00	3808	1.0	9.70	629.1	TOT = 10257 Feet
10/26/2014	05:40:38	4453	1.0	9.70	629.7	
10/26/2014	05:41:38	4960	0.0	9.71	630.5	
10/26/2014	05:42:38	4877	0.0	9.70	630.5	
10/26/2014	05:43:38	4807	0.0	9.71	630.5	
10/26/2014	05:44:38	4746	0.0	9.71	630.5	
10/26/2014	05:45:38	4692	0.0	9.71	630.5	
10/26/2014	05:46:38	4643	0.0	9.71	630.5	
10/26/2014	05:47:38	4601	0.0	9.71	630.5	
10/26/2014	05:48:38	4561	0.0	9.70	630.5	
10/26/2014	05:49:38	4524	0.0	9.71	630.5	
10/26/2014	05:50:38	4491	0.0	9.71	630.5	
10/26/2014	05:51:38	4460	0.0	9.71	630.5	
10/26/2014	05:52:38	4472	0.0	9.71	630.5	
10/26/2014	05:53:38	4442	0.0	9.71	630.5	
10/26/2014	05:54:38	4415	0.0	9.71	630.5	

Well ie Federal 5300 41-18 12 TX Kline Federal 5300 41-18 12 TX			Field Baker		Job Start Oct/26/2014		Customer Oasis		Job Number 2008051	
Date	Time 24-hr clock	Treating Pressure PSI	Flow Rate B/M	Density LB/G	Volume BBL	Message				
10/26/2014	05:56:02	2698	0.0	9.71	630.5	Floats Held 4 Bbl Back				
10/26/2014	05:56:22	292	0.0	9.71	630.5	Reset Total, Vol = 255.51 bbl				
10/26/2014	05:56:30	-14	0.0	9.71	630.5	End Job				

Post Job Summary

Average Pump Rates, bbl/min					Volume of Fluid Injected, bbl			
Slurry 3.2	N2	Mud	Maximum Rate 6.0	Total Slurry 244.0	Mud 0.0	Spacer 19.4	N2	
Treating Pressure Summary, psi					Breakdown Fluid			
Maximum 5338	Final 0	Average 1318	Bump Plug to	Breakdown	Type	Volume bbl	Density lb/gal	
Avg. N2 Percent %		Designed Slurry Volume 244.0 bbl	Displacement 255.5 bbl	Mix Water Temp 85 degF	Cement Circulated to Surface? ☐	Volume bbl		
					Washed Thru Perfs ☐	To ft		
Customer or Authorized Representative Mark Lawlar			Schlumberger Supervisor Charles Pitcher / Blaine Buxton			Circulation Lost ☐	Job Completed ☐	
						-	-	

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2008)



Well File No.
28658

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date June 2, 2015
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-61.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Change well status to CONFIDENTIAL

Well Name and Number Kline Federal 5300 41-18 12TX					
Footages	Qtr-Qtr	Section	Township	Range	
434 F S L 237 F W L	LOT4	18	163 N	100 W	
Field Baker	Pool BAKKEN	County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Effective Immediately, we request **CONFIDENTIAL STATUS** for the above referenced well.

This well has not been completed

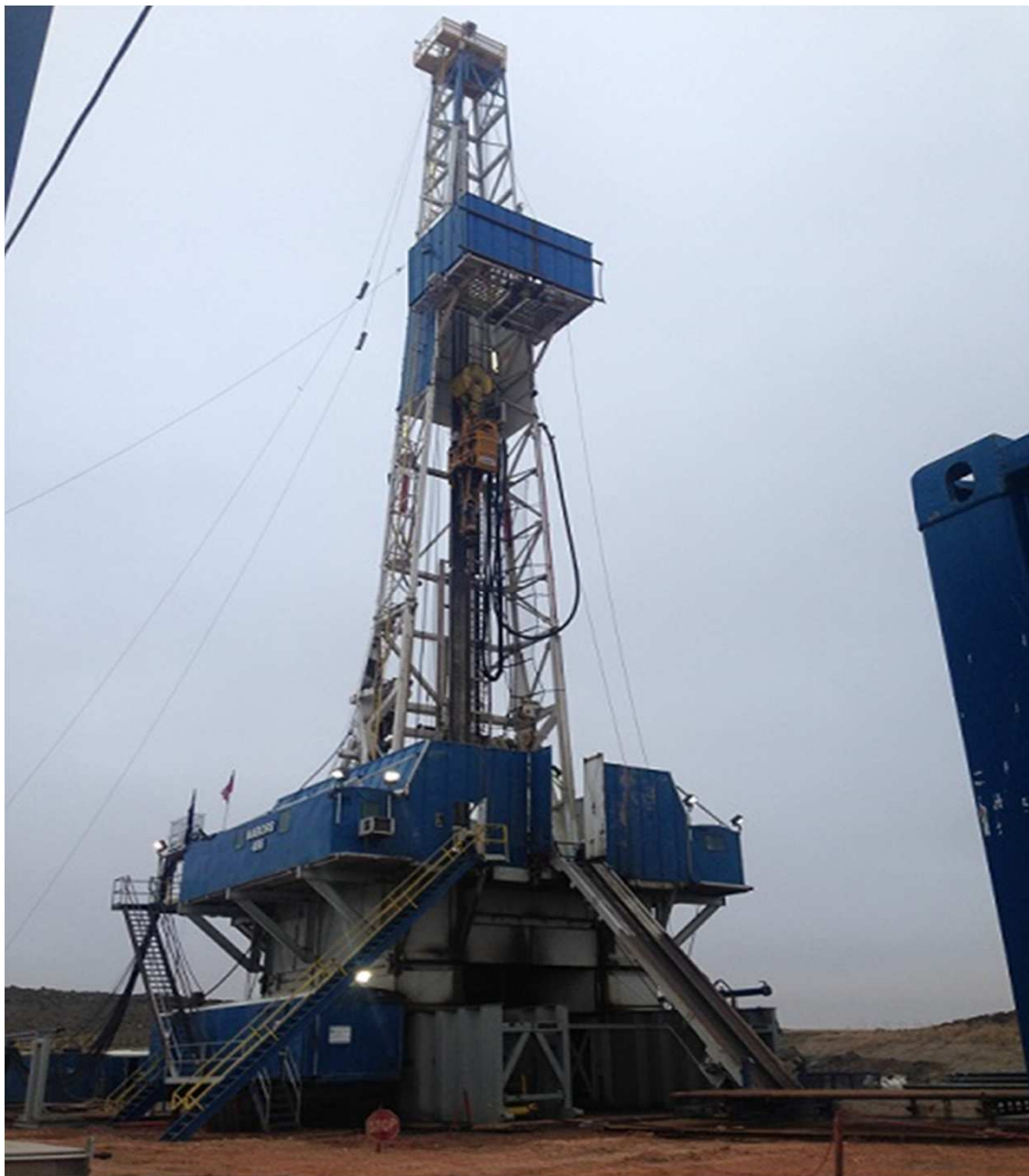
OFF CONFIDENTIAL 12/02/15.

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9436	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature 		Printed Name Jennifer Swenson	
Title Regulatory Specialist		Date June 2, 2015	
Email Address jswenson@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date	6/03/15
By	
Title	Engineering Technician

Oasis Petroleum North America

Kline Federal 5300 41-18 12TX



Services Performed For:

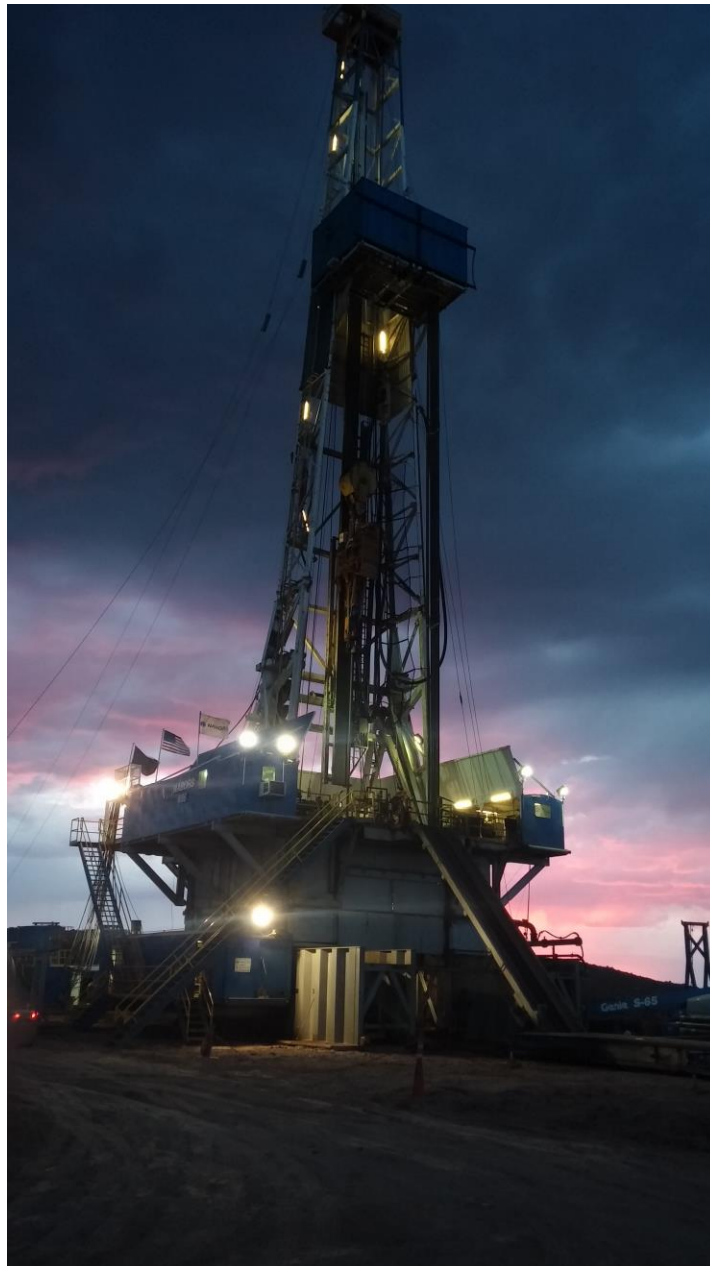
Andrew Nelson, Brendan Hargrove
Oasis Petroleum North America
1001 Fannin Suite 202
Houston, TX 77002

Onsite Geology Performed by:

Annika Tostengard, Bob Witmer
RPM Geologic
geology@rpmconsultinginc.com
(303) 595-7625

WELL EVALUATION

Oasis Petroleum
Kline Federal 5300 41-18 12TX
SW SW Sec. 18, T153N, R100W
Williams County, North Dakota



Synopsis

Oasis Petroleum *Kline Federal 5300 41-18 12TX* [SW SW Sec. 18 T153N R100W] is located ~6 miles Northwest of the town of Williston, North Dakota. *Kline Federal 5300 41-18 12TX* is situated within the Baker Field. A single lateral leg trending 90° was proposed to be drilled from the SW SW corner of Section 18 to the SE SE corner of Section 17 targeting the porous dolostone in the first bench of the **Three Forks**.

Control Wells

Two completed wells were used as the control offsets to the *Kline Federal 5300 41-18 12TX*.

The primary offset considered was the *Kline Federal 5300 41-18 13T2X* [SW SW Sec. 18 T153N, R100W] which is a horizontal well drilled by Oasis Petroleum. ***Kline Federal 5300 41-18 13T2X*** is located approximately 33 feet south of *Kline Federal 5300 41-18 12TX*.

The other offset considered was the *Kline Federal 5300 41-18 14BX* [SW SW Sec. 18 T153N, R100W] which is a horizontal well drilled by Oasis Petroleum. *Kline Federal 5300 41-18 14BX* is located approximately 66 feet south of *Kline Federal 5300 41-18 12TX*.

During the curve, the gamma ray that was produced was constantly compared to the *Kline Federal 5300 41-18 13T2X* and the *Kline Federal 5300 41-18 14BX* wells. Gamma ray was used to help determine a proper landing depth.

To assist in the choosing of the landing target, an isopach table (Table 1) was constructed to measure the distance of formation tops to the target depth determined from the *Kline Federal 5300 41-18 13T2X* and *Kline Federal 5300 41-18 14BX* offset wells. Comparison of gamma ray markers showed the most consistent isopach for up-hole formations to be the **Middle Bakken**. The same information was coalesced into a single chart (Table 2) displaying offset tops and an average of the estimated landing point based on the offset wells. This chart was used to corroborate the landing point, but isopach thickness was the main consideration in choosing a landing point.

<u>Control Well</u>				
Operator:	Oasis Petroleum			
Well Name:	<i>Kline Federal 5300 41-18 13T2X</i>			
Location:	SW SW Sec. 18 T153N, R100W McKenzie Co., ND Approximately 33 ft South			
Elevation:	KB: 2,082'			
Formation/ Zone	Est. TVD	MSL Datum	Thickness	Dist to target
Kibbey Lime	8,382'	-8,382'	178'	-8,382'
Charles Salt	8,560'	-8,560'	646'	-8,560'
Base Last Salt	9,206'	-9,206'	222'	-9,206'
Mission Canyon	9,428'	-9,428'	547'	-9,428'
Lodgepole	9,975'	-9,975'	730'	-9,975'
False Bakken	10,705'	-10,705'	7'	-10,705'
Upper Bakken Shale	10,712'	-10,712'	18'	-10,712'
Middle Bakken	10,730'	-10,730'	40'	-10,730'
Lower Bakken	10,770'	-10,770'	12'	-10,770'
Pronghorn	10,782'	-10,782'	17'	-10,782'
Three Forks	10,799'	-10,799'	10,799'	-10,799'

<u>Control Well</u>				
Operator:	Oasis Petroleum			
Well Name:	<i>Kline Federal 5300 41-18 14BX</i>			
Location:	SW SW Sec. 18 T153N, R100W McKenzie Co., ND Approximately 213 feet South			
Elevation:	KB: 2,082'			
Formation/ Zone	Est. TVD	MSL Datum	Thickness	Dist to target
Kibbey Lime	8,382	-6,300'	182'	2,435'
Charles Salt	8,564	-6,482'	641'	2,253'
Base Last Salt	9,205	-7,123'	223'	1,612'
Mission Canyon	9,428	-7,346'	548'	1,389'
Lodgepole	9,976	-7,894'	728'	841'
False Bakken	10,704	-8,622'	8'	113'
Upper Bakken Shale	10,712	-8,630'	16'	105'
Middle Bakken	10,728	-8,646'	27'	89'

Table 1: Distance from Formation tops in the control wells were utilized to determine an initial drilling target.

Kline Federal 12TX Target Landing Chart										
Operator: Well Name: Location:	Oasis Petroleum Kline Federal 5300 41-18 12TX SW SW Sec. 18 T153N, R100W McKenzie County, ND			Oasis Petroleum Kline Federal 5300 41-18 13TX SW SW Sec. 18 T153N, R100W McKenzie County, ND			Oasis Petroleum Kline Federal 5300 41-18 14BX SW SW Sec. 18 T153N, R100W McKenzie County, ND			
Elevation:	KB: 2,082'			KB: 2,082'			KB: 2,082'			
Formation / Zone	Offset Tops TVD (gamma)	Offset Tops MSL	Thickness	Offset Tops TVD (gamma)	Offset Tops MSL	Thickness	Primary Tops TVD (gamma)	Primary Tops MSL	Thickness	Proj. Target Landing
Kibbey	8,384'	-6,302'	172'	8,382'	-6,300'	178'	8,382'	-6,300'	182'	10,819'
Charles	8,556'	-6,474'	650'	8,560'	-6,478'	646'	8,564'	-6,482'	641'	10,826'
Base Last Salt	9,206'	-7,124'	222'	9,206'	-7,124'	222'	9,205'	-7,123'	223'	10,819'
Mission Canyon	9,428'	-7,346'	547'	9,428'	-7,346'	547'	9,428'	-7,346'	548'	10,820'
Lodgepole	9,975'	-7,893'	115'	9,975'	-7,893'	112'	9,976'	-7,894'	114'	10,821'
LPA	10,090'	-8,008'	75'	10,087'	-8,005'	70'	10,090'	-8,008'	74'	10,822'
LPB	10,165'	-8,083'	46'	10,157'	-8,075'	54'	10,164'	-8,082'	45'	10,823'
LPC	10,211'	-8,129'	188'	10,211'	-8,129'	186'	10,209'	-8,127'	194'	10,818'
LPD	10,399'	-8,317'	161'	10,397'	-8,315'	162'	10,403'	-8,321'	154'	10,825'
LPE	10,560'	-8,478'	65'	10,559'	-8,477'	67'	10,557'	-8,475'	68'	10,818'
LPF	10,625'	-8,543'	79'	10,626'	-8,544'	79'	10,625'	-8,543'	77'	10,820'
False Bakken	10,704'	-8,622'	9'	10,705'	-8,623'	7'	10,702'	-8,620'	10'	10,818'
Upper Bakken Shale	10,713'	-8,631'	17'	10,712'	-8,630'	18'	10,712'	-8,630'	16'	10,820'
Middle Bakken	10,730'	-8,648'	41'	10,730'	-8,648'	-	10,728'	-8,646'	-	10,818'
Lower Bakken	10,771'	-8,689'	12'	10,770'	-	-	-	-	-	-
Pronghorn	10,783'	-8,701'	16'	10,771'	-	-	-	-	-	-
Three Forks	10,799'	-8,717'	21'	10,799'	-	-	-	-	-	-
Target	10,820'	-8,738'	-10,820'	10,820'	-8,738'	-	10,817'	-8,735'	-	10,817'
									Average -->	10,820'

Table 2: Estimated landing point based on offset well formation tops.

Geologic Assessment

Methods

Geologic supervision of Kline Federal 5300 41-18 12TX was provided by experienced RPM well site geologists. Gas and chromatograph levels were measured using NOV WellSight Gas Detection real time gas detector and chromatograph system. The WellSight gas detection system uses non-dispersive infrared and chemical sensor gas detection to evaluate gases liberated from the formation by the drill bit and carried to the surface by the drilling fluid.

The gas detector was interfaced with a RigWatch electronic data recorder system. RigWatch provided drill rate, on-off bottom and pump strokes to the gas detector and received total gas information from the gas detector for viewing on location and remotely.

Under the direction of RPM well site geologists, rig crews were instructed to catch lagged drill cutting samples at 30' intervals in the vertical hole and curve build from 8,350' to 11,190' MD and 50' samples from 11,190' to TD (20,583').

Sampled drill cuttings were examined wet and dry under a binocular microscope using both plain (broad spectrum) and transmitted light. Cuttings were evaluated for hydrocarbon "cut" by immersion in Acetone and inspection under a UV fluoroscope. 10% hydrochloric acid and alizarin red were used to determine the calcareous and dolomitic content of rocks and cementing.

RPM Geologic, Inc. (RPM) well site geologists also closely examined MWD gamma-ray information and penetration rates to aid in steering decisions and dip degree estimations.

Vertical Operations

Overview

The *Kline Federal 5300 41-18 12TX* was spud for surface drilling on 13 September 2014 utilizing Nabors 486 as the drilling contractor. Prior to commencement of RPM mud logging services, a 17 1/2" hole was drilled with fresh water to depth of 2,160' and isolated with 13 3/8" 36# J-55 casing cemented from 2,157' to surface. Due to uphole stability issues a second set of casing was set. Two 12 1/4" bits were used to drill to a depth of 6,138' and 9 5/8 40# HCL 80 casing was set @ 6,118'

RPM well site geologists began logging the vertical interval at 8,350'.

One PDC bit was used to drill out of secondary casing to vertical TD. PDC bit #4 (8 3/4" Security MM65D) was used to drill from 6,135' to 10,300' and averaged an ROP of 62.16 ft/hr in 67.0 hours of use.

Diesel invert drilling fluid with a mud weight ranging from 10.00-11.60 ppg was used for the remainder of the vertical hole and in the curve builds sections. The vertical interval was drilled to a kick off point (KOP) of 10,300', at which point vertical operations were ceased.

Lithology

The top of the **Kibbey Lime** [Mississippian Madison Group] was picked at 8,384' (-6,302'), 2' low to the *Kline Federal 5300 41-18 13T2X*. Samples from this interval (Figure 1) were described as:

LIMESTONE: wackestone, medium gray, friable to hard, finely crystalline, blocky, earthy to sucrosic, trace visible porosity



*Figure 1: Photograph of Limestone seen in sample from the **Kibby Limestone** formation.*

The top of the **Charles Salt** [Mississippian Madison Group] was logged at 8,556' (-6,474'), 6' low to the *Kline Federal 5300 41-18 13T2X*. The **Base Last Salt** was drilled at 9,206' (-7,124'), flat to the *Kline Federal 5300 41-18 13T2X*. Samples from this interval (Figure 2) were described as:

SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain

LIMESTONE: mudstone, light gray to medium gray, off white to cream, microcrystalline, soft to fair, chalky to crystalline texture, trace intercrystalline porosity, slightly light brown oil stain

ANHYDRITE: off white, cream, soft to firm, amorphous texture, no visible porosity



*Figure 2: Photograph of Salt seen in sample from the **Charles** formation.*

The top of the **Mission Canyon** formation of the Madison Group [Mississippian] was reached at 9,428' (-7,346'), flat to the *Kline Federal 5300 41-18 13T2X*. Samples from this interval (Figure 3) were described as:

LIMESTONE: mudstone, light gray to medium gray, occasional off white to cream, firm to hard, microcrystalline, chalky to earthy texture, trace sparry calcite, possible intercrystalline porosity

DOLOMITE LIMESTONE: mudstone, dark gray to medium gray, off white in part, firm to hard, microcrystalline, occasional sparry calcite, trace intercrystalline porosity, slightly light brown spotty oil stain



Figure 3: Photograph of limestone and argillaceous limestone from the Mission Canyon formation.

The top of the **Lodgepole** formation of the Madison Group (Mississippian) was logged at 9,975' (-7,893'), flat to the *Kline Federal 5300 41-18 13T2X*. Samples from this interval (Figure 4) were described as:

ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcite, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain



Figure 4: Photograph of argillaceous limestone from Lodgepole formation.

Directional Operations

Gyrodata provided equipment and personnel for MWD teams, while RPM Consulting provided directional personnel. RPM geologists worked closely with RPM directional as well as Gyrodata MWD teams to formulate steering decisions to maximize the footage of borehole in the pay zone.

Curve Build

Overview

One PDC bit was used to build the curve section. Bit #5 was an 8 ¾” Security MMD55L bit used to drill the curve from a measured depth of 10,300’ to 11,190’ in 32.0 hours, averaging 27.65 ft/hr. 7” P-110 32 lbs casing was set to 11,189’ MD upon completion of the curve.

A Chart was constructed for the curve (Chart 1) that compared the isopach of certain easily recognizable gamma markers from the Kibbey Lime to the Three Forks on the control well *Kline Federal 5300 41-18 13T2X* to the *Kline Federal 5300 41-18 12TX* as marker picks were made. This chart shows the Middle Bakken top being the closest in thickness to target based on the offset well, and was the main point used to make changes to the target landing depth during the curve. The curve was completed at a measured depth of 11,190’ and TVD of 10,812’, ~12 feet below the base of the Three Forks.

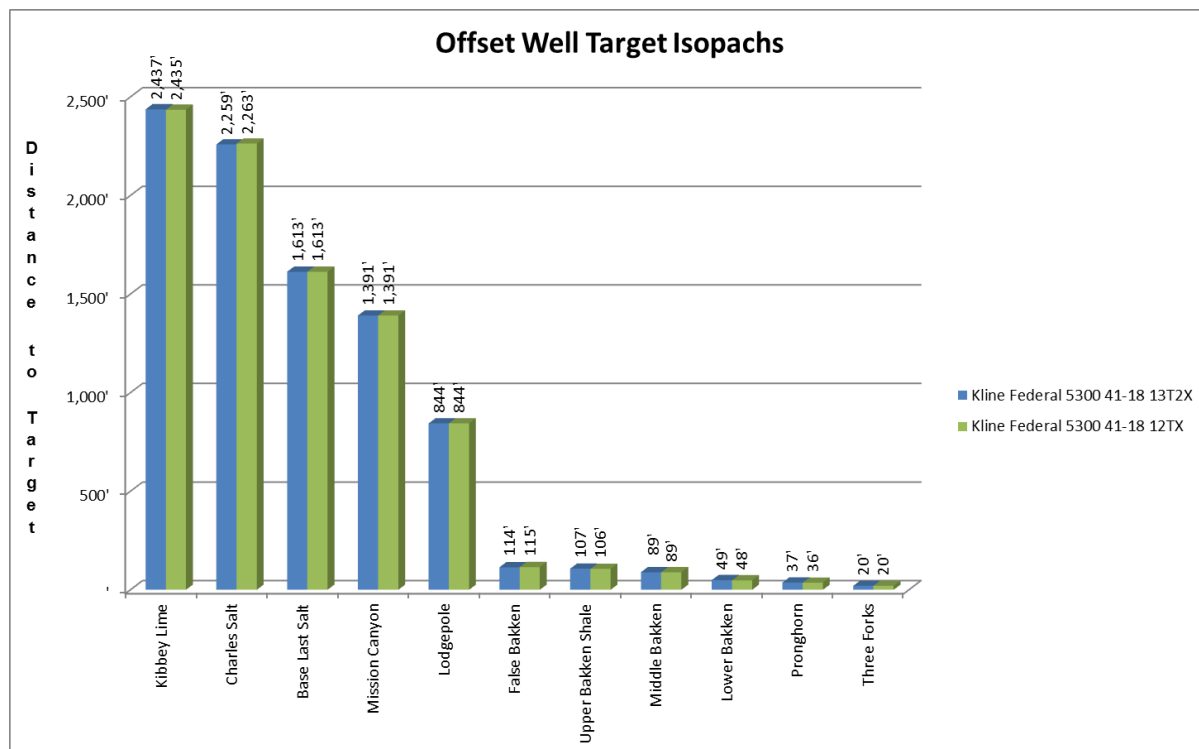


Chart 1: Comparing distances from gamma markers on the offset wells to markers in gamma in the curve.

Lithology

The top of the “**False Bakken**” was drilled at a measured depth of 10,781’ (TVD 10,704’) (-8,622’) 1’ high to the *Kline Federal 5300 41-18 13T2X*.

The *Upper Shale* of the **Bakken** formation [Mississippian – Devonian] was drilled at 10,798’ MD (10,713’ TVD) (-8,631’) with sample returns of black, carbonaceous, and petroliferous shale (Figure 5) characterized by gamma ray values in excess of 255 API counts. The *Upper Shale* was 2’ low to the *Kline Federal 5300 41-18 13T2X*. Samples were described as:

SHALE: black, firm, earthy to waxy texture, pyrite, carbonaceous, platy, visible fracture porosity



*Figure 5: Photograph sample from the **Upper Bakken Shale**.*

The *Middle Member* of the **Bakken** formation was penetrated at 10,829’ MD, (10,730’ TVD) (-8,648’), flat to the *Kline Federal 5300 41-18 13T2X*. The *Middle Member* contained layers of sandstone and silty sandstones with varying amounts of porosity and oil stain (Figure 6). Samples from the *Middle Member* were described as:

SLTY SANDSTONE: common salt and pepper, occasional light gray to medium gray, visible white, very fine grained, soft to firm, moderately to well sorted, rounded to sub rounded, moderately to poorly cemented, visible calcareous filled fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty to even oil stain



*Figure 6: Photograph of sandstone and silty sandstone from the middle member of the **Bakken** formation.*

The *Lower Member* of the **Bakken** formation was penetrated at 10,918' MD, (10,771' TVD) (-8,698'), 1' low to the *Kline Federal 5300 41-18 13T2X*. The *Lower Member* consists of a dark, petroliferous shale (Figure 7). Samples from the *Lower Member* were described as:

SHALE: black, dark gray black, firm, sub-blocky, earthy, calcareous cement, petro, carbonaceous, abundant fine disseminated pyrite, fine laminated



*Figure 7: Photograph of sandstone and silty sandstone from the lower member of the **Bakken** formation.
8 3/4" Bit & Diesel Invert Drilling Fluid*

The **Pronghorn** formation was drilled at 10,952' MD, (10,783' TVD) (-8,701'), 1' low to the *Kline Federal 5300 41-18 13T2X*.

The **Three Forks** formation was drilled at 11,011' MD, (10,799' TVD) (-8,717'), flat to the *Kline Federal 5300 41-18 13T2X*. The **Three Forks** consists of a light colored, pyritic, slightly oil stained dolostone (Figure 8). Samples from the **Three Forks** were described as:

DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain



*Figure 8: Photograph of dolostone from the **Three Forks** formation.
8 3/4" Bit & Diesel Invert Drilling Fluid*

Lateral

Overview

Drilling fluid consisting of open system salt water brine (9.8-9.9 ppg) was used while drilling the lateral section up to the TD of 20,584'. One 6" PDC bit was used to drill the lateral.

Bit #6 (Varel VM613DUX) was the PDC bit used to drill the start of the entire lateral from 11,190' to 20,584' in 202 hours of drilling for averaged an ROP of 46.5 ft/hr

Drilling operations were concluded at 06:15 on January 26, 2015 at a bottom hole location of 235.00' S & 9,994.22' E of surface location or approximately 199.00' FSL & 257.93' FEL of SE SE Sec. 17 T153N, R100W.

Lithology

To aid in communications while drilling the lateral, a type log was created using a common gamma signature from the original offset well for the pad, the *Kline Federal 5300 41-18 13T2X* (Diagram 1). The cool gamma “C” marker (30 api – 50 api) as well as the warm gamma signature characteristic of the claystone appeared to be reliable and served as our main markers for steering.

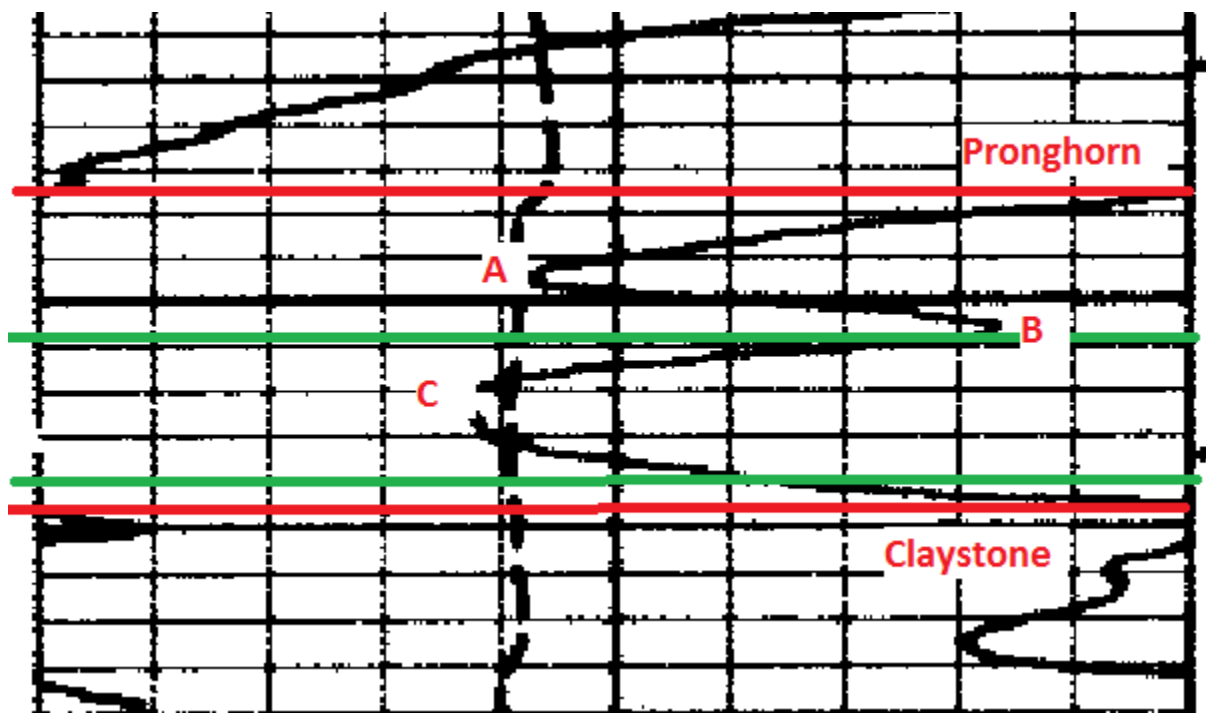


Diagram 1: Type log created from gamma log of the offset well Kline Federal 5300 41-18 13T2X

The target Zone of the **Three Forks** consisted of a dolostone with very apparent intergranular porosity, moderate gas, and moderate to abundant oil show (Figure 9). Samples from the target Zone were generally described as:

DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain



*Figure 9: Photograph of sandstone from "Target Zone" on Diagram 1
6" Bit & Salt Water Brine*

Gas and hydrocarbon Shows

It was found that the gas and hydrocarbon shows were a fair indicator of placement in Zone throughout the lateral. While drilling in the target Dolomite gas levels maintained average levels of 1000 to 2200 units with peaks over approaching 4000 units. When the bit dropped in vertical position into the claystone gas levels dropped to under 500 units with peaks slightly over 1000 units.

The Bloodhound gas chromatograph showed components of all C1-C4 present through the lateral while drilling in the target zone. C1 was primarily present while drilling in the claystone below. Sample cuttings throughout the lateral showed oil stain and a fast white diffuse cut when submerged in Acetone solution and observed with a UV fluoroscope. Trace oil was observed in the sample wash water.

Summary

- 1) The Kline Federal 5300 41-18 12TX was initially spud for surface drilling on 13th of September 2014 in the Indian Hills field of McKenzie County, North Dakota. The well was spud by Nabors 486 and the vertical hole was drilled to a total depth (KOP) of 10,300'.
- 2) A mud program consisted of diesel invert (10.3-11.4 ppg) after surface casing through the curve build sections. The drilling mud was successful in maintaining stable hole conditions and minimizing washout through the salt intervals. Brine water with a weight of 9.8-9.9 ppg was used during lateral drilling operations to maintain hydrostatic balance until a TD of 20,548'.
- 3) Two 13 1/4" Security PDC bits were used to drill to a depth of 6,138. 9 5/8 40# casing was set to isolate out the upper vertical hole and maintain hole stability during curve and lateral operations
- 4) One 8 3/4" Security PDC bit was used to drill out of secondary casing casing to a KOP of 10,300'. One 8 3/4" PDC bits were used to drill the curve to 11,190' MD. One 6" PDC bits were used to drill the lateral to 20,584' TD.
- 5) The lateral was targeting the **Three Forks** member ranging from about 8' below the base of the **Pronghorn** down to approximately 1' above the **Claystone**.
- 6) Gas levels between 500 and 4000 units were maintained through the lateral drilling operations while in the **Three Forks**.
- 7) The **Three Forks** target consisted of 3 observed gross lithology. The Upper **Three Forks (Pronghorn)** was observed to be a gray immature claystone. The Middle **Three Forks** was observed to be a dolostone with lots of disseminated pyrite and apparent intergranular porosity. The Lower Three Forks was an off white to gray claystone
- 8) Trace to moderate oil stain was observed in zone associated with a fast, white, diffuse cut. Trace oil washed out of the samples into the wash water when they were cleaned.
- 9) Drilling operations were concluded at 06:15 on January 26th, 2015 at a bottom hole location of 235.00' S & 9994.22' E of surface location or approximately 199.00' FSL & 256.93' FEL of SW SE Sec. 17, T153N-R100W.

Respectfully submitted,
Devin Wood
RPM Geologic LLC

Well Information

Operator: Oasis Petroleum North America

Address: 1001 Fannin Suite 1500
Houston, TX 77002

Well Name: Kline Federal 5300 41-18 12TX

Field/ Prospect: Indin Hill

Elevation: GL: 2,056'
KB: 2,082'

Spud Date: 13-Sep-14

API #: 33-053-06030

NDIC Well File #: 28658

Surface Location: SW SW Sec. 18 T153N, R100W

Footages: 434' FSL & 237' FWL

County, State: McKenzie Co., ND

Basin: Williston

Well Type: Horizontal Three Forks B1

Contractor: Nabors 486

Toolpushers: Logan Penhollow, Tyler Schultz

Field Supervisors: Ryan Romans/Ty Tschacher

Directional Drilling RPM Consulting
Ryan Romans
Ty Tschacher

Chemical Company Halliburton

Mud Engineer Tim Vaira

H₂S MONITORING: NOV WellSight Gas Watch

MWD Gyro Data
Beau Watt, Josh Elliott

Wellsite Geologist Devin Wood
Caleb Sperry

Rock Sampling: 30' from 8,350' to 11,200' MD
50' 11,200' MD to 20,583' (TD)

Prospect Geologist Brendan Hargrove

Gas Detector NOV WellSight Gas Watch

Sample Examination: Binocular microscope & fluoroscope

Sample Cuts: Acetone

Horizontal Target Three Forks B1

Key Offset Wells:

Oasis Petroleum.	Kline Federal 41-18 13T2X	SE SE Sec. 10 T153N R100W	McKenzie Co., ND
Oasis Petroleum	Kline Federal 41-18 14BX	SE SE Sec. 10 T153N R100W	McKenzie Co., ND

Pumps: #1 & #2: HHP 1600 Triplex - 12" Stroke length Output: 0.0838 bbl/stk

Mud Type: Diesel invert mud 2,160'-11,190' (Curve TD)
Salt Water Brine, 11,190' - 20,584' (TD)

Casing: Surface: 13 3/8" 54# J-55 @ 2,157'
Intermediate: 9 5/8" 40# HCL80 @ 6,118', 7" 32# P-110 @ 11,189'

Hole Size: 17 1/2" to 2,160' MD, 12 1/2" to 6,138', 8 3/4" to 11,190' MD 6" to 20,584'(TD)

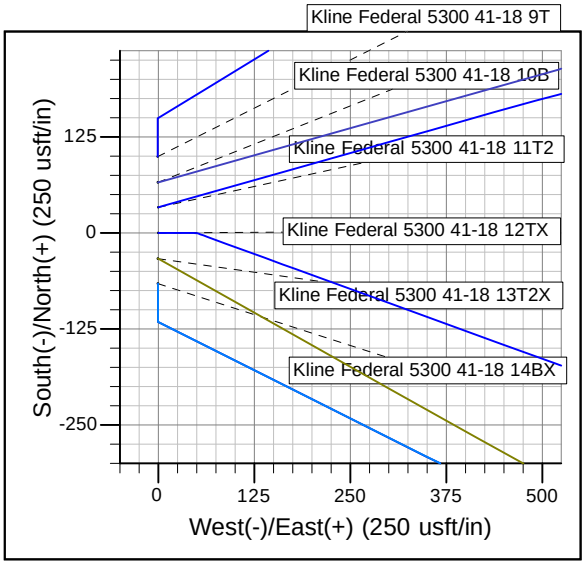
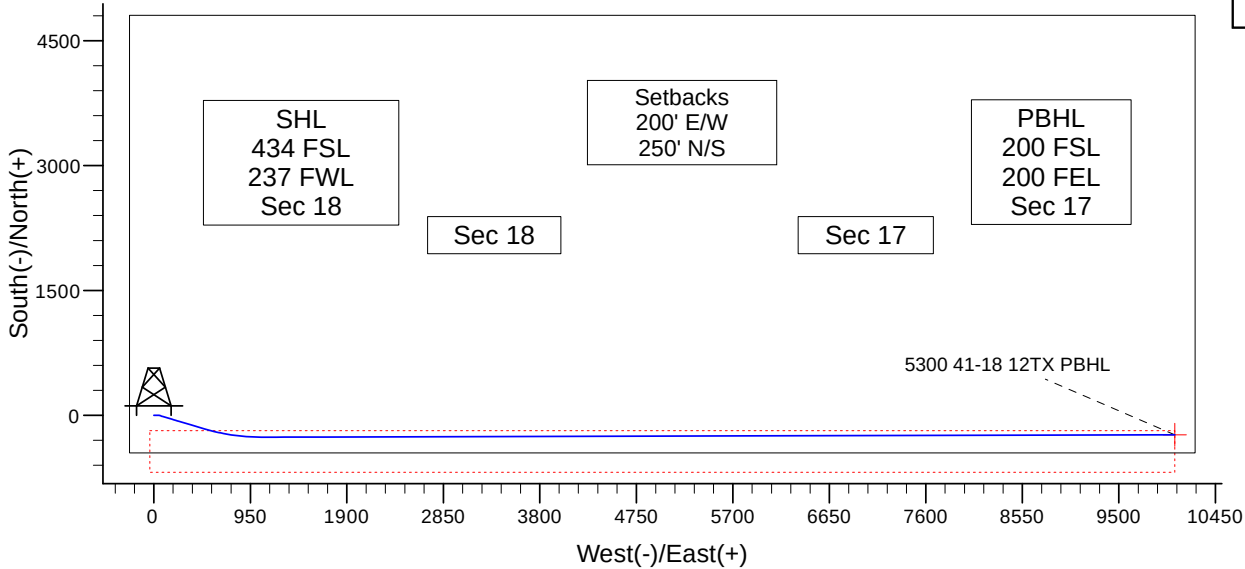
Total Drilling Days: 23

<u>Horizontal Target:</u>	<i>Three Forks B1</i>	<u>BOTTOM HOLE LOCATION:</u>
		<i>235.00' S & 9994.22' E</i>
<u>Kick-Off Point / Date:</u>	<i>10,300' on October 23, 2014</i>	<i>of surface, or approximately</i>
		<i>199.00' FSL & 256.93' FEL</i>
<u>Total Depth/ Date:</u>	<i>20,583' on January 26, 2015</i>	<i>SE SE Section 17, T153N, R100W</i>
<u>Ending Vertical Section</u>	<i>9994.22'</i>	
<u>Ending Azimuth</u>	<i>89.31°</i>	
<u>Status of Well:</u>	<i>Awaiting completion</i>	

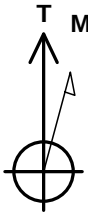
Project: Indian Hills
Site: 153N-100W-17/18
Well: Kline Federal 5300 41-18 12TX
Wellbore: Kline Federal 5300 41-18 12TX
Design: Design #2



WELL DETAILS: Kline Federal 5300 41-18 12TX			
Ground Level: 2057.0			
Northing	Easting	Latitude	Longitude
405136.64	1209968.21	48° 4' 7.550 N	103° 36' 11.950 W

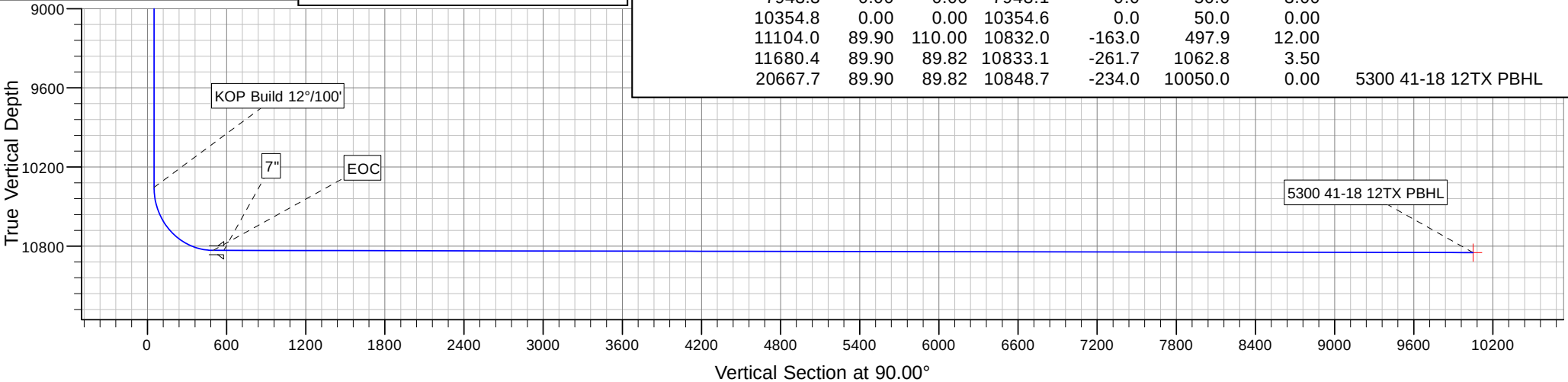


CASING DETAILS			
TVD	MD	Name	Size
10832.2	11189.0	7"	7"



Azimuths to True North
Magnetic North: 8.27°
Magnetic Field
Strength: 56437.1snT
Dip Angle: 72.99°
Date: 5/23/2014
Model: IGRF2010

SECTION DETAILS							
MD	Inc	Azi	TVD	+N/-S	+E/-W	Dleg	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	
2200.0	0.00	0.00	2200.0	0.0	0.0	0.00	
2216.7	0.50	90.00	2216.7	0.0	0.1	3.00	
7926.7	0.50	90.00	7926.4	0.0	49.9	0.00	
7943.3	0.00	0.00	7943.1	0.0	50.0	3.00	
10354.8	0.00	0.00	10354.6	0.0	50.0	0.00	
11104.0	89.90	110.00	10832.0	-163.0	497.9	12.00	
11680.4	89.90	89.82	10833.1	-261.7	1062.8	3.50	
20667.7	89.90	89.82	10848.7	-234.0	10050.0	0.00	5300 41-18 12TX PBHL



Daily Activity

Day	Date 2014	Depth 0600 Hrs	24 Hr Footage	Bit #	WOB (Klbs) Rotate	WOB (Klbs) Slide	RPM (RT)	Pump Pressure	SPM 1	SPM 2	GPM	24 Hr Activity	Formation
1	16-Oct	2,160'	0'	-	-	-	-	-	-	-	-	skid rig, rig up, nipple up bops, test bops and casing, p/u bha, wait on bit from another rig, TIH, change rotating head/rubber, TIH w/ rock bit to mill broken pdc blade downhole, displace to oil base, cut drilling line, guide plate over	Surface
2	17-Oct	3,656'	1,496'	3	-	-	-	-	-	-	-	TIH, service rig, TIH, ream/wash 2052-2353, drill 2353-2522, change rotating head, drill 2522-3656	Pierre
3	18-Oct	6,138'	2,482'	3	-	-	-	-	-	-	-	drill 3656-5201, service rig, drill 5201-6138, circulate and condition, TOOH	Dakota
4	19-Oct	6,138'	0'	3	-	-	-	-	-	-	-	circulate and condition, TOH Lat down BHA, Rig up casing crew. Run casing, Cement TIH	Dakota
5	20-Oct	7,443'	1,305'	4	22	0	50	3100	65	65	451	TIH, Pressure test shoe, Reaming, Drill cement FIT test, Drill F/6138-7437	Opeche Salt
6	21-Oct	8,856'	1,413'	4	7.5	0	40	3150	65	65	451	Drill F/7443-8195, Rig service, Drill F/8195-8844	Charles Salt
7	22-Oct	10,015'	1,159'	4	31	42	40	3590	77	77	545	Drill F/8856-9421, Rig Service, Drill F/9421-10010	Lodgepole
8	23-Oct	10,300'	285'	4	35	0	45	3500	77	77	545	Drill F/10015-10300, Circulate and condition, Rig Service, TOOH, Install/remove wear bushing and install flow nipple, TOOH with HWDP and collars, Lay down BHA, Rig up loggers, Lay down tools and rig down, Pick up BHA, TIH	Lodgepole
9	24-Oct	10,860'	560'	5	24	28	30	2923	67	67	441	TIH ream and wash through salts, Drill F/10300-10545, Rig Service, Frill F/10545-10880	Middle Bakken

Daily Activity

Day	Date 2014	Depth 0600 Hrs	24 Hr Footage	Bit #	WOB (Klbs) Rotate	WOB (Klbs) Slide	RPM (RT)	Pump Pressure	SPM 1	SPM 2	GPM	24 Hr Activity	Formation
10	25-Oct	11,190'	330'	5	24	28	30	2923	67	67	441	Drill F/10860-11190, Service Rig and top drive, Circulate and condition, TOH, Fix top drive HYD transducer, TIH, Lay down BHA, Rig service, rig up casers, run casing	Three Forks
11	26-Oct	11,190'	0'	-	-	-	-	-	-	-	-	Run casing, Work stuck pipe, Rig down casers, primary cementing	Three Forks
12	27-Oct	11,190'	0'	-	-	-	-	-	-	-	-	Rig up/down casers and cementers, Nipple down BOPS, Install well head cut casing install night cap. Realesed rig at 1200 hours. Skidding rig and rigging up	Three Forks
13	16-Jan	11,190'	0'	-	-	-	-	-	-	-	-	Cement, Rig down, skid over to the 12TX, Presure test	Three Forks
14	17-Jan	11,190'	0'	5	-	-	-	-	-	-	-	TIH and tag bottom @ 7195 // Drill cement // Circulate bottoms up and condition	Three Forks
15	18-Jan	11,505'	315'	6	15	35	10	2000	75	0	314	Trip out of hole // Pick up BHA // Service Rig // Change rotating head/rubber // Trip in hole // Drill actual F/11,190' - 11,505'	Three Forks
16	19-Jan	12,470'	965'	6	13	42	45	2150	0	65	272	Drill actual F/ 11,502' - 12,470' // Service Rig	Three Forks
17	20-Jan	13,603'	1,133'	6	18	55	45	2600	0	65	272	Drill actual F/12,481 - 13,603 // Service Rig	Three Forks
18	21-Jan	14,894'	1,291'	6	15	42	45	2700	68	0	272	Drill actual F/13,610 - 14,894' // Service Rig	Three Forks
19	22-Jan	16,153'	1,259'	6	16	47	45	2700	0	71	297	Drill actual F/14,910 - 16,153' // Service Rig	Three Forks

Daily Activity

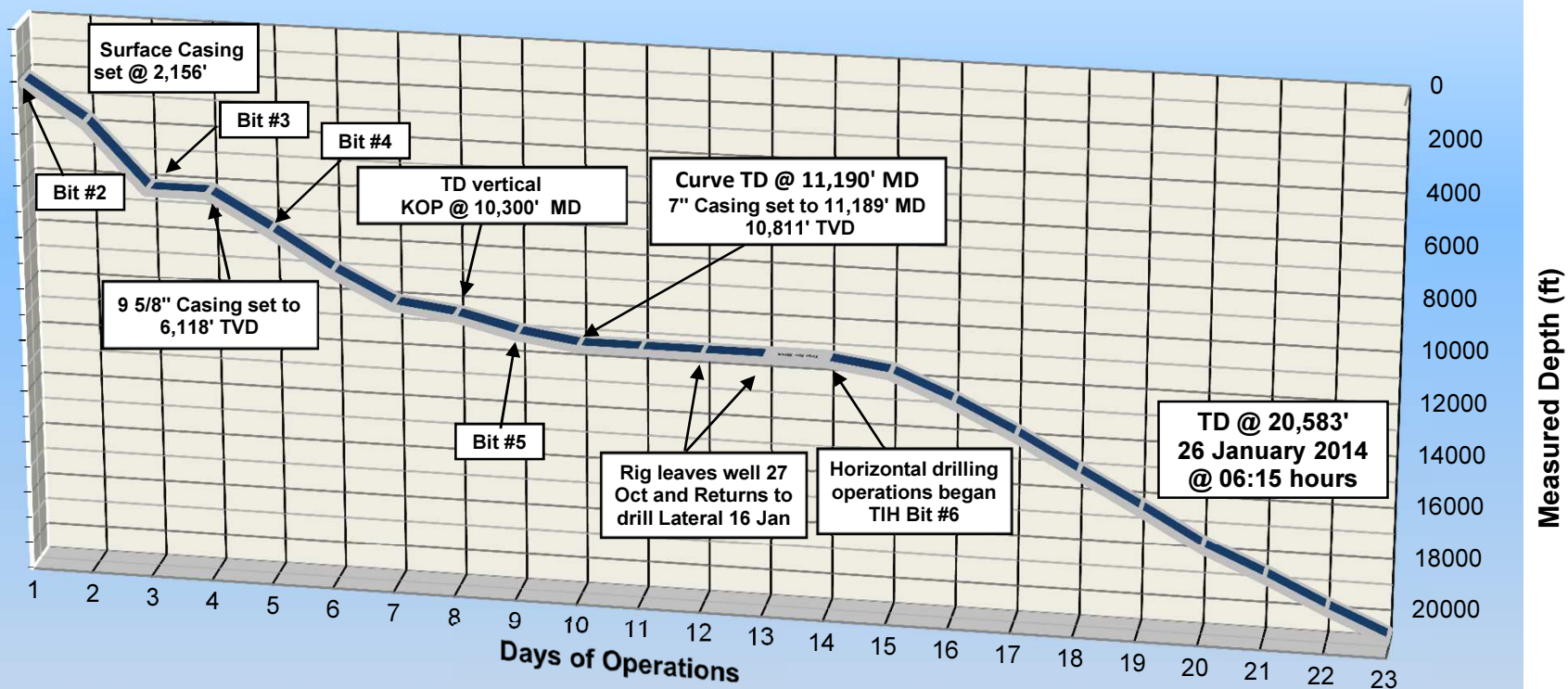
Day	Date 2014	Depth 0600 Hrs	24 Hr Footage	Bit #	WOB (Klbs) Rotate	WOB (Klbs) Slide	RPM (RT)	Pump Pressure	SPM 1	SPM 2	GPM	24 Hr Activity	Formation
20	23-Jan	17,476'	1,323'	6	20	47	45	3250	0	71	297	Drill actual F/16,175 - 17,476' // Service Rig	Three Forks
21	24-Jan	18,481'	1,005'	6	20	50	45	3250	0	71	297	Drill actual F/17,475 - 18,481' // Service Rig	Three Forks
22	25-Jan	19,578'	1,097'	6	20	45	45	2900	71	0	297	Drill actual F/18,475 - 19,578' // Service Rig	Three Forks
23	26-Jan	20,584'	1,006'	6	20	45	45	2900	71	0	297	Drill actual F/19,578 - 20,584' // Service Rig	Three Forks



Daily Progress

Oasis Petroleum
Kline Federal 5300 41-18 12TX

Spud 13 September 2014



Bit Record

<i>Bit #</i>	<i>Size</i>	<i>Make</i>	<i>Model</i>	<i>Serial #</i>	<i>Jets</i>	<i>Depth In</i>	<i>Depth Out</i>	<i>Footage</i>	<i>Hours</i>	<i>Mean ROP (ft/hr)</i>	<i>Accum. Hours</i>
1	17 1/2	NOV	TFR6195	E163630	12x18	105'	2,160'	1,321'	11.5	114.87	11.5
2	12 1/4	Haliburton	Security	P92215	6x20	2,160'	2,353'	193'	2	96.50	13.5
3	12 1/4	Haliburton	Security	12138204	6x20	2,353'	6,138'	3,785'	30.5	124.10	44.0
4	8 3/4	Halliburton	Security	12529681	6x14	6,135'	10,300'	4,165'	285	14.61	298.5
5	8 3/4	Security DBS	Security	12560304	6x18	10,300'	11,190'	890'	32	27.81	330.5
6	6	Varel	V613DUX	4008030	6x20	11,190'	20,584'	9,394'	202	46.50	532.5

Daily Mud Data

Day	Date 2014	Depth (0600 Hrs)	Mud WT (ppg)	VIS (sec)	PV (cP)	YP (lbs/ 100 ft²)	Gels (lbs/ 100 ft²)	600/ 300	HTHP (cc/30min)	NAP/ H₂O (ratio)	Cake (APT/ HT)	Solid s (%)	pH	Alk	Cl- (mg/l)	CaCl₂	ES (v)	Loss (Bbls)
1	16-Oct	2,150'	11.40	56	21	22	10/12/16	64/43	9	76.6/23.4	2	14.5	-	-	33K	-	740	-
2	17-Oct	3,656'	11.40	56	22	23	10/12/17	64/44	9	76.6/23.5	2	14.6	-	-	33K	-	741	-
3	18-Oct	6,138'	11.40	58	23	24	8/9/14	64/45	9	76.6/23.6	2	10.5	-	-	25K	-	885	-
4	19-Oct	6,138'	11.60	56	25	23	12/13/18	73/48	7	76.5/23.5	2	14.70	-	-	38K	-	740	-
5	20-Oct	7,425'	11.60	57	26	23	12/13/18	73/48	7	76.5/23.5	2	14.70	-	-	38K	-	740	-
6	21-Oct	8,844'	10.10	50	18	15	7/8/11	21/33	7	77.8/22.2	2	10.20	-	-	34K	-	710	-
7	22-Oct	10,010'	10.00	44	15	12	6/7/9	42/27	6	79.5/20.5	2	9.90	-	-	29K	-	773	-
8	23-Oct	10,300'	10.05	46	19	13	7/8/10	51/32	4	78.8/21.2	2	10.70	-	-	37K	-	780	-
9	24-Oct	10,880'	10.20	46	19	14	7/8/10	52/33	3	78.2/21.8	2	10.60	-	-	39K	-	800	-
10	25-Oct	11,190'	10.30	45	19	13	6/7/9	5/32	3	78.7/21.3	2	10.80	-	-	40K	-	820	-
11	17-Jan	11,190'	10.30	45	19	13	6/7/9	5/32	3	78.7/21.3	2	10.80	-	-	40K	-	820	-
12	18-Jan	11,190'	10.30	45	19	13	6/7/9	5/32	3	78.7/21.3	2	10.80	-	-	40K	-	820	-
13	19-Jan	12,966'	9.90	28	2	1	0/1/1	4/3	-	0/89.3	-	0.70	10	-	168k	-	-	-
14	20-Jan	12,966'	9.90	28	2	1	0/1/1	4/3	-	0/89.3	-	0.70	10	-	168k	-	-	-
15	21-Jan	15,554'	9.85	28	2	1	0/1/1	4/3	-	0/89.6	-	0.50	10	-	166k	-	-	-
16	22-Jan	15,554'	9.85	28	2	1	0/1/1	4/3	-	0/89.6	-	0.50	10	-	166k	-	-	-
17	23-Jan	17,964'	9.90	28	2	1	0/1/1	5/3	-	0/89.2	-	0.50	9	-	172k	-	-	-
18	24-Jan	17,964'	9.90	28	2	1	0/1/1	5/3	-	0/89.2	-	0.50	9	-	172k	-	-	-
19	25-Jan	17,964'	9.90	28	2	1	0/1/1	5/3	-	0/89.2	-	0.50	9	-	172k	-	-	-
20	26-Jan	17,964'	9.90	28	2	1	0/1/1	5/3	-	0/89.2	-	0.50	9	-	172k	-	-	-

DRILLING PLAN																																																																																																															
OPERATOR		Oasis Petroleum			COUNTY/STATE		McKenzie Co., ND																																																																																																								
WELL NAME		Kline Federal 5300 41-18 12TX			RIG		Nabors B22																																																																																																								
WELL TYPE		Horizontal Upper Three Forks																																																																																																													
LOCATION		SWSW 18-153N-100W			Surface Location (survey plat): 434' fsj		237' fwl																																																																																																								
EST. T.D.		20,668'			GROUND ELEV:		2057' Finished Pad Elev.		Sub Height: 25'																																																																																																						
TOTAL LATERA		9,479'			KB ELEV:		2082'																																																																																																								
PROGNOSIS: Based on 2,082' KB(est)					LOGS:																																																																																																										
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DEVIATION:					Surf: 3 deg. max., 1 deg / 100'; srvy every 500' Prod: 5 deg. max., 1 deg / 100'; srvy every 100'																																																																																																										
DST'S:					None planned																																																																																																										
CORES:					None planned																																																																																																										
MUDLOGGING:					Two-Man: 8,342' ~200' above the Charles (Kibbey) to Casing point; Casing point to TD 30' samples at direction of wellsite geologist; 10' through target @ curve land																																																																																																										
BOP:					11" 5000 psi blind, pipe & annular																																																																																																										
Dip Rate: -0.1					Surface Formation: Glacial till																																																																																																										
Max. Anticipated BHP: 4692																																																																																																															
MUD:		Interval			Type		WT		Vis																																																																																																						
Surface:		0' - 2,072'			FW/Gel - Lime Sweeps		8.4-9.0		28-32																																																																																																						
Intermediate:		2,072' - 11,189'			Invert		9.5-10.4		40-50																																																																																																						
Lateral:		11,189' - 20,668'			Salt Water		9.8-10.2		28-32																																																																																																						
CASING:		Size		Wt ppf	Hole	Depth	Cement	WOC	Remarks																																																																																																						
Surface:		9-5/8"		36#	13-1/2"	2,072'	To Surface	12	100' into Pierre																																																																																																						
Intermediate:		7"		29/32#	8-3/4"	11,189'	3971	24	1500' above Dakota																																																																																																						
Production Liner:		4.5"		11.6#	6"	20,668'	TOL @ 10,305'		50' above KOP																																																																																																						
PROBABLE PLUGS, IF REQ'D:																																																																																																															
OTHER:		MD	TVD	FNL/FSL	FEL/FWL	S-T-R	AZI																																																																																																								
Surface:		2,072	2,072	434' FSL	237' FWL	SEC 18-T153N-R100W	Survey Company:																																																																																																								
KOP:		10,355'	10,355'	434' FSL	287' FWL	SEC 18-T153N-R100W	Build Rate: 12 deg /100'																																																																																																								
EOC:		11,104'	10,832'	306' FSL	746' FWL	SEC 18-T153N-R100W	110.00																																																																																																								
Casing Point:		11,189'	10,832'	244' FSL	816' FWL	SEC 18-T153N-R100W	107.02																																																																																																								
Upper Three Forks Lateral TD:		20,668'	10,849'	200' FSL	200' FEL	SEC 17-T153N-R100W	89.82																																																																																																								
Comments: Request a Sundry for an Open Hole Log Waiver Exception well: Oasis Petroleum's Kline Federal 5300 11-18H (153N 100W 18 NW NW) 35 packers, 35 sleeves, no frac string Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations. 68334-30-5 (Primary Name: Fuels, diesel) 68476-34-6 (Primary Name: Fuels, diesel, No. 2) 68476-30-2 (Primary Name: Fuel oil No. 2) 68476-31-3 (Primary Name: Fuel oil, No. 4) 8008-20-6 (Primary Name: Kerosene)																																																																																																															
																																																																																																															
Geology: M.Steed 4/23/2014					Engineering: hlbader rpm 6/17/14																																																																																																										

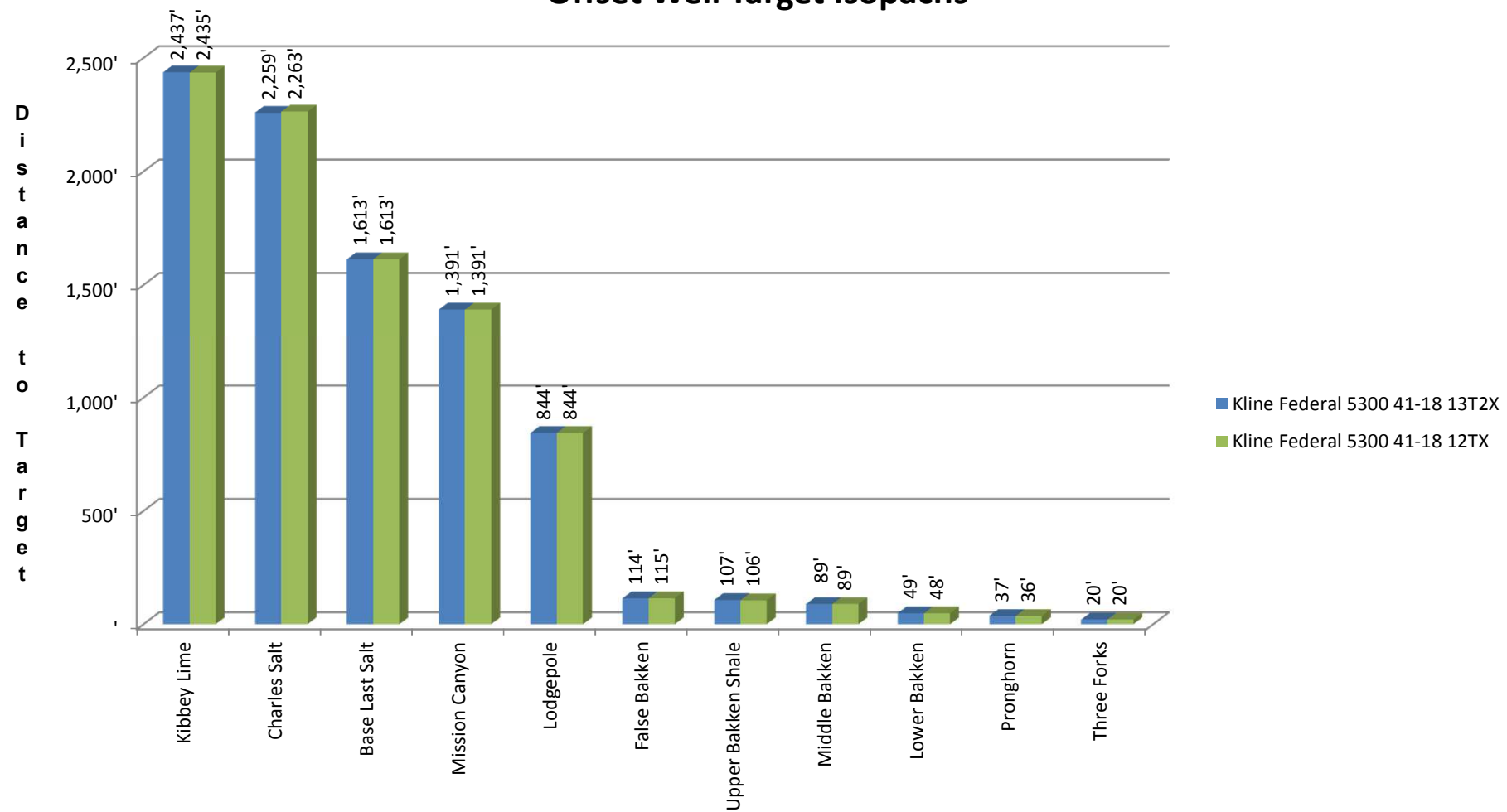
Formation Tops

Operator: Well Name: Location: Elevation:	Subject Well: Oasis Petroleum North America <i>Kline Federal 5300 41-18 12TX</i> 434' FSL &237' FWL SW SW Sec. 18 T153N, R100W 2,057 Sub: 25' KB: 2,082'								
	Formation/ Zone	Prog. Top	Prog. MSL Datum	Est. MD Top (Gamma)	TVD Top (Gamma)	Est. MLS Datum	Thickness	Dip To <i>Kline Federal 5300 41-18 13T2X</i>	Dist to Target
	Kibbey Lime	8,382	-6,300'	8,384'	8,384'	-6,302'	172'	2' Low	2,435'
	Charles Salt	8,560	-6,478'	8,556'	8,556'	-6,474'	650'	4' Low	2,263'
Base Last Salt	9,206	-7,124'	9,206'	9,206'	-7,124'	222'	Flat	1,613'	
Mission Canyon	9,428	-7,346'	9,428'	9,428'	-7,346'	547'	Flat	1,391'	
Lodgepole	9,975	-7,893'	9,975'	9,975'	-7,893'	729'	Flat	844'	
False Bakken	10,703	-8,621'	10,781'	10,704'	-8,622'	9'	1' High	115'	
Upper Bakken Shale	10,712	-8,630'	10,798'	10,713'	-8,631'	17'	1' Low	106'	
Middle Bakken	10,729	-8,647'	10,829'	10,730'	-8,648'	41'	Flat	89'	
Lower Bakken	10,770	-8,688'	10,918'	10,771'	-8,689'	12'	1' Low	48'	
Pronghorn	10,781	-8,699'	10,952'	10,783'	-8,701'	16'	1' Low	36'	
Three Forks	10,797	-8,715'	11,011'	10,799'	-8,717'	18'	Flat	20'	
Target	10,817	-8,735'		10819	-8737		2' Low	'	

Control Well

<u><i>Control Well</i></u>								
Operator: Well Name: Location: Elevation:	Oasis Petroleum <i>Kline Federal 5300 41-18 13T2X</i> SW SW Sec. 18 T153N, R100W McKenzie Co., ND Approximately 33 ft South KB: 2,082'				Oasis Petroleum <i>Kline Federal 5300 41-18 14BX</i> SW SW Sec. 18 T153N, R100W McKenzie Co., ND Approximately 213 feet South KB: 2,082'			
Formation/ Zone	Est. TVD	MSL Datum	Thickness	Dist to Target	Est. TVD	MSL Datum	Thickness	Dist to target
Kibbey Lime	8,382'	-6,300'	178'	2,437'	8,382	-6,300'	182'	2,435'
Charles Salt	8,560'	-6,478'	646'	2,259'	8,564	-6,482'	641'	2,253'
Base Last Salt	9,206'	-7,124'	222'	1,613'	9,205	-7,123'	223'	1,612'
Mission Canyon	9,428'	-7,346'	547'	1,391'	9,428	-7,346'	548'	1,389'
Lodgepole	9,975'	-7,893'	730'	844'	9,976	-7,894'	728'	841'
False Bakken	10,705'	-8,623'	7'	114'	10,704	-8,622'	8'	113'
Upper Bakken Shale	10,712'	-8,630'	18'	107'	10,712	-8,630'	16'	105'
Middle Bakken	10,730'	-8,648'	40'	89'	10,728	-8,646'	27'	89'
Lower Bakken	10,770'	-8,688'	12'	49'				
Pronghorn	10,782'	-8,700'	17'	37'				
Three Forks	10,799'	-8,717'	20'	20'				
Target	10,819'	-8,737'	20'	'				

Offset Well Target Isopachs

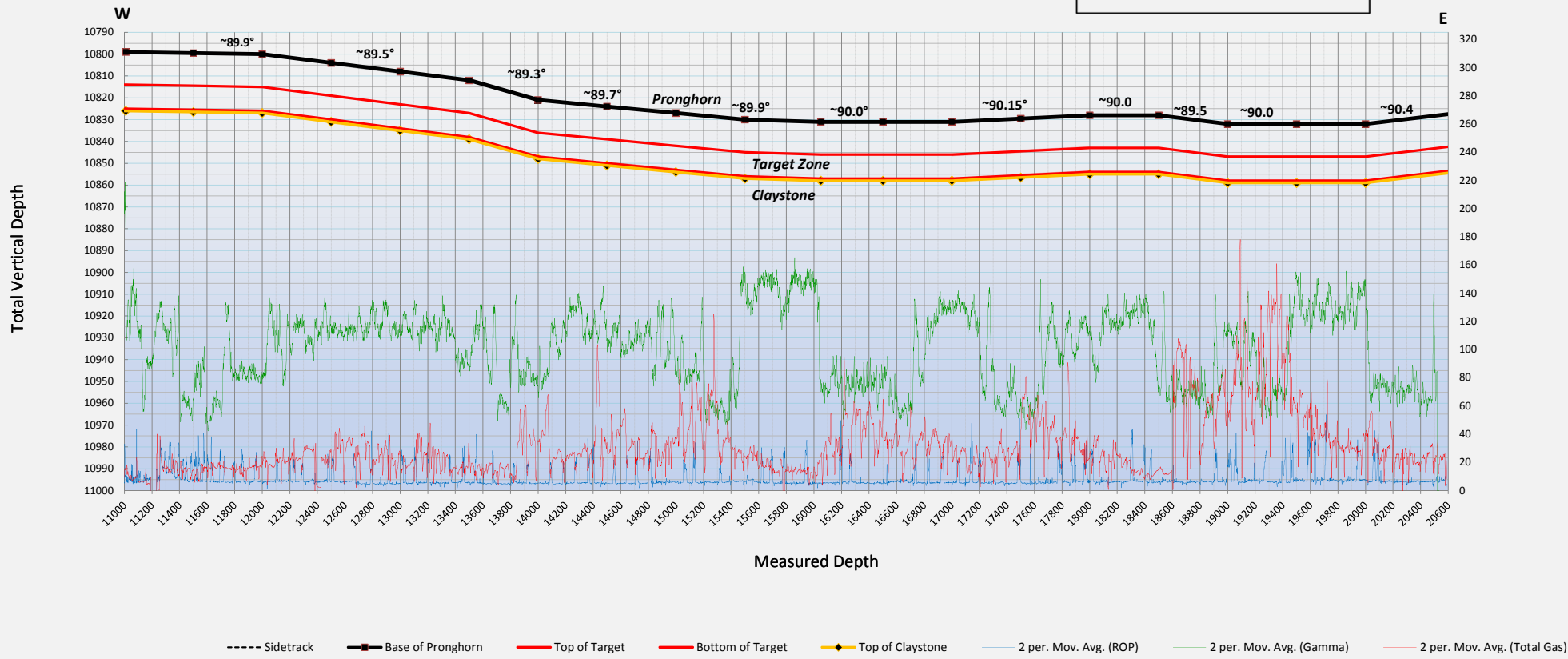


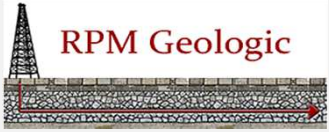


Surface Location:
434' FSL & 237' FWL
SW SW Sec. 18, T153N, R100W
McKenzie County, ND

Kline Federal 5300 41-18 12TX
Gamma Cross Section

Bottom Hole Location:
235.00' S & 9994.22' E
of surface location or approx.
199.00' FSL & 256.93' FEL
SE SE Sec. 17, T153N, R100W

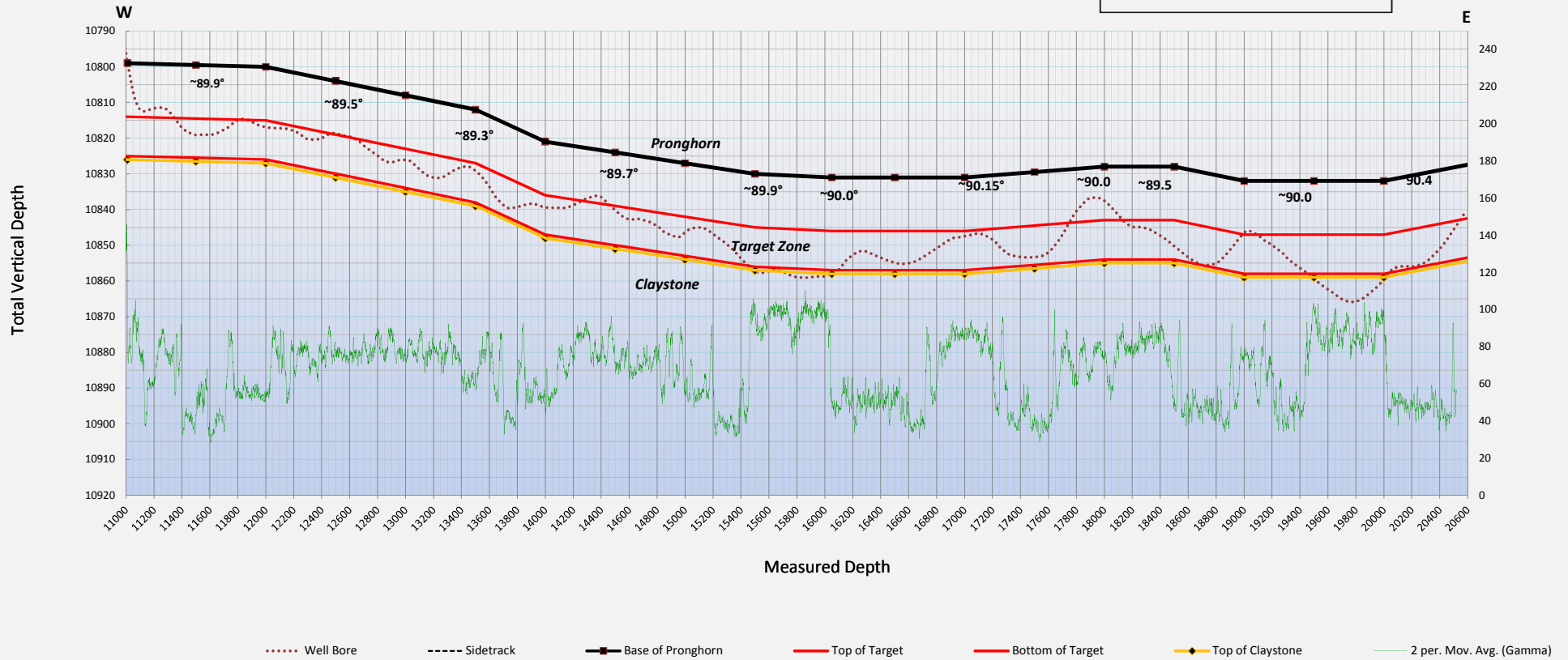




Surface Location:
434' FSL & 237' FWL
SW SW Sec. 18, T153N, R100W
McKenzie County, ND

Kline Federal 5300 41-18 12TX
Gamma Cross Section

Bottom Hole Location:
235.00' S & 9994.22' E
of surface location or approx.
199.00' FSL & 256.93' FEL
SE SE Sec. 17, T153N, R100W



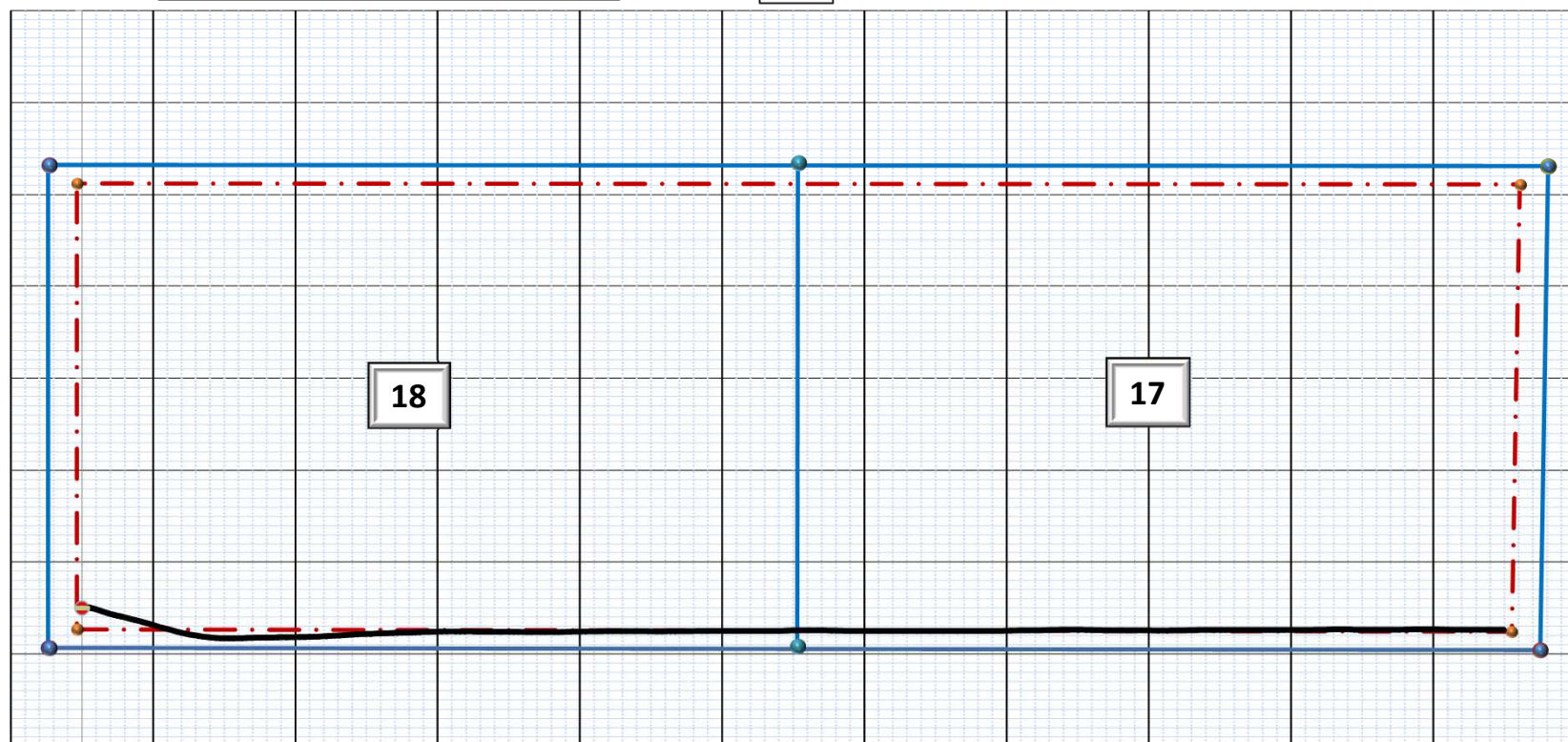


Vertical Section View

Surface Location:
434' FSL & 237' FWL
SW SW Sec. 18, T153N, R100W
McKenzie County, ND

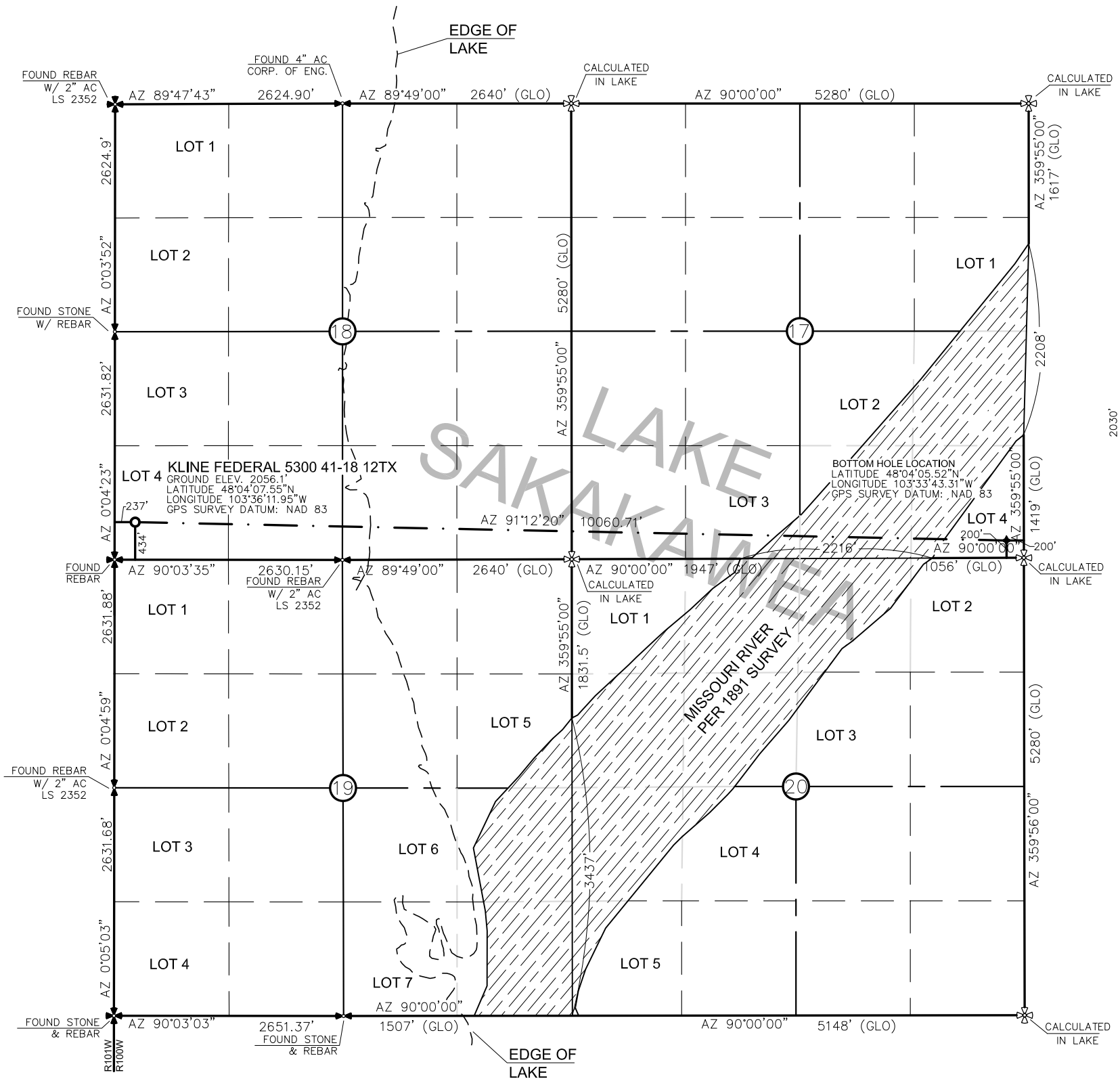
Bottom Hole Location
235.00' S & 9994.22' E
of surface location or approx.
199.00' FSL & 256.93' FEL
SE SE Sec. 17 T153N, R100W

E-W



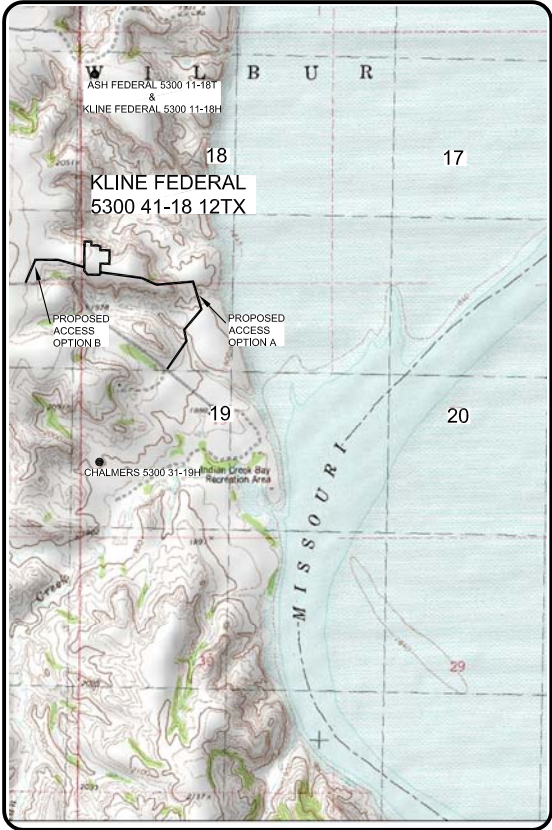
N-S

WELL LOCATION PLAT
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 41-18 12TX"
434 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



- ⊕ ⊗ - MONUMENT - RECOVERED
⊕ ⊗ - MONUMENT - NOT RECOVERED

VICINITY MAP



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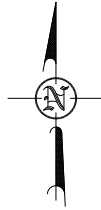
STAKED ON 4/15/14
VERTICAL CONTROL DATUM WAS BASED UPON
CONTROL POINT 4 WITH AN ELEVATION OF 2090.8'

THIS SURVEY AND PLAT IS BEING PROVIDED AT THE REQUEST
OF ERIC BAYES OF OASIS PETROLEUM. I CERTIFY THAT THIS
PLAT CORRECTLY REPRESENTS WORK PERFORMED BY ME OR
UNDER MY SUPERVISION AND IS TRUE AND CORRECT TO THE
BEST OF MY KNOWLEDGE AND BELIEF.

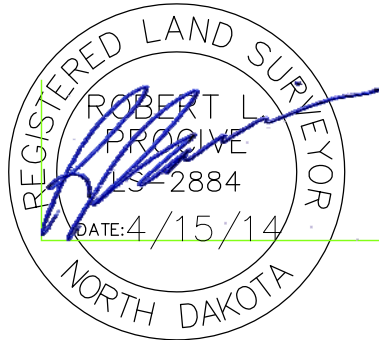
Robert L. Procive

ROBERT L. PROCIVE 2884LS

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0 1500
1" = 1500'



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OASIS PETROLEUM NORTH AMERICA, LLC
WELL LOCATION PLAT
SECTION 18, T153N, R100W

MCKENZIE COUNTY, NORTH DAKOTA

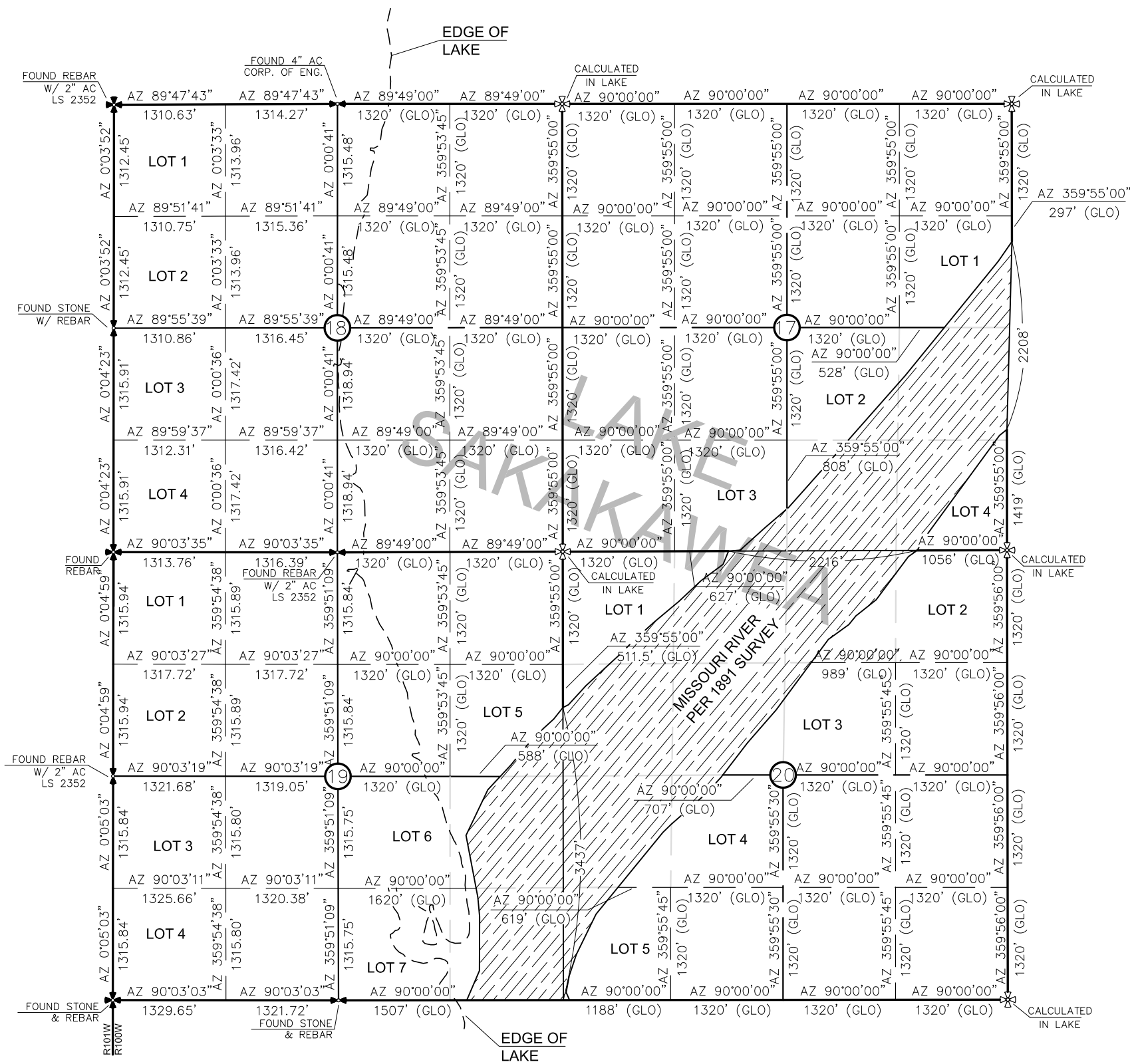
Drawn By: B.H.H. Project No.: S14-09-081.02

Checked By: R.L.P. Date: APRIL 2014

Revision No.	Date	By	Description

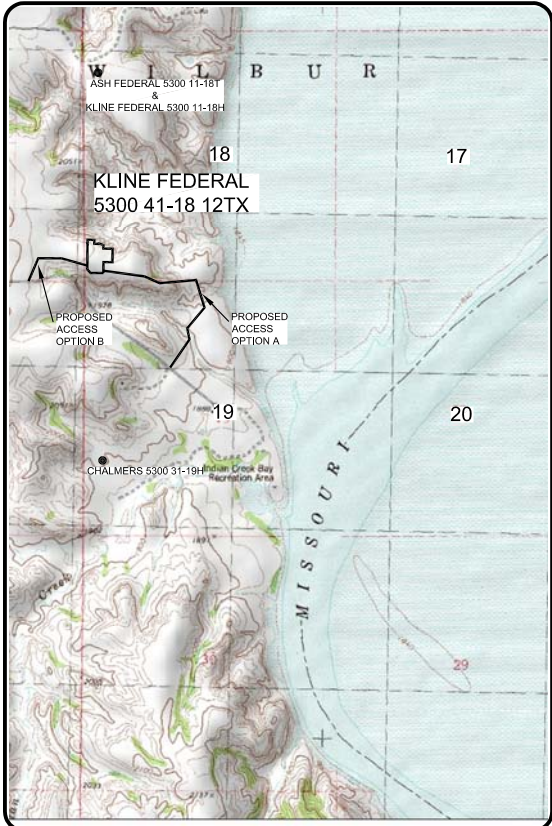
© 2014 S14-09-081.02 Oasis Petroleum - Kline Federal 5300 41-18 12TX
Location\CAD\Kline Federal 5300 41-18 12TX.dwg - 4/15/2014 2:51 PM joah schmierer

SECTION BREAKDOWN
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 41-18 12TX"
434 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE
SECTIONS 17, 18, 19, & 20, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



- ⛶ — MONUMENT — RECOVERED
⛶ — MONUMENT — NOT RECOVERED

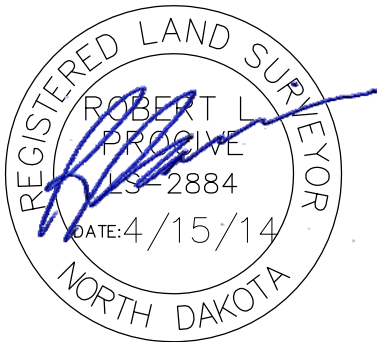
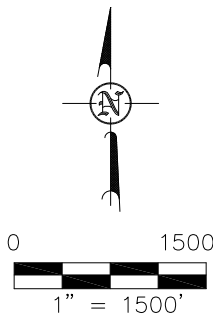
VICINITY MAP



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ALL AZIMUTHS ARE BASED ON G.P.S.
OBSERVATIONS. THE ORIGINAL SURVEY OF THIS
AREA FOR THE GENERAL LAND OFFICE (G.L.O.)
WAS 1891. THE CORNERS FOUND ARE AS
INDICATED AND ALL OTHERS ARE COMPUTED FROM
THOSE CORNERS FOUND AND BASED ON G.L.O.
DATA. THE MAPPING ANGLE FOR THIS AREA IS
APPROXIMATELY 0°03'.

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Fax (406) 433-5618
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Other offices in Minnesota, North Dakota and South Dakota

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SECTION BREAKDOWN
SECTIONS 17, 18, 19, & 20, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

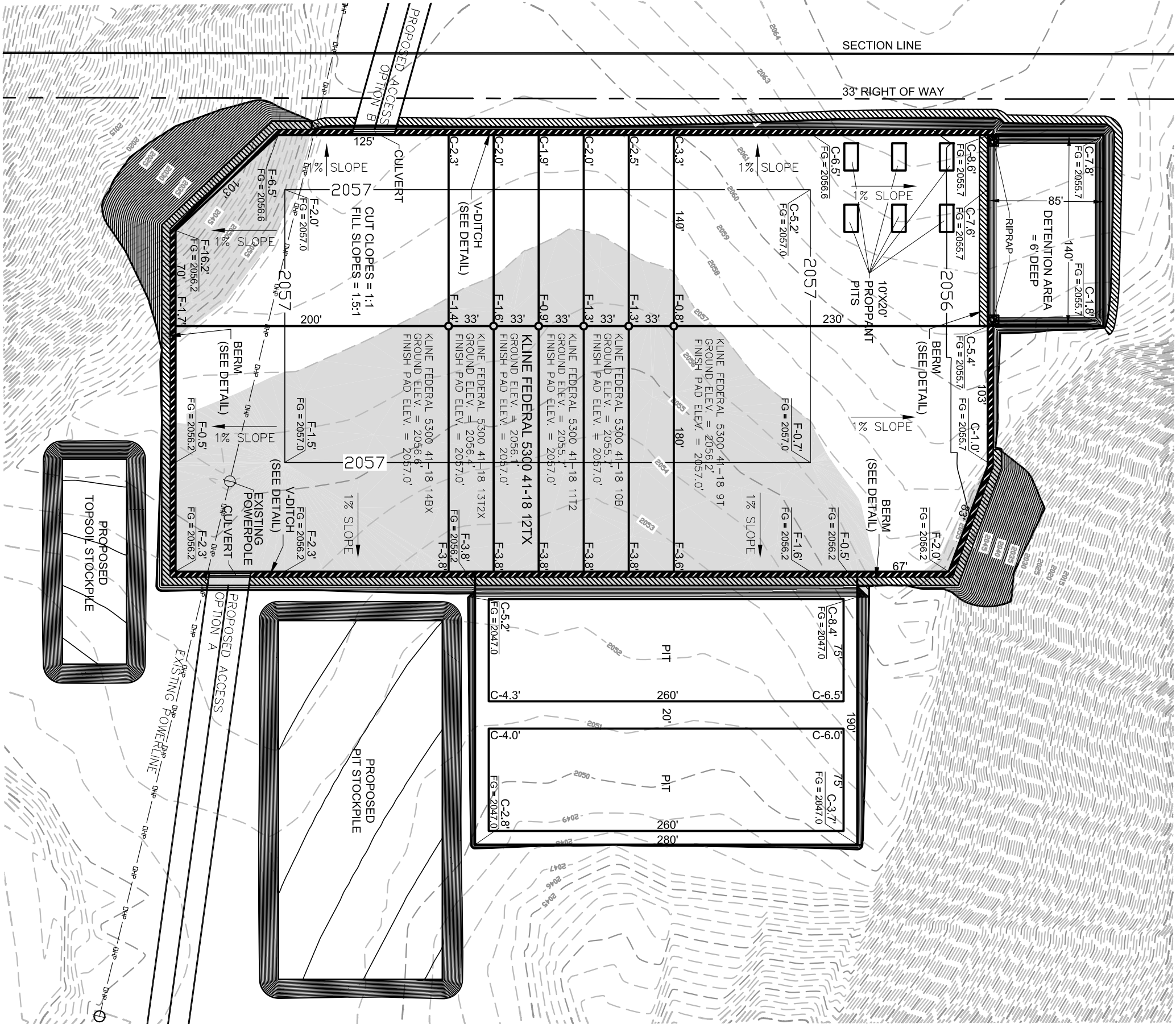
Drawn By: B.H.H. Project No.: S14-09-081.02
Checked By: R.L.P. Date: APRIL 2014

Revision No.	Date	By	Description

6:2014/S14-09-081.02 Oasis Petroleum - Kline Federal 5300 41-18 12TX
Location: CAD/Kline Federal 5300 41-18 12TX.dwg - 4/15/2014 2:51 PM john schmiere

PAD LAYOUT

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 41-18 12TX"
434 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

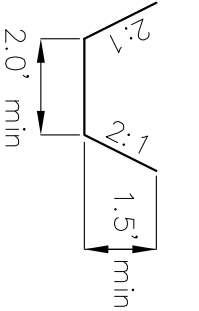


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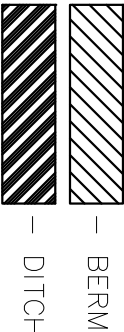
NOTE: Pad dimensions shown are to usable area, the v-ditch and berm areas shall be built to the outside of the pad dimensions.



V-DITCH DETAIL



Proposed Contours
Original Contours



NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

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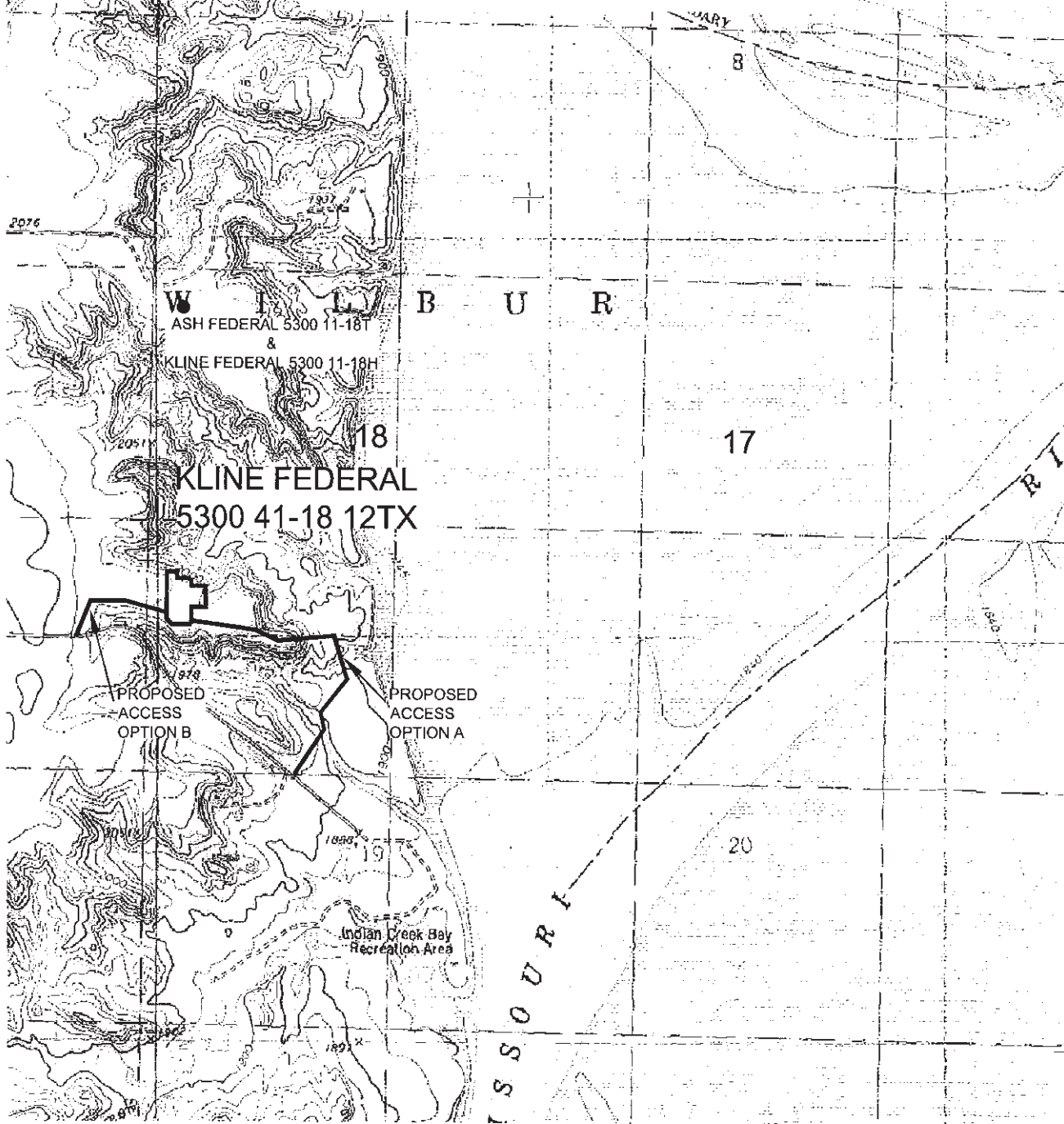
OASIS PETROLEUM NORTH AMERICA, LLC
PAD LAYOUT
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA
Drawn By: B.H.H.
Checked By: D.D.K.
Project No.: S14-09-08.102
Date: APRIL 2014

Revision No.	Date	By	Description

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LIANS CO
MCKENZIE CO

OASIS PETROLEUM NORTH AMERICA, LLC
KLINE FEDERAL 5300 41-18 12TX
434' FSL/237' FWL
QUAD LOCATION MAP
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA



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OASIS PETROLEUM NORTH AMERICA, LLC
QUAD LOCATION MAP
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No: S14-09-081.02
Checked By: R.L.P. Date: APRIL 2014

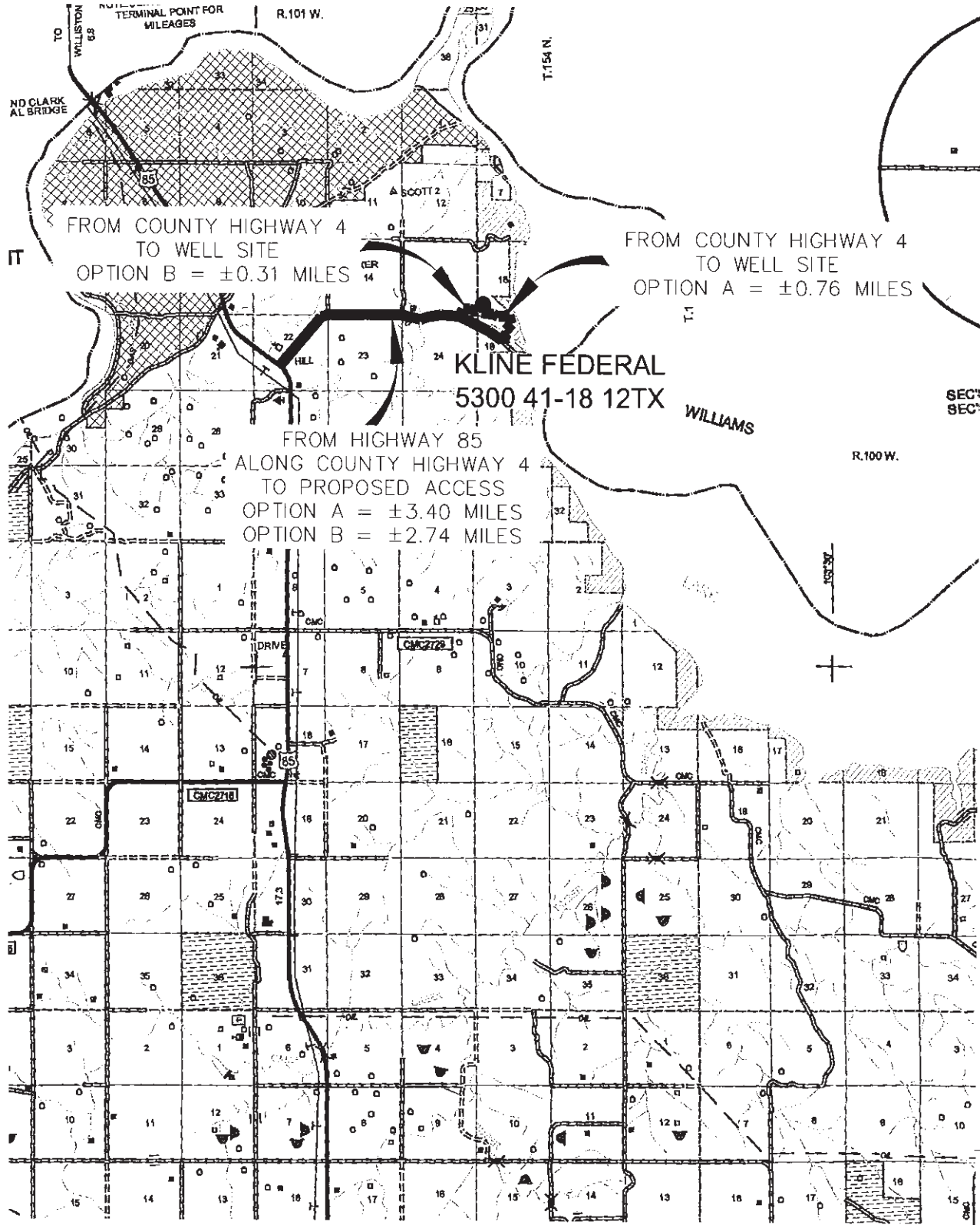
Revision No.	Date	By	Description

COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

"KLINE FEDERAL 5300 41-18 12TX"

434 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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SCALE: 1" = 2 MILE

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OASIS PETROLEUM NORTH AMERICA, LLC
COUNTY ROAD MAP
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S14-09-081.02
Checked By: R.L.P. Date: APRIL 2014

Revision No.	Date	By	Description

Co:	RPM Consulting Inc	Units:	Feet, °, °/100ft	VS Az:	90.00
Drillers:	Ceynar	Elevation:	2082.00	Method:	Minimum Curvature
Well Name:	Kline Federal 5300 41-18 12TX	Northing:	405137.45	Latitude:	48.068764
Location:	153N-100W-17/18	Easting:	1209970.63	Longitude:	-103.603319

Indian Hills: 153N-100W-17/18: Kline Fed 5300 41-18 12TX: Kline Fed 12TX Surveys

No.	MD	CL	Inc.	Azi.	TVD	VS	+N/S-	+E/W-	BR	WR	DLS
1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			
2	140.00	140.00	0.35	146.08	140.00	0.24	-0.35	0.24	0.25	104.34	0.25
3	234.00	94.00	0.66	143.09	234.00	0.72	-1.03	0.72	0.33	-3.18	0.33
4	327.00	93.00	0.57	172.80	326.99	1.10	-1.91	1.10	-0.10	31.95	0.35
5	421.00	94.00	0.48	154.78	420.99	1.33	-2.73	1.33	-0.10	-19.17	0.20
6	516.00	95.00	0.40	139.75	515.98	1.71	-3.35	1.71	-0.08	-15.82	0.15
7	607.00	91.00	0.79	150.04	606.98	2.23	-4.13	2.23	0.43	11.31	0.44
8	699.00	92.00	0.44	178.51	698.97	2.56	-5.04	2.56	-0.38	30.95	0.49
9	791.00	92.00	0.66	158.91	790.97	2.76	-5.88	2.76	0.24	-21.30	0.31
10	882.00	91.00	1.41	157.07	881.95	3.38	-7.40	3.38	0.82	-2.02	0.82
11	974.00	92.00	0.40	19.78	973.95	3.93	-8.14	3.93	-1.10	-149.23	1.88
12	1066.00	92.00	0.22	101.87	1065.94	4.21	-7.88	4.21	-0.20	89.23	0.47
13	1158.00	92.00	0.22	177.28	1157.94	4.40	-8.09	4.40	0.00	81.97	0.29
14	1250.00	92.00	0.44	325.29	1249.94	4.20	-7.98	4.20	0.24	160.88	0.69
15	1342.00	92.00	0.88	351.66	1341.94	3.90	-6.99	3.90	0.48	28.66	0.57
16	1433.00	91.00	0.92	4.14	1432.93	3.85	-5.57	3.85	0.04	13.71	0.22
17	1528.00	95.00	1.10	337.60	1527.91	3.56	-3.96	3.56	0.19	-27.94	0.52
18	1623.00	95.00	0.84	354.73	1622.90	3.15	-2.43	3.15	-0.27	18.03	0.41
19	1719.00	96.00	1.10	346.74	1718.88	2.87	-0.83	2.87	0.27	-8.32	0.30
20	1813.00	94.00	0.84	354.56	1812.87	2.60	0.74	2.60	-0.28	8.32	0.31
21	1908.00	95.00	0.63	245.43	1907.87	2.06	1.21	2.06	-0.22	-114.87	1.27
22	2003.00	95.00	0.78	267.58	2002.86	0.94	0.97	0.94	0.16	23.32	0.32
23	2097.00	94.00	0.75	283.27	2096.85	-0.30	1.08	-0.30	-0.03	16.69	0.22
24	2189.00	92.00	0.88	358.43	2188.84	-0.91	1.93	-0.91	0.14	81.70	1.09
25	2284.00	95.00	1.01	338.83	2283.83	-1.23	3.44	-1.23	0.14	-20.63	0.36
26	2378.00	94.00	0.92	101.87	2377.82	-0.79	4.05	-0.79	-0.10	-252.09	1.81
27	2471.00	93.00	1.01	107.41	2470.81	0.72	3.65	0.72	0.10	5.96	0.14
28	2566.00	95.00	1.05	106.62	2565.80	2.36	3.16	2.36	0.04	-0.83	0.04

29	2661.00	95.00	0.57	101.79	2660.79	3.65	2.81	3.65	-0.51	-5.08	0.51
30	2756.00	95.00	0.35	110.14	2755.78	4.39	2.61	4.39	-0.23	8.79	0.24
31	2851.00	95.00	0.62	130.61	2850.78	5.05	2.18	5.05	0.28	21.55	0.33
32	2944.00	93.00	0.44	137.82	2943.78	5.67	1.59	5.67	-0.19	7.75	0.21
33	3038.00	94.00	0.53	107.94	3037.77	6.33	1.19	6.33	0.10	-31.79	0.28
34	3133.00	95.00	0.44	86.40	3132.77	7.11	1.07	7.11	-0.09	-22.67	0.21
35	3227.00	94.00	0.35	115.50	3226.77	7.73	0.97	7.73	-0.10	30.96	0.23
36	3322.00	95.00	0.35	93.70	3321.77	8.28	0.83	8.28	0.00	-22.95	0.14
37	3416.00	94.00	0.26	105.21	3415.76	8.77	0.75	8.77	-0.10	12.24	0.12
38	3512.00	96.00	0.35	135.36	3511.76	9.19	0.49	9.19	0.09	31.41	0.19
39	3606.00	94.00	0.53	117.25	3605.76	9.78	0.08	9.78	0.19	-19.27	0.24
40	3700.00	94.00	0.35	125.78	3699.76	10.40	-0.28	10.40	-0.19	9.07	0.20
41	3795.00	95.00	0.44	106.88	3794.76	10.98	-0.56	10.98	0.09	-19.89	0.17
42	3889.00	94.00	0.44	123.67	3888.75	11.63	-0.86	11.63	0.00	17.86	0.14
43	3983.00	94.00	0.31	119.01	3982.75	12.15	-1.19	12.15	-0.14	-4.96	0.14
44	4078.00	95.00	0.31	103.46	4077.75	12.63	-1.37	12.63	0.00	-16.37	0.09
45	4170.00	92.00	0.13	121.47	4169.75	12.96	-1.48	12.96	-0.20	19.58	0.21
46	4265.00	95.00	0.09	117.52	4264.75	13.12	-1.57	13.12	-0.04	-4.16	0.04
47	4360.00	95.00	0.09	177.20	4359.75	13.19	-1.68	13.19	0.00	62.82	0.09
48	4455.00	95.00	0.22	137.65	4454.75	13.31	-1.89	13.31	0.14	-41.63	0.17
49	4550.00	95.00	0.04	153.11	4549.75	13.45	-2.06	13.45	-0.19	16.27	0.19
50	4644.00	94.00	0.35	25.94	4643.75	13.59	-1.83	13.59	0.33	-135.29	0.40
51	4739.00	95.00	0.40	53.80	4738.74	13.98	-1.37	13.98	0.05	29.33	0.20
52	4834.00	95.00	0.44	45.89	4833.74	14.51	-0.92	14.51	0.04	-8.33	0.07
53	4930.00	96.00	0.31	1.15	4929.74	14.78	-0.41	14.78	-0.14	-46.60	0.32
54	5024.00	94.00	0.66	10.73	5023.74	14.89	0.38	14.89	0.37	10.19	0.38
55	5119.00	95.00	0.92	49.31	5118.73	15.57	1.42	15.57	0.27	40.61	0.61
56	5213.00	94.00	0.88	65.75	5212.72	16.80	2.20	16.80	-0.04	17.49	0.28
57	5308.00	95.00	1.01	71.46	5307.70	18.26	2.77	18.26	0.14	6.01	0.17
58	5402.00	94.00	0.70	58.98	5401.69	19.54	3.33	19.54	-0.33	-13.28	0.38
59	5497.00	95.00	1.10	62.06	5496.68	20.84	4.06	20.84	0.42	3.24	0.42
60	5591.00	94.00	0.97	85.88	5590.67	22.43	4.54	22.43	-0.14	25.34	0.47
61	5686.00	95.00	0.88	81.75	5685.65	23.96	4.70	23.96	-0.09	-4.35	0.12
62	5781.00	95.00	0.88	74.54	5780.64	25.38	5.00	25.38	0.00	-7.59	0.12

63	5876.00	95.00	0.75	74.54	5875.63	26.68	5.36	26.68	-0.14	0.00	0.14
64	5970.00	94.00	0.62	74.36	5969.63	27.77	5.66	27.77	-0.14	-0.19	0.14
65	6064.00	94.00	0.66	61.62	6063.62	28.73	6.05	28.73	0.04	-13.55	0.16
66	6087.00	23.00	0.70	40.88	6086.62	28.94	6.22	28.94	0.17	-90.17	1.08
67	6161.00	74.00	0.48	42.81	6160.62	29.45	6.79	29.45	-0.30	2.61	0.30
68	6255.00	94.00	0.75	118.48	6254.61	30.26	6.79	30.26	0.29	80.50	0.83
69	6350.00	95.00	0.57	109.43	6349.60	31.25	6.33	31.25	-0.19	-9.53	0.22
70	6445.00	95.00	0.26	85.53	6444.60	31.91	6.19	31.91	-0.33	-25.16	0.37
71	6539.00	94.00	0.44	3.79	6538.60	32.14	6.57	32.14	0.19	-86.96	0.51
72	6634.00	95.00	0.53	119.72	6633.60	32.55	6.72	32.55	0.09	122.03	0.87
73	6727.00	93.00	0.26	30.51	6726.60	33.03	6.69	33.03	-0.29	-95.92	0.63
74	6823.00	96.00	0.66	12.49	6822.59	33.26	7.41	33.26	0.42	-18.77	0.44
75	6916.00	93.00	0.79	102.84	6915.59	34.00	7.79	34.00	0.14	97.15	1.11
76	7012.00	96.00	0.53	94.84	7011.58	35.09	7.61	35.09	-0.27	-8.33	0.29
77	7105.00	93.00	0.97	184.23	7104.58	35.46	6.79	35.46	0.47	96.12	1.18
78	7199.00	94.00	1.14	195.92	7198.56	35.15	5.09	35.15	0.18	12.44	0.29
79	7294.00	95.00	1.36	184.14	7293.54	34.81	3.06	34.81	0.23	-12.40	0.35
80	7388.00	94.00	0.40	72.52	7387.53	35.04	2.05	35.04	-1.02	-118.74	1.65
81	7482.00	94.00	0.53	80.69	7481.53	35.78	2.22	35.78	0.14	8.69	0.15
82	7577.00	95.00	0.62	67.95	7576.52	36.69	2.48	36.69	0.09	-13.41	0.16
83	7671.00	94.00	0.53	63.90	7670.52	37.55	2.86	37.55	-0.10	-4.31	0.11
84	7766.00	95.00	0.48	50.72	7765.51	38.25	3.31	38.25	-0.05	-13.87	0.13
85	7858.00	92.00	0.22	50.11	7857.51	38.69	3.66	38.69	-0.28	-0.66	0.28
86	7953.00	95.00	0.18	136.15	7952.51	38.93	3.67	38.93	-0.04	90.57	0.29
87	8047.00	94.00	0.04	243.55	8046.51	39.00	3.55	39.00	-0.15	114.26	0.21
88	8141.00	94.00	0.44	288.73	8140.51	38.63	3.65	38.63	0.43	48.06	0.44
89	8236.00	95.00	1.14	100.64	8235.51	39.22	3.60	39.22	0.74	-197.99	1.66
90	8330.00	94.00	1.36	98.97	8329.48	41.24	3.25	41.24	0.23	-1.78	0.24
91	8424.00	94.00	0.53	255.86	8423.48	41.92	2.97	41.92	-0.88	166.90	1.98
92	8518.00	94.00	0.62	258.76	8517.47	41.00	2.77	41.00	0.10	3.08	0.10
93	8613.00	95.00	0.57	276.95	8612.47	40.02	2.72	40.02	-0.05	19.15	0.20
94	8707.00	94.00	0.66	279.68	8706.46	39.03	2.87	39.03	0.10	2.90	0.10
95	8802.00	95.00	0.57	264.12	8801.46	38.02	2.91	38.02	-0.09	-16.38	0.20
96	8897.00	95.00	0.48	271.68	8896.45	37.15	2.88	37.15	-0.09	7.96	0.12

97	8990.00	93.00	0.92	65.31	8989.45	37.44	3.20	37.44	0.47	165.19	1.47
98	9084.00	94.00	0.88	67.07	9083.44	38.79	3.80	38.79	-0.04	1.87	0.05
99	9179.00	95.00	0.70	70.41	9178.43	40.01	4.28	40.01	-0.19	3.52	0.20
100	9273.00	94.00	0.70	67.86	9272.42	41.08	4.68	41.08	0.00	-2.71	0.03
101	9367.00	94.00	0.75	70.50	9366.41	42.19	5.11	42.19	0.05	2.81	0.06
102	9462.00	95.00	0.66	59.95	9461.41	43.25	5.59	43.25	-0.09	-11.11	0.17
103	9555.00	93.00	0.66	51.69	9554.40	44.14	6.19	44.14	0.00	-8.88	0.10
104	9649.00	94.00	0.97	117.17	9648.39	45.27	6.16	45.27	0.33	69.66	0.98
105	9743.00	94.00	0.84	119.19	9742.38	46.58	5.46	46.58	-0.14	2.15	0.14
106	9837.00	94.00	0.79	117.43	9836.37	47.75	4.83	47.75	-0.05	-1.87	0.06
107	9931.00	94.00	0.88	107.59	9930.36	49.02	4.31	49.02	0.10	-10.47	0.18
108	10024.00	93.00	0.84	101.08	10023.35	50.37	3.96	50.37	-0.04	-7.00	0.11
109	10119.00	95.00	0.62	82.19	10118.34	51.56	3.90	51.56	-0.23	-19.88	0.34
110	10214.00	95.00	0.40	77.70	10213.34	52.39	4.04	52.39	-0.23	-4.73	0.24
111	10245.00	31.00	0.40	69.44	10244.34	52.60	4.10	52.60	0.00	-26.65	0.19



Operator: Oasis Petroleum	
Well : KLINE FEDERAL 5300 41-18 12TX	
MWD Providers	Gyrodatta
Directional Supervision:	RPM

Section:	18	QQ:	SW SW	County:	McKenzie	State:	ND
Township:	153	N/S:	N	Footages:	434	FN/SL:	S
Range:	100	E/W:	W		237	FE/WL:	W

Vertical Section Plane:	90.00
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#	MD	Inc.	Azm.	T.V.D.	Ver. Sect.	Coordinates		DLS
						+N / -S	+E / -W	
Tie	10245.00	0.40	69.44	10244.34	52.60	4.10	52.60	0.19
1	10277.00	0.53	87.37	10276.34	52.85	4.15	52.85	0.61
2	10309.00	2.99	112.42	10308.32	53.77	3.83	53.77	7.87
3	10340.00	8.26	114.44	10339.16	56.55	2.60	56.55	17.01
4	10372.00	12.79	113.91	10370.61	61.88	0.22	61.88	14.16
5	10404.00	16.84	111.89	10401.54	69.42	-2.95	69.42	12.76
6	10436.00	21.63	110.57	10431.75	79.25	-6.75	79.25	15.03
7	10467.00	23.78	111.89	10460.35	90.40	-11.09	90.40	7.13
8	10500.00	26.29	114.88	10490.24	103.21	-16.65	103.21	8.52
9	10531.00	29.05	118.84	10517.70	116.03	-23.17	116.03	10.69
10	10563.00	32.26	120.24	10545.22	130.22	-31.22	130.22	10.28
11	10594.00	35.16	118.66	10571.01	145.20	-39.67	145.20	9.77
12	10626.00	39.21	117.34	10596.50	162.28	-48.74	162.28	12.90
13	10657.00	43.16	114.09	10619.83	180.67	-57.57	180.67	14.49
14	10689.00	45.45	113.30	10642.73	201.14	-66.55	201.14	7.36
15	10720.00	46.59	109.52	10664.26	221.90	-74.68	221.90	9.51
16	10752.00	50.11	109.34	10685.52	244.45	-82.63	244.45	11.01
17	10783.00	54.20	109.96	10704.53	267.50	-90.87	267.50	13.29
18	10815.00	58.11	110.22	10722.35	292.45	-99.99	292.45	12.24
19	10847.00	61.63	111.28	10738.41	318.33	-109.80	318.33	11.37
20	10877.00	61.75	111.45	10752.64	342.92	-119.42	342.92	0.64
21	10909.00	65.14	112.16	10766.94	369.49	-130.06	369.49	10.78
22	10940.00	69.67	112.68	10778.85	395.94	-140.97	395.94	14.69
23	10972.00	73.58	113.30	10788.94	423.89	-152.83	423.89	12.36
24	11003.00	76.62	114.27	10796.91	451.30	-164.91	451.30	10.26
25	11035.00	78.07	114.00	10803.92	479.80	-177.68	479.80	4.61
26	11067.00	82.42	114.71	10809.34	508.52	-190.68	508.52	13.77
27	11112.00	89.89	114.18	10812.36	549.36	-209.25	549.36	16.64
28	11179.00	90.90	110.67	10811.89	611.29	-234.80	611.29	5.45
29	11211.00	90.29	112.96	10811.56	640.99	-246.69	640.99	7.41
30	11242.00	90.15	111.81	10811.44	669.65	-258.50	669.65	3.74
31	11273.00	89.32	109.44	10811.59	698.66	-269.42	698.66	8.10

32	11304.00	87.74	108.74	10812.38	727.95	-279.55	727.95	5.57
33	11336.00	87.08	106.45	10813.83	758.42	-289.22	758.42	7.44
34	11366.00	86.99	104.25	10815.38	787.31	-297.15	787.31	7.33
35	11397.00	87.30	103.64	10816.92	817.36	-304.61	817.36	2.21
36	11428.00	88.90	101.97	10817.95	847.57	-311.47	847.57	7.46
37	11459.00	88.79	99.68	10818.58	878.01	-317.29	878.01	7.39
38	11490.00	89.41	97.57	10819.06	908.65	-321.94	908.65	7.09
39	11521.00	90.37	96.17	10819.12	939.43	-325.65	939.43	5.48
40	11552.00	90.11	94.32	10818.99	970.30	-328.48	970.30	6.03
41	11584.00	89.76	92.39	10819.03	1002.24	-330.36	1002.24	6.13
42	11615.00	90.37	90.72	10818.99	1033.23	-331.20	1033.23	5.74
43	11646.00	90.81	88.61	10818.67	1064.22	-331.02	1064.22	6.95
44	11708.00	91.38	86.41	10817.49	1126.15	-328.32	1126.15	3.66
45	11740.00	91.47	87.12	10816.69	1158.09	-326.52	1158.09	2.24
46	11771.00	92.35	86.59	10815.66	1189.02	-324.82	1189.02	3.31
47	11802.00	90.55	88.96	10814.88	1219.99	-323.62	1219.99	9.60
48	11833.00	90.50	89.00	10814.59	1250.98	-323.06	1250.98	0.21
49	11864.00	88.75	89.05	10814.79	1281.97	-322.54	1281.97	5.65
50	11927.00	89.10	89.66	10815.98	1344.96	-321.83	1344.96	1.12
51	11958.00	89.05	87.29	10816.48	1375.94	-321.00	1375.94	7.65
52	12020.00	89.85	88.87	10817.07	1437.90	-318.93	1437.90	2.86
53	12113.00	89.93	89.14	10817.25	1530.89	-317.31	1530.89	0.30
54	12145.00	89.76	87.99	10817.34	1562.88	-316.51	1562.88	3.63
55	12207.00	88.92	88.35	10818.05	1624.84	-314.53	1624.84	1.47
56	12238.00	88.70	85.62	10818.70	1655.79	-312.90	1655.79	8.83
57	12301.00	88.44	85.18	10820.27	1718.56	-307.85	1718.56	0.81
58	12393.00	91.65	83.95	10820.20	1810.14	-299.14	1810.14	3.74
59	12424.00	91.43	83.34	10819.36	1840.93	-295.71	1840.93	2.09
60	12486.00	90.02	84.30	10818.58	1902.57	-289.03	1902.57	2.75
61	12518.00	89.01	87.12	10818.85	1934.47	-286.64	1934.47	9.36
62	12580.00	89.76	86.74	10819.51	1996.38	-283.32	1996.38	1.36
63	12612.00	89.49	86.59	10819.72	2028.32	-281.46	2028.32	0.97
64	12675.00	87.96	86.06	10821.13	2091.18	-277.42	2091.18	2.57
65	12707.00	87.96	86.59	10822.26	2123.09	-275.37	2123.09	1.66
66	12769.00	88.75	87.47	10824.04	2184.98	-272.16	2184.98	1.91
67	12801.00	88.13	88.61	10824.92	2216.95	-271.07	2216.95	4.05
68	12833.00	88.22	87.56	10825.93	2248.92	-270.00	2248.92	3.29
69	12864.00	89.23	86.85	10826.62	2279.87	-268.49	2279.87	3.98
70	12927.00	90.77	88.17	10826.62	2342.81	-265.75	2342.81	3.22
71	12959.00	90.95	87.64	10826.14	2374.78	-264.58	2374.78	1.75
72	13022.00	88.92	86.59	10826.22	2437.70	-261.41	2437.70	3.63
73	13053.00	88.00	87.29	10827.05	2468.64	-259.75	2468.64	3.73
74	13116.00	87.91	89.23	10829.30	2531.57	-257.84	2531.57	3.08
75	13148.00	88.48	88.43	10830.30	2563.55	-257.19	2563.55	3.07

76	13211.00	89.98	90.54	10831.15	2626.53	-256.62	2626.53	4.11
77	13242.00	90.29	90.46	10831.08	2657.53	-256.89	2657.53	1.03
78	13274.00	90.81	90.28	10830.77	2689.53	-257.10	2689.53	1.72
79	13337.00	91.96	90.90	10829.25	2752.51	-257.75	2752.51	2.07
80	13368.00	90.51	91.51	10828.58	2783.49	-258.40	2783.49	5.07
81	13431.00	90.73	90.28	10827.90	2846.48	-259.38	2846.48	1.98
82	13463.00	88.40	90.81	10828.14	2878.48	-259.69	2878.48	7.47
83	13526.00	88.22	91.16	10830.00	2941.44	-260.77	2941.44	0.62
84	13588.00	86.37	89.58	10832.93	3003.36	-261.17	3003.36	3.92
85	13620.00	86.33	90.54	10834.96	3035.30	-261.21	3035.30	3.00
86	13651.00	86.90	90.37	10836.79	3066.24	-261.45	3066.24	1.92
87	13714.00	88.75	91.07	10839.18	3129.19	-262.24	3129.19	3.14
88	13745.00	89.63	90.19	10839.62	3160.18	-262.58	3160.18	4.01
89	13808.00	91.12	89.75	10839.21	3223.18	-262.55	3223.18	2.47
90	13840.00	90.37	88.96	10838.79	3255.18	-262.19	3255.18	3.40
91	13903.00	90.42	90.02	10838.36	3318.17	-261.63	3318.17	1.68
92	13935.00	88.79	89.75	10838.58	3350.17	-261.57	3350.17	5.16
93	13998.00	89.76	88.08	10839.38	3413.15	-260.37	3413.15	3.07
94	14092.00	90.07	87.99	10839.52	3507.09	-257.15	3507.09	0.34
95	14124.00	89.58	89.31	10839.61	3539.08	-256.40	3539.08	4.40
96	14187.00	91.52	88.61	10839.01	3602.07	-255.25	3602.07	3.27
97	14281.00	90.73	88.43	10837.16	3696.02	-252.82	3696.02	0.86
98	14313.00	90.99	89.66	10836.68	3728.01	-252.29	3728.01	3.93
99	14375.00	90.07	89.49	10836.11	3790.00	-251.83	3790.00	1.51
100	14407.00	87.74	90.28	10836.72	3821.99	-251.77	3821.99	7.69
101	14470.00	87.91	91.07	10839.11	3884.94	-252.51	3884.94	1.28
102	14565.00	88.31	90.98	10842.25	3979.88	-254.21	3979.88	0.43
103	14628.00	90.77	90.63	10842.75	4042.86	-255.09	4042.86	3.94
104	14659.00	89.85	89.40	10842.58	4073.86	-255.10	4073.86	4.95
105	14722.00	89.05	88.61	10843.19	4136.85	-254.01	4136.85	1.78
106	14754.00	89.41	89.14	10843.62	4168.84	-253.38	4168.84	2.00
107	14817.00	87.96	89.66	10845.06	4231.82	-252.72	4231.82	2.45
108	14848.00	87.82	89.49	10846.21	4262.80	-252.49	4262.80	0.71
109	14943.00	90.33	88.79	10847.74	4357.76	-251.06	4357.76	2.74
110	14974.00	91.30	89.23	10847.30	4388.76	-250.53	4388.76	3.44
111	15037.00	92.04	88.52	10845.46	4451.72	-249.29	4451.72	1.63
112	15131.00	88.31	90.81	10845.18	4545.69	-248.74	4545.69	4.66
113	15225.00	88.79	90.54	10847.55	4639.66	-249.85	4639.66	0.59
114	15320.00	87.74	89.79	10850.43	4734.61	-250.12	4734.61	1.36
115	15414.00	87.60	89.66	10854.25	4828.53	-249.67	4828.53	0.20
116	15477.00	88.18	89.31	10856.57	4891.48	-249.11	4891.48	1.08
117	15509.00	88.79	87.73	10857.42	4923.46	-248.28	4923.46	5.29
118	15540.00	89.85	88.08	10857.79	4954.44	-247.15	4954.44	3.60
119	15603.00	91.21	87.82	10857.20	5017.39	-244.89	5017.39	2.20

120	15635.00	90.20	88.26	10856.81	5049.37	-243.80	5049.37	3.44
121	15698.00	88.66	88.96	10857.44	5112.35	-242.27	5112.35	2.69
122	15761.00	89.27	90.19	10858.57	5175.33	-241.80	5175.33	2.18
123	15792.00	89.58	91.95	10858.89	5206.33	-242.38	5206.33	5.76
124	15855.00	89.98	93.44	10859.13	5269.25	-245.34	5269.25	2.45
125	15887.00	90.55	91.95	10858.98	5301.22	-246.85	5301.22	4.99
126	15950.00	89.85	91.07	10858.76	5364.19	-248.51	5364.19	1.78
127	15981.00	90.07	90.19	10858.78	5395.19	-248.85	5395.19	2.93
128	16013.00	90.51	90.28	10858.62	5427.19	-248.98	5427.19	1.40
129	16076.00	91.38	89.93	10857.58	5490.18	-249.10	5490.18	1.49
130	16107.00	92.35	89.58	10856.57	5521.16	-248.96	5521.16	3.33
131	16170.00	93.01	90.54	10853.63	5584.09	-249.03	5584.09	1.85
132	16265.00	89.45	90.37	10851.59	5679.05	-249.78	5679.05	3.75
133	16359.00	89.05	90.46	10852.82	5773.04	-250.46	5773.04	0.44
134	16454.00	89.14	89.75	10854.32	5868.03	-250.64	5868.03	0.75
135	16548.00	89.76	90.72	10855.22	5962.02	-251.02	5962.02	1.22
136	16643.00	91.03	89.49	10854.56	6057.02	-251.20	6057.02	1.86
137	16737.00	91.52	90.19	10852.47	6150.99	-250.94	6150.99	0.91
138	16831.00	91.47	88.96	10850.02	6244.96	-250.24	6244.96	1.31
139	16895.00	91.43	89.49	10848.40	6308.93	-249.37	6308.93	0.83
140	16927.00	90.42	89.49	10847.88	6340.93	-249.09	6340.93	3.16
141	16958.00	89.93	90.10	10847.79	6371.92	-248.98	6371.92	2.52
142	17021.00	90.95	90.28	10847.31	6434.92	-249.19	6434.92	1.64
143	17053.00	89.93	88.08	10847.06	6466.91	-248.73	6466.91	7.58
144	17115.00	90.59	88.43	10846.78	6528.88	-246.84	6528.88	1.20
145	17210.00	87.16	88.43	10848.64	6623.82	-244.24	6623.82	3.61
146	17241.00	87.38	89.40	10850.12	6654.78	-243.65	6654.78	3.21
147	17304.00	88.79	88.79	10852.23	6717.73	-242.66	6717.73	2.44
148	17336.00	89.19	88.70	10852.79	6749.72	-241.96	6749.72	1.28
149	17399.00	89.98	87.82	10853.25	6812.69	-240.04	6812.69	1.88
150	17493.00	89.89	88.70	10853.35	6906.64	-237.19	6906.64	0.94
151	17588.00	91.30	91.60	10852.37	7001.62	-237.44	7001.62	3.39
152	17682.00	93.85	91.69	10848.14	7095.48	-240.13	7095.48	2.71
153	17777.00	93.85	90.28	10841.76	7190.25	-241.76	7190.25	1.48
154	17808.00	92.26	90.37	10840.11	7221.21	-241.94	7221.21	5.14
155	17871.00	92.62	90.02	10837.43	7284.15	-242.15	7284.15	0.80
156	17902.00	89.98	90.90	10836.73	7315.14	-242.40	7315.14	8.98
157	17933.00	90.24	91.25	10836.67	7346.13	-242.98	7346.13	1.41
158	17965.00	89.27	90.10	10836.80	7378.13	-243.36	7378.13	4.70
159	17996.00	88.62	90.63	10837.37	7409.12	-243.56	7409.12	2.71
160	18059.00	87.12	91.07	10839.72	7472.07	-244.49	7472.07	2.48
161	18153.00	87.96	91.16	10843.75	7565.96	-246.32	7565.96	0.90
162	18215.00	89.85	87.99	10844.94	7627.94	-245.86	7627.94	5.95
163	18247.00	90.37	87.99	10844.87	7659.92	-244.74	7659.92	1.63

164	18310.00	88.75	88.35	10845.36	7722.88	-242.73	7722.88	2.63
165	18341.00	89.01	87.56	10845.96	7753.86	-241.62	7753.86	2.68
166	18435.00	88.26	89.49	10848.20	7847.79	-239.20	7847.79	2.20
167	18530.00	88.13	90.10	10851.20	7942.75	-238.86	7942.75	0.66
168	18593.00	88.31	90.28	10853.15	8005.72	-239.07	8005.72	0.40
169	18624.00	88.48	90.02	10854.02	8036.70	-239.15	8036.70	1.00
170	18687.00	89.14	90.54	10855.33	8099.69	-239.46	8099.69	1.33
171	18718.00	89.80	89.66	10855.62	8130.69	-239.51	8130.69	3.55
172	18812.00	91.21	90.90	10854.79	8224.68	-239.97	8224.68	2.00
173	18906.00	93.41	88.87	10851.00	8318.59	-239.78	8318.59	3.18
174	18937.00	92.40	89.58	10849.43	8349.55	-239.37	8349.55	3.98
175	18968.00	93.14	89.84	10847.93	8380.51	-239.21	8380.51	2.53
176	19000.00	91.96	88.43	10846.51	8412.47	-238.73	8412.47	5.74
177	19031.00	90.11	90.02	10845.95	8443.46	-238.31	8443.46	7.87
178	19094.00	87.87	89.93	10847.06	8506.45	-238.28	8506.45	3.56
179	19157.00	88.57	91.25	10849.01	8569.41	-238.93	8569.41	2.37
180	19188.00	89.23	89.75	10849.61	8600.41	-239.20	8600.41	5.29
181	19282.00	87.34	87.91	10852.42	8694.34	-237.28	8694.34	2.81
182	19313.00	87.69	87.56	10853.77	8725.28	-236.06	8725.28	1.60
183	19376.00	88.79	87.03	10855.70	8788.18	-233.09	8788.18	1.94
184	19470.00	88.09	91.77	10858.26	8882.11	-232.10	8882.11	5.10
185	19501.00	87.82	93.53	10859.37	8913.06	-233.53	8913.06	5.74
186	19533.00	88.22	92.92	10860.47	8944.99	-235.33	8944.99	2.28
187	19564.00	88.35	93.27	10861.40	8975.93	-237.01	8975.93	1.20
188	19596.00	88.57	91.16	10862.26	9007.89	-238.24	9007.89	6.63
189	19659.00	88.31	88.35	10863.98	9070.86	-237.97	9070.86	4.48
190	19690.00	88.44	88.79	10864.86	9101.84	-237.20	9101.84	1.48
191	19721.00	89.27	89.23	10865.47	9132.83	-236.66	9132.83	3.03
192	19753.00	89.63	89.49	10865.78	9164.83	-236.31	9164.83	1.39
193	19784.00	90.20	90.02	10865.83	9195.83	-236.17	9195.83	2.51
194	19847.00	91.30	88.87	10865.00	9258.82	-235.56	9258.82	2.53
195	19941.00	92.35	89.66	10862.01	9352.76	-234.36	9352.76	1.40
196	19973.00	92.09	90.02	10860.77	9384.73	-234.27	9384.73	1.39
197	20004.00	92.57	90.19	10859.51	9415.71	-234.33	9415.71	1.64
198	20035.00	93.45	89.14	10857.88	9446.66	-234.14	9446.66	4.42
199	20067.00	91.38	90.02	10856.53	9478.63	-233.91	9478.63	7.03
200	20129.00	89.76	90.72	10855.92	9540.63	-234.31	9540.63	2.85
201	20193.00	90.11	91.07	10855.99	9604.62	-235.31	9604.62	0.77
202	20224.00	90.33	90.81	10855.87	9635.61	-235.82	9635.61	1.10
203	20287.00	91.30	90.81	10854.97	9698.60	-236.71	9698.60	1.54
204	20318.00	91.65	90.10	10854.18	9729.59	-236.96	9729.59	2.55
205	20413.00	92.92	89.66	10850.39	9824.51	-236.76	9824.51	1.41
206	20507.00	93.41	89.31	10845.20	9918.36	-235.91	9918.36	0.64
Proj	20583.00	93.41	89.31	10840.68	9994.22	-235.00	9994.22	0.00

Kline Federal 5300 41-18 12TX LITHOLOGY

Geologists caught lagged samples in 30' intervals in the vertical hole, from 8,350' to 10,300' MD; 30' intervals in the curve build from 10,300' to 11,190' MD; and 50' samples in the lateral from 11,190' to TD at 20,583'

Electric geophysical log, sample and/or MWD gamma ray markers and tops are included in the sample descriptions below for reference. Samples were examined wet and dry under a binocular microscope in approximately 30' to 50' intervals. Sample descriptions begin in the **Kibbey** formation. The drilling fluid was diesel invert in the vertical, curve, and lateral.

Drilling vertical hole in Dakota Formation

8350-8384 SILTSTONE: reddish brown, occasional, gray to light gray, brown to light brown gray, soft, sub-blocky, earthy texture, no visible porosity, no visible oil stain

Kibbey Lime

8,384' MD, 8,384' TVD (-6,302')

8384-8410 SILTSTONE: reddish brown, occasional, gray to light gray, brown to light brown gray, soft, sub-blocky, earthy texture, no visible porosity, no visible oil stain; LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

8410-8440 SILTY SANDSTONE: medium gray, fine grained, firm to friable, rounded, well sorted, moderately cemented, calcite cement, possible intergranular porosity, no visible oil stain; SILTSTONE: reddish brown, occasional, gray to light gray, brown to light brown gray, soft, sub-blocky, earthy texture, no visible porosity, no visible oil stain

8440-8470 SILTY SANDSTONE: medium gray, fine grained, firm to friable, rounded, well sorted, moderately cemented, calcite cement, possible intergranular porosity, no visible oil stain; SILTSTONE: reddish brown, occasional, gray to light gray, brown to light brown gray, soft, sub-blocky, earthy texture, no visible porosity, no visible oil stain

8470-8500 SILTY SANDSTONE: medium gray, fine grained, firm to friable, rounded, well sorted, moderately cemented, calcite cement, possible intergranular porosity, no visible oil stain; SILTSTONE: reddish brown, occasional, gray to light gray, brown to light brown gray, soft, sub-blocky, earthy texture, no visible porosity, no visible oil stain

8500-8530 SILTY SANDSTONE: medium gray, fine grained, firm to friable, rounded, well sorted, moderately cemented, calcite cement, possible intergranular porosity, no visible oil stain; SILTSTONE: reddish brown, occasional, gray to light gray, brown to light brown gray, soft, sub-blocky, earthy texture, no visible porosity, no visible oil stain

8530-8556 SILTY SANDSTONE: medium gray, fine grained, firm to friable, rounded, well sorted, moderately cemented, calcite cement, possible intergranular porosity, no visible oil stain; SILTSTONE: reddish brown, occasional, gray to light gray, brown to light brown gray, soft, sub-blocky, earthy texture, no visible porosity, no visible oil stain

8556-8590 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain

8590-8620 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain

8620-8650 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

8650-8680 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain

8680-8710 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain

8710-8740 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

8740-8770 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain

8770-8800 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain

8800-8830 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

8830-8860 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

8860-8890 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

88900-8920 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

8920-8950 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

8950-8980 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

8980-9010 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9010-9040 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9040-9070 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9070-9100 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9100-9130 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9130-91760 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9160-9190 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9190-9206 SALT: translucent, trace off white, crystalline, firm, crystalline texture, no visible porosity, no visible oil stain, LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

Base Charles Salt

9,206' MD, 9,206' TVD (-7,124')

9206-9250 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9250-9280 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9280-9310 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9310-9340 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9340-9370 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9370-9400 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9400-9428 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

Mission Canyon

9,428' MD, 9,428' TVD (-7,346')

9428-9460 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9460-9490 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9490-9520 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9520-9550 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9550-9580 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9580-9610 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9610-9640 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9640-9670 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9670-9700 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9700-9730 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9730-9760 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9760-9790 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9790-9820 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9820-9850 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9850-9880 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9880-9910 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9910-9940 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

9940-9975 LIMESTONE: wackestone, light gray to medium gray, light gray brown to medium gray brown, occasional off white friable to firm, crystalline to microcrystalline

Lodgepole

9,975' MD, 9,975' TVD (-7,893')

9975-10000 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10000-10030 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10030-10060 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10060-10090 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10090-10120 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10120-10150 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10150-10180 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10180-10210 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10210-10240 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10240-10270 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10270-10300 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10,300' Vertical TD reached @ 11:00 hours CST, 22 October 2014

Drilling curve build in the Lodgepole Formation

10300-10330 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10330-10360 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10360-10390 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10390-10420 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10420-10450 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10450-10480 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10480-10510 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10510-10540 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10540-10570 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10570-10600 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10600-10630 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10630-10660 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10660-10690 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10690-10720 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10720-10750 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

10750-10781 ARGILLACEOUS LIMESTONE: wackestone, medium gray brown to light gray brown, firm, crystalline texture, trace sparry calcareous, carbonaceous, possible intercrystalline porosity, possible slightly light brown spotty oil stain

False Bakken **10,781' MD, 10,704' TVD (-8,622')**

10781-10798 SHALE: black, dark gray black, firm, sub-blocky, earthy, calcareous cement, petro, carbonaceous, abundant fine disseminated pyrite, fine laminated

Upper Bakken Shale **10,798' MD, 10,713' TVD (-8,631')**

10798 – 10829 SHALE: black, dark gray black, firm, sub-blocky, earthy, calcareous cement, petro, carbonaceous, abundant fine disseminated pyrite, fine laminated

Middle Bakken **10,829' MD, 10,730' TVD (-8,648')**

10829-10840 SILTY SANDSTONE: common salt and pepper, occasional light gray to medium gray, visible white, very fine grained, soft to firm, moderately to well sorted, rounded to subrounded, moderately to poorly cemented, visible calcareous filled fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty to even oil stain

10840-10870 SILTY SANDSTONE: common salt and pepper, occasional light gray to medium gray, visible white, very fine grained, soft to firm, moderately to well sorted, rounded to subrounded, moderately to poorly cemented, visible calcareous filled fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty to even oil stain

10870-10918 SILTY SANDSTONE: common salt and pepper, occasional light gray to medium gray, visible white, very fine grained, soft to firm, moderately to well sorted, rounded to subrounded, moderately to poorly cemented, visible calcareous filled fractures, possible fracture porosity, common intergranular porosity, visible light brown spotty to even oil stain

Lower Bakken Shale

10,918' MD, 10,771' TVD (-8,689')

10918-10952 SHALE: black, dark gray black, firm, sub-blocky, earthy, calcareous cement, petro, carbonaceous, abundant fine disseminated pyrite, fine laminated

Pronghorn

10,952' MD, 10,783' TVD (-8,701')

10952-10960 SILTSTONE: medium gray, medium gray brown, friable to firm, sub-blocky to sub platy, dolomite cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity

10960-10990 SILTSTONE: medium gray, medium gray brown, friable to firm, sub-blocky to sub platy, dolomite cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity

10990-11011 SILTSTONE: medium gray, medium gray brown, friable to firm, sub-blocky to sub platy, dolomite cement, moderately cemented, trace disseminated pyrite, possible intergranular porosity

Three Forks

11,011' MD, 10,799' TVD (-8,717')

11011-11050 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

11050-11080 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

11080-11110 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

11110-11140 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

11140-11170 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

11170-11190 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

Curve TD of 11,190' @ 15:00 CST 24 October-2014

Intermediate 7" Casing Set @ 11,189' MD

11190-11250 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

11850-11900 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

12500-12550 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

13150-13200 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

13800-13850 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

14450-14500 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

15100-15150 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

15150-15200 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

15200-15250 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

15250-15300 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

15300-15350 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

15350-15400 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

15400-15450 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

15450-15500 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain; CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

15500-15550 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

15550-15600 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

15600-15650 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

15650-15700 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

15700-15750 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

15750-15800 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

15800-15850 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

15850-15900 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

15900-15950 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

15950-16000 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

16000-16050 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite; DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

16050-16100 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

16100-16150 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

16150-16200 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

16200-16250 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

16250-16300 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

16300-16350 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

16350-16400 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

16400-16450 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

16450-16500 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

16500-16550 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

17150-17200 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

17800-17850 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

18450-18500 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

19100-19150 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

19150-19200 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

19200-19250 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

19250-19300 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

19300-19350 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

19350-19400 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

19400-19450 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite; DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

19450-19500 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

19500-19550 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

19550-19600 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

19600-19650 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

19650-19700 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

19700-19750 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

19750-19800 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

19800-19850 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

19850-19900 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

19900-19950 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

19950-20000 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite

20000-20050 CLAYSTONE: off white to light gray, light mint green, moderately soft, sub-blocky to sub-angular, very slightly calcareous, trace silt crystalline porosity, trace disseminated and nodular pyrite; DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

20050-20100 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

20100-20150 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

20150-20200 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

20200-20250 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

20250-20300 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

20300-20350 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

20350-20400 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

20400-20450 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

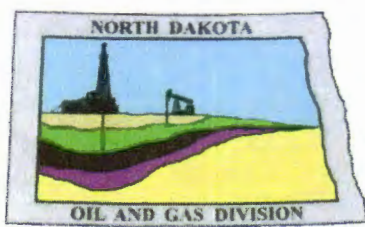
20450-20500 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

20500-20550 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

20550-20583 DOLOMITE: mudstone, light to medium gray, very fine crystalline, firm, earthy texture, trace silty, trace very fine sand, trace intercrystalline porosity, trace disseminated and nodular pyrite, trace spotty light brown oil stain

20,583' TD reached @ 06:15 hours CST, 26 January 2015





Oil and Gas Division

28658

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.dmr.nd.gov/oilgas

January 8, 2016

Michael Kukuk
OASIS PETRO NO AMER
1001 FANNIN ST. SUITE 1500
HOUSTON, TX 77002

RE: See Attached List of Wells

List Reference WF# 28658

Dear Michael Kukuk:

We have not received the geological reports on the referenced well list. Please submit one paper copy and one digital pdf file of this report for each well.

If you have any questions, please contact Richard Suggs at (701) 328-8020.

Sincerely,

Taylor Roth
Engineering Technician



Gyrodata Incorporated
301 Thelma Dr. #413
Casper, Wyoming 82601

307-472-0182
Fax: 307-472-0204

DIRECTIONAL SURVEY CERTIFICATION

RE: _____ Oasis Petroleum _____ (Operator)
_____ Kline Federal #5300 41-18 12TX _____ (Well Name & No.)
_____ 8.27 _____ (Magnetic Declination)

Surveyor: Richard Bradbury

I, Kristopher Bertsch, having personal knowledge of all the facts, hereby certify that the attached directional survey run from measured depth of 140 feet to a measured depth of 20507 feet and is projected to 20584 feet, is true and correct as determined from all available records.


Signature

MWD Coordinator
Title

Gyrodata, Inc.
Company

*All measurements recorded with a manned MWD Survey Tool.

A Gyrodata Directional Survey

FINAL DEFINITIVE COPY

for

OASIS PETROLEUM

Lease: Kline Federal Well: 5300 41-18 12TX, 8 3/4" Case, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M180

Run Date: 03 Mar 2015

Surveyor: Richard Bradbury

Calculation Method: MINIMUM CURVATURE

Survey Latitude: 48.068791 deg. N Longitude: 103.602346 deg. W

Magnetic Declination: 8.2700 deg

Total Correction: 8.2700 deg

Proposed Well Direction: 90.000 deg

Depth Reference: Rotary Table

Air Gap (RKB to Ground / RKB to MSL): 25.00

Ground Elevation (Ground to MSL / MSL to Mudline): 2057.00

Vertical Section Calculated from Well Head Location

Closure Calculated from Well Head Location

Horizontal Coordinates Calculated from Well Head Location

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 12TX, 8 3/4" Case, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M180

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
---------------------------	--------------	-----------------	--	---------------------------	---------------------------------------	-----------------------------------

0.00	0.00	0.00	0.00	0.00	0.0 0.0	0.00 N 0.00 E
------	------	------	------	------	---------	---------------

NO TIE IN DATA PROVIDED. TIE INTO SURFACE.

ALL AZIMUTHS ARE REFERENCED TO TRUE NORTH.

ALL MEASURED DEPTHS AND COORDINATES REFERENCED TO NABORS 486 R.K.B. AT 25 FT.

140-20507 FT. MD ARE MAGNETIC MEASUREMENT WHILE DRILLING SURVEYS.

STRAIGHT LINE PROJECTION TO BIT DEPTH AT 20584 FT. MD.

140.00	0.35	146.08	0.25	140.00	0.4 146.1	0.35 S 0.24 E
234.00	0.66	143.09	0.33	234.00	1.3 144.8	1.03 S 0.72 E
327.00	0.57	172.80	0.35	326.99	2.2 150.0	1.91 S 1.10 E
421.00	0.48	154.78	0.20	420.99	3.0 154.1	2.73 S 1.33 E
516.00	0.40	139.75	0.15	515.98	3.8 152.9	3.35 S 1.71 E
607.00	0.79	150.04	0.44	606.98	4.7 151.6	4.13 S 2.23 E
699.00	0.44	178.51	0.49	698.97	5.6 153.1	5.03 S 2.56 E
791.00	0.66	158.91	0.31	790.97	6.5 154.9	5.88 S 2.76 E
882.00	1.41	157.07	0.82	881.95	8.1 155.4	7.40 S 3.38 E
974.00	0.40	19.78	1.88	973.95	9.0 154.2	8.14 S 3.93 E
1066.00	0.22	101.87	0.47	1065.94	8.9 151.9	7.88 S 4.21 E
1158.00	0.22	177.28	0.29	1157.94	9.2 151.5	8.09 S 4.40 E
1250.00	0.44	325.29	0.69	1249.94	9.0 152.2	7.98 S 4.20 E
1342.00	0.88	351.66	0.57	1341.94	8.0 150.8	6.99 S 3.90 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 12TX, 8 3/4" Case, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M180

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. feet	AZIMUTH deg.	HORIZONTAL COORDINATES feet	
1433.00	0.92	4.14	0.22	1432.93	6.8	145.3	5.57 S	3.85 E
1528.00	1.10	337.60	0.52	1527.91	5.3	138.1	3.96 S	3.56 E
1623.00	0.84	354.73	0.41	1622.90	4.0	127.6	2.43 S	3.15 E
1719.00	1.10	346.74	0.30	1718.88	3.0	106.1	0.83 S	2.87 E
1813.00	0.84	354.56	0.31	1812.87	2.7	74.2	0.74 N	2.60 E
1908.00	0.63	245.43	1.27	1907.87	2.4	59.5	1.21 N	2.06 E
2003.00	0.78	267.58	0.32	2002.86	1.3	44.1	0.97 N	0.94 E
2097.00	0.75	283.27	0.22	2096.85	1.1	344.4	1.08 N	0.30 W
2189.00	0.88	358.43	1.09	2188.84	2.1	334.8	1.93 N	0.91 W
2284.00	1.01	338.83	0.36	2283.83	3.6	340.3	3.44 N	1.23 W
2378.00	0.92	101.87	1.81	2377.82	4.1	349.0	4.05 N	0.79 W
2471.00	1.01	107.41	0.14	2470.81	3.7	11.2	3.65 N	0.72 E
2566.00	1.05	106.62	0.04	2565.80	3.9	36.7	3.15 N	2.36 E
2661.00	0.57	101.79	0.51	2660.79	4.6	52.4	2.81 N	3.65 E
2756.00	0.35	110.14	0.24	2755.78	5.1	59.2	2.61 N	4.39 E
2851.00	0.62	130.61	0.33	2850.78	5.5	66.7	2.18 N	5.05 E
2944.00	0.44	137.82	0.21	2943.78	5.9	74.4	1.59 N	5.67 E
3038.00	0.53	107.94	0.28	3037.77	6.4	79.4	1.19 N	6.33 E
3133.00	0.44	86.40	0.21	3132.77	7.2	81.4	1.07 N	7.11 E
3227.00	0.35	115.50	0.23	3226.77	7.8	82.8	0.97 N	7.73 E
3322.00	0.35	93.70	0.14	3321.77	8.3	84.3	0.83 N	8.28 E
3416.00	0.26	105.21	0.12	3415.76	8.8	85.1	0.75 N	8.77 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 12TX, 8 3/4" Case, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M180

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. feet	AZIMUTH deg.	HORIZONTAL COORDINATES feet	
3512.00	0.35	135.36	0.19	3511.76	9.2	87.0	0.49 N	9.19 E
3606.00	0.53	117.25	0.24	3605.76	9.8	89.5	0.08 N	9.78 E
3700.00	0.35	125.78	0.20	3699.76	10.4	91.6	0.28 S	10.40 E
3795.00	0.44	106.88	0.17	3794.76	11.0	92.9	0.56 S	10.98 E
3889.00	0.44	123.67	0.14	3888.75	11.7	94.2	0.86 S	11.63 E
3983.00	0.31	119.01	0.14	3982.75	12.2	95.6	1.19 S	12.15 E
4078.00	0.31	103.46	0.09	4077.75	12.7	96.2	1.37 S	12.62 E
4170.00	0.13	121.47	0.21	4169.75	13.0	96.5	1.48 S	12.96 E
4265.00	0.09	117.52	0.04	4264.75	13.2	96.8	1.57 S	13.11 E
4360.00	0.09	177.20	0.09	4359.75	13.3	97.3	1.68 S	13.18 E
4455.00	0.22	137.65	0.17	4454.75	13.4	98.1	1.89 S	13.31 E
4550.00	0.04	153.11	0.19	4549.75	13.6	98.7	2.06 S	13.45 E
4644.00	0.35	25.94	0.40	4643.75	13.7	97.7	1.83 S	13.59 E
4739.00	0.40	53.80	0.20	4738.74	14.0	95.6	1.37 S	13.98 E
4834.00	0.44	45.89	0.07	4833.74	14.5	93.6	0.92 S	14.51 E
4930.00	0.31	1.15	0.32	4929.74	14.8	91.6	0.41 S	14.78 E
5024.00	0.66	10.73	0.38	5023.74	14.9	88.5	0.38 N	14.89 E
5119.00	0.92	49.31	0.61	5118.73	15.6	84.8	1.42 N	15.57 E
5213.00	0.88	65.75	0.28	5212.72	16.9	82.5	2.20 N	16.80 E
5308.00	1.01	71.46	0.17	5307.70	18.5	81.4	2.77 N	18.26 E
5402.00	0.70	58.98	0.38	5401.69	19.8	80.3	3.33 N	19.53 E
5497.00	1.10	62.06	0.42	5496.68	21.2	79.0	4.06 N	20.84 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 12TX, 8 3/4" Case, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M180

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. feet	AZIMUTH deg.	HORIZONTAL COORDINATES feet
5591.00	0.97	85.88	0.47	5590.67	22.9	78.6	4.54 N 22.43 E
5686.00	0.88	81.75	0.12	5685.65	24.4	78.9	4.70 N 23.95 E
5781.00	0.88	74.54	0.12	5780.64	25.9	78.9	5.00 N 25.38 E
5876.00	0.75	74.54	0.14	5875.63	27.2	78.6	5.36 N 26.68 E
5970.00	0.62	74.36	0.14	5969.63	28.3	78.5	5.66 N 27.76 E
6064.00	0.66	61.62	0.16	6063.62	29.4	78.1	6.05 N 28.73 E
6087.00	0.70	40.88	1.08	6086.62	29.6	77.9	6.22 N 28.94 E
6161.00	0.48	42.81	0.30	6160.62	30.2	77.0	6.79 N 29.44 E
6255.00	0.75	118.48	0.83	6254.61	31.0	77.4	6.79 N 30.25 E
6350.00	0.57	109.43	0.22	6349.60	31.9	78.5	6.33 N 31.24 E
6445.00	0.26	85.53	0.37	6444.60	32.5	79.0	6.19 N 31.90 E
6539.00	0.44	3.79	0.51	6538.60	32.8	78.4	6.57 N 32.14 E
6634.00	0.53	119.72	0.87	6633.60	33.2	78.3	6.72 N 32.55 E
6727.00	0.26	30.51	0.63	6726.60	33.7	78.6	6.68 N 33.03 E
6823.00	0.66	12.49	0.44	6822.59	34.1	77.4	7.41 N 33.26 E
6916.00	0.79	102.84	1.11	6915.59	34.9	77.1	7.79 N 34.00 E
7012.00	0.53	94.84	0.29	7011.58	35.9	77.8	7.61 N 35.09 E
7105.00	0.97	184.23	1.18	7104.58	36.1	79.2	6.79 N 35.46 E
7199.00	1.14	195.92	0.29	7198.56	35.5	81.8	5.09 N 35.14 E
7294.00	1.36	184.14	0.35	7293.54	34.9	85.0	3.06 N 34.80 E
7388.00	0.40	72.52	1.65	7387.53	35.1	86.7	2.05 N 35.03 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 12TX, 8 3/4" Case, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M180

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. feet	AZIMUTH deg.	HORIZONTAL COORDINATES feet
7482.00	0.53	80.69	0.15	7481.53	35.8	86.5	2.22 N 35.78 E
7577.00	0.62	67.95	0.16	7576.52	36.8	86.1	2.48 N 36.69 E
7671.00	0.53	63.90	0.10	7670.52	37.7	85.6	2.86 N 37.55 E
7766.00	0.48	50.72	0.13	7765.51	38.4	85.1	3.31 N 38.25 E
7858.00	0.22	50.11	0.28	7857.51	38.9	84.6	3.66 N 38.68 E
7953.00	0.18	136.15	0.29	7952.51	39.1	84.6	3.67 N 38.93 E
8047.00	0.04	243.55	0.21	8046.51	39.2	84.8	3.55 N 39.00 E
8141.00	0.44	288.73	0.44	8140.51	38.8	84.6	3.65 N 38.63 E
8236.00	1.14	100.64	1.66	8235.51	39.4	84.8	3.60 N 39.21 E
8330.00	1.36	98.97	0.24	8329.48	41.4	85.5	3.25 N 41.23 E
8424.00	0.53	255.86	1.98	8423.48	42.0	85.9	2.97 N 41.91 E
8518.00	0.62	258.76	0.10	8517.47	41.1	86.1	2.76 N 40.99 E
8613.00	0.57	276.95	0.20	8612.47	40.1	86.1	2.72 N 40.02 E
8707.00	0.66	279.68	0.10	8706.46	39.1	85.8	2.87 N 39.02 E
8802.00	0.57	264.12	0.20	8801.46	38.1	85.6	2.91 N 38.01 E
8897.00	0.48	271.68	0.12	8896.45	37.3	85.6	2.88 N 37.14 E
8990.00	0.92	65.31	1.47	8989.45	37.6	85.1	3.20 N 37.43 E
9084.00	0.88	67.07	0.05	9083.44	39.0	84.4	3.80 N 38.78 E
9179.00	0.70	70.41	0.20	9178.43	40.2	83.9	4.27 N 40.00 E
9273.00	0.70	67.86	0.03	9272.42	41.3	83.5	4.68 N 41.08 E
9367.00	0.75	70.50	0.06	9366.41	42.5	83.1	5.10 N 42.19 E
9462.00	0.66	59.95	0.17	9461.41	43.6	82.6	5.59 N 43.25 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 12TX, 8 3/4" Case, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M180

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
9555.00	0.66	51.69	0.10	9554.40	44.6 82.0	6.19 N 44.13 E
9649.00	0.97	117.17	0.98	9648.39	45.7 82.3	6.16 N 45.26 E
9743.00	0.84	119.19	0.14	9742.38	46.9 83.3	5.46 N 46.57 E
9837.00	0.79	117.43	0.06	9836.37	48.0 84.2	4.83 N 47.75 E
9931.00	0.88	107.59	0.18	9930.36	49.2 85.0	4.31 N 49.01 E
10024.00	0.84	101.08	0.11	10023.35	50.5 85.5	3.96 N 50.36 E
10119.00	0.62	82.19	0.34	10118.34	51.7 85.7	3.90 N 51.56 E
10214.00	0.40	77.70	0.24	10213.34	52.5 85.6	4.04 N 52.39 E
10246.00	0.40	69.44	0.18	10245.34	52.8 85.5	4.10 N 52.60 E
10277.00	0.53	87.37	0.62	10276.34	53.0 85.5	4.15 N 52.85 E
10309.00	2.99	112.42	7.87	10308.32	53.9 85.9	3.83 N 53.77 E
10340.00	8.26	114.44	17.01	10339.16	56.6 87.4	2.60 N 56.54 E
10372.00	12.79	113.91	14.16	10370.61	61.9 89.8	0.22 N 61.88 E
10404.00	16.84	111.89	12.76	10401.54	69.5 92.4	2.95 S 69.42 E
10436.00	21.63	110.57	15.03	10431.75	79.5 94.9	6.75 S 79.25 E
10467.00	23.78	111.89	7.13	10460.34	91.1 97.0	11.09 S 90.40 E
10500.00	26.29	114.88	8.52	10490.24	104.5 99.2	16.65 S 103.20 E
10531.00	29.05	118.84	10.69	10517.70	118.3 101.3	23.17 S 116.03 E
10563.00	32.26	120.24	10.28	10545.22	133.9 103.5	31.22 S 130.22 E
10594.00	35.16	118.66	9.77	10571.01	150.5 105.3	39.67 S 145.20 E
10626.00	39.21	117.34	12.90	10596.50	169.4 106.7	48.74 S 162.28 E
10657.00	43.16	114.09	14.49	10619.82	189.6 107.7	57.57 S 180.67 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 12TX, 8 3/4" Case, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M180

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
10689.00	45.45	113.30	7.36	10642.72	211.9 108.3	66.55 S 201.13 E
10720.00	46.59	109.52	9.51	10664.25	234.1 108.6	74.68 S 221.90 E
10752.00	50.11	109.34	11.01	10685.52	258.0 108.7	82.63 S 244.44 E
10783.00	54.20	109.96	13.29	10704.53	282.5 108.8	90.87 S 267.49 E
10815.00	58.11	110.22	12.24	10722.35	309.1 108.9	99.99 S 292.45 E
10847.00	61.63	111.28	11.37	10738.41	336.7 109.0	109.80 S 318.32 E
10877.00	61.76	111.45	0.66	10752.64	363.1 109.2	119.42 S 342.92 E
10909.00	65.14	112.16	10.75	10766.94	391.7 109.4	130.06 S 369.49 E
10940.00	69.67	112.68	14.69	10778.85	420.3 109.6	140.97 S 395.94 E
10972.00	73.58	113.30	12.36	10788.93	450.6 109.8	152.83 S 423.89 E
11003.00	76.62	114.27	10.26	10796.90	480.5 110.1	164.92 S 451.30 E
11035.00	78.07	114.00	4.61	10803.91	511.6 110.3	177.68 S 479.79 E
11067.00	82.42	114.71	13.77	10809.33	543.1 110.6	190.69 S 508.52 E
11112.00	89.89	114.18	16.64	10812.35	587.9 110.9	209.25 S 549.36 E
11179.00	90.90	110.67	5.45	10811.89	654.8 111.0	234.81 S 611.28 E
11211.00	90.29	112.96	7.41	10811.55	686.8 111.1	246.70 S 640.99 E
11242.00	90.15	111.81	3.74	10811.44	717.8 111.1	258.50 S 669.65 E
11273.00	89.32	109.44	8.10	10811.58	748.8 111.1	269.42 S 698.66 E
11304.00	87.74	108.74	5.57	10812.37	779.8 111.0	279.55 S 727.95 E
11336.00	87.08	106.45	7.44	10813.82	811.7 110.9	289.22 S 758.41 E
11366.00	86.99	104.25	7.33	10815.37	841.5 110.7	297.15 S 787.30 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 12TX, 8 3/4" Case, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M180

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
11397.00	87.30	103.64	2.21	10816.92	872.3 110.4	304.61 S 817.35 E
11428.00	88.90	101.97	7.46	10817.94	903.0 110.2	311.48 S 847.56 E
11459.00	88.79	99.68	7.39	10818.57	933.6 109.9	317.30 S 878.00 E
11490.00	89.41	97.57	7.09	10819.06	964.0 109.5	321.94 S 908.65 E
11521.00	90.37	96.17	5.48	10819.12	994.3 109.1	325.65 S 939.42 E
11552.00	90.11	94.32	6.03	10818.99	1024.4 108.7	328.49 S 970.29 E
11584.00	89.76	92.39	6.13	10819.02	1055.3 108.2	330.36 S 1002.24 E
11615.00	90.37	90.72	5.74	10818.99	1085.0 107.8	331.20 S 1033.22 E
11646.00	90.81	88.61	6.95	10818.67	1114.5 107.3	331.02 S 1064.22 E
11708.00	91.38	86.41	3.66	10817.48	1173.0 106.3	328.33 S 1126.14 E
11740.00	91.38	87.47	3.31	10816.71	1203.3 105.8	326.62 S 1158.09 E
11771.00	92.35	86.59	4.22	10815.70	1232.7 105.3	325.01 S 1189.03 E
11802.00	90.55	88.96	9.60	10814.92	1262.2 104.9	323.81 S 1219.99 E
11833.00	90.55	88.79	0.55	10814.62	1292.1 104.5	323.20 S 1250.99 E
11864.00	88.75	89.05	5.87	10814.81	1322.0 104.1	322.62 S 1281.98 E
11927.00	89.10	89.66	1.12	10815.99	1383.0 103.5	321.91 S 1344.96 E
11958.00	89.05	87.29	7.65	10816.49	1412.9 103.1	321.08 S 1375.95 E
12020.00	89.85	88.87	2.86	10817.09	1472.9 102.5	319.01 S 1437.91 E
12113.00	89.93	89.14	0.30	10817.27	1563.4 101.7	317.39 S 1530.89 E
12145.00	89.76	87.99	3.63	10817.35	1594.6 101.5	316.59 S 1562.88 E
12207.00	88.92	88.35	1.47	10818.07	1655.0 101.0	314.61 S 1624.85 E
12238.00	88.70	85.62	8.83	10818.71	1685.1 100.7	312.98 S 1655.79 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 12TX, 8 3/4" Case, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M180

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
12301.00	88.44	85.18	0.81	10820.28	1745.9 100.2	307.93 S 1718.57 E
12393.00	91.65	83.95	3.74	10820.21	1834.7 99.4	299.22 S 1810.14 E
12424.00	91.43	83.34	2.09	10819.38	1864.6 99.1	295.79 S 1840.94 E
12486.00	90.02	84.30	2.75	10818.59	1924.4 98.6	289.11 S 1902.57 E
12518.00	89.01	87.12	9.36	10818.86	1955.6 98.4	286.72 S 1934.48 E
12580.00	89.76	86.94	1.24	10819.53	2016.4 98.1	283.51 S 1996.39 E
12612.00	89.49	86.59	1.38	10819.74	2047.8 97.9	281.70 S 2028.34 E
12675.00	87.96	86.06	2.57	10821.14	2109.5 97.6	277.66 S 2091.19 E
12707.00	87.96	86.59	1.66	10822.28	2140.9 97.4	275.61 S 2123.11 E
12769.00	88.75	87.47	1.91	10824.06	2201.9 97.1	272.40 S 2185.00 E
12801.00	88.13	88.61	4.05	10824.93	2233.5 97.0	271.31 S 2216.97 E
12833.00	88.22	87.56	3.29	10825.95	2265.1 96.9	270.24 S 2248.93 E
12864.00	89.23	86.85	3.98	10826.64	2295.7 96.7	268.73 S 2279.89 E
12927.00	90.77	88.17	3.22	10826.64	2357.9 96.5	265.99 S 2342.82 E
12959.00	90.95	87.64	1.75	10826.16	2389.5 96.4	264.82 S 2374.80 E
13022.00	88.92	86.59	3.63	10826.23	2451.7 96.1	261.65 S 2437.71 E
13053.00	88.00	87.29	3.73	10827.06	2482.3 96.0	260.00 S 2468.66 E
13116.00	87.91	89.23	3.08	10829.31	2544.7 95.8	258.09 S 2531.59 E
13148.00	88.48	88.43	3.07	10830.32	2576.5 95.7	257.43 S 2563.56 E
13211.00	89.98	90.54	4.11	10831.17	2639.1 95.6	256.87 S 2626.55 E
13242.00	90.29	90.46	1.03	10831.09	2670.0 95.5	257.14 S 2657.55 E
13274.00	90.81	90.28	1.72	10830.79	2701.8 95.5	257.34 S 2689.55 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 12TX, 8 3/4" Case, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M180

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
13337.00	91.96	90.90	2.07	10829.26	2764.6 95.4	257.99 S 2752.52 E
13368.00	90.51	91.51	5.07	10828.60	2795.5 95.3	258.64 S 2783.51 E
13431.00	90.73	90.28	1.98	10827.91	2858.3 95.2	259.63 S 2846.49 E
13463.00	88.40	90.81	7.47	10828.16	2890.2 95.2	259.93 S 2878.49 E
13526.00	88.22	91.16	0.62	10830.02	2953.0 95.1	261.02 S 2941.45 E
13588.00	86.37	89.58	3.92	10832.94	3014.7 95.0	261.42 S 3003.38 E
13620.00	86.33	90.54	3.00	10834.98	3046.6 94.9	261.45 S 3035.31 E
13651.00	86.90	90.37	1.92	10836.81	3077.4 94.9	261.69 S 3066.26 E
13714.00	88.75	91.07	3.14	10839.20	3140.2 94.8	262.49 S 3129.20 E
13745.00	89.63	90.19	4.01	10839.64	3171.1 94.8	262.83 S 3160.20 E
13808.00	91.12	89.75	2.47	10839.23	3233.9 94.7	262.79 S 3223.20 E
13840.00	90.37	88.96	3.40	10838.81	3265.8 94.6	262.43 S 3255.19 E
13903.00	90.42	90.02	1.68	10838.38	3328.5 94.5	261.87 S 3318.19 E
13935.00	88.79	89.75	5.16	10838.60	3360.4 94.5	261.81 S 3350.18 E
13998.00	89.76	88.08	3.07	10839.39	3423.1 94.4	260.62 S 3413.16 E
14092.00	90.07	87.99	0.34	10839.53	3516.5 94.2	257.39 S 3507.11 E
14124.00	89.58	89.31	4.40	10839.63	3548.4 94.1	256.64 S 3539.10 E
14187.00	91.52	88.61	3.27	10839.03	3611.1 94.1	255.50 S 3602.08 E
14281.00	90.73	88.43	0.86	10837.18	3704.7 93.9	253.07 S 3696.03 E
14313.00	90.99	89.66	3.93	10836.70	3736.6 93.9	252.54 S 3728.02 E
14375.00	90.07	89.49	1.51	10836.13	3798.4 93.8	252.08 S 3790.02 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 12TX, 8 3/4" Case, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M180

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
14407.00	87.74	90.28	7.69	10836.74	3830.3 93.8	252.01 S 3822.01 E
14470.00	87.91	91.07	1.28	10839.13	3893.2 93.7	252.75 S 3884.96 E
14565.00	88.31	90.98	0.43	10842.26	3988.0 93.7	254.45 S 3979.89 E
14628.00	90.77	90.63	3.94	10842.77	4050.9 93.6	255.34 S 4042.88 E
14659.00	89.85	89.40	4.95	10842.60	4081.9 93.6	255.34 S 4073.88 E
14722.00	89.05	88.61	1.78	10843.20	4144.7 93.5	254.25 S 4136.86 E
14754.00	89.41	89.14	2.00	10843.63	4176.6 93.5	253.62 S 4168.85 E
14817.00	87.96	89.66	2.45	10845.08	4239.4 93.4	252.96 S 4231.83 E
14848.00	87.82	89.49	0.71	10846.22	4270.3 93.4	252.73 S 4262.81 E
14943.00	90.33	88.79	2.74	10847.76	4365.0 93.3	251.31 S 4357.78 E
14974.00	91.30	89.23	3.44	10847.31	4395.9 93.3	250.77 S 4388.77 E
15037.00	92.04	88.52	1.63	10845.48	4458.7 93.2	249.53 S 4451.73 E
15131.00	88.31	90.81	4.66	10845.19	4552.5 93.1	248.99 S 4545.71 E
15225.00	88.79	90.54	0.59	10847.57	4646.4 93.1	250.09 S 4639.67 E
15320.00	87.74	89.79	1.36	10850.45	4741.2 93.0	250.37 S 4734.62 E
15414.00	87.60	89.66	0.20	10854.27	4835.0 93.0	249.92 S 4828.55 E
15477.00	88.18	89.31	1.08	10856.59	4897.9 92.9	249.35 S 4891.50 E
15509.00	88.79	87.73	5.29	10857.43	4929.7 92.9	248.52 S 4923.48 E
15540.00	89.85	88.08	3.60	10857.80	4960.6 92.9	247.39 S 4954.45 E
15603.00	91.21	87.82	2.20	10857.22	5023.4 92.8	245.14 S 5017.41 E
15635.00	90.20	88.26	3.44	10856.82	5055.3 92.8	244.04 S 5049.39 E
15698.00	88.66	88.96	2.69	10857.45	5118.1 92.7	242.51 S 5112.36 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 12TX, 8 3/4" Case, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M180

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
15761.00	89.27	90.19	2.18	10858.59	5181.0 92.7	242.05 S 5175.35 E
15792.00	89.58	91.95	5.76	10858.90	5212.0 92.7	242.63 S 5206.34 E
15855.00	89.98	93.44	2.45	10859.14	5275.0 92.7	245.59 S 5269.27 E
15887.00	90.55	91.95	4.99	10858.99	5307.0 92.7	247.09 S 5301.23 E
15950.00	89.85	91.07	1.78	10858.77	5370.0 92.7	248.75 S 5364.21 E
15981.00	90.07	90.19	2.93	10858.80	5401.0 92.6	249.09 S 5395.21 E
16013.00	90.51	90.28	1.40	10858.63	5432.9 92.6	249.22 S 5427.21 E
16076.00	91.38	89.93	1.49	10857.60	5495.9 92.6	249.34 S 5490.20 E
16107.00	92.35	89.58	3.33	10856.59	5526.8 92.6	249.21 S 5521.18 E
16170.00	93.01	90.54	1.85	10853.64	5589.7 92.6	249.27 S 5584.11 E
16265.00	89.45	90.37	3.75	10851.60	5684.6 92.5	250.03 S 5679.07 E
16359.00	89.05	90.46	0.44	10852.83	5778.5 92.5	250.71 S 5773.06 E
16454.00	89.14	89.75	0.75	10854.33	5873.4 92.4	250.88 S 5868.05 E
16548.00	89.76	90.72	1.22	10855.24	5967.3 92.4	251.27 S 5962.04 E
16643.00	91.03	89.49	1.86	10854.58	6062.3 92.4	251.44 S 6057.03 E
16737.00	91.52	90.19	0.91	10852.49	6156.1 92.3	251.18 S 6151.01 E
16831.00	91.47	88.96	1.31	10850.04	6250.0 92.3	250.48 S 6244.97 E
16895.00	91.43	89.49	0.83	10848.42	6313.9 92.3	249.62 S 6308.95 E
16927.00	90.42	89.49	3.16	10847.90	6345.8 92.3	249.33 S 6340.94 E
16958.00	89.93	90.10	2.52	10847.81	6376.8 92.2	249.22 S 6371.94 E
17021.00	90.95	90.28	1.64	10847.32	6439.8 92.2	249.43 S 6434.94 E
17053.00	89.93	88.08	7.58	10847.08	6471.7 92.2	248.97 S 6466.93 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 12TX, 8 3/4" Case, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M180

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
17115.00	90.59	88.43	1.20	10846.79	6533.6 92.2	247.08 S 6528.90 E
17210.00	87.16	88.43	3.61	10848.66	6628.3 92.1	244.48 S 6623.83 E
17241.00	87.38	89.40	3.21	10850.14	6659.3 92.1	243.90 S 6654.79 E
17304.00	88.79	88.79	2.44	10852.24	6722.1 92.1	242.90 S 6717.75 E
17336.00	89.19	88.70	1.28	10852.81	6754.1 92.1	242.20 S 6749.73 E
17399.00	89.98	87.82	1.88	10853.26	6816.9 92.0	240.29 S 6812.70 E
17493.00	89.89	88.70	0.94	10853.37	6910.7 92.0	237.43 S 6906.66 E
17588.00	91.30	91.60	3.39	10852.38	7005.7 91.9	237.68 S 7001.64 E
17682.00	93.85	91.69	2.71	10848.16	7099.6 91.9	240.38 S 7095.50 E
17777.00	93.85	90.28	1.48	10841.78	7194.3 91.9	242.01 S 7190.27 E
17808.00	92.26	90.37	5.14	10840.13	7225.3 91.9	242.18 S 7221.22 E
17871.00	92.62	90.02	0.80	10837.45	7288.2 91.9	242.40 S 7284.16 E
17902.00	89.98	90.90	8.98	10836.74	7319.2 91.9	242.65 S 7315.15 E
17933.00	90.24	91.25	1.41	10836.68	7350.2 91.9	243.23 S 7346.15 E
17965.00	89.27	90.10	4.70	10836.82	7382.2 91.9	243.60 S 7378.14 E
17996.00	88.62	90.63	2.71	10837.39	7413.1 91.9	243.80 S 7409.14 E
18059.00	87.12	91.07	2.48	10839.73	7476.1 91.9	244.74 S 7472.08 E
18153.00	87.96	91.16	0.90	10843.77	7570.0 91.9	246.56 S 7565.98 E
18215.00	89.85	87.99	5.95	10844.95	7631.9 91.8	246.10 S 7627.95 E
18247.00	90.37	87.99	1.63	10844.89	7663.9 91.8	244.98 S 7659.93 E
18310.00	88.75	88.35	2.63	10845.37	7726.7 91.8	242.97 S 7722.90 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 12TX, 8 3/4" Case, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M180

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
18341.00	89.01	87.56	2.68	10845.98	7757.6 91.8	241.86 S 7753.87 E
18435.00	88.26	89.49	2.20	10848.22	7851.5 91.7	239.44 S 7847.81 E
18530.00	88.13	90.10	0.66	10851.21	7946.4 91.7	239.10 S 7942.76 E
18593.00	88.31	90.28	0.40	10853.17	8009.3 91.7	239.31 S 8005.73 E
18624.00	88.48	90.02	1.00	10854.04	8040.3 91.7	239.39 S 8036.72 E
18687.00	89.14	90.54	1.33	10855.34	8103.2 91.7	239.70 S 8099.70 E
18718.00	89.80	89.66	3.55	10855.63	8134.2 91.7	239.76 S 8130.70 E
18812.00	91.21	90.90	2.00	10854.80	8228.2 91.7	240.22 S 8224.69 E
18906.00	93.41	88.87	3.18	10851.01	8322.1 91.7	240.03 S 8318.61 E
18937.00	92.40	89.58	3.98	10849.44	8353.0 91.6	239.61 S 8349.56 E
18968.00	93.14	89.84	2.53	10847.95	8383.9 91.6	239.45 S 8380.53 E
19000.00	91.96	88.43	5.74	10846.52	8415.9 91.6	238.97 S 8412.49 E
19031.00	90.11	90.02	7.87	10845.96	8446.8 91.6	238.55 S 8443.48 E
19094.00	87.87	89.93	3.56	10847.07	8509.8 91.6	238.52 S 8506.46 E
19157.00	88.57	91.25	2.37	10849.03	8572.8 91.6	239.17 S 8569.43 E
19188.00	89.23	89.75	5.29	10849.62	8603.8 91.6	239.44 S 8600.42 E
19282.00	87.34	87.91	2.81	10852.44	8697.6 91.6	237.52 S 8694.35 E
19313.00	87.69	87.56	1.60	10853.78	8728.5 91.6	236.30 S 8725.30 E
19376.00	88.79	87.03	1.94	10855.72	8791.3 91.5	233.33 S 8788.20 E
19470.00	88.09	91.77	5.10	10858.28	8885.2 91.5	232.34 S 8882.13 E
19501.00	87.82	93.53	5.74	10859.38	8916.1 91.5	233.78 S 8913.08 E
19533.00	88.22	92.92	2.28	10860.49	8948.1 91.5	235.58 S 8945.01 E

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 12TX, 8 3/4" Case, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M180

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
19564.00	88.35	93.27	1.20	10861.42	8979.1 91.5	237.25 S 8975.95 E
19596.00	88.57	91.16	6.63	10862.28	9011.1 91.5	238.48 S 9007.91 E
19659.00	88.31	88.35	4.48	10863.99	9074.0 91.5	238.22 S 9070.88 E
19690.00	88.44	88.79	1.48	10864.87	9105.0 91.5	237.44 S 9101.86 E
19721.00	89.27	89.23	3.03	10865.49	9135.9 91.5	236.91 S 9132.85 E
19753.00	89.63	89.49	1.39	10865.80	9167.9 91.5	236.55 S 9164.84 E
19784.00	90.20	90.02	2.51	10865.84	9198.9 91.5	236.42 S 9195.84 E
19847.00	91.30	88.87	2.53	10865.02	9261.8 91.5	235.81 S 9258.83 E
19941.00	92.35	89.66	1.40	10862.03	9355.7 91.4	234.60 S 9352.77 E
19973.00	92.09	90.02	1.39	10860.79	9387.7 91.4	234.51 S 9384.75 E
20004.00	92.57	90.19	1.64	10859.53	9418.6 91.4	234.57 S 9415.72 E
20035.00	93.45	89.14	4.42	10857.90	9449.6 91.4	234.39 S 9446.68 E
20067.00	91.38	90.02	7.03	10856.55	9481.5 91.4	234.15 S 9478.65 E
20129.00	89.76	90.72	2.85	10855.93	9543.5 91.4	234.55 S 9540.64 E
20193.00	90.11	91.07	0.77	10856.01	9607.5 91.4	235.55 S 9604.63 E
20224.00	90.33	90.81	1.10	10855.89	9638.5 91.4	236.06 S 9635.63 E
20287.00	91.30	90.81	1.54	10854.99	9701.5 91.4	236.95 S 9698.62 E
20318.00	91.65	90.10	2.55	10854.19	9732.5 91.4	237.20 S 9729.60 E
20413.00	92.92	89.66	1.41	10850.40	9827.4 91.4	237.00 S 9824.53 E
20507.00	93.41	89.31	0.64	10845.21	9921.2 91.4	236.16 S 9918.38 E

STRAIGHT LINE PROJECTION TO BIT DEPTH AT 20584 FT. MD.

A Gyrodata Directional Survey

Oasis Petroleum

Lease: Kline Federal Well: 5300 41-18 12TX, 8 3/4" Case, 6" Hole

Location: Nabors #486, McKenzie County, North Dakota

Job Number: RD0814M180

MEASURED DEPTH feet	INCL deg.	AZIMUTH deg.	DOGLEG SEVERITY deg./ 100 ft.	VERTICAL DEPTH feet	CLOSURE DIST. AZIMUTH feet deg.	HORIZONTAL COORDINATES feet
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20584.00	93.41	89.31	0.00	10840.63	9998.0 91.3	235.23 S 9995.24 E
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Final Station Closure: Distance: 9998.00 ft Az: 91.35 deg.

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
28658

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date December 15, 2014
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☐ Other	Waiver from tubing/packer requirement

Well Name and Number Kline Federal 5300 41-18 12TX					
Footages	Qtr-Qtr	Section	Township	Range	
434 F S L 237 F W L	LOT 4	18	153 N	100 W	
Field BAKER	Pool Bakken	County MCKENZIE			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum North America LLC requests a variance to NDAC 43-02-03-21 for the tubing/packer requirement: Casing, tubing, and cementing requirements during the completion period immediately following the upcoming fracture stimulation.

The following assurances apply:

1. the well is equipped with new 29# and 32# casing at surface with an API burst rating of 11,220 psi;
2. The Frac design will use a safety factor of 0.85 API burst rating to determine the maximum pressure;
3. Damage to the casing during the frac would be detected immediately by monitoring equipment;
4. The casing is exposed to significantly lower rates and pressures during flowback than during the frac job;
5. The frac fluid and formation fluids have very low corrosion and erosion rates;
6. Production equipment will be installed as soon as possible after the well ceases flowing;
7. A 300# gauge will be installed on the surface casing during the flowback period

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9436	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature 	Printed Name Jennifer Swenson		
Title Regulatory Assistant	Date December 15, 2014		
Email Address jswenson@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date Dec 22, 2014	
By 	
Title PETROLEUM ENGINEER	



SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4
INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)



Well File No.
28658

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date July 28, 2014
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☒ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☐ Other Change casing	

Well Name and Number Kline Federal 5300 41-18 12TX					
Footages 434 F S L 237 F W L		Qtr-Qtr SWSW	Section 18	Township 153 N	Range 100 W
Field Baker	Pool Bakken	County McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum respectfully requests permission to make the following changes to the above referenced well:

Surface Casing: 13-3/8, 54.5#, 17-1/2" Hole, 2,072' MD
Dakota Contingency: 9-5/8, 40#, 12-1/4" Hole, 6,101' MD

Attached are revised plats, drill plan, well summary, directional plan and plot

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9563	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature <i>Heather McCowan</i>	Printed Name Heather McCowan		
Title Regulatory Assistant	Date July 28, 2014		
Email Address hmccowan@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 8-5-14	
By <i>Anthony Eubels</i>	
Title Petroleum Resource Specialist	

DRILLING PLAN									
OPERATOR	Oasis Petroleum				COUNTY/STATE	McKenzie Co., ND			
WELL NAME	Kline Federal 5300 41-18 12TX				RIG	Nabors B22			
WELL TYPE	Horizontal Upper Three Forks								
LOCATION	SWSW 18-153N-100W				Surface Location (survey plat):	434' fsl		237' fwl	
EST. T.D.	20,655'				GROUND ELEV:	2057' Finished Pad Elev.		Sub Height: 25'	
TOTAL LATERS		9,465'		KB ELEV:		2082			
PROGNOSIS: Based on 2,082' KB (est)					LOGS:				
MARKER	DEPTH (Surf Loc)	DATUM (Surf Loc)		Type OH Logs: Triple Combo KOP to Kibby (or min run of 1800' whichever is greater); GR/Res to BSC; GR to surf; CND through the Dakota CBL/GR: Above top of cement/GR to base of casing MWD GR: KOP to lateral TD Interval					
Pierre	NDIC MAP	1,972	110'	DEVIATION: Surf: 3 deg max, 1 deg / 100'; sry every 500' Prod: 5 deg max, 1 deg / 100'; sry every 100' DST'S: None planned CORES: None planned MUDLOGGING: Two-Man: 8,342' ~200' above the Charles (Kibbey) to Casing point; Casing point to TD 30' samples at direction of wellsite geologist; 10' through target @ curve land BOP: 11" 5000 psi blind, pipe & annular					
Greenhorn		4,646	-2,564'						
Mowry		5,057	-2,975'						
Dakota		5,471	-3,389'						
Rierdon		6,486	-4,404'						
Dunham Salt		6,814	-4,732'						
Dunham Salt Base		6,929	-4,847'						
Spearfish		7,024	-4,942'						
Pine Salt		7,283	-5,201'						
Pine Salt Base		7,318	-5,236'						
Opeche Salt		7,374	-5,292'						
Opeche Salt Base		7,403	-5,321'						
Broom Creek (Top of Minnelusa Gp.)		7,605	-5,523'						
Amsden		7,684	-5,602'						
Tyler		7,853	-5,771'						
Otter (Base of Minnelusa Gp.)		8,047	-5,965'						
Kibbey Lime		8,390	-6,308'						
Charles Salt		8,542	-6,460'						
UB		9,159	-7,077'						
Base Last Salt		9,238	-7,156'						
Ratcliffe		9,301	-7,219'						
Mission Canyon		9,454	-7,372'						
Lodgepole		10,022	-7,940'						
Lodgepole Fracture Zone		10,209	-8,127'						
False Bakken		10,726	-8,644'						
Upper Bakken		10,735	-8,653'						
Middle Bakken		10,751	-8,669'						
Lower Bakken		10,785	-8,703'						
Pronghorn		10,791	-8,709'						
Three Forks		10,813	-8,731'						
TF Target Top		10,827	-8,745'						
TF Target Base		10,837	-8,755'						
Claystone		10,838	-8,756'						
Dip Rate: 0.1									
Max. Anticipated BHP: 4692					Surface Formation: Glacial till				
MUD:	Interval	Type	WT	Vis	WL	Remarks			
Surface:	0' -	2,072' FW/Gel - Lime Sweeps	8.4-9.0	28-32	NC	Circ Mud Tanks			
Intermediate:	2,072' -	11,189' Invert	9.5-10.4	40-50	30+HHP	Circ Mud Tanks			
Lateral:	11,189' -	20,655' Salt Water	9.8-10.2	28-32	NC	Circ Mud Tanks			
CASING:	Size	WT ppi	Hole	Depth	Cement	WOC	Remarks		
Surface:	13-3/8"	54.5#	17-1/2"	2,072'	To Surface	12	100' into Pierre		
Dakota Contingency:	9-5/8"	40#	12-1/4"	6,101'	To Surface	12	Below Dakota		
Intermediate:	7"	29/32#	8-3/4"	11,189'	3971	24	1500' above Dakota		
Production Liner:	4.5"	13.5#	6"	20,655'	TOL @ 10,305'		50' above KOP		
PROBABLE PLUGS, IF REQ'D:									
OTHER:	MD	TVD	FNL/FSL	FEL/FWL	S-T-R	AZI			
Surface:	2,072'	2,072'	434' FSL	237' FWL	SEC 18-T153N-R100W	Survey Company:			
KOP:	10,355'	10,355'	434' FSL	287' FWL	SEC 18-T153N-R100W	Build Rate:	12 deg / 100'		
EOC:	11,104'	10,832'	306' FSL	748' FWL	SEC 18-T153N-R100W	110.00			
Casing Point:	11,189'	10,832'	244' FSL	818' FWL	SEC 18-T153N-R100W	107.02			
Upper Three Forks Lateral TD:	20,655'	10,849'	200' FSL	200' FEL	SEC 17-T153N-R100W	89.82			
Comments:									
Request a Sundry for an Open Hole Log Waiver									
Exception well: Oasis Petroleum's Kline Federal 5300 11-18H (153N 100W 18 NW NW)									
35 packers, 35 sleeves, no frac string									
Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.									
68334-30-5 (Primary Name: Fuels, diesel) 68476-34-6 (Primary Name: Fuels, diesel, No. 2) 68476-30-2 (Primary Name: Fuel oil No. 2)									
68476-31-3 (Primary Name: Fuel oil, No. 4) 8008-20-6 (Primary Name: Kerosene)									
									
Geology: M.Steed 4/23/2014					Engineering: hlbader rpm 7/18/14				



Oasis Petroleum

Indian Hills

153N-100W-17/18

Kline Federal 5300 41-18 12TX

Kline Federal 5300 41-18 12TX

Plan: Design #2

Standard Planning Report

17 June, 2014



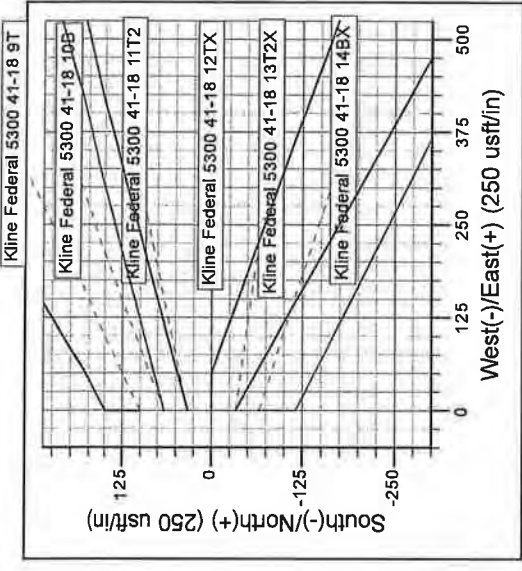
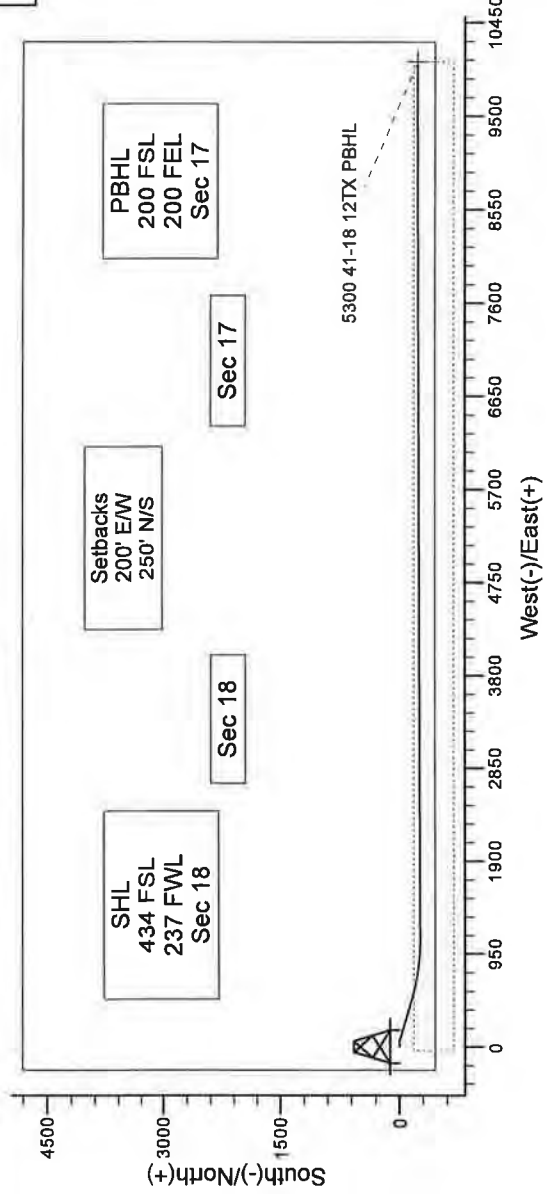
**RYAN DIRECTIONAL
SERVICES**
A NABORS COMPANY



Project: Indian Hills
Site: 153N-100W-17/18
Well: Kline Federal 5300 41-18 12TX
Wellbore: Kline Federal 5300 41-18 12TX
Design: Design #2

WELL DETAILS: Kline Federal 5300 41-18 12TX

Ground Level: 2057.0
Northing: 405136.64
Easting: 1209968.21
Latitude: 48° 4' 7.550 N
Longitude: 103° 36' 11.950 W



SECTION DETAILS

MD 0.0
2200.0
2216.7
7926.7
7943.3
10354.8
11104.0
11680.4
20667.7

Azi 0.00
0.00
90.00
90.00
0.00
0.00
110.00
89.82
89.82

Inc 0.00
0.00
0.50
0.50
0.00
0.00
89.90
89.90
89.90

TVD 0.0
2200.0
2216.7
7926.4
7943.1
10354.6
10832.0
10833.1
10848.7

+N/-S 0.0
0.0
0.0
0.0
0.0
0.0
-163.0
-261.7
-234.0

+E/-W 0.0
0.0
0.1
49.9
50.0
50.0
497.9
1062.8
10050.0

Dleg 0.00
0.00
3.00
0.00
3.00
3.00
12.00
3.50
0.00

Target 5300 41-18 12TX PBHL

CASING DETAILS

MD 11189.0
Name 7" 7
Size 7"

MD 0.0
2200.0
2216.7
7926.7
7943.3
10354.8
11104.0
11680.4
20667.7

Azi 0.00
0.00
90.00
90.00
0.00
0.00
110.00
89.82
89.82

Inc 0.00
0.00
0.50
0.50
0.00
0.00
89.90
89.90
89.90

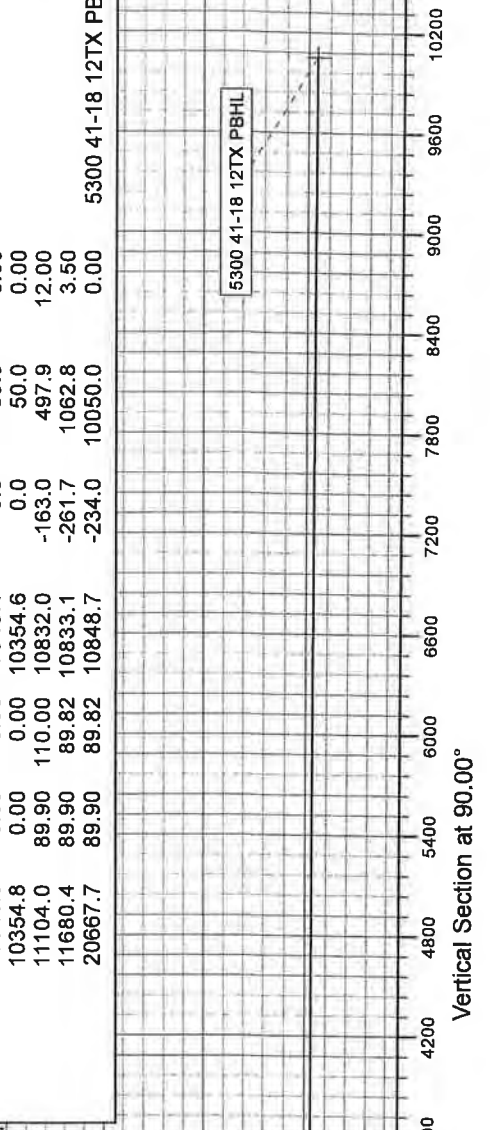
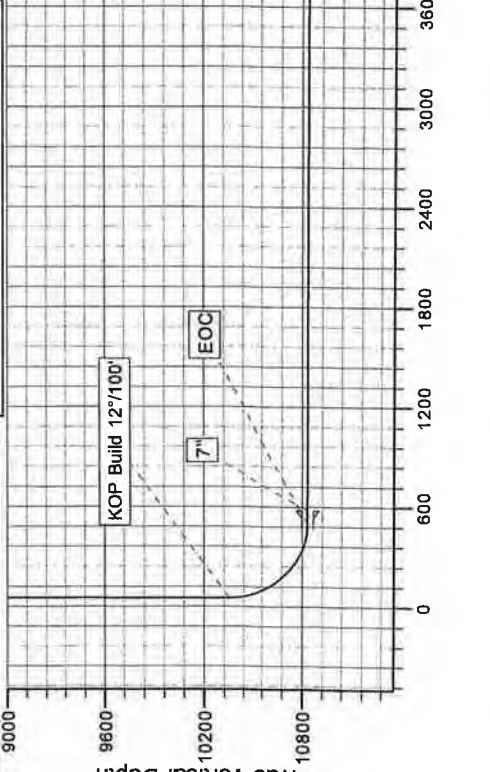
TVD 0.0
2200.0
2216.7
7926.4
7943.1
10354.6
10832.0
10833.1
10848.7

+N/-S 0.0
0.0
0.0
0.0
0.0
0.0
-163.0
-261.7
-234.0

+E/-W 0.0
0.0
0.1
49.9
50.0
50.0
497.9
1062.8
10050.0

Dleg 0.00
0.00
3.00
0.00
3.00
3.00
12.00
3.50
0.00

Target 5300 41-18 12TX PBHL





Ryan Directional Services Planning Report



Database: EDM 5000.1 Single User Db
Company: Oasis Petroleum
Project: Indian Hills
Site: 153N-100W-17/18
Well: Kline Federal 5300 41-18 12TX
Wellbore: Kline Federal 5300 41-18 12TX
Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature

Project	Indian Hills		
Map System:	US State Plane 1983	System Datum:	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		

Site	153N-100W-17/18		
Site Position:		Northing:	408,962.44 usft
From:	Lat/Long	Easting:	1,210,229.18 usft
Position Uncertainty:	0.0 usft	Slot Radius:	13-3/16 "
		Latitude:	48° 4' 45.380 N
		Longitude:	103° 36' 10.380 W
		Grid Convergence:	-2.31 °

Well	Kline Federal 5300 41-18 12TX		
Well Position	+N/-S	-3,833.2 usft	Northing: 405,136.63 usft
	+E/-W	-106.6 usft	Easting: 1,209,968.21 usft
Position Uncertainty	2.0 usft	Wellhead Elevation:	Latitude: 48° 4' 7.550 N
			Longitude: 103° 36' 11.950 W
			Ground Level: 2,057.0 usft

Wellbore	Kline Federal 5300 41-18 12TX				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	IGRF2010	5/23/2014	8.27	72.99	56,437

Design	Design #2			
Audit Notes:				
Version:	Phase:	PROTOTYPE	Tie On Depth:	0.0
Vertical Section:	Depth From (TVD) (usft)	+N/-S (usft)	+E/-W (usft)	Direction (°)
	0.0	0.0	0.0	90.00

Plan Sections										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)	TFO (°)	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,200.0	0.00	0.00	2,200.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,216.7	0.50	90.00	2,216.7	0.0	0.1	3.00	3.00	0.00	90.00	
7,926.7	0.50	90.00	7,926.4	0.0	49.9	0.00	0.00	0.00	0.00	
7,943.3	0.00	0.00	7,943.1	0.0	50.0	3.00	-3.00	0.00	180.00	
10,354.8	0.00	0.00	10,354.6	0.0	50.0	0.00	0.00	0.00	0.00	
11,104.0	89.90	110.00	10,832.0	-163.0	497.9	12.00	12.00	0.00	110.00	
11,680.4	89.90	89.82	10,833.1	-261.7	1,062.8	3.50	0.00	-3.50	-90.02	
20,667.7	89.90	89.82	10,848.7	-234.0	10,050.0	0.00	0.00	0.00	0.00	5300 41-18 12TX PBI



Ryan Directional Services Planning Report



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Project: Indian Hills
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Design: Design #2

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MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N-S (usft)	+E-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
2,200.0	0.00	90.00	2,200.0	0.0	0.0	0.0	0.00	0.00	0.00
2,216.7	0.50	90.00	2,216.7	0.0	0.1	0.1	3.00	3.00	0.00
2,300.0	0.50	90.00	2,300.0	0.0	0.8	0.8	0.00	0.00	0.00
2,400.0	0.50	90.00	2,400.0	0.0	1.7	1.7	0.00	0.00	0.00
2,500.0	0.50	90.00	2,500.0	0.0	2.5	2.5	0.00	0.00	0.00
2,600.0	0.50	90.00	2,600.0	0.0	3.4	3.4	0.00	0.00	0.00
2,700.0	0.50	90.00	2,700.0	0.0	4.3	4.3	0.00	0.00	0.00
2,800.0	0.50	90.00	2,800.0	0.0	5.2	5.2	0.00	0.00	0.00
2,900.0	0.50	90.00	2,900.0	0.0	6.0	6.0	0.00	0.00	0.00
3,000.0	0.50	90.00	3,000.0	0.0	6.9	6.9	0.00	0.00	0.00
3,100.0	0.50	90.00	3,100.0	0.0	7.8	7.8	0.00	0.00	0.00
3,200.0	0.50	90.00	3,200.0	0.0	8.7	8.7	0.00	0.00	0.00
3,300.0	0.50	90.00	3,300.0	0.0	9.5	9.5	0.00	0.00	0.00
3,400.0	0.50	90.00	3,400.0	0.0	10.4	10.4	0.00	0.00	0.00
3,500.0	0.50	90.00	3,500.0	0.0	11.3	11.3	0.00	0.00	0.00
3,600.0	0.50	90.00	3,599.9	0.0	12.1	12.1	0.00	0.00	0.00
3,700.0	0.50	90.00	3,699.9	0.0	13.0	13.0	0.00	0.00	0.00
3,800.0	0.50	90.00	3,799.9	0.0	13.9	13.9	0.00	0.00	0.00
3,900.0	0.50	90.00	3,899.9	0.0	14.8	14.8	0.00	0.00	0.00
4,000.0	0.50	90.00	3,999.9	0.0	15.6	15.6	0.00	0.00	0.00
4,100.0	0.50	90.00	4,099.9	0.0	16.5	16.5	0.00	0.00	0.00
4,200.0	0.50	90.00	4,199.9	0.0	17.4	17.4	0.00	0.00	0.00
4,300.0	0.50	90.00	4,299.9	0.0	18.3	18.3	0.00	0.00	0.00
4,400.0	0.50	90.00	4,399.9	0.0	19.1	19.1	0.00	0.00	0.00
4,500.0	0.50	90.00	4,499.9	0.0	20.0	20.0	0.00	0.00	0.00
4,600.0	0.50	90.00	4,599.9	0.0	20.9	20.9	0.00	0.00	0.00
4,646.1	0.50	90.00	4,646.0	0.0	21.3	21.3	0.00	0.00	0.00
Greenhorn									
4,700.0	0.50	90.00	4,699.9	0.0	21.7	21.7	0.00	0.00	0.00
4,800.0	0.50	90.00	4,799.9	0.0	22.6	22.6	0.00	0.00	0.00
4,900.0	0.50	90.00	4,899.9	0.0	23.5	23.5	0.00	0.00	0.00
5,000.0	0.50	90.00	4,999.9	0.0	24.4	24.4	0.00	0.00	0.00
5,057.1	0.50	90.00	5,057.0	0.0	24.9	24.9	0.00	0.00	0.00
Mowry									
5,100.0	0.50	90.00	5,099.9	0.0	25.2	25.2	0.00	0.00	0.00
5,200.0	0.50	90.00	5,199.9	0.0	26.1	26.1	0.00	0.00	0.00
5,300.0	0.50	90.00	5,299.9	0.0	27.0	27.0	0.00	0.00	0.00
5,400.0	0.50	90.00	5,399.9	0.0	27.9	27.9	0.00	0.00	0.00
5,471.1	0.50	90.00	5,471.0	0.0	28.5	28.5	0.00	0.00	0.00
Dakota									
5,500.0	0.50	90.00	5,499.9	0.0	28.7	28.7	0.00	0.00	0.00
5,600.0	0.50	90.00	5,599.9	0.0	29.6	29.6	0.00	0.00	0.00
5,700.0	0.50	90.00	5,699.9	0.0	30.5	30.5	0.00	0.00	0.00
5,800.0	0.50	90.00	5,799.9	0.0	31.3	31.3	0.00	0.00	0.00
5,900.0	0.50	90.00	5,899.9	0.0	32.2	32.2	0.00	0.00	0.00
6,000.0	0.50	90.00	5,999.9	0.0	33.1	33.1	0.00	0.00	0.00
6,100.0	0.50	90.00	6,099.9	0.0	34.0	34.0	0.00	0.00	0.00
6,200.0	0.50	90.00	6,199.8	0.0	34.8	34.8	0.00	0.00	0.00
6,300.0	0.50	90.00	6,299.8	0.0	35.7	35.7	0.00	0.00	0.00
6,400.0	0.50	90.00	6,399.8	0.0	36.6	36.6	0.00	0.00	0.00
6,486.2	0.50	90.00	6,486.0	0.0	37.3	37.3	0.00	0.00	0.00
Rierdon									
6,500.0	0.50	90.00	6,499.8	0.0	37.5	37.5	0.00	0.00	0.00



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Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
6,600.0	0.50	90.00	6,599.8	0.0	38.3	38.3	0.00	0.00	0.00
6,700.0	0.50	90.00	6,699.8	0.0	39.2	39.2	0.00	0.00	0.00
6,800.0	0.50	90.00	6,799.8	0.0	40.1	40.1	0.00	0.00	0.00
6,814.2	0.50	90.00	6,814.0	0.0	40.2	40.2	0.00	0.00	0.00
Dunham Salt									
6,900.0	0.50	90.00	6,899.8	0.0	40.9	40.9	0.00	0.00	0.00
6,929.2	0.50	90.00	6,929.0	0.0	41.2	41.2	0.00	0.00	0.00
Dunham Salt Base									
7,000.0	0.50	90.00	6,999.8	0.0	41.8	41.8	0.00	0.00	0.00
7,024.2	0.50	90.00	7,024.0	0.0	42.0	42.0	0.00	0.00	0.00
Spearfish									
7,100.0	0.50	90.00	7,099.8	0.0	42.7	42.7	0.00	0.00	0.00
7,200.0	0.50	90.00	7,199.8	0.0	43.6	43.6	0.00	0.00	0.00
7,283.2	0.50	90.00	7,283.0	0.0	44.3	44.3	0.00	0.00	0.00
Pine Salt									
7,300.0	0.50	90.00	7,299.8	0.0	44.4	44.4	0.00	0.00	0.00
7,318.2	0.50	90.00	7,318.0	0.0	44.6	44.6	0.00	0.00	0.00
Pine Salt Base									
7,374.2	0.50	90.00	7,374.0	0.0	45.1	45.1	0.00	0.00	0.00
Opeche Salt									
7,400.0	0.50	90.00	7,399.8	0.0	45.3	45.3	0.00	0.00	0.00
7,403.2	0.50	90.00	7,403.0	0.0	45.3	45.3	0.00	0.00	0.00
Opeche Salt Base									
7,500.0	0.50	90.00	7,499.8	0.0	46.2	46.2	0.00	0.00	0.00
7,600.0	0.50	90.00	7,599.8	0.0	47.1	47.1	0.00	0.00	0.00
7,605.2	0.50	90.00	7,605.0	0.0	47.1	47.1	0.00	0.00	0.00
Broom Creek (Top of Minnelusa Gp.)									
7,684.2	0.50	90.00	7,684.0	0.0	47.8	47.8	0.00	0.00	0.00
Amsden									
7,700.0	0.50	90.00	7,699.8	0.0	47.9	47.9	0.00	0.00	0.00
7,800.0	0.50	90.00	7,799.8	0.0	48.8	48.8	0.00	0.00	0.00
7,853.2	0.50	90.00	7,853.0	0.0	49.3	49.3	0.00	0.00	0.00
Tyler									
7,900.0	0.50	90.00	7,899.8	0.0	49.7	49.7	0.00	0.00	0.00
7,926.7	0.50	90.00	7,926.4	0.0	49.9	49.9	0.00	0.00	0.00
7,943.3	0.00	0.00	7,943.1	0.0	50.0	50.0	3.00	-3.00	0.00
8,000.0	0.00	0.00	7,999.8	0.0	50.0	50.0	0.00	0.00	0.00
8,047.2	0.00	0.00	8,047.0	0.0	50.0	50.0	0.00	0.00	0.00
Otter (Base of Minnelusa Gp.)									
8,100.0	0.00	0.00	8,099.8	0.0	50.0	50.0	0.00	0.00	0.00
8,200.0	0.00	0.00	8,199.8	0.0	50.0	50.0	0.00	0.00	0.00
8,300.0	0.00	0.00	8,299.8	0.0	50.0	50.0	0.00	0.00	0.00
8,390.2	0.00	0.00	8,390.0	0.0	50.0	50.0	0.00	0.00	0.00
Kibbey Lime									
8,400.0	0.00	0.00	8,399.8	0.0	50.0	50.0	0.00	0.00	0.00
8,500.0	0.00	0.00	8,499.8	0.0	50.0	50.0	0.00	0.00	0.00
8,542.2	0.00	0.00	8,542.0	0.0	50.0	50.0	0.00	0.00	0.00
Charles Salt									
8,600.0	0.00	0.00	8,599.8	0.0	50.0	50.0	0.00	0.00	0.00
8,700.0	0.00	0.00	8,699.8	0.0	50.0	50.0	0.00	0.00	0.00
8,800.0	0.00	0.00	8,799.8	0.0	50.0	50.0	0.00	0.00	0.00
8,900.0	0.00	0.00	8,899.8	0.0	50.0	50.0	0.00	0.00	0.00



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Design: Design #2

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MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
9,000.0	0.00	0.00	8,999.8	0.0	50.0	50.0	0.00	0.00	0.00
9,100.0	0.00	0.00	9,099.8	0.0	50.0	50.0	0.00	0.00	0.00
9,159.2	0.00	0.00	9,159.0	0.0	50.0	50.0	0.00	0.00	0.00
UB									
9,200.0	0.00	0.00	9,199.8	0.0	50.0	50.0	0.00	0.00	0.00
9,238.2	0.00	0.00	9,238.0	0.0	50.0	50.0	0.00	0.00	0.00
Base Last Salt									
9,300.0	0.00	0.00	9,299.8	0.0	50.0	50.0	0.00	0.00	0.00
9,301.2	0.00	0.00	9,301.0	0.0	50.0	50.0	0.00	0.00	0.00
Ratcliffe									
9,400.0	0.00	0.00	9,399.8	0.0	50.0	50.0	0.00	0.00	0.00
9,454.2	0.00	0.00	9,454.0	0.0	50.0	50.0	0.00	0.00	0.00
Mission Canyon									
9,500.0	0.00	0.00	9,499.8	0.0	50.0	50.0	0.00	0.00	0.00
9,600.0	0.00	0.00	9,599.8	0.0	50.0	50.0	0.00	0.00	0.00
9,700.0	0.00	0.00	9,699.8	0.0	50.0	50.0	0.00	0.00	0.00
9,800.0	0.00	0.00	9,799.8	0.0	50.0	50.0	0.00	0.00	0.00
9,900.0	0.00	0.00	9,899.8	0.0	50.0	50.0	0.00	0.00	0.00
10,000.0	0.00	0.00	9,999.8	0.0	50.0	50.0	0.00	0.00	0.00
10,022.2	0.00	0.00	10,022.0	0.0	50.0	50.0	0.00	0.00	0.00
Lodgepole									
10,100.0	0.00	0.00	10,099.8	0.0	50.0	50.0	0.00	0.00	0.00
10,200.0	0.00	0.00	10,199.8	0.0	50.0	50.0	0.00	0.00	0.00
10,209.2	0.00	0.00	10,209.0	0.0	50.0	50.0	0.00	0.00	0.00
Lodgepole Fracture Zone									
10,300.0	0.00	0.00	10,299.8	0.0	50.0	50.0	0.00	0.00	0.00
10,354.8	0.00	0.00	10,354.6	0.0	50.0	50.0	0.00	0.00	0.00
KOP Build 12°/100'									
10,375.0	2.42	110.00	10,374.8	-0.1	50.4	50.4	12.00	12.00	0.00
10,400.0	5.42	110.00	10,399.7	-0.7	52.0	52.0	12.00	12.00	0.00
10,425.0	8.42	110.00	10,424.5	-1.8	54.8	54.8	12.00	12.00	0.00
10,450.0	11.42	110.00	10,449.2	-3.2	58.9	58.9	12.00	12.00	0.00
10,475.0	14.42	110.00	10,473.5	-5.1	64.1	64.1	12.00	12.00	0.00
10,500.0	17.42	110.00	10,497.6	-7.5	70.6	70.6	12.00	12.00	0.00
10,525.0	20.42	110.00	10,521.2	-10.3	78.2	78.2	12.00	12.00	0.00
10,550.0	23.42	110.00	10,544.4	-13.5	86.9	86.9	12.00	12.00	0.00
10,575.0	26.42	110.00	10,567.1	-17.1	96.8	96.8	12.00	12.00	0.00
10,600.0	29.42	110.00	10,589.1	-21.1	107.8	107.8	12.00	12.00	0.00
10,625.0	32.42	110.00	10,610.6	-25.5	119.9	119.9	12.00	12.00	0.00
10,650.0	35.42	110.00	10,631.3	-30.2	133.0	133.0	12.00	12.00	0.00
10,675.0	38.42	110.00	10,651.3	-35.4	147.1	147.1	12.00	12.00	0.00
10,700.0	41.42	110.00	10,670.5	-40.9	162.2	162.2	12.00	12.00	0.00
10,725.0	44.42	110.00	10,688.8	-46.7	178.2	178.2	12.00	12.00	0.00
10,750.0	47.42	110.00	10,706.2	-52.8	195.1	195.1	12.00	12.00	0.00
10,775.0	50.42	110.00	10,722.6	-59.3	212.8	212.8	12.00	12.00	0.00
10,780.4	51.07	110.00	10,726.0	-60.7	216.7	216.7	12.00	12.00	0.00
False Bakken									
10,795.0	52.82	110.00	10,735.0	-64.6	227.5	227.5	12.00	12.00	0.00
Upper Bakken									
10,800.0	53.42	110.00	10,738.0	-66.0	231.3	231.3	12.00	12.00	0.00
10,822.5	56.13	110.00	10,751.0	-72.3	248.6	248.6	12.00	12.00	0.00
Middle Bakken									
10,825.0	56.42	110.00	10,752.4	-73.0	250.5	250.5	12.00	12.00	0.00



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Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
10,850.0	59.42	110.00	10,765.7	-80.2	270.4	270.4	12.00	12.00	0.00
10,875.0	62.42	110.00	10,777.8	-87.7	290.9	290.9	12.00	12.00	0.00
10,891.1	64.35	110.00	10,785.0	-92.6	304.4	304.4	12.00	12.00	0.00
Lower Bakken									
10,900.0	65.42	110.00	10,788.8	-95.4	312.0	312.0	12.00	12.00	0.00
10,905.4	66.07	110.00	10,791.0	-97.1	316.6	316.6	12.00	12.00	0.00
Pronghorn									
10,925.0	68.42	110.00	10,798.6	-103.3	333.7	333.7	12.00	12.00	0.00
10,950.0	71.42	110.00	10,807.2	-111.3	355.7	355.7	12.00	12.00	0.00
10,969.5	73.76	110.00	10,813.0	-117.6	373.2	373.2	12.00	12.00	0.00
Three Forks									
10,975.0	74.42	110.00	10,814.5	-119.5	378.2	378.2	12.00	12.00	0.00
11,000.0	77.42	110.00	10,820.6	-127.7	401.0	401.0	12.00	12.00	0.00
11,025.0	80.42	110.00	10,825.4	-136.1	424.0	424.0	12.00	12.00	0.00
11,035.3	81.66	110.00	10,827.0	-139.6	433.6	433.6	12.00	12.00	0.00
TF Target Top									
11,050.0	83.42	110.00	10,828.9	-144.6	447.3	447.3	12.00	12.00	0.00
11,075.0	86.42	110.00	10,831.1	-153.1	470.7	470.7	12.00	12.00	0.00
11,100.0	89.42	110.00	10,832.0	-161.7	494.1	494.1	12.00	12.00	0.00
11,104.0	89.90	110.00	10,832.0	-163.0	497.9	497.9	11.90	11.90	0.00
EOC									
11,189.0	89.90	107.02	10,832.2	-190.0	578.5	578.5	3.50	0.00	-3.50
7"									
11,200.0	89.90	106.64	10,832.2	-193.2	589.0	589.0	3.50	0.00	-3.50
11,300.0	89.90	103.14	10,832.4	-218.9	685.6	685.6	3.50	0.00	-3.50
11,400.0	89.90	99.64	10,832.6	-238.6	783.7	783.7	3.50	0.00	-3.50
11,500.0	89.90	96.14	10,832.7	-252.4	882.7	882.7	3.50	0.00	-3.50
11,600.0	89.90	92.64	10,832.9	-260.0	982.4	982.4	3.50	0.00	-3.50
11,680.4	89.90	89.82	10,833.1	-261.7	1,062.8	1,062.8	3.50	0.00	-3.50
11,700.0	89.90	89.82	10,833.1	-261.7	1,082.4	1,082.4	0.00	0.00	0.00
11,800.0	89.90	89.82	10,833.3	-261.4	1,182.4	1,182.4	0.00	0.00	0.00
11,900.0	89.90	89.82	10,833.4	-261.1	1,282.4	1,282.4	0.00	0.00	0.00
12,000.0	89.90	89.82	10,833.6	-260.7	1,382.4	1,382.4	0.00	0.00	0.00
12,100.0	89.90	89.82	10,833.8	-260.4	1,482.4	1,482.4	0.00	0.00	0.00
12,200.0	89.90	89.82	10,834.0	-260.1	1,582.4	1,582.4	0.00	0.00	0.00
12,300.0	89.90	89.82	10,834.1	-259.8	1,682.4	1,682.4	0.00	0.00	0.00
12,400.0	89.90	89.82	10,834.3	-259.5	1,782.4	1,782.4	0.00	0.00	0.00
12,500.0	89.90	89.82	10,834.5	-259.2	1,882.4	1,882.4	0.00	0.00	0.00
12,600.0	89.90	89.82	10,834.7	-258.9	1,982.4	1,982.4	0.00	0.00	0.00
12,700.0	89.90	89.82	10,834.8	-258.6	2,082.4	2,082.4	0.00	0.00	0.00
12,800.0	89.90	89.82	10,835.0	-258.3	2,182.4	2,182.4	0.00	0.00	0.00
12,900.0	89.90	89.82	10,835.2	-258.0	2,282.4	2,282.4	0.00	0.00	0.00
13,000.0	89.90	89.82	10,835.4	-257.7	2,382.4	2,382.4	0.00	0.00	0.00
13,100.0	89.90	89.82	10,835.5	-257.4	2,482.4	2,482.4	0.00	0.00	0.00
13,200.0	89.90	89.82	10,835.7	-257.0	2,582.3	2,582.3	0.00	0.00	0.00
13,300.0	89.90	89.82	10,835.9	-256.7	2,682.3	2,682.3	0.00	0.00	0.00
13,400.0	89.90	89.82	10,836.1	-256.4	2,782.3	2,782.3	0.00	0.00	0.00
13,500.0	89.90	89.82	10,836.2	-256.1	2,882.3	2,882.3	0.00	0.00	0.00
13,600.0	89.90	89.82	10,836.4	-255.8	2,982.3	2,982.3	0.00	0.00	0.00
13,700.0	89.90	89.82	10,836.6	-255.5	3,082.3	3,082.3	0.00	0.00	0.00
13,800.0	89.90	89.82	10,836.7	-255.2	3,182.3	3,182.3	0.00	0.00	0.00
13,900.0	89.90	89.82	10,836.9	-254.9	3,282.3	3,282.3	0.00	0.00	0.00
13,943.9	89.90	89.82	10,837.0	-254.7	3,326.2	3,326.2	0.00	0.00	0.00



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TF Target Base									
14,000.0	89.90	89.82	10,837.1	-254.6	3,382.3	3,382.3	0.00	0.00	0.00
14,100.0	89.90	89.82	10,837.3	-254.3	3,482.3	3,482.3	0.00	0.00	0.00
14,200.0	89.90	89.82	10,837.4	-254.0	3,582.3	3,582.3	0.00	0.00	0.00
14,300.0	89.90	89.82	10,837.6	-253.7	3,682.3	3,682.3	0.00	0.00	0.00
14,400.0	89.90	89.82	10,837.8	-253.3	3,782.3	3,782.3	0.00	0.00	0.00
14,500.0	89.90	89.82	10,838.0	-253.0	3,882.3	3,882.3	0.00	0.00	0.00
14,518.6	89.90	89.82	10,838.0	-253.0	3,900.9	3,900.9	0.00	0.00	0.00
Claystone									
14,600.0	89.90	89.82	10,838.1	-252.7	3,982.3	3,982.3	0.00	0.00	0.00
14,700.0	89.90	89.82	10,838.3	-252.4	4,082.3	4,082.3	0.00	0.00	0.00
14,800.0	89.90	89.82	10,838.5	-252.1	4,182.3	4,182.3	0.00	0.00	0.00
14,900.0	89.90	89.82	10,838.7	-251.8	4,282.3	4,282.3	0.00	0.00	0.00
15,000.0	89.90	89.82	10,838.8	-251.5	4,382.3	4,382.3	0.00	0.00	0.00
15,100.0	89.90	89.82	10,839.0	-251.2	4,482.3	4,482.3	0.00	0.00	0.00
15,200.0	89.90	89.82	10,839.2	-250.9	4,582.3	4,582.3	0.00	0.00	0.00
15,300.0	89.90	89.82	10,839.4	-250.6	4,682.3	4,682.3	0.00	0.00	0.00
15,400.0	89.90	89.82	10,839.5	-250.3	4,782.3	4,782.3	0.00	0.00	0.00
15,500.0	89.90	89.82	10,839.7	-249.9	4,882.3	4,882.3	0.00	0.00	0.00
15,600.0	89.90	89.82	10,839.9	-249.6	4,982.3	4,982.3	0.00	0.00	0.00
15,700.0	89.90	89.82	10,840.1	-249.3	5,082.3	5,082.3	0.00	0.00	0.00
15,800.0	89.90	89.82	10,840.2	-249.0	5,182.3	5,182.3	0.00	0.00	0.00
15,900.0	89.90	89.82	10,840.4	-248.7	5,282.3	5,282.3	0.00	0.00	0.00
16,000.0	89.90	89.82	10,840.6	-248.4	5,382.3	5,382.3	0.00	0.00	0.00
16,100.0	89.90	89.82	10,840.8	-248.1	5,482.3	5,482.3	0.00	0.00	0.00
16,200.0	89.90	89.82	10,840.9	-247.8	5,582.3	5,582.3	0.00	0.00	0.00
16,300.0	89.90	89.82	10,841.1	-247.5	5,682.3	5,682.3	0.00	0.00	0.00
16,400.0	89.90	89.82	10,841.3	-247.2	5,782.3	5,782.3	0.00	0.00	0.00
16,500.0	89.90	89.82	10,841.4	-246.9	5,882.3	5,882.3	0.00	0.00	0.00
16,600.0	89.90	89.82	10,841.6	-246.6	5,982.3	5,982.3	0.00	0.00	0.00
16,700.0	89.90	89.82	10,841.8	-246.2	6,082.3	6,082.3	0.00	0.00	0.00
16,800.0	89.90	89.82	10,842.0	-245.9	6,182.3	6,182.3	0.00	0.00	0.00
16,900.0	89.90	89.82	10,842.1	-245.6	6,282.3	6,282.3	0.00	0.00	0.00
17,000.0	89.90	89.82	10,842.3	-245.3	6,382.3	6,382.3	0.00	0.00	0.00
17,100.0	89.90	89.82	10,842.5	-245.0	6,482.3	6,482.3	0.00	0.00	0.00
17,200.0	89.90	89.82	10,842.7	-244.7	6,582.3	6,582.3	0.00	0.00	0.00
17,300.0	89.90	89.82	10,842.8	-244.4	6,682.3	6,682.3	0.00	0.00	0.00
17,400.0	89.90	89.82	10,843.0	-244.1	6,782.3	6,782.3	0.00	0.00	0.00
17,500.0	89.90	89.82	10,843.2	-243.8	6,882.3	6,882.3	0.00	0.00	0.00
17,600.0	89.90	89.82	10,843.4	-243.5	6,982.3	6,982.3	0.00	0.00	0.00
17,700.0	89.90	89.82	10,843.5	-243.2	7,082.3	7,082.3	0.00	0.00	0.00
17,800.0	89.90	89.82	10,843.7	-242.8	7,182.3	7,182.3	0.00	0.00	0.00
17,900.0	89.90	89.82	10,843.9	-242.5	7,282.3	7,282.3	0.00	0.00	0.00
18,000.0	89.90	89.82	10,844.1	-242.2	7,382.3	7,382.3	0.00	0.00	0.00
18,100.0	89.90	89.82	10,844.2	-241.9	7,482.3	7,482.3	0.00	0.00	0.00
18,200.0	89.90	89.82	10,844.4	-241.6	7,582.3	7,582.3	0.00	0.00	0.00
18,300.0	89.90	89.82	10,844.6	-241.3	7,682.3	7,682.3	0.00	0.00	0.00
18,400.0	89.90	89.82	10,844.8	-241.0	7,782.3	7,782.3	0.00	0.00	0.00
18,500.0	89.90	89.82	10,844.9	-240.7	7,882.3	7,882.3	0.00	0.00	0.00
18,600.0	89.90	89.82	10,845.1	-240.4	7,982.3	7,982.3	0.00	0.00	0.00
18,700.0	89.90	89.82	10,845.3	-240.1	8,082.3	8,082.3	0.00	0.00	0.00
18,800.0	89.90	89.82	10,845.4	-239.8	8,182.3	8,182.3	0.00	0.00	0.00
18,900.0	89.90	89.82	10,845.6	-239.5	8,282.3	8,282.3	0.00	0.00	0.00
19,000.0	89.90	89.82	10,845.8	-239.1	8,382.3	8,382.3	0.00	0.00	0.00



Ryan Directional Services Planning Report



Database: EDM 5000.1 Single User Db
Company: Oasis Petroleum
Project: Indian Hills
Site: 153N-100W-17/18
Well: Kline Federal 5300 41-18 12TX
Wellbore: Kline Federal 5300 41-18 12TX
Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature

Planned Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
19,100.0	89.90	89.82	10,846.0	-238.8	8,482.3	8,482.3	0.00	0.00	0.00
19,200.0	89.90	89.82	10,846.1	-238.5	8,582.3	8,582.3	0.00	0.00	0.00
19,300.0	89.90	89.82	10,846.3	-238.2	8,682.3	8,682.3	0.00	0.00	0.00
19,400.0	89.90	89.82	10,846.5	-237.9	8,782.3	8,782.3	0.00	0.00	0.00
19,500.0	89.90	89.82	10,846.7	-237.6	8,882.3	8,882.3	0.00	0.00	0.00
19,600.0	89.90	89.82	10,846.8	-237.3	8,982.3	8,982.3	0.00	0.00	0.00
19,700.0	89.90	89.82	10,847.0	-237.0	9,082.3	9,082.3	0.00	0.00	0.00
19,800.0	89.90	89.82	10,847.2	-236.7	9,182.3	9,182.3	0.00	0.00	0.00
19,900.0	89.90	89.82	10,847.4	-236.4	9,282.3	9,282.3	0.00	0.00	0.00
20,000.0	89.90	89.82	10,847.5	-236.1	9,382.3	9,382.3	0.00	0.00	0.00
20,100.0	89.90	89.82	10,847.7	-235.8	9,482.3	9,482.3	0.00	0.00	0.00
20,200.0	89.90	89.82	10,847.9	-235.4	9,582.3	9,582.3	0.00	0.00	0.00
20,300.0	89.90	89.82	10,848.1	-235.1	9,682.3	9,682.3	0.00	0.00	0.00
20,400.0	89.90	89.82	10,848.2	-234.8	9,782.3	9,782.3	0.00	0.00	0.00
20,500.0	89.90	89.82	10,848.4	-234.5	9,882.3	9,882.3	0.00	0.00	0.00
20,600.0	89.90	89.82	10,848.6	-234.2	9,982.3	9,982.3	0.00	0.00	0.00
20,667.7	89.90	89.82	10,848.7	-234.0	10,050.0	10,050.0	0.00	0.00	0.00

5300 41-18 12TX PBHL

Design Targets

Target Name	Dip Angle (°)	Dip Dir. (°)	TVD (usft)	+N/-S (usft)	+E/-W (usft)	Northing (usft)	Easting (usft)	Latitude	Longitude
5300 41-18 12TX PBHL	0.00	0.00	10,848.7	-234.0	10,050.0	404,497.87	1,220,000.62	48° 4' 5.214 N	103° 33' 43.972 W
- hit/miss target									
- Shape									
- plan hits target center									
- Point									

Casing Points

Measured Depth (usft)	Vertical Depth (usft)	Name	Casing Diameter (")	Hole Diameter (")
11,189.0	10,832.2	7"	7	8-3/4



Ryan Directional Services Planning Report



Database: EDM 5000.1 Single User Db
Company: Oasis Petroleum
Project: Indian Hills
Site: 153N-100W-17/18
Well: Kline Federal 5300 41-18 12TX
Wellbore: Kline Federal 5300 41-18 12TX
Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature

Formations

Measured Depth (usft)	Vertical Depth (usft)	Name	Lithology	Dip (°)	Dip Direction (°)
1,972.0	1,972.0	Pierre			
4,646.1	4,646.0	Greenhorn			
5,057.1	5,057.0	Mowry			
5,471.1	5,471.0	Dakota			
6,486.2	6,486.0	Rierdon			
6,814.2	6,814.0	Dunham Salt			
6,929.2	6,929.0	Dunham Salt Base			
7,024.2	7,024.0	Spearfish			
7,283.2	7,283.0	Pine Salt			
7,318.2	7,318.0	Pine Salt Base			
7,374.2	7,374.0	Opeche Salt			
7,403.2	7,403.0	Opeche Salt Base			
7,605.2	7,605.0	Broom Creek (Top of Minnelusa Gp.)			
7,684.2	7,684.0	Amsden			
7,853.2	7,853.0	Tyler			
8,047.2	8,047.0	Otter (Base of Minnelusa Gp.)			
8,390.2	8,390.0	Kibbey Lime			
8,542.2	8,542.0	Charles Salt			
9,159.2	9,159.0	UB			
9,238.2	9,238.0	Base Last Salt			
9,301.2	9,301.0	Ratcliffe			
9,454.2	9,454.0	Mission Canyon			
10,022.2	10,022.0	Lodgepole			
10,209.2	10,209.0	Lodgepole Fracture Zone			
10,780.4	10,726.0	False Bakken			
10,795.0	10,735.0	Upper Bakken			
10,822.5	10,751.0	Middle Bakken			
10,891.1	10,785.0	Lower Bakken			
10,905.4	10,791.0	Pronghorn			
10,969.5	10,813.0	Three Forks			
11,035.3	10,827.0	TF Target Top			
13,943.9	10,837.0	TF Target Base			
14,518.6	10,838.0	Claystone			

Plan Annotations

Measured Depth (usft)	Vertical Depth (usft)	Local Coordinates		Comment
		+N/-S (usft)	+E/-W (usft)	
10,354.8	10,354.6	0.0	50.0	KOP Build 12"/100'
11,104.0	10,832.0	-163.0	497.9	EOC



Oasis Petroleum

Indian Hills

153N-100W-17/18

Kline Federal 5300 41-18 12TX

Kline Federal 5300 41-18 12TX

Design #2

Anticollision Report

17 June, 2014



**RYAN DIRECTIONAL
SERVICES**
A NABORS COMPANY



Ryan Directional Services Anticollision Report



Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Reference	Design #2		
Filter type:	NO GLOBAL FILTER: Using user defined selection & filtering criteria		
Interpolation Method:	Stations	Error Model:	ISCWSA
Depth Range:	Unlimited	Scan Method:	Closest Approach 3D
Results Limited by:	Maximum center-center distance of 10,000.0 usft	Error Surface:	Elliptical Conic
Warning Levels Evaluated at:	2.00 Sigma	Casing Method:	Not applied

Survey Tool Program	Date	6/17/2014		
From (usft)	To (usft)	Survey (Wellbore)	Tool Name	Description
0.0	20,667.7	Design #2 (Kline Federal 5300 41-18 12TX	MWD	MWD - Standard

Summary

Site Name	Reference Measured Depth (usft)	Offset Measured Depth (usft)	Distance Between Centres (usft)	Distance Between Ellipses (usft)	Separation Factor	Warning
Offset Well - Wellbore - Design						
153N-100W-17/18						
Kline Federal 5300 41-18 10B - Kline Federal 5300 41-18	2,200.0	2,200.0	65.9	55.5	6.328 CC	
Kline Federal 5300 41-18 10B - Kline Federal 5300 41-18	20,667.7	20,590.0	553.9	-27.2	0.953 Level 1, ES, SF	
Kline Federal 5300 41-18 11T2 - Kline Federal 5300 41-1	2,200.0	2,200.0	33.4	23.0	3.213 CC	
Kline Federal 5300 41-18 11T2 - Kline Federal 5300 41-1	20,667.7	20,743.0	506.9	-77.4	0.868 Level 1, ES, SF	
Kline Federal 5300 41-18 13T2X - Kline Federal 5300 41	2,200.0	2,200.0	33.4	23.0	3.213 CC	
Kline Federal 5300 41-18 13T2X - Kline Federal 5300 41	20,667.7	20,839.5	402.6	-181.2	0.690 Level 1, ES, SF	
Kline Federal 5300 41-18 14BX - Kline Federal 5300 41-1	2,200.0	2,200.0	65.9	55.5	6.328 CC	
Kline Federal 5300 41-18 14BX - Kline Federal 5300 41-1	20,667.7	20,608.5	433.3	-110.3	0.797 Level 1, ES, SF	
Kline Federal 5300 41-18 14BX - Kline Federal 5300 41-1	2,200.0	2,200.0	65.9	55.5	6.328 CC	
Kline Federal 5300 41-18 14BX - Kline Federal 5300 41-1	20,667.7	20,708.4	405.5	-173.0	0.701 Level 1, ES, SF	
Kline Federal 5300 41-18 9T - Kline Federal 5300 41-18	2,200.0	2,200.0	99.3	88.9	9.540 CC, ES	
Kline Federal 5300 41-18 9T - Kline Federal 5300 41-18	20,667.7	20,824.5	1,050.8	467.1	1.800 SF	

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 10B - Kline Federal 5300 41-18 10B - Design #1											Offset Site Error:	0.0 usft		
Survey Program: 0-MWD											Offset Well Error:	2.0 usft		
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Semi Major Axis Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Distance Between Centres (usft)	Distance Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning	
0.0	0.0	0.0	0.0	2.0	2.0	0.01	65.9	0.0	65.9					
100.0	100.0	100.0	100.0	2.0	2.0	0.01	65.9	0.0	65.9	61.9	4.00	16.450		
200.0	200.0	200.0	200.0	2.0	2.0	0.01	65.9	0.0	65.9	61.8	4.05	16.272		
300.0	300.0	300.0	300.0	2.1	2.1	0.01	65.9	0.0	65.9	61.7	4.14	15.908		
400.0	400.0	400.0	400.0	2.1	2.1	0.01	65.9	0.0	65.9	61.6	4.28	15.395		
500.0	500.0	500.0	500.0	2.2	2.2	0.01	65.9	0.0	65.9	61.4	4.46	14.775		
600.0	600.0	600.0	600.0	2.3	2.3	0.01	65.9	0.0	65.9	61.2	4.67	14.093		
700.0	700.0	700.0	700.0	2.5	2.5	0.01	65.9	0.0	65.9	60.9	4.92	13.384		
800.0	800.0	800.0	800.0	2.6	2.6	0.01	65.9	0.0	65.9	60.7	5.20	12.677		
900.0	900.0	900.0	900.0	2.7	2.7	0.01	65.9	0.0	65.9	60.4	5.49	11.989		
1,000.0	1,000.0	1,000.0	1,000.0	2.9	2.9	0.01	65.9	0.0	65.9	60.0	5.81	11.335		
1,100.0	1,100.0	1,100.0	1,100.0	3.1	3.1	0.01	65.9	0.0	65.9	59.7	6.14	10.719		
1,200.0	1,200.0	1,200.0	1,200.0	3.2	3.2	0.01	65.9	0.0	65.9	59.4	6.49	10.144		
1,300.0	1,300.0	1,300.0	1,300.0	3.4	3.4	0.01	65.9	0.0	65.9	59.0	6.85	9.612		
1,400.0	1,400.0	1,400.0	1,400.0	3.6	3.6	0.01	65.9	0.0	65.9	58.6	7.22	9.120		
1,500.0	1,500.0	1,500.0	1,500.0	3.8	3.8	0.01	65.9	0.0	65.9	58.3	7.60	8.686		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services

Anticollision Report



Company: Oasis Petroleum
Project: Indian Hills
Reference Site: 153N-100W-17/18
Site Error: 0.0 usft
Reference Well: Kline Federal 5300 41-18 12TX
Well Error: 2.0 usft
Reference Wellbore: Kline Federal 5300 41-18 12TX
Reference Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at: 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 10B - Kline Federal 5300 41-18 10B - Design #1													Offset Site Error: 0.0 usft	
Survey Program: 0-MWD													Offset Well Error: 2.0 usft	
Reference		Offset		Semi Major Axis			Distance						Warning	
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	Offset Wellbore Centre +E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
1,600.0	1,600.0	1,600.0	1,600.0	4.0	4.0	0.01	65.9	0.0	65.9	57.9	7.99	8.247		
1,700.0	1,700.0	1,700.0	1,700.0	4.2	4.2	0.01	65.9	0.0	65.9	57.5	8.38	7.861	CC	
1,800.0	1,800.0	1,800.0	1,800.0	4.4	4.4	0.01	65.9	0.0	65.9	57.1	8.78	7.505		
1,900.0	1,900.0	1,900.0	1,900.0	4.6	4.6	0.01	65.9	0.0	65.9	56.7	9.18	7.176		
2,000.0	2,000.0	2,000.0	2,000.0	4.8	4.8	0.01	65.9	0.0	65.9	56.3	9.58	6.872		
2,100.0	2,100.0	2,100.0	2,100.0	5.0	5.0	0.01	65.9	0.0	65.9	55.9	9.99	6.589		
2,200.0	2,200.0	2,200.0	2,200.0	5.2	5.2	0.01	65.9	0.0	65.9	55.5	10.41	6.328	CC	
2,216.7	2,216.7	2,216.5	2,216.5	5.2	5.2	-90.00	65.9	0.1	65.9	55.4	10.47	6.290		
2,300.0	2,300.0	2,299.8	2,299.8	5.4	5.4	-90.03	66.1	0.8	66.1	55.3	10.78	6.127		
2,400.0	2,400.0	2,399.8	2,399.8	5.6	5.6	-90.05	66.3	1.6	66.3	55.2	11.16	5.941		
2,500.0	2,500.0	2,499.8	2,499.8	5.8	5.8	-90.08	66.5	2.4	66.5	55.0	11.54	5.765		
2,600.0	2,600.0	2,599.8	2,599.8	6.0	6.0	-90.11	66.8	3.3	66.8	54.9	11.93	5.597		
2,700.0	2,700.0	2,699.8	2,699.8	6.1	6.2	-90.14	67.0	4.1	67.0	54.7	12.32	5.438		
2,800.0	2,800.0	2,799.8	2,799.8	6.3	6.4	-90.17	67.3	5.0	67.3	54.5	12.72	5.288		
2,900.0	2,900.0	2,899.8	2,899.8	6.5	6.6	-90.19	67.5	5.8	67.5	54.4	13.12	5.145		
3,000.0	3,000.0	2,999.8	2,999.8	6.7	6.8	-90.22	67.7	6.6	67.7	54.2	13.52	5.009		
3,100.0	3,100.0	3,099.8	3,099.8	6.9	7.0	-90.25	68.0	7.5	68.0	54.0	13.93	4.880		
3,200.0	3,200.0	3,199.8	3,199.8	7.1	7.2	-90.27	68.2	8.3	68.2	53.9	14.34	4.758		
3,300.0	3,300.0	3,299.8	3,299.8	7.3	7.4	-90.30	68.4	9.2	68.4	53.7	14.75	4.641		
3,400.0	3,400.0	3,399.8	3,399.8	7.6	7.6	-90.33	68.7	10.0	68.7	53.5	15.16	4.531		
3,500.0	3,500.0	3,499.8	3,499.8	7.8	7.8	-90.35	68.9	10.8	68.9	53.3	15.57	4.425		
3,600.0	3,599.9	3,599.8	3,599.8	8.0	8.0	-90.38	69.2	11.7	69.2	53.2	15.99	4.324		
3,700.0	3,699.9	3,699.8	3,699.8	8.2	8.2	-90.40	69.4	12.5	69.4	53.0	16.41	4.229		
3,800.0	3,799.9	3,799.8	3,799.8	8.4	8.5	-90.43	69.6	13.4	69.6	52.8	16.83	4.137		
3,900.0	3,899.9	3,899.8	3,899.8	8.6	8.7	-90.45	69.9	14.2	69.9	52.6	17.25	4.050		
4,000.0	3,999.9	3,999.8	3,999.8	8.8	8.9	-90.48	70.1	15.0	70.1	52.4	17.68	3.966		
4,100.0	4,099.9	4,099.8	4,099.8	9.0	9.1	-90.50	70.3	15.9	70.3	52.2	18.10	3.886		
4,200.0	4,199.9	4,199.8	4,199.8	9.2	9.3	-90.53	70.6	16.7	70.6	52.1	18.53	3.809		
4,300.0	4,299.9	4,299.8	4,299.8	9.4	9.5	-90.55	70.8	17.6	70.8	51.9	18.95	3.736		
4,400.0	4,399.9	4,399.8	4,399.8	9.7	9.7	-90.58	71.0	18.4	71.1	51.7	19.38	3.666		
4,500.0	4,499.9	4,499.8	4,499.8	9.9	10.0	-90.60	71.3	19.2	71.3	51.5	19.81	3.598		
4,600.0	4,599.9	4,599.8	4,599.7	10.1	10.2	-90.63	71.5	20.1	71.5	51.3	20.24	3.534		
4,700.0	4,699.9	4,699.8	4,699.7	10.3	10.4	-90.65	71.8	20.9	71.8	51.1	20.67	3.472		
4,800.0	4,799.9	4,799.8	4,799.7	10.5	10.6	-90.68	72.0	21.8	72.0	50.9	21.10	3.412		
4,900.0	4,899.9	4,899.8	4,899.7	10.7	10.8	-90.70	72.2	22.6	72.2	50.7	21.54	3.354		
5,000.0	4,999.9	4,999.8	4,999.7	10.9	11.0	-90.72	72.5	23.4	72.5	50.5	21.97	3.299		
5,100.0	5,099.9	5,099.8	5,099.7	11.2	11.3	-90.75	72.7	24.3	72.7	50.3	22.40	3.246		
5,200.0	5,199.9	5,199.8	5,199.7	11.4	11.5	-90.77	72.9	25.1	73.0	50.1	22.84	3.194		
5,300.0	5,299.9	5,299.8	5,299.7	11.6	11.7	-90.79	73.2	26.0	73.2	49.9	23.27	3.145		
5,400.0	5,399.9	5,399.8	5,399.7	11.8	11.9	-90.82	73.4	26.8	73.4	49.7	23.71	3.097		
5,500.0	5,499.9	5,499.8	5,499.7	12.0	12.1	-90.84	73.7	27.6	73.7	49.5	24.15	3.051		
5,600.0	5,599.9	5,599.8	5,599.7	12.2	12.3	-90.86	73.9	28.5	73.9	49.3	24.58	3.006		
5,700.0	5,699.9	5,699.8	5,699.7	12.5	12.6	-90.88	74.1	29.3	74.1	49.1	25.02	2.963		
5,800.0	5,799.9	5,799.8	5,799.7	12.7	12.8	-90.91	74.4	30.2	74.4	48.9	25.46	2.921		
5,900.0	5,899.9	5,899.8	5,899.7	12.9	13.0	-90.93	74.6	31.0	74.6	48.7	25.90	2.881		
6,000.0	5,999.9	5,999.8	5,999.7	13.1	13.2	-90.95	74.8	31.8	74.9	48.5	26.34	2.842		
6,100.0	6,099.9	6,099.8	6,099.7	13.3	13.4	-90.97	75.1	32.7	75.1	48.3	26.77	2.804		
6,200.0	6,199.8	6,199.8	6,199.7	13.6	13.7	-90.99	75.3	33.5	75.3	48.1	27.21	2.768		
6,300.0	6,299.8	6,299.8	6,299.7	13.8	13.9	-91.02	75.6	34.4	75.6	47.9	27.65	2.732		
6,400.0	6,399.8	6,399.8	6,399.7	14.0	14.1	-91.04	75.8	35.2	75.8	47.7	28.09	2.698		
6,500.0	6,499.8	6,499.8	6,499.7	14.2	14.3	-91.06	76.0	36.0	76.0	47.5	28.53	2.665		
6,600.0	6,599.8	6,599.8	6,599.7	14.4	14.5	-91.08	76.3	36.9	76.3	47.3	28.97	2.632		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services
Anticollision Report



Company: Oasis Petroleum
Project: Indian Hills
Reference Site: 153N-100W-17/18
Site Error: 0.0 usft
Reference Well: Kline Federal 5300 41-18 12TX
Well Error: 2.0 usft
Reference Wellbore: Kline Federal 5300 41-18 12TX
Reference Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at: 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 10B - Kline Federal 5300 41-18 10B - Design #1														Offset Site Error:	0.0 usft
Survey Program: 0-MWD														Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis			Distance							Warning	
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	Centre +E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor			
6,700.0	6,699.8	6,699.8	6,699.7	14.7	14.8	-91.10	76.5	37.7	76.5	47.1	29.42	2.601			
6,800.0	6,799.8	6,799.8	6,799.7	14.9	15.0	-91.12	76.7	38.6	76.7	46.9	29.86	2.571			
6,900.0	6,899.8	6,899.8	6,899.7	15.1	15.2	-91.14	77.0	39.4	77.0	46.7	30.30	2.541			
7,000.0	6,999.8	6,999.8	6,999.6	15.3	15.4	-91.16	77.2	40.2	77.2	46.5	30.74	2.512			
7,100.0	7,099.8	7,099.8	7,099.6	15.5	15.7	-91.19	77.4	41.1	77.5	46.3	31.18	2.484			
7,200.0	7,199.8	7,199.8	7,199.6	15.8	15.9	-91.21	77.7	41.9	77.7	46.1	31.63	2.457			
7,300.0	7,299.8	7,299.8	7,299.6	16.0	16.1	-91.23	77.9	42.8	77.9	45.9	32.07	2.430			
7,400.0	7,399.8	7,399.8	7,399.6	16.2	16.3	-91.25	78.2	43.6	78.2	45.7	32.51	2.405			
7,500.0	7,499.8	7,499.8	7,499.6	16.4	16.5	-91.27	78.4	44.4	78.4	45.5	32.95	2.379			
7,600.0	7,599.8	7,599.8	7,599.6	16.6	16.8	-91.29	78.6	45.3	78.6	45.3	33.40	2.355			
7,700.0	7,699.8	7,699.8	7,699.6	16.9	17.0	-91.31	78.9	46.1	78.9	45.0	33.84	2.331			
7,800.0	7,799.8	7,799.8	7,799.6	17.1	17.2	-91.33	79.1	47.0	79.1	44.8	34.28	2.308			
7,900.0	7,899.8	7,899.8	7,899.6	17.3	17.4	-91.35	79.3	47.8	79.4	44.6	34.73	2.285			
7,926.7	7,926.4	7,926.5	7,926.3	17.4	17.5	-91.35	79.4	48.0	79.4	44.6	34.85	2.279			
7,943.3	7,943.1	7,943.3	7,943.1	17.4	17.5	-1.35	79.4	48.1	79.4	44.6	34.87	2.278			
8,000.0	7,999.8	8,000.0	7,999.8	17.5	17.6	-1.35	79.4	48.1	79.4	44.4	35.10	2.264			
8,100.0	8,099.8	8,100.0	8,099.8	17.7	17.8	-1.35	79.4	48.1	79.4	43.9	35.51	2.237			
8,200.0	8,199.8	8,200.0	8,199.8	17.9	18.1	-1.35	79.4	48.1	79.4	43.5	35.94	2.211			
8,300.0	8,299.8	8,300.0	8,299.8	18.1	18.3	-1.35	79.4	48.1	79.4	43.1	36.36	2.185			
8,400.0	8,399.8	8,400.0	8,399.8	18.4	18.5	-1.35	79.4	48.1	79.4	42.7	36.78	2.160			
8,500.0	8,499.8	8,500.0	8,499.8	18.6	18.7	-1.35	79.4	48.1	79.4	42.2	37.20	2.136			
8,600.0	8,599.8	8,600.0	8,599.8	18.8	18.9	-1.35	79.4	48.1	79.4	41.8	37.62	2.112			
8,700.0	8,699.8	8,700.0	8,699.8	19.0	19.1	-1.35	79.4	48.1	79.4	41.4	38.05	2.088			
8,800.0	8,799.8	8,800.0	8,799.8	19.2	19.3	-1.35	79.4	48.1	79.4	41.0	38.47	2.065			
8,900.0	8,899.8	8,900.0	8,899.8	19.4	19.6	-1.35	79.4	48.1	79.4	40.6	38.89	2.043			
9,000.0	8,999.8	9,000.0	8,999.8	19.6	19.8	-1.35	79.4	48.1	79.4	40.1	39.32	2.021			
9,100.0	9,099.8	9,100.0	9,099.8	19.8	20.0	-1.35	79.4	48.1	79.4	39.7	39.75	1.999			
9,200.0	9,199.8	9,200.0	9,199.8	20.0	20.2	-1.35	79.4	48.1	79.4	39.3	40.17	1.978			
9,300.0	9,299.8	9,300.0	9,299.8	20.2	20.4	-1.35	79.4	48.1	79.4	38.8	40.60	1.957			
9,400.0	9,399.8	9,400.0	9,399.8	20.5	20.6	-1.35	79.4	48.1	79.4	38.4	41.03	1.936			
9,500.0	9,499.8	9,500.0	9,499.8	20.7	20.8	-1.35	79.4	48.1	79.4	38.0	41.45	1.917			
9,600.0	9,599.8	9,600.0	9,599.8	20.9	21.1	-1.35	79.4	48.1	79.4	37.6	41.88	1.897			
9,700.0	9,699.8	9,700.0	9,699.8	21.1	21.3	-1.35	79.4	48.1	79.4	37.1	42.31	1.878			
9,800.0	9,799.8	9,800.0	9,799.8	21.3	21.5	-1.35	79.4	48.1	79.4	36.7	42.74	1.859			
9,900.0	9,899.8	9,900.0	9,899.8	21.5	21.7	-1.35	79.4	48.1	79.4	36.3	43.17	1.840			
10,000.0	9,999.8	10,000.0	9,999.8	21.7	21.9	-1.35	79.4	48.1	79.4	35.8	43.60	1.822			
10,100.0	10,099.8	10,100.0	10,099.8	21.9	22.1	-1.35	79.4	48.1	79.4	35.4	44.03	1.804			
10,200.0	10,199.8	10,200.0	10,199.8	22.2	22.4	-1.35	79.4	48.1	79.4	35.0	44.46	1.787			
10,262.1	10,261.8	10,262.1	10,261.8	22.3	22.5	-1.35	79.4	48.1	79.4	34.7	44.73	1.776			
10,300.0	10,299.8	10,299.5	10,299.3	22.4	22.6	-1.25	79.5	48.2	79.5	34.6	44.89	1.771			
10,354.8	10,354.6	10,351.8	10,351.3	22.5	22.7	1.59	80.6	52.2	80.7	35.6	45.12	1.788			
10,375.0	10,374.8	10,370.7	10,370.1	22.5	22.7	-106.60	81.4	55.0	81.8	36.5	45.27	1.807			
10,400.0	10,399.7	10,394.1	10,393.0	22.6	22.8	-104.41	82.6	59.4	84.0	36.6	45.37	1.850			
10,425.0	10,424.5	10,417.2	10,415.4	22.7	22.9	-102.32	84.2	64.9	87.0	41.5	45.48	1.912			
10,450.0	10,449.2	10,440.0	10,437.2	22.7	22.9	-100.38	86.0	71.3	90.8	45.2	45.60	1.992			
10,475.0	10,473.5	10,462.6	10,458.5	22.8	23.0	-98.80	88.0	78.5	95.5	49.7	45.71	2.088			
10,500.0	10,497.6	10,484.9	10,479.1	22.8	23.0	-96.98	90.3	86.7	100.8	55.0	45.84	2.200			
10,525.0	10,521.2	10,506.9	10,499.1	22.9	23.1	-95.53	92.8	95.6	106.9	60.9	45.97	2.325			
10,550.0	10,544.4	10,528.6	10,518.3	23.0	23.2	-94.21	95.6	105.3	113.6	67.5	46.10	2.484			
10,575.0	10,567.1	10,550.0	10,536.8	23.0	23.2	-93.03	98.5	115.7	120.9	74.7	46.24	2.615			
10,600.0	10,566.1	10,571.1	10,554.5	23.1	23.3	-91.98	101.6	126.6	128.8	82.4	46.39	2.777			
10,625.0	10,610.6	10,591.8	10,571.5	23.2	23.4	-90.97	104.8	138.1	137.2	90.7	46.54	2.949			

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services
Anticollision Report



Company: Oasis Petroleum
Project: Indian Hills
Reference Site: 153N-100W-17/18
Site Error: 0.0 usft
Reference Well: Kline Federal 5300 41-18 12TX
Well Error: 2.0 usft
Reference Wellbore: Kline Federal 5300 41-18 12TX
Reference Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at: 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 10B - Kline Federal 5300 41-18 10B - Design #1														Offset Site Error:	0.0 usft
Survey Program: 0-MWD														Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis			Distance							Warning	
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N-S (usft)	+E-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor			
10,650.0	10,631.3	10,612.3	10,587.6	23.3	23.4	-90.05	108.2	150.2	146.2	99.5	46.70	3.130			
10,675.0	10,651.3	10,632.4	10,603.0	23.4	23.5	-89.19	111.7	162.7	156.6	108.7	46.86	3.321			
10,700.0	10,670.5	10,652.2	10,617.6	23.5	23.6	-88.37	115.4	175.6	165.5	116.4	47.03	3.518			
10,725.0	10,688.8	10,671.7	10,631.5	23.6	23.7	-87.57	119.1	188.8	175.7	128.5	47.21	3.723			
10,750.0	10,706.2	10,691.0	10,644.5	23.7	23.8	-86.79	122.9	202.4	186.4	139.0	47.40	3.933			
10,775.0	10,722.6	10,709.9	10,656.8	23.9	23.9	-86.03	126.9	216.3	197.4	149.8	47.59	4.148			
10,800.0	10,738.0	10,728.7	10,668.4	24.0	24.0	-85.26	130.9	230.5	208.8	161.0	47.80	4.368			
10,825.0	10,752.4	10,747.1	10,679.3	24.2	24.1	-84.49	134.9	244.8	220.5	172.4	48.01	4.592			
10,850.0	10,765.7	10,765.4	10,689.4	24.4	24.2	-83.72	139.0	259.4	232.4	184.2	48.24	4.817			
10,875.0	10,777.8	10,783.4	10,698.9	24.6	24.3	-82.95	143.2	274.2	244.6	196.1	48.48	5.046			
10,900.0	10,788.8	10,800.0	10,707.0	24.8	24.5	-82.10	147.1	288.1	257.0	208.3	48.71	5.276			
10,925.0	10,798.6	10,818.9	10,715.7	25.0	24.6	-81.37	151.7	304.2	269.7	220.7	48.98	5.505			
10,950.0	10,807.2	10,836.4	10,723.2	25.3	24.8	-80.57	156.0	319.5	282.5	233.2	49.25	5.735			
10,975.0	10,814.5	10,853.8	10,730.0	25.6	24.9	-79.77	160.3	334.9	295.4	245.9	49.53	5.964			
11,000.0	10,820.6	10,871.1	10,736.2	25.8	25.1	-78.96	164.7	350.4	308.5	258.7	49.83	6.192			
11,025.0	10,825.4	10,888.3	10,741.8	26.1	25.2	-78.16	169.1	366.0	321.7	271.6	50.13	6.417			
11,050.0	10,828.9	10,905.4	10,746.8	26.5	25.4	-77.35	173.5	381.8	335.0	284.6	50.45	6.640			
11,075.0	10,831.1	10,922.5	10,751.2	26.8	25.6	-76.55	178.0	397.7	348.4	297.6	50.79	6.860			
11,100.0	10,832.0	10,939.6	10,755.0	27.1	25.8	-75.76	182.6	413.8	361.8	310.6	51.13	7.075			
11,104.0	10,832.0	10,942.3	10,755.5	27.2	25.8	-75.64	183.3	416.3	363.9	312.7	51.19	7.109			
11,200.0	10,832.2	11,010.8	10,764.3	28.8	26.6	-79.00	201.7	481.8	414.8	361.1	53.76	7.716			
11,300.0	10,832.4	11,103.8	10,765.2	30.3	27.9	-80.68	226.2	571.4	464.4	407.7	56.74	8.185			
11,400.0	10,832.6	11,213.0	10,765.4	32.2	29.6	-81.82	251.3	677.6	505.8	445.5	60.30	8.888			
11,500.0	10,832.7	11,328.4	10,765.6	34.1	31.7	-82.57	273.5	790.8	538.0	473.7	64.35	9.361			
11,600.0	10,832.9	11,448.8	10,765.8	36.2	34.1	-83.03	291.7	909.7	560.6	491.7	68.81	8.147			
11,680.4	10,833.1	11,547.8	10,765.9	38.0	36.2	-83.23	303.0	1,008.2	571.3	498.7	72.60	7.870			
11,700.0	10,833.1	11,572.1	10,766.0	38.4	36.7	-83.26	305.2	1,032.4	573.1	499.5	73.56	7.791			
11,800.0	10,833.3	11,696.8	10,766.2	40.7	39.5	-83.35	313.5	1,156.8	579.3	500.7	78.62	7.369			
11,900.0	10,833.4	11,822.0	10,766.4	43.0	42.5	-83.38	316.4	1,282.0	581.3	497.4	83.91	6.928			
12,000.0	10,833.6	11,922.5	10,766.6	45.4	44.9	-83.38	316.4	1,382.5	581.0	492.2	88.75	6.546			
12,100.0	10,833.8	12,022.5	10,766.8	47.9	47.3	-83.37	316.3	1,482.5	580.7	487.0	93.69	6.198			
12,200.0	10,834.0	12,122.5	10,766.9	50.4	49.8	-83.37	316.3	1,582.5	580.4	481.6	98.73	5.878			
12,300.0	10,834.1	12,222.5	10,767.1	53.0	52.4	-83.36	316.3	1,682.5	580.0	476.2	103.85	5.586			
12,400.0	10,834.3	12,322.5	10,767.3	55.5	55.0	-83.36	316.3	1,782.5	579.7	470.7	109.04	5.317			
12,500.0	10,834.5	12,422.5	10,767.5	58.2	57.6	-83.36	316.3	1,882.5	579.4	465.1	114.29	5.070			
12,600.0	10,834.7	12,522.5	10,767.6	60.8	60.2	-83.35	316.3	1,982.5	579.1	469.5	119.59	4.842			
12,700.0	10,834.8	12,622.5	10,767.8	63.5	62.9	-83.35	316.3	2,082.5	578.8	463.9	124.94	4.633			
12,800.0	10,835.0	12,722.5	10,768.0	66.2	65.6	-83.35	316.3	2,182.5	578.5	448.1	130.33	4.439			
12,900.0	10,835.2	12,822.5	10,768.2	68.9	68.3	-83.34	316.3	2,282.5	578.2	442.4	135.76	4.259			
13,000.0	10,835.4	12,922.5	10,768.3	71.7	71.0	-83.34	316.3	2,382.5	577.9	436.8	141.22	4.092			
13,100.0	10,835.5	13,022.5	10,768.5	74.4	73.8	-83.34	316.3	2,482.5	577.5	430.8	146.70	3.937			
13,200.0	10,835.7	13,122.5	10,768.7	77.2	76.5	-83.33	316.3	2,582.5	577.2	425.0	152.22	3.792			
13,300.0	10,835.9	13,222.5	10,768.9	79.9	79.3	-83.33	316.3	2,682.5	576.9	419.2	157.75	3.657			
13,400.0	10,836.1	13,322.5	10,769.0	82.7	82.1	-83.33	316.3	2,782.5	576.6	413.3	163.31	3.531			
13,500.0	10,836.2	13,422.5	10,769.2	85.5	84.9	-83.32	316.3	2,882.5	576.3	407.4	168.88	3.412			
13,600.0	10,836.4	13,522.5	10,769.4	88.3	87.7	-83.32	316.3	2,982.5	576.0	401.5	174.47	3.301			
13,700.0	10,836.6	13,622.5	10,769.6	91.1	90.5	-83.31	316.3	3,082.5	575.7	395.6	180.07	3.197			
13,800.0	10,836.7	13,722.5	10,769.7	94.0	93.3	-83.31	316.3	3,182.5	575.4	389.7	185.69	3.099			
13,900.0	10,836.9	13,822.5	10,769.9	96.8	96.1	-83.31	316.2	3,282.5	575.0	383.7	191.32	3.008			
14,000.0	10,837.1	13,922.5	10,770.1	99.6	99.0	-83.30	316.2	3,382.5	574.7	377.8	196.96	2.918			
14,100.0	10,837.3	14,022.5	10,770.3	102.5	101.8	-83.30	316.2	3,482.5	574.4	371.8	202.61	2.835			
14,200.0	10,837.4	14,122.5	10,770.4	105.3	104.7	-83.30	316.2	3,582.5	574.1	365.8	208.27	2.757			

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services Anticollision Report



Company: Oasis Petroleum
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Site Error: 0.0 usft
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Well Error: 2.0 usft
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Reference Design: Design #2

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TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at: 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 10B - Kline Federal 5300 41-18 10B - Design #1											Offset Site Error: 0.0 usft		
Survey Program: 0-MWD											Offset Well Error: 2.0 usft		
Reference		Offset		Semi Major Axis			Distance				Warning		
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)		Minimum Separation (usft)	Separation Factor
14,300.0	10,837.6	14,222.5	10,770.6	108.2	107.5	-83.29	316.2	3,682.5	573.8	359.9	213.94	2.682	
14,400.0	10,837.8	14,322.5	10,770.8	111.0	110.4	-83.29	316.2	3,782.5	573.5	353.9	219.82	2.611	
14,500.0	10,838.0	14,422.5	10,771.0	113.9	113.2	-83.29	316.2	3,882.5	573.2	347.9	225.31	2.544	
14,600.0	10,838.1	14,522.5	10,771.1	116.7	116.1	-83.28	316.2	3,982.5	572.9	341.9	231.00	2.480	
14,700.0	10,838.3	14,622.5	10,771.3	119.6	118.9	-83.28	316.2	4,082.5	572.6	335.9	236.70	2.419	
14,800.0	10,838.5	14,722.5	10,771.5	122.5	121.8	-83.27	316.2	4,182.5	572.2	329.8	242.40	2.361	
14,900.0	10,838.7	14,822.5	10,771.7	125.3	124.7	-83.27	316.2	4,282.5	571.9	323.8	248.11	2.305	
15,000.0	10,838.8	14,922.5	10,771.8	128.2	127.5	-83.27	316.2	4,382.5	571.6	317.8	253.82	2.252	
15,100.0	10,839.0	15,022.5	10,772.0	131.1	130.4	-83.26	316.2	4,482.5	571.3	311.8	259.54	2.201	
15,200.0	10,839.2	15,122.5	10,772.2	133.9	133.3	-83.26	316.2	4,582.5	571.0	305.7	265.27	2.153	
15,300.0	10,839.4	15,222.5	10,772.3	136.8	136.2	-83.26	316.2	4,682.5	570.7	299.7	271.00	2.106	
15,400.0	10,839.5	15,322.5	10,772.5	139.7	139.0	-83.25	316.2	4,782.5	570.4	293.6	276.73	2.061	
15,500.0	10,839.7	15,422.5	10,772.7	142.6	141.9	-83.25	316.2	4,882.5	570.1	287.6	282.46	2.018	
15,600.0	10,839.9	15,522.5	10,772.9	145.5	144.8	-83.25	316.1	4,982.5	569.7	281.5	288.20	1.977	
15,700.0	10,840.1	15,622.5	10,773.0	148.4	147.7	-83.24	316.1	5,082.5	569.4	275.5	293.95	1.937	
15,800.0	10,840.2	15,722.5	10,773.2	151.3	150.6	-83.24	316.1	5,182.5	569.1	269.4	299.69	1.899	
15,900.0	10,840.4	15,822.5	10,773.4	154.1	153.5	-83.23	316.1	5,282.5	568.8	263.4	305.44	1.862	
16,000.0	10,840.6	15,922.5	10,773.6	157.0	156.4	-83.23	316.1	5,382.5	568.5	257.3	311.19	1.827	
16,100.0	10,840.8	16,022.5	10,773.7	159.9	159.3	-83.23	316.1	5,482.5	568.2	251.2	316.94	1.793	
16,200.0	10,840.9	16,122.5	10,773.9	162.8	162.2	-83.22	316.1	5,582.5	567.9	245.2	322.70	1.760	
16,300.0	10,841.1	16,222.5	10,774.1	165.7	165.1	-83.22	316.1	5,682.5	567.6	239.1	328.46	1.728	
16,400.0	10,841.3	16,322.5	10,774.3	168.6	168.0	-83.22	316.1	5,782.5	567.2	233.0	334.22	1.697	
16,500.0	10,841.4	16,422.5	10,774.4	171.5	170.9	-83.21	316.1	5,882.5	566.9	227.0	339.98	1.668	
16,600.0	10,841.6	16,522.5	10,774.6	174.4	173.8	-83.21	316.1	5,982.5	566.6	220.9	345.75	1.639	
16,700.0	10,841.8	16,622.5	10,774.8	177.3	176.7	-83.20	316.1	6,082.5	566.3	214.8	351.51	1.611	
16,800.0	10,842.0	16,722.5	10,775.0	180.2	179.6	-83.20	316.1	6,182.5	566.0	208.7	357.28	1.584	
16,900.0	10,842.1	16,822.5	10,775.1	183.1	182.5	-83.20	316.1	6,282.5	565.7	202.6	363.05	1.558	
17,000.0	10,842.3	16,922.5	10,775.3	186.0	185.4	-83.19	316.1	6,382.5	565.4	196.6	368.82	1.533	
17,100.0	10,842.5	17,022.5	10,775.5	188.9	188.3	-83.19	316.1	6,482.5	565.1	190.5	374.59	1.508	
17,200.0	10,842.7	17,122.5	10,775.7	191.8	191.2	-83.19	316.1	6,582.5	564.7	184.4	380.37	1.485 Level 3	
17,300.0	10,842.8	17,222.5	10,775.8	194.7	194.1	-83.18	316.1	6,682.5	564.4	178.3	386.14	1.462 Level 3	
17,400.0	10,843.0	17,322.5	10,776.0	197.6	197.0	-83.18	316.0	6,782.5	564.1	172.2	391.92	1.439 Level 3	
17,500.0	10,843.2	17,422.5	10,776.2	200.6	199.9	-83.17	316.0	6,882.5	563.8	166.1	397.70	1.418 Level 3	
17,600.0	10,843.4	17,522.5	10,776.4	203.5	202.8	-83.17	316.0	6,982.5	563.5	160.0	403.48	1.397 Level 3	
17,700.0	10,843.5	17,622.5	10,776.5	206.4	205.7	-83.17	316.0	7,082.5	563.2	153.9	409.26	1.376 Level 3	
17,800.0	10,843.7	17,722.5	10,776.7	209.3	208.6	-83.16	316.0	7,182.5	562.9	147.8	415.04	1.356 Level 3	
17,900.0	10,843.9	17,822.5	10,776.9	212.2	211.5	-83.16	316.0	7,282.5	562.6	141.7	420.82	1.337 Level 3	
18,000.0	10,844.1	17,922.5	10,777.1	215.1	214.4	-83.16	316.0	7,382.5	562.3	135.6	426.60	1.318 Level 3	
18,100.0	10,844.2	18,022.5	10,777.2	218.0	217.3	-83.15	316.0	7,482.5	561.9	129.5	432.39	1.300 Level 3	
18,200.0	10,844.4	18,122.5	10,777.4	220.9	220.3	-83.15	316.0	7,582.5	561.6	123.5	438.17	1.282 Level 3	
18,300.0	10,844.6	18,222.5	10,777.6	223.8	223.2	-83.14	316.0	7,682.5	561.3	117.4	443.96	1.264 Level 3	
18,400.0	10,844.8	18,322.5	10,777.7	226.7	226.1	-83.14	316.0	7,782.5	561.0	111.3	449.75	1.247 Level 2	
18,500.0	10,844.9	18,422.5	10,777.9	229.7	229.0	-83.14	316.0	7,882.5	560.7	105.2	455.53	1.231 Level 2	
18,600.0	10,845.1	18,522.5	10,778.1	232.6	231.9	-83.13	316.0	7,982.5	560.4	99.1	461.32	1.215 Level 2	
18,700.0	10,845.3	18,622.5	10,778.3	235.5	234.8	-83.13	316.0	8,082.5	560.1	93.0	467.11	1.199 Level 2	
18,800.0	10,845.4	18,722.5	10,778.4	238.4	237.7	-83.12	316.0	8,182.5	559.8	86.9	472.90	1.184 Level 2	
18,900.0	10,845.6	18,822.5	10,778.6	241.3	240.7	-83.12	316.0	8,282.5	559.4	80.8	478.69	1.169 Level 2	
19,000.0	10,845.8	18,922.5	10,778.8	244.2	243.6	-83.12	316.0	8,382.5	559.1	74.6	484.48	1.164 Level 2	
19,100.0	10,846.0	19,022.5	10,779.0	247.1	246.5	-83.11	315.9	8,482.5	558.8	68.5	490.27	1.140 Level 2	
19,200.0	10,846.1	19,122.5	10,779.1	250.1	249.4	-83.11	315.9	8,582.5	558.5	62.4	496.07	1.126 Level 2	
19,300.0	10,846.3	19,222.5	10,779.3	253.0	252.3	-83.11	315.9	8,682.5	558.2	56.3	501.86	1.112 Level 2	
19,400.0	10,846.5	19,322.6	10,779.5	255.9	255.2	-83.10	315.9	8,782.5	557.9	50.2	507.65	1.099 Level 2	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services

Anticollision Report



Company: Oasis Petroleum
Project: Indian Hills
Reference Site: 153N-100W-17/18
Site Error: 0.0 usft
Reference Well: Kline Federal 5300 41-18 12TX
Well Error: 2.0 usft
Reference Wellbore: Kline Federal 5300 41-18 12TX
Reference Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 10B - Kline Federal 5300 41-18 10B - Design #1													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis		Highside Toolface (°)	Distance		Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning	
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)		Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)						
19,500.0	10,846.7	19,422.5	10,779.7	258.8	258.1	-83.10	315.9	8,862.5	557.6	44.1	513.45	1.086	Level 2	
19,600.0	10,846.8	19,522.5	10,779.8	261.7	261.1	-83.09	315.9	8,982.5	557.3	38.0	519.24	1.073	Level 2	
19,700.0	10,847.0	19,622.5	10,780.0	264.8	264.0	-83.09	315.9	9,082.5	556.9	31.9	525.04	1.061	Level 2	
19,800.0	10,847.2	19,722.5	10,780.2	267.6	266.9	-83.09	315.9	9,182.5	556.6	25.8	530.83	1.049	Level 2	
19,900.0	10,847.4	19,822.5	10,780.4	270.5	269.8	-83.08	315.9	9,282.5	556.3	19.7	536.63	1.037	Level 2	
20,000.0	10,847.5	19,922.5	10,780.5	273.4	272.7	-83.08	315.9	9,382.5	556.0	13.6	542.42	1.025	Level 2	
20,100.0	10,847.7	20,022.5	10,780.7	276.3	275.7	-83.07	315.9	9,482.5	555.7	7.5	548.22	1.014	Level 2	
20,200.0	10,847.9	20,122.5	10,780.9	279.2	278.6	-83.07	315.9	9,582.5	555.4	1.4	554.02	1.002	Level 2	
20,300.0	10,848.1	20,222.5	10,781.1	282.1	281.5	-83.07	315.9	9,682.5	555.1	-4.7	559.81	0.992	Level 1	
20,400.0	10,848.2	20,322.5	10,781.2	285.1	284.4	-83.06	315.9	9,782.5	554.8	-10.8	565.61	0.981	Level 1	
20,500.0	10,848.4	20,422.5	10,781.4	288.0	287.3	-83.06	315.9	9,882.5	554.4	-17.0	571.41	0.970	Level 1	
20,600.0	10,848.6	20,522.5	10,781.6	290.9	290.3	-83.06	315.9	9,982.5	554.1	-23.1	577.21	0.960	Level 1	
20,667.7	10,848.7	20,590.0	10,781.7	292.9	292.2	-83.05	315.9	10,050.0	553.9	-27.2	581.13	0.953	Level 1, ES, SF	



Ryan Directional Services

Anticollision Report



Company: Oasis Petroleum
Project: Indian Hills
Reference Site: 153N-100W-17/18
Site Error: 0.0 usft
Reference Well: Kline Federal 5300 41-18 12TX
Well Error: 2.0 usft
Reference Wellbore: Kline Federal 5300 41-18 12TX
Reference Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 11T2 - Kline Federal 5300 41-18 11T2 - Design #1													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis			Distance							Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
0.0	0.0	0.0	0.0	2.0	2.0	0.00	33.4	0.0	33.4					
100.0	100.0	100.0	100.0	2.0	2.0	0.00	33.4	0.0	33.4	29.4	4.00	8.352		
200.0	200.0	200.0	200.0	2.0	2.0	0.00	33.4	0.0	33.4	29.4	4.05	8.261		
300.0	300.0	300.0	300.0	2.1	2.1	0.00	33.4	0.0	33.4	29.3	4.14	8.077		
400.0	400.0	400.0	400.0	2.1	2.1	0.00	33.4	0.0	33.4	29.2	4.28	7.816		
500.0	500.0	500.0	500.0	2.2	2.2	0.00	33.4	0.0	33.4	29.0	4.46	7.502		
600.0	600.0	600.0	600.0	2.3	2.3	0.00	33.4	0.0	33.4	28.8	4.67	7.155		
700.0	700.0	700.0	700.0	2.5	2.5	0.00	33.4	0.0	33.4	28.5	4.92	6.795		
800.0	800.0	800.0	800.0	2.6	2.6	0.00	33.4	0.0	33.4	28.2	5.20	6.436		
900.0	900.0	900.0	900.0	2.7	2.7	0.00	33.4	0.0	33.4	27.9	5.49	6.087		
1,000.0	1,000.0	1,000.0	1,000.0	2.9	2.9	0.00	33.4	0.0	33.4	27.6	5.81	5.755		
1,100.0	1,100.0	1,100.0	1,100.0	3.1	3.1	0.00	33.4	0.0	33.4	27.3	6.14	5.442		
1,200.0	1,200.0	1,200.0	1,200.0	3.2	3.2	0.00	33.4	0.0	33.4	26.9	6.49	5.151		
1,300.0	1,300.0	1,300.0	1,300.0	3.4	3.4	0.00	33.4	0.0	33.4	26.6	6.85	4.880		
1,400.0	1,400.0	1,400.0	1,400.0	3.6	3.6	0.00	33.4	0.0	33.4	26.2	7.22	4.630		
1,500.0	1,500.0	1,500.0	1,500.0	3.8	3.8	0.00	33.4	0.0	33.4	25.8	7.60	4.400		
1,600.0	1,600.0	1,600.0	1,600.0	4.0	4.0	0.00	33.4	0.0	33.4	25.5	7.99	4.187		
1,700.0	1,700.0	1,700.0	1,700.0	4.2	4.2	0.00	33.4	0.0	33.4	25.1	8.38	3.991		
1,800.0	1,800.0	1,800.0	1,800.0	4.4	4.4	0.00	33.4	0.0	33.4	24.7	8.78	3.810		
1,900.0	1,900.0	1,900.0	1,900.0	4.6	4.6	0.00	33.4	0.0	33.4	24.3	9.18	3.643		
2,000.0	2,000.0	2,000.0	2,000.0	4.8	4.8	0.00	33.4	0.0	33.4	23.9	9.58	3.489		
2,100.0	2,100.0	2,100.0	2,100.0	5.0	5.0	0.00	33.4	0.0	33.4	23.4	9.99	3.346		
2,200.0	2,200.0	2,200.0	2,200.0	5.2	5.2	0.00	33.4	0.0	33.4	23.0	10.41	3.213		
2,200.0	2,200.0	2,200.0	2,200.0	5.2	5.2	0.00	33.4	0.0	33.4	23.0	10.41	3.213 CC		
2,216.7	2,216.7	2,216.7	2,216.7	5.2	5.2	-90.12	33.4	0.0	33.4	23.0	10.48	3.192		
2,300.0	2,300.0	2,300.0	2,300.0	5.4	5.4	-91.37	33.4	0.0	33.4	22.6	10.80	3.097		
2,400.0	2,400.0	2,400.0	2,400.0	5.6	5.6	-92.88	33.4	0.0	33.5	22.3	11.20	2.990		
2,500.0	2,500.0	2,500.0	2,500.0	5.8	5.8	-94.35	33.4	0.0	33.5	21.9	11.59	2.892		
2,600.0	2,600.0	2,600.0	2,600.0	6.0	6.0	-95.84	33.4	0.0	33.6	21.6	12.00	2.802		
2,700.0	2,700.0	2,700.0	2,700.0	6.1	6.3	-97.31	33.4	0.0	33.7	21.3	12.40	2.718		
2,800.0	2,800.0	2,800.0	2,800.0	6.3	6.5	-98.78	33.4	0.0	33.8	21.0	12.81	2.641		
2,900.0	2,900.0	2,900.0	2,900.0	6.5	6.7	-100.23	33.4	0.0	34.0	20.8	13.22	2.570		
3,000.0	3,000.0	3,000.0	3,000.0	6.7	6.9	-101.67	33.4	0.0	34.1	20.5	13.64	2.504		
3,100.0	3,100.0	3,100.0	3,100.0	6.9	7.1	-103.10	33.4	0.0	34.3	20.3	14.05	2.443		
3,200.0	3,200.0	3,200.0	3,200.0	7.1	7.3	-104.51	33.4	0.0	34.5	20.1	14.47	2.387		
3,300.0	3,300.0	3,300.0	3,300.0	7.3	7.5	-105.90	33.4	0.0	34.8	19.9	14.89	2.335		
3,400.0	3,400.0	3,400.0	3,400.0	7.6	7.8	-107.27	33.4	0.0	35.0	19.7	15.31	2.287		
3,500.0	3,500.0	3,500.0	3,500.0	7.8	8.0	-108.63	33.4	0.0	35.3	19.6	15.73	2.243		
3,600.0	3,599.9	3,599.9	3,599.9	8.0	8.2	-109.96	33.4	0.0	35.6	19.4	16.16	2.202		
3,700.0	3,699.9	3,699.9	3,699.9	8.2	8.4	-111.27	33.4	0.0	35.9	19.3	16.58	2.164		
3,800.0	3,799.9	3,799.9	3,799.9	8.4	8.6	-112.56	33.4	0.0	36.2	19.2	17.01	2.129		
3,900.0	3,899.9	3,899.9	3,899.9	8.6	8.9	-113.82	33.4	0.0	36.6	19.1	17.44	2.096		
4,000.0	3,899.9	3,899.9	3,899.9	8.8	9.1	-115.08	33.4	0.0	36.9	19.0	17.87	2.066		
4,100.0	4,099.9	4,099.9	4,099.9	9.0	9.3	-116.27	33.4	0.0	37.3	19.0	18.30	2.038		
4,200.0	4,199.9	4,199.9	4,199.9	9.2	9.5	-117.46	33.4	0.0	37.7	19.0	18.73	2.012		
4,300.0	4,299.9	4,299.9	4,299.9	9.4	9.7	-118.63	33.4	0.0	38.1	18.9	19.16	1.986		
4,400.0	4,399.9	4,399.9	4,399.9	9.7	10.0	-119.77	33.4	0.0	38.5	18.9	19.59	1.966		
4,500.0	4,499.9	4,499.9	4,499.9	9.9	10.2	-120.88	33.4	0.0	39.0	18.9	20.03	1.946		
4,600.0	4,599.9	4,599.9	4,599.9	10.1	10.4	-121.97	33.4	0.0	39.4	19.0	20.46	1.927		
4,700.0	4,699.9	4,699.9	4,699.9	10.3	10.6	-123.03	33.4	0.0	39.9	19.0	20.89	1.909		
4,800.0	4,799.9	4,799.9	4,799.9	10.5	10.8	-124.07	33.4	0.0	40.4	19.0	21.33	1.893		
4,900.0	4,899.9	4,899.9	4,899.9	10.7	11.1	-125.09	33.4	0.0	40.9	19.1	21.76	1.878		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services
Anticollision Report



Company: Oasis Petroleum
Project: Indian Hills
Reference Site: 153N-100W-17/18
Site Error: 0.0 usft
Reference Well: Kline Federal 5300 41-18 12TX
Well Error: 2.0 usft
Reference Wellbore: Kline Federal 5300 41-18 12TX
Reference Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at: 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 11T2 - Kline Federal 5300 41-18 11T2 - Design #1														Offset Site Error:	0.0 usft
Survey Program: 0-MWD														Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis		Highside Toolface (°)	Distance		Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)										
5,000.0	4,999.9	4,999.9	4,999.9	10.9	11.3	-126.07	33.4	0.0	41.4	19.2	22.20	1.864			
5,100.0	5,099.9	5,099.9	5,099.9	11.2	11.5	-127.04	33.4	0.0	41.9	19.3	22.64	1.851			
5,200.0	5,199.9	5,199.9	5,199.9	11.4	11.7	-127.98	33.4	0.0	42.4	19.3	23.07	1.839			
5,300.0	5,299.9	5,299.9	5,299.9	11.6	11.9	-128.90	33.4	0.0	43.0	19.5	23.51	1.827			
5,400.0	5,399.9	5,399.9	5,399.9	11.8	12.2	-129.79	33.4	0.0	43.5	19.6	23.95	1.817			
5,500.0	5,499.9	5,499.9	5,499.9	12.0	12.4	-130.66	33.4	0.0	44.1	19.7	24.39	1.808			
5,600.0	5,599.9	5,599.9	5,599.9	12.2	12.6	-131.51	33.4	0.0	44.7	19.8	24.83	1.799			
5,700.0	5,699.9	5,699.9	5,699.9	12.5	12.8	-132.34	33.4	0.0	45.2	20.0	25.26	1.791			
5,800.0	5,799.9	5,799.9	5,799.9	12.7	13.0	-133.15	33.4	0.0	45.8	20.1	25.70	1.783			
5,900.0	5,899.9	5,899.9	5,899.9	12.9	13.3	-133.93	33.4	0.0	46.4	20.3	26.14	1.776			
6,000.0	5,999.9	5,999.9	5,999.9	13.1	13.5	-134.70	33.4	0.0	47.0	20.5	26.58	1.770			
6,100.0	6,099.9	6,099.9	6,099.9	13.3	13.7	-135.44	33.4	0.0	47.7	20.6	27.02	1.764			
6,200.0	6,199.9	6,199.9	6,199.9	13.6	13.9	-136.17	33.4	0.0	48.3	20.8	27.47	1.758			
6,300.0	6,299.9	6,299.9	6,299.9	13.8	14.2	-136.88	33.4	0.0	48.9	21.0	27.91	1.753			
6,400.0	6,399.9	6,399.9	6,399.9	14.0	14.4	-137.57	33.4	0.0	49.6	21.2	28.35	1.748			
6,500.0	6,499.9	6,499.9	6,499.9	14.2	14.6	-138.24	33.4	0.0	50.2	21.4	28.79	1.744			
6,600.0	6,599.9	6,599.9	6,599.9	14.4	14.8	-138.89	33.4	0.0	50.9	21.6	29.23	1.740			
6,700.0	6,699.9	6,699.9	6,699.9	14.7	15.1	-139.53	33.4	0.0	51.5	21.8	29.67	1.736			
6,800.0	6,799.9	6,799.9	6,799.9	14.9	15.3	-140.15	33.4	0.0	52.2	22.1	30.11	1.733			
6,900.0	6,899.9	6,899.9	6,899.9	15.1	15.5	-140.76	33.4	0.0	52.9	22.3	30.56	1.730			
7,000.0	6,999.9	6,999.9	6,999.9	15.3	15.7	-141.35	33.4	0.0	53.5	22.5	31.00	1.727			
7,100.0	7,099.9	7,099.9	7,099.9	15.5	15.9	-141.93	33.4	0.0	54.2	22.8	31.44	1.725			
7,200.0	7,199.9	7,199.9	7,199.9	15.8	16.2	-142.49	33.4	0.0	54.9	23.0	31.88	1.722			
7,300.0	7,299.9	7,299.9	7,299.9	16.0	16.4	-143.04	33.4	0.0	55.6	23.3	32.33	1.720			
7,400.0	7,399.9	7,399.9	7,399.9	16.2	16.6	-143.57	33.4	0.0	56.3	23.5	32.77	1.718			
7,500.0	7,499.9	7,499.9	7,499.9	16.4	16.8	-144.09	33.4	0.0	57.0	23.8	33.21	1.717			
7,600.0	7,599.9	7,599.9	7,599.9	16.6	17.1	-144.60	33.4	0.0	57.7	24.1	33.66	1.715			
7,700.0	7,699.9	7,699.9	7,699.9	16.9	17.3	-145.08	33.4	0.0	58.4	24.3	34.10	1.714			
7,800.0	7,799.9	7,799.9	7,799.9	17.1	17.5	-145.58	33.4	0.0	59.2	24.6	34.55	1.712			
7,900.0	7,899.9	7,899.9	7,899.9	17.3	17.7	-146.05	33.4	0.0	59.9	24.9	34.99	1.711			
7,926.7	7,926.4	7,926.4	7,926.4	17.4	17.8	-146.17	33.4	0.0	60.1	25.0	35.11	1.711			
7,943.3	7,943.1	7,943.1	7,943.1	17.4	17.8	-56.21	33.4	0.0	60.1	24.9	35.23	1.707			
8,000.0	7,999.9	7,999.9	7,999.9	17.5	18.0	-56.21	33.4	0.0	60.1	24.7	35.47	1.696			
8,100.0	8,099.9	8,099.9	8,099.9	17.7	18.2	-56.21	33.4	0.0	60.1	24.2	35.90	1.675			
8,200.0	8,199.9	8,199.9	8,199.9	17.9	18.4	-56.21	33.4	0.0	60.1	23.8	36.33	1.655			
8,300.0	8,299.9	8,299.9	8,299.9	18.1	18.6	-56.21	33.4	0.0	60.1	23.4	36.76	1.636			
8,400.0	8,399.9	8,399.9	8,399.9	18.4	18.8	-56.21	33.4	0.0	60.1	22.9	37.19	1.617			
8,500.0	8,499.9	8,499.9	8,499.9	18.6	19.1	-56.21	33.4	0.0	60.1	22.5	37.63	1.598			
8,600.0	8,599.9	8,599.9	8,599.9	18.8	19.3	-56.21	33.4	0.0	60.1	22.1	38.06	1.580			
8,700.0	8,699.9	8,699.9	8,699.9	19.0	19.5	-56.21	33.4	0.0	60.1	21.6	38.49	1.562			
8,800.0	8,799.9	8,799.9	8,799.9	19.2	19.7	-56.21	33.4	0.0	60.1	21.2	38.93	1.545			
8,900.0	8,899.9	8,899.9	8,899.9	19.4	20.0	-56.21	33.4	0.0	60.1	20.8	39.36	1.528			
9,000.0	8,999.9	8,999.9	8,999.9	19.6	20.2	-56.21	33.4	0.0	60.1	20.3	39.80	1.511			
9,100.0	9,099.9	9,099.9	9,099.9	19.8	20.4	-56.21	33.4	0.0	60.1	19.9	40.23	1.495 Level 3			
9,200.0	9,199.9	9,199.9	9,199.9	20.0	20.6	-56.21	33.4	0.0	60.1	19.5	40.67	1.479 Level 3			
9,300.0	9,299.9	9,299.9	9,299.9	20.2	20.9	-56.21	33.4	0.0	60.1	19.0	41.10	1.463 Level 3			
9,400.0	9,399.9	9,399.9	9,399.9	20.5	21.1	-56.21	33.4	0.0	60.1	18.8	41.54	1.448 Level 3			
9,500.0	9,499.9	9,499.9	9,499.9	20.7	21.3	-56.21	33.4	0.0	60.1	18.2	41.97	1.433 Level 3			
9,600.0	9,599.9	9,599.9	9,599.9	20.9	21.5	-56.21	33.4	0.0	60.1	17.7	42.41	1.418 Level 3			
9,700.0	9,699.9	9,699.9	9,699.9	21.1	21.8	-56.21	33.4	0.0	60.1	17.3	42.85	1.403 Level 3			
9,800.0	9,799.9	9,799.9	9,799.9	21.3	22.0	-56.21	33.4	0.0	60.1	16.8	43.28	1.389 Level 3			
9,900.0	9,899.9	9,899.9	9,899.9	21.5	22.2	-56.21	33.4	0.0	60.1	16.4	43.72	1.375 Level 3			

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services Anticollision Report



Company: Oasis Petroleum
Project: Indian Hills
Reference Site: 153N-100W-17/18
Site Error: 0.0 usft
Reference Well: Kline Federal 5300 41-18 12TX
Well Error: 2.0 usft
Reference Wellbore: Kline Federal 5300 41-18 12TX
Reference Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at: 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 11T2 - Kline Federal 5300 41-18 11T2 - Design #1													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis		Hightide Toolface (°)	Distance		Minimum Separation (usft)	Separation Factor	Warning			
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)		Between Centres (usft)	Between Ellipses (usft)				Offset +N/-S (usft)	Offset +E/-W (usft)	
10,000.0	9,999.8	9,999.8	9,999.8	21.7	22.4	-56.21	33.4	0.0	60.1	16.0	1.362 Level 3			
10,100.0	10,099.8	10,099.8	10,099.8	21.9	22.6	-56.21	33.4	0.0	60.1	15.5	1.348 Level 3			
10,200.0	10,199.8	10,199.8	10,199.8	22.2	22.9	-56.21	33.4	0.0	60.1	15.1	1.335 Level 3			
10,300.0	10,299.8	10,299.8	10,299.8	22.4	23.1	-56.21	33.4	0.0	60.1	14.7	1.322 Level 3			
10,354.8	10,354.6	10,354.6	10,354.6	22.5	23.2	-56.21	33.4	0.0	60.1	14.4	1.315 Level 3			
10,375.0	10,374.8	10,374.8	10,374.8	22.5	23.3	-166.30	33.4	0.0	60.5	14.8	1.324 Level 3			
10,400.0	10,399.7	10,400.4	10,400.4	22.6	23.3	-166.59	33.5	0.1	62.2	16.4	1.359 Level 3			
10,425.0	10,424.5	10,427.6	10,427.6	22.7	23.4	-166.38	33.8	1.2	64.4	18.7	1.410 Level 3			
10,450.0	10,449.2	10,454.9	10,454.8	22.7	23.4	-165.45	34.5	3.9	66.9	21.4	1.471 Level 3			
10,475.0	10,473.5	10,482.2	10,481.6	22.8	23.5	-163.88	35.7	8.1	68.8	24.6	1.543			
10,500.0	10,497.6	10,509.3	10,508.2	22.8	23.6	-161.78	37.3	13.7	73.2	28.2	1.627			
10,525.0	10,521.2	10,536.4	10,534.2	22.9	23.6	-159.25	39.3	20.7	77.0	32.3	1.724			
10,550.0	10,544.4	10,563.2	10,559.6	23.0	23.7	-156.41	41.7	29.1	81.4	37.0	1.836			
10,575.0	10,567.1	10,589.8	10,584.3	23.0	23.7	-153.35	44.4	38.7	86.3	42.3	1.962			
10,600.0	10,589.1	10,616.2	10,608.1	23.1	23.8	-150.17	47.4	49.5	92.0	48.2	2.103			
10,625.0	10,610.6	10,642.3	10,631.1	23.2	23.9	-146.93	50.8	61.5	98.3	54.8	2.259			
10,650.0	10,631.3	10,668.1	10,653.0	23.3	24.0	-143.69	54.5	74.4	105.3	61.9	2.428			
10,675.0	10,651.3	10,693.5	10,674.0	23.4	24.0	-140.50	58.4	88.3	113.0	69.7	2.610			
10,700.0	10,670.5	10,718.6	10,693.9	23.5	24.1	-137.38	62.6	103.1	121.4	78.1	2.801			
10,725.0	10,688.8	10,743.4	10,712.7	23.6	24.2	-134.35	67.0	118.6	130.6	87.1	3.002			
10,750.0	10,706.2	10,767.8	10,730.4	23.7	24.3	-131.42	71.8	134.8	140.3	96.6	3.209			
10,775.0	10,722.6	10,791.9	10,746.9	23.9	24.4	-128.58	76.3	151.6	150.7	106.7	3.421			
10,800.0	10,738.0	10,815.7	10,762.4	24.0	24.5	-125.83	81.2	169.9	161.7	117.3	3.635			
10,825.0	10,752.4	10,839.1	10,776.8	24.2	24.6	-123.17	86.3	186.7	173.3	128.3	3.852			
10,850.0	10,765.7	10,862.2	10,790.0	24.4	24.8	-120.59	91.4	205.0	185.3	139.8	4.068			
10,875.0	10,777.8	10,885.1	10,802.2	24.6	24.9	-118.08	96.7	223.6	197.8	151.7	4.284			
10,900.0	10,788.8	10,907.7	10,813.4	24.8	25.1	-115.64	102.1	242.5	210.8	163.9	4.499			
10,925.0	10,798.6	10,930.1	10,823.5	25.0	25.2	-113.25	107.5	261.7	224.1	176.5	4.712			
10,950.0	10,807.2	10,952.2	10,832.6	25.3	25.4	-110.92	113.0	281.1	237.7	189.4	4.923			
10,975.0	10,814.5	10,974.2	10,840.6	25.6	25.6	-108.64	118.6	300.8	251.6	202.6	5.132			
11,000.0	10,820.6	10,996.1	10,847.7	25.8	25.8	-106.40	124.2	320.7	265.8	216.0	5.340			
11,025.0	10,825.4	11,017.9	10,853.8	26.1	26.0	-104.22	129.9	340.8	280.2	229.8	5.545			
11,050.0	10,828.9	11,039.6	10,858.9	26.5	26.2	-102.06	135.6	361.1	294.7	243.4	5.750			
11,075.0	10,831.1	11,061.2	10,863.1	26.8	26.4	-99.99	141.4	381.6	308.4	257.4	5.953			
11,100.0	10,832.0	11,082.9	10,866.3	27.1	26.7	-97.85	147.3	402.3	324.1	271.4	6.154			
11,104.0	10,832.0	11,086.4	10,868.7	27.2	26.7	-97.83	148.2	405.5	328.5	273.7	6.186			
11,200.0	10,832.2	11,174.3	10,870.0	28.6	27.8	-96.77	171.9	490.1	380.2	325.0	6.889			
11,300.0	10,832.4	11,276.1	10,870.2	30.3	29.3	-95.89	198.8	588.8	428.5	370.2	7.351			
11,400.0	10,832.8	11,384.8	10,870.4	32.2	31.1	-94.99	219.4	695.1	468.1	406.2	7.562			
11,500.0	10,832.7	11,499.3	10,870.6	34.1	33.2	-94.53	238.9	807.9	498.3	432.4	7.553			
11,600.0	10,832.9	11,618.2	10,870.8	36.2	35.6	-94.25	254.3	925.8	518.8	448.3	7.361			
11,680.4	10,833.1	11,715.9	10,871.0	38.0	37.7	-94.14	263.3	1,023.1	527.9	453.6	7.107			
11,700.0	10,833.1	11,739.8	10,871.0	38.4	38.2	-94.12	265.0	1,047.0	529.3	454.0	7.033			
11,800.0	10,833.3	11,882.4	10,871.2	40.7	41.0	-94.08	270.8	1,169.4	533.5	453.2	6.644			
11,900.0	10,833.4	11,975.3	10,871.4	43.0	43.6	-94.08	271.6	1,282.3	533.9	448.6	6.258			
12,000.0	10,833.6	12,075.3	10,871.6	45.4	46.0	-94.08	271.6	1,382.3	533.6	443.4	5.918			
12,100.0	10,833.8	12,175.3	10,871.8	47.9	48.4	-94.08	271.5	1,482.3	533.2	438.1	5.607			
12,200.0	10,834.0	12,275.3	10,872.0	50.4	50.9	-94.09	271.5	1,582.3	532.9	432.8	5.321			
12,300.0	10,834.1	12,375.3	10,872.1	53.0	53.5	-94.09	271.5	1,682.3	532.6	427.3	5.059			
12,400.0	10,834.3	12,475.3	10,872.3	55.5	56.0	-94.09	271.5	1,782.3	532.3	421.8	4.818			
12,500.0	10,834.5	12,575.3	10,872.5	58.2	58.6	-94.09	271.5	1,882.3	532.0	416.3	4.596			
12,600.0	10,834.7	12,675.3	10,872.6	60.8	61.3	-94.10	271.5	1,982.3	531.7	410.6	4.392			

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Ryan Directional Services Anticollision Report



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Well Error: 2.0 usft
Reference Wellbore: Kline Federal 5300 41-18 12TX
Reference Design: Design #2

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TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at: 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 11T2 - Kline Federal 5300 41-18 11T2 - Design #1														Offset Site Error: 0.0 usft	
Survey Program: 0-MWD														Offset Well Error: 2.0 usft	
Reference		Offset		Semi Major Axis			Distance								
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning		
12,700.0	10,834.8	12,775.3	10,872.8	63.5	64.0	-94.10	271.5	2,082.3	531.4	405.0	126.44	4.203			
12,800.0	10,835.0	12,875.3	10,873.0	66.2	66.6	-94.10	271.5	2,182.3	531.1	399.3	131.84	4.028			
12,900.0	10,835.2	12,975.3	10,873.2	68.9	69.4	-94.10	271.5	2,282.3	530.8	393.5	137.28	3.866			
13,000.0	10,835.4	13,075.3	10,873.3	71.7	72.1	-94.11	271.5	2,382.3	530.5	387.7	142.76	3.716			
13,100.0	10,835.5	13,175.3	10,873.5	74.4	74.8	-94.11	271.5	2,482.3	530.2	381.9	148.26	3.576			
13,200.0	10,835.7	13,275.3	10,873.7	77.2	77.6	-94.11	271.5	2,582.3	529.9	376.1	153.78	3.445			
13,300.0	10,835.9	13,375.3	10,873.9	79.9	80.3	-94.11	271.5	2,682.3	529.6	370.2	159.33	3.324			
13,400.0	10,836.1	13,475.3	10,874.0	82.7	83.1	-94.12	271.5	2,782.3	529.2	364.3	164.90	3.209			
13,500.0	10,836.2	13,575.3	10,874.2	85.5	85.9	-94.12	271.5	2,882.3	528.9	358.4	170.49	3.102			
13,600.0	10,836.4	13,675.3	10,874.4	88.3	88.7	-94.12	271.5	2,982.3	528.6	352.5	176.10	3.002			
13,700.0	10,836.6	13,775.3	10,874.6	91.1	91.5	-94.12	271.5	3,082.3	528.3	346.6	181.72	2.907			
13,800.0	10,836.7	13,875.3	10,874.7	94.0	94.3	-94.13	271.5	3,182.3	528.0	340.7	187.35	2.818			
13,900.0	10,836.9	13,975.3	10,874.9	96.8	97.2	-94.13	271.5	3,282.3	527.7	334.7	193.00	2.734			
14,000.0	10,837.1	14,075.3	10,875.1	99.6	100.0	-94.13	271.5	3,382.3	527.4	328.7	198.66	2.655			
14,100.0	10,837.3	14,175.3	10,875.3	102.5	102.8	-94.13	271.5	3,482.3	527.1	322.8	204.33	2.580			
14,200.0	10,837.4	14,275.3	10,875.4	105.3	105.7	-94.14	271.5	3,582.3	526.8	316.8	210.01	2.508			
14,300.0	10,837.6	14,375.3	10,875.6	108.2	108.5	-94.14	271.5	3,682.3	526.5	310.8	215.69	2.441			
14,400.0	10,837.8	14,475.3	10,875.8	111.0	111.3	-94.14	271.5	3,782.3	526.2	304.8	221.39	2.377			
14,500.0	10,838.0	14,575.3	10,876.0	113.9	114.2	-94.14	271.5	3,882.3	525.9	298.8	227.09	2.316			
14,600.0	10,838.1	14,675.3	10,876.1	116.7	117.1	-94.15	271.5	3,982.3	525.6	292.7	232.80	2.257			
14,700.0	10,838.3	14,775.3	10,876.3	119.6	119.9	-94.15	271.5	4,082.3	525.2	286.7	238.52	2.202			
14,800.0	10,838.5	14,875.3	10,876.5	122.5	122.8	-94.15	271.5	4,182.3	524.9	280.7	244.24	2.149			
14,900.0	10,838.7	14,975.3	10,876.7	125.3	125.6	-94.15	271.5	4,282.3	524.6	274.7	249.97	2.099			
15,000.0	10,838.8	15,075.3	10,876.8	128.2	128.5	-94.16	271.5	4,382.3	524.3	268.6	255.71	2.050			
15,100.0	10,839.0	15,175.3	10,877.0	131.1	131.4	-94.16	271.5	4,482.3	524.0	262.6	261.44	2.004			
15,200.0	10,839.2	15,275.3	10,877.2	133.9	134.3	-94.16	271.5	4,582.3	523.7	256.5	267.19	1.960			
15,300.0	10,839.4	15,375.3	10,877.4	136.8	137.1	-94.16	271.5	4,682.3	523.4	250.5	272.94	1.918			
15,400.0	10,839.5	15,475.3	10,877.5	139.7	140.0	-94.17	271.5	4,782.3	523.1	244.4	278.69	1.877			
15,500.0	10,839.7	15,575.3	10,877.7	142.6	142.9	-94.17	271.5	4,882.3	522.8	238.3	284.44	1.838			
15,600.0	10,839.9	15,675.3	10,877.9	145.5	145.8	-94.17	271.5	4,982.3	522.5	232.3	290.20	1.800			
15,700.0	10,840.1	15,775.3	10,878.1	148.4	148.7	-94.17	271.5	5,082.3	522.2	226.2	295.96	1.764			
15,800.0	10,840.2	15,875.3	10,878.2	151.3	151.6	-94.18	271.5	5,182.3	521.9	220.1	301.73	1.730			
15,900.0	10,840.4	15,975.3	10,878.4	154.1	154.4	-94.18	271.5	5,282.3	521.6	214.1	307.50	1.696			
16,000.0	10,840.6	16,075.3	10,878.6	157.0	157.3	-94.18	271.5	5,382.3	521.2	208.0	313.27	1.664			
16,100.0	10,840.8	16,175.3	10,878.8	159.9	160.2	-94.18	271.5	5,482.3	520.9	201.9	319.04	1.633			
16,200.0	10,840.9	16,275.3	10,878.9	162.8	163.1	-94.19	271.5	5,582.3	520.6	195.8	324.82	1.603			
16,300.0	10,841.1	16,375.3	10,879.1	165.7	166.0	-94.19	271.5	5,682.3	520.3	189.7	330.60	1.574			
16,400.0	10,841.3	16,475.3	10,879.3	168.6	168.9	-94.19	271.5	5,782.3	520.0	183.6	336.38	1.546			
16,500.0	10,841.4	16,575.3	10,879.5	171.5	171.8	-94.19	271.5	5,882.3	519.7	177.5	342.16	1.519			
16,600.0	10,841.6	16,675.3	10,879.6	174.4	174.7	-94.20	271.5	5,982.3	519.4	171.5	347.95	1.493 Level 3			
16,700.0	10,841.8	16,775.3	10,879.8	177.3	177.6	-94.20	271.5	6,082.3	519.1	165.4	353.73	1.467 Level 3			
16,800.0	10,842.0	16,875.3	10,880.0	180.2	180.5	-94.20	271.5	6,182.3	518.8	159.3	359.52	1.443 Level 3			
16,900.0	10,842.1	16,975.3	10,880.2	183.1	183.4	-94.20	271.5	6,282.3	518.5	153.2	365.31	1.419 Level 3			
17,000.0	10,842.3	17,075.3	10,880.3	186.0	186.3	-94.21	271.5	6,382.3	518.2	147.1	371.11	1.398 Level 3			
17,100.0	10,842.5	17,175.3	10,880.5	188.9	189.2	-94.21	271.5	6,482.3	517.9	141.0	376.90	1.374 Level 3			
17,200.0	10,842.7	17,275.3	10,880.7	191.8	192.1	-94.21	271.5	6,582.3	517.6	134.9	382.70	1.352 Level 3			
17,300.0	10,842.8	17,375.3	10,880.9	194.7	195.0	-94.21	271.5	6,682.3	517.2	128.8	388.49	1.331 Level 3			
17,400.0	10,843.0	17,475.3	10,881.0	197.6	197.9	-94.22	271.5	6,782.3	516.9	122.6	394.29	1.311 Level 3			
17,500.0	10,843.2	17,575.3	10,881.2	200.5	200.8	-94.22	271.5	6,882.3	516.6	116.5	400.09	1.291 Level 3			
17,600.0	10,843.4	17,675.3	10,881.4	203.5	203.7	-94.22	271.5	6,982.3	516.3	110.4	405.89	1.272 Level 3			
17,700.0	10,843.5	17,775.3	10,881.6	206.4	206.6	-94.22	271.5	7,082.3	516.0	104.3	411.69	1.253 Level 3			
17,800.0	10,843.7	17,875.3	10,881.7	209.3	209.6	-94.23	271.5	7,182.3	515.7	98.2	417.50	1.235 Level 2			

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services

Anticollision Report



Company: Oasis Petroleum
Project: Indian Hills
Reference Site: 153N-100W-17/18
Site Error: 0.0 usft
Reference Well: Kline Federal 5300 41-18 12TX
Well Error: 2.0 usft
Reference Wellbore: Kline Federal 5300 41-18 12TX
Reference Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at: 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 11T2 - Kline Federal 5300 41-18 11T2 - Design #1													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis			Distance			Minimum Separation (usft)	Separation Factor	Warning		
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N-S (usft)	Offset Wellbore Centre +E-W (usft)	Between Centres (usft)				Between Ellipses (usft)	
17,900.0	10,843.9	17,975.3	10,881.9	212.2	212.5	-94.23	271.5	7,282.3	515.4	92.1	423.30	1.218 Level 2		
18,000.0	10,844.1	18,075.3	10,882.1	215.1	215.4	-94.23	271.5	7,382.3	515.1	86.0	429.11	1.200 Level 2		
18,100.0	10,844.2	18,175.3	10,882.2	218.0	218.3	-94.23	271.5	7,482.3	514.8	79.9	434.91	1.184 Level 2		
18,200.0	10,844.4	18,275.3	10,882.4	220.9	221.2	-94.24	271.5	7,582.3	514.5	73.8	440.72	1.167 Level 2		
18,300.0	10,844.6	18,375.3	10,882.6	223.8	224.1	-94.24	271.5	7,682.3	514.2	67.6	446.53	1.151 Level 2		
18,400.0	10,844.8	18,475.3	10,882.8	226.7	227.0	-94.24	271.5	7,782.3	513.9	61.5	452.34	1.136 Level 2		
18,500.0	10,844.9	18,575.3	10,882.9	229.7	229.9	-94.25	271.5	7,882.3	513.6	55.4	458.15	1.121 Level 2		
18,600.0	10,845.1	18,675.3	10,883.1	232.6	232.8	-94.25	271.5	7,982.2	513.2	49.3	463.96	1.106 Level 2		
18,700.0	10,845.3	18,775.3	10,883.3	235.5	235.8	-94.25	271.5	8,082.2	512.9	43.2	469.77	1.092 Level 2		
18,800.0	10,845.4	18,875.3	10,883.5	238.4	238.7	-94.25	271.5	8,182.2	512.6	37.0	475.58	1.078 Level 2		
18,900.0	10,845.6	18,975.3	10,883.6	241.3	241.6	-94.26	271.5	8,282.2	512.3	30.9	481.39	1.064 Level 2		
19,000.0	10,845.8	19,075.3	10,883.8	244.2	244.5	-94.26	271.5	8,382.2	512.0	24.8	487.21	1.051 Level 2		
19,100.0	10,846.0	19,175.3	10,884.0	247.1	247.4	-94.26	271.5	8,482.2	511.7	18.7	493.02	1.038 Level 2		
19,200.0	10,846.1	19,275.3	10,884.2	250.1	250.3	-94.26	271.5	8,582.2	511.4	12.6	498.84	1.025 Level 2		
19,300.0	10,846.3	19,375.3	10,884.3	253.0	253.2	-94.27	271.5	8,682.2	511.1	6.4	504.65	1.013 Level 2		
19,400.0	10,846.5	19,475.3	10,884.5	255.9	256.2	-94.27	271.5	8,782.2	510.8	0.3	510.47	1.001 Level 2		
19,500.0	10,846.7	19,575.3	10,884.7	258.8	259.1	-94.27	271.5	8,882.2	510.5	-5.8	516.29	0.989 Level 1		
19,600.0	10,846.8	19,675.3	10,884.9	261.7	262.0	-94.27	271.5	8,982.2	510.2	-11.9	522.10	0.977 Level 1		
19,700.0	10,847.0	19,775.3	10,885.0	264.6	264.9	-94.28	271.5	9,082.2	509.9	-18.1	527.92	0.966 Level 1		
19,800.0	10,847.2	19,875.3	10,885.2	267.6	267.8	-94.28	271.5	9,182.2	509.6	-24.2	533.74	0.955 Level 1		
19,900.0	10,847.4	19,975.3	10,885.4	270.5	270.7	-94.28	271.5	9,282.2	509.2	-30.3	539.56	0.944 Level 1		
20,000.0	10,847.5	20,075.3	10,885.6	273.4	273.7	-94.28	271.5	9,382.2	508.9	-36.4	545.38	0.933 Level 1		
20,100.0	10,847.7	20,175.3	10,885.7	276.3	276.6	-94.29	271.5	9,482.2	508.6	-42.6	551.20	0.923 Level 1		
20,200.0	10,847.9	20,275.3	10,885.9	279.2	279.5	-94.29	271.5	9,582.2	508.3	-48.7	557.02	0.913 Level 1		
20,300.0	10,848.1	20,375.3	10,886.1	282.1	282.4	-94.29	271.5	9,682.2	508.0	-54.8	562.84	0.903 Level 1		
20,400.0	10,848.2	20,475.3	10,886.3	285.1	285.3	-94.30	271.5	9,782.2	507.7	-61.0	568.66	0.893 Level 1		
20,500.0	10,848.4	20,575.3	10,886.4	288.0	288.3	-94.30	271.5	9,882.2	507.4	-67.1	574.48	0.883 Level 1		
20,600.0	10,848.6	20,675.3	10,886.6	290.9	291.2	-94.30	271.5	9,982.2	507.1	-73.2	580.30	0.874 Level 1		
20,667.7	10,848.7	20,743.0	10,886.7	292.9	293.2	-94.30	271.5	10,049.9	506.9	-77.4	584.25	0.866 Level 1, ES, SF		



Ryan Directional Services Anticollision Report



Company: Oasis Petroleum
Project: Indian Hills
Reference Site: 153N-100W-17/18
Site Error: 0.0 usft
Reference Well: Kline Federal 5300 41-18 12TX
Well Error: 2.0 usft
Reference Wellbore: Kline Federal 5300 41-18 12TX
Reference Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at: 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 13T2X - Kline Federal 5300 41-18 13T2X - Design #1													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis		Highside Toolface (°)	Distance		Minimum Separation (usft)	Separation Factor	Warning			
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)		Between Centres (usft)	Between Ellipses (usft)						
0.0	0.0	0.0	0.0	2.0	2.0	-180.00	-33.4	0.0	33.4					
100.0	100.0	100.0	100.0	2.0	2.0	-180.00	-33.4	0.0	33.4	29.4	4.00	8.352		
200.0	200.0	200.0	200.0	2.0	2.0	-180.00	-33.4	0.0	33.4	29.4	4.05	8.261		
300.0	300.0	300.0	300.0	2.1	2.1	-180.00	-33.4	0.0	33.4	29.3	4.14	8.077		
400.0	400.0	400.0	400.0	2.1	2.1	-180.00	-33.4	0.0	33.4	29.2	4.28	7.816		
500.0	500.0	500.0	500.0	2.2	2.2	-180.00	-33.4	0.0	33.4	29.0	4.48	7.602		
600.0	600.0	600.0	600.0	2.3	2.3	-180.00	-33.4	0.0	33.4	28.8	4.67	7.155		
700.0	700.0	700.0	700.0	2.5	2.5	-180.00	-33.4	0.0	33.4	28.5	4.92	6.795		
800.0	800.0	800.0	800.0	2.6	2.6	-180.00	-33.4	0.0	33.4	28.2	5.20	6.436		
900.0	900.0	900.0	900.0	2.7	2.7	-180.00	-33.4	0.0	33.4	27.9	5.49	6.087		
1,000.0	1,000.0	1,000.0	1,000.0	2.9	2.9	-180.00	-33.4	0.0	33.4	27.6	5.81	5.755		
1,100.0	1,100.0	1,100.0	1,100.0	3.1	3.1	-180.00	-33.4	0.0	33.4	27.3	6.14	5.442		
1,200.0	1,200.0	1,200.0	1,200.0	3.2	3.2	-180.00	-33.4	0.0	33.4	26.9	6.49	5.151		
1,300.0	1,300.0	1,300.0	1,300.0	3.4	3.4	-180.00	-33.4	0.0	33.4	26.6	6.85	4.880		
1,400.0	1,400.0	1,400.0	1,400.0	3.6	3.6	-180.00	-33.4	0.0	33.4	26.2	7.22	4.630		
1,500.0	1,500.0	1,500.0	1,500.0	3.8	3.8	-180.00	-33.4	0.0	33.4	25.8	7.60	4.400		
1,600.0	1,600.0	1,600.0	1,600.0	4.0	4.0	-180.00	-33.4	0.0	33.4	25.5	7.99	4.187		
1,700.0	1,700.0	1,700.0	1,700.0	4.2	4.2	-180.00	-33.4	0.0	33.4	25.1	8.38	3.991		
1,800.0	1,800.0	1,800.0	1,800.0	4.4	4.4	-180.00	-33.4	0.0	33.4	24.7	8.78	3.810		
1,900.0	1,900.0	1,900.0	1,900.0	4.6	4.6	-180.00	-33.4	0.0	33.4	24.3	9.18	3.643		
2,000.0	2,000.0	2,000.0	2,000.0	4.8	4.8	-180.00	-33.4	0.0	33.4	23.9	9.58	3.489		
2,100.0	2,100.0	2,100.0	2,100.0	5.0	5.0	-180.00	-33.4	0.0	33.4	23.4	9.99	3.346		
2,200.0	2,200.0	2,200.0	2,200.0	5.2	5.2	-180.00	-33.4	0.0	33.4	23.0	10.41	3.213		
2,200.0	2,200.0	2,200.0	2,200.0	5.2	5.2	-180.00	-33.4	0.0	33.4	23.0	10.41	3.213 CC		
2,216.7	2,216.7	2,216.7	2,216.7	5.2	5.2	90.12	-33.4	0.0	33.4	23.0	10.48	3.192		
2,300.0	2,300.0	2,300.0	2,300.0	5.4	5.4	91.37	-33.4	0.0	33.4	22.6	10.80	3.097		
2,400.0	2,400.0	2,400.0	2,400.0	5.6	5.6	92.86	-33.4	0.0	33.5	22.3	11.20	2.990		
2,500.0	2,500.0	2,500.0	2,500.0	5.8	5.8	94.35	-33.4	0.0	33.5	21.9	11.59	2.892		
2,600.0	2,600.0	2,600.0	2,600.0	6.0	6.0	95.84	-33.4	0.0	33.6	21.6	12.00	2.802		
2,700.0	2,700.0	2,700.0	2,700.0	6.1	6.3	97.31	-33.4	0.0	33.7	21.3	12.40	2.718		
2,800.0	2,800.0	2,800.0	2,800.0	6.3	6.5	98.78	-33.4	0.0	33.8	21.0	12.81	2.641		
2,900.0	2,900.0	2,900.0	2,900.0	6.5	6.7	100.23	-33.4	0.0	34.0	20.8	13.22	2.569		
3,000.0	3,000.0	3,000.0	3,000.0	6.7	6.9	101.67	-33.4	0.0	34.1	20.5	13.64	2.504		
3,100.0	3,100.0	3,100.0	3,100.0	6.9	7.1	103.10	-33.4	0.0	34.3	20.3	14.05	2.443		
3,200.0	3,200.0	3,200.0	3,200.0	7.1	7.3	104.51	-33.4	0.0	34.5	20.1	14.47	2.387		
3,300.0	3,300.0	3,300.0	3,300.0	7.3	7.5	105.90	-33.4	0.0	34.8	19.9	14.89	2.335		
3,400.0	3,400.0	3,400.0	3,400.0	7.6	7.8	107.27	-33.4	0.0	35.0	19.7	15.31	2.287		
3,500.0	3,500.0	3,500.0	3,500.0	7.8	8.0	108.63	-33.4	0.0	35.3	19.5	15.74	2.242		
3,600.0	3,599.9	3,599.9	3,599.9	8.0	8.2	109.96	-33.4	0.0	35.6	19.4	16.16	2.201		
3,700.0	3,699.9	3,699.9	3,699.9	8.2	8.4	111.27	-33.4	0.0	35.9	19.3	16.59	2.163		
3,800.0	3,799.9	3,799.9	3,799.9	8.4	8.6	112.56	-33.4	0.0	36.2	19.2	17.02	2.128		
3,900.0	3,899.9	3,899.9	3,899.9	8.6	8.9	113.82	-33.4	0.0	36.6	19.1	17.44	2.095		
4,000.0	3,899.9	3,999.9	3,999.9	8.8	9.1	115.06	-33.4	0.0	36.9	19.0	17.87	2.065		
4,100.0	4,099.9	4,099.9	4,099.9	9.0	9.3	116.27	-33.4	0.0	37.3	19.0	18.30	2.037		
4,200.0	4,199.9	4,199.9	4,199.9	9.2	9.5	117.46	-33.4	0.0	37.7	18.9	18.74	2.011		
4,300.0	4,299.9	4,299.9	4,299.9	9.4	9.7	118.63	-33.4	0.0	38.1	18.9	19.17	1.987		
4,400.0	4,399.9	4,399.9	4,399.9	9.7	10.0	119.77	-33.4	0.0	38.5	18.9	19.60	1.965		
4,500.0	4,499.9	4,499.9	4,499.9	9.9	10.2	120.88	-33.4	0.0	39.0	18.9	20.04	1.945		
4,600.0	4,599.9	4,599.9	4,599.9	10.1	10.4	121.97	-33.4	0.0	39.4	18.9	20.47	1.926		
4,700.0	4,699.9	4,699.9	4,699.9	10.3	10.6	123.03	-33.4	0.0	39.9	19.0	20.90	1.908		
4,800.0	4,799.9	4,799.9	4,799.9	10.5	10.8	124.07	-33.4	0.0	40.4	19.0	21.34	1.892		
4,900.0	4,899.9	4,899.9	4,899.9	10.7	11.1	125.09	-33.4	0.0	40.9	19.1	21.78	1.876		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services

Anticollision Report



Company: Oasis Petroleum
Project: Indian Hills
Reference Site: 153N-100W-17/18
Site Error: 0.0 usft
Reference Well: Kline Federal 5300 41-18 12TX
Well Error: 2.0 usft
Reference Wellbore: Kline Federal 5300 41-18 12TX
Reference Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at: 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 13T2X - Kline Federal 5300 41-18 13T2X - Design #1													Offset Site Error: 0.0 usft	
Survey Program: 0-MWD													Offset Well Error: 2.0 usft	
Reference		Offset		Semi Major Axis		Distance								Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
5,000.0	4,999.9	4,999.9	4,999.9	10.9	11.3	126.07	-33.4	0.0	41.4	19.2	22.21	1.862		
5,100.0	5,099.9	5,099.9	5,099.9	11.2	11.5	127.04	-33.4	0.0	41.9	19.2	22.65	1.849		
5,200.0	5,199.9	5,199.9	5,199.9	11.4	11.7	127.98	-33.4	0.0	42.4	19.3	23.09	1.837		
5,300.0	5,299.9	5,299.9	5,299.9	11.6	11.9	128.90	-33.4	0.0	43.0	19.4	23.53	1.826		
5,400.0	5,399.9	5,399.9	5,399.9	11.8	12.2	129.79	-33.4	0.0	43.5	19.6	23.97	1.816		
5,500.0	5,499.9	5,499.9	5,499.9	12.0	12.4	130.86	-33.4	0.0	44.1	19.7	24.41	1.808		
5,600.0	5,599.9	5,599.9	5,599.9	12.2	12.6	131.51	-33.4	0.0	44.7	19.8	24.85	1.797		
5,700.0	5,699.9	5,699.9	5,699.9	12.5	12.8	132.34	-33.4	0.0	45.2	20.0	25.28	1.789		
5,800.0	5,799.9	5,799.9	5,799.9	12.7	13.0	133.15	-33.4	0.0	45.8	20.1	25.73	1.782		
5,900.0	5,899.9	5,899.9	5,899.9	12.9	13.3	133.93	-33.4	0.0	46.4	20.3	26.17	1.775		
6,000.0	5,999.9	5,999.9	5,999.9	13.1	13.5	134.70	-33.4	0.0	47.0	20.4	26.61	1.768		
6,100.0	6,099.9	6,099.9	6,099.9	13.3	13.7	135.44	-33.4	0.0	47.7	20.6	27.05	1.762		
6,200.0	6,199.8	6,199.8	6,199.8	13.6	13.9	136.17	-33.4	0.0	48.3	20.8	27.49	1.757		
6,300.0	6,299.8	6,299.8	6,299.8	13.8	14.2	136.88	-33.4	0.0	48.9	21.0	27.93	1.751		
6,400.0	6,399.8	6,399.8	6,399.8	14.0	14.4	137.57	-33.4	0.0	49.6	21.2	28.37	1.747		
6,500.0	6,499.8	6,499.8	6,499.8	14.2	14.6	138.24	-33.4	0.0	50.2	21.4	28.81	1.742		
6,600.0	6,599.8	6,599.8	6,599.8	14.4	14.8	138.89	-33.4	0.0	50.9	21.6	29.26	1.738		
6,700.0	6,699.8	6,699.8	6,699.8	14.7	15.1	139.53	-33.4	0.0	51.5	21.8	29.70	1.735		
6,800.0	6,799.8	6,799.8	6,799.8	14.9	15.3	140.15	-33.4	0.0	52.2	22.0	30.14	1.731		
6,900.0	6,899.8	6,899.8	6,899.8	15.1	15.5	140.76	-33.4	0.0	52.9	22.3	30.59	1.728		
7,000.0	6,999.8	6,999.8	6,999.8	15.3	15.7	141.35	-33.4	0.0	53.5	22.5	31.03	1.726		
7,100.0	7,099.8	7,099.8	7,099.8	15.5	15.9	141.93	-33.4	0.0	54.2	22.8	31.47	1.723		
7,200.0	7,199.8	7,199.8	7,199.8	15.8	16.2	142.49	-33.4	0.0	54.9	23.0	31.92	1.721		
7,300.0	7,299.8	7,299.8	7,299.8	16.0	16.4	143.04	-33.4	0.0	55.6	23.2	32.36	1.718		
7,400.0	7,399.8	7,399.8	7,399.8	16.2	16.6	143.57	-33.4	0.0	56.3	23.5	32.80	1.717		
7,500.0	7,499.8	7,499.8	7,499.8	16.4	16.8	144.09	-33.4	0.0	57.0	23.8	33.25	1.715		
7,600.0	7,599.8	7,599.8	7,599.8	16.6	17.1	144.60	-33.4	0.0	57.7	24.0	33.69	1.713		
7,700.0	7,699.8	7,699.8	7,699.8	16.9	17.3	145.09	-33.4	0.0	58.4	24.3	34.14	1.712		
7,800.0	7,799.8	7,799.8	7,799.8	17.1	17.5	145.58	-33.4	0.0	59.2	24.6	34.58	1.711		
7,900.0	7,899.8	7,899.8	7,899.8	17.3	17.7	146.05	-33.4	0.0	59.9	24.9	35.02	1.710		
7,926.7	7,926.4	7,926.4	7,926.4	17.4	17.8	146.17	-33.4	0.0	60.1	24.9	35.14	1.709		
7,943.3	7,943.1	7,943.1	7,943.1	17.4	17.8	-123.79	-33.4	0.0	60.1	24.9	35.18	1.709		
8,000.0	7,999.8	7,999.8	7,999.8	17.5	18.0	-123.79	-33.4	0.0	60.1	24.7	35.42	1.698		
8,100.0	8,099.8	8,099.8	8,099.8	17.7	18.2	-123.79	-33.4	0.0	60.1	24.3	35.85	1.677		
8,200.0	8,199.8	8,199.8	8,199.8	17.9	18.4	-123.79	-33.4	0.0	60.1	23.8	36.28	1.657		
8,300.0	8,299.8	8,299.8	8,299.8	18.1	18.6	-123.79	-33.4	0.0	60.1	23.4	36.72	1.638		
8,400.0	8,399.8	8,399.8	8,399.8	18.4	18.8	-123.79	-33.4	0.0	60.1	23.0	37.15	1.619		
8,500.0	8,499.8	8,499.8	8,499.8	18.6	19.1	-123.79	-33.4	0.0	60.1	22.5	37.58	1.600		
8,600.0	8,599.8	8,599.8	8,599.8	18.8	19.3	-123.79	-33.4	0.0	60.1	22.1	38.01	1.582		
8,700.0	8,699.8	8,699.8	8,699.8	19.0	19.5	-123.79	-33.4	0.0	60.1	21.7	38.45	1.564		
8,800.0	8,799.8	8,799.8	8,799.8	19.2	19.7	-123.79	-33.4	0.0	60.1	21.2	38.88	1.546		
8,900.0	8,899.8	8,899.8	8,899.8	19.4	20.0	-123.79	-33.4	0.0	60.1	20.8	39.32	1.529		
9,000.0	8,999.8	8,999.8	8,999.8	19.6	20.2	-123.79	-33.4	0.0	60.1	20.4	39.75	1.513		
9,100.0	9,099.8	9,099.8	9,099.8	19.8	20.4	-123.79	-33.4	0.0	60.1	19.9	40.19	1.496 Level 3		
9,200.0	9,199.8	9,199.8	9,199.8	20.0	20.6	-123.79	-33.4	0.0	60.1	19.5	40.62	1.480 Level 3		
9,300.0	9,299.8	9,299.8	9,299.8	20.2	20.9	-123.79	-33.4	0.0	60.1	19.1	41.06	1.465 Level 3		
9,400.0	9,399.8	9,399.8	9,399.8	20.5	21.1	-123.79	-33.4	0.0	60.1	18.6	41.49	1.449 Level 3		
9,500.0	9,499.8	9,499.8	9,499.8	20.7	21.3	-123.79	-33.4	0.0	60.1	18.2	41.93	1.434 Level 3		
9,600.0	9,599.8	9,599.8	9,599.8	20.9	21.5	-123.79	-33.4	0.0	60.1	17.8	42.37	1.419 Level 3		
9,700.0	9,699.8	9,699.8	9,699.8	21.1	21.8	-123.79	-33.4	0.0	60.1	17.3	42.80	1.405 Level 3		
9,800.0	9,799.8	9,799.8	9,799.8	21.3	22.0	-123.79	-33.4	0.0	60.1	16.9	43.24	1.391 Level 3		
9,900.0	9,899.8	9,899.8	9,899.8	21.5	22.2	-123.79	-33.4	0.0	60.1	16.5	43.68	1.377 Level 3		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services Anticollision Report



Company: Oasis Petroleum
Project: Indian Hills
Reference Site: 153N-100W-17/18
Site Error: 0.0 usft
Reference Well: Kline Federal 5300 41-18 12TX
Well Error: 2.0 usft
Reference Wellbore: Kline Federal 5300 41-18 12TX
Reference Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at: 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 13T2X - Kline Federal 5300 41-18 13T2X - Design #1													Offset Site Error:
Survey Program: 0-MWD													0.0 usft
Reference		Offset		Semi Major Axis		Distance							
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
10,000.0	9,999.8	9,999.8	9,999.8	21.7	22.4	-123.79	-33.4	0.0	60.1	16.0	44.11	1.363	Level 3
10,100.0	10,099.8	10,099.8	10,099.8	21.9	22.6	-123.79	-33.4	0.0	60.1	15.6	44.55	1.350	Level 3
10,200.0	10,199.8	10,199.8	10,199.8	22.2	22.9	-123.79	-33.4	0.0	60.1	15.1	44.99	1.336	Level 3
10,300.0	10,299.8	10,299.8	10,299.8	22.4	23.1	-123.79	-33.4	0.0	60.1	14.7	45.43	1.324	Level 3
10,354.8	10,354.6	10,354.6	10,354.6	22.5	23.2	-123.79	-33.4	0.0	60.1	14.5	45.67	1.317	Level 3
10,375.0	10,374.8	10,374.8	10,374.8	22.5	23.3	126.52	-33.4	0.0	60.4	14.6	45.78	1.319	Level 3
10,400.0	10,399.7	10,400.2	10,400.2	22.6	23.3	127.68	-33.5	0.1	61.4	15.5	45.84	1.339	Level 3
10,425.0	10,424.6	10,426.8	10,426.8	22.7	23.4	128.94	-34.0	1.1	62.7	16.9	45.88	1.388	Level 3
10,450.0	10,449.2	10,453.5	10,453.3	22.7	23.4	129.94	-35.3	3.4	64.2	18.4	45.84	1.401	Level 3
10,475.0	10,473.5	10,480.3	10,479.8	22.8	23.5	130.70	-37.4	7.0	65.9	20.1	45.79	1.439	Level 3
10,500.0	10,497.6	10,507.2	10,506.1	22.8	23.5	131.22	-40.2	12.0	67.6	22.0	45.69	1.460	Level 3
10,525.0	10,521.2	10,534.3	10,532.2	22.9	23.6	131.51	-43.7	18.2	69.5	24.0	45.56	1.526	
10,550.0	10,544.4	10,561.3	10,557.8	23.0	23.6	131.58	-47.9	25.8	71.5	26.1	45.41	1.575	
10,575.0	10,567.1	10,588.5	10,583.0	23.0	23.7	131.45	-52.9	34.6	73.6	28.4	45.24	1.627	
10,600.0	10,589.1	10,615.7	10,607.7	23.1	23.8	131.14	-58.6	44.7	75.8	30.7	45.07	1.681	
10,625.0	10,610.6	10,643.0	10,631.7	23.2	23.8	130.66	-64.9	56.0	78.0	33.1	44.90	1.738	
10,650.0	10,631.3	10,670.3	10,654.9	23.3	23.9	130.02	-71.9	68.5	80.4	36.6	44.76	1.796	
10,675.0	10,651.3	10,697.7	10,677.3	23.4	24.0	129.25	-79.6	82.1	82.8	38.2	44.64	1.855	
10,700.0	10,670.5	10,725.0	10,698.8	23.5	24.1	128.34	-87.9	96.9	85.3	40.8	44.57	1.914	
10,725.0	10,688.8	10,752.4	10,719.3	23.6	24.2	127.33	-96.8	112.7	87.9	43.4	44.56	1.973	
10,750.0	10,708.2	10,779.8	10,738.7	23.7	24.3	126.22	-106.3	129.6	90.6	46.0	44.62	2.031	
10,775.0	10,722.6	10,807.2	10,757.0	23.9	24.4	125.03	-116.3	147.3	93.4	48.6	44.74	2.087	
10,800.0	10,738.0	10,834.6	10,774.1	24.0	24.5	123.76	-126.8	166.0	96.2	51.3	44.95	2.140	
10,825.0	10,752.4	10,861.9	10,789.9	24.2	24.7	122.44	-137.7	185.5	99.1	53.9	45.24	2.191	
10,850.0	10,766.7	10,889.3	10,804.4	24.4	24.9	121.06	-149.1	205.7	102.1	56.5	45.62	2.239	
10,875.0	10,777.8	10,916.6	10,817.5	24.6	25.0	119.64	-160.8	226.5	105.2	59.1	46.08	2.283	
10,900.0	10,788.8	10,943.8	10,829.2	24.8	25.2	118.20	-172.9	248.0	108.4	61.8	46.62	2.325	
10,925.0	10,798.6	10,971.1	10,839.5	25.0	25.5	116.73	-185.2	269.9	111.6	64.4	47.24	2.363	
10,950.0	10,807.2	10,998.2	10,848.3	25.3	25.7	115.24	-197.8	292.3	114.9	67.0	47.92	2.398	
10,975.0	10,814.5	11,025.3	10,855.7	25.6	26.0	113.75	-210.6	315.1	118.3	69.6	48.66	2.430	
11,000.0	10,820.6	11,052.4	10,861.6	25.8	26.2	112.25	-223.6	338.1	121.7	72.2	49.46	2.460	
11,025.0	10,825.4	11,079.4	10,865.8	26.1	26.5	110.76	-236.6	361.3	125.1	74.8	50.29	2.488	
11,050.0	10,828.9	11,108.3	10,868.6	26.5	26.9	109.28	-249.8	384.7	128.6	77.5	51.16	2.514	
11,075.0	10,831.1	11,133.2	10,869.9	26.8	27.2	107.81	-262.9	408.1	132.2	80.1	52.05	2.539	
11,100.0	10,832.0	11,159.3	10,870.0	27.1	27.5	106.51	-275.7	430.9	135.8	82.9	52.93	2.566	
11,104.0	10,832.0	11,163.4	10,870.0	27.2	27.6	106.36	-277.7	434.5	136.4	83.3	53.08	2.570	
11,200.0	10,832.2	11,262.9	10,870.2	28.6	29.0	104.72	-324.2	522.4	151.8	95.8	56.01	2.711	
11,300.0	10,832.4	11,366.8	10,870.4	30.3	30.7	103.16	-369.6	615.6	170.4	111.0	58.35	2.870	
11,400.0	10,832.6	11,470.8	10,870.5	32.2	32.6	101.77	-411.6	711.0	191.4	128.5	62.89	3.043	
11,500.0	10,832.7	11,575.2	10,870.7	34.1	34.7	100.54	-450.2	807.9	214.9	148.4	66.51	3.231	
11,600.0	10,832.9	11,679.9	10,870.9	36.2	36.9	99.47	-486.4	908.5	240.8	170.6	70.16	3.432	
11,680.4	10,833.1	11,764.4	10,871.1	38.0	38.7	98.70	-511.1	987.0	263.4	190.3	73.08	3.605	
11,700.0	10,833.1	11,785.0	10,871.1	38.4	39.2	98.50	-517.0	1,008.7	269.0	196.1	73.94	3.638	
11,800.0	10,833.3	11,891.7	10,871.3	40.7	41.6	97.86	-545.4	1,109.6	296.7	217.0	78.82	3.761	
11,900.0	10,833.4	12,000.6	10,871.5	43.0	44.2	97.04	-570.4	1,215.6	318.7	235.2	83.51	3.817	
12,000.0	10,833.6	12,111.3	10,871.7	45.4	46.9	96.59	-591.6	1,324.2	338.1	249.5	88.60	3.816	
12,100.0	10,833.8	12,223.5	10,871.9	47.9	49.6	96.26	-608.9	1,435.1	353.7	259.8	93.85	3.788	
12,200.0	10,834.0	12,337.0	10,872.1	50.4	52.4	96.03	-621.9	1,547.8	365.4	266.1	99.24	3.682	
12,300.0	10,834.1	12,451.3	10,872.3	53.0	55.3	95.89	-630.4	1,661.8	373.1	268.4	104.75	3.582	
12,400.0	10,834.3	12,566.2	10,872.5	55.5	58.2	95.82	-634.5	1,776.7	376.9	266.6	110.33	3.416	
12,500.0	10,834.5	12,671.6	10,872.7	58.2	60.9	95.81	-634.8	1,892.3	377.5	261.8	115.73	3.262	
12,600.0	10,834.7	12,771.8	10,872.9	60.6	63.4	95.80	-634.8	1,962.3	377.8	256.8	121.04	3.121	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services
Anticollision Report



Company: Oasis Petroleum
Project: Indian Hills
Reference Site: 153N-100W-17/18
Site Error: 0.0 usft
Reference Well: Kline Federal 5300 41-18 12TX
Well Error: 2.0 usft
Reference Wellbore: Kline Federal 5300 41-18 12TX
Reference Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at: 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 13T2X - Kline Federal 5300 41-18 13T2X - Design #1													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis		Highside Toolface (°)	Distance		Minimum Separation (usft)	Separation Factor	Warning			
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)		Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)						
12,700.0	10,834.8	12,871.8	10,873.0	63.5	66.0	95.80	-634.8	2,082.3	378.1	251.7	126.40	2.991		
12,800.0	10,835.0	12,971.8	10,873.2	66.2	68.7	95.79	-634.8	2,182.3	378.4	246.6	131.80	2.871		
12,900.0	10,835.2	13,071.8	10,873.4	68.9	71.3	95.79	-634.8	2,282.3	378.7	241.5	137.24	2.760		
13,000.0	10,835.4	13,171.8	10,873.6	71.7	74.0	95.79	-634.8	2,382.3	379.0	236.3	142.70	2.656		
13,100.0	10,835.5	13,271.8	10,873.7	74.4	76.7	95.78	-634.8	2,482.3	379.3	231.2	148.20	2.560		
13,200.0	10,835.7	13,371.8	10,873.9	77.2	79.4	95.78	-634.8	2,582.3	379.7	225.9	153.72	2.470		
13,300.0	10,835.9	13,471.8	10,874.1	79.9	82.1	95.77	-634.8	2,682.3	380.0	220.7	159.26	2.386		
13,400.0	10,836.1	13,571.8	10,874.3	82.7	84.8	95.77	-634.8	2,782.3	380.3	215.4	164.82	2.307		
13,500.0	10,836.2	13,671.8	10,874.4	85.5	87.6	95.76	-634.8	2,882.3	380.6	210.2	170.40	2.233		
13,600.0	10,836.4	13,771.8	10,874.6	88.3	90.3	95.76	-634.8	2,982.3	380.9	204.9	176.00	2.164		
13,700.0	10,836.6	13,871.8	10,874.8	91.1	93.1	95.75	-634.8	3,082.3	381.2	199.6	181.51	2.099		
13,800.0	10,836.7	13,971.8	10,875.0	94.0	95.9	95.75	-634.8	3,182.3	381.5	194.3	187.24	2.037		
13,900.0	10,836.9	14,071.8	10,875.1	96.8	98.7	95.74	-634.8	3,282.3	381.8	188.9	192.88	1.979		
14,000.0	10,837.1	14,171.8	10,875.3	99.6	101.5	95.74	-634.8	3,382.3	382.1	183.6	198.53	1.925		
14,100.0	10,837.3	14,271.8	10,875.5	102.5	104.3	95.74	-634.8	3,482.3	382.4	178.2	204.19	1.873		
14,200.0	10,837.4	14,371.8	10,875.7	105.3	107.1	95.73	-634.8	3,582.3	382.7	172.9	209.86	1.824		
14,300.0	10,837.6	14,471.8	10,875.8	108.2	109.9	95.73	-634.8	3,682.3	383.0	167.5	215.54	1.777		
14,400.0	10,837.8	14,571.8	10,876.0	111.0	112.7	95.72	-634.8	3,782.3	383.3	162.1	221.23	1.733		
14,500.0	10,838.0	14,671.8	10,876.2	113.9	115.6	95.72	-634.8	3,882.3	383.6	156.7	226.93	1.691		
14,600.0	10,838.1	14,771.8	10,876.4	116.7	118.4	95.71	-634.8	3,982.3	384.0	151.3	232.63	1.650		
14,700.0	10,838.3	14,871.8	10,876.5	119.6	121.2	95.71	-634.8	4,082.3	384.3	145.9	238.34	1.612		
14,800.0	10,838.5	14,971.8	10,876.7	122.5	124.1	95.70	-634.8	4,182.3	384.6	140.5	244.06	1.576		
14,900.0	10,838.7	15,071.8	10,876.9	125.3	126.9	95.70	-634.8	4,282.3	384.9	135.1	249.78	1.541		
15,000.0	10,838.8	15,171.8	10,877.1	128.2	129.8	95.69	-634.8	4,382.3	385.2	129.7	255.51	1.508		
15,100.0	10,839.0	15,271.8	10,877.2	131.1	132.6	95.69	-634.8	4,482.3	385.5	124.3	261.24	1.476 Level 3		
15,200.0	10,839.2	15,371.8	10,877.4	133.9	135.5	95.69	-634.8	4,582.3	385.8	118.8	266.98	1.445 Level 3		
15,300.0	10,839.4	15,471.8	10,877.6	136.8	138.3	95.68	-634.8	4,682.3	386.1	113.4	272.72	1.416 Level 3		
15,400.0	10,839.6	15,571.8	10,877.8	139.7	141.2	95.68	-634.8	4,782.3	386.4	108.0	278.46	1.388 Level 3		
15,500.0	10,839.7	15,671.8	10,877.9	142.6	144.1	95.67	-634.8	4,882.3	386.7	102.5	284.21	1.361 Level 3		
15,600.0	10,839.9	15,771.8	10,878.1	145.5	146.9	95.67	-634.8	4,982.3	387.0	97.1	289.96	1.335 Level 3		
15,700.0	10,840.1	15,871.8	10,878.3	148.4	149.8	95.66	-634.8	5,082.3	387.3	91.6	295.72	1.310 Level 3		
15,800.0	10,840.2	15,971.8	10,878.5	151.3	152.7	95.66	-634.8	5,182.3	387.6	86.2	301.48	1.286 Level 3		
15,900.0	10,840.4	16,071.8	10,878.6	154.1	155.5	95.65	-634.8	5,282.3	387.9	80.7	307.24	1.263 Level 3		
16,000.0	10,840.6	16,171.8	10,878.8	157.0	158.4	95.65	-634.8	5,382.3	388.3	75.2	313.01	1.240 Level 2		
16,100.0	10,840.8	16,271.8	10,879.0	159.9	161.3	95.65	-634.8	5,482.3	388.6	69.8	318.77	1.219 Level 2		
16,200.0	10,840.9	16,371.8	10,879.1	162.8	164.2	95.64	-634.8	5,582.3	388.9	64.3	324.54	1.198 Level 2		
16,300.0	10,841.1	16,471.8	10,879.3	165.7	167.1	95.64	-634.8	5,682.3	389.2	58.9	330.32	1.178 Level 2		
16,400.0	10,841.3	16,571.8	10,879.5	168.6	169.9	95.63	-634.8	5,782.3	389.5	53.4	336.09	1.159 Level 2		
16,500.0	10,841.4	16,671.8	10,879.7	171.5	172.8	95.63	-634.8	5,882.3	389.8	47.9	341.87	1.140 Level 2		
16,600.0	10,841.6	16,771.8	10,879.8	174.4	175.7	95.62	-634.8	5,982.3	390.1	42.4	347.65	1.122 Level 2		
16,700.0	10,841.8	16,871.8	10,880.0	177.3	178.6	95.62	-634.8	6,082.3	390.4	37.0	353.43	1.105 Level 2		
16,800.0	10,842.0	16,971.8	10,880.2	180.2	181.5	95.62	-634.8	6,182.3	390.7	31.5	359.22	1.088 Level 2		
16,900.0	10,842.1	17,071.8	10,880.4	183.1	184.4	95.61	-634.8	6,282.3	391.0	26.0	365.00	1.071 Level 2		
17,000.0	10,842.3	17,171.8	10,880.5	186.0	187.3	95.61	-634.8	6,382.3	391.3	20.5	370.79	1.055 Level 2		
17,100.0	10,842.5	17,271.8	10,880.7	188.9	190.2	95.60	-634.8	6,482.3	391.6	15.1	376.58	1.040 Level 2		
17,200.0	10,842.7	17,371.8	10,880.9	191.8	193.1	95.60	-634.8	6,582.3	391.9	9.6	382.37	1.025 Level 2		
17,300.0	10,842.8	17,471.8	10,881.1	194.7	196.0	95.59	-634.8	6,682.3	392.2	4.1	388.16	1.011 Level 2		
17,400.0	10,843.0	17,571.8	10,881.2	197.6	198.9	95.59	-634.8	6,782.3	392.6	-1.4	393.95	0.998 Level 1		
17,500.0	10,843.2	17,671.8	10,881.4	200.6	201.8	95.58	-634.8	6,882.3	392.9	-6.9	399.75	0.983 Level 1		
17,600.0	10,843.4	17,771.8	10,881.6	203.5	204.7	95.58	-634.8	6,982.3	393.2	-12.4	405.55	0.969 Level 1		
17,700.0	10,843.5	17,871.8	10,881.8	206.4	207.6	95.58	-634.8	7,082.3	393.5	-17.9	411.34	0.957 Level 1		
17,800.0	10,843.7	17,971.8	10,881.9	209.3	210.5	95.57	-634.8	7,182.3	393.8	-23.4	417.14	0.944 Level 1		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services
Anticollision Report



Company: Oasis Petroleum
Project: Indian Hills
Reference Site: 153N-100W-17/18
Site Error: 0.0 usft
Reference Well: Kline Federal 5300 41-18 12TX
Well Error: 2.0 usft
Reference Wellbore: Kline Federal 5300 41-18 12TX
Reference Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at: 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 13T2X - Kline Federal 5300 41-18 13T2X - Design #1													Offset Site Error: 0.0 usft
Survey Program: 0-MWD													Offset Well Error: 2.0 usft
Reference		Offset		Semi Major Axis		Highside Toolface (°)	Distance		Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)		Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)					
17,900.0	10,843.9	18,071.8	10,882.1	212.2	213.4	95.57	-634.8	7,282.3	394.1	-28.9	422.94	0.932 Level 1	
18,000.0	10,844.1	18,171.8	10,882.3	215.1	216.3	95.56	-634.8	7,382.3	394.4	-34.3	428.74	0.920 Level 1	
18,100.0	10,844.2	18,271.8	10,882.5	218.0	219.2	95.56	-634.8	7,482.3	394.7	-39.8	434.55	0.908 Level 1	
18,200.0	10,844.4	18,371.8	10,882.6	220.9	222.1	95.55	-634.8	7,582.3	395.0	-45.3	440.35	0.897 Level 1	
18,300.0	10,844.6	18,471.8	10,882.8	223.8	225.0	95.55	-634.8	7,682.3	395.3	-50.8	446.15	0.886 Level 1	
18,400.0	10,844.8	18,571.8	10,883.0	226.7	227.9	95.55	-634.8	7,782.3	395.6	-56.3	451.98	0.875 Level 1	
18,500.0	10,844.9	18,671.8	10,883.2	229.7	230.8	95.54	-634.8	7,882.2	395.9	-61.8	457.77	0.865 Level 1	
18,600.0	10,845.1	18,771.8	10,883.3	232.6	233.7	95.54	-634.8	7,982.2	396.2	-67.3	463.57	0.855 Level 1	
18,700.0	10,845.3	18,871.8	10,883.5	235.5	236.6	95.53	-634.8	8,082.2	396.5	-72.8	469.38	0.845 Level 1	
18,800.0	10,845.4	18,971.8	10,883.7	238.4	239.5	95.53	-634.8	8,182.2	396.9	-78.3	475.19	0.835 Level 1	
18,900.0	10,845.6	19,071.8	10,883.9	241.3	242.4	95.53	-634.8	8,282.2	397.2	-83.8	481.00	0.826 Level 1	
19,000.0	10,845.8	19,171.8	10,884.0	244.2	245.3	95.52	-634.8	8,382.2	397.5	-89.3	486.81	0.816 Level 1	
19,100.0	10,846.0	19,271.8	10,884.2	247.1	248.2	95.52	-634.8	8,482.2	397.8	-94.8	492.62	0.807 Level 1	
19,200.0	10,846.1	19,371.8	10,884.4	250.1	251.1	95.51	-634.8	8,582.2	398.1	-100.4	498.44	0.799 Level 1	
19,300.0	10,846.3	19,471.8	10,884.6	253.0	254.1	95.51	-634.8	8,682.2	398.4	-105.9	504.25	0.790 Level 1	
19,400.0	10,846.5	19,571.8	10,884.7	255.9	257.0	95.50	-634.8	8,782.2	398.7	-111.4	510.06	0.782 Level 1	
19,500.0	10,846.7	19,671.8	10,884.9	258.8	259.9	95.50	-634.8	8,882.2	399.0	-116.9	515.88	0.773 Level 1	
19,600.0	10,846.8	19,771.8	10,885.1	261.7	262.8	95.50	-634.8	8,982.2	399.3	-122.4	521.69	0.765 Level 1	
19,700.0	10,847.0	19,871.8	10,885.3	264.6	265.7	95.49	-634.8	9,082.2	399.6	-127.9	527.51	0.758 Level 1	
19,800.0	10,847.2	19,971.8	10,885.4	267.6	268.6	95.49	-634.8	9,182.2	399.9	-133.4	533.32	0.750 Level 1	
19,900.0	10,847.4	20,071.8	10,885.6	270.5	271.5	95.48	-634.8	9,282.2	400.2	-138.9	539.14	0.742 Level 1	
20,000.0	10,847.5	20,171.8	10,885.8	273.4	274.4	95.48	-634.8	9,382.2	400.5	-144.4	544.96	0.735 Level 1	
20,100.0	10,847.7	20,271.8	10,886.0	276.3	277.4	95.48	-634.8	9,482.2	400.9	-149.9	550.78	0.728 Level 1	
20,200.0	10,847.9	20,371.8	10,886.1	279.2	280.3	95.47	-634.8	9,582.2	401.2	-155.4	556.60	0.721 Level 1	
20,300.0	10,848.1	20,471.8	10,886.3	282.1	283.2	95.47	-634.8	9,682.2	401.5	-161.0	562.41	0.714 Level 1	
20,400.0	10,848.2	20,571.8	10,886.5	285.1	286.1	95.46	-634.8	9,782.2	401.8	-166.5	568.23	0.707 Level 1	
20,500.0	10,848.4	20,671.8	10,886.7	288.0	289.0	95.46	-634.8	9,882.2	402.1	-172.0	574.05	0.700 Level 1	
20,600.0	10,848.6	20,771.8	10,886.8	290.9	291.9	95.45	-634.8	9,982.2	402.4	-177.5	579.88	0.694 Level 1	
20,667.7	10,848.7	20,839.5	10,886.9	292.9	293.9	95.45	-634.8	10,049.9	402.6	-181.2	583.82	0.690 Level 1, ES, SF	



Ryan Directional Services Anticollision Report



Company: Oasis Petroleum
Project: Indian Hills
Reference Site: 153N-100W-17/18
Site Error: 0.0 usft
Reference Well: Kline Federal 5300 41-18 12TX
Well Error: 2.0 usft
Reference Wellbore: Kline Federal 5300 41-18 12TX
Reference Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at: 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 14BX - Kline Federal 5300 41-18 14BX - Design #1												Offset Site Error:	0.0 usft
Survey Program: 0-MWD												Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis		Highside Toolface (°)	Distance		Minimum Separation (usft)	Separation Factor	Warning		
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)		Between Centres (usft)	Between Ellipses (usft)					
0.0	0.0	0.0	0.0	2.0	2.0	-180.00	-65.9	0.0	65.9				
100.0	100.0	100.0	100.0	2.0	2.0	-180.00	-65.9	0.0	65.9	61.8	4.00	16.452	
200.0	200.0	200.0	200.0	2.0	2.0	-180.00	-65.9	0.0	65.9	61.8	4.05	16.273	
300.0	300.0	300.0	300.0	2.1	2.1	-180.00	-65.9	0.0	65.9	61.7	4.14	15.909	
400.0	400.0	400.0	400.0	2.1	2.1	-180.00	-65.9	0.0	65.9	61.6	4.28	15.398	
500.0	500.0	500.0	500.0	2.2	2.2	-180.00	-65.9	0.0	65.9	61.4	4.48	14.777	
600.0	600.0	600.0	600.0	2.3	2.3	-180.00	-65.9	0.0	65.9	61.2	4.87	14.094	
700.0	700.0	700.0	700.0	2.5	2.5	-180.00	-65.9	0.0	65.9	60.9	4.92	13.385	
800.0	800.0	800.0	800.0	2.6	2.6	-180.00	-65.9	0.0	65.9	60.7	5.20	12.678	
900.0	900.0	900.0	900.0	2.7	2.7	-180.00	-65.9	0.0	65.9	60.4	5.49	11.990	
1,000.0	1,000.0	1,000.0	1,000.0	2.9	2.9	-180.00	-65.9	0.0	65.9	60.1	5.81	11.336	
1,100.0	1,100.0	1,100.0	1,100.0	3.1	3.1	-180.00	-65.9	0.0	65.9	59.7	6.14	10.720	
1,200.0	1,200.0	1,200.0	1,200.0	3.2	3.2	-180.00	-65.9	0.0	65.9	59.4	6.49	10.145	
1,300.0	1,300.0	1,300.0	1,300.0	3.4	3.4	-180.00	-65.9	0.0	65.9	59.0	6.85	9.613	
1,400.0	1,400.0	1,400.0	1,400.0	3.6	3.6	-180.00	-65.9	0.0	65.9	58.6	7.22	9.121	
1,500.0	1,500.0	1,500.0	1,500.0	3.8	3.8	-180.00	-65.9	0.0	65.9	58.3	7.60	8.667	
1,600.0	1,600.0	1,600.0	1,600.0	4.0	4.0	-180.00	-65.9	0.0	65.9	57.9	7.99	8.248	
1,700.0	1,700.0	1,700.0	1,700.0	4.2	4.2	-180.00	-65.9	0.0	65.9	57.5	8.38	7.862	
1,800.0	1,800.0	1,800.0	1,800.0	4.4	4.4	-180.00	-65.9	0.0	65.9	57.1	8.78	7.506	
1,900.0	1,900.0	1,900.0	1,900.0	4.6	4.6	-180.00	-65.9	0.0	65.9	56.7	9.18	7.177	
2,000.0	2,000.0	2,000.0	2,000.0	4.8	4.8	-180.00	-65.9	0.0	65.9	56.3	9.58	6.872	
2,100.0	2,100.0	2,100.0	2,100.0	5.0	5.0	-180.00	-65.9	0.0	65.9	55.9	9.99	6.590	
2,200.0	2,200.0	2,200.0	2,200.0	5.2	5.2	-180.00	-65.9	0.0	65.9	55.5	10.41	6.328 CC	
2,216.7	2,216.7	2,216.1	2,216.1	5.2	5.2	90.06	-65.9	0.0	65.9	55.5	10.47	6.298	
2,300.0	2,300.0	2,299.4	2,299.4	5.4	5.4	90.68	-66.7	0.0	66.7	55.0	10.75	6.201	
2,400.0	2,400.0	2,399.4	2,399.4	5.6	5.5	91.41	-67.5	0.0	67.6	54.5	11.10	6.088	
2,500.0	2,500.0	2,499.4	2,499.4	5.8	5.7	92.13	-68.4	0.0	68.5	57.0	11.45	5.979	
2,600.0	2,600.0	2,599.4	2,599.4	6.0	5.9	92.82	-69.3	0.0	69.4	57.6	11.81	5.874	
2,700.0	2,700.0	2,699.4	2,699.4	6.1	6.0	93.50	-70.1	0.0	70.3	58.1	12.17	5.773	
2,800.0	2,800.0	2,799.4	2,799.4	6.3	6.2	94.15	-71.0	0.0	71.2	58.7	12.54	5.677	
2,900.0	2,900.0	2,899.4	2,899.3	6.5	6.4	94.79	-71.9	0.0	72.2	59.2	12.92	5.584	
3,000.0	3,000.0	2,999.4	2,999.3	6.7	6.6	95.42	-72.8	0.0	73.1	59.8	13.30	5.496	
3,100.0	3,100.0	3,099.4	3,099.3	6.9	6.7	96.03	-73.6	0.0	74.1	60.4	13.69	5.411	
3,200.0	3,200.0	3,199.3	3,199.3	7.1	6.9	96.62	-74.5	0.0	75.0	60.9	14.07	5.330	
3,300.0	3,300.0	3,299.3	3,299.3	7.3	7.1	97.20	-75.4	0.0	76.0	61.5	14.47	5.253	
3,400.0	3,400.0	3,399.3	3,399.3	7.6	7.3	97.76	-76.3	0.0	77.0	62.1	14.86	5.179	
3,500.0	3,500.0	3,499.3	3,499.3	7.8	7.5	98.31	-77.1	0.0	78.0	62.7	15.26	5.108	
3,600.0	3,599.9	3,599.3	3,599.3	8.0	7.7	98.84	-78.0	0.0	78.9	63.3	15.68	5.041	
3,700.0	3,699.9	3,699.3	3,699.3	8.2	7.9	99.37	-78.9	0.0	79.9	63.9	16.07	4.976	
3,800.0	3,799.9	3,799.3	3,799.2	8.4	8.1	99.88	-79.7	0.0	81.0	64.5	16.47	4.916	
3,900.0	3,899.9	3,899.3	3,899.2	8.6	8.3	100.37	-80.6	0.0	82.0	65.1	16.88	4.856	
4,000.0	3,999.9	3,999.3	3,999.2	8.8	8.5	100.86	-81.5	0.0	83.0	65.7	17.29	4.799	
4,100.0	4,099.9	4,099.3	4,099.2	9.0	8.7	101.33	-82.4	0.0	84.0	66.3	17.70	4.745	
4,200.0	4,199.9	4,199.3	4,199.2	9.2	8.9	101.79	-83.2	0.0	85.0	66.9	18.12	4.694	
4,300.0	4,299.9	4,299.3	4,299.2	9.4	9.1	102.24	-84.1	0.0	86.1	67.5	18.53	4.644	
4,400.0	4,399.9	4,399.3	4,399.2	9.7	9.3	102.68	-85.0	0.0	87.1	68.2	18.95	4.597	
4,500.0	4,499.9	4,499.3	4,499.2	9.9	9.5	103.11	-85.9	0.0	88.2	68.8	19.37	4.551	
4,600.0	4,599.9	4,599.2	4,599.2	10.1	9.7	103.53	-86.7	0.0	89.2	69.4	19.79	4.507	
4,700.0	4,699.9	4,699.2	4,699.1	10.3	9.9	103.94	-87.6	0.0	90.3	70.0	20.21	4.468	
4,800.0	4,799.9	4,799.2	4,799.1	10.5	10.1	104.33	-88.5	0.0	91.3	70.7	20.64	4.425	
4,900.0	4,899.9	4,899.2	4,899.1	10.7	10.3	104.72	-89.3	0.0	92.4	71.3	21.06	4.387	
5,000.0	4,999.9	4,999.2	4,999.1	10.9	10.6	105.11	-90.2	0.0	93.5	72.0	21.49	4.350	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services
Anticollision Report



Company: Oasis Petroleum
Project: Indian Hills
Reference Site: 153N-100W-17/18
Site Error: 0.0 usft
Reference Well: Kline Federal 5300 41-18 12TX
Well Error: 2.0 usft
Reference Wellbore: Kline Federal 5300 41-18 12TX
Reference Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 14BX - Kline Federal 5300 41-18 14BX - Design #1											Offset Site Error: 0.0 usft		
Survey Program: 0-MWD											Offset Well Error: 2.0 usft		
Reference		Offset		Semi Major Axis			Distance					Warning	
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Tooface (")	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)		Separation Factor
5,100.0	5,099.9	5,099.2	5,099.1	11.2	10.8	105.48	-91.1	0.0	94.5	72.6	21.91	4.314	
5,200.0	5,199.9	5,199.2	5,199.1	11.4	11.0	105.84	-92.0	0.0	95.6	73.3	22.34	4.280	
5,300.0	5,299.9	5,299.2	5,299.1	11.6	11.2	106.20	-92.8	0.0	96.7	73.9	22.77	4.246	
5,400.0	5,399.9	5,399.2	5,399.1	11.8	11.4	106.55	-93.7	0.0	97.8	74.6	23.20	4.215	
5,500.0	5,499.9	5,499.2	5,499.0	12.0	11.6	106.89	-94.6	0.0	98.9	75.2	23.63	4.184	
5,600.0	5,599.9	5,599.2	5,599.0	12.2	11.8	107.22	-95.5	0.0	99.9	75.9	24.06	4.154	
5,700.0	5,699.9	5,699.2	5,699.0	12.5	12.0	107.55	-96.3	0.0	101.0	76.5	24.49	4.126	
5,800.0	5,799.9	5,799.2	5,799.0	12.7	12.3	107.87	-97.2	0.0	102.1	77.2	24.92	4.098	
5,900.0	5,899.9	5,899.1	5,899.0	12.9	12.5	108.18	-98.1	0.0	103.2	77.9	25.35	4.072	
6,000.0	5,999.9	5,999.1	5,999.0	13.1	12.7	108.49	-98.9	0.0	104.3	78.5	25.79	4.046	
6,100.0	6,099.9	6,099.1	6,099.0	13.3	12.9	108.79	-99.8	0.0	105.4	79.2	26.22	4.021	
6,200.0	6,199.8	6,199.1	6,199.0	13.6	13.1	109.08	-100.7	0.0	106.5	79.9	26.65	3.997	
6,300.0	6,299.8	6,299.1	6,299.0	13.8	13.3	109.37	-101.6	0.0	107.7	80.6	27.09	3.974	
6,400.0	6,399.8	6,399.1	6,399.0	14.0	13.6	109.65	-102.4	0.0	108.8	81.2	27.53	3.952	
6,500.0	6,499.8	6,499.1	6,498.9	14.2	13.8	109.92	-103.3	0.0	109.9	81.9	27.96	3.930	
6,600.0	6,599.8	6,599.1	6,598.9	14.4	14.0	110.19	-104.2	0.0	111.0	82.6	28.40	3.909	
6,700.0	6,699.8	6,699.1	6,698.9	14.7	14.2	110.46	-105.1	0.0	112.1	83.3	28.83	3.889	
6,800.0	6,799.8	6,799.1	6,798.9	14.9	14.4	110.72	-105.9	0.0	113.3	84.0	29.27	3.869	
6,900.0	6,899.8	6,899.1	6,898.9	15.1	14.6	110.97	-106.8	0.0	114.4	84.7	29.71	3.850	
7,000.0	6,999.8	6,999.1	6,998.9	15.3	14.9	111.22	-107.7	0.0	115.5	85.4	30.15	3.831	
7,100.0	7,099.8	7,099.1	7,098.9	15.5	15.1	111.48	-108.5	0.0	116.6	86.1	30.59	3.813	
7,200.0	7,199.8	7,199.0	7,198.9	15.8	15.3	111.70	-109.4	0.0	117.8	86.7	31.02	3.796	
7,300.0	7,299.8	7,299.0	7,298.8	16.0	15.5	111.94	-110.3	0.0	118.9	87.4	31.46	3.779	
7,400.0	7,399.8	7,399.0	7,398.8	16.2	15.7	112.17	-111.2	0.0	120.0	88.1	31.90	3.763	
7,500.0	7,499.8	7,499.0	7,498.8	16.4	16.0	112.40	-112.0	0.0	121.2	88.8	32.34	3.747	
7,600.0	7,599.8	7,599.0	7,598.8	16.6	16.2	112.62	-112.9	0.0	122.3	89.5	32.78	3.731	
7,700.0	7,699.8	7,699.0	7,698.8	16.9	16.4	112.84	-113.8	0.0	123.5	90.2	33.22	3.716	
7,800.0	7,799.8	7,799.0	7,798.8	17.1	16.6	113.05	-114.7	0.0	124.6	90.9	33.66	3.701	
7,900.0	7,899.8	7,899.0	7,898.8	17.3	16.8	113.26	-115.5	0.0	125.8	91.6	34.11	3.687	
7,926.7	7,926.4	7,925.7	7,925.4	17.4	16.9	113.32	-115.8	0.0	126.1	91.9	34.22	3.683	
7,943.3	7,943.1	7,943.3	7,943.1	17.4	16.9	-156.66	-115.8	0.0	126.2	91.9	34.30	3.679	
8,000.0	7,999.8	8,000.0	7,999.8	17.5	17.1	-156.66	-115.8	0.0	126.2	91.6	34.52	3.654	
8,100.0	8,099.8	8,100.0	8,099.8	17.7	17.2	-156.66	-115.8	0.0	126.2	91.2	34.82	3.613	
8,200.0	8,199.8	8,200.0	8,199.8	17.9	17.4	-156.66	-115.8	0.0	126.2	90.8	35.32	3.572	
8,300.0	8,299.8	8,300.0	8,299.8	18.1	17.6	-156.66	-115.8	0.0	126.2	90.4	35.72	3.532	
8,400.0	8,399.8	8,400.0	8,399.8	18.4	17.8	-156.66	-115.8	0.0	126.2	90.0	36.12	3.493	
8,500.0	8,499.8	8,500.0	8,499.8	18.6	18.0	-156.66	-115.8	0.0	126.2	89.6	36.52	3.454	
8,600.0	8,599.8	8,600.0	8,599.8	18.8	18.2	-156.66	-115.8	0.0	126.2	89.2	36.93	3.416	
8,700.0	8,699.8	8,700.0	8,699.8	19.0	18.4	-156.66	-115.8	0.0	126.2	88.8	37.33	3.379	
8,800.0	8,799.8	8,800.0	8,799.8	19.2	18.6	-156.66	-115.8	0.0	126.2	88.4	37.74	3.343	
8,900.0	8,899.8	8,900.0	8,899.8	19.4	18.8	-156.66	-115.8	0.0	126.2	88.0	38.14	3.307	
9,000.0	8,999.8	9,000.0	8,999.8	19.6	19.0	-156.66	-115.8	0.0	126.2	87.6	38.56	3.273	
9,100.0	9,099.8	9,100.0	9,099.8	19.8	19.2	-156.66	-115.8	0.0	126.2	87.2	38.96	3.238	
9,200.0	9,199.8	9,200.0	9,199.8	20.0	19.4	-156.66	-115.8	0.0	126.2	86.8	39.37	3.205	
9,300.0	9,299.8	9,300.0	9,299.8	20.2	19.6	-156.66	-115.8	0.0	126.2	86.4	39.78	3.172	
9,400.0	9,399.8	9,400.0	9,399.8	20.5	19.8	-156.66	-115.8	0.0	126.2	86.0	40.19	3.139	
9,500.0	9,499.8	9,500.0	9,499.8	20.7	20.0	-156.66	-115.8	0.0	126.2	85.6	40.60	3.107	
9,600.0	9,599.8	9,600.0	9,599.8	20.9	20.2	-156.66	-115.8	0.0	126.2	85.1	41.01	3.076	
9,700.0	9,699.8	9,700.0	9,699.8	21.1	20.4	-156.66	-115.8	0.0	126.2	84.7	41.43	3.045	
9,800.0	9,799.8	9,800.0	9,799.8	21.3	20.6	-156.66	-115.8	0.0	126.2	84.3	41.84	3.015	
9,900.0	9,899.8	9,900.0	9,899.8	21.5	20.8	-156.66	-115.8	0.0	126.2	83.9	42.26	2.986	
10,000.0	9,999.8	10,000.0	9,999.8	21.7	21.0	-156.66	-115.8	0.0	126.2	83.5	42.67	2.956	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services Anticollision Report



Company: Oasis Petroleum
Project: Indian Hills
Reference Site: 153N-100W-17/18
Site Error: 0.0 usft
Reference Well: Kline Federal 5300 41-18 12TX
Well Error: 2.0 usft
Reference Wellbore: Kline Federal 5300 41-18 12TX
Reference Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at: 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 14BX - Kline Federal 5300 41-18 14BX - Design #1													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis		Highside Toolface (°)	Distance		Minimum Separation (usft)	Separation Factor	Warning			
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)		Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)				Between Centres (usft)	Between Ellipses (usft)	Between Centres (usft)
10,100.0	10,099.8	10,100.0	10,099.8	21.9	21.2	-156.86	-115.8	0.0	126.2	83.1	43.09	2.928		
10,162.1	10,161.9	10,162.1	10,161.9	22.1	21.3	-156.86	-115.8	0.0	126.2	82.8	43.35	2.910		
10,200.0	10,199.8	10,199.8	10,199.8	22.2	21.4	-156.73	-115.9	0.1	126.2	82.7	43.50	2.900		
10,300.0	10,299.8	10,296.4	10,296.2	22.4	21.6	-162.19	-121.4	11.0	127.5	83.6	43.93	2.903		
10,354.8	10,354.6	10,346.2	10,343.1	22.5	21.7	-168.17	-127.5	23.3	130.8	86.6	44.17	2.961		
10,375.0	10,374.8	10,363.9	10,359.7	22.5	21.8	-79.20	-130.2	28.7	132.7	88.5	44.24	3.000		
10,400.0	10,399.7	10,385.5	10,379.7	22.6	21.8	76.08	-133.9	36.1	135.6	91.3	44.34	3.059		
10,425.0	10,424.5	10,406.9	10,399.1	22.7	21.9	73.14	-138.0	44.1	139.0	94.6	44.42	3.129		
10,450.0	10,449.2	10,428.0	10,417.8	22.7	21.9	70.38	-142.4	52.9	142.8	98.3	44.49	3.209		
10,475.0	10,473.5	10,450.0	10,438.8	22.8	22.0	67.70	-147.3	62.7	146.9	102.3	44.53	3.298		
10,500.0	10,497.6	10,469.5	10,453.2	22.8	22.0	65.43	-152.0	72.1	151.2	105.7	44.53	3.395		
10,525.0	10,521.2	10,489.9	10,469.9	22.9	22.1	63.24	-157.3	82.6	155.8	111.3	44.51	3.500		
10,550.0	10,544.4	10,510.1	10,486.0	23.0	22.2	61.22	-162.8	93.6	160.5	116.1	44.44	3.612		
10,575.0	10,567.1	10,530.2	10,501.4	23.0	22.2	59.39	-168.6	105.1	165.4	121.0	44.32	3.731		
10,600.0	10,589.1	10,550.0	10,516.0	23.1	22.3	57.72	-174.6	117.0	170.3	126.1	44.17	3.855		
10,625.0	10,610.6	10,569.8	10,530.1	23.2	22.4	56.20	-180.8	129.4	175.2	131.2	43.98	3.983		
10,650.0	10,631.3	10,589.3	10,543.4	23.3	22.5	54.83	-187.2	142.2	180.1	136.4	43.78	4.117		
10,675.0	10,651.3	10,608.8	10,556.1	23.4	22.6	53.80	-193.8	155.4	185.1	141.5	43.51	4.253		
10,700.0	10,670.5	10,628.1	10,568.1	23.5	22.7	52.49	-200.6	168.9	189.9	146.7	43.23	4.392		
10,725.0	10,688.8	10,647.2	10,579.3	23.6	22.8	51.51	-207.6	182.8	194.7	151.7	42.94	4.533		
10,750.0	10,706.2	10,666.3	10,589.9	23.7	22.9	50.84	-214.7	197.0	199.3	156.7	42.65	4.673		
10,775.0	10,722.6	10,685.3	10,599.8	23.9	23.1	49.87	-221.9	211.4	203.8	161.5	42.35	4.813		
10,800.0	10,738.0	10,704.2	10,609.0	24.0	23.2	49.20	-228.3	226.2	208.2	166.1	42.07	4.950		
10,825.0	10,752.4	10,725.0	10,618.4	24.2	23.4	48.60	-237.7	242.8	212.4	170.6	41.81	5.082		
10,850.0	10,765.7	10,741.7	10,625.3	24.4	23.5	48.13	-244.5	256.4	216.5	174.9	41.58	5.208		
10,875.0	10,777.8	10,760.3	10,632.4	24.6	23.7	47.72	-252.2	271.8	220.4	179.0	41.37	5.327		
10,900.0	10,788.8	10,778.9	10,638.9	24.8	23.9	47.39	-260.1	287.4	224.1	182.8	41.22	5.436		
10,925.0	10,798.6	10,800.0	10,645.3	25.0	24.1	47.14	-269.0	305.3	227.6	186.4	41.15	5.530		
10,950.0	10,807.2	10,816.0	10,649.6	25.3	24.3	46.94	-276.0	319.1	230.9	189.7	41.11	5.616		
10,975.0	10,814.5	10,834.5	10,653.9	25.6	24.5	46.82	-284.0	335.1	233.9	192.8	41.16	5.684		
11,000.0	10,820.6	10,850.0	10,657.0	25.8	24.6	46.72	-290.8	348.7	236.8	195.6	41.24	5.742		
11,025.0	10,825.4	10,871.3	10,660.4	26.1	24.9	46.77	-300.3	367.5	239.5	198.0	41.49	5.772		
11,050.0	10,828.9	10,889.7	10,662.6	26.5	25.1	46.83	-308.5	383.8	241.9	200.1	41.78	5.790		
11,075.0	10,831.1	10,908.1	10,664.1	26.8	25.4	46.96	-316.7	400.2	244.1	202.0	42.18	5.791		
11,100.0	10,832.0	10,926.0	10,664.9	27.1	25.6	47.11	-324.3	415.3	246.2	203.6	42.60	5.778		
11,104.0	10,832.0	10,929.4	10,665.0	27.2	25.7	47.18	-326.2	419.2	246.4	203.7	42.71	5.770		
11,200.0	10,832.2	11,027.9	10,665.2	28.6	27.1	48.96	-369.1	507.9	255.8	210.3	45.46	5.826		
11,300.0	10,832.4	11,134.2	10,665.4	30.3	28.9	51.02	-412.0	605.2	267.7	218.8	48.83	5.482		
11,400.0	10,832.6	11,240.7	10,665.6	32.2	30.9	53.17	-451.2	704.2	281.8	229.2	52.57	5.361		
11,500.0	10,832.7	11,347.5	10,665.8	34.1	33.1	55.36	-486.9	804.8	298.2	241.7	56.57	5.272		
11,600.0	10,832.9	11,454.6	10,666.0	36.2	35.4	57.51	-518.9	907.0	317.1	256.4	60.73	5.222		
11,680.4	10,833.1	11,541.0	10,666.1	38.0	37.3	59.20	-541.8	990.3	334.1	269.9	64.11	5.210		
11,700.0	10,833.1	11,562.1	10,666.2	38.4	37.8	59.67	-547.1	1,010.7	338.3	273.2	65.13	5.195		
11,800.0	10,833.3	11,671.0	10,666.4	40.7	40.4	61.73	-571.7	1,116.9	358.4	288.1	70.34	5.095		
11,900.0	10,833.4	11,781.8	10,666.6	43.0	43.1	63.28	-592.6	1,225.7	375.5	299.9	75.60	4.986		
12,000.0	10,833.6	11,894.1	10,666.8	45.4	45.8	64.43	-609.4	1,336.7	389.2	308.3	80.90	4.811		
12,100.0	10,833.8	12,007.7	10,667.0	47.9	48.7	65.23	-622.0	1,449.5	399.6	313.4	86.21	4.635		
12,200.0	10,834.0	12,122.0	10,667.2	50.4	51.6	65.74	-630.2	1,563.6	405.3	314.8	91.52	4.440		
12,300.0	10,834.1	12,236.9	10,667.4	53.0	54.5	65.98	-633.8	1,678.4	409.5	312.7	98.78	4.231		
12,400.0	10,834.3	12,341.1	10,667.5	55.5	57.2	65.99	-634.0	1,782.6	409.9	308.2	101.72	4.030		
12,500.0	10,834.5	12,441.1	10,667.7	58.2	59.7	66.01	-634.0	1,882.6	410.2	303.6	106.81	3.848		
12,600.0	10,834.7	12,541.1	10,667.9	60.8	62.4	66.03	-634.0	1,982.6	410.5	298.9	111.55	3.680		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services

Anticollision Report



Company: Oasis Petroleum
Project: Indian Hills
Reference Site: 153N-100W-17/18
Site Error: 0.0 usft
Reference Well: Kline Federal 5300 41-18 12TX
Well Error: 2.0 usft
Reference Wellbore: Kline Federal 5300 41-18 12TX
Reference Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 14BX - Kline Federal 5300 41-18 14BX - Design #1														Offset Site Error:	0.0 usft
Survey Program: 0-MWD														Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis		Highside Toolface (°)	Distance		Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)										
12,700.0	10,834.8	12,641.1	10,668.1	63.5	65.0	66.05	-634.0	2,082.6	410.8	294.2	116.53	3.525			
12,800.0	10,835.0	12,741.1	10,668.2	66.2	67.7	66.06	-634.0	2,182.6	411.0	289.5	121.54	3.382			
12,900.0	10,835.2	12,841.1	10,668.4	66.9	70.4	66.08	-634.0	2,282.6	411.3	284.7	126.59	3.249			
13,000.0	10,835.4	12,941.1	10,668.6	71.7	73.1	66.10	-634.0	2,382.6	411.6	279.9	131.67	3.128			
13,100.0	10,835.5	13,041.1	10,668.8	74.4	75.8	66.12	-634.0	2,482.6	411.9	275.1	136.77	3.012			
13,200.0	10,835.7	13,141.1	10,668.9	77.2	78.5	66.13	-634.0	2,582.6	412.2	270.3	141.80	2.905			
13,300.0	10,835.9	13,241.1	10,669.1	79.9	81.3	66.15	-634.0	2,682.6	412.5	265.4	147.05	2.805			
13,400.0	10,836.1	13,341.1	10,669.3	82.7	84.1	66.17	-634.0	2,782.6	412.7	260.5	152.21	2.712			
13,500.0	10,836.2	13,441.1	10,669.5	85.5	86.8	66.19	-634.0	2,882.6	413.0	255.6	157.40	2.624			
13,600.0	10,836.4	13,541.1	10,669.6	88.3	89.6	66.20	-634.0	2,982.6	413.3	250.7	162.60	2.542			
13,700.0	10,836.6	13,641.1	10,669.8	91.1	92.4	66.22	-634.0	3,082.6	413.6	245.8	167.81	2.465			
13,800.0	10,836.7	13,741.1	10,670.0	94.0	95.2	66.24	-634.0	3,182.6	413.9	240.8	173.04	2.392			
13,900.0	10,836.9	13,841.1	10,670.2	96.8	98.0	66.25	-634.0	3,282.6	414.1	235.9	178.28	2.323			
14,000.0	10,837.1	13,941.1	10,670.3	99.6	100.9	66.27	-634.0	3,382.6	414.4	230.9	183.53	2.258			
14,100.0	10,837.3	14,041.1	10,670.5	102.5	103.7	66.29	-634.0	3,482.6	414.7	225.9	188.79	2.197			
14,200.0	10,837.4	14,141.1	10,670.7	105.3	106.5	66.31	-634.0	3,582.6	415.0	220.9	194.06	2.138			
14,300.0	10,837.6	14,241.1	10,670.9	108.2	109.3	66.32	-634.0	3,682.6	415.3	215.9	199.35	2.083			
14,400.0	10,837.8	14,341.1	10,671.0	111.0	112.2	66.34	-634.0	3,782.6	415.6	210.9	204.63	2.031			
14,500.0	10,838.0	14,441.1	10,671.2	113.9	115.0	66.36	-634.0	3,882.6	415.8	205.9	209.93	1.981			
14,600.0	10,838.1	14,541.1	10,671.4	116.7	117.9	66.38	-634.0	3,982.6	416.1	200.9	215.24	1.933			
14,700.0	10,838.3	14,641.1	10,671.6	119.6	120.7	66.39	-634.0	4,082.6	416.4	195.9	220.55	1.888			
14,800.0	10,838.5	14,741.1	10,671.7	122.5	123.6	66.41	-634.0	4,182.6	416.7	190.8	225.86	1.845			
14,900.0	10,838.7	14,841.1	10,671.9	125.3	126.4	66.43	-634.0	4,282.6	417.0	185.8	231.19	1.804			
15,000.0	10,838.8	14,941.1	10,672.1	128.2	129.3	66.44	-634.0	4,382.6	417.3	180.7	236.52	1.764			
15,100.0	10,839.0	15,041.1	10,672.3	131.1	132.2	66.46	-634.0	4,482.6	417.5	175.7	241.86	1.726			
15,200.0	10,839.2	15,141.1	10,672.4	133.9	135.0	66.48	-634.0	4,582.6	417.8	170.6	247.19	1.680			
15,300.0	10,839.4	15,241.1	10,672.6	136.8	137.9	66.49	-634.0	4,682.6	418.1	165.6	252.54	1.656			
15,400.0	10,839.5	15,341.1	10,672.8	139.7	140.8	66.51	-634.0	4,782.6	418.4	160.5	257.89	1.622			
15,500.0	10,839.7	15,441.1	10,673.0	142.6	143.7	66.53	-634.0	4,882.6	418.7	155.4	263.24	1.590			
15,600.0	10,839.9	15,541.1	10,673.1	145.5	146.6	66.54	-634.0	4,982.6	419.0	150.3	268.60	1.560			
15,700.0	10,840.1	15,641.1	10,673.3	148.4	149.4	66.56	-634.0	5,082.6	419.2	145.3	273.97	1.530			
15,800.0	10,840.2	15,741.1	10,673.5	151.3	152.3	66.58	-634.0	5,182.6	419.5	140.2	279.33	1.502			
15,900.0	10,840.4	15,841.1	10,673.6	154.1	155.2	66.60	-634.0	5,282.6	419.8	135.1	284.70	1.475 Level 3			
16,000.0	10,840.6	15,941.1	10,673.8	157.0	158.1	66.61	-634.0	5,382.6	420.1	130.0	290.08	1.448 Level 3			
16,100.0	10,840.8	16,041.1	10,674.0	159.9	161.0	66.63	-634.0	5,482.6	420.4	124.9	295.46	1.423 Level 3			
16,200.0	10,840.9	16,141.1	10,674.2	162.8	163.9	66.65	-634.0	5,582.6	420.6	119.8	300.84	1.398 Level 3			
16,300.0	10,841.1	16,241.1	10,674.3	165.7	166.8	66.66	-634.0	5,682.6	420.9	114.7	306.22	1.375 Level 3			
16,400.0	10,841.3	16,341.1	10,674.5	168.6	169.8	66.68	-634.0	5,782.6	421.2	109.6	311.61	1.352 Level 3			
16,500.0	10,841.4	16,441.1	10,674.7	171.5	172.5	66.70	-634.0	5,882.6	421.5	104.5	317.00	1.330 Level 3			
16,600.0	10,841.6	16,541.1	10,674.9	174.4	175.4	66.71	-634.0	5,982.6	421.8	99.4	322.40	1.308 Level 3			
16,700.0	10,841.8	16,641.1	10,675.0	177.3	178.3	66.73	-634.0	6,082.6	422.1	94.3	327.80	1.288 Level 3			
16,800.0	10,842.0	16,741.1	10,675.2	180.2	181.2	66.75	-634.0	6,182.6	422.3	89.2	333.20	1.268 Level 3			
16,900.0	10,842.1	16,841.1	10,675.4	183.1	184.1	66.76	-634.0	6,282.6	422.6	84.0	338.60	1.248 Level 2			
17,000.0	10,842.3	16,941.1	10,675.6	186.0	187.0	66.78	-634.0	6,382.6	422.9	78.9	344.00	1.229 Level 2			
17,100.0	10,842.5	17,041.1	10,675.7	188.9	189.9	66.80	-634.0	6,482.6	423.2	73.8	349.41	1.211 Level 2			
17,200.0	10,842.7	17,141.1	10,675.9	191.8	192.8	66.81	-634.0	6,582.6	423.5	68.7	354.82	1.194 Level 2			
17,300.0	10,842.8	17,241.1	10,676.1	194.7	195.7	66.83	-634.0	6,682.6	423.8	63.5	360.23	1.176 Level 2			
17,400.0	10,843.0	17,341.1	10,676.3	197.6	198.6	66.84	-634.0	6,782.6	424.0	58.4	365.65	1.160 Level 2			
17,500.0	10,843.2	17,441.1	10,676.4	200.5	201.5	66.86	-634.0	6,882.6	424.3	53.3	371.07	1.144 Level 2			
17,600.0	10,843.4	17,541.1	10,676.6	203.5	204.4	66.88	-634.0	6,982.6	424.6	48.1	376.49	1.128 Level 2			
17,700.0	10,843.5	17,641.1	10,676.8	206.4	207.3	66.89	-634.0	7,082.6	424.9	43.0	381.91	1.113 Level 2			
17,800.0	10,843.7	17,741.1	10,677.0	209.3	210.3	66.91	-634.0	7,182.6	425.2	37.8	387.33	1.098 Level 2			

CC - Min centre to center distance or covergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services

Anticollision Report



Company: Oasis Petroleum
Project: Indian Hills
Reference Site: 153N-100W-17/18
Site Error: 0.0 usft
Reference Well: Kline Federal 5300 41-18 12TX
Well Error: 2.0 usft
Reference Wellbore: Kline Federal 5300 41-18 12TX
Reference Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at: 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 14BX - Kline Federal 5300 41-18 14BX - Design #1													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis		Highside Toolface (°)	Distance		Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning	
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)		Offset Wellbore Centre +N/-S (usft)	Offset Wellbore Centre +E/-W (usft)						
17,900.0	10,843.9	17,841.1	10,677.1	212.2	213.2	66.93	-634.0	7,282.6	425.5	32.7	392.76	1.083	Level 2	
18,000.0	10,844.1	17,941.1	10,677.3	215.1	216.1	66.94	-634.0	7,382.6	425.8	27.6	398.19	1.089	Level 2	
18,100.0	10,844.2	18,041.1	10,677.5	218.0	219.0	66.96	-634.0	7,482.6	426.0	22.4	403.62	1.056	Level 2	
18,200.0	10,844.4	18,141.1	10,677.7	220.9	221.9	66.98	-634.0	7,582.6	426.3	17.3	409.05	1.042	Level 2	
18,300.0	10,844.6	18,241.1	10,677.8	223.8	224.8	66.99	-634.0	7,682.6	426.6	12.1	414.49	1.029	Level 2	
18,400.0	10,844.8	18,341.1	10,678.0	226.7	227.7	67.01	-634.0	7,782.6	426.9	7.0	419.92	1.017	Level 2	
18,500.0	10,844.9	18,441.1	10,678.2	229.7	230.6	67.02	-634.0	7,882.6	427.2	1.8	425.36	1.004	Level 2	
18,600.0	10,845.1	18,541.1	10,678.4	232.6	233.5	67.04	-634.0	7,982.6	427.5	-3.3	430.80	0.992	Level 1	
18,700.0	10,845.3	18,641.1	10,678.5	235.5	236.4	67.06	-634.0	8,082.6	427.7	-8.5	436.24	0.981	Level 1	
18,800.0	10,845.4	18,741.1	10,678.7	238.4	239.3	67.07	-634.0	8,182.6	428.0	-13.7	441.69	0.969	Level 1	
18,900.0	10,845.6	18,841.1	10,678.9	241.3	242.3	67.09	-634.0	8,282.6	428.3	-18.8	447.13	0.958	Level 1	
19,000.0	10,845.8	18,941.1	10,679.1	244.2	245.2	67.11	-634.0	8,382.6	428.6	-24.0	452.58	0.947	Level 1	
19,100.0	10,846.0	19,041.1	10,679.2	247.1	248.1	67.12	-634.0	8,482.6	428.9	-29.2	458.03	0.936	Level 1	
19,200.0	10,846.1	19,141.1	10,679.4	250.1	251.0	67.14	-634.0	8,582.6	429.2	-34.3	463.48	0.926	Level 1	
19,300.0	10,846.3	19,241.1	10,679.6	253.0	253.9	67.15	-634.0	8,682.6	429.4	-39.5	468.93	0.916	Level 1	
19,400.0	10,846.5	19,341.1	10,679.8	255.9	256.8	67.17	-634.0	8,782.6	429.7	-44.7	474.39	0.906	Level 1	
19,500.0	10,846.7	19,441.1	10,679.9	258.8	259.7	67.19	-634.0	8,882.6	430.0	-49.8	479.84	0.896	Level 1	
19,600.0	10,846.8	19,541.1	10,680.1	261.7	262.7	67.20	-634.0	8,982.6	430.3	-55.0	485.30	0.887	Level 1	
19,700.0	10,847.0	19,641.1	10,680.3	264.6	265.6	67.22	-634.0	9,082.6	430.6	-60.2	490.76	0.877	Level 1	
19,800.0	10,847.2	19,741.1	10,680.5	267.6	268.5	67.23	-634.0	9,182.6	430.9	-65.4	496.22	0.868	Level 1	
19,900.0	10,847.4	19,841.1	10,680.6	270.5	271.4	67.25	-634.0	9,282.6	431.1	-70.5	501.68	0.859	Level 1	
20,000.0	10,847.5	19,941.1	10,680.8	273.4	274.3	67.27	-634.0	9,382.6	431.4	-75.7	507.14	0.851	Level 1	
20,100.0	10,847.7	20,041.1	10,681.0	276.3	277.2	67.29	-634.0	9,482.6	431.7	-80.9	512.61	0.842	Level 1	
20,200.0	10,847.9	20,141.1	10,681.2	279.2	280.2	67.30	-634.0	9,582.6	432.0	-86.1	518.08	0.834	Level 1	
20,300.0	10,848.1	20,241.1	10,681.3	282.1	283.1	67.31	-634.0	9,682.6	432.3	-91.3	523.54	0.826	Level 1	
20,400.0	10,848.2	20,341.1	10,681.5	285.1	286.0	67.33	-634.0	9,782.6	432.6	-96.4	529.01	0.818	Level 1	
20,500.0	10,848.4	20,441.1	10,681.7	288.0	288.9	67.34	-634.0	9,882.6	432.9	-101.6	534.49	0.810	Level 1	
20,600.0	10,848.6	20,541.1	10,681.9	290.9	291.8	67.36	-634.0	9,982.6	433.1	-106.8	539.96	0.802	Level 1	
20,687.7	10,848.7	20,608.5	10,682.0	292.9	293.8	67.37	-634.0	10,050.0	433.3	-110.3	543.66	0.797	Level 1, ES, SF	



Ryan Directional Services Anticollision Report



Company: Oasis Petroleum
Project: Indian Hills
Reference Site: 153N-100W-17/18
Site Error: 0.0 usft
Reference Well: Kline Federal 5300 41-18 12TX
Well Error: 2.0 usft
Reference Wellbore: Kline Federal 5300 41-18 12TX
Reference Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at: 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 14BX - Kline Federal 5300 41-18 14BX - Design #2													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis		Highside Toolface (°)	Distance		Minimum Separation (usft)	Separation Factor	Warning			
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)		Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)				Between Centres (usft)	Between Ellipses (usft)	
0.0	0.0	0.0	0.0	2.0	2.0	-180.00	-65.9	0.0	65.9					
100.0	100.0	100.0	100.0	2.0	2.0	-180.00	-65.9	0.0	65.9	61.9		4.00	16.452	
200.0	200.0	200.0	200.0	2.0	2.0	-180.00	-65.9	0.0	65.9	61.8		4.05	16.273	
300.0	300.0	300.0	300.0	2.1	2.1	-180.00	-65.9	0.0	65.9	61.7		4.14	15.909	
400.0	400.0	400.0	400.0	2.1	2.1	-180.00	-65.9	0.0	65.9	61.6		4.28	15.396	
500.0	500.0	500.0	500.0	2.2	2.2	-180.00	-65.9	0.0	65.9	61.4		4.48	14.777	
600.0	600.0	600.0	600.0	2.3	2.3	-180.00	-65.9	0.0	65.9	61.2		4.67	14.094	
700.0	700.0	700.0	700.0	2.5	2.5	-180.00	-65.9	0.0	65.9	60.9		4.92	13.385	
800.0	800.0	800.0	800.0	2.6	2.6	-180.00	-65.9	0.0	65.9	60.7		5.20	12.678	
900.0	900.0	900.0	900.0	2.7	2.7	-180.00	-65.9	0.0	65.9	60.4		5.49	11.980	
1,000.0	1,000.0	1,000.0	1,000.0	2.9	2.9	-180.00	-65.9	0.0	65.9	60.1		5.81	11.336	
1,100.0	1,100.0	1,100.0	1,100.0	3.1	3.1	-180.00	-65.9	0.0	65.9	59.7		6.14	10.720	
1,200.0	1,200.0	1,200.0	1,200.0	3.2	3.2	-180.00	-65.9	0.0	65.9	59.4		6.49	10.146	
1,300.0	1,300.0	1,300.0	1,300.0	3.4	3.4	-180.00	-65.9	0.0	65.9	59.0		6.85	9.613	
1,400.0	1,400.0	1,400.0	1,400.0	3.6	3.6	-180.00	-65.9	0.0	65.9	58.6		7.22	9.121	
1,500.0	1,500.0	1,500.0	1,500.0	3.8	3.8	-180.00	-65.9	0.0	65.9	58.3		7.60	8.667	
1,600.0	1,600.0	1,600.0	1,600.0	4.0	4.0	-180.00	-65.9	0.0	65.9	57.9		7.99	8.248	
1,700.0	1,700.0	1,700.0	1,700.0	4.2	4.2	-180.00	-65.9	0.0	65.9	57.5		8.38	7.862	
1,800.0	1,800.0	1,800.0	1,800.0	4.4	4.4	-180.00	-65.9	0.0	65.9	57.1		8.78	7.506	
1,900.0	1,900.0	1,900.0	1,900.0	4.6	4.6	-180.00	-65.9	0.0	65.9	56.7		9.18	7.177	
2,000.0	2,000.0	2,000.0	2,000.0	4.8	4.8	-180.00	-65.9	0.0	65.9	56.3		9.58	6.872	
2,100.0	2,100.0	2,100.0	2,100.0	5.0	5.0	-180.00	-65.9	0.0	65.9	55.9		9.99	6.590	
2,166.7	2,166.7	2,166.7	2,166.7	5.1	5.1	-180.00	-65.9	0.0	65.9	55.6		10.27	6.413	
2,200.0	2,200.0	2,200.0	2,200.0	5.2	5.2	-180.00	-65.9	0.0	65.9	55.5		10.41	6.328 CC	
2,216.7	2,216.7	2,216.1	2,216.1	5.2	5.2	90.06	-65.9	0.0	65.9	55.5		10.47	6.298	
2,300.0	2,300.0	2,299.4	2,299.4	5.4	5.4	90.88	-66.7	0.0	66.7	55.9		10.75	6.201	
2,400.0	2,400.0	2,399.4	2,399.4	5.6	5.5	91.41	-67.5	0.0	67.6	56.5		11.10	6.088	
2,500.0	2,500.0	2,499.4	2,499.4	5.8	5.7	92.13	-68.4	0.0	68.5	57.0		11.45	5.979	
2,600.0	2,600.0	2,599.4	2,599.4	6.0	5.9	92.82	-69.3	0.0	69.4	57.6		11.81	5.874	
2,700.0	2,700.0	2,699.4	2,699.4	6.1	6.0	93.50	-70.1	0.0	70.3	58.1		12.17	5.773	
2,800.0	2,800.0	2,799.4	2,799.4	6.3	6.2	94.15	-71.0	0.0	71.2	58.7		12.54	5.677	
2,900.0	2,900.0	2,899.4	2,899.3	6.5	6.4	94.79	-71.9	0.0	72.2	59.2		12.92	5.584	
3,000.0	3,000.0	2,999.4	2,999.3	6.7	6.6	95.42	-72.8	0.0	73.1	59.8		13.30	5.496	
3,100.0	3,100.0	3,099.4	3,099.3	6.9	6.7	96.03	-73.6	0.0	74.1	60.4		13.69	5.411	
3,200.0	3,200.0	3,199.3	3,199.3	7.1	6.9	96.62	-74.5	0.0	75.0	60.9		14.07	5.330	
3,300.0	3,300.0	3,299.3	3,299.3	7.3	7.1	97.20	-75.4	0.0	76.0	61.5		14.47	5.253	
3,400.0	3,400.0	3,399.3	3,399.3	7.6	7.3	97.76	-76.3	0.0	77.0	62.1		14.86	5.179	
3,500.0	3,500.0	3,499.3	3,499.3	7.8	7.5	98.31	-77.1	0.0	78.0	62.7		15.26	5.108	
3,600.0	3,599.9	3,599.3	3,599.3	8.0	7.7	98.84	-78.0	0.0	78.9	63.3		15.68	5.041	
3,700.0	3,699.9	3,699.3	3,699.3	8.2	7.9	99.37	-78.9	0.0	79.9	63.9		16.07	4.976	
3,800.0	3,799.9	3,799.3	3,799.2	8.4	8.1	99.88	-79.7	0.0	81.0	64.5		16.47	4.915	
3,900.0	3,899.9	3,899.3	3,899.2	8.6	8.3	100.37	-80.6	0.0	82.0	65.1		16.88	4.856	
4,000.0	3,999.9	3,999.3	3,999.2	8.8	8.5	100.86	-81.5	0.0	83.0	65.7		17.29	4.799	
4,100.0	4,099.9	4,099.3	4,099.2	9.0	8.7	101.33	-82.4	0.0	84.0	66.3		17.70	4.745	
4,200.0	4,199.9	4,199.3	4,199.2	9.2	8.9	101.79	-83.2	0.0	85.0	66.9		18.12	4.694	
4,300.0	4,299.9	4,299.3	4,299.2	9.4	9.1	102.24	-84.1	0.0	86.1	67.5		18.53	4.644	
4,400.0	4,399.9	4,399.3	4,399.2	9.7	9.3	102.68	-85.0	0.0	87.1	68.2		18.95	4.597	
4,500.0	4,499.9	4,499.3	4,499.2	9.9	9.5	103.11	-85.9	0.0	88.2	68.8		19.37	4.551	
4,600.0	4,599.9	4,599.2	4,599.2	10.1	9.7	103.53	-86.7	0.0	89.2	69.4		19.79	4.507	
4,700.0	4,699.9	4,699.2	4,699.1	10.3	9.9	103.94	-87.6	0.0	90.3	70.0		20.21	4.466	
4,800.0	4,799.9	4,799.2	4,799.1	10.5	10.1	104.33	-88.5	0.0	91.3	70.7		20.64	4.425	
4,900.0	4,899.9	4,899.2	4,899.1	10.7	10.3	104.72	-89.3	0.0	92.4	71.3		21.06	4.387	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services Anticollision Report



Company: Oasis Petroleum
Project: Indian Hills
Reference Site: 153N-100W-17/18
Site Error: 0.0 usft
Reference Well: Kline Federal 5300 41-18 12TX
Well Error: 2.0 usft
Reference Wellbore: Kline Federal 5300 41-18 12TX
Reference Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at: 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 14BX - Kline Federal 5300 41-18 14BX - Design #2													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis			Distance				Separation Factor	Warning		
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)			Minimum Separation (usft)	
5,000.0	4,999.9	4,999.2	4,999.1	10.9	10.6	105.11	-90.2	0.0	93.5	72.0	21.49	4.350		
5,100.0	5,099.9	5,099.2	5,099.1	11.2	10.8	105.48	-91.1	0.0	94.5	72.6	21.91	4.314		
5,200.0	5,199.9	5,199.2	5,199.1	11.4	11.0	105.84	-92.0	0.0	95.6	73.3	22.34	4.280		
5,300.0	5,299.9	5,299.2	5,299.1	11.6	11.2	106.20	-92.8	0.0	96.7	73.9	22.77	4.246		
5,400.0	5,399.9	5,399.2	5,399.1	11.8	11.4	106.55	-93.7	0.0	97.8	74.6	23.20	4.215		
5,500.0	5,499.9	5,499.2	5,499.0	12.0	11.8	106.89	-94.6	0.0	98.9	75.2	23.63	4.184		
5,600.0	5,599.9	5,599.2	5,599.0	12.2	11.8	107.22	-95.5	0.0	99.9	75.9	24.06	4.154		
5,700.0	5,699.9	5,699.2	5,699.0	12.5	12.0	107.56	-96.3	0.0	101.0	76.5	24.49	4.126		
5,800.0	5,799.9	5,799.2	5,799.0	12.7	12.3	107.87	-97.2	0.0	102.1	77.2	24.92	4.098		
5,900.0	5,899.9	5,899.1	5,899.0	12.9	12.5	108.18	-98.1	0.0	103.2	77.9	25.35	4.072		
6,000.0	5,999.9	5,999.1	5,999.0	13.1	12.7	108.49	-98.9	0.0	104.3	78.5	25.79	4.046		
6,100.0	6,099.9	6,099.1	6,099.0	13.3	12.9	108.79	-99.8	0.0	105.4	79.2	26.22	4.021		
6,200.0	6,199.9	6,199.1	6,199.0	13.6	13.1	109.09	-100.7	0.0	106.5	79.9	26.65	3.997		
6,300.0	6,299.9	6,299.1	6,299.0	13.8	13.3	109.37	-101.6	0.0	107.7	80.6	27.09	3.974		
6,400.0	6,399.9	6,399.1	6,399.0	14.0	13.6	109.65	-102.4	0.0	108.8	81.2	27.53	3.952		
6,500.0	6,499.9	6,499.1	6,499.0	14.2	13.8	109.92	-103.3	0.0	109.9	81.9	27.96	3.930		
6,600.0	6,599.9	6,599.1	6,599.0	14.4	14.0	110.19	-104.2	0.0	111.0	82.6	28.40	3.909		
6,700.0	6,699.9	6,699.1	6,699.0	14.7	14.2	110.46	-105.1	0.0	112.1	83.3	28.83	3.889		
6,800.0	6,799.9	6,799.1	6,799.0	14.9	14.4	110.72	-105.9	0.0	113.3	84.0	29.27	3.869		
6,900.0	6,899.9	6,899.1	6,899.0	15.1	14.6	110.97	-106.8	0.0	114.4	84.7	29.71	3.850		
7,000.0	6,999.9	6,999.1	6,999.0	15.3	14.9	111.22	-107.7	0.0	115.5	85.4	30.15	3.831		
7,100.0	7,099.9	7,099.1	7,099.0	15.5	15.1	111.46	-108.5	0.0	116.6	86.1	30.59	3.813		
7,200.0	7,199.9	7,199.0	7,199.0	15.8	15.3	111.70	-109.4	0.0	117.8	86.7	31.02	3.796		
7,300.0	7,299.9	7,299.0	7,299.0	16.0	15.5	111.94	-110.3	0.0	118.9	87.4	31.46	3.779		
7,400.0	7,399.9	7,399.0	7,399.0	16.2	15.7	112.17	-111.2	0.0	120.0	88.1	31.90	3.763		
7,500.0	7,499.9	7,499.0	7,499.0	16.4	16.0	112.40	-112.0	0.0	121.2	88.8	32.34	3.747		
7,600.0	7,599.9	7,599.0	7,599.0	16.6	16.2	112.62	-112.9	0.0	122.3	89.5	32.78	3.731		
7,700.0	7,699.9	7,699.0	7,699.0	16.9	16.4	112.84	-113.8	0.0	123.5	90.2	33.22	3.716		
7,800.0	7,799.9	7,799.0	7,799.0	17.1	16.6	113.05	-114.7	0.0	124.6	90.9	33.66	3.701		
7,900.0	7,899.9	7,899.0	7,899.0	17.3	16.8	113.26	-115.5	0.0	125.8	91.6	34.11	3.687		
7,926.7	7,926.4	7,926.7	7,926.4	17.4	16.9	113.32	-115.8	0.0	126.1	91.8	34.22	3.683		
7,943.3	7,943.1	7,943.3	7,943.1	17.4	16.9	-156.66	-115.8	0.0	126.2	91.9	34.30	3.679		
8,000.0	7,999.9	8,000.0	7,999.9	17.5	17.1	-156.66	-115.8	0.0	126.2	91.6	34.52	3.664		
8,100.0	8,099.9	8,100.0	8,099.9	17.7	17.2	-156.66	-115.8	0.0	126.2	91.2	34.92	3.613		
8,200.0	8,199.9	8,200.0	8,199.9	17.9	17.4	-156.66	-115.8	0.0	126.2	90.8	35.32	3.572		
8,300.0	8,299.9	8,300.0	8,299.9	18.1	17.6	-156.66	-115.8	0.0	126.2	90.4	35.72	3.532		
8,400.0	8,399.9	8,400.0	8,399.9	18.4	17.8	-156.66	-115.8	0.0	126.2	90.0	36.12	3.493		
8,500.0	8,499.9	8,500.0	8,499.9	18.6	18.0	-156.66	-115.8	0.0	126.2	89.6	36.52	3.454		
8,600.0	8,599.9	8,600.0	8,599.9	18.8	18.2	-156.66	-115.8	0.0	126.2	89.2	36.93	3.416		
8,700.0	8,699.9	8,700.0	8,699.9	19.0	18.4	-156.66	-115.8	0.0	126.2	88.8	37.33	3.379		
8,800.0	8,799.9	8,800.0	8,799.9	19.2	18.6	-156.66	-115.8	0.0	126.2	88.4	37.74	3.343		
8,900.0	8,899.9	8,900.0	8,899.9	19.4	18.8	-156.66	-115.8	0.0	126.2	88.0	38.14	3.307		
9,000.0	8,999.9	9,000.0	8,999.9	19.6	19.0	-156.66	-115.8	0.0	126.2	87.6	38.55	3.273		
9,100.0	9,099.9	9,100.0	9,099.9	19.8	19.2	-156.66	-115.8	0.0	126.2	87.2	38.96	3.238		
9,200.0	9,199.9	9,200.0	9,199.9	20.0	19.4	-156.66	-115.8	0.0	126.2	86.8	39.37	3.206		
9,300.0	9,299.9	9,300.0	9,299.9	20.2	19.6	-156.66	-115.8	0.0	126.2	86.4	39.78	3.172		
9,400.0	9,399.9	9,400.0	9,399.9	20.5	19.8	-156.66	-115.8	0.0	126.2	86.0	40.19	3.139		
9,500.0	9,499.9	9,500.0	9,499.9	20.7	20.0	-156.66	-115.8	0.0	126.2	85.6	40.60	3.107		
9,600.0	9,599.9	9,600.0	9,599.9	20.9	20.2	-156.66	-115.8	0.0	126.2	85.1	41.01	3.076		
9,700.0	9,699.9	9,700.0	9,699.9	21.1	20.4	-156.66	-115.8	0.0	126.2	84.7	41.43	3.045		
9,800.0	9,799.9	9,800.0	9,799.9	21.3	20.6	-156.66	-115.8	0.0	126.2	84.3	41.84	3.015		
9,900.0	9,899.9	9,900.0	9,899.9	21.5	20.8	-156.66	-115.8	0.0	126.2	83.9	42.26	2.986		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services Anticollision Report



Company: Oasis Petroleum
Project: Indian Hills
Reference Site: 153N-100W-17/18
Site Error: 0.0 usft
Reference Well: Kline Federal 5300 41-18 12TX
Well Error: 2.0 usft
Reference Wellbore: Kline Federal 5300 41-18 12TX
Reference Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 14BX - Kline Federal 5300 41-18 14BX - Design #2														Offset Site Error:	0.0 usft
Survey Program: O-MWD														Offset Well Error:	2.0 usft
Reference		Offset		Semi-Major Axis		Highside Toolface (°)	Distance		Minimum Separation (usft)	Separation Factor	Warning				
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)		Offset Wellbore Centre +N-S (usft)	+E-W (usft)				Between Centres (usft)	Between Ellipses (usft)		
10,000.0	9,999.8	10,000.0	9,999.8	21.7	21.0	-156.86	-115.8	0.0	126.2	83.5	42.67	2.956			
10,100.0	10,099.8	10,100.0	10,099.8	21.9	21.2	-156.66	-115.8	0.0	126.2	83.1	43.09	2.928			
10,200.0	10,199.8	10,200.0	10,199.8	22.2	21.4	-156.66	-115.8	0.0	126.2	82.7	43.50	2.900			
10,282.1	10,261.8	10,262.1	10,261.8	22.3	21.5	-156.66	-115.8	0.0	120.2	82.4	43.76	2.883			
10,300.0	10,299.8	10,299.8	10,299.6	22.4	21.6	-156.73	-115.9	0.1	126.2	82.2	43.92	2.873			
10,354.8	10,354.6	10,353.4	10,352.9	22.5	21.7	-158.70	-117.9	4.0	126.5	82.4	44.18	2.865			
10,375.0	10,374.8	10,372.8	10,372.2	22.5	21.7	90.06	-119.2	6.8	126.8	82.6	44.26	2.866			
10,400.0	10,399.7	10,396.8	10,395.6	22.6	21.8	88.58	-121.4	11.1	127.5	83.1	44.38	2.873			
10,425.0	10,424.5	10,420.6	10,418.7	22.7	21.9	87.07	-124.1	16.4	128.3	83.9	44.47	2.886			
10,450.0	10,449.2	10,444.2	10,441.2	22.7	21.9	85.63	-127.2	22.7	128.4	84.8	44.58	2.903			
10,475.0	10,473.5	10,467.7	10,463.3	22.8	22.0	84.23	-130.9	30.0	130.7	86.0	44.69	2.924			
10,500.0	10,497.6	10,491.1	10,484.8	22.8	22.0	82.88	-135.0	38.1	132.2	87.4	44.79	2.950			
10,525.0	10,521.2	10,514.3	10,505.7	22.9	22.1	81.59	-139.5	47.1	133.8	88.9	44.90	2.981			
10,550.0	10,544.4	10,537.4	10,525.9	23.0	22.1	80.37	-144.5	57.0	135.6	90.6	44.99	3.014			
10,575.0	10,567.1	10,560.3	10,545.6	23.0	22.2	79.20	-149.8	67.7	137.6	92.5	45.09	3.052			
10,600.0	10,589.1	10,583.2	10,564.5	23.1	22.3	78.11	-155.5	79.1	139.7	94.5	45.18	3.093			
10,625.0	10,610.6	10,605.9	10,582.7	23.2	22.3	77.09	-161.7	91.3	141.9	96.7	45.26	3.138			
10,650.0	10,631.3	10,628.5	10,600.1	23.3	22.4	76.13	-168.1	104.2	144.3	98.9	45.35	3.182			
10,675.0	10,651.3	10,650.0	10,616.0	23.4	22.5	75.27	-174.6	117.1	146.7	101.3	45.43	3.230			
10,700.0	10,670.5	10,673.5	10,632.6	23.5	22.6	74.43	-182.0	131.9	149.3	103.7	45.52	3.279			
10,725.0	10,688.8	10,695.8	10,647.7	23.6	22.7	73.69	-189.4	146.8	151.9	108.2	45.61	3.329			
10,750.0	10,706.2	10,718.1	10,661.9	23.7	22.8	73.01	-197.1	161.9	154.5	108.8	45.72	3.380			
10,775.0	10,722.6	10,740.2	10,675.3	23.8	23.0	72.40	-205.0	177.7	157.2	111.4	45.83	3.430			
10,800.0	10,738.0	10,762.3	10,687.7	24.0	23.1	71.85	-213.2	194.0	160.0	114.0	45.97	3.480			
10,825.0	10,752.4	10,784.4	10,699.3	24.2	23.3	71.37	-221.6	210.8	162.7	116.6	46.13	3.528			
10,850.0	10,765.7	10,806.3	10,710.0	24.4	23.4	70.94	-230.2	227.9	165.5	119.2	46.32	3.574			
10,875.0	10,777.8	10,828.3	10,719.8	24.6	23.6	70.58	-239.0	245.5	168.3	121.8	46.54	3.617			
10,900.0	10,788.8	10,850.0	10,728.5	24.8	23.8	70.26	-247.9	263.2	171.1	124.3	46.79	3.658			
10,925.0	10,798.6	10,872.0	10,736.5	25.0	24.0	70.01	-257.1	281.6	173.9	126.8	47.08	3.694			
10,950.0	10,807.2	10,893.8	10,743.4	25.3	24.2	69.80	-266.4	300.0	176.7	129.3	47.42	3.727			
10,975.0	10,814.5	10,915.6	10,749.4	25.6	24.4	69.64	-275.8	318.7	179.5	131.7	47.79	3.755			
11,000.0	10,820.6	10,937.3	10,754.5	25.8	24.7	69.54	-285.3	337.7	182.2	134.0	48.21	3.779			
11,025.0	10,825.4	10,959.1	10,758.5	26.1	24.9	69.47	-294.9	356.8	184.9	136.2	48.68	3.799			
11,050.0	10,828.9	10,980.8	10,761.6	26.5	25.2	69.46	-304.5	376.0	187.6	138.4	49.19	3.814			
11,075.0	10,831.1	11,002.6	10,763.7	26.8	25.5	69.48	-314.3	395.4	190.2	140.5	49.74	3.824			
11,100.0	10,832.0	11,025.0	10,764.8	27.1	25.8	69.55	-324.3	415.4	192.8	142.5	50.35	3.829			
11,104.0	10,832.0	11,027.8	10,764.9	27.2	25.8	69.55	-325.5	417.9	193.2	142.8	50.43	3.831			
11,200.0	10,832.2	11,127.7	10,765.1	28.6	27.3	70.73	-369.1	507.8	205.0	151.7	53.31	3.845			
11,300.0	10,832.4	11,233.9	10,765.3	30.3	29.1	72.00	-411.9	605.0	219.7	163.0	56.67	3.876			
11,400.0	10,832.6	11,340.4	10,765.5	32.2	31.0	73.27	-451.2	704.0	236.7	178.4	60.29	3.925			
11,500.0	10,832.7	11,447.2	10,765.7	34.1	33.2	74.50	-486.8	804.6	256.1	192.0	64.07	3.997			
11,600.0	10,832.9	11,554.3	10,765.9	36.2	35.5	75.67	-516.8	906.8	277.8	209.9	67.90	4.091			
11,680.4	10,833.1	11,640.7	10,766.1	38.0	37.5	76.58	-541.8	990.1	297.0	226.0	70.97	4.185			
11,700.0	10,833.1	11,661.8	10,766.1	38.4	38.0	76.80	-547.0	1,010.5	301.8	228.9	71.90	4.197			
11,800.0	10,833.3	11,770.8	10,766.3	40.7	40.5	77.83	-571.7	1,116.7	324.2	247.4	76.78	4.222			
11,800.0	10,833.4	11,881.6	10,766.5	43.0	43.2	78.59	-592.5	1,225.5	342.9	261.1	81.84	4.190			
12,000.0	10,833.6	11,993.9	10,766.7	45.4	45.9	79.14	-609.4	1,336.5	358.0	270.9	87.07	4.111			
12,100.0	10,833.8	12,107.4	10,766.9	47.9	48.8	79.52	-622.0	1,449.3	369.2	276.8	92.43	3.994			
12,200.0	10,834.0	12,221.8	10,767.1	50.4	51.7	79.76	-630.2	1,563.4	378.5	278.8	97.89	3.846			
12,300.0	10,834.1	12,336.6	10,767.3	53.0	54.8	79.87	-633.8	1,678.2	379.9	276.5	103.42	3.673			
12,400.0	10,834.3	12,440.9	10,767.5	55.5	57.2	79.88	-634.0	1,782.5	380.4	271.7	108.72	3.499			
12,500.0	10,834.5	12,540.9	10,767.6	58.2	59.8	79.89	-634.0	1,882.5	380.7	266.7	113.97	3.340			

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services Anticollision Report



Company: Oasis Petroleum
Project: Indian Hills
Reference Site: 153N-100W-17/18
Site Error: 0.0 usft
Reference Well: Kline Federal 5300 41-18 12TX
Well Error: 2.0 usft
Reference Wellbore: Kline Federal 5300 41-18 12TX
Reference Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at: 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 14BX - Kline Federal 5300 41-18 14BX - Design #2													Offset Site Error: 0.0 usft	
Survey Program: 0-MWD													Offset Well Error: 2.0 usft	
Reference		Offset		Semi-Major Axis			Distance						Warning	
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
12,600.0	10,834.7	12,640.9	10,767.8	60.8	82.4	79.90	-634.0	1,982.5	381.0	261.7	119.27	3.194		
12,700.0	10,834.8	12,740.9	10,768.0	63.5	85.1	79.90	-634.0	2,082.5	381.3	256.7	124.60	3.060		
12,800.0	10,835.0	12,840.9	10,768.2	66.2	87.7	79.91	-634.0	2,182.5	381.6	251.6	129.98	2.936		
12,900.0	10,835.2	12,940.9	10,768.3	68.9	90.4	79.92	-634.0	2,282.5	381.9	246.5	135.39	2.821		
13,000.0	10,835.4	13,040.9	10,768.5	71.7	93.1	79.93	-634.0	2,382.5	382.2	241.4	140.83	2.714		
13,100.0	10,835.5	13,140.9	10,768.7	74.4	95.9	79.94	-634.0	2,482.5	382.5	236.2	146.29	2.615		
13,200.0	10,835.7	13,240.9	10,768.9	77.2	98.6	79.95	-634.0	2,582.5	382.8	231.0	151.78	2.522		
13,300.0	10,835.9	13,340.9	10,769.0	79.9	101.3	79.95	-634.0	2,682.5	383.1	225.8	157.29	2.436		
13,400.0	10,836.1	13,440.9	10,769.2	82.7	104.1	79.96	-634.0	2,782.5	383.4	220.6	162.82	2.355		
13,500.0	10,836.2	13,540.9	10,769.4	85.5	106.9	79.97	-634.0	2,882.5	383.7	215.4	168.36	2.279		
13,600.0	10,836.4	13,640.9	10,769.6	88.3	109.7	79.98	-634.0	2,982.5	384.0	210.1	173.93	2.208		
13,700.0	10,836.6	13,740.9	10,769.7	91.1	112.5	79.99	-634.0	3,082.5	384.3	204.8	179.50	2.141		
13,800.0	10,836.7	13,840.9	10,769.9	94.0	115.3	79.99	-634.0	3,182.5	384.6	199.5	185.09	2.078		
13,900.0	10,836.9	13,940.9	10,770.1	96.8	118.1	80.00	-634.0	3,282.5	384.9	194.2	190.69	2.019		
14,000.0	10,837.1	14,040.9	10,770.3	99.6	120.9	80.01	-634.0	3,382.5	385.2	188.9	196.30	1.963		
14,100.0	10,837.3	14,140.9	10,770.4	102.5	123.7	80.02	-634.0	3,482.5	385.5	183.6	201.92	1.909		
14,200.0	10,837.4	14,240.9	10,770.6	105.3	126.5	80.03	-634.0	3,582.5	385.8	178.3	207.55	1.859		
14,300.0	10,837.6	14,340.9	10,770.8	108.2	129.4	80.03	-634.0	3,682.5	386.1	173.0	213.19	1.811		
14,400.0	10,837.8	14,440.9	10,771.0	111.0	132.2	80.04	-634.0	3,782.5	386.5	167.6	218.83	1.766		
14,500.0	10,838.0	14,540.9	10,771.1	113.9	135.1	80.05	-634.0	3,882.5	386.8	162.3	224.48	1.723		
14,600.0	10,838.1	14,640.9	10,771.3	116.7	137.9	80.06	-634.0	3,982.5	387.1	156.9	230.14	1.682		
14,700.0	10,838.3	14,740.9	10,771.5	119.6	140.8	80.07	-634.0	4,082.5	387.4	151.6	235.81	1.643		
14,800.0	10,838.5	14,840.9	10,771.7	122.5	143.6	80.07	-634.0	4,182.5	387.7	146.2	241.48	1.605		
14,900.0	10,838.7	14,940.9	10,771.8	125.3	146.5	80.08	-634.0	4,282.5	388.0	140.8	247.15	1.570		
15,000.0	10,838.8	15,040.9	10,772.0	128.2	149.3	80.09	-634.0	4,382.5	388.3	135.4	252.84	1.536		
15,100.0	10,839.0	15,140.9	10,772.2	131.1	152.2	80.10	-634.0	4,482.5	388.6	130.1	258.52	1.503		
15,200.0	10,839.2	15,240.9	10,772.4	133.9	155.1	80.11	-634.0	4,582.5	388.9	124.7	264.21	1.472 Level 3		
15,300.0	10,839.4	15,340.9	10,772.5	136.8	157.9	80.11	-634.0	4,682.5	389.2	119.3	269.91	1.442 Level 3		
15,400.0	10,839.5	15,440.9	10,772.7	139.7	160.8	80.12	-634.0	4,782.5	389.5	113.9	275.61	1.413 Level 3		
15,500.0	10,839.7	15,540.9	10,772.9	142.6	163.7	80.13	-634.0	4,882.5	389.8	108.5	281.31	1.386 Level 3		
15,600.0	10,839.9	15,640.9	10,773.1	145.5	166.6	80.14	-634.0	4,982.5	390.1	103.1	287.02	1.359 Level 3		
15,700.0	10,840.1	15,740.9	10,773.2	148.4	169.5	80.14	-634.0	5,082.5	390.4	97.7	292.73	1.334 Level 3		
15,800.0	10,840.2	15,840.9	10,773.4	151.3	172.3	80.15	-634.0	5,182.5	390.7	92.3	298.44	1.309 Level 3		
15,900.0	10,840.4	15,940.9	10,773.6	154.1	175.2	80.16	-634.0	5,282.4	391.0	86.9	304.15	1.286 Level 3		
16,000.0	10,840.6	16,040.9	10,773.8	157.0	178.1	80.17	-634.0	5,382.4	391.3	81.4	309.87	1.263 Level 3		
16,100.0	10,840.8	16,140.9	10,773.9	159.9	181.0	80.18	-634.0	5,482.4	391.6	76.0	315.59	1.241 Level 2		
16,200.0	10,840.9	16,240.9	10,774.1	162.8	183.9	80.18	-634.0	5,582.4	391.9	70.6	321.32	1.220 Level 2		
16,300.0	10,841.1	16,340.9	10,774.3	165.7	186.8	80.19	-634.0	5,682.4	392.2	65.2	327.04	1.199 Level 2		
16,400.0	10,841.3	16,440.9	10,774.5	168.6	189.7	80.20	-634.0	5,782.4	392.5	59.8	332.77	1.180 Level 2		
16,500.0	10,841.4	16,540.9	10,774.6	171.5	192.6	80.21	-634.0	5,882.4	392.8	54.3	338.50	1.161 Level 2		
16,600.0	10,841.6	16,640.9	10,774.8	174.4	195.5	80.21	-634.0	5,982.4	393.1	48.9	344.23	1.142 Level 2		
16,700.0	10,841.8	16,740.9	10,775.0	177.3	198.4	80.22	-634.0	6,082.4	393.4	43.5	349.97	1.124 Level 2		
16,800.0	10,842.0	16,840.9	10,775.2	180.2	201.2	80.23	-634.0	6,182.4	393.7	38.0	355.70	1.107 Level 2		
16,900.0	10,842.1	16,940.9	10,775.3	183.1	204.1	80.24	-634.0	6,282.4	394.1	32.6	361.44	1.090 Level 2		
17,000.0	10,842.3	17,040.9	10,775.5	186.0	207.0	80.25	-634.0	6,382.4	394.4	27.2	367.18	1.074 Level 2		
17,100.0	10,842.5	17,140.9	10,775.7	188.9	209.9	80.25	-634.0	6,482.4	394.7	21.7	372.93	1.058 Level 2		
17,200.0	10,842.7	17,240.9	10,775.9	191.8	212.8	80.26	-634.0	6,582.4	395.0	16.3	378.67	1.043 Level 2		
17,300.0	10,842.8	17,340.9	10,776.0	194.7	215.7	80.27	-634.0	6,682.4	395.3	10.9	384.41	1.028 Level 2		
17,400.0	10,843.0	17,440.9	10,776.2	197.6	218.6	80.28	-634.0	6,782.4	395.6	5.4	390.16	1.014 Level 2		
17,500.0	10,843.2	17,540.9	10,776.4	200.6	221.5	80.28	-634.0	6,882.4	395.9	0.0	395.91	1.000 Level 1		
17,600.0	10,843.4	17,640.9	10,776.5	203.5	224.4	80.29	-634.0	6,982.4	396.2	-5.5	401.66	0.986 Level 1		
17,700.0	10,843.5	17,740.9	10,776.7	206.4	227.3	80.30	-634.0	7,082.4	396.5	-10.9	407.41	0.973 Level 1		

CC - Min centre to center distance or covergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services Anticollision Report



Company: Oasis Petroleum
Project: Indian Hills
Reference Site: 153N-100W-17/18
Site Error: 0.0 usft
Reference Well: Kline Federal 5300 41-18 12TX
Well Error: 2.0 usft
Reference Wellbore: Kline Federal 5300 41-18 12TX
Reference Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at: 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 14BX - Kline Federal 5300 41-18 14BX - Design #2										Offset Site Error: 0.0 usft			
Survey Program: 0-MWD										Offset Well Error: 2.0 usft			
Reference		Offset		Semi Major Axis		Highside Toolface (°)	Offset Wellbore Centre		Distance		Minimum Separation (usft)	Separation Factor	Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)		+N-S (usft)	+E-W (usft)	Between Centres (usft)	Between Ellipses (usft)			
17,800.0	10,843.7	17,840.9	10,776.9	209.3	210.3	80.31	-634.0	7,182.4	398.8	-16.4	413.16	0.960 Level 1	
17,900.0	10,843.9	17,940.9	10,777.1	212.2	213.2	80.31	-634.0	7,282.4	397.1	-21.8	418.91	0.948 Level 1	
18,000.0	10,844.1	18,040.9	10,777.2	215.1	216.1	80.32	-634.0	7,382.4	397.4	-27.3	424.67	0.936 Level 1	
18,100.0	10,844.2	18,140.9	10,777.4	218.0	219.0	80.33	-634.0	7,482.4	397.7	-32.7	430.42	0.924 Level 1	
18,200.0	10,844.4	18,240.9	10,777.6	220.9	221.9	80.34	-634.0	7,582.4	398.0	-38.2	436.18	0.912 Level 1	
18,300.0	10,844.6	18,340.9	10,777.8	223.8	224.8	80.34	-634.0	7,682.4	398.3	-43.6	441.94	0.901 Level 1	
18,400.0	10,844.8	18,440.9	10,777.9	226.7	227.7	80.35	-634.0	7,782.4	398.6	-49.1	447.70	0.890 Level 1	
18,500.0	10,844.9	18,540.9	10,778.1	229.7	230.6	80.36	-634.0	7,882.4	398.9	-54.5	453.46	0.880 Level 1	
18,600.0	10,845.1	18,640.9	10,778.3	232.6	233.5	80.37	-634.0	7,982.4	399.2	-60.0	459.22	0.869 Level 1	
18,700.0	10,845.3	18,740.9	10,778.5	235.5	236.5	80.37	-634.0	8,082.4	399.5	-65.5	464.98	0.859 Level 1	
18,800.0	10,845.4	18,840.9	10,778.6	238.4	239.4	80.38	-634.0	8,182.4	399.8	-70.9	470.74	0.849 Level 1	
18,900.0	10,845.6	18,940.9	10,778.8	241.3	242.3	80.39	-634.0	8,282.4	400.1	-76.4	476.51	0.840 Level 1	
19,000.0	10,845.8	19,040.9	10,779.0	244.2	245.2	80.40	-634.0	8,382.4	400.4	-81.8	482.27	0.830 Level 1	
19,100.0	10,846.0	19,140.9	10,779.2	247.1	248.1	80.40	-634.0	8,482.4	400.7	-87.3	488.04	0.821 Level 1	
19,200.0	10,846.1	19,240.9	10,779.3	250.1	251.0	80.41	-634.0	8,582.4	401.0	-92.8	493.81	0.812 Level 1	
19,300.0	10,846.3	19,340.9	10,779.5	253.0	253.9	80.42	-634.0	8,682.4	401.3	-98.2	499.57	0.803 Level 1	
19,400.0	10,846.5	19,440.9	10,779.7	255.9	256.8	80.43	-634.0	8,782.4	401.7	-103.7	505.34	0.795 Level 1	
19,500.0	10,846.7	19,540.9	10,779.9	258.8	259.8	80.43	-634.0	8,882.4	402.0	-109.2	511.11	0.786 Level 1	
19,600.0	10,846.8	19,640.9	10,780.0	261.7	262.7	80.44	-634.0	8,982.4	402.3	-114.6	516.88	0.778 Level 1	
19,700.0	10,847.0	19,740.9	10,780.2	264.6	265.6	80.45	-634.0	9,082.4	402.6	-120.1	522.65	0.770 Level 1	
19,800.0	10,847.2	19,840.9	10,780.4	267.6	268.5	80.46	-634.0	9,182.4	402.9	-125.6	528.42	0.762 Level 1	
19,900.0	10,847.4	19,940.9	10,780.6	270.5	271.4	80.46	-634.0	9,282.4	403.2	-131.0	534.19	0.755 Level 1	
20,000.0	10,847.5	20,040.9	10,780.7	273.4	274.3	80.47	-634.0	9,382.4	403.5	-136.5	539.97	0.747 Level 1	
20,100.0	10,847.7	20,140.9	10,780.9	276.3	277.3	80.48	-634.0	9,482.4	403.8	-142.0	545.74	0.740 Level 1	
20,200.0	10,847.9	20,240.9	10,781.1	279.2	280.2	80.48	-634.0	9,582.4	404.1	-147.4	551.51	0.733 Level 1	
20,300.0	10,848.1	20,340.9	10,781.3	282.1	283.1	80.49	-634.0	9,682.4	404.4	-152.9	557.29	0.726 Level 1	
20,400.0	10,848.2	20,440.9	10,781.4	285.1	286.0	80.50	-634.0	9,782.4	404.7	-158.4	563.06	0.719 Level 1	
20,500.0	10,848.4	20,540.9	10,781.6	288.0	288.9	80.51	-634.0	9,882.4	405.0	-163.8	568.84	0.712 Level 1	
20,600.0	10,848.6	20,640.9	10,781.8	290.9	291.8	80.51	-634.0	9,982.4	405.3	-169.3	574.61	0.705 Level 1	
20,667.7	10,848.7	20,708.4	10,781.9	292.9	293.8	80.52	-634.0	10,050.0	405.5	-173.0	578.52	0.701 Level 1, ES, SF	



Ryan Directional Services

Anticollision Report



Company: Oasis Petroleum
Project: Indian Hills
Reference Site: 153N-100W-17/18
Site Error: 0.0 usft
Reference Well: Kline Federal 5300 41-18 12TX
Well Error: 2.0 usft
Reference Wellbore: Kline Federal 5300 41-18 12TX
Reference Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 9T - Kline Federal 5300 41-18 9T - Design #1													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	2.0 usft
Reference														
Measured		Vertical		Measured		Vertical		Semi Major Axis		Highside		Offset Wellbore Centre		Warning
Depth	(usft)	Depth	(usft)	Depth	(usft)	Depth	(usft)	Reference	Offset	Toolface	(°)	+N/-S	+E/-W	
(usft)	(usft)	(usft)	(usft)	(usft)	(usft)	(usft)	(usft)	(usft)	(usft)	(usft)	(usft)	(usft)	(usft)	
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	2.0	2.0	-0.39		99.3	-0.7	99.3
100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	2.0	2.0	-0.39		99.3	-0.7	99.3
200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	2.0	2.0	-0.39		99.3	-0.7	99.3
300.0	300.0	300.0	300.0	300.0	300.0	300.0	300.0	2.1	2.1	-0.39		99.3	-0.7	99.3
400.0	400.0	400.0	400.0	400.0	400.0	400.0	400.0	2.1	2.1	-0.39		99.3	-0.7	99.3
500.0	500.0	500.0	500.0	500.0	500.0	500.0	500.0	2.2	2.2	-0.39		99.3	-0.7	99.3
600.0	600.0	600.0	600.0	600.0	600.0	600.0	600.0	2.3	2.3	-0.39		99.3	-0.7	99.3
700.0	700.0	700.0	700.0	700.0	700.0	700.0	700.0	2.5	2.5	-0.39		99.3	-0.7	99.3
800.0	800.0	800.0	800.0	800.0	800.0	800.0	800.0	2.6	2.6	-0.39		99.3	-0.7	99.3
900.0	900.0	900.0	900.0	900.0	900.0	900.0	900.0	2.7	2.7	-0.39		99.3	-0.7	99.3
1,000.0	1,000.0	1,000.0	1,000.0	1,000.0	1,000.0	1,000.0	1,000.0	2.9	2.9	-0.39		99.3	-0.7	99.3
1,100.0	1,100.0	1,100.0	1,100.0	1,100.0	1,100.0	1,100.0	1,100.0	3.1	3.1	-0.39		99.3	-0.7	99.3
1,200.0	1,200.0	1,200.0	1,200.0	1,200.0	1,200.0	1,200.0	1,200.0	3.2	3.2	-0.39		99.3	-0.7	99.3
1,300.0	1,300.0	1,300.0	1,300.0	1,300.0	1,300.0	1,300.0	1,300.0	3.4	3.4	-0.39		99.3	-0.7	99.3
1,400.0	1,400.0	1,400.0	1,400.0	1,400.0	1,400.0	1,400.0	1,400.0	3.6	3.6	-0.39		99.3	-0.7	99.3
1,500.0	1,500.0	1,500.0	1,500.0	1,500.0	1,500.0	1,500.0	1,500.0	3.8	3.8	-0.39		99.3	-0.7	99.3
1,600.0	1,600.0	1,600.0	1,600.0	1,600.0	1,600.0	1,600.0	1,600.0	4.0	4.0	-0.39		99.3	-0.7	99.3
1,700.0	1,700.0	1,700.0	1,700.0	1,700.0	1,700.0	1,700.0	1,700.0	4.2	4.2	-0.39		99.3	-0.7	99.3
1,800.0	1,800.0	1,800.0	1,800.0	1,800.0	1,800.0	1,800.0	1,800.0	4.4	4.4	-0.39		99.3	-0.7	99.3
1,900.0	1,900.0	1,900.0	1,900.0	1,900.0	1,900.0	1,900.0	1,900.0	4.6	4.6	-0.39		99.3	-0.7	99.3
2,000.0	2,000.0	2,000.0	2,000.0	2,000.0	2,000.0	2,000.0	2,000.0	4.8	4.8	-0.39		99.3	-0.7	99.3
2,100.0	2,100.0	2,100.0	2,100.0	2,100.0	2,100.0	2,100.0	2,100.0	5.0	5.0	-0.39		99.3	-0.7	99.3
2,200.0	2,200.0	2,200.0	2,200.0	2,200.0	2,200.0	2,200.0	2,200.0	5.2	5.2	-0.39		99.3	-0.7	99.3
2,216.7	2,216.7	2,216.7	2,216.7	2,216.7	2,216.7	2,216.7	2,216.7	5.2	5.2	-90.43		99.4	-0.7	99.4
2,300.0	2,300.0	2,299.1	2,299.1	2,299.1	2,299.1	2,299.1	2,299.1	5.4	5.4	-80.84		100.1	-0.7	100.1
2,400.0	2,400.0	2,399.1	2,399.1	2,399.1	2,399.1	2,399.1	2,399.1	5.6	5.6	-91.33		101.0	-0.7	101.0
2,500.0	2,500.0	2,499.1	2,499.1	2,499.1	2,499.1	2,499.1	2,499.1	5.8	5.8	-91.81		101.8	-0.7	101.8
2,600.0	2,600.0	2,599.1	2,599.1	2,599.1	2,599.1	2,599.1	2,599.1	6.0	6.0	-92.28		102.7	-0.7	102.8
2,700.0	2,700.0	2,699.1	2,699.1	2,699.1	2,699.1	2,699.1	2,699.1	6.1	6.3	-92.74		103.6	-0.7	103.7
2,800.0	2,800.0	2,799.1	2,799.1	2,799.1	2,799.1	2,799.1	2,799.1	6.3	6.5	-93.20		104.4	-0.7	104.6
2,900.0	2,900.0	2,899.1	2,899.1	2,899.1	2,899.1	2,899.1	2,899.1	6.5	6.7	-93.64		105.3	-0.7	105.5
3,000.0	3,000.0	2,999.1	2,999.1	2,999.1	2,999.1	2,999.1	2,999.1	6.7	6.9	-94.08		106.2	-0.7	106.5
3,100.0	3,100.0	3,099.1	3,099.1	3,099.1	3,099.1	3,099.1	3,099.1	6.9	7.1	-94.51		107.1	-0.7	107.4
3,200.0	3,200.0	3,199.1	3,199.1	3,199.1	3,199.1	3,199.1	3,199.1	7.1	7.3	-94.94		107.9	-0.7	108.3
3,300.0	3,300.0	3,299.1	3,299.1	3,299.1	3,299.1	3,299.1	3,299.1	7.3	7.5	-95.35		108.8	-0.7	109.3
3,400.0	3,400.0	3,399.1	3,399.1	3,399.1	3,399.1	3,399.1	3,399.1	7.6	7.8	-95.78		109.7	-0.7	110.2
3,500.0	3,500.0	3,499.1	3,499.1	3,499.1	3,499.1	3,499.1	3,499.1	7.8	8.0	-96.17		110.6	-0.7	111.2
3,600.0	3,599.9	3,599.1	3,599.1	3,599.1	3,599.1	3,599.1	3,599.1	8.0	8.2	-96.58		111.4	-0.7	112.2
3,700.0	3,699.9	3,699.1	3,699.1	3,699.1	3,699.1	3,699.1	3,699.1	8.2	8.4	-96.95		112.3	-0.7	113.1
3,800.0	3,799.9	3,799.1	3,799.1	3,799.1	3,799.1	3,799.1	3,799.1	8.4	8.6	-97.33		113.2	-0.7	114.1
3,900.0	3,899.9	3,899.1	3,899.1	3,899.1	3,899.1	3,899.1	3,899.1	8.6	8.9	-97.71		114.0	-0.7	115.1
4,000.0	3,999.9	3,999.1	3,999.1	3,999.1	3,999.1	3,999.1	3,999.1	8.8	9.1	-98.08		114.9	-0.7	116.1
4,100.0	4,099.9	4,099.1	4,099.1	4,099.1	4,099.1	4,099.1	4,099.1	9.0	9.3	-98.44		115.8	-0.7	117.1
4,200.0	4,199.9	4,199.1	4,199.1	4,199.1	4,199.1	4,199.1	4,199.1	9.2	9.5	-98.79		116.7	-0.7	118.1
4,300.0	4,299.9	4,299.1	4,299.1	4,299.1	4,299.1	4,299.1	4,299.1	9.4	9.7	-99.15		117.5	-0.7	119.1
4,400.0	4,399.9	4,399.1	4,399.1	4,399.1	4,399.1	4,399.1	4,399.1	9.7	10.0	-99.49		118.4	-0.7	120.1
4,500.0	4,499.9	4,499.1	4,499.1	4,499.1	4,499.1	4,499.1	4,499.1	9.9	10.2	-99.83		119.3	-0.7	121.1
4,600.0	4,599.9	4,599.1	4,599.1	4,599.1	4,599.1	4,599.1	4,599.1	10.1	10.4	-100.16		120.2	-0.7	122.1
4,700.0	4,699.9	4,699.1	4,699.1	4,699.1	4,699.1	4,699.1	4,699.1	10.3	10.6	-100.49		121.0	-0.7	123.1
4,800.0	4,799.9	4,799.1	4,799.1	4,799.1	4,799.1	4,799.1	4,799.1	10.5	10.8	-100.81		121.9	-0.7	124.1
4,900.0	4,899.9	4,899.1	4,899.1	4,899.1	4,899.1	4,899.1	4,899.1	10.7	11.1	-101.13		122.8	-0.7	125.1
5,000.0	4,999.9	4,999.1	4,999.1	4,999.1	4,999.1	4,999.1	4,999.1	10.9	11.3	-101.44		123.6	-0.7	126.2

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services Anticollision Report



Company: Oasis Petroleum
Project: Indian Hills
Reference Site: 153N-100W-17/18
Site Error: 0.0 usft
Reference Well: Kline Federal 5300 41-18 12TX
Well Error: 2.0 usft
Reference Wellbore: Kline Federal 5300 41-18 12TX
Reference Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at: 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 9 T - Kline Federal 5300 41-18 9 T - Design #1														Offset Site Error:	0.0 usft
Survey Program: 0-MWD														Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis		Highside Toolface (°)	Distance		Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning		
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)		+N/-S (usft)	+E/-W (usft)							
5,100.0	5,099.9	5,099.9	5,099.8	11.2	11.5	-101.75	124.5	-0.7	127.2	104.5	22.64	5.617			
5,200.0	5,199.9	5,199.9	5,199.8	11.4	11.7	-102.05	125.4	-0.7	128.2	105.1	23.08	5.555			
5,300.0	5,299.9	5,299.9	5,299.8	11.6	11.9	-102.35	126.3	-0.7	129.3	105.7	23.52	5.496			
5,400.0	5,399.9	5,399.9	5,399.8	11.8	12.2	-102.64	127.1	-0.7	130.3	106.3	23.95	5.439			
5,500.0	5,499.9	5,499.9	5,499.8	12.0	12.4	-102.93	128.0	-0.7	131.3	106.9	24.40	5.384			
5,600.0	5,599.9	5,599.9	5,599.7	12.2	12.6	-103.22	128.9	-0.7	132.4	107.6	24.84	5.330			
5,700.0	5,699.9	5,699.9	5,699.7	12.5	12.8	-103.49	129.8	-0.7	133.4	108.2	25.28	5.279			
5,800.0	5,799.9	5,799.9	5,799.7	12.7	13.0	-103.77	130.8	-0.7	134.5	108.8	25.72	5.230			
5,900.0	5,899.9	5,899.9	5,899.7	12.9	13.3	-104.04	131.5	-0.7	135.6	109.4	26.16	5.182			
6,000.0	5,999.9	5,999.9	5,999.7	13.1	13.5	-104.31	132.4	-0.7	136.6	110.0	26.60	5.136			
6,100.0	6,099.9	6,099.9	6,099.7	13.3	13.7	-104.57	133.2	-0.7	137.7	110.6	27.04	5.092			
6,200.0	6,199.8	6,199.8	6,199.7	13.6	13.9	-104.83	134.1	-0.7	138.7	111.3	27.48	5.049			
6,300.0	6,299.8	6,299.8	6,299.7	13.8	14.2	-105.08	135.0	-0.7	139.8	111.9	27.92	5.007			
6,400.0	6,399.8	6,399.8	6,399.7	14.0	14.4	-105.33	135.9	-0.7	140.9	112.5	28.36	4.967			
6,500.0	6,499.8	6,499.8	6,499.6	14.2	14.6	-105.58	136.7	-0.7	142.0	113.2	28.81	4.928			
6,600.0	6,599.8	6,599.8	6,599.6	14.4	14.8	-105.82	137.6	-0.7	143.0	113.8	29.25	4.890			
6,700.0	6,699.8	6,699.8	6,699.6	14.7	15.1	-106.06	138.5	-0.7	144.1	114.4	29.69	4.854			
6,800.0	6,799.8	6,799.8	6,799.6	14.9	15.3	-106.30	139.4	-0.7	145.2	115.1	30.13	4.818			
6,900.0	6,899.8	6,899.8	6,899.6	15.1	15.5	-106.53	140.2	-0.7	146.3	115.7	30.58	4.784			
7,000.0	6,999.8	6,999.8	6,999.6	15.3	15.7	-106.76	141.1	-0.7	147.4	116.3	31.02	4.751			
7,100.0	7,099.8	7,099.8	7,099.6	15.5	15.9	-106.98	142.0	-0.7	148.5	117.0	31.46	4.718			
7,200.0	7,199.8	7,199.8	7,199.6	15.8	16.2	-107.20	142.8	-0.7	149.5	117.6	31.91	4.687			
7,300.0	7,299.8	7,299.7	7,299.6	16.0	16.4	-107.42	143.7	-0.7	150.6	118.3	32.35	4.657			
7,400.0	7,399.8	7,399.7	7,399.5	16.2	16.6	-107.64	144.6	-0.7	151.7	118.9	32.79	4.627			
7,500.0	7,499.8	7,499.7	7,499.5	16.4	16.8	-107.85	145.5	-0.7	152.8	119.6	33.24	4.598			
7,600.0	7,599.8	7,599.7	7,599.5	16.6	17.1	-108.06	146.3	-0.7	153.9	120.2	33.68	4.570			
7,700.0	7,699.8	7,699.7	7,699.5	16.9	17.3	-108.27	147.2	-0.7	155.0	120.9	34.12	4.543			
7,800.0	7,799.8	7,799.7	7,799.5	17.1	17.5	-108.47	148.1	-0.7	156.1	121.6	34.57	4.517			
7,900.0	7,899.8	7,899.7	7,899.5	17.3	17.7	-108.67	149.0	-0.7	157.2	122.2	35.01	4.491			
7,926.7	7,926.4	7,926.4	7,925.1	17.4	17.8	-108.72	149.2	-0.7	157.5	122.4	35.13	4.484			
7,943.3	7,943.1	7,943.3	7,943.1	17.4	17.8	-18.74	149.3	-0.7	157.6	122.4	35.21	4.477			
8,000.0	7,999.8	8,000.0	7,999.8	17.5	17.9	-18.74	149.3	-0.7	157.6	122.2	35.43	4.449			
8,100.0	8,099.8	8,100.0	8,099.8	17.7	18.2	-18.74	149.3	-0.7	157.6	121.8	35.86	4.395			
8,200.0	8,199.8	8,200.0	8,199.8	17.9	18.4	-18.74	149.3	-0.7	157.6	121.3	36.29	4.343			
8,300.0	8,299.8	8,300.0	8,299.8	18.1	18.6	-18.74	149.3	-0.7	157.6	120.9	36.72	4.292			
8,400.0	8,399.8	8,400.0	8,399.8	18.4	18.8	-18.74	149.3	-0.7	157.6	120.5	37.16	4.242			
8,500.0	8,499.8	8,500.0	8,499.8	18.6	19.1	-18.74	149.3	-0.7	157.6	120.0	37.59	4.194			
8,600.0	8,599.8	8,600.0	8,599.8	18.8	19.3	-18.74	149.3	-0.7	157.6	119.6	38.02	4.146			
8,700.0	8,699.8	8,700.0	8,699.8	19.0	19.5	-18.74	149.3	-0.7	157.6	119.2	38.45	4.099			
8,800.0	8,799.8	8,800.0	8,799.8	19.2	19.7	-18.74	149.3	-0.7	157.6	118.7	38.89	4.053			
8,900.0	8,899.8	8,900.0	8,899.8	19.4	19.9	-18.74	149.3	-0.7	157.6	118.3	39.32	4.009			
9,000.0	8,999.8	9,000.0	8,999.8	19.6	20.2	-18.74	149.3	-0.7	157.6	117.9	39.76	3.965			
9,100.0	9,099.8	9,100.0	9,099.8	19.8	20.4	-18.74	149.3	-0.7	157.6	117.4	40.19	3.922			
9,200.0	9,199.8	9,200.0	9,199.8	20.0	20.6	-18.74	149.3	-0.7	157.6	117.0	40.63	3.880			
9,300.0	9,299.8	9,300.0	9,299.8	20.2	20.8	-18.74	149.3	-0.7	157.6	116.6	41.06	3.839			
9,400.0	9,399.8	9,400.0	9,399.8	20.5	21.1	-18.74	149.3	-0.7	157.6	116.1	41.50	3.799			
9,500.0	9,499.8	9,500.0	9,499.8	20.7	21.3	-18.74	149.3	-0.7	157.6	115.7	41.93	3.759			
9,600.0	9,599.8	9,600.0	9,599.8	20.9	21.5	-18.74	149.3	-0.7	157.6	115.3	42.37	3.720			
9,700.0	9,699.8	9,700.0	9,699.8	21.1	21.7	-18.74	149.3	-0.7	157.6	114.8	42.80	3.683			
9,800.0	9,799.8	9,800.0	9,799.8	21.3	22.0	-18.74	149.3	-0.7	157.6	114.4	43.24	3.645			
9,900.0	9,899.8	9,900.0	9,899.8	21.5	22.2	-18.74	149.3	-0.7	157.6	113.9	43.68	3.609			
10,000.0	9,999.8	10,000.0	9,999.8	21.7	22.4	-18.74	149.3	-0.7	157.6	113.5	44.11	3.573			

CC - Min centre to center distance or covergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services Anticollision Report



Company: Oasis Petroleum
Project: Indian Hills
Reference Site: 153N-100W-17/18
Site Error: 0.0 usft
Reference Well: Kline Federal 5300 41-18 12TX
Well Error: 2.0 usft
Reference Wellbore: Kline Federal 5300 41-18 12TX
Reference Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 9T - Kline Federal 5300 41-18 9T - Design #1														Offset Site Error:	0.0 usft
Survey Program: 0-MWD														Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis			Distance			Minimum Separation			Warning		
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor			
10,100.0	10,099.8	10,100.0	10,099.8	21.9	22.6	-18.74	149.3	-0.7	157.6	113.1	44.55	3.538			
10,200.0	10,199.8	10,200.0	10,199.8	22.2	22.9	-18.74	149.3	-0.7	157.6	112.6	44.99	3.504			
10,300.0	10,299.8	10,300.0	10,299.8	22.4	23.1	-18.74	149.3	-0.7	157.6	112.2	45.43	3.470			
10,354.8	10,354.6	10,354.8	10,354.6	22.5	23.2	-18.74	149.3	-0.7	157.6	112.0	45.67	3.452			
10,375.0	10,374.8	10,373.6	10,373.4	22.5	23.2	-128.69	149.5	-0.4	158.0	112.2	45.74	3.454			
10,400.0	10,399.7	10,396.9	10,396.6	22.6	23.3	-128.49	150.2	0.9	159.4	113.6	45.80	3.480			
10,425.0	10,424.5	10,420.0	10,419.6	22.7	23.3	-128.13	151.6	3.1	161.9	116.1	45.83	3.533			
10,450.0	10,449.2	10,443.0	10,442.3	22.7	23.4	-127.82	153.5	6.3	165.5	119.6	45.83	3.610			
10,475.0	10,473.5	10,465.7	10,464.5	22.8	23.4	-126.98	156.0	10.3	170.1	124.3	45.81	3.713			
10,500.0	10,497.6	10,488.2	10,486.3	22.8	23.5	-126.21	158.9	15.1	175.8	130.0	45.78	3.839			
10,525.0	10,521.2	10,510.3	10,507.4	22.9	23.5	-125.34	162.3	20.8	182.4	136.7	45.73	3.989			
10,550.0	10,544.4	10,532.1	10,527.8	23.0	23.6	-124.35	166.2	27.1	190.1	144.4	45.69	4.161			
10,575.0	10,567.1	10,553.5	10,547.6	23.0	23.6	-123.28	170.5	34.1	198.7	153.1	45.66	4.353			
10,600.0	10,569.1	10,575.0	10,567.1	23.1	23.7	-122.11	175.2	41.9	208.3	162.6	45.64	4.564			
10,625.0	10,610.6	10,594.9	10,584.7	23.2	23.8	-120.87	180.0	49.8	218.7	173.1	45.64	4.792			
10,650.0	10,631.3	10,614.9	10,602.0	23.3	23.8	-119.54	185.3	58.3	230.0	184.3	45.67	5.035			
10,675.0	10,651.3	10,634.4	10,618.5	23.4	23.9	-118.14	190.7	67.2	242.0	196.3	45.74	5.291			
10,700.0	10,670.5	10,653.4	10,634.1	23.5	23.9	-116.86	196.3	76.5	254.8	209.0	45.85	5.557			
10,725.0	10,688.8	10,672.0	10,648.9	23.6	24.0	-115.11	202.2	86.0	268.3	222.3	46.01	5.832			
10,750.0	10,706.2	10,690.1	10,663.0	23.7	24.0	-113.47	208.1	95.7	282.5	236.3	46.22	6.113			
10,775.0	10,722.6	10,707.7	10,676.2	23.9	24.1	-111.76	214.1	105.6	297.3	250.8	46.47	6.397			
10,800.0	10,738.0	10,725.0	10,688.8	24.0	24.2	-109.98	220.3	115.8	312.6	265.9	46.76	6.684			
10,825.0	10,752.4	10,741.5	10,700.4	24.2	24.2	-108.11	226.5	125.8	328.5	281.4	47.13	6.971			
10,850.0	10,765.7	10,757.7	10,711.4	24.4	24.3	-106.17	232.7	136.0	344.9	297.3	47.52	7.257			
10,875.0	10,777.8	10,775.0	10,722.6	24.6	24.4	-104.20	239.5	147.2	361.6	313.7	47.84	7.544			
10,900.0	10,788.8	10,789.0	10,731.4	24.8	24.5	-102.05	245.2	156.6	378.8	330.4	48.38	7.829			
10,925.0	10,798.6	10,804.1	10,740.4	25.0	24.5	-99.89	251.5	166.8	396.3	347.4	48.84	8.114			
10,950.0	10,807.2	10,818.8	10,748.9	25.3	24.6	-97.67	257.8	177.1	414.1	364.8	49.30	8.400			
10,975.0	10,814.5	10,833.2	10,756.9	25.6	24.7	-95.39	264.0	187.4	432.2	382.4	49.74	8.688			
11,000.0	10,820.6	10,847.3	10,764.3	25.8	24.8	-93.06	270.2	197.8	450.5	400.3	50.17	8.980			
11,025.0	10,825.4	10,861.1	10,771.2	26.1	24.9	-90.70	276.5	207.8	469.0	418.4	50.55	9.276			
11,050.0	10,828.9	10,875.0	10,777.8	26.5	25.0	-88.34	282.8	218.2	487.6	436.7	50.90	9.579			
11,075.0	10,831.1	10,888.0	10,783.7	26.8	25.1	-85.92	288.9	228.1	506.4	455.2	51.20	9.890			
11,100.0	10,832.0	10,900.0	10,788.8	27.1	25.1	-83.43	294.5	237.4	525.3	473.8	51.44	10.211			
11,104.0	10,832.0	10,903.2	10,790.1	27.2	25.2	-83.14	296.0	239.8	528.3	476.8	51.48	10.260			
11,200.0	10,832.2	10,957.3	10,809.4	28.6	25.6	-86.90	322.3	283.0	600.0	546.6	53.38	11.240			
11,300.0	10,832.4	11,022.3	10,824.9	30.3	26.2	-89.21	355.2	336.9	671.8	616.3	55.48	12.109			
11,400.0	10,832.6	11,094.8	10,831.9	32.2	27.0	-90.02	382.8	398.4	739.7	681.8	57.83	12.791			
11,500.0	10,832.7	11,198.8	10,832.2	34.1	28.2	-90.03	445.6	487.9	801.8	741.1	60.68	13.214			
11,600.0	10,832.9	11,317.2	10,832.4	36.2	30.0	-90.02	501.7	592.2	855.9	791.8	64.10	13.352			
11,680.4	10,833.1	11,420.0	10,832.6	38.0	31.7	-90.01	547.0	684.5	892.8	825.5	67.25	13.276			
11,700.0	10,833.1	11,445.9	10,832.7	38.4	32.1	-90.01	557.8	708.0	900.9	832.9	68.07	13.235			
11,800.0	10,833.3	11,581.6	10,832.9	40.7	34.7	-90.02	611.2	832.7	940.0	867.5	72.51	12.965			
11,800.0	10,833.4	11,722.5	10,833.1	43.0	37.5	-90.02	660.4	964.7	974.7	897.3	77.41	12.591			
12,000.0	10,833.6	11,868.3	10,833.4	45.4	40.6	-90.02	704.4	1,103.7	1,004.6	921.8	82.73	12.142			
12,100.0	10,833.8	12,018.4	10,833.7	47.9	44.0	-90.02	742.3	1,249.0	1,029.5	941.0	88.44	11.640			
12,200.0	10,834.0	12,172.3	10,834.0	50.4	47.6	-90.02	773.1	1,399.7	1,049.2	954.7	94.61	11.102			
12,300.0	10,834.1	12,329.1	10,834.2	53.0	51.4	-90.02	798.1	1,554.8	1,063.6	982.8	100.87	10.545			
12,400.0	10,834.3	12,487.9	10,834.5	55.5	55.3	-90.02	810.8	1,712.9	1,072.8	985.1	107.48	9.981			
12,500.0	10,834.5	12,647.9	10,834.8	58.2	59.3	-90.02	816.7	1,872.7	1,075.9	961.7	114.24	9.418			
12,600.0	10,834.7	12,757.5	10,835.0	60.8	62.0	-90.02	816.8	1,982.4	1,075.7	955.9	119.79	8.990			
12,700.0	10,834.8	12,857.5	10,835.2	63.5	64.5	-90.02	816.8	2,082.4	1,075.4	950.3	125.13	8.594			

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services

Anticollision Report



Company: Oasis Petroleum
Project: Indian Hills
Reference Site: 153N-100W-17/18
Site Error: 0.0 usft
Reference Well: Kline Federal 5300 41-18 12TX
Well Error: 2.0 usft
Reference Wellbore: Kline Federal 5300 41-18 12TX
Reference Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at: 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 9T - Kline Federal 5300 41-18 9T - Design #1											Offset Site Error:	0.0 usft
Survey Program: 0-MWD											Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis		Highside Toolface (°)	Distance				Warning	
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)		Offset Wellbore +N/-S (usft)	Centre +E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)		Minimum Separation (usft)
12,800.0	10,835.0	12,957.5	10,835.4	66.2	67.1	-90.02	816.8	2,182.4	1,075.1	944.6	130.52	8.237
12,900.0	10,835.2	13,057.5	10,835.5	68.9	69.7	-90.02	816.8	2,282.4	1,074.8	938.8	135.94	7.906
13,000.0	10,835.4	13,157.5	10,835.7	71.7	72.3	-90.02	816.8	2,382.4	1,074.5	933.1	141.40	7.599
13,100.0	10,835.5	13,257.5	10,835.9	74.4	75.0	-90.02	816.8	2,482.4	1,074.2	927.3	146.89	7.313
13,200.0	10,835.7	13,357.5	10,836.0	77.2	77.6	-90.02	816.8	2,582.3	1,073.9	921.4	152.41	7.046
13,300.0	10,835.9	13,457.5	10,836.2	79.9	80.3	-90.02	816.8	2,682.3	1,073.5	915.6	157.95	6.797
13,400.0	10,836.1	13,557.5	10,836.4	82.7	83.0	-90.02	816.8	2,782.3	1,073.2	909.7	163.52	6.563
13,500.0	10,836.2	13,657.5	10,836.6	85.5	85.7	-90.02	816.8	2,882.3	1,072.9	903.8	169.11	6.345
13,600.0	10,836.4	13,757.5	10,836.7	88.3	88.4	-90.02	816.8	2,982.3	1,072.6	897.9	174.71	6.139
13,700.0	10,836.6	13,857.5	10,836.9	91.1	91.2	-90.02	816.8	3,082.3	1,072.3	892.0	180.33	5.946
13,800.0	10,836.7	13,957.5	10,837.1	94.0	93.9	-90.02	816.8	3,182.3	1,072.0	886.0	185.97	5.765
13,900.0	10,836.9	14,057.5	10,837.3	96.8	96.7	-90.02	816.8	3,282.3	1,071.7	880.1	191.61	5.593
14,000.0	10,837.1	14,157.5	10,837.4	99.8	99.5	-90.02	816.8	3,382.3	1,071.4	874.1	197.28	5.431
14,100.0	10,837.3	14,257.5	10,837.6	102.5	102.2	-90.02	816.8	3,482.3	1,071.1	868.1	202.95	5.278
14,200.0	10,837.4	14,357.5	10,837.8	105.3	105.0	-90.02	816.8	3,582.3	1,070.8	862.1	208.63	5.132
14,300.0	10,837.6	14,457.5	10,838.0	108.2	107.8	-90.02	816.8	3,682.3	1,070.5	856.1	214.32	4.995
14,400.0	10,837.8	14,557.5	10,838.1	111.0	110.6	-90.02	816.8	3,782.3	1,070.2	850.1	220.03	4.864
14,500.0	10,838.0	14,657.5	10,838.3	113.9	113.5	-90.02	816.8	3,882.3	1,069.8	844.1	225.73	4.739
14,600.0	10,838.1	14,757.5	10,838.5	116.7	116.3	-90.02	816.8	3,982.3	1,069.5	838.1	231.45	4.621
14,700.0	10,838.3	14,857.5	10,838.7	119.6	119.1	-90.02	816.8	4,082.3	1,069.2	832.0	237.18	4.508
14,800.0	10,838.5	14,957.5	10,838.8	122.5	121.9	-90.02	816.8	4,182.3	1,068.9	826.0	242.91	4.401
14,900.0	10,838.7	15,057.5	10,839.0	125.3	124.7	-90.02	816.8	4,282.3	1,068.6	820.0	248.64	4.298
15,000.0	10,838.8	15,157.5	10,839.2	128.2	127.6	-90.02	816.8	4,382.3	1,068.3	813.9	254.39	4.199
15,100.0	10,839.0	15,257.5	10,839.4	131.1	130.4	-90.02	816.8	4,482.3	1,068.0	807.9	260.14	4.106
15,200.0	10,839.2	15,357.5	10,839.5	133.9	133.3	-90.02	816.8	4,582.3	1,067.7	801.8	265.89	4.016
15,300.0	10,839.4	15,457.5	10,839.7	136.8	136.1	-90.02	816.8	4,682.3	1,067.4	795.7	271.65	3.929
15,400.0	10,839.5	15,557.5	10,839.9	139.7	139.0	-90.02	816.8	4,782.3	1,067.1	789.7	277.41	3.847
15,500.0	10,839.7	15,657.5	10,840.1	142.6	141.8	-90.02	816.8	4,882.3	1,066.8	783.6	283.18	3.767
15,600.0	10,839.9	15,757.5	10,840.2	145.5	144.7	-90.02	816.8	4,982.3	1,066.4	777.5	288.95	3.691
15,700.0	10,840.1	15,857.5	10,840.4	148.4	147.5	-90.02	816.8	5,082.3	1,066.1	771.4	294.72	3.617
15,800.0	10,840.2	15,957.5	10,840.6	151.3	150.4	-90.02	816.8	5,182.3	1,065.8	765.3	300.50	3.547
15,900.0	10,840.4	16,057.5	10,840.8	154.1	153.3	-90.02	816.8	5,282.3	1,065.5	759.2	306.28	3.479
16,000.0	10,840.6	16,157.5	10,840.9	157.0	156.1	-90.02	816.8	5,382.3	1,065.2	753.2	312.06	3.413
16,100.0	10,840.8	16,257.5	10,841.1	159.9	159.0	-90.02	816.8	5,482.3	1,064.9	747.1	317.85	3.350
16,200.0	10,840.9	16,357.5	10,841.3	162.8	161.9	-90.02	816.8	5,582.3	1,064.6	741.0	323.63	3.290
16,300.0	10,841.1	16,457.5	10,841.5	165.7	164.8	-90.02	816.8	5,682.3	1,064.3	734.9	329.43	3.231
16,400.0	10,841.3	16,557.5	10,841.8	168.6	167.6	-90.02	816.8	5,782.3	1,064.0	728.8	335.22	3.174
16,500.0	10,841.4	16,657.5	10,841.8	171.5	170.5	-90.02	816.8	5,882.3	1,063.7	722.7	341.02	3.119
16,600.0	10,841.5	16,757.5	10,842.0	174.4	173.4	-90.02	816.8	5,982.3	1,063.4	716.5	346.81	3.066
16,700.0	10,841.8	16,857.5	10,842.2	177.3	176.3	-90.02	816.8	6,082.3	1,063.1	710.4	352.62	3.015
16,800.0	10,842.0	16,957.5	10,842.3	180.2	179.2	-90.02	816.8	6,182.3	1,062.7	704.3	358.42	2.965
16,900.0	10,842.1	17,057.5	10,842.5	183.1	182.1	-90.02	816.8	6,282.3	1,062.4	698.2	364.22	2.917
17,000.0	10,842.3	17,157.5	10,842.7	186.0	184.9	-90.02	816.8	6,382.3	1,062.1	692.1	370.03	2.870
17,100.0	10,842.5	17,257.5	10,842.9	188.9	187.8	-90.02	816.8	6,482.3	1,061.8	686.0	375.84	2.825
17,200.0	10,842.7	17,357.5	10,843.0	191.8	190.7	-90.02	816.8	6,582.3	1,061.5	679.9	381.65	2.781
17,300.0	10,842.8	17,457.5	10,843.2	194.7	193.6	-90.02	816.8	6,682.3	1,061.2	673.7	387.46	2.739
17,400.0	10,843.0	17,557.4	10,843.4	197.6	196.5	-90.02	816.8	6,782.3	1,060.9	667.6	393.27	2.698
17,500.0	10,843.2	17,657.4	10,843.6	200.6	199.4	-90.02	816.8	6,882.3	1,060.6	661.5	399.08	2.658
17,600.0	10,843.4	17,757.4	10,843.7	203.5	202.3	-90.02	816.8	6,982.3	1,060.3	655.4	404.90	2.619
17,700.0	10,843.5	17,857.4	10,843.9	206.4	205.2	-90.02	816.8	7,082.3	1,060.0	649.3	410.72	2.581
17,800.0	10,843.7	17,957.4	10,844.1	209.3	208.1	-90.02	816.8	7,182.3	1,059.7	643.1	416.54	2.544
17,900.0	10,843.9	18,057.4	10,844.3	212.2	211.0	-90.02	816.8	7,282.3	1,059.4	637.0	422.35	2.508

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services Anticollision Report



Company: Oasis Petroleum
Project: Indian Hills
Reference Site: 153N-100W-17/18
Site Error: 0.0 usft
Reference Well: Kline Federal 5300 41-18 12TX
Well Error: 2.0 usft
Reference Wellbore: Kline Federal 5300 41-18 12TX
Reference Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at: 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 9T - Kline Federal 5300 41-18 9T - Design #1													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis			Distance							Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
18,000.0	10,844.1	18,157.4	10,844.4	215.1	213.9	-90.02	816.8	7,382.3	1,059.0	630.9	428.17	2.473		
18,100.0	10,844.2	18,257.4	10,844.6	218.0	216.8	-90.02	816.8	7,482.3	1,058.7	624.7	434.00	2.439		
18,200.0	10,844.4	18,357.4	10,844.8	220.9	219.7	-90.02	816.8	7,582.3	1,058.4	618.6	439.82	2.406		
18,300.0	10,844.6	18,457.4	10,845.0	223.8	222.6	-90.02	816.8	7,682.3	1,058.1	612.5	445.84	2.374		
18,400.0	10,844.8	18,557.4	10,845.1	226.7	225.5	-90.02	816.8	7,782.3	1,057.8	606.3	451.47	2.343		
18,500.0	10,844.9	18,657.4	10,845.3	229.7	228.4	-90.02	816.8	7,882.3	1,057.5	600.2	457.29	2.313		
18,600.0	10,845.1	18,757.4	10,845.5	232.6	231.3	-90.02	816.8	7,982.3	1,057.2	594.1	463.12	2.283		
18,700.0	10,845.3	18,857.4	10,845.8	235.5	234.2	-90.02	816.8	8,082.3	1,056.9	587.9	468.95	2.254		
18,800.0	10,845.4	18,957.4	10,845.8	238.4	237.1	-90.02	816.8	8,182.3	1,056.6	581.8	474.78	2.225		
18,900.0	10,845.6	19,057.4	10,846.0	241.3	240.0	-90.02	816.8	8,282.3	1,056.3	575.7	480.60	2.198		
19,000.0	10,845.8	19,157.4	10,846.2	244.2	242.9	-90.02	816.8	8,382.3	1,056.0	569.5	486.43	2.171		
19,100.0	10,846.0	19,257.4	10,846.3	247.1	245.8	-90.02	816.8	8,482.3	1,055.6	563.4	492.26	2.144		
19,200.0	10,846.1	19,357.4	10,846.5	250.1	248.7	-90.02	816.8	8,582.3	1,055.3	557.2	498.10	2.119		
19,300.0	10,846.3	19,457.4	10,846.7	253.0	251.6	-90.02	816.8	8,682.3	1,055.0	551.1	503.93	2.094		
19,400.0	10,846.5	19,557.4	10,846.9	255.9	254.5	-90.02	816.8	8,782.3	1,054.7	545.0	509.76	2.069		
19,500.0	10,846.7	19,657.4	10,847.0	258.8	257.4	-90.02	816.8	8,882.3	1,054.4	538.8	515.59	2.045		
19,600.0	10,846.8	19,757.4	10,847.2	261.7	260.4	-90.02	816.8	8,982.3	1,054.1	532.7	521.43	2.022		
19,700.0	10,847.0	19,857.4	10,847.4	264.6	263.3	-90.02	816.8	9,082.3	1,053.8	526.5	527.26	1.999		
19,800.0	10,847.2	19,957.4	10,847.6	267.6	266.2	-90.02	816.8	9,182.3	1,053.5	520.4	533.10	1.976		
19,900.0	10,847.4	20,057.4	10,847.7	270.5	269.1	-90.02	816.8	9,282.3	1,053.2	514.2	538.93	1.954		
20,000.0	10,847.5	20,157.4	10,847.9	273.4	272.0	-90.02	816.8	9,382.3	1,052.9	508.1	544.77	1.933		
20,100.0	10,847.7	20,257.4	10,848.1	276.3	274.9	-90.02	816.8	9,482.3	1,052.6	502.0	550.61	1.912		
20,200.0	10,847.9	20,357.4	10,848.3	279.2	277.8	-90.02	816.8	9,582.3	1,052.3	495.8	556.45	1.891		
20,300.0	10,848.1	20,457.4	10,848.4	282.1	280.7	-90.02	816.8	9,682.3	1,051.9	489.7	562.28	1.871		
20,400.0	10,848.2	20,557.4	10,848.6	285.1	283.6	-90.02	816.8	9,782.3	1,051.6	483.5	568.12	1.851		
20,500.0	10,848.4	20,657.4	10,848.8	288.0	286.6	-90.02	816.8	9,882.3	1,051.3	477.4	573.96	1.832		
20,600.0	10,848.6	20,757.4	10,849.0	290.9	289.5	-90.02	816.8	9,982.3	1,051.0	471.2	579.80	1.813		
20,667.7	10,848.7	20,824.5	10,849.1	292.9	291.4	-90.02	816.8	10,049.3	1,050.8	467.1	583.73	1.800 SF		



Ryan Directional Services Anticollision Report



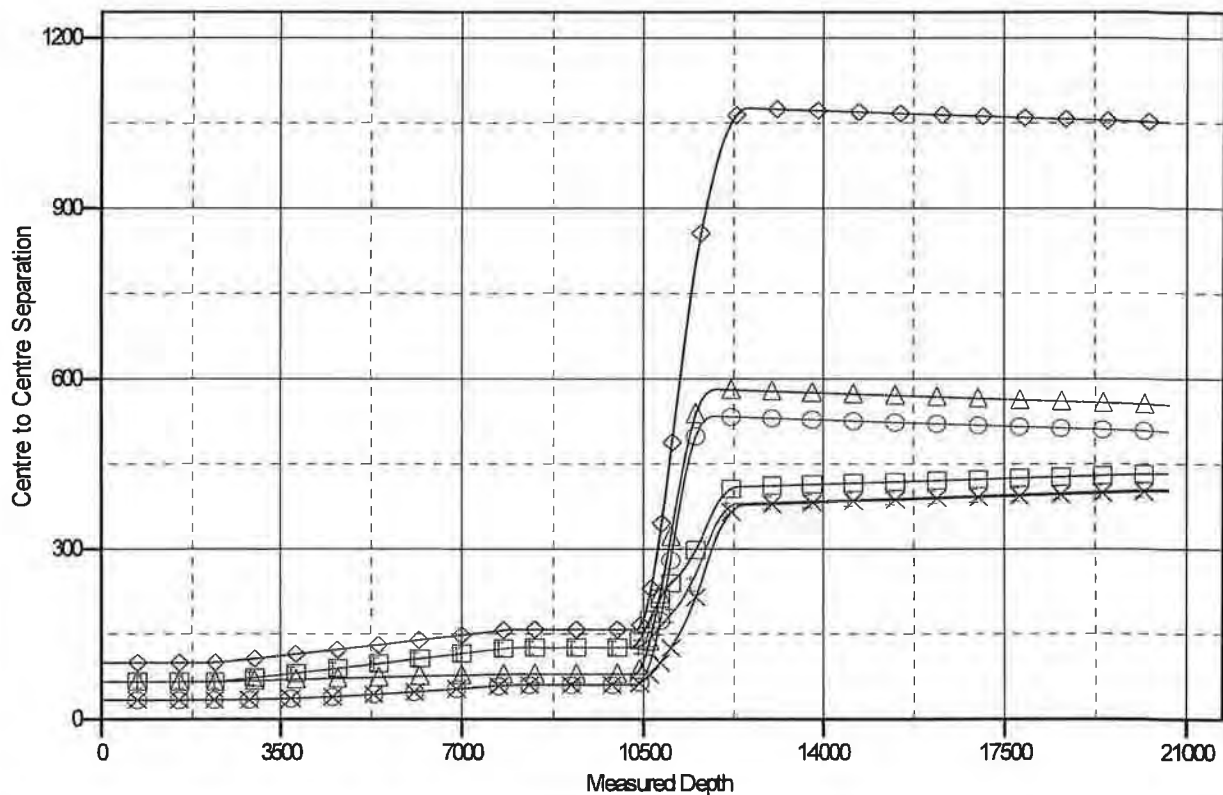
Company: Oasis Petroleum
Project: Indian Hills
Reference Site: 153N-100W-17/18
Site Error: 0.0 usft
Reference Well: Kline Federal 5300 41-18 12TX
Well Error: 2.0 usft
Reference Wellbore: Kline Federal 5300 41-18 12TX
Reference Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at: 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Reference Depths are relative to WELL @ 2082.0usft (Original Well Ele)
Offset Depths are relative to Offset Datum
Central Meridian is 100° 30' 0.000 W

Coordinates are relative to: Kline Federal 5300 41-18 12TX
Coordinate System is US State Plane 1983, North Dakota Northern Zone
Grid Convergence at Surface is: -2.31°

Ladder Plot



LEGEND

5300 41-18 9T, Design #1 V0 Kline Federal 5300 41-18 11T2, Kline Federal 5300 41-18 11T2, Design #1 V0 Kline Federal 5300 41-18 14BX
5300 41-18 10B, Design #1 V0 Kline Federal 5300 41-18 13T2X, Kline Federal 5300 41-18 13T2X, Design #1 V0 Kline Federal 5300 41-18 14BX



Ryan Directional Services Anticollision Report



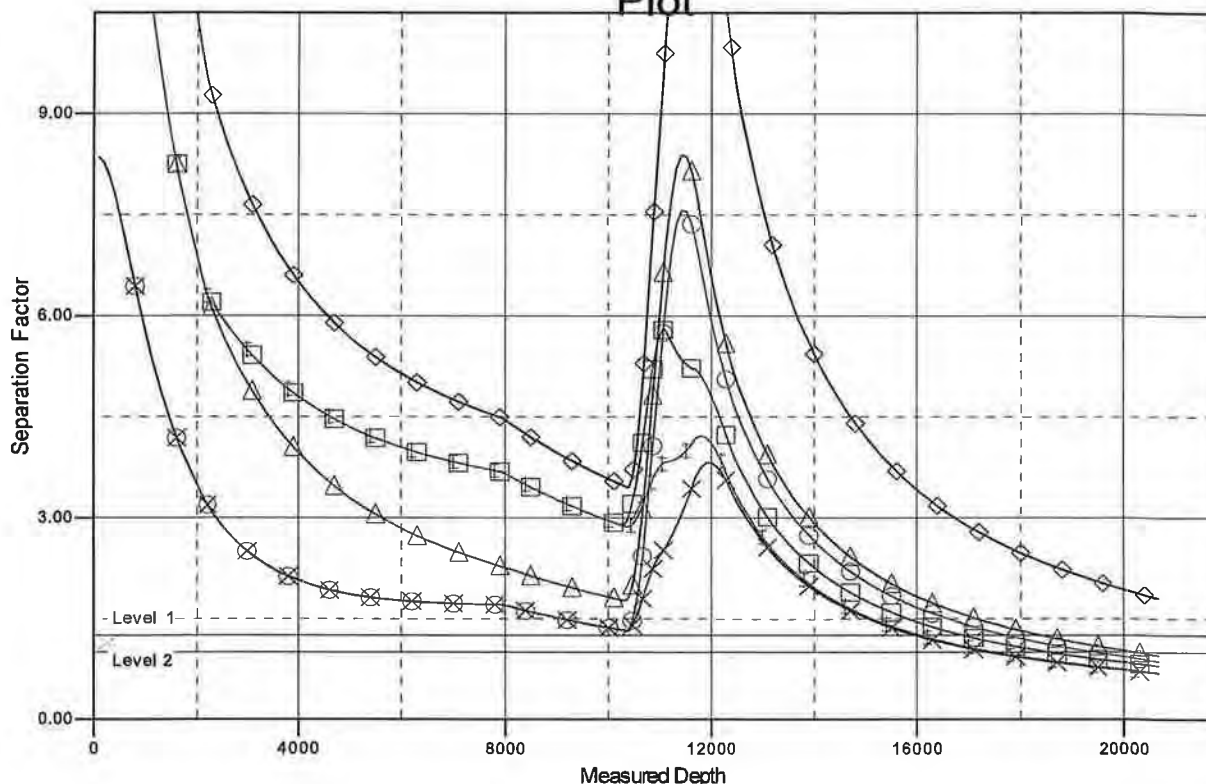
Company: Oasis Petroleum
Project: Indian Hills
Reference Site: 153N-100W-17/18
Site Error: 0.0 usft
Reference Well: Kline Federal 5300 41-18 12TX
Well Error: 2.0 usft
Reference Wellbore: Kline Federal 5300 41-18 12TX
Reference Design: Design #2

Local Co-ordinate Reference: Well Kline Federal 5300 41-18 12TX
TVD Reference: WELL @ 2082.0usft (Original Well Elev)
MD Reference: WELL @ 2082.0usft (Original Well Elev)
North Reference: True
Survey Calculation Method: Minimum Curvature
Output errors are at 2.00 sigma
Database: EDM 5000.1 Single User Db
Offset TVD Reference: Offset Datum

Reference Depths are relative to WELL @ 2082.0usft (Original Well Ele
Offset Depths are relative to Offset Datum
Central Meridian is 100° 30' 0.000 W

Coordinates are relative to: Kline Federal 5300 41-18 12TX
Coordinate System is US State Plane 1983, North Dakota Northern Zone
Grid Convergence at Surface is: -2.31"

Separation Factor Plot



LEGEND

5300 41-18 9T, Design #1 V0
5300 41-18 10B, Design #1 V0

—◇— Kline Federal 5300 41-18 11T2, Kline Federal 5300 41-18 11T2, Design #1 V0
—x— Kline Federal 5300 41-18 13T2X, Kline Federal 5300 41-18 13T2X, Design #1 V0

—□— Kline Federal 5300 41-18 14B2X
—◆— Kline Federal 5300 41-18 14B2X

**Oasis Petroleum
Well Summary
Kline Federal 5300 41-18 12TX
Section 18 T153N R100W
McKenzie County, ND**

SURFACE CASING AND CEMENT DESIGN

							Make-up Torque (ft-lbs)		
Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Minimum	Optimum	Max
13-3/8"	0' to 2,072'	54.5	J-55	STC	12.615"	12.459"	4,100	5,470	6,840

Interval	Description	Collapse	Burst	Tension
		(psi) a	(psi) b	(1000 lbs) c
0' to 2,072'	13-3/8", 54.5#, J-55, STC, 8rd	1130 / 1.17	2730 / 2.82	514 / 2.61

API Rating & Safety Factor

- a) Based on full casing evacuation with 9 ppg fluid on backside (2,072' setting depth).
- b) Burst pressure based on 9 ppg fluid with no fluid on backside (2,072' setting depth).
- c) Based on string weight in 9 ppg fluid at 2,072' TVD plus 100k# overpull. (Buoyed weight equals 97k lbs.)

Cement volumes are based on 13-3/8" casing set in 17-1/2" hole with 50% excess to circulate cement back to surface.
Mix and pump the following slurry.

Pre-flush (Spacer): **20 bbls** fresh water

Lead Slurry: **635 sks** (328 bbls) 2.9 yield conventional system with 94 lb/sk cement, .25 lb/sk D130 Lost Circulation Control Agent, 2% CaCl₂, 4% D079 Extender and 2% D053 Expanding Agent.

Tail Slurry: **349 sks** (72 bbls) 1.16 yield conventional system with 94 lb/sk cement, .25% CaCl₂ and 0.25 lb/sk Lost Circulation Control Agent

**Oasis Petroleum
Well Summary
Kline Federal 5300 31-18 12TX
Section 18 T153N R100W
McKenzie County, ND**

CONTINGENCY SURFACE CASING AND CEMENT DESIGN

							Make-up Torque (ft-lbs)		
Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Minimum	Optimum	Max
9-5/8"	0' to 6,101'	40	HCL-80	LTC	8.835"	8.75"	5,450	7,270	9,090

Interval	Description	Collapse	Burst	Tension
		(psi) a	(psi) b	(1000 lbs) c
0' to 6,101'	9-5/8", 40#, HCL-80, LTC, 8rd	4230 / 2.13	5750 / 3.72	837 / 2.78

API Rating & Safety Factor

- a) Collapse pressure based on 11.5 ppg fluid on the backside and 9 ppg fluid inside of casing.
- b) Burst pressure calculated from a gas kick coming from the production zone (Bakken Pool) at 9,000 psi and a subsequent breakdown at the 9-5/8" shoe, based on a 13.5#/ft fracture gradient. Backup of 9 ppg fluid.
- c) Yield based on string weight in 10 ppg fluid, (207k lbs buoyed weight) plus 100k lbs overpull.

Cement volumes are based on 9-5/8" casing set in 12-1/4" hole with 10% excess in OH and 0% excess inside surface casing. TOC at surface.

Pre-flush (Spacer): **20 bbls** Chem wash

Lead Slurry: **598 sks** (309 bbls) Conventional system with 75 lb/sk cement, 0.5 lb/sk lost circulation, 10% expanding agent, 2% extender, 2% CaCl₂, 0.2% anti-foam and 0.4% fluid loss agent.

Tail Slurry: **349 sks** (72 bbls) Conventional system with 94 lb/sk cement, 0.3% anti-settling agent, 0.3% fluid loss agent, 0.3 lb/sk lost circulation control agent, 0.2% anti-foam and 0.1% retarder.

**Oasis Petroleum
Well Summary
Kline Federal 5300 41-18 12TX
Section 18 T153N R100W
McKenzie County, ND**

INTERMEDIATE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
7"	0' – 6,664'	29	P-110	LTC	6.184"	6.059"	5,980	7,970	8,770
7"	6,664' – 10,355'	32	HCP-110	LTC	6.094"	6.000***	6,730	8,970	9,870
7"	10,355' – 11,189'	29	P-110	LTC	6.184"	6.059"	5,980	7,970	8,770

***Special drift

Interval	Length	Description	Collapse	Burst	Tension
			(psi) a	(psi) b	(1000 lbs) c
0' – 6,664'	6,664'	7", 29#, P-110, LTC, 8rd	8530 / 2.46*	11220 / 1.19	797 / 2.07
6,664' – 10,355'	3,691'	7", 32#, HCP-110, LTC, 8rd	11820 / 2.19*	12460 / 1.29	
6,664' – 10,355'	3,691'	7", 32#, HCP-110, LTC, 8rd	11820 / 1.07**	12460 / 1.29	
10,355' – 11,189'	834'	7", 29 lb, P-110, LTC, 8rd	8530 / 1.51*	11220 / 1.15	

API Rating & Safety Factor

- a) *Assume full casing evacuation with 10 ppg fluid on backside. **Assume full casing evacuation with 1.2 psi/ft equivalent fluid gradient across salt intervals.
- b) Burst pressure based on 9000 psig max press for stimulation plus 10.2 ppg fluid in casing and 9 ppg fluid on backside-to 10,832' TVD.
- c) Based on string weight in 10 ppg fluid, (284k lbs buoyed weight) plus 100k

Cement volumes are estimates based on 7" casing set in an 8-3/4" hole with 30% excess.

Pre-flush (Spacer): **50 bbls Saltwater**
40 bbls Weighted MudPush Express

Lead Slurry: **157 sks** (72 bbls) 2.21 yield conventional system with 47 lb/sk cement, 37 lb/sk D035 extender, 3.0% KCl, 3.0% D154 extender, 0.3% D208 viscosifier, 0.07% retarder, 0.2% anti-foam, 0.5 lb/sk, D130 LCM.

Tail Slurry: **627 sks** (173 bbls) 1.54 yield conventional system with 94 lb/sk cement, 3.0% KCl, 35.0% Silica, 0.5% retarder, 0.2% fluid loss, 0.2% anti-foam and 0.5 lb/sk LCM.

**Oasis Petroleum
Well Summary
Kline Federal 5300 41-18 12TX
Section 18 T153N R100W
McKenzie County, ND**

PRODUCTION LINER

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Estimated Torque
4-1/2"	10,305' – 20,655'	13.5	P-110	BTC	3.92"	3.795"	4,500

Interval	Description	Collapse	Burst	Tension
		(psi) a	(psi) b	(1000 lbs) c
10,305' – 20,655'	4-1/2", 13.5 lb, P-110, BTC, 8rd	10680 / 1.99	12410 / 1.28	443 / 2.02

API Rating & Safety Factor

- a) Based on full casing evacuation with 9.5 ppg fluid on backside @ 10,849' TVD.
Burst pressure based on 9000 psi treating pressure with 10.2 ppg internal fluid gradient and 9 ppg external
- b) fluid gradient @ 10,849' TVD.
- c) Based on string weight in 9.5 ppg fluid (Buoyed weight: 119k lbs.) plus 100k lbs overpull.

Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.

68334-30-5 (Primary Name: Fuels, diesel)
68476-34-6 (Primary Name: Fuels, diesel, No. 2)
68476-30-2 (Primary Name: Fuel oil No. 2)
68476-31-3 (Primary Name: Fuel oil, No. 4)
8008-20-6 (Primary Name: Kerosene)



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.dmr.nd.gov/oilgas

28658

BRANDI TERRY
OASIS PETROLEUM NORTH AMERICA LLC
1001 FANNIN STE 1500
HOUSTON, TX 77002 USA

Date: 6/23/2014

RE: CORES AND SAMPLES

Well Name: **KLINE FEDERAL 5300 41-18 12TX** Well File No.: **28658**
Location: **LOT4 18-153-100** County: **MCKENZIE**
Permit Type: **Development - HORIZONTAL**
Field: **BAKER** Target Horizon: **THREE FORKS B1**

Dear BRANDI TERRY:

North Dakota Century Code Section 38-08-04 provides for the preservation of cores and samples and their shipment to the State Geologist when requested. The following is required on the above referenced well:

- 1) All cores, core chips and samples must be submitted to the State Geologist as provided for under North Dakota Century Code: Section 38-08-04 and North Dakota Administrative Code: Section 43-02-03-38.1.
- 2) Samples: The Operator is to begin collecting sample drill cuttings no lower than the:
Base of the Last Charles Salt
 - Sample cuttings shall be collected at:
 - o 30' maximum intervals through all vertical and build sections.
 - o 100' maximum intervals through any horizontal sections.
 - Samples must be washed, dried, placed in standard sample envelopes (3" x 4.5"), packed in the correct order into standard sample boxes (3.5" x 5.25" x 15.25").
 - Samples boxes are to be carefully identified with a label that indicates the operator, well name, well file number, American Petroleum Institute (API) number, location and depth of samples; and forwarded in to the state core and sample library within 30 days of the completion of drilling operations.
- 3) Cores: Any cores cut shall be preserved in correct order, boxed in standard core boxes (4.5", 4.5", 35.75"), and the entire core forwarded to the state core and samples library within 180 days of completion of drilling operations. Any extension of time must have approval on a Form 4 Sundry Notice.

All cores, core chips, and samples must be shipped, prepaid, to the state core and samples library at the following address:

**ND Geological Survey Core Library
2835 Campus Road, Stop 8156
Grand Forks, ND 58202**

North Dakota Century Code Section 38-08-16 allows for a civil penalty for any violation of Chapter 38 08 not to exceed \$12,500 for each offense, and each day's violation is a separate offense.

Sincerely

Stephen Fried
Geologist

**SUNDRY NOTICES AND REPORTS ON WELLS FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2008)



Well File No.
28658

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date October 1, 2014	☐ Drilling Prognosis	☐ Spill Report
☐ Report of Work Done	Date Work Completed	☐ Redrilling or Repair	☐ Shooting
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date	☐ Casing or Liner	☐ Acidizing
		☐ Plug Well	☐ Fracture Treatment
		☐ Supplemental History	☐ Change Production Method
		☐ Temporarily Abandon	☐ Reclamation
		☒ Other	Waiver to rule Rule 43-02-03-31

Well Name and Number Kline Federal 5300 41-18 12TX							
Footages		Qtr-Qtr	Section	Township	Range		
434 F S L 237 F W L		Lot 4	18	153 N	100 W		
Field		Pool	County				
		Bakken	McKenzie				

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s)			
Address	City	State	Zip Code

DETAILS OF WORK

Oasis Petroleum respectfully requests a waiver to Rule 43-02-03-31 in regards to running open hole logs for the above referenced well. Justification for this request is as follows:

The Oasis Petroleum's Kline Federal 5300 11-18H (153N 100W 18 NW NW) located within a mile of the subject well

If this exception is approved, Oasis Petroleum will run a CBL on the intermediate string, and we will also run GR to surface. Oasis Petroleum will also submit two digital copies of each cased hole log and a copy of the mud log containing MWD gamma ray.

Company Oasis Petroleum North America LLC		Telephone Number 281-404-9563	
Address 1001 Fannin, Suite 1500			
City Houston		State TX	Zip Code 77002
Signature <i>Heather McCowan</i>		Printed Name Heather McCowan	
Title Regulatory Assistant		Date May 29, 2014	
Email Address hmccowan@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date 6-17-2014	
By <i>Stephen Fried</i>	
Title Stephen Fried Geologist	

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 5749 (09-2006)

Well File No.
28658

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☒ Notice of Intent	Approximate Start Date October 1, 2014
☐ Report of Work Done	Date Work Completed
☐ Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.	Approximate Start Date

☐ Drilling Prognosis	☐ Spill Report
☐ Redrilling or Repair	☐ Shooting
☐ Casing or Liner	☐ Acidizing
☐ Plug Well	☐ Fracture Treatment
☐ Supplemental History	☐ Change Production Method
☐ Temporarily Abandon	☐ Reclamation
☒ Other	Suspension of Drilling

Well Name and Number Kline Federal 5300 41-18 12TX					
Footages	Qtr-Qtr	Section	Township	Range	
434 F S L 237 F WL	SWSW	18	153 N	100 W	
Field	Pool	County			
	Bakken	McKenzie			

24-HOUR PRODUCTION RATE			
Before		After	
Oil	Bbls	Oil	Bbls
Water	Bbls	Water	Bbls
Gas	MCF	Gas	MCF

Name of Contractor(s) Advanced Energy Services			
Address	City	State	Zip Code

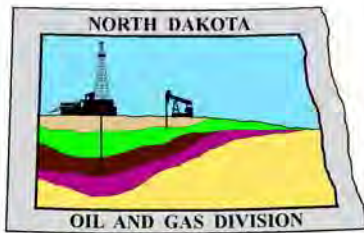
DETAILS OF WORK

Oasis Petroleum North America LLC requests permission for suspension of drilling for up to 90 days for the referenced well under NDAC 43-02-03-55. Oasis Petroleum North America LLC intends to drill the surface hole with freshwater based drilling mud and set surface casing with a small drilling rig and move off within 3 to 5 days. The casing will be set at a depth pre-approved by the NDIC per the Application for Permit to Drill NDAC 43-02-03-21. No saltwater will be used in the drilling and cementing operations of the surface casing. Once the surface casing is cemented, a plug or mechanical seal will be placed at the top of the casing to prevent any foreign matter from getting into the well. A rig capable of drilling to TD will move onto the location within the 90 days previously outlined to complete the drilling and casing plan as per the APD. The undersigned states that this request for suspension of drilling operations in accordance with the Subsection 4 of Section 43-02-03-55 of the NDAC, is being requested to take advantage of the cost savings and time savings of using an initial rig that is smaller than the rig necessary to drill a well to total depth but is not intended to alter or extend the terms and conditions of, or suspend any obligation under, any oil and gas lease with acreage in or under the spacing or drilling unit for the above-referenced well. Oasis Petroleum North America LLC understands NDAC 43-02-03-31 requirements regarding confidentiality pertaining to this permit. The drilling pit will be fenced immediately after construction if the well pad is located in a pasture (NDAC 43-02-03-19 & 19.1). Oasis Petroleum North America LL will plug and abandon the well and reclaim the well site if the well is not drilled by the larger rotary rig within 90 days after spudding the well with the smaller drilling rig.

Notify NDIC inspector **Richard Dunn** at 701-770-3554 with spud and TD info.

Company Oasis Petroleum North America LLC		Telephone Number (281) 404-9563	
Address 1001 Fannin, Suite 1500			
City Houston	State TX	Zip Code 77002	
Signature 	Printed Name Heather McCowan		
Title Regulatory Assistant	Date May 29, 2014		
Email Address hmccowan@oasispetroleum.com			

FOR STATE USE ONLY	
☐ Received	☒ Approved
Date	6/17/14
By	Nathaniel Erbele
Title	Petroleum Resource Specialist



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.oilgas.nd.gov

June 17, 2014

Heather McCowan
Regulatory Assistant
OASIS PETROLEUM NORTH AMERICA LLC
1001 Fannin Suite 1500
Houston, TX 77002

**RE: HORIZONTAL WELL
KLINE FEDERAL 5300 41-18 12TX
LOT4 Section 18-153N-100W
McKenzie County
Well File # 28658**

Dear Heather:

Pursuant to Commission Order No. 23752, approval to drill the above captioned well is hereby given. The approval is granted on the condition that all portions of the well bore not isolated by cement shall be located approximately down the east-west axis (**within 250' of the section line** per Commission policy) and no closer than **200 feet** to the east or west boundary within the 2560 acre spacing unit consisting of Sections 17, 18, 19, & 20 T153N-R100W.

PERMIT STIPULATIONS: Due to the proximity of Lake Sakakawea to the well site, a dike is required surrounding the entire location. Effective June 1, 2014, a covered leak-proof container (with placard) for filter sock disposal must be maintained on the well site beginning when the well is spud, and must remain on-site during clean-out, completion, and flow-back whenever filtration operations are conducted. OASIS PETRO NO AMER must contact NDIC Field Inspector Richard Dunn at 701-770-3554 prior to location construction.

Drilling pit

NDAC 43-02-03-19.4 states that "a pit may be utilized to bury drill cuttings and solids generated during well drilling and completion operations, providing the pit can be constructed, used and reclaimed in a manner that will prevent pollution of the land surface and freshwaters. Reserve and circulation of mud system through earthen pits are prohibited. All pits shall be inspected by an authorized representative of the director prior to lining and use. Drill cuttings and solids must be stabilized in a manner approved by the director prior to placement in a cuttings pit."

Form 1 Changes & Hard Lines

Any changes, shortening of casing point or lengthening at Total Depth must have prior approval by the NDIC. The proposed directional plan is at a legal location. The minimum legal coordinate from the well head at casing point is: 184' south. Also, based on the azimuth of the proposed lateral the maximum legal coordinate from the well head is: 10052' east.

Location Construction Commencement (Three Day Waiting Period)

Operators shall not commence operations on a drill site until the 3rd business day following publication of the approved drilling permit on the NDIC - OGD Daily Activity Report. If circumstances require operations to commence before the 3rd business day following publication on the Daily Activity Report, the waiting period may be waived by the Director. Application for a waiver must be by sworn affidavit providing the information necessary to evaluate the extenuating circumstances, the factors of NDAC 43-02-03-16.2 (1), (a)-(f), and any other information that would allow the Director to conclude that in the event another owner seeks revocation of the drilling permit, the applicant should retain the permit.

Permit Fee & Notification

Payment was received in the amount of \$100 via credit card .The permit fee has been received. It is requested that notification be given immediately upon the spudding of the well. This information should be relayed to the Oil & Gas Division, Bismarck, via telephone. The following information must be included: Well name, legal location, permit number, drilling contractor, company representative, date and time of spudding. Office hours are 8:00 a.m. to 12:00 p.m. and 1:00 p.m. to 5:00 p.m. Central Time. Our telephone number is (701) 328-8020, leave a message if after hours or on the weekend.

Survey Requirements for Horizontal, Horizontal Re-entry, and Directional Wells

NDAC Section 43-02-03-25 (Deviation Tests and Directional Surveys) states in part (that) the survey contractor shall file a certified copy of all surveys with the director free of charge within thirty days of completion. Surveys must be submitted as one electronic copy, or in a form approved by the director. However, the director may require the directional survey to be filed immediately after completion if the survey is needed to conduct the operation of the director's office in a timely manner. Certified surveys must be submitted via email in one adobe document, with a certification cover page to certsurvey@nd.gov.

Survey points shall be of such frequency to accurately determine the entire location of the well bore.

Specifically, the Horizontal and Directional well survey frequency is 100 feet in the vertical, 30 feet in the curve (or when sliding) and 90 feet in the lateral.

Surface casing cement

Tail cement utilized on surface casing must have a minimum compressive strength of 500 psi within 12 hours, and tail cement utilized on production casing must have a minimum compressive strength of 500 psi before drilling the plug or initiating tests.

Logs

NDAC Section 43-02-03-31 requires the running of (1) a suite of open hole logs from which formation tops and porosity zones can be determined, (2) a Gamma Ray Log run from total depth to ground level elevation of the well bore, and (3) a log from which the presence and quality of cement can be determined (Standard CBL or Ultrasonic cement evaluation log) in every well in which production or intermediate casing has been set, this log must be run prior to completing the well. All logs run must be submitted free of charge, as one digital TIFF (tagged image file format) copy and one digital LAS (log ASCII) formatted copy. Digital logs may be submitted on a standard CD, DVD, or attached to an email sent to digitallogs@nd.gov

Thank you for your cooperation.

Sincerely,

Nathaniel Erbele
Petroleum Resource Specialist



APPLICATION FOR PERMIT TO DRILL HORIZONTAL WELL - FORM 1H

INDUSTRIAL COMMISSION OF NORTH DAKOTA
OIL AND GAS DIVISION
600 EAST BOULEVARD DEPT 405
BISMARCK, ND 58505-0840
SFN 54269 (08-2005)

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

Type of Work New Location	Type of Well Oil & Gas	Approximate Date Work Will Start 10 / 1 / 2013	Confidential Status No
Operator OASIS PETROLEUM NORTH AMERICA LLC			Telephone Number 281-404-9563
Address 1001 Fannin Suite 1500		City Houston	State TX Zip Code 77002

☒ Notice has been provided to the owner of any permanently occupied dwelling within 1,320 feet.

☒ This well is not located within five hundred feet of an occupied dwelling.

WELL INFORMATION (If more than one lateral proposed, enter data for additional laterals on page 2)

Well Name KLINE FEDERAL				Well Number 5300 41-18 12TX			
Surface Footages 434 F S L 237 F W L		Qtr-Qtr LOT4	Section 18	Township 153 N	Range 100 W	County McKenzie	
Longstring Casing Point Footages 244 F S L 816 F W L		Qtr-Qtr LOT4	Section 18	Township 153 N	Range 100 W	County McKenzie	
Longstring Casing Point Coordinates From Well Head 190 S From WH 579 E From WH		Azimuth 105.6 °	Longstring Total Depth 11189 Feet MD 10832 Feet TVD				
Bottom Hole Footages From Nearest Section Line 200 F S L 202 F E L		Qtr-Qtr LOT4	Section 17	Township 153 N	Range 100 W	County McKenzie	
Bottom Hole Coordinates From Well Head 234 S From WH 10050 E From WH		KOP Lateral 1 10355 Feet MD	Azimuth Lateral 1 90 °		Estimated Total Depth Lateral 1 20668 Feet MD 10849 Feet TVD		
Latitude of Well Head 48 ° 04 ' 7.55 "		Longitude of Well Head -103 ° 36 ' 11.95 "		NAD Reference NAD83	Description of (Subject to NDIC Approval) Spacing Unit: Sec 17,18,19,&20 T153N-R100W		
Ground Elevation 2056 Feet Above S.L.	Acres in Spacing/Drilling Unit 2560	Spacing/Drilling Unit Setback Requirement 0 Feet N/S 200 Feet E/W			Industrial Commission Order 23752		
North Line of Spacing/Drilling Unit 10544 Feet	South Line of Spacing/Drilling Unit 9306 Feet	East Line of Spacing/Drilling Unit 10527 Feet			West Line of Spacing/Drilling Unit 10520 Feet		
Objective Horizons Three Forks B1					Pierre Shale Top 1972		
Proposed Surface Casing	Size 9 - 5/8 "	Weight 36 Lb./Ft.	Depth 2072 Feet	Cement Volume 749 Sacks	NOTE: Surface hole must be drilled with fresh water and surface casing must be cemented back to surface.		
Proposed Longstring Casing	Size 7 - "	Weight(s) 29/32 Lb./Ft.	Longstring Total Depth 11189 Feet MD 10832 Feet TVD		Cement Volume 760 Sacks	Cement Top 3971 Feet	Top Dakota Sand 5471 Feet
Base Last Charles Salt (If Applicable) 9238 Feet		NOTE: Intermediate or longstring casing string must be cemented above the top Dakota Group Sand.					
Proposed Logs Triple Combo: KOP to Kibby GR/Res to BSC GR to surf CND through the Dakota							
Drilling Mud Type (Vertical Hole - Below Surface Casing) Invert				Drilling Mud Type (Lateral) Salt Water Gel			
Survey Type in Vertical Portion of Well MWD Every 100 Feet		Survey Frequency: Build Section 30 Feet		Survey Frequency: Lateral 90 Feet		Survey Contractor Ryan	

NOTE: A Gamma Ray log must be run to ground surface and a CBL must be run on intermediate or longstring casing string if set.

Surveys are required at least every 30 feet in the build section and every 90 feet in the lateral section of a horizontal well. Measurement inaccuracies are not considered when determining compliance with the spacing/drilling unit boundary setback requirement except in the following scenarios: 1) When the angle between the well bore and the respective boundary is 10 degrees or less; or 2) If Industry standard methods and equipment are not utilized. Consult the applicable field order for exceptions.

If measurement inaccuracies are required to be considered, a 2° MWD measurement inaccuracy will be applied to the horizontal portion of the well bore. This measurement inaccuracy is applied to the well bore from KOP to TD.

REQUIRED ATTACHMENTS: Certified surveyor's plat, horizontal section plat, estimated geological tops, proposed mud/cementing plan, directional plot/plan, \$100 fee.
See Page 2 for Comments section and signature block.

COMMENTS, ADDITIONAL INFORMATION, AND/OR LIST OF ATTACHMENTS

Documents forwarded by email: Drill plan with drilling fluids, Well Summary with casing/cement plans, Directional Plan & Plot, Plats

Lateral 2

KOP Lateral 2 Feet MD	Azimuth Lateral 2 °	Estimated Total Depth Lateral 2 Feet MD	Feet TVD	KOP Coordinates From Well Head From WH		From WH
Formation Entry Point Coordinates From Well Head From WH		Bottom Hole Coordinates From Well Head From WH				
KOP Footages From Nearest Section Line F L	F L	Qtr-Qtr	Section	Township N	Range W	County
Bottom Hole Footages From Nearest Section Line F L	F L	Qtr-Qtr	Section	Township N	Range W	County

Lateral 3

KOP Lateral 3 Feet MD	Azimuth Lateral 3 °	Estimated Total Depth Lateral 3 Feet MD	Feet TVD	KOP Coordinates From Well Head From WH		From WH
Formation Entry Point Coordinates From Well Head From WH		Bottom Hole Coordinates From Well Head From WH				
KOP Footages From Nearest Section Line F L	F L	Qtr-Qtr	Section	Township N	Range W	County
Bottom Hole Footages From Nearest Section Line F L	F L	Qtr-Qtr	Section	Township N	Range W	County

Lateral 4

KOP Lateral 4 Feet MD	Azimuth Lateral 4 °	Estimated Total Depth Lateral 4 Feet MD	Feet TVD	KOP Coordinates From Well Head From WH		From WH
Formation Entry Point Coordinates From Well Head From WH		Bottom Hole Coordinates From Well Head From WH				
KOP Footages From Nearest Section Line F L	F L	Qtr-Qtr	Section	Township N	Range W	County
Bottom Hole Footages From Nearest Section Line F L	F L	Qtr-Qtr	Section	Township N	Range W	County

Lateral 5

KOP Lateral 5 Feet MD	Azimuth Lateral 5 °	Estimated Total Depth Lateral 5 Feet MD	Feet TVD	KOP Coordinates From Well Head From WH		From WH
Formation Entry Point Coordinates From Well Head From WH		Bottom Hole Coordinates From Well Head From WH				
KOP Footages From Nearest Section Line F L	F L	Qtr-Qtr	Section	Township N	Range W	County
Bottom Hole Footages From Nearest Section Line F L	F L	Qtr-Qtr	Section	Township N	Range W	County

I hereby swear or affirm the information provided is true, complete and correct as determined from all available records.

Date

5 / 29 / 2014

ePermit

Printed Name

Heather McCowan

Title

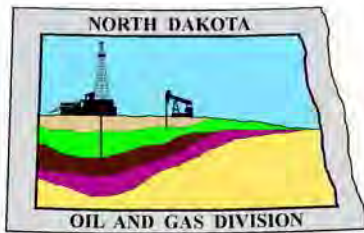
Regulatory Assistant

FOR STATE USE ONLY

Permit and File Number 28658	API Number 33 - 053 - 06030
Field BAKER	
Pool BAKKEN	Permit Type DEVELOPMENT

FOR STATE USE ONLY

Date Approved 6 / 17 / 2014
By Nathaniel Erbele
Title Petroleum Resource Specialist



Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

www.oilgas.nd.gov

April 9, 2014

**RE: Filter Socks and Other Filter Media
Leakproof Container Required
Oil and Gas Wells**

Dear Operator,

North Dakota Administrative Code Section 43-02-03-19.2 states in part that all waste material associated with exploration or production of oil and gas must be properly disposed of in an authorized facility in accord with all applicable local, state, and federal laws and regulations.

Filtration systems are commonly used during oil and gas operations in North Dakota. The Commission is very concerned about the proper disposal of used filters (including filter socks) used by the oil and gas industry.

Effective June 1, 2014, a container must be maintained on each well drilled in North Dakota beginning when the well is spud and must remain on-site during clean-out, completion, and flow-back whenever filtration operations are conducted. The on-site container must be used to store filters until they can be properly disposed of in an authorized facility. Such containers must be:

- leakproof to prevent any fluids from escaping the container
- covered to prevent precipitation from entering the container
- placard to indicate only filters are to be placed in the container

If the operator will not utilize a filtration system, a waiver to the container requirement will be considered, but only upon the operator submitting a Sundry Notice (Form 4) justifying their request.

As previously stated in our March 13, 2014 letter, North Dakota Administrative Code Section 33-20-02.1-01 states in part that every person who transports solid waste (which includes oil and gas exploration and production wastes) is required to have a valid permit issued by the North Dakota Department of Health, Division of Waste Management. Please contact the Division of Waste Management at (701) 328-5166 with any questions on the solid waste program. Note oil and gas exploration and production wastes include produced water, drilling mud, invert mud, tank bottom sediment, pipe scale, filters, and fly ash.

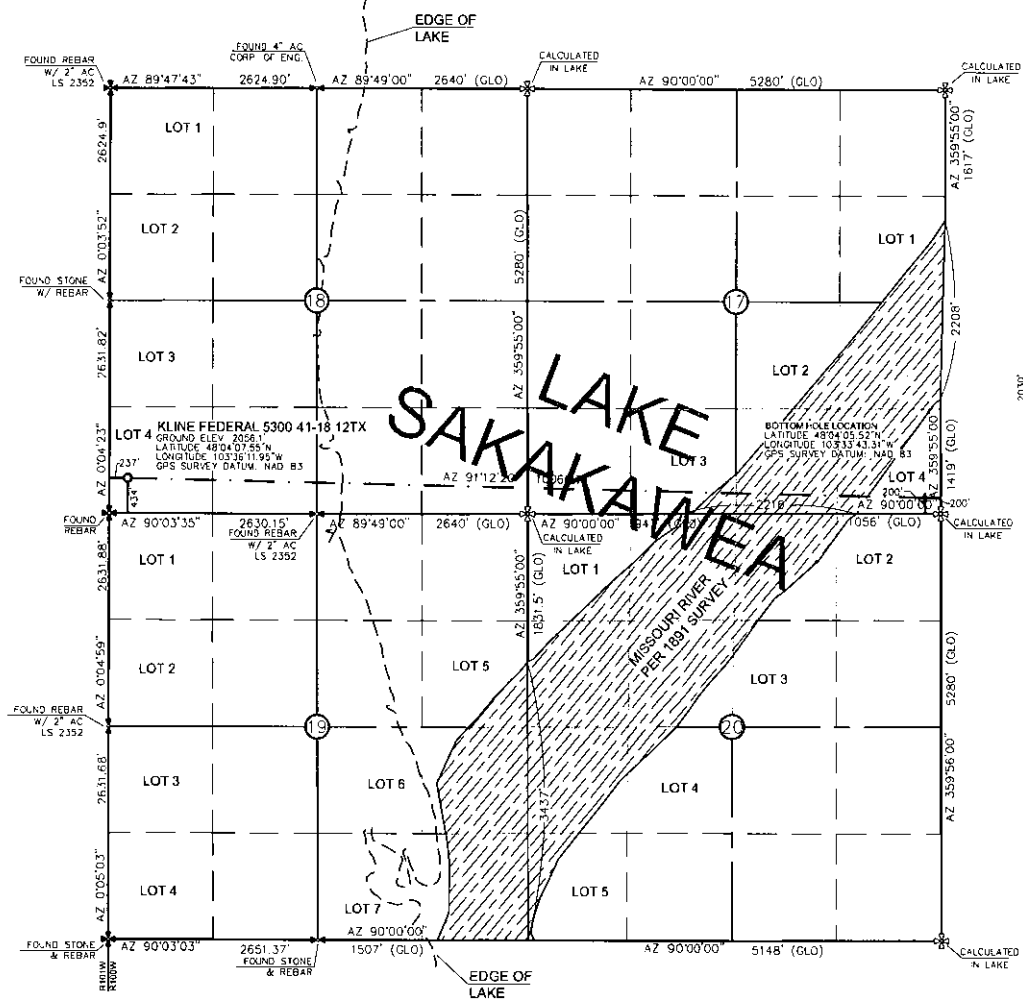
Thank you for your cooperation.

Sincerely,

Bruce E. Hicks

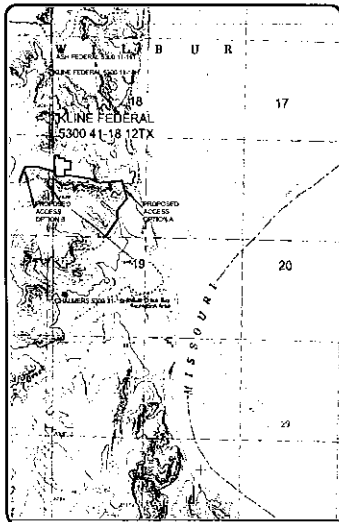
Assistant Director

WELL LOCATION PLAT
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "KLINE FEDERAL 5300 41-18 12TX"
 434 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE
 SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



- MONUMENT - RECOVERED
 - MONUMENT - NOT RECOVERED

VICINITY MAP



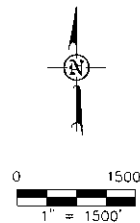
THIS DOCUMENT WAS ORIGINALLY ISSUED
 AND SEALED BY ROBERT L. PROCIWE,
 PLS, REGISTRATION NUMBER 2884 ON
 4/15/14 AND THE ORIGINAL
 DOCUMENTS ARE STORED AT THE
 OFFICES OF INTERSTATE ENGINEERING,
 INC.

STAKED ON 4/15/14
 VERTICAL CONTROL DATUM WAS BASED UPON
 CONTROL POINT 4 WITH AN ELEVATION OF 2090.8'

THIS SURVEY AND PLAT IS BEING PROVIDED AT THE REQUEST
 OF ERIC BAYES OF OASIS PETROLEUM. I CERTIFY THAT THIS
 PLAT CORRECTLY REPRESENTS WORK PERFORMED BY ME OR
 UNDER MY SUPERVISION AND IS TRUE AND CORRECT TO THE
 BEST OF MY KNOWLEDGE AND BELIEF.

Robert L. Prociwe
 ROBERT L. PROCIWE 2884LS

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1/8
 SHEET NO.



Interstate Engineering, Inc.
 P.O. Box 648
 425 East Main Street
 Sidney, Montana 59270
 Ph: (406) 433-5617
 Fax: (406) 433-5618
 www.interstateeng.com
 Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
 WELL LOCATION PLAT
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA
 Drawn By: B.H.M.
 Checked By: R.L.P.
 Project No.: 314-09-081-02
 Date: APRIL 2014

Revision No.	Date	By	Description

DRILLING PLAN							
OPERATOR	Oasis Petroleum		COUNTY/STATE	McKenzie Co., ND			
WELL NAME	Kline Federal 5300 41-18 12TX		RIG	Nabors B22			
WELL TYPE	Horizontal Upper Three Forks						
LOCATION	SWSW 18-153N-100W		Surface Location (survey plat):	434' fsl 237' fwl			
EST. T.D.	20,668'		GROUND ELEV:	2057 Finished Pad Elev. Sub Height: 25			
TOTAL LATERA		9,479'	KB ELEV:	2082			
PROGNOSIS:		Based on 2,082' KB(est)					
MARKER		DEPTH (Surf Loc)	DATUM (Surf Loc)	LOGS: Type Interval OH Logs: Triple Combo KOP to Kibby (or min run of 1800' whichever is greater); GR/Res to BSC; GR to surf; CND through the Dakota CBL/GR: Above top of cement/GR to base of casing MWD GR: KOP to lateral TD			
Pierre		NDIC MAP	1,972 110'				
Greenhorn			4,646 -2,564'	DEVIATION: Surf: 3 deg. max., 1 deg / 100'; srvy every 500' Prod: 5 deg. max., 1 deg / 100'; srvy every 100'			
Mowry			5,057 -2,975'				
Dakota			5,471 -3,389'	DST'S: None planned			
Rierdon			6,486 -4,404'				
Dunham Salt			6,814 -4,732'	CORES: None planned			
Dunham Salt Base			6,929 -4,847'				
Spearfish			7,024 -4,942'	MUDLOGGING: Two-Man: 8,342' ~200' above the Charles (Kibbey) to Casing point; Casing point to TD 30' samples at direction of wellsite geologist; 10' through target @ curve land			
Pine Salt			7,283 -5,201'				
Pine Salt Base			7,318 -5,236'	BOP: 11" 5000 psi blind, pipe & annular			
Opeche Salt			7,374 -5,292'				
Opeche Salt Base			7,403 -5,321'	Dip Rate: -0.1			
Broom Creek (Top of Minnelusa Gp.)			7,605 -5,523'				
Amsden			7,684 -5,602'				
Tyler			7,853 -5,771'				
Otter (Base of Minnelusa Gp.)			8,047 -5,965'				
Kibbey Lime			8,390 -6,308'				
Charles Salt			8,542 -6,460'				
UB			9,159 -7,077'				
Base Last Salt			9,238 -7,156'				
Ratcliffe			9,301 -7,219'				
Mission Canyon			9,454 -7,372'				
Lodgepole			10,022 -7,940'				
Lodgepole Fracture Zone			10,209 -8,127'				
False Bakken			10,726 -8,644'				
Upper Bakken			10,735 -8,653'				
Middle Bakken			10,751 -8,669'				
Lower Bakken			10,785 -8,703'				
Pronghorn			10,791 -8,709'				
Three Forks			10,813 -8,731'				
TF Target Top			10,827 -8,745'				
TF Target Base			10,837 -8,755'				
Claystone			10,838 -8,756'				
Max. Anticipated BHP:		4692					
Surface Formation:		Glacial till					
MUD:	Interval	Type	WT	Vis	WL	Remarks	
Surface:	0' -	2,072'	FW/Gel - Lime Sweeps	8.4-9.0	28-32	NC	Circ Mud Tanks
Intermediate:	2,072' -	11,189'	Invert	9.5-10.4	40-50	30+HtHp	Circ Mud Tanks
Lateral:	11,189' -	20,668'	Salt Water	9.8-10.2	28-32	NC	Circ Mud Tanks
CASING:	Size	Wt ppf	Hole	Depth	Cement	WOC	Remarks
Surface:	9-5/8"	36#	13-1/2"	2,072'	To Surface	12	100' into Pierre
Intermediate:	7"	29/32#	8-3/4"	11,189'	3971	24	1500' above Dakota
Production Liner:	4.5"	11.6#	6"	20,668'	TOL @ 10,305'		50' above KOP
PROBABLE PLUGS, IF REQ'D:							
OTHER:	MD	TVD	FNL/FSL	FEL/FWL	S-T-R	AZI	
Surface:	2,072	2,072	434' FSL	237' FWL	SEC 18-T153N-R100W		Survey Company:
KOP:	10,355'	10,355'	434' FSL	287' FWL	SEC 18-T153N-R100W		Build Rate: 12 deg /100'
EOC:	11,104'	10,832'	306' FSL	746' FWL	SEC 18-T153N-R100W	110.00	
Casing Point:	11,189'	10,832'	244' FSL	816' FWL	SEC 18-T153N-R100W	107.02	
Upper Three Forks Lateral TD:	20,668'	10,849'	200' FSL	200' FEL	SEC 17-T153N-R100W	89.82	
Comments:							
Request a Sundry for an Open Hole Log Waiver							
Exception well: Oasis Petroleum's Kline Federal 5300 11-18H (153N 100W 18 NW NW)							
35 packers, 35 sleeves, no frac string							
Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.							
68334-30-5 (Primary Name: Fuels, diesel) 68476-34-6 (Primary Name: Fuels, diesel, No. 2)68476-30-2 (Primary Name: Fuel oil No. 2)							
68476-31-3 (Primary Name: Fuel oil, No. 4)8008-20-6 (Primary Name: Kerosene)							
OASIS PETROLEUM							
Geology: M.Steed 4/23/2014				Engineering: hlbader rpm 6/17/14			

**Oasis Petroleum
Well Summary
Kline Federal 5300 41-18 12TX
Section 18 T153N R100W
McKenzie County, ND**

SURFACE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
9-5/8"	0' to 2,072'	36	J-55	LTC	8.921"	8.765"	3400	4530	5660

Interval	Description	Collapse	Burst	Tension	Cost per ft
		(psi) a	(psi) b	(1000 lbs) c	
0' to 2,072'	9-5/8", 36#, J-55, LTC, 8rd	2020 / 2.08	3520 / 3.63	453 / 2.76	

API Rating & Safety Factor

- a) Based on full casing evacuation with 9 ppg fluid on backside (2,072' setting depth).
- b) Burst pressure based on 9 ppg fluid with no fluid on backside (2,072' setting depth).
- c) Based on string weight in 9 ppg fluid at 2,072' TVD plus 100k# overpull. (Buoyed weight equals 64k lbs.)

Cement volumes are based on 9-5/8" casing set in 13-1/2" hole with 60% excess to circulate cement back to surface. Mix and pump the following slurry.

Pre-flush (Spacer): **20 bbls** fresh water

Lead Slurry: **449 sks** (232 bbls) Conventional system with 94 lb/sk cement, 4% extender, 2% expanding agent, 2% CaCl₂ and 0.25 lb/sk lost circulation control agent

Tail Slurry: **300 sks** (62 bbls) Conventional system with 94 lb/sk cement, 3% NaCl, and 0.25 lb/sk lost circulation control agent

**Oasis Petroleum
Well Summary
Kline Federal 5300 41-18 12TX
Section 18 T153N R100W
McKenzie County, ND**

INTERMEDIATE CASING AND CEMENT DESIGN

Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Make-up Torque (ft-lbs)		
							Minimum	Optimum	Max
7"	0' – 6,614'	29	P-110	LTC	6.184"	6.059"	5,980	7,970	8,770
7"	6,614' – 10,355'	32	HCP-110	LTC	6.094"	6.000***	6,730	8,970	9,870
7"	10,355' – 11,189'	29	P-110	LTC	6.184"	6.059"	5,980	7,970	8,770

***Special drift

Interval	Length	Description	Collapse	Burst	Tension
			(psi) a	(psi) b	(1000 lbs) c
0' – 6,614'	6,614'	7", 29#, P-110, LTC, 8rd	8530 / 2.48*	11220 / 1.19	797 / 2.08
6,614' – 10,355'	3,741'	7", 32#, HCP-110, LTC, 8rd	11820 / 2.19*	12460 / 1.29	
6,614' – 10,355'	3,741'	7", 32#, HCP-110, LTC, 8rd	11820 / 1.07**	12460 / 1.29	
10,355' – 11,189'	834'	7", 29 lb, P-110, LTC, 8rd	8530 / 1.51*	11220 / 1.16	

API Rating & Safety Factor

- a) *Assume full casing evacuation with 10 ppg fluid on backside
 **Assume full casing evacuation with 1.2 psi/ft equivalent fluid gradient across salt intervals. (Bottom of last salt 9,238' TVD)
- b) Burst pressure based on 9000 psig max press for stimulation plus 10.2 ppg fluid in casing and 9 ppg fluid on backside-to 10,832' TVD.
- c) Based on string weight in 10 ppg fluid, (284k lbs buoyed weight) plus 100k

Cement volumes are estimates based on 7" casing set in an 8-3/4" hole with 30% excess.

Pre-flush (Spacer): **170 bbls** Saltwater
 20 bbls CW8 System
 10 bbls Fresh Water

Lead Slurry: **177 sks** (82 bbls) Conventional system with 47 lb/sk cement, 10% NaCl, 34 lb/sk extender, 10% D020 extender, 1% D079 extender, 1% anti-settling agent, 1% fluid loss agent, 0.2% anti-foam agent, 0.7% retarder, 0.125 lb/sk lost circulation control agent, and 0.3% dispersant

Tail Slurry: **593 sks** (173 bbls) Conventional system with 94 lb/sk cement, 10% NaCl, 35% Silica, 0.2% fluid loss agent, 0.8% dispersant, 0.125 lb/sk lost circulation control agent and 0.3% retarder

**Oasis Petroleum
Well Summary
Kline Federal 5300 41-18 12TX
Section 18 T153N R100W
McKenzie County, ND**

PRODUCTION LINER

							Make-up Torque (ft-lbs)		
Size	Interval	Weight	Grade	Coupling	I.D.	Drift	Minimum	Optimum	Max
4-1/2"	10,305' – 20,668'	11.6	P-110	BTC	4.000"	3.875"			

Interval	Description	Collapse	Burst	Tension	Cost per ft
		(psi) a	(psi) b	(1000 lbs) c	
10,305' – 20,668'	4-1/2", 11.6 lb, P-110, BTC, 8rd	7560 / 1.41	10690 / 1.10	385 / 1.90	

API Rating & Safety Factor

- a) Based on full casing evacuation with 9.5 ppg fluid on backside @ 10,832' TVD.
Burst pressure based on 9000 psi treating pressure with 10.2 ppg internal fluid gradient and 9 ppg external
- b) fluid gradient @ 10,832' TVD.
- c) Based on string weight in 9.5 ppg fluid (Buoyed weight: 103k lbs.) plus 100k lbs overpull.



Oasis Petroleum

Indian Hills

153N-100W-17/18

Kline Federal 5300 41-18 12TX

Kline Federal 5300 41-18 12TX

Plan: Design #2

Standard Planning Report

17 June, 2014

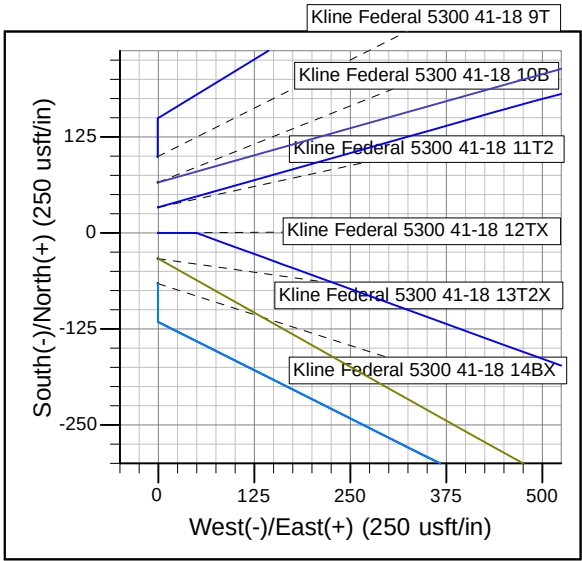
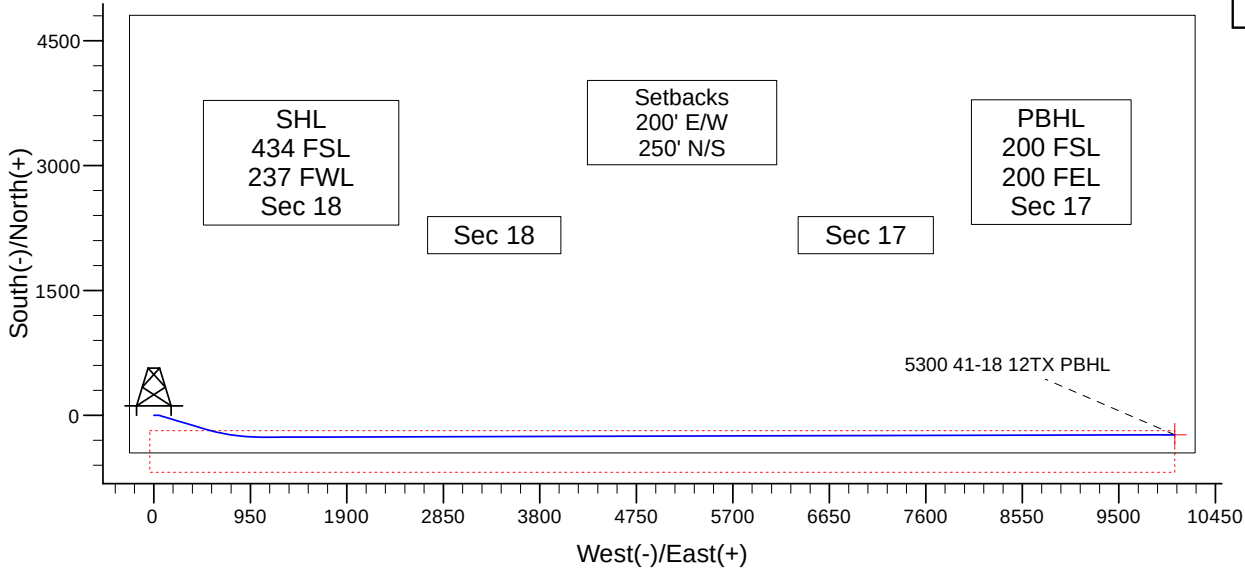


**RYAN DIRECTIONAL
SERVICES**
A NABORS COMPANY

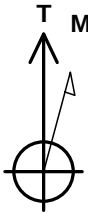
Project: Indian Hills
Site: 153N-100W-17/18
Well: Kline Federal 5300 41-18 12TX
Wellbore: Kline Federal 5300 41-18 12TX
Design: Design #2



WELL DETAILS: Kline Federal 5300 41-18 12TX			
Ground Level: 2057.0			
Northing	Easting	Latitude	Longitude
405136.64	1209968.21	48° 4' 7.550 N	103° 36' 11.950 W

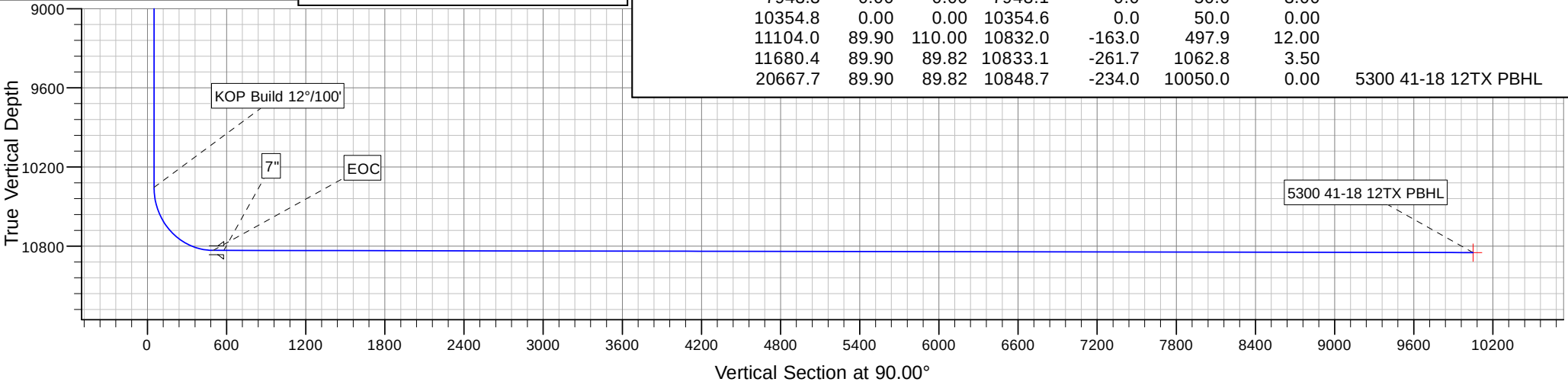


CASING DETAILS			
TVD	MD	Name	Size
10832.2	11189.0	7"	7"



Azimuths to True North
Magnetic North: 8.27°
Magnetic Field
Strength: 56437.1snT
Dip Angle: 72.99°
Date: 5/23/2014
Model: IGRF2010

SECTION DETAILS							
MD	Inc	Azi	TVD	+N/-S	+E/-W	Dleg	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	
2200.0	0.00	0.00	2200.0	0.0	0.0	0.00	
2216.7	0.50	90.00	2216.7	0.0	0.1	3.00	
7926.7	0.50	90.00	7926.4	0.0	49.9	0.00	
7943.3	0.00	0.00	7943.1	0.0	50.0	3.00	
10354.8	0.00	0.00	10354.6	0.0	50.0	0.00	
11104.0	89.90	110.00	10832.0	-163.0	497.9	12.00	
11680.4	89.90	89.82	10833.1	-261.7	1062.8	3.50	
20667.7	89.90	89.82	10848.7	-234.0	10050.0	0.00	5300 41-18 12TX PBHL



Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 41-18 12TX		
Design:	Design #2		

Project	Indian Hills		
Map System:	US State Plane 1983	System Datum:	Mean Sea Level
Geo Datum:	North American Datum 1983		
Map Zone:	North Dakota Northern Zone		

Site		153N-100W-17/18			
Site Position:		Northing:	408,962.44 usft	Latitude:	48° 4' 45.380 N
From:	Lat/Long	Easting:	1,210,229.18 usft	Longitude:	103° 36' 10.380 W
Position Uncertainty:	0.0 usft	Slot Radius:	13-3/16 "	Grid Convergence:	-2.31 °

Well	Kline Federal 5300 41-18 12TX					
Well Position	+N/-S	-3,833.2 usft	Northing:	405,136.63 usft	Latitude:	48° 4' 7.550 N
	+E/-W	-106.6 usft	Easting:	1,209,968.21 usft	Longitude:	103° 36' 11.950 W
Position Uncertainty	2.0 usft		Wellhead Elevation:		Ground Level:	2,057.0 usft

Wellbore	Kline Federal 5300 41-18 12TX				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	IGRF2010	5/23/2014	8.27	72.99	56,437

Design	Design #2			
Audit Notes:				
Version:	Phase:	PROTOTYPE	Tie On Depth:	0.0
Vertical Section:	Depth From (TVD) (usft)	+N/-S (usft)	+E/-W (usft)	Direction (°)
	0.0	0.0	0.0	90.00

Plan Sections										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)	TFO (°)	Target
0.0	0.00	0.00	0.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,200.0	0.00	0.00	2,200.0	0.0	0.0	0.00	0.00	0.00	0.00	
2,216.7	0.50	90.00	2,216.7	0.0	0.1	3.00	3.00	0.00	90.00	
7,926.7	0.50	90.00	7,926.4	0.0	49.9	0.00	0.00	0.00	0.00	
7,943.3	0.00	0.00	7,943.1	0.0	50.0	3.00	-3.00	0.00	180.00	
10,354.8	0.00	0.00	10,354.6	0.0	50.0	0.00	0.00	0.00	0.00	
11,104.0	89.90	110.00	10,832.0	-163.0	497.9	12.00	12.00	0.00	110.00	
11,680.4	89.90	89.82	10,833.1	-261.7	1,062.8	3.50	0.00	-3.50	-90.02	
20,667.7	89.90	89.82	10,848.7	-234.0	10,050.0	0.00	0.00	0.00	0.00	5300 41-18 12TX PBI

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 41-18 12TX		
Design:	Design #2		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
2,200.0	0.00	0.00	2,200.0	0.0	0.0	0.0	0.00	0.00	0.00
2,216.7	0.50	90.00	2,216.7	0.0	0.1	0.1	3.00	3.00	0.00
2,300.0	0.50	90.00	2,300.0	0.0	0.8	0.8	0.00	0.00	0.00
2,400.0	0.50	90.00	2,400.0	0.0	1.7	1.7	0.00	0.00	0.00
2,500.0	0.50	90.00	2,500.0	0.0	2.5	2.5	0.00	0.00	0.00
2,600.0	0.50	90.00	2,600.0	0.0	3.4	3.4	0.00	0.00	0.00
2,700.0	0.50	90.00	2,700.0	0.0	4.3	4.3	0.00	0.00	0.00
2,800.0	0.50	90.00	2,800.0	0.0	5.2	5.2	0.00	0.00	0.00
2,900.0	0.50	90.00	2,900.0	0.0	6.0	6.0	0.00	0.00	0.00
3,000.0	0.50	90.00	3,000.0	0.0	6.9	6.9	0.00	0.00	0.00
3,100.0	0.50	90.00	3,100.0	0.0	7.8	7.8	0.00	0.00	0.00
3,200.0	0.50	90.00	3,200.0	0.0	8.7	8.7	0.00	0.00	0.00
3,300.0	0.50	90.00	3,300.0	0.0	9.5	9.5	0.00	0.00	0.00
3,400.0	0.50	90.00	3,400.0	0.0	10.4	10.4	0.00	0.00	0.00
3,500.0	0.50	90.00	3,500.0	0.0	11.3	11.3	0.00	0.00	0.00
3,600.0	0.50	90.00	3,599.9	0.0	12.1	12.1	0.00	0.00	0.00
3,700.0	0.50	90.00	3,699.9	0.0	13.0	13.0	0.00	0.00	0.00
3,800.0	0.50	90.00	3,799.9	0.0	13.9	13.9	0.00	0.00	0.00
3,900.0	0.50	90.00	3,899.9	0.0	14.8	14.8	0.00	0.00	0.00
4,000.0	0.50	90.00	3,999.9	0.0	15.6	15.6	0.00	0.00	0.00
4,100.0	0.50	90.00	4,099.9	0.0	16.5	16.5	0.00	0.00	0.00
4,200.0	0.50	90.00	4,199.9	0.0	17.4	17.4	0.00	0.00	0.00
4,300.0	0.50	90.00	4,299.9	0.0	18.3	18.3	0.00	0.00	0.00
4,400.0	0.50	90.00	4,399.9	0.0	19.1	19.1	0.00	0.00	0.00
4,500.0	0.50	90.00	4,499.9	0.0	20.0	20.0	0.00	0.00	0.00
4,600.0	0.50	90.00	4,599.9	0.0	20.9	20.9	0.00	0.00	0.00
4,646.1	0.50	90.00	4,646.0	0.0	21.3	21.3	0.00	0.00	0.00
Greenhorn									
4,700.0	0.50	90.00	4,699.9	0.0	21.7	21.7	0.00	0.00	0.00
4,800.0	0.50	90.00	4,799.9	0.0	22.6	22.6	0.00	0.00	0.00
4,900.0	0.50	90.00	4,899.9	0.0	23.5	23.5	0.00	0.00	0.00
5,000.0	0.50	90.00	4,999.9	0.0	24.4	24.4	0.00	0.00	0.00
5,057.1	0.50	90.00	5,057.0	0.0	24.9	24.9	0.00	0.00	0.00
Mowry									
5,100.0	0.50	90.00	5,099.9	0.0	25.2	25.2	0.00	0.00	0.00
5,200.0	0.50	90.00	5,199.9	0.0	26.1	26.1	0.00	0.00	0.00
5,300.0	0.50	90.00	5,299.9	0.0	27.0	27.0	0.00	0.00	0.00
5,400.0	0.50	90.00	5,399.9	0.0	27.9	27.9	0.00	0.00	0.00
5,471.1	0.50	90.00	5,471.0	0.0	28.5	28.5	0.00	0.00	0.00
Dakota									
5,500.0	0.50	90.00	5,499.9	0.0	28.7	28.7	0.00	0.00	0.00
5,600.0	0.50	90.00	5,599.9	0.0	29.6	29.6	0.00	0.00	0.00
5,700.0	0.50	90.00	5,699.9	0.0	30.5	30.5	0.00	0.00	0.00
5,800.0	0.50	90.00	5,799.9	0.0	31.3	31.3	0.00	0.00	0.00
5,900.0	0.50	90.00	5,899.9	0.0	32.2	32.2	0.00	0.00	0.00
6,000.0	0.50	90.00	5,999.9	0.0	33.1	33.1	0.00	0.00	0.00
6,100.0	0.50	90.00	6,099.9	0.0	34.0	34.0	0.00	0.00	0.00
6,200.0	0.50	90.00	6,199.8	0.0	34.8	34.8	0.00	0.00	0.00
6,300.0	0.50	90.00	6,299.8	0.0	35.7	35.7	0.00	0.00	0.00
6,400.0	0.50	90.00	6,399.8	0.0	36.6	36.6	0.00	0.00	0.00
6,486.2	0.50	90.00	6,486.0	0.0	37.3	37.3	0.00	0.00	0.00
Rierdon									
6,500.0	0.50	90.00	6,499.8	0.0	37.5	37.5	0.00	0.00	0.00

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 41-18 12TX		
Design:	Design #2		

Planned Survey										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)	
6,600.0	0.50	90.00	6,599.8	0.0	38.3	38.3	0.00	0.00	0.00	
6,700.0	0.50	90.00	6,699.8	0.0	39.2	39.2	0.00	0.00	0.00	
6,800.0	0.50	90.00	6,799.8	0.0	40.1	40.1	0.00	0.00	0.00	
6,814.2	0.50	90.00	6,814.0	0.0	40.2	40.2	0.00	0.00	0.00	
Dunham Salt										
6,900.0	0.50	90.00	6,899.8	0.0	40.9	40.9	0.00	0.00	0.00	
6,929.2	0.50	90.00	6,929.0	0.0	41.2	41.2	0.00	0.00	0.00	
Dunham Salt Base										
7,000.0	0.50	90.00	6,999.8	0.0	41.8	41.8	0.00	0.00	0.00	
7,024.2	0.50	90.00	7,024.0	0.0	42.0	42.0	0.00	0.00	0.00	
Spearfish										
7,100.0	0.50	90.00	7,099.8	0.0	42.7	42.7	0.00	0.00	0.00	
7,200.0	0.50	90.00	7,199.8	0.0	43.6	43.6	0.00	0.00	0.00	
7,283.2	0.50	90.00	7,283.0	0.0	44.3	44.3	0.00	0.00	0.00	
Pine Salt										
7,300.0	0.50	90.00	7,299.8	0.0	44.4	44.4	0.00	0.00	0.00	
7,318.2	0.50	90.00	7,318.0	0.0	44.6	44.6	0.00	0.00	0.00	
Pine Salt Base										
7,374.2	0.50	90.00	7,374.0	0.0	45.1	45.1	0.00	0.00	0.00	
Opeche Salt										
7,400.0	0.50	90.00	7,399.8	0.0	45.3	45.3	0.00	0.00	0.00	
7,403.2	0.50	90.00	7,403.0	0.0	45.3	45.3	0.00	0.00	0.00	
Opeche Salt Base										
7,500.0	0.50	90.00	7,499.8	0.0	46.2	46.2	0.00	0.00	0.00	
7,600.0	0.50	90.00	7,599.8	0.0	47.1	47.1	0.00	0.00	0.00	
7,605.2	0.50	90.00	7,605.0	0.0	47.1	47.1	0.00	0.00	0.00	
Broom Creek (Top of Minnelusa Gp.)										
7,684.2	0.50	90.00	7,684.0	0.0	47.8	47.8	0.00	0.00	0.00	
Amsden										
7,700.0	0.50	90.00	7,699.8	0.0	47.9	47.9	0.00	0.00	0.00	
7,800.0	0.50	90.00	7,799.8	0.0	48.8	48.8	0.00	0.00	0.00	
7,853.2	0.50	90.00	7,853.0	0.0	49.3	49.3	0.00	0.00	0.00	
Tyler										
7,900.0	0.50	90.00	7,899.8	0.0	49.7	49.7	0.00	0.00	0.00	
7,926.7	0.50	90.00	7,926.4	0.0	49.9	49.9	0.00	0.00	0.00	
7,943.3	0.00	0.00	7,943.1	0.0	50.0	50.0	3.00	-3.00	0.00	
8,000.0	0.00	0.00	7,999.8	0.0	50.0	50.0	0.00	0.00	0.00	
8,047.2	0.00	0.00	8,047.0	0.0	50.0	50.0	0.00	0.00	0.00	
Otter (Base of Minnelusa Gp.)										
8,100.0	0.00	0.00	8,099.8	0.0	50.0	50.0	0.00	0.00	0.00	
8,200.0	0.00	0.00	8,199.8	0.0	50.0	50.0	0.00	0.00	0.00	
8,300.0	0.00	0.00	8,299.8	0.0	50.0	50.0	0.00	0.00	0.00	
8,390.2	0.00	0.00	8,390.0	0.0	50.0	50.0	0.00	0.00	0.00	
Kibbey Lime										
8,400.0	0.00	0.00	8,399.8	0.0	50.0	50.0	0.00	0.00	0.00	
8,500.0	0.00	0.00	8,499.8	0.0	50.0	50.0	0.00	0.00	0.00	
8,542.2	0.00	0.00	8,542.0	0.0	50.0	50.0	0.00	0.00	0.00	
Charles Salt										
8,600.0	0.00	0.00	8,599.8	0.0	50.0	50.0	0.00	0.00	0.00	
8,700.0	0.00	0.00	8,699.8	0.0	50.0	50.0	0.00	0.00	0.00	
8,800.0	0.00	0.00	8,799.8	0.0	50.0	50.0	0.00	0.00	0.00	
8,900.0	0.00	0.00	8,899.8	0.0	50.0	50.0	0.00	0.00	0.00	

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 41-18 12TX		
Design:	Design #2		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
9,000.0	0.00	0.00	8,999.8	0.0	50.0	50.0	0.00	0.00	0.00
9,100.0	0.00	0.00	9,099.8	0.0	50.0	50.0	0.00	0.00	0.00
9,159.2	0.00	0.00	9,159.0	0.0	50.0	50.0	0.00	0.00	0.00
UB									
9,200.0	0.00	0.00	9,199.8	0.0	50.0	50.0	0.00	0.00	0.00
9,238.2	0.00	0.00	9,238.0	0.0	50.0	50.0	0.00	0.00	0.00
Base Last Salt									
9,300.0	0.00	0.00	9,299.8	0.0	50.0	50.0	0.00	0.00	0.00
9,301.2	0.00	0.00	9,301.0	0.0	50.0	50.0	0.00	0.00	0.00
Ratcliffe									
9,400.0	0.00	0.00	9,399.8	0.0	50.0	50.0	0.00	0.00	0.00
9,454.2	0.00	0.00	9,454.0	0.0	50.0	50.0	0.00	0.00	0.00
Mission Canyon									
9,500.0	0.00	0.00	9,499.8	0.0	50.0	50.0	0.00	0.00	0.00
9,600.0	0.00	0.00	9,599.8	0.0	50.0	50.0	0.00	0.00	0.00
9,700.0	0.00	0.00	9,699.8	0.0	50.0	50.0	0.00	0.00	0.00
9,800.0	0.00	0.00	9,799.8	0.0	50.0	50.0	0.00	0.00	0.00
9,900.0	0.00	0.00	9,899.8	0.0	50.0	50.0	0.00	0.00	0.00
10,000.0	0.00	0.00	9,999.8	0.0	50.0	50.0	0.00	0.00	0.00
10,022.2	0.00	0.00	10,022.0	0.0	50.0	50.0	0.00	0.00	0.00
Lodgepole									
10,100.0	0.00	0.00	10,099.8	0.0	50.0	50.0	0.00	0.00	0.00
10,200.0	0.00	0.00	10,199.8	0.0	50.0	50.0	0.00	0.00	0.00
10,209.2	0.00	0.00	10,209.0	0.0	50.0	50.0	0.00	0.00	0.00
Lodgepole Fracture Zone									
10,300.0	0.00	0.00	10,299.8	0.0	50.0	50.0	0.00	0.00	0.00
10,354.8	0.00	0.00	10,354.6	0.0	50.0	50.0	0.00	0.00	0.00
KOP Build 12°/100'									
10,375.0	2.42	110.00	10,374.8	-0.1	50.4	50.4	12.00	12.00	0.00
10,400.0	5.42	110.00	10,399.7	-0.7	52.0	52.0	12.00	12.00	0.00
10,425.0	8.42	110.00	10,424.5	-1.8	54.8	54.8	12.00	12.00	0.00
10,450.0	11.42	110.00	10,449.2	-3.2	58.9	58.9	12.00	12.00	0.00
10,475.0	14.42	110.00	10,473.5	-5.1	64.1	64.1	12.00	12.00	0.00
10,500.0	17.42	110.00	10,497.6	-7.5	70.6	70.6	12.00	12.00	0.00
10,525.0	20.42	110.00	10,521.2	-10.3	78.2	78.2	12.00	12.00	0.00
10,550.0	23.42	110.00	10,544.4	-13.5	86.9	86.9	12.00	12.00	0.00
10,575.0	26.42	110.00	10,567.1	-17.1	96.8	96.8	12.00	12.00	0.00
10,600.0	29.42	110.00	10,589.1	-21.1	107.8	107.8	12.00	12.00	0.00
10,625.0	32.42	110.00	10,610.6	-25.5	119.9	119.9	12.00	12.00	0.00
10,650.0	35.42	110.00	10,631.3	-30.2	133.0	133.0	12.00	12.00	0.00
10,675.0	38.42	110.00	10,651.3	-35.4	147.1	147.1	12.00	12.00	0.00
10,700.0	41.42	110.00	10,670.5	-40.9	162.2	162.2	12.00	12.00	0.00
10,725.0	44.42	110.00	10,688.8	-46.7	178.2	178.2	12.00	12.00	0.00
10,750.0	47.42	110.00	10,706.2	-52.8	195.1	195.1	12.00	12.00	0.00
10,775.0	50.42	110.00	10,722.6	-59.3	212.8	212.8	12.00	12.00	0.00
10,780.4	51.07	110.00	10,726.0	-60.7	216.7	216.7	12.00	12.00	0.00
False Bakken									
10,795.0	52.82	110.00	10,735.0	-64.6	227.5	227.5	12.00	12.00	0.00
Upper Bakken									
10,800.0	53.42	110.00	10,738.0	-66.0	231.3	231.3	12.00	12.00	0.00
10,822.5	56.13	110.00	10,751.0	-72.3	248.6	248.6	12.00	12.00	0.00
Middle Bakken									
10,825.0	56.42	110.00	10,752.4	-73.0	250.5	250.5	12.00	12.00	0.00

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 41-18 12TX		
Design:	Design #2		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
10,850.0	59.42	110.00	10,765.7	-80.2	270.4	270.4	12.00	12.00	0.00
10,875.0	62.42	110.00	10,777.8	-87.7	290.9	290.9	12.00	12.00	0.00
10,891.1	64.35	110.00	10,785.0	-92.6	304.4	304.4	12.00	12.00	0.00
Lower Bakken									
10,900.0	65.42	110.00	10,788.8	-95.4	312.0	312.0	12.00	12.00	0.00
10,905.4	66.07	110.00	10,791.0	-97.1	316.6	316.6	12.00	12.00	0.00
Pronghorn									
10,925.0	68.42	110.00	10,798.6	-103.3	333.7	333.7	12.00	12.00	0.00
10,950.0	71.42	110.00	10,807.2	-111.3	355.7	355.7	12.00	12.00	0.00
10,969.5	73.76	110.00	10,813.0	-117.6	373.2	373.2	12.00	12.00	0.00
Three Forks									
10,975.0	74.42	110.00	10,814.5	-119.5	378.2	378.2	12.00	12.00	0.00
11,000.0	77.42	110.00	10,820.6	-127.7	401.0	401.0	12.00	12.00	0.00
11,025.0	80.42	110.00	10,825.4	-136.1	424.0	424.0	12.00	12.00	0.00
11,035.3	81.66	110.00	10,827.0	-139.6	433.6	433.6	12.00	12.00	0.00
TF Target Top									
11,050.0	83.42	110.00	10,828.9	-144.6	447.3	447.3	12.00	12.00	0.00
11,075.0	86.42	110.00	10,831.1	-153.1	470.7	470.7	12.00	12.00	0.00
11,100.0	89.42	110.00	10,832.0	-161.7	494.1	494.1	12.00	12.00	0.00
11,104.0	89.90	110.00	10,832.0	-163.0	497.9	497.9	11.90	11.90	0.00
EOC									
11,189.0	89.90	107.02	10,832.2	-190.0	578.5	578.5	3.50	0.00	-3.50
7"									
11,200.0	89.90	106.64	10,832.2	-193.2	589.0	589.0	3.50	0.00	-3.50
11,300.0	89.90	103.14	10,832.4	-218.9	685.6	685.6	3.50	0.00	-3.50
11,400.0	89.90	99.64	10,832.6	-238.6	783.7	783.7	3.50	0.00	-3.50
11,500.0	89.90	96.14	10,832.7	-252.4	882.7	882.7	3.50	0.00	-3.50
11,600.0	89.90	92.64	10,832.9	-260.0	982.4	982.4	3.50	0.00	-3.50
11,680.4	89.90	89.82	10,833.1	-261.7	1,062.8	1,062.8	3.50	0.00	-3.50
11,700.0	89.90	89.82	10,833.1	-261.7	1,082.4	1,082.4	0.00	0.00	0.00
11,800.0	89.90	89.82	10,833.3	-261.4	1,182.4	1,182.4	0.00	0.00	0.00
11,900.0	89.90	89.82	10,833.4	-261.1	1,282.4	1,282.4	0.00	0.00	0.00
12,000.0	89.90	89.82	10,833.6	-260.7	1,382.4	1,382.4	0.00	0.00	0.00
12,100.0	89.90	89.82	10,833.8	-260.4	1,482.4	1,482.4	0.00	0.00	0.00
12,200.0	89.90	89.82	10,834.0	-260.1	1,582.4	1,582.4	0.00	0.00	0.00
12,300.0	89.90	89.82	10,834.1	-259.8	1,682.4	1,682.4	0.00	0.00	0.00
12,400.0	89.90	89.82	10,834.3	-259.5	1,782.4	1,782.4	0.00	0.00	0.00
12,500.0	89.90	89.82	10,834.5	-259.2	1,882.4	1,882.4	0.00	0.00	0.00
12,600.0	89.90	89.82	10,834.7	-258.9	1,982.4	1,982.4	0.00	0.00	0.00
12,700.0	89.90	89.82	10,834.8	-258.6	2,082.4	2,082.4	0.00	0.00	0.00
12,800.0	89.90	89.82	10,835.0	-258.3	2,182.4	2,182.4	0.00	0.00	0.00
12,900.0	89.90	89.82	10,835.2	-258.0	2,282.4	2,282.4	0.00	0.00	0.00
13,000.0	89.90	89.82	10,835.4	-257.7	2,382.4	2,382.4	0.00	0.00	0.00
13,100.0	89.90	89.82	10,835.5	-257.4	2,482.4	2,482.4	0.00	0.00	0.00
13,200.0	89.90	89.82	10,835.7	-257.0	2,582.3	2,582.3	0.00	0.00	0.00
13,300.0	89.90	89.82	10,835.9	-256.7	2,682.3	2,682.3	0.00	0.00	0.00
13,400.0	89.90	89.82	10,836.1	-256.4	2,782.3	2,782.3	0.00	0.00	0.00
13,500.0	89.90	89.82	10,836.2	-256.1	2,882.3	2,882.3	0.00	0.00	0.00
13,600.0	89.90	89.82	10,836.4	-255.8	2,982.3	2,982.3	0.00	0.00	0.00
13,700.0	89.90	89.82	10,836.6	-255.5	3,082.3	3,082.3	0.00	0.00	0.00
13,800.0	89.90	89.82	10,836.7	-255.2	3,182.3	3,182.3	0.00	0.00	0.00
13,900.0	89.90	89.82	10,836.9	-254.9	3,282.3	3,282.3	0.00	0.00	0.00
13,943.9	89.90	89.82	10,837.0	-254.7	3,326.2	3,326.2	0.00	0.00	0.00

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Project:	Indian Hills	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 41-18 12TX		
Design:	Design #2		

Planned Survey									
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
TF Target Base									
14,000.0	89.90	89.82	10,837.1	-254.6	3,382.3	3,382.3	0.00	0.00	0.00
14,100.0	89.90	89.82	10,837.3	-254.3	3,482.3	3,482.3	0.00	0.00	0.00
14,200.0	89.90	89.82	10,837.4	-254.0	3,582.3	3,582.3	0.00	0.00	0.00
14,300.0	89.90	89.82	10,837.6	-253.7	3,682.3	3,682.3	0.00	0.00	0.00
14,400.0	89.90	89.82	10,837.8	-253.3	3,782.3	3,782.3	0.00	0.00	0.00
14,500.0	89.90	89.82	10,838.0	-253.0	3,882.3	3,882.3	0.00	0.00	0.00
14,518.6	89.90	89.82	10,838.0	-253.0	3,900.9	3,900.9	0.00	0.00	0.00
Claystone									
14,600.0	89.90	89.82	10,838.1	-252.7	3,982.3	3,982.3	0.00	0.00	0.00
14,700.0	89.90	89.82	10,838.3	-252.4	4,082.3	4,082.3	0.00	0.00	0.00
14,800.0	89.90	89.82	10,838.5	-252.1	4,182.3	4,182.3	0.00	0.00	0.00
14,900.0	89.90	89.82	10,838.7	-251.8	4,282.3	4,282.3	0.00	0.00	0.00
15,000.0	89.90	89.82	10,838.8	-251.5	4,382.3	4,382.3	0.00	0.00	0.00
15,100.0	89.90	89.82	10,839.0	-251.2	4,482.3	4,482.3	0.00	0.00	0.00
15,200.0	89.90	89.82	10,839.2	-250.9	4,582.3	4,582.3	0.00	0.00	0.00
15,300.0	89.90	89.82	10,839.4	-250.6	4,682.3	4,682.3	0.00	0.00	0.00
15,400.0	89.90	89.82	10,839.5	-250.3	4,782.3	4,782.3	0.00	0.00	0.00
15,500.0	89.90	89.82	10,839.7	-249.9	4,882.3	4,882.3	0.00	0.00	0.00
15,600.0	89.90	89.82	10,839.9	-249.6	4,982.3	4,982.3	0.00	0.00	0.00
15,700.0	89.90	89.82	10,840.1	-249.3	5,082.3	5,082.3	0.00	0.00	0.00
15,800.0	89.90	89.82	10,840.2	-249.0	5,182.3	5,182.3	0.00	0.00	0.00
15,900.0	89.90	89.82	10,840.4	-248.7	5,282.3	5,282.3	0.00	0.00	0.00
16,000.0	89.90	89.82	10,840.6	-248.4	5,382.3	5,382.3	0.00	0.00	0.00
16,100.0	89.90	89.82	10,840.8	-248.1	5,482.3	5,482.3	0.00	0.00	0.00
16,200.0	89.90	89.82	10,840.9	-247.8	5,582.3	5,582.3	0.00	0.00	0.00
16,300.0	89.90	89.82	10,841.1	-247.5	5,682.3	5,682.3	0.00	0.00	0.00
16,400.0	89.90	89.82	10,841.3	-247.2	5,782.3	5,782.3	0.00	0.00	0.00
16,500.0	89.90	89.82	10,841.4	-246.9	5,882.3	5,882.3	0.00	0.00	0.00
16,600.0	89.90	89.82	10,841.6	-246.6	5,982.3	5,982.3	0.00	0.00	0.00
16,700.0	89.90	89.82	10,841.8	-246.2	6,082.3	6,082.3	0.00	0.00	0.00
16,800.0	89.90	89.82	10,842.0	-245.9	6,182.3	6,182.3	0.00	0.00	0.00
16,900.0	89.90	89.82	10,842.1	-245.6	6,282.3	6,282.3	0.00	0.00	0.00
17,000.0	89.90	89.82	10,842.3	-245.3	6,382.3	6,382.3	0.00	0.00	0.00
17,100.0	89.90	89.82	10,842.5	-245.0	6,482.3	6,482.3	0.00	0.00	0.00
17,200.0	89.90	89.82	10,842.7	-244.7	6,582.3	6,582.3	0.00	0.00	0.00
17,300.0	89.90	89.82	10,842.8	-244.4	6,682.3	6,682.3	0.00	0.00	0.00
17,400.0	89.90	89.82	10,843.0	-244.1	6,782.3	6,782.3	0.00	0.00	0.00
17,500.0	89.90	89.82	10,843.2	-243.8	6,882.3	6,882.3	0.00	0.00	0.00
17,600.0	89.90	89.82	10,843.4	-243.5	6,982.3	6,982.3	0.00	0.00	0.00
17,700.0	89.90	89.82	10,843.5	-243.2	7,082.3	7,082.3	0.00	0.00	0.00
17,800.0	89.90	89.82	10,843.7	-242.8	7,182.3	7,182.3	0.00	0.00	0.00
17,900.0	89.90	89.82	10,843.9	-242.5	7,282.3	7,282.3	0.00	0.00	0.00
18,000.0	89.90	89.82	10,844.1	-242.2	7,382.3	7,382.3	0.00	0.00	0.00
18,100.0	89.90	89.82	10,844.2	-241.9	7,482.3	7,482.3	0.00	0.00	0.00
18,200.0	89.90	89.82	10,844.4	-241.6	7,582.3	7,582.3	0.00	0.00	0.00
18,300.0	89.90	89.82	10,844.6	-241.3	7,682.3	7,682.3	0.00	0.00	0.00
18,400.0	89.90	89.82	10,844.8	-241.0	7,782.3	7,782.3	0.00	0.00	0.00
18,500.0	89.90	89.82	10,844.9	-240.7	7,882.3	7,882.3	0.00	0.00	0.00
18,600.0	89.90	89.82	10,845.1	-240.4	7,982.3	7,982.3	0.00	0.00	0.00
18,700.0	89.90	89.82	10,845.3	-240.1	8,082.3	8,082.3	0.00	0.00	0.00
18,800.0	89.90	89.82	10,845.4	-239.8	8,182.3	8,182.3	0.00	0.00	0.00
18,900.0	89.90	89.82	10,845.6	-239.5	8,282.3	8,282.3	0.00	0.00	0.00
19,000.0	89.90	89.82	10,845.8	-239.1	8,382.3	8,382.3	0.00	0.00	0.00

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 41-18 12TX		
Design:	Design #2		

Planned Survey										
Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)	
19,100.0	89.90	89.82	10,846.0	-238.8	8,482.3	8,482.3	0.00	0.00	0.00	
19,200.0	89.90	89.82	10,846.1	-238.5	8,582.3	8,582.3	0.00	0.00	0.00	
19,300.0	89.90	89.82	10,846.3	-238.2	8,682.3	8,682.3	0.00	0.00	0.00	
19,400.0	89.90	89.82	10,846.5	-237.9	8,782.3	8,782.3	0.00	0.00	0.00	
19,500.0	89.90	89.82	10,846.7	-237.6	8,882.3	8,882.3	0.00	0.00	0.00	
19,600.0	89.90	89.82	10,846.8	-237.3	8,982.3	8,982.3	0.00	0.00	0.00	
19,700.0	89.90	89.82	10,847.0	-237.0	9,082.3	9,082.3	0.00	0.00	0.00	
19,800.0	89.90	89.82	10,847.2	-236.7	9,182.3	9,182.3	0.00	0.00	0.00	
19,900.0	89.90	89.82	10,847.4	-236.4	9,282.3	9,282.3	0.00	0.00	0.00	
20,000.0	89.90	89.82	10,847.5	-236.1	9,382.3	9,382.3	0.00	0.00	0.00	
20,100.0	89.90	89.82	10,847.7	-235.8	9,482.3	9,482.3	0.00	0.00	0.00	
20,200.0	89.90	89.82	10,847.9	-235.4	9,582.3	9,582.3	0.00	0.00	0.00	
20,300.0	89.90	89.82	10,848.1	-235.1	9,682.3	9,682.3	0.00	0.00	0.00	
20,400.0	89.90	89.82	10,848.2	-234.8	9,782.3	9,782.3	0.00	0.00	0.00	
20,500.0	89.90	89.82	10,848.4	-234.5	9,882.3	9,882.3	0.00	0.00	0.00	
20,600.0	89.90	89.82	10,848.6	-234.2	9,982.3	9,982.3	0.00	0.00	0.00	
20,667.7	89.90	89.82	10,848.7	-234.0	10,050.0	10,050.0	0.00	0.00	0.00	
5300 41-18 12TX PBHL										

Design Targets									
Target Name	Dip Angle (°)	Dip Dir. (°)	TVD (usft)	+N/-S (usft)	+E/-W (usft)	Northing (usft)	Easting (usft)	Latitude	Longitude
5300 41-18 12TX PBHL	0.00	0.00	10,848.7	-234.0	10,050.0	404,497.87	1,220,000.62	48° 4' 5.214 N	103° 33' 43.972 W
- hit/miss target									
- Shape									
- plan hits target center									
- Point									

Casing Points					
Measured Depth (usft)	Vertical Depth (usft)	Name	Casing Diameter (")	Hole Diameter (")	
11,189.0	10,832.2	7"	7	8-3/4	

Database:	EDM 5000.1 Single User Db	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Company:	Oasis Petroleum	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Project:	Indian Hills	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site:	153N-100W-17/18	North Reference:	True
Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Wellbore:	Kline Federal 5300 41-18 12TX		
Design:	Design #2		

Formations						
Measured Depth (usft)	Vertical Depth (usft)	Name	Lithology	Dip (°)	Dip Direction (°)	
1,972.0	1,972.0	Pierre				
4,646.1	4,646.0	Greenhorn				
5,057.1	5,057.0	Mowry				
5,471.1	5,471.0	Dakota				
6,486.2	6,486.0	Rierdon				
6,814.2	6,814.0	Dunham Salt				
6,929.2	6,929.0	Dunham Salt Base				
7,024.2	7,024.0	Spearfish				
7,283.2	7,283.0	Pine Salt				
7,318.2	7,318.0	Pine Salt Base				
7,374.2	7,374.0	Opeche Salt				
7,403.2	7,403.0	Opeche Salt Base				
7,605.2	7,605.0	Broom Creek (Top of Minnelusa Gp.)				
7,684.2	7,684.0	Amsden				
7,853.2	7,853.0	Tyler				
8,047.2	8,047.0	Otter (Base of Minnelusa Gp.)				
8,390.2	8,390.0	Kibbey Lime				
8,542.2	8,542.0	Charles Salt				
9,159.2	9,159.0	UB				
9,238.2	9,238.0	Base Last Salt				
9,301.2	9,301.0	Ratcliffe				
9,454.2	9,454.0	Mission Canyon				
10,022.2	10,022.0	Lodgepole				
10,209.2	10,209.0	Lodgepole Fracture Zone				
10,780.4	10,726.0	False Bakken				
10,795.0	10,735.0	Upper Bakken				
10,822.5	10,751.0	Middle Bakken				
10,891.1	10,785.0	Lower Bakken				
10,905.4	10,791.0	Pronghorn				
10,969.5	10,813.0	Three Forks				
11,035.3	10,827.0	TF Target Top				
13,943.9	10,837.0	TF Target Base				
14,518.6	10,838.0	Claystone				

Plan Annotations				
Measured Depth (usft)	Vertical Depth (usft)	Local Coordinates		Comment
		+N/-S (usft)	+E/-W (usft)	
10,354.8	10,354.6	0.0	50.0	KOP Build 12°/100'
11,104.0	10,832.0	-163.0	497.9	EOC



Oasis Petroleum

Indian Hills

153N-100W-17/18

Kline Federal 5300 41-18 12TX

Kline Federal 5300 41-18 12TX

Design #2

Anticollision Report

17 June, 2014



**RYAN DIRECTIONAL
SERVICES**
A NABORS COMPANY



Ryan Directional Services

Anticollision Report



Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Reference	Design #2		
Filter type:	NO GLOBAL FILTER: Using user defined selection & filtering criteria		
Interpolation Method:	Stations	Error Model:	ISCWSA
Depth Range:	Unlimited	Scan Method:	Closest Approach 3D
Results Limited by:	Maximum center-center distance of 10,000.0 usft	Error Surface:	Elliptical Conic
Warning Levels Evaluated at:	2.00 Sigma	Casing Method:	Not applied

Survey Tool Program		Date	6/17/2014		
From (usft)	To (usft)	Survey (Wellbore)	Tool Name	Description	
0.0	20,667.7	Design #2 (Kline Federal 5300 41-18 12TX	MWD	MWD - Standard	

Summary						
Site Name	Reference Measured Depth (usft)	Offset Measured Depth (usft)	Distance Between Centres (usft)	Distance Between Ellipses (usft)	Separation Factor	Warning
Offset Well - Wellbore - Design						
153N-100W-17/18						
Kline Federal 5300 41-18 10B - Kline Federal 5300 41-18	2,200.0	2,200.0	65.9	55.5	6.328	CC
Kline Federal 5300 41-18 10B - Kline Federal 5300 41-18	20,667.7	20,590.0	553.9	-27.2	0.953	Level 1, ES, SF
Kline Federal 5300 41-18 11T2 - Kline Federal 5300 41-1	2,200.0	2,200.0	33.4	23.0	3.213	CC
Kline Federal 5300 41-18 11T2 - Kline Federal 5300 41-1	20,667.7	20,743.0	506.9	-77.4	0.868	Level 1, ES, SF
Kline Federal 5300 41-18 13T2X - Kline Federal 5300 41	2,200.0	2,200.0	33.4	23.0	3.213	CC
Kline Federal 5300 41-18 13T2X - Kline Federal 5300 41	20,667.7	20,839.5	402.6	-181.2	0.690	Level 1, ES, SF
Kline Federal 5300 41-18 14BX - Kline Federal 5300 41-1	2,200.0	2,200.0	65.9	55.5	6.328	CC
Kline Federal 5300 41-18 14BX - Kline Federal 5300 41-1	20,667.7	20,608.5	433.3	-110.3	0.797	Level 1, ES, SF
Kline Federal 5300 41-18 14BX - Kline Federal 5300 41-1	2,200.0	2,200.0	65.9	55.5	6.328	CC
Kline Federal 5300 41-18 14BX - Kline Federal 5300 41-1	20,667.7	20,708.4	405.5	-173.0	0.701	Level 1, ES, SF
Kline Federal 5300 41-18 9T - Kline Federal 5300 41-18	2,200.0	2,200.0	99.3	88.9	9.540	CC, ES
Kline Federal 5300 41-18 9T - Kline Federal 5300 41-18	20,667.7	20,824.5	1,050.8	467.1	1.800	SF

Offset Design		153N-100W-17/18 - Kline Federal 5300 41-18 10B - Kline Federal 5300 41-18 10B - Design #1										Offset Site Error:		0.0 usft
Survey Program:		0-MWD										Offset Well Error:		2.0 usft
Reference		Offset		Semi Major Axis			Distance							
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning	
0.0	0.0	0.0	0.0	2.0	2.0	0.01	65.9	0.0	65.9					
100.0	100.0	100.0	100.0	2.0	2.0	0.01	65.9	0.0	65.9	61.9	4.00	16.450		
200.0	200.0	200.0	200.0	2.0	2.0	0.01	65.9	0.0	65.9	61.8	4.05	16.272		
300.0	300.0	300.0	300.0	2.1	2.1	0.01	65.9	0.0	65.9	61.7	4.14	15.908		
400.0	400.0	400.0	400.0	2.1	2.1	0.01	65.9	0.0	65.9	61.6	4.28	15.395		
500.0	500.0	500.0	500.0	2.2	2.2	0.01	65.9	0.0	65.9	61.4	4.46	14.775		
600.0	600.0	600.0	600.0	2.3	2.3	0.01	65.9	0.0	65.9	61.2	4.67	14.093		
700.0	700.0	700.0	700.0	2.5	2.5	0.01	65.9	0.0	65.9	60.9	4.92	13.384		
800.0	800.0	800.0	800.0	2.6	2.6	0.01	65.9	0.0	65.9	60.7	5.20	12.677		
900.0	900.0	900.0	900.0	2.7	2.7	0.01	65.9	0.0	65.9	60.4	5.49	11.989		
1,000.0	1,000.0	1,000.0	1,000.0	2.9	2.9	0.01	65.9	0.0	65.9	60.0	5.81	11.335		
1,100.0	1,100.0	1,100.0	1,100.0	3.1	3.1	0.01	65.9	0.0	65.9	59.7	6.14	10.719		
1,200.0	1,200.0	1,200.0	1,200.0	3.2	3.2	0.01	65.9	0.0	65.9	59.4	6.49	10.144		
1,300.0	1,300.0	1,300.0	1,300.0	3.4	3.4	0.01	65.9	0.0	65.9	59.0	6.85	9.612		
1,400.0	1,400.0	1,400.0	1,400.0	3.6	3.6	0.01	65.9	0.0	65.9	58.6	7.22	9.120		
1,500.0	1,500.0	1,500.0	1,500.0	3.8	3.8	0.01	65.9	0.0	65.9	58.3	7.60	8.666		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 10B - Kline Federal 5300 41-18 10B - Design #1													Offset Site Error: 0.0 usft
Survey Program: 0-MWD													Offset Well Error: 2.0 usft
Reference		Offset		Semi Major Axis			Distance						Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	
1,600.0	1,600.0	1,600.0	1,600.0	4.0	4.0	0.01	65.9	0.0	65.9	57.9	7.99	8.247	
1,700.0	1,700.0	1,700.0	1,700.0	4.2	4.2	0.01	65.9	0.0	65.9	57.5	8.38	7.861	
1,800.0	1,800.0	1,800.0	1,800.0	4.4	4.4	0.01	65.9	0.0	65.9	57.1	8.78	7.505	
1,900.0	1,900.0	1,900.0	1,900.0	4.6	4.6	0.01	65.9	0.0	65.9	56.7	9.18	7.176	
2,000.0	2,000.0	2,000.0	2,000.0	4.8	4.8	0.01	65.9	0.0	65.9	56.3	9.58	6.872	
2,100.0	2,100.0	2,100.0	2,100.0	5.0	5.0	0.01	65.9	0.0	65.9	55.9	9.99	6.589	
2,200.0	2,200.0	2,200.0	2,200.0	5.2	5.2	0.01	65.9	0.0	65.9	55.5	10.41	6.328 CC	
2,216.7	2,216.7	2,216.5	2,216.5	5.2	5.2	-90.00	65.9	0.1	65.9	55.4	10.47	6.290	
2,300.0	2,300.0	2,299.8	2,299.8	5.4	5.4	-90.03	66.1	0.8	66.1	55.3	10.78	6.127	
2,400.0	2,400.0	2,399.8	2,399.8	5.6	5.6	-90.05	66.3	1.6	66.3	55.2	11.16	5.941	
2,500.0	2,500.0	2,499.8	2,499.8	5.8	5.8	-90.08	66.5	2.4	66.5	55.0	11.54	5.765	
2,600.0	2,600.0	2,599.8	2,599.8	6.0	6.0	-90.11	66.8	3.3	66.8	54.9	11.93	5.597	
2,700.0	2,700.0	2,699.8	2,699.8	6.1	6.2	-90.14	67.0	4.1	67.0	54.7	12.32	5.438	
2,800.0	2,800.0	2,799.8	2,799.8	6.3	6.4	-90.17	67.3	5.0	67.3	54.5	12.72	5.288	
2,900.0	2,900.0	2,899.8	2,899.8	6.5	6.6	-90.19	67.5	5.8	67.5	54.4	13.12	5.145	
3,000.0	3,000.0	2,999.8	2,999.8	6.7	6.8	-90.22	67.7	6.6	67.7	54.2	13.52	5.009	
3,100.0	3,100.0	3,099.8	3,099.8	6.9	7.0	-90.25	68.0	7.5	68.0	54.0	13.93	4.880	
3,200.0	3,200.0	3,199.8	3,199.8	7.1	7.2	-90.27	68.2	8.3	68.2	53.9	14.34	4.758	
3,300.0	3,300.0	3,299.8	3,299.8	7.3	7.4	-90.30	68.4	9.2	68.4	53.7	14.75	4.641	
3,400.0	3,400.0	3,399.8	3,399.8	7.6	7.6	-90.33	68.7	10.0	68.7	53.5	15.16	4.531	
3,500.0	3,500.0	3,499.8	3,499.8	7.8	7.8	-90.35	68.9	10.8	68.9	53.3	15.57	4.425	
3,600.0	3,599.9	3,599.8	3,599.8	8.0	8.0	-90.38	69.2	11.7	69.2	53.2	15.99	4.324	
3,700.0	3,699.9	3,699.8	3,699.8	8.2	8.2	-90.40	69.4	12.5	69.4	53.0	16.41	4.229	
3,800.0	3,799.9	3,799.8	3,799.8	8.4	8.5	-90.43	69.6	13.4	69.6	52.8	16.83	4.137	
3,900.0	3,899.9	3,899.8	3,899.8	8.6	8.7	-90.45	69.9	14.2	69.9	52.6	17.25	4.050	
4,000.0	3,999.9	3,999.8	3,999.8	8.8	8.9	-90.48	70.1	15.0	70.1	52.4	17.68	3.966	
4,100.0	4,099.9	4,099.8	4,099.8	9.0	9.1	-90.50	70.3	15.9	70.3	52.2	18.10	3.886	
4,200.0	4,199.9	4,199.8	4,199.8	9.2	9.3	-90.53	70.6	16.7	70.6	52.1	18.53	3.809	
4,300.0	4,299.9	4,299.8	4,299.8	9.4	9.5	-90.55	70.8	17.6	70.8	51.9	18.95	3.736	
4,400.0	4,399.9	4,399.8	4,399.8	9.7	9.7	-90.58	71.0	18.4	71.1	51.7	19.38	3.666	
4,500.0	4,499.9	4,499.8	4,499.8	9.9	10.0	-90.60	71.3	19.2	71.3	51.5	19.81	3.598	
4,600.0	4,599.9	4,599.8	4,599.7	10.1	10.2	-90.63	71.5	20.1	71.5	51.3	20.24	3.534	
4,700.0	4,699.9	4,699.8	4,699.7	10.3	10.4	-90.65	71.8	20.9	71.8	51.1	20.67	3.472	
4,800.0	4,799.9	4,799.8	4,799.7	10.5	10.6	-90.68	72.0	21.8	72.0	50.9	21.10	3.412	
4,900.0	4,899.9	4,899.8	4,899.7	10.7	10.8	-90.70	72.2	22.6	72.2	50.7	21.54	3.354	
5,000.0	4,999.9	4,999.8	4,999.7	10.9	11.0	-90.72	72.5	23.4	72.5	50.5	21.97	3.299	
5,100.0	5,099.9	5,099.8	5,099.7	11.2	11.3	-90.75	72.7	24.3	72.7	50.3	22.40	3.246	
5,200.0	5,199.9	5,199.8	5,199.7	11.4	11.5	-90.77	72.9	25.1	73.0	50.1	22.84	3.194	
5,300.0	5,299.9	5,299.8	5,299.7	11.6	11.7	-90.79	73.2	26.0	73.2	49.9	23.27	3.145	
5,400.0	5,399.9	5,399.8	5,399.7	11.8	11.9	-90.82	73.4	26.8	73.4	49.7	23.71	3.097	
5,500.0	5,499.9	5,499.8	5,499.7	12.0	12.1	-90.84	73.7	27.6	73.7	49.5	24.15	3.051	
5,600.0	5,599.9	5,599.8	5,599.7	12.2	12.3	-90.86	73.9	28.5	73.9	49.3	24.58	3.006	
5,700.0	5,699.9	5,699.8	5,699.7	12.5	12.6	-90.88	74.1	29.3	74.1	49.1	25.02	2.963	
5,800.0	5,799.9	5,799.8	5,799.7	12.7	12.8	-90.91	74.4	30.2	74.4	48.9	25.46	2.921	
5,900.0	5,899.9	5,899.8	5,899.7	12.9	13.0	-90.93	74.6	31.0	74.6	48.7	25.90	2.881	
6,000.0	5,999.9	5,999.8	5,999.7	13.1	13.2	-90.95	74.8	31.8	74.9	48.5	26.34	2.842	
6,100.0	6,099.9	6,099.8	6,099.7	13.3	13.4	-90.97	75.1	32.7	75.1	48.3	26.77	2.804	
6,200.0	6,199.8	6,199.8	6,199.7	13.6	13.7	-90.99	75.3	33.5	75.3	48.1	27.21	2.768	
6,300.0	6,299.8	6,299.8	6,299.7	13.8	13.9	-91.02	75.6	34.4	75.6	47.9	27.65	2.732	
6,400.0	6,399.8	6,399.8	6,399.7	14.0	14.1	-91.04	75.8	35.2	75.8	47.7	28.09	2.698	
6,500.0	6,499.8	6,499.8	6,499.7	14.2	14.3	-91.06	76.0	36.0	76.0	47.5	28.53	2.665	
6,600.0	6,599.8	6,599.8	6,599.7	14.4	14.5	-91.08	76.3	36.9	76.3	47.3	28.97	2.632	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services

Anticollision Report



Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 10B - Kline Federal 5300 41-18 10B - Design #1													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis			Distance							Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
6,700.0	6,699.8	6,699.8	6,699.7	14.7	14.8	-91.10	76.5	37.7	76.5	47.1	29.42	2.601		
6,800.0	6,799.8	6,799.8	6,799.7	14.9	15.0	-91.12	76.7	38.6	76.7	46.9	29.86	2.571		
6,900.0	6,899.8	6,899.8	6,899.7	15.1	15.2	-91.14	77.0	39.4	77.0	46.7	30.30	2.541		
7,000.0	6,999.8	6,999.8	6,999.6	15.3	15.4	-91.16	77.2	40.2	77.2	46.5	30.74	2.512		
7,100.0	7,099.8	7,099.8	7,099.6	15.5	15.7	-91.19	77.4	41.1	77.5	46.3	31.18	2.484		
7,200.0	7,199.8	7,199.8	7,199.6	15.8	15.9	-91.21	77.7	41.9	77.7	46.1	31.63	2.457		
7,300.0	7,299.8	7,299.8	7,299.6	16.0	16.1	-91.23	77.9	42.8	77.9	45.9	32.07	2.430		
7,400.0	7,399.8	7,399.8	7,399.6	16.2	16.3	-91.25	78.2	43.6	78.2	45.7	32.51	2.405		
7,500.0	7,499.8	7,499.8	7,499.6	16.4	16.5	-91.27	78.4	44.4	78.4	45.5	32.95	2.379		
7,600.0	7,599.8	7,599.8	7,599.6	16.6	16.8	-91.29	78.6	45.3	78.6	45.3	33.40	2.355		
7,700.0	7,699.8	7,699.8	7,699.6	16.9	17.0	-91.31	78.9	46.1	78.9	45.0	33.84	2.331		
7,800.0	7,799.8	7,799.8	7,799.6	17.1	17.2	-91.33	79.1	47.0	79.1	44.8	34.28	2.308		
7,900.0	7,899.8	7,899.8	7,899.6	17.3	17.4	-91.35	79.3	47.8	79.4	44.6	34.73	2.285		
7,926.7	7,926.4	7,926.5	7,926.3	17.4	17.5	-91.35	79.4	48.0	79.4	44.6	34.85	2.279		
7,943.3	7,943.1	7,943.3	7,943.1	17.4	17.5	-1.35	79.4	48.1	79.4	44.6	34.87	2.278		
8,000.0	7,999.8	8,000.0	7,999.8	17.5	17.6	-1.35	79.4	48.1	79.4	44.4	35.10	2.264		
8,100.0	8,099.8	8,100.0	8,099.8	17.7	17.8	-1.35	79.4	48.1	79.4	43.9	35.51	2.237		
8,200.0	8,199.8	8,200.0	8,199.8	17.9	18.1	-1.35	79.4	48.1	79.4	43.5	35.94	2.211		
8,300.0	8,299.8	8,300.0	8,299.8	18.1	18.3	-1.35	79.4	48.1	79.4	43.1	36.36	2.185		
8,400.0	8,399.8	8,400.0	8,399.8	18.4	18.5	-1.35	79.4	48.1	79.4	42.7	36.78	2.160		
8,500.0	8,499.8	8,500.0	8,499.8	18.6	18.7	-1.35	79.4	48.1	79.4	42.2	37.20	2.136		
8,600.0	8,599.8	8,600.0	8,599.8	18.8	18.9	-1.35	79.4	48.1	79.4	41.8	37.62	2.112		
8,700.0	8,699.8	8,700.0	8,699.8	19.0	19.1	-1.35	79.4	48.1	79.4	41.4	38.05	2.088		
8,800.0	8,799.8	8,800.0	8,799.8	19.2	19.3	-1.35	79.4	48.1	79.4	41.0	38.47	2.065		
8,900.0	8,899.8	8,900.0	8,899.8	19.4	19.6	-1.35	79.4	48.1	79.4	40.6	38.89	2.043		
9,000.0	8,999.8	9,000.0	8,999.8	19.6	19.8	-1.35	79.4	48.1	79.4	40.1	39.32	2.021		
9,100.0	9,099.8	9,100.0	9,099.8	19.8	20.0	-1.35	79.4	48.1	79.4	39.7	39.75	1.999		
9,200.0	9,199.8	9,200.0	9,199.8	20.0	20.2	-1.35	79.4	48.1	79.4	39.3	40.17	1.978		
9,300.0	9,299.8	9,300.0	9,299.8	20.2	20.4	-1.35	79.4	48.1	79.4	38.8	40.60	1.957		
9,400.0	9,399.8	9,400.0	9,399.8	20.5	20.6	-1.35	79.4	48.1	79.4	38.4	41.03	1.936		
9,500.0	9,499.8	9,500.0	9,499.8	20.7	20.8	-1.35	79.4	48.1	79.4	38.0	41.45	1.917		
9,600.0	9,599.8	9,600.0	9,599.8	20.9	21.1	-1.35	79.4	48.1	79.4	37.6	41.88	1.897		
9,700.0	9,699.8	9,700.0	9,699.8	21.1	21.3	-1.35	79.4	48.1	79.4	37.1	42.31	1.878		
9,800.0	9,799.8	9,800.0	9,799.8	21.3	21.5	-1.35	79.4	48.1	79.4	36.7	42.74	1.859		
9,900.0	9,899.8	9,900.0	9,899.8	21.5	21.7	-1.35	79.4	48.1	79.4	36.3	43.17	1.840		
10,000.0	9,999.8	10,000.0	9,999.8	21.7	21.9	-1.35	79.4	48.1	79.4	35.8	43.60	1.822		
10,100.0	10,099.8	10,100.0	10,099.8	21.9	22.1	-1.35	79.4	48.1	79.4	35.4	44.03	1.804		
10,200.0	10,199.8	10,200.0	10,199.8	22.2	22.4	-1.35	79.4	48.1	79.4	35.0	44.46	1.787		
10,262.1	10,261.8	10,262.1	10,261.8	22.3	22.5	-1.35	79.4	48.1	79.4	34.7	44.73	1.776		
10,300.0	10,299.8	10,299.5	10,299.3	22.4	22.6	-1.25	79.5	48.2	79.5	34.6	44.89	1.771		
10,354.8	10,354.6	10,351.8	10,351.3	22.5	22.7	1.59	80.6	52.2	80.7	35.6	45.12	1.788		
10,375.0	10,374.8	10,370.7	10,370.1	22.5	22.7	-106.60	81.4	55.0	81.8	36.5	45.27	1.807		
10,400.0	10,399.7	10,394.1	10,393.0	22.6	22.8	-104.41	82.6	59.4	84.0	38.6	45.37	1.850		
10,425.0	10,424.5	10,417.2	10,415.4	22.7	22.9	-102.32	84.2	64.9	87.0	41.5	45.48	1.912		
10,450.0	10,449.2	10,440.0	10,437.2	22.7	22.9	-100.38	86.0	71.3	90.8	45.2	45.60	1.992		
10,475.0	10,473.5	10,462.6	10,458.5	22.8	23.0	-98.60	88.0	78.5	95.5	49.7	45.71	2.088		
10,500.0	10,497.6	10,484.9	10,479.1	22.8	23.0	-96.98	90.3	86.7	100.8	55.0	45.84	2.200		
10,525.0	10,521.2	10,506.9	10,499.1	22.9	23.1	-95.53	92.8	95.6	106.9	60.9	45.97	2.325		
10,550.0	10,544.4	10,528.6	10,518.3	23.0	23.2	-94.21	95.6	105.3	113.6	67.5	46.10	2.464		
10,575.0	10,567.1	10,550.0	10,536.8	23.0	23.2	-93.03	98.5	115.7	120.9	74.7	46.24	2.615		
10,600.0	10,589.1	10,571.1	10,554.5	23.1	23.3	-91.96	101.6	126.6	128.8	82.4	46.39	2.777		
10,625.0	10,610.6	10,591.8	10,571.5	23.2	23.4	-90.97	104.8	138.1	137.2	90.7	46.54	2.949		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services

Anticollision Report



Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 10B - Kline Federal 5300 41-18 10B - Design #1													Offset Site Error: 0.0 usft
Survey Program: 0-MWD													Offset Well Error: 2.0 usft
Reference		Offset		Semi Major Axis			Distance						Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	
10,650.0	10,631.3	10,612.3	10,587.6	23.3	23.4	-90.05	108.2	150.2	146.2	99.5	46.70	3.130	
10,675.0	10,651.3	10,632.4	10,603.0	23.4	23.5	-89.19	111.7	162.7	155.6	108.7	46.86	3.321	
10,700.0	10,670.5	10,652.2	10,617.6	23.5	23.6	-88.37	115.4	175.6	165.5	118.4	47.03	3.518	
10,725.0	10,688.8	10,671.7	10,631.5	23.6	23.7	-87.57	119.1	188.8	175.7	128.5	47.21	3.723	
10,750.0	10,706.2	10,691.0	10,644.5	23.7	23.8	-86.79	122.9	202.4	186.4	139.0	47.40	3.933	
10,775.0	10,722.6	10,709.9	10,656.8	23.9	23.9	-86.03	126.9	216.3	197.4	149.8	47.59	4.148	
10,800.0	10,738.0	10,728.7	10,668.4	24.0	24.0	-85.26	130.9	230.5	208.8	161.0	47.80	4.368	
10,825.0	10,752.4	10,747.1	10,679.3	24.2	24.1	-84.49	134.9	244.8	220.5	172.4	48.01	4.592	
10,850.0	10,765.7	10,765.4	10,689.4	24.4	24.2	-83.72	139.0	259.4	232.4	184.2	48.24	4.817	
10,875.0	10,777.8	10,783.4	10,698.9	24.6	24.3	-82.95	143.2	274.2	244.6	196.1	48.48	5.046	
10,900.0	10,788.8	10,800.0	10,707.0	24.8	24.5	-82.10	147.1	288.1	257.0	208.3	48.71	5.276	
10,925.0	10,798.6	10,818.9	10,715.7	25.0	24.6	-81.37	151.7	304.2	269.7	220.7	48.98	5.505	
10,950.0	10,807.2	10,836.4	10,723.2	25.3	24.8	-80.57	156.0	319.5	282.5	233.2	49.25	5.735	
10,975.0	10,814.5	10,853.8	10,730.0	25.6	24.9	-79.77	160.3	334.9	295.4	245.9	49.53	5.964	
11,000.0	10,820.6	10,871.1	10,736.2	25.8	25.1	-78.96	164.7	350.4	308.5	258.7	49.83	6.192	
11,025.0	10,825.4	10,888.3	10,741.8	26.1	25.2	-78.16	169.1	366.0	321.7	271.6	50.13	6.417	
11,050.0	10,828.9	10,905.4	10,746.8	26.5	25.4	-77.35	173.5	381.8	335.0	284.6	50.45	6.640	
11,075.0	10,831.1	10,922.5	10,751.2	26.8	25.6	-76.55	178.0	397.7	348.4	297.6	50.79	6.860	
11,100.0	10,832.0	10,939.6	10,755.0	27.1	25.8	-75.76	182.6	413.8	361.8	310.6	51.13	7.075	
11,104.0	10,832.0	10,942.3	10,755.5	27.2	25.8	-75.64	183.3	416.3	363.9	312.7	51.19	7.109	
11,200.0	10,832.2	11,010.8	10,764.3	28.6	26.6	-79.00	201.7	481.6	414.8	361.1	53.76	7.716	
11,300.0	10,832.4	11,103.8	10,765.2	30.3	27.9	-80.68	226.2	571.4	464.4	407.7	56.74	8.185	
11,400.0	10,832.6	11,213.0	10,765.4	32.2	29.6	-81.82	251.3	677.6	505.8	445.5	60.30	8.388	
11,500.0	10,832.7	11,328.4	10,765.6	34.1	31.7	-82.57	273.5	790.8	538.0	473.7	64.35	8.361	
11,600.0	10,832.9	11,448.6	10,765.8	36.2	34.1	-83.03	291.7	909.7	560.6	491.7	68.81	8.147	
11,680.4	10,833.1	11,547.8	10,765.9	38.0	36.2	-83.23	303.0	1,008.2	571.3	498.7	72.60	7.870	
11,700.0	10,833.1	11,572.1	10,766.0	38.4	36.7	-83.26	305.2	1,032.4	573.1	499.5	73.56	7.791	
11,800.0	10,833.3	11,696.8	10,766.2	40.7	39.5	-83.35	313.5	1,156.8	579.3	500.7	78.62	7.369	
11,900.0	10,833.4	11,822.0	10,766.4	43.0	42.5	-83.38	316.4	1,282.0	581.3	497.4	83.91	6.928	
12,000.0	10,833.6	11,922.5	10,766.6	45.4	44.9	-83.38	316.4	1,382.5	581.0	492.2	88.75	6.546	
12,100.0	10,833.8	12,022.5	10,766.8	47.9	47.3	-83.37	316.3	1,482.5	580.7	487.0	93.69	6.198	
12,200.0	10,834.0	12,122.5	10,766.9	50.4	49.8	-83.37	316.3	1,582.5	580.4	481.6	98.73	5.878	
12,300.0	10,834.1	12,222.5	10,767.1	53.0	52.4	-83.36	316.3	1,682.5	580.0	476.2	103.85	5.586	
12,400.0	10,834.3	12,322.5	10,767.3	55.5	55.0	-83.36	316.3	1,782.5	579.7	470.7	109.04	5.317	
12,500.0	10,834.5	12,422.5	10,767.5	58.2	57.6	-83.36	316.3	1,882.5	579.4	465.1	114.29	5.070	
12,600.0	10,834.7	12,522.5	10,767.6	60.8	60.2	-83.35	316.3	1,982.5	579.1	459.5	119.59	4.842	
12,700.0	10,834.8	12,622.5	10,767.8	63.5	62.9	-83.35	316.3	2,082.5	578.8	453.9	124.94	4.633	
12,800.0	10,835.0	12,722.5	10,768.0	66.2	65.6	-83.35	316.3	2,182.5	578.5	448.1	130.33	4.439	
12,900.0	10,835.2	12,822.5	10,768.2	68.9	68.3	-83.34	316.3	2,282.5	578.2	442.4	135.76	4.259	
13,000.0	10,835.4	12,922.5	10,768.3	71.7	71.0	-83.34	316.3	2,382.5	577.9	436.6	141.22	4.092	
13,100.0	10,835.5	13,022.5	10,768.5	74.4	73.8	-83.34	316.3	2,482.5	577.5	430.8	146.70	3.937	
13,200.0	10,835.7	13,122.5	10,768.7	77.2	76.5	-83.33	316.3	2,582.5	577.2	425.0	152.22	3.792	
13,300.0	10,835.9	13,222.5	10,768.9	79.9	79.3	-83.33	316.3	2,682.5	576.9	419.2	157.75	3.657	
13,400.0	10,836.1	13,322.5	10,769.0	82.7	82.1	-83.33	316.3	2,782.5	576.6	413.3	163.31	3.531	
13,500.0	10,836.2	13,422.5	10,769.2	85.5	84.9	-83.32	316.3	2,882.5	576.3	407.4	168.88	3.412	
13,600.0	10,836.4	13,522.5	10,769.4	88.3	87.7	-83.32	316.3	2,982.5	576.0	401.5	174.47	3.301	
13,700.0	10,836.6	13,622.5	10,769.6	91.1	90.5	-83.31	316.3	3,082.5	575.7	395.6	180.07	3.197	
13,800.0	10,836.7	13,722.5	10,769.7	94.0	93.3	-83.31	316.3	3,182.5	575.4	389.7	185.69	3.099	
13,900.0	10,836.9	13,822.5	10,769.9	96.8	96.1	-83.31	316.2	3,282.5	575.0	383.7	191.32	3.006	
14,000.0	10,837.1	13,922.5	10,770.1	99.6	99.0	-83.30	316.2	3,382.5	574.7	377.8	196.96	2.918	
14,100.0	10,837.3	14,022.5	10,770.3	102.5	101.8	-83.30	316.2	3,482.5	574.4	371.8	202.61	2.835	
14,200.0	10,837.4	14,122.5	10,770.4	105.3	104.7	-83.30	316.2	3,582.5	574.1	365.8	208.27	2.757	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services

Anticollision Report



Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 10B - Kline Federal 5300 41-18 10B - Design #1													Offset Site Error: 0.0 usft
Survey Program: 0-MWD													Offset Well Error: 2.0 usft
Reference		Offset		Semi Major Axis			Distance						Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	
14,300.0	10,837.6	14,222.5	10,770.6	108.2	107.5	-83.29	316.2	3,682.5	573.8	359.9	213.94	2.682	
14,400.0	10,837.8	14,322.5	10,770.8	111.0	110.4	-83.29	316.2	3,782.5	573.5	353.9	219.62	2.611	
14,500.0	10,838.0	14,422.5	10,771.0	113.9	113.2	-83.29	316.2	3,882.5	573.2	347.9	225.31	2.544	
14,600.0	10,838.1	14,522.5	10,771.1	116.7	116.1	-83.28	316.2	3,982.5	572.9	341.9	231.00	2.480	
14,700.0	10,838.3	14,622.5	10,771.3	119.6	118.9	-83.28	316.2	4,082.5	572.6	335.9	236.70	2.419	
14,800.0	10,838.5	14,722.5	10,771.5	122.5	121.8	-83.27	316.2	4,182.5	572.2	329.8	242.40	2.361	
14,900.0	10,838.7	14,822.5	10,771.7	125.3	124.7	-83.27	316.2	4,282.5	571.9	323.8	248.11	2.305	
15,000.0	10,838.8	14,922.5	10,771.8	128.2	127.5	-83.27	316.2	4,382.5	571.6	317.8	253.82	2.252	
15,100.0	10,839.0	15,022.5	10,772.0	131.1	130.4	-83.26	316.2	4,482.5	571.3	311.8	259.54	2.201	
15,200.0	10,839.2	15,122.5	10,772.2	133.9	133.3	-83.26	316.2	4,582.5	571.0	305.7	265.27	2.153	
15,300.0	10,839.4	15,222.5	10,772.3	136.8	136.2	-83.26	316.2	4,682.5	570.7	299.7	271.00	2.106	
15,400.0	10,839.5	15,322.5	10,772.5	139.7	139.0	-83.25	316.2	4,782.5	570.4	293.6	276.73	2.061	
15,500.0	10,839.7	15,422.5	10,772.7	142.6	141.9	-83.25	316.2	4,882.5	570.1	287.6	282.46	2.018	
15,600.0	10,839.9	15,522.5	10,772.9	145.5	144.8	-83.25	316.1	4,982.5	569.7	281.5	288.20	1.977	
15,700.0	10,840.1	15,622.5	10,773.0	148.4	147.7	-83.24	316.1	5,082.5	569.4	275.5	293.95	1.937	
15,800.0	10,840.2	15,722.5	10,773.2	151.3	150.6	-83.24	316.1	5,182.5	569.1	269.4	299.69	1.899	
15,900.0	10,840.4	15,822.5	10,773.4	154.1	153.5	-83.23	316.1	5,282.5	568.8	263.4	305.44	1.862	
16,000.0	10,840.6	15,922.5	10,773.6	157.0	156.4	-83.23	316.1	5,382.5	568.5	257.3	311.19	1.827	
16,100.0	10,840.8	16,022.5	10,773.7	159.9	159.3	-83.23	316.1	5,482.5	568.2	251.2	316.94	1.793	
16,200.0	10,840.9	16,122.5	10,773.9	162.8	162.2	-83.22	316.1	5,582.5	567.9	245.2	322.70	1.760	
16,300.0	10,841.1	16,222.5	10,774.1	165.7	165.1	-83.22	316.1	5,682.5	567.6	239.1	328.46	1.728	
16,400.0	10,841.3	16,322.5	10,774.3	168.6	168.0	-83.22	316.1	5,782.5	567.2	233.0	334.22	1.697	
16,500.0	10,841.4	16,422.5	10,774.4	171.5	170.9	-83.21	316.1	5,882.5	566.9	227.0	339.98	1.668	
16,600.0	10,841.6	16,522.5	10,774.6	174.4	173.8	-83.21	316.1	5,982.5	566.6	220.9	345.75	1.639	
16,700.0	10,841.8	16,622.5	10,774.8	177.3	176.7	-83.20	316.1	6,082.5	566.3	214.8	351.51	1.611	
16,800.0	10,842.0	16,722.5	10,775.0	180.2	179.6	-83.20	316.1	6,182.5	566.0	208.7	357.28	1.584	
16,900.0	10,842.1	16,822.5	10,775.1	183.1	182.5	-83.20	316.1	6,282.5	565.7	202.6	363.05	1.558	
17,000.0	10,842.3	16,922.5	10,775.3	186.0	185.4	-83.19	316.1	6,382.5	565.4	196.6	368.82	1.533	
17,100.0	10,842.5	17,022.5	10,775.5	188.9	188.3	-83.19	316.1	6,482.5	565.1	190.5	374.59	1.508	
17,200.0	10,842.7	17,122.5	10,775.7	191.8	191.2	-83.19	316.1	6,582.5	564.7	184.4	380.37	1.485 Level 3	
17,300.0	10,842.8	17,222.5	10,775.8	194.7	194.1	-83.18	316.1	6,682.5	564.4	178.3	386.14	1.462 Level 3	
17,400.0	10,843.0	17,322.5	10,776.0	197.6	197.0	-83.18	316.0	6,782.5	564.1	172.2	391.92	1.439 Level 3	
17,500.0	10,843.2	17,422.5	10,776.2	200.6	199.9	-83.17	316.0	6,882.5	563.8	166.1	397.70	1.418 Level 3	
17,600.0	10,843.4	17,522.5	10,776.4	203.5	202.8	-83.17	316.0	6,982.5	563.5	160.0	403.48	1.397 Level 3	
17,700.0	10,843.5	17,622.5	10,776.5	206.4	205.7	-83.17	316.0	7,082.5	563.2	153.9	409.26	1.376 Level 3	
17,800.0	10,843.7	17,722.5	10,776.7	209.3	208.6	-83.16	316.0	7,182.5	562.9	147.8	415.04	1.356 Level 3	
17,900.0	10,843.9	17,822.5	10,776.9	212.2	211.5	-83.16	316.0	7,282.5	562.6	141.7	420.82	1.337 Level 3	
18,000.0	10,844.1	17,922.5	10,777.1	215.1	214.4	-83.16	316.0	7,382.5	562.3	135.6	426.60	1.318 Level 3	
18,100.0	10,844.2	18,022.5	10,777.2	218.0	217.3	-83.15	316.0	7,482.5	561.9	129.5	432.39	1.300 Level 3	
18,200.0	10,844.4	18,122.5	10,777.4	220.9	220.3	-83.15	316.0	7,582.5	561.6	123.5	438.17	1.282 Level 3	
18,300.0	10,844.6	18,222.5	10,777.6	223.8	223.2	-83.14	316.0	7,682.5	561.3	117.4	443.96	1.264 Level 3	
18,400.0	10,844.8	18,322.5	10,777.7	226.7	226.1	-83.14	316.0	7,782.5	561.0	111.3	449.75	1.247 Level 2	
18,500.0	10,844.9	18,422.5	10,777.9	229.7	229.0	-83.14	316.0	7,882.5	560.7	105.2	455.53	1.231 Level 2	
18,600.0	10,845.1	18,522.5	10,778.1	232.6	231.9	-83.13	316.0	7,982.5	560.4	99.1	461.32	1.215 Level 2	
18,700.0	10,845.3	18,622.5	10,778.3	235.5	234.8	-83.13	316.0	8,082.5	560.1	93.0	467.11	1.199 Level 2	
18,800.0	10,845.4	18,722.5	10,778.4	238.4	237.7	-83.12	316.0	8,182.5	559.8	86.9	472.90	1.184 Level 2	
18,900.0	10,845.6	18,822.5	10,778.6	241.3	240.7	-83.12	316.0	8,282.5	559.4	80.8	478.69	1.169 Level 2	
19,000.0	10,845.8	18,922.5	10,778.8	244.2	243.6	-83.12	316.0	8,382.5	559.1	74.6	484.48	1.154 Level 2	
19,100.0	10,846.0	19,022.5	10,779.0	247.1	246.5	-83.11	315.9	8,482.5	558.8	68.5	490.27	1.140 Level 2	
19,200.0	10,846.1	19,122.5	10,779.1	250.1	249.4	-83.11	315.9	8,582.5	558.5	62.4	496.07	1.126 Level 2	
19,300.0	10,846.3	19,222.5	10,779.3	253.0	252.3	-83.11	315.9	8,682.5	558.2	56.3	501.86	1.112 Level 2	
19,400.0	10,846.5	19,322.5	10,779.5	255.9	255.2	-83.10	315.9	8,782.5	557.9	50.2	507.65	1.099 Level 2	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services

Anticollision Report



Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 10B - Kline Federal 5300 41-18 10B - Design #1												Offset Site Error:	0.0 usft
Survey Program: 0-MWD												Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis			Distance						
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
19,500.0	10,846.7	19,422.5	10,779.7	258.8	258.1	-83.10	315.9	8,882.5	557.6	44.1	513.45	1.086	Level 2
19,600.0	10,846.8	19,522.5	10,779.8	261.7	261.1	-83.09	315.9	8,982.5	557.3	38.0	519.24	1.073	Level 2
19,700.0	10,847.0	19,622.5	10,780.0	264.6	264.0	-83.09	315.9	9,082.5	556.9	31.9	525.04	1.061	Level 2
19,800.0	10,847.2	19,722.5	10,780.2	267.6	266.9	-83.09	315.9	9,182.5	556.6	25.8	530.83	1.049	Level 2
19,900.0	10,847.4	19,822.5	10,780.4	270.5	269.8	-83.08	315.9	9,282.5	556.3	19.7	536.63	1.037	Level 2
20,000.0	10,847.5	19,922.5	10,780.5	273.4	272.7	-83.08	315.9	9,382.5	556.0	13.6	542.42	1.025	Level 2
20,100.0	10,847.7	20,022.5	10,780.7	276.3	275.7	-83.07	315.9	9,482.5	555.7	7.5	548.22	1.014	Level 2
20,200.0	10,847.9	20,122.5	10,780.9	279.2	278.6	-83.07	315.9	9,582.5	555.4	1.4	554.02	1.002	Level 2
20,300.0	10,848.1	20,222.5	10,781.1	282.1	281.5	-83.07	315.9	9,682.5	555.1	-4.7	559.81	0.992	Level 1
20,400.0	10,848.2	20,322.5	10,781.2	285.1	284.4	-83.06	315.9	9,782.5	554.8	-10.8	565.61	0.981	Level 1
20,500.0	10,848.4	20,422.5	10,781.4	288.0	287.3	-83.06	315.9	9,882.5	554.4	-17.0	571.41	0.970	Level 1
20,600.0	10,848.6	20,522.5	10,781.6	290.9	290.3	-83.06	315.9	9,982.5	554.1	-23.1	577.21	0.960	Level 1
20,667.7	10,848.7	20,590.0	10,781.7	292.9	292.2	-83.05	315.9	10,050.0	553.9	-27.2	581.13	0.953	Level 1, ES, SF



Ryan Directional Services

Anticollision Report



Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 11T2 - Kline Federal 5300 41-18 11T2 - Design #1												Offset Site Error:	0.0 usft
Survey Program: 0-MWD												Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis		Highside Toolface (°)	Distance		Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)		Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)					
0.0	0.0	0.0	0.0	2.0	2.0	0.00	33.4	0.0	33.4				
100.0	100.0	100.0	100.0	2.0	2.0	0.00	33.4	0.0	33.4	29.4	4.00	8.352	
200.0	200.0	200.0	200.0	2.0	2.0	0.00	33.4	0.0	33.4	29.4	4.05	8.261	
300.0	300.0	300.0	300.0	2.1	2.1	0.00	33.4	0.0	33.4	29.3	4.14	8.077	
400.0	400.0	400.0	400.0	2.1	2.1	0.00	33.4	0.0	33.4	29.2	4.28	7.816	
500.0	500.0	500.0	500.0	2.2	2.2	0.00	33.4	0.0	33.4	29.0	4.46	7.502	
600.0	600.0	600.0	600.0	2.3	2.3	0.00	33.4	0.0	33.4	28.8	4.67	7.155	
700.0	700.0	700.0	700.0	2.5	2.5	0.00	33.4	0.0	33.4	28.5	4.92	6.795	
800.0	800.0	800.0	800.0	2.6	2.6	0.00	33.4	0.0	33.4	28.2	5.20	6.436	
900.0	900.0	900.0	900.0	2.7	2.7	0.00	33.4	0.0	33.4	27.9	5.49	6.087	
1,000.0	1,000.0	1,000.0	1,000.0	2.9	2.9	0.00	33.4	0.0	33.4	27.6	5.81	5.755	
1,100.0	1,100.0	1,100.0	1,100.0	3.1	3.1	0.00	33.4	0.0	33.4	27.3	6.14	5.442	
1,200.0	1,200.0	1,200.0	1,200.0	3.2	3.2	0.00	33.4	0.0	33.4	26.9	6.49	5.151	
1,300.0	1,300.0	1,300.0	1,300.0	3.4	3.4	0.00	33.4	0.0	33.4	26.6	6.85	4.880	
1,400.0	1,400.0	1,400.0	1,400.0	3.6	3.6	0.00	33.4	0.0	33.4	26.2	7.22	4.630	
1,500.0	1,500.0	1,500.0	1,500.0	3.8	3.8	0.00	33.4	0.0	33.4	25.8	7.60	4.400	
1,600.0	1,600.0	1,600.0	1,600.0	4.0	4.0	0.00	33.4	0.0	33.4	25.5	7.99	4.187	
1,700.0	1,700.0	1,700.0	1,700.0	4.2	4.2	0.00	33.4	0.0	33.4	25.1	8.38	3.991	
1,800.0	1,800.0	1,800.0	1,800.0	4.4	4.4	0.00	33.4	0.0	33.4	24.7	8.78	3.810	
1,900.0	1,900.0	1,900.0	1,900.0	4.6	4.6	0.00	33.4	0.0	33.4	24.3	9.18	3.643	
2,000.0	2,000.0	2,000.0	2,000.0	4.8	4.8	0.00	33.4	0.0	33.4	23.9	9.58	3.489	
2,100.0	2,100.0	2,100.0	2,100.0	5.0	5.0	0.00	33.4	0.0	33.4	23.4	9.99	3.346	
2,200.0	2,200.0	2,200.0	2,200.0	5.2	5.2	0.00	33.4	0.0	33.4	23.0	10.41	3.213	
2,200.0	2,200.0	2,200.0	2,200.0	5.2	5.2	0.00	33.4	0.0	33.4	23.0	10.41	3.213 CC	
2,216.7	2,216.7	2,216.7	2,216.7	5.2	5.2	-90.12	33.4	0.0	33.4	23.0	10.48	3.192	
2,300.0	2,300.0	2,300.0	2,300.0	5.4	5.4	-91.37	33.4	0.0	33.4	22.6	10.80	3.097	
2,400.0	2,400.0	2,400.0	2,400.0	5.6	5.6	-92.86	33.4	0.0	33.5	22.3	11.20	2.990	
2,500.0	2,500.0	2,500.0	2,500.0	5.8	5.8	-94.35	33.4	0.0	33.5	21.9	11.59	2.892	
2,600.0	2,600.0	2,600.0	2,600.0	6.0	6.0	-95.84	33.4	0.0	33.6	21.6	12.00	2.802	
2,700.0	2,700.0	2,700.0	2,700.0	6.1	6.3	-97.31	33.4	0.0	33.7	21.3	12.40	2.718	
2,800.0	2,800.0	2,800.0	2,800.0	6.3	6.5	-98.78	33.4	0.0	33.8	21.0	12.81	2.641	
2,900.0	2,900.0	2,900.0	2,900.0	6.5	6.7	-100.23	33.4	0.0	34.0	20.8	13.22	2.570	
3,000.0	3,000.0	3,000.0	3,000.0	6.7	6.9	-101.67	33.4	0.0	34.1	20.5	13.64	2.504	
3,100.0	3,100.0	3,100.0	3,100.0	6.9	7.1	-103.10	33.4	0.0	34.3	20.3	14.05	2.443	
3,200.0	3,200.0	3,200.0	3,200.0	7.1	7.3	-104.51	33.4	0.0	34.5	20.1	14.47	2.387	
3,300.0	3,300.0	3,300.0	3,300.0	7.3	7.5	-105.90	33.4	0.0	34.8	19.9	14.89	2.335	
3,400.0	3,400.0	3,400.0	3,400.0	7.6	7.8	-107.27	33.4	0.0	35.0	19.7	15.31	2.287	
3,500.0	3,500.0	3,500.0	3,500.0	7.8	8.0	-108.63	33.4	0.0	35.3	19.6	15.73	2.243	
3,600.0	3,599.9	3,599.9	3,599.9	8.0	8.2	-109.96	33.4	0.0	35.6	19.4	16.16	2.202	
3,700.0	3,699.9	3,699.9	3,699.9	8.2	8.4	-111.27	33.4	0.0	35.9	19.3	16.58	2.164	
3,800.0	3,799.9	3,799.9	3,799.9	8.4	8.6	-112.56	33.4	0.0	36.2	19.2	17.01	2.129	
3,900.0	3,899.9	3,899.9	3,899.9	8.6	8.9	-113.82	33.4	0.0	36.6	19.1	17.44	2.096	
4,000.0	3,999.9	3,999.9	3,999.9	8.8	9.1	-115.06	33.4	0.0	36.9	19.0	17.87	2.066	
4,100.0	4,099.9	4,099.9	4,099.9	9.0	9.3	-116.27	33.4	0.0	37.3	19.0	18.30	2.038	
4,200.0	4,199.9	4,199.9	4,199.9	9.2	9.5	-117.46	33.4	0.0	37.7	19.0	18.73	2.012	
4,300.0	4,299.9	4,299.9	4,299.9	9.4	9.7	-118.63	33.4	0.0	38.1	18.9	19.16	1.988	
4,400.0	4,399.9	4,399.9	4,399.9	9.7	10.0	-119.77	33.4	0.0	38.5	18.9	19.59	1.966	
4,500.0	4,499.9	4,499.9	4,499.9	9.9	10.2	-120.88	33.4	0.0	39.0	18.9	20.03	1.946	
4,600.0	4,599.9	4,599.9	4,599.9	10.1	10.4	-121.97	33.4	0.0	39.4	19.0	20.46	1.927	
4,700.0	4,699.9	4,699.9	4,699.9	10.3	10.6	-123.03	33.4	0.0	39.9	19.0	20.89	1.909	
4,800.0	4,799.9	4,799.9	4,799.9	10.5	10.8	-124.07	33.4	0.0	40.4	19.0	21.33	1.893	
4,900.0	4,899.9	4,899.9	4,899.9	10.7	11.1	-125.09	33.4	0.0	40.9	19.1	21.76	1.878	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services

Anticollision Report



Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 11T2 - Kline Federal 5300 41-18 11T2 - Design #1													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis			Distance							Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
5,000.0	4,999.9	4,999.9	4,999.9	10.9	11.3	-126.07	33.4	0.0	41.4	19.2	22.20	1.864		
5,100.0	5,099.9	5,099.9	5,099.9	11.2	11.5	-127.04	33.4	0.0	41.9	19.3	22.64	1.851		
5,200.0	5,199.9	5,199.9	5,199.9	11.4	11.7	-127.98	33.4	0.0	42.4	19.3	23.07	1.839		
5,300.0	5,299.9	5,299.9	5,299.9	11.6	11.9	-128.90	33.4	0.0	43.0	19.5	23.51	1.827		
5,400.0	5,399.9	5,399.9	5,399.9	11.8	12.2	-129.79	33.4	0.0	43.5	19.6	23.95	1.817		
5,500.0	5,499.9	5,499.9	5,499.9	12.0	12.4	-130.66	33.4	0.0	44.1	19.7	24.39	1.808		
5,600.0	5,599.9	5,599.9	5,599.9	12.2	12.6	-131.51	33.4	0.0	44.7	19.8	24.83	1.799		
5,700.0	5,699.9	5,699.9	5,699.9	12.5	12.8	-132.34	33.4	0.0	45.2	20.0	25.26	1.791		
5,800.0	5,799.9	5,799.9	5,799.9	12.7	13.0	-133.15	33.4	0.0	45.8	20.1	25.70	1.783		
5,900.0	5,899.9	5,899.9	5,899.9	12.9	13.3	-133.93	33.4	0.0	46.4	20.3	26.14	1.776		
6,000.0	5,999.9	5,999.9	5,999.9	13.1	13.5	-134.70	33.4	0.0	47.0	20.5	26.58	1.770		
6,100.0	6,099.9	6,099.9	6,099.9	13.3	13.7	-135.44	33.4	0.0	47.7	20.6	27.02	1.764		
6,200.0	6,199.8	6,199.8	6,199.8	13.6	13.9	-136.17	33.4	0.0	48.3	20.8	27.47	1.758		
6,300.0	6,299.8	6,299.8	6,299.8	13.8	14.2	-136.88	33.4	0.0	48.9	21.0	27.91	1.753		
6,400.0	6,399.8	6,399.8	6,399.8	14.0	14.4	-137.57	33.4	0.0	49.6	21.2	28.35	1.748		
6,500.0	6,499.8	6,499.8	6,499.8	14.2	14.6	-138.24	33.4	0.0	50.2	21.4	28.79	1.744		
6,600.0	6,599.8	6,599.8	6,599.8	14.4	14.8	-138.89	33.4	0.0	50.9	21.6	29.23	1.740		
6,700.0	6,699.8	6,699.8	6,699.8	14.7	15.1	-139.53	33.4	0.0	51.5	21.8	29.67	1.736		
6,800.0	6,799.8	6,799.8	6,799.8	14.9	15.3	-140.15	33.4	0.0	52.2	22.1	30.11	1.733		
6,900.0	6,899.8	6,899.8	6,899.8	15.1	15.5	-140.76	33.4	0.0	52.9	22.3	30.56	1.730		
7,000.0	6,999.8	6,999.8	6,999.8	15.3	15.7	-141.35	33.4	0.0	53.5	22.5	31.00	1.727		
7,100.0	7,099.8	7,099.8	7,099.8	15.5	15.9	-141.93	33.4	0.0	54.2	22.8	31.44	1.725		
7,200.0	7,199.8	7,199.8	7,199.8	15.8	16.2	-142.49	33.4	0.0	54.9	23.0	31.88	1.722		
7,300.0	7,299.8	7,299.8	7,299.8	16.0	16.4	-143.04	33.4	0.0	55.6	23.3	32.33	1.720		
7,400.0	7,399.8	7,399.8	7,399.8	16.2	16.6	-143.57	33.4	0.0	56.3	23.5	32.77	1.718		
7,500.0	7,499.8	7,499.8	7,499.8	16.4	16.8	-144.09	33.4	0.0	57.0	23.8	33.21	1.717		
7,600.0	7,599.8	7,599.8	7,599.8	16.6	17.1	-144.60	33.4	0.0	57.7	24.1	33.66	1.715		
7,700.0	7,699.8	7,699.8	7,699.8	16.9	17.3	-145.09	33.4	0.0	58.4	24.3	34.10	1.714		
7,800.0	7,799.8	7,799.8	7,799.8	17.1	17.5	-145.58	33.4	0.0	59.2	24.6	34.55	1.712		
7,900.0	7,899.8	7,899.8	7,899.8	17.3	17.7	-146.05	33.4	0.0	59.9	24.9	34.99	1.711		
7,926.7	7,926.4	7,926.4	7,926.4	17.4	17.8	-146.17	33.4	0.0	60.1	25.0	35.11	1.711		
7,943.3	7,943.1	7,943.1	7,943.1	17.4	17.8	-56.21	33.4	0.0	60.1	24.9	35.23	1.707		
8,000.0	7,999.8	7,999.8	7,999.8	17.5	18.0	-56.21	33.4	0.0	60.1	24.7	35.47	1.695		
8,100.0	8,099.8	8,099.8	8,099.8	17.7	18.2	-56.21	33.4	0.0	60.1	24.2	35.90	1.675		
8,200.0	8,199.8	8,199.8	8,199.8	17.9	18.4	-56.21	33.4	0.0	60.1	23.8	36.33	1.655		
8,300.0	8,299.8	8,299.8	8,299.8	18.1	18.6	-56.21	33.4	0.0	60.1	23.4	36.76	1.636		
8,400.0	8,399.8	8,399.8	8,399.8	18.4	18.8	-56.21	33.4	0.0	60.1	22.9	37.19	1.617		
8,500.0	8,499.8	8,499.8	8,499.8	18.6	19.1	-56.21	33.4	0.0	60.1	22.5	37.63	1.598		
8,600.0	8,599.8	8,599.8	8,599.8	18.8	19.3	-56.21	33.4	0.0	60.1	22.1	38.06	1.580		
8,700.0	8,699.8	8,699.8	8,699.8	19.0	19.5	-56.21	33.4	0.0	60.1	21.6	38.49	1.562		
8,800.0	8,799.8	8,799.8	8,799.8	19.2	19.7	-56.21	33.4	0.0	60.1	21.2	38.93	1.545		
8,900.0	8,899.8	8,899.8	8,899.8	19.4	20.0	-56.21	33.4	0.0	60.1	20.8	39.36	1.528		
9,000.0	8,999.8	8,999.8	8,999.8	19.6	20.2	-56.21	33.4	0.0	60.1	20.3	39.80	1.511		
9,100.0	9,099.8	9,099.8	9,099.8	19.8	20.4	-56.21	33.4	0.0	60.1	19.9	40.23	1.495 Level 3		
9,200.0	9,199.8	9,199.8	9,199.8	20.0	20.6	-56.21	33.4	0.0	60.1	19.5	40.67	1.479 Level 3		
9,300.0	9,299.8	9,299.8	9,299.8	20.2	20.9	-56.21	33.4	0.0	60.1	19.0	41.10	1.463 Level 3		
9,400.0	9,399.8	9,399.8	9,399.8	20.5	21.1	-56.21	33.4	0.0	60.1	18.6	41.54	1.448 Level 3		
9,500.0	9,499.8	9,499.8	9,499.8	20.7	21.3	-56.21	33.4	0.0	60.1	18.2	41.97	1.433 Level 3		
9,600.0	9,599.8	9,599.8	9,599.8	20.9	21.5	-56.21	33.4	0.0	60.1	17.7	42.41	1.418 Level 3		
9,700.0	9,699.8	9,699.8	9,699.8	21.1	21.8	-56.21	33.4	0.0	60.1	17.3	42.85	1.403 Level 3		
9,800.0	9,799.8	9,799.8	9,799.8	21.3	22.0	-56.21	33.4	0.0	60.1	16.8	43.28	1.389 Level 3		
9,900.0	9,899.8	9,899.8	9,899.8	21.5	22.2	-56.21	33.4	0.0	60.1	16.4	43.72	1.375 Level 3		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 11T2 - Kline Federal 5300 41-18 11T2 - Design #1													Offset Site Error: 0.0 usft	
Survey Program: 0-MWD													Offset Well Error: 2.0 usft	
Reference		Offset		Semi Major Axis			Distance							Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
10,000.0	9,999.8	9,999.8	9,999.8	21.7	22.4	-56.21	33.4	0.0	60.1	16.0	44.16	1.362	Level 3	
10,100.0	10,099.8	10,099.8	10,099.8	21.9	22.6	-56.21	33.4	0.0	60.1	15.5	44.60	1.348	Level 3	
10,200.0	10,199.8	10,199.8	10,199.8	22.2	22.9	-56.21	33.4	0.0	60.1	15.1	45.04	1.335	Level 3	
10,300.0	10,299.8	10,299.8	10,299.8	22.4	23.1	-56.21	33.4	0.0	60.1	14.7	45.47	1.322	Level 3	
10,354.8	10,354.6	10,354.6	10,354.6	22.5	23.2	-56.21	33.4	0.0	60.1	14.4	45.71	1.315	Level 3	
10,375.0	10,374.8	10,374.8	10,374.8	22.5	23.3	-166.30	33.4	0.0	60.5	14.8	45.73	1.324	Level 3	
10,400.0	10,399.7	10,400.4	10,400.4	22.6	23.3	-166.59	33.5	0.1	62.2	16.4	45.73	1.359	Level 3	
10,425.0	10,424.5	10,427.6	10,427.6	22.7	23.4	-166.38	33.8	1.2	64.4	18.7	45.65	1.410	Level 3	
10,450.0	10,449.2	10,454.9	10,454.8	22.7	23.4	-165.45	34.5	3.9	66.9	21.4	45.49	1.471	Level 3	
10,475.0	10,473.5	10,482.2	10,481.6	22.8	23.5	-163.88	35.7	8.1	69.8	24.6	45.26	1.543		
10,500.0	10,497.6	10,509.3	10,508.2	22.8	23.6	-161.78	37.3	13.7	73.2	28.2	44.98	1.627		
10,525.0	10,521.2	10,536.4	10,534.2	22.9	23.6	-159.25	39.3	20.7	77.0	32.3	44.66	1.724		
10,550.0	10,544.4	10,563.2	10,559.6	23.0	23.7	-156.41	41.7	29.1	81.4	37.0	44.33	1.836		
10,575.0	10,567.1	10,589.8	10,584.3	23.0	23.7	-153.35	44.4	38.7	86.3	42.3	44.01	1.962		
10,600.0	10,589.1	10,616.2	10,608.1	23.1	23.8	-150.17	47.4	49.5	92.0	48.2	43.73	2.103		
10,625.0	10,610.6	10,642.3	10,631.1	23.2	23.9	-146.93	50.8	61.5	98.3	54.8	43.51	2.259		
10,650.0	10,631.3	10,668.1	10,653.0	23.3	24.0	-143.69	54.5	74.4	105.3	61.9	43.36	2.428		
10,675.0	10,651.3	10,693.5	10,674.0	23.4	24.0	-140.50	58.4	88.3	113.0	69.7	43.31	2.610		
10,700.0	10,670.5	10,718.6	10,693.9	23.5	24.1	-137.38	62.6	103.1	121.4	78.1	43.35	2.801		
10,725.0	10,688.8	10,743.4	10,712.7	23.6	24.2	-134.35	67.0	118.6	130.6	87.1	43.49	3.002		
10,750.0	10,706.2	10,767.8	10,730.4	23.7	24.3	-131.42	71.6	134.8	140.3	96.6	43.73	3.209		
10,775.0	10,722.6	10,791.9	10,746.9	23.9	24.4	-128.58	76.3	151.6	150.7	106.7	44.07	3.421		
10,800.0	10,738.0	10,815.7	10,762.4	24.0	24.5	-125.83	81.2	168.9	161.7	117.3	44.49	3.635		
10,825.0	10,752.4	10,839.1	10,776.8	24.2	24.6	-123.17	86.3	186.7	173.3	128.3	44.99	3.852		
10,850.0	10,765.7	10,862.2	10,790.0	24.4	24.8	-120.59	91.4	205.0	185.3	139.8	45.56	4.068		
10,875.0	10,777.8	10,885.1	10,802.2	24.6	24.9	-118.08	96.7	223.6	197.8	151.7	46.18	4.284		
10,900.0	10,788.8	10,907.7	10,813.4	24.8	25.1	-115.64	102.1	242.5	210.8	163.9	46.85	4.499		
10,925.0	10,798.6	10,930.1	10,823.5	25.0	25.2	-113.25	107.5	261.7	224.1	176.5	47.56	4.712		
10,950.0	10,807.2	10,952.2	10,832.6	25.3	25.4	-110.92	113.0	281.1	237.7	189.4	48.29	4.923		
10,975.0	10,814.5	10,974.2	10,840.6	25.6	25.6	-108.64	118.6	300.8	251.6	202.6	49.03	5.132		
11,000.0	10,820.6	10,996.1	10,847.7	25.8	25.8	-106.40	124.2	320.7	265.8	216.0	49.78	5.340		
11,025.0	10,825.4	11,017.9	10,853.8	26.1	26.0	-104.22	129.9	340.8	280.2	229.6	50.52	5.545		
11,050.0	10,828.9	11,039.6	10,858.9	26.5	26.2	-102.08	135.6	361.1	294.7	243.4	51.25	5.750		
11,075.0	10,831.1	11,061.2	10,863.1	26.8	26.4	-99.99	141.4	381.6	309.4	257.4	51.97	5.953		
11,100.0	10,832.0	11,082.9	10,866.3	27.1	26.7	-97.95	147.3	402.3	324.1	271.4	52.67	6.154		
11,104.0	10,832.0	11,086.4	10,866.7	27.2	26.7	-97.63	148.2	405.5	326.5	273.7	52.78	6.186		
11,200.0	10,832.2	11,174.3	10,870.0	28.6	27.8	-96.77	171.9	490.1	380.2	325.0	55.19	6.889		
11,300.0	10,832.4	11,276.1	10,870.2	30.3	29.3	-95.69	196.8	588.8	428.5	370.2	58.29	7.351		
11,400.0	10,832.6	11,384.8	10,870.4	32.2	31.1	-94.99	219.4	695.1	468.1	406.2	61.90	7.562		
11,500.0	10,832.7	11,499.3	10,870.6	34.1	33.2	-94.53	238.9	807.9	498.3	432.4	65.98	7.553		
11,600.0	10,832.9	11,618.2	10,870.8	36.2	35.6	-94.25	254.3	925.8	518.8	448.3	70.48	7.361		
11,680.4	10,833.1	11,715.9	10,871.0	38.0	37.7	-94.14	263.3	1,023.1	527.9	453.6	74.28	7.107		
11,700.0	10,833.1	11,739.8	10,871.0	38.4	38.2	-94.12	265.0	1,047.0	529.3	454.0	75.25	7.033		
11,800.0	10,833.3	11,862.4	10,871.2	40.7	41.0	-94.08	270.6	1,169.4	533.5	453.2	80.30	6.644		
11,900.0	10,833.4	11,975.3	10,871.4	43.0	43.6	-94.08	271.5	1,282.3	533.9	448.6	85.31	6.258		
12,000.0	10,833.6	12,075.3	10,871.6	45.4	46.0	-94.08	271.5	1,382.3	533.6	443.4	90.15	5.918		
12,100.0	10,833.8	12,175.3	10,871.8	47.9	48.4	-94.08	271.5	1,482.3	533.2	438.1	95.11	5.607		
12,200.0	10,834.0	12,275.3	10,872.0	50.4	50.9	-94.09	271.5	1,582.3	532.9	432.8	100.16	5.321		
12,300.0	10,834.1	12,375.3	10,872.1	53.0	53.5	-94.09	271.5	1,682.3	532.6	427.3	105.29	5.059		
12,400.0	10,834.3	12,475.3	10,872.3	55.5	56.0	-94.09	271.5	1,782.3	532.3	421.8	110.49	4.818		
12,500.0	10,834.5	12,575.3	10,872.5	58.2	58.6	-94.09	271.5	1,882.3	532.0	416.3	115.75	4.596		
12,600.0	10,834.7	12,675.3	10,872.6	60.8	61.3	-94.10	271.5	1,982.3	531.7	410.6	121.07	4.392		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services

Anticollision Report



Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 11T2 - Kline Federal 5300 41-18 11T2 - Design #1													Offset Site Error: 0.0 usft
Survey Program: 0-MWD													Offset Well Error: 2.0 usft
Reference		Offset		Semi Major Axis			Distance						Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	
12,700.0	10,834.8	12,775.3	10,872.8	63.5	64.0	-94.10	271.5	2,082.3	531.4	405.0	126.44	4.203	
12,800.0	10,835.0	12,875.3	10,873.0	66.2	66.6	-94.10	271.5	2,182.3	531.1	399.3	131.84	4.028	
12,900.0	10,835.2	12,975.3	10,873.2	68.9	69.4	-94.10	271.5	2,282.3	530.8	393.5	137.28	3.866	
13,000.0	10,835.4	13,075.3	10,873.3	71.7	72.1	-94.11	271.5	2,382.3	530.5	387.7	142.76	3.716	
13,100.0	10,835.5	13,175.3	10,873.5	74.4	74.8	-94.11	271.5	2,482.3	530.2	381.9	148.26	3.576	
13,200.0	10,835.7	13,275.3	10,873.7	77.2	77.6	-94.11	271.5	2,582.3	529.9	376.1	153.78	3.445	
13,300.0	10,835.9	13,375.3	10,873.9	79.9	80.3	-94.11	271.5	2,682.3	529.6	370.2	159.33	3.324	
13,400.0	10,836.1	13,475.3	10,874.0	82.7	83.1	-94.12	271.5	2,782.3	529.2	364.3	164.90	3.209	
13,500.0	10,836.2	13,575.3	10,874.2	85.5	85.9	-94.12	271.5	2,882.3	528.9	358.4	170.49	3.102	
13,600.0	10,836.4	13,675.3	10,874.4	88.3	88.7	-94.12	271.5	2,982.3	528.6	352.5	176.10	3.002	
13,700.0	10,836.6	13,775.3	10,874.6	91.1	91.5	-94.12	271.5	3,082.3	528.3	346.6	181.72	2.907	
13,800.0	10,836.7	13,875.3	10,874.7	94.0	94.3	-94.13	271.5	3,182.3	528.0	340.7	187.35	2.818	
13,900.0	10,836.9	13,975.3	10,874.9	96.8	97.2	-94.13	271.5	3,282.3	527.7	334.7	193.00	2.734	
14,000.0	10,837.1	14,075.3	10,875.1	99.6	100.0	-94.13	271.5	3,382.3	527.4	328.7	198.66	2.655	
14,100.0	10,837.3	14,175.3	10,875.3	102.5	102.8	-94.13	271.5	3,482.3	527.1	322.8	204.33	2.580	
14,200.0	10,837.4	14,275.3	10,875.4	105.3	105.7	-94.14	271.5	3,582.3	526.8	316.8	210.01	2.508	
14,300.0	10,837.6	14,375.3	10,875.6	108.2	108.5	-94.14	271.5	3,682.3	526.5	310.8	215.69	2.441	
14,400.0	10,837.8	14,475.3	10,875.8	111.0	111.3	-94.14	271.5	3,782.3	526.2	304.8	221.39	2.377	
14,500.0	10,838.0	14,575.3	10,876.0	113.9	114.2	-94.14	271.5	3,882.3	525.9	298.8	227.09	2.316	
14,600.0	10,838.1	14,675.3	10,876.1	116.7	117.1	-94.15	271.5	3,982.3	525.6	292.7	232.80	2.257	
14,700.0	10,838.3	14,775.3	10,876.3	119.6	119.9	-94.15	271.5	4,082.3	525.2	286.7	238.52	2.202	
14,800.0	10,838.5	14,875.3	10,876.5	122.5	122.8	-94.15	271.5	4,182.3	524.9	280.7	244.24	2.149	
14,900.0	10,838.7	14,975.3	10,876.7	125.3	125.6	-94.15	271.5	4,282.3	524.6	274.7	249.97	2.099	
15,000.0	10,838.8	15,075.3	10,876.8	128.2	128.5	-94.16	271.5	4,382.3	524.3	268.6	255.71	2.050	
15,100.0	10,839.0	15,175.3	10,877.0	131.1	131.4	-94.16	271.5	4,482.3	524.0	262.6	261.44	2.004	
15,200.0	10,839.2	15,275.3	10,877.2	133.9	134.3	-94.16	271.5	4,582.3	523.7	256.5	267.19	1.960	
15,300.0	10,839.4	15,375.3	10,877.4	136.8	137.1	-94.16	271.5	4,682.3	523.4	250.5	272.94	1.918	
15,400.0	10,839.5	15,475.3	10,877.5	139.7	140.0	-94.17	271.5	4,782.3	523.1	244.4	278.69	1.877	
15,500.0	10,839.7	15,575.3	10,877.7	142.6	142.9	-94.17	271.5	4,882.3	522.8	238.3	284.44	1.838	
15,600.0	10,839.9	15,675.3	10,877.9	145.5	145.8	-94.17	271.5	4,982.3	522.5	232.3	290.20	1.800	
15,700.0	10,840.1	15,775.3	10,878.1	148.4	148.7	-94.17	271.5	5,082.3	522.2	226.2	295.96	1.764	
15,800.0	10,840.2	15,875.3	10,878.2	151.3	151.6	-94.18	271.5	5,182.3	521.9	220.1	301.73	1.730	
15,900.0	10,840.4	15,975.3	10,878.4	154.1	154.4	-94.18	271.5	5,282.3	521.6	214.1	307.50	1.696	
16,000.0	10,840.6	16,075.3	10,878.6	157.0	157.3	-94.18	271.5	5,382.3	521.2	208.0	313.27	1.664	
16,100.0	10,840.8	16,175.3	10,878.8	159.9	160.2	-94.18	271.5	5,482.3	520.9	201.9	319.04	1.633	
16,200.0	10,840.9	16,275.3	10,878.9	162.8	163.1	-94.19	271.5	5,582.3	520.6	195.8	324.82	1.603	
16,300.0	10,841.1	16,375.3	10,879.1	165.7	166.0	-94.19	271.5	5,682.3	520.3	189.7	330.60	1.574	
16,400.0	10,841.3	16,475.3	10,879.3	168.6	168.9	-94.19	271.5	5,782.3	520.0	183.6	336.38	1.546	
16,500.0	10,841.4	16,575.3	10,879.5	171.5	171.8	-94.19	271.5	5,882.3	519.7	177.5	342.16	1.519	
16,600.0	10,841.6	16,675.3	10,879.6	174.4	174.7	-94.20	271.5	5,982.3	519.4	171.5	347.95	1.493 Level 3	
16,700.0	10,841.8	16,775.3	10,879.8	177.3	177.6	-94.20	271.5	6,082.3	519.1	165.4	353.73	1.467 Level 3	
16,800.0	10,842.0	16,875.3	10,880.0	180.2	180.5	-94.20	271.5	6,182.3	518.8	159.3	359.52	1.443 Level 3	
16,900.0	10,842.1	16,975.3	10,880.2	183.1	183.4	-94.20	271.5	6,282.3	518.5	153.2	365.31	1.419 Level 3	
17,000.0	10,842.3	17,075.3	10,880.3	186.0	186.3	-94.21	271.5	6,382.3	518.2	147.1	371.11	1.396 Level 3	
17,100.0	10,842.5	17,175.3	10,880.5	188.9	189.2	-94.21	271.5	6,482.3	517.9	141.0	376.90	1.374 Level 3	
17,200.0	10,842.7	17,275.3	10,880.7	191.8	192.1	-94.21	271.5	6,582.3	517.6	134.9	382.70	1.352 Level 3	
17,300.0	10,842.8	17,375.3	10,880.9	194.7	195.0	-94.21	271.5	6,682.3	517.2	128.8	388.49	1.331 Level 3	
17,400.0	10,843.0	17,475.3	10,881.0	197.6	197.9	-94.22	271.5	6,782.3	516.9	122.6	394.29	1.311 Level 3	
17,500.0	10,843.2	17,575.3	10,881.2	200.6	200.8	-94.22	271.5	6,882.3	516.6	116.5	400.09	1.291 Level 3	
17,600.0	10,843.4	17,675.3	10,881.4	203.5	203.7	-94.22	271.5	6,982.3	516.3	110.4	405.89	1.272 Level 3	
17,700.0	10,843.5	17,775.3	10,881.6	206.4	206.6	-94.22	271.5	7,082.3	516.0	104.3	411.69	1.253 Level 3	
17,800.0	10,843.7	17,875.3	10,881.7	209.3	209.6	-94.23	271.5	7,182.3	515.7	98.2	417.50	1.235 Level 2	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services

Anticollision Report



Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 11T2 - Kline Federal 5300 41-18 11T2 - Design #1													Offset Site Error: 0.0 usft
Survey Program: 0-MWD													Offset Well Error: 2.0 usft
Reference		Offset		Semi Major Axis			Distance						Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	
17,900.0	10,843.9	17,975.3	10,881.9	212.2	212.5	-94.23	271.5	7,282.3	515.4	92.1	423.30	1.218 Level 2	
18,000.0	10,844.1	18,075.3	10,882.1	215.1	215.4	-94.23	271.5	7,382.3	515.1	86.0	429.11	1.200 Level 2	
18,100.0	10,844.2	18,175.3	10,882.2	218.0	218.3	-94.23	271.5	7,482.3	514.8	79.9	434.91	1.184 Level 2	
18,200.0	10,844.4	18,275.3	10,882.4	220.9	221.2	-94.24	271.5	7,582.3	514.5	73.8	440.72	1.167 Level 2	
18,300.0	10,844.6	18,375.3	10,882.6	223.8	224.1	-94.24	271.5	7,682.3	514.2	67.6	446.53	1.151 Level 2	
18,400.0	10,844.8	18,475.3	10,882.8	226.7	227.0	-94.24	271.5	7,782.3	513.9	61.5	452.34	1.136 Level 2	
18,500.0	10,844.9	18,575.3	10,882.9	229.7	229.9	-94.25	271.5	7,882.3	513.6	55.4	458.15	1.121 Level 2	
18,600.0	10,845.1	18,675.3	10,883.1	232.6	232.8	-94.25	271.5	7,982.2	513.2	49.3	463.96	1.106 Level 2	
18,700.0	10,845.3	18,775.3	10,883.3	235.5	235.8	-94.25	271.5	8,082.2	512.9	43.2	469.77	1.092 Level 2	
18,800.0	10,845.4	18,875.3	10,883.5	238.4	238.7	-94.25	271.5	8,182.2	512.6	37.0	475.58	1.078 Level 2	
18,900.0	10,845.6	18,975.3	10,883.6	241.3	241.6	-94.26	271.5	8,282.2	512.3	30.9	481.39	1.064 Level 2	
19,000.0	10,845.8	19,075.3	10,883.8	244.2	244.5	-94.26	271.5	8,382.2	512.0	24.8	487.21	1.051 Level 2	
19,100.0	10,846.0	19,175.3	10,884.0	247.1	247.4	-94.26	271.5	8,482.2	511.7	18.7	493.02	1.038 Level 2	
19,200.0	10,846.1	19,275.3	10,884.2	250.1	250.3	-94.26	271.5	8,582.2	511.4	12.6	498.84	1.025 Level 2	
19,300.0	10,846.3	19,375.3	10,884.3	253.0	253.2	-94.27	271.5	8,682.2	511.1	6.4	504.65	1.013 Level 2	
19,400.0	10,846.5	19,475.3	10,884.5	255.9	256.2	-94.27	271.5	8,782.2	510.8	0.3	510.47	1.001 Level 2	
19,500.0	10,846.7	19,575.3	10,884.7	258.8	259.1	-94.27	271.5	8,882.2	510.5	-5.8	516.29	0.989 Level 1	
19,600.0	10,846.8	19,675.3	10,884.9	261.7	262.0	-94.27	271.5	8,982.2	510.2	-11.9	522.10	0.977 Level 1	
19,700.0	10,847.0	19,775.3	10,885.0	264.6	264.9	-94.28	271.5	9,082.2	509.9	-18.1	527.92	0.966 Level 1	
19,800.0	10,847.2	19,875.3	10,885.2	267.6	267.8	-94.28	271.5	9,182.2	509.6	-24.2	533.74	0.955 Level 1	
19,900.0	10,847.4	19,975.3	10,885.4	270.5	270.7	-94.28	271.5	9,282.2	509.2	-30.3	539.56	0.944 Level 1	
20,000.0	10,847.5	20,075.3	10,885.6	273.4	273.7	-94.28	271.5	9,382.2	508.9	-36.4	545.38	0.933 Level 1	
20,100.0	10,847.7	20,175.3	10,885.7	276.3	276.6	-94.29	271.5	9,482.2	508.6	-42.6	551.20	0.923 Level 1	
20,200.0	10,847.9	20,275.3	10,885.9	279.2	279.5	-94.29	271.5	9,582.2	508.3	-48.7	557.02	0.913 Level 1	
20,300.0	10,848.1	20,375.3	10,886.1	282.1	282.4	-94.29	271.5	9,682.2	508.0	-54.8	562.84	0.903 Level 1	
20,400.0	10,848.2	20,475.3	10,886.3	285.1	285.3	-94.30	271.5	9,782.2	507.7	-61.0	568.66	0.893 Level 1	
20,500.0	10,848.4	20,575.3	10,886.4	288.0	288.3	-94.30	271.5	9,882.2	507.4	-67.1	574.48	0.883 Level 1	
20,600.0	10,848.6	20,675.3	10,886.6	290.9	291.2	-94.30	271.5	9,982.2	507.1	-73.2	580.30	0.874 Level 1	
20,667.7	10,848.7	20,743.0	10,886.7	292.9	293.2	-94.30	271.5	10,049.9	506.9	-77.4	584.25	0.868 Level 1, ES, SF	

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 13T2X - Kline Federal 5300 41-18 13T2X - Design #1													Offset Site Error: 0.0 usft
Survey Program: 0-MWD													Offset Well Error: 2.0 usft
Reference		Offset		Semi Major Axis		Distance							
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
0.0	0.0	0.0	0.0	2.0	2.0	-180.00	-33.4	0.0	33.4				
100.0	100.0	100.0	100.0	2.0	2.0	-180.00	-33.4	0.0	33.4	29.4	4.00	8.352	
200.0	200.0	200.0	200.0	2.0	2.0	-180.00	-33.4	0.0	33.4	29.4	4.05	8.261	
300.0	300.0	300.0	300.0	2.1	2.1	-180.00	-33.4	0.0	33.4	29.3	4.14	8.077	
400.0	400.0	400.0	400.0	2.1	2.1	-180.00	-33.4	0.0	33.4	29.2	4.28	7.816	
500.0	500.0	500.0	500.0	2.2	2.2	-180.00	-33.4	0.0	33.4	29.0	4.46	7.502	
600.0	600.0	600.0	600.0	2.3	2.3	-180.00	-33.4	0.0	33.4	28.8	4.67	7.155	
700.0	700.0	700.0	700.0	2.5	2.5	-180.00	-33.4	0.0	33.4	28.5	4.92	6.795	
800.0	800.0	800.0	800.0	2.6	2.6	-180.00	-33.4	0.0	33.4	28.2	5.20	6.436	
900.0	900.0	900.0	900.0	2.7	2.7	-180.00	-33.4	0.0	33.4	27.9	5.49	6.087	
1,000.0	1,000.0	1,000.0	1,000.0	2.9	2.9	-180.00	-33.4	0.0	33.4	27.6	5.81	5.755	
1,100.0	1,100.0	1,100.0	1,100.0	3.1	3.1	-180.00	-33.4	0.0	33.4	27.3	6.14	5.442	
1,200.0	1,200.0	1,200.0	1,200.0	3.2	3.2	-180.00	-33.4	0.0	33.4	26.9	6.49	5.151	
1,300.0	1,300.0	1,300.0	1,300.0	3.4	3.4	-180.00	-33.4	0.0	33.4	26.6	6.85	4.880	
1,400.0	1,400.0	1,400.0	1,400.0	3.6	3.6	-180.00	-33.4	0.0	33.4	26.2	7.22	4.630	
1,500.0	1,500.0	1,500.0	1,500.0	3.8	3.8	-180.00	-33.4	0.0	33.4	25.8	7.60	4.400	
1,600.0	1,600.0	1,600.0	1,600.0	4.0	4.0	-180.00	-33.4	0.0	33.4	25.5	7.99	4.187	
1,700.0	1,700.0	1,700.0	1,700.0	4.2	4.2	-180.00	-33.4	0.0	33.4	25.1	8.38	3.991	
1,800.0	1,800.0	1,800.0	1,800.0	4.4	4.4	-180.00	-33.4	0.0	33.4	24.7	8.78	3.810	
1,900.0	1,900.0	1,900.0	1,900.0	4.6	4.6	-180.00	-33.4	0.0	33.4	24.3	9.18	3.643	
2,000.0	2,000.0	2,000.0	2,000.0	4.8	4.8	-180.00	-33.4	0.0	33.4	23.9	9.58	3.489	
2,100.0	2,100.0	2,100.0	2,100.0	5.0	5.0	-180.00	-33.4	0.0	33.4	23.4	9.99	3.346	
2,200.0	2,200.0	2,200.0	2,200.0	5.2	5.2	-180.00	-33.4	0.0	33.4	23.0	10.41	3.213	
2,200.0	2,200.0	2,200.0	2,200.0	5.2	5.2	-180.00	-33.4	0.0	33.4	23.0	10.41	3.213 CC	
2,216.7	2,216.7	2,216.7	2,216.7	5.2	5.2	90.12	-33.4	0.0	33.4	23.0	10.48	3.192	
2,300.0	2,300.0	2,300.0	2,300.0	5.4	5.4	91.37	-33.4	0.0	33.4	22.6	10.80	3.097	
2,400.0	2,400.0	2,400.0	2,400.0	5.6	5.6	92.86	-33.4	0.0	33.5	22.3	11.20	2.990	
2,500.0	2,500.0	2,500.0	2,500.0	5.8	5.8	94.35	-33.4	0.0	33.5	21.9	11.59	2.892	
2,600.0	2,600.0	2,600.0	2,600.0	6.0	6.0	95.84	-33.4	0.0	33.6	21.6	12.00	2.802	
2,700.0	2,700.0	2,700.0	2,700.0	6.1	6.3	97.31	-33.4	0.0	33.7	21.3	12.40	2.718	
2,800.0	2,800.0	2,800.0	2,800.0	6.3	6.5	98.78	-33.4	0.0	33.8	21.0	12.81	2.641	
2,900.0	2,900.0	2,900.0	2,900.0	6.5	6.7	100.23	-33.4	0.0	34.0	20.8	13.22	2.569	
3,000.0	3,000.0	3,000.0	3,000.0	6.7	6.9	101.67	-33.4	0.0	34.1	20.5	13.64	2.504	
3,100.0	3,100.0	3,100.0	3,100.0	6.9	7.1	103.10	-33.4	0.0	34.3	20.3	14.05	2.443	
3,200.0	3,200.0	3,200.0	3,200.0	7.1	7.3	104.51	-33.4	0.0	34.5	20.1	14.47	2.387	
3,300.0	3,300.0	3,300.0	3,300.0	7.3	7.5	105.90	-33.4	0.0	34.8	19.9	14.89	2.335	
3,400.0	3,400.0	3,400.0	3,400.0	7.6	7.8	107.27	-33.4	0.0	35.0	19.7	15.31	2.287	
3,500.0	3,500.0	3,500.0	3,500.0	7.8	8.0	108.63	-33.4	0.0	35.3	19.5	15.74	2.242	
3,600.0	3,599.9	3,599.9	3,599.9	8.0	8.2	109.96	-33.4	0.0	35.6	19.4	16.16	2.201	
3,700.0	3,699.9	3,699.9	3,699.9	8.2	8.4	111.27	-33.4	0.0	35.9	19.3	16.59	2.163	
3,800.0	3,799.9	3,799.9	3,799.9	8.4	8.6	112.56	-33.4	0.0	36.2	19.2	17.02	2.128	
3,900.0	3,899.9	3,899.9	3,899.9	8.6	8.9	113.82	-33.4	0.0	36.6	19.1	17.44	2.095	
4,000.0	3,999.9	3,999.9	3,999.9	8.8	9.1	115.06	-33.4	0.0	36.9	19.0	17.87	2.065	
4,100.0	4,099.9	4,099.9	4,099.9	9.0	9.3	116.27	-33.4	0.0	37.3	19.0	18.30	2.037	
4,200.0	4,199.9	4,199.9	4,199.9	9.2	9.5	117.46	-33.4	0.0	37.7	18.9	18.74	2.011	
4,300.0	4,299.9	4,299.9	4,299.9	9.4	9.7	118.63	-33.4	0.0	38.1	18.9	19.17	1.987	
4,400.0	4,399.9	4,399.9	4,399.9	9.7	10.0	119.77	-33.4	0.0	38.5	18.9	19.60	1.965	
4,500.0	4,499.9	4,499.9	4,499.9	9.9	10.2	120.88	-33.4	0.0	39.0	18.9	20.04	1.945	
4,600.0	4,599.9	4,599.9	4,599.9	10.1	10.4	121.97	-33.4	0.0	39.4	18.9	20.47	1.926	
4,700.0	4,699.9	4,699.9	4,699.9	10.3	10.6	123.03	-33.4	0.0	39.9	19.0	20.90	1.908	
4,800.0	4,799.9	4,799.9	4,799.9	10.5	10.8	124.07	-33.4	0.0	40.4	19.0	21.34	1.892	
4,900.0	4,899.9	4,899.9	4,899.9	10.7	11.1	125.09	-33.4	0.0	40.9	19.1	21.78	1.876	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services

Anticollision Report



Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 13T2X - Kline Federal 5300 41-18 13T2X - Design #1													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis		Highside Toolface (°)	Distance		Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning	
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)		Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)						
5,000.0	4,999.9	4,999.9	4,999.9	10.9	11.3	126.07	-33.4	0.0	41.4	19.2	22.21	1.862		
5,100.0	5,099.9	5,099.9	5,099.9	11.2	11.5	127.04	-33.4	0.0	41.9	19.2	22.65	1.849		
5,200.0	5,199.9	5,199.9	5,199.9	11.4	11.7	127.98	-33.4	0.0	42.4	19.3	23.09	1.837		
5,300.0	5,299.9	5,299.9	5,299.9	11.6	11.9	128.90	-33.4	0.0	43.0	19.4	23.53	1.826		
5,400.0	5,399.9	5,399.9	5,399.9	11.8	12.2	129.79	-33.4	0.0	43.5	19.6	23.97	1.816		
5,500.0	5,499.9	5,499.9	5,499.9	12.0	12.4	130.66	-33.4	0.0	44.1	19.7	24.41	1.806		
5,600.0	5,599.9	5,599.9	5,599.9	12.2	12.6	131.51	-33.4	0.0	44.7	19.8	24.85	1.797		
5,700.0	5,699.9	5,699.9	5,699.9	12.5	12.8	132.34	-33.4	0.0	45.2	20.0	25.28	1.789		
5,800.0	5,799.9	5,799.9	5,799.9	12.7	13.0	133.15	-33.4	0.0	45.8	20.1	25.73	1.782		
5,900.0	5,899.9	5,899.9	5,899.9	12.9	13.3	133.93	-33.4	0.0	46.4	20.3	26.17	1.775		
6,000.0	5,999.9	5,999.9	5,999.9	13.1	13.5	134.70	-33.4	0.0	47.0	20.4	26.61	1.768		
6,100.0	6,099.9	6,099.9	6,099.9	13.3	13.7	135.44	-33.4	0.0	47.7	20.6	27.05	1.762		
6,200.0	6,199.8	6,199.8	6,199.8	13.6	13.9	136.17	-33.4	0.0	48.3	20.8	27.49	1.757		
6,300.0	6,299.8	6,299.8	6,299.8	13.8	14.2	136.88	-33.4	0.0	48.9	21.0	27.93	1.751		
6,400.0	6,399.8	6,399.8	6,399.8	14.0	14.4	137.57	-33.4	0.0	49.6	21.2	28.37	1.747		
6,500.0	6,499.8	6,499.8	6,499.8	14.2	14.6	138.24	-33.4	0.0	50.2	21.4	28.81	1.742		
6,600.0	6,599.8	6,599.8	6,599.8	14.4	14.8	138.89	-33.4	0.0	50.9	21.6	29.26	1.738		
6,700.0	6,699.8	6,699.8	6,699.8	14.7	15.1	139.53	-33.4	0.0	51.5	21.8	29.70	1.735		
6,800.0	6,799.8	6,799.8	6,799.8	14.9	15.3	140.15	-33.4	0.0	52.2	22.0	30.14	1.731		
6,900.0	6,899.8	6,899.8	6,899.8	15.1	15.5	140.76	-33.4	0.0	52.9	22.3	30.59	1.728		
7,000.0	6,999.8	6,999.8	6,999.8	15.3	15.7	141.35	-33.4	0.0	53.5	22.5	31.03	1.726		
7,100.0	7,099.8	7,099.8	7,099.8	15.5	15.9	141.93	-33.4	0.0	54.2	22.8	31.47	1.723		
7,200.0	7,199.8	7,199.8	7,199.8	15.8	16.2	142.49	-33.4	0.0	54.9	23.0	31.92	1.721		
7,300.0	7,299.8	7,299.8	7,299.8	16.0	16.4	143.04	-33.4	0.0	55.6	23.2	32.36	1.718		
7,400.0	7,399.8	7,399.8	7,399.8	16.2	16.6	143.57	-33.4	0.0	56.3	23.5	32.80	1.717		
7,500.0	7,499.8	7,499.8	7,499.8	16.4	16.8	144.09	-33.4	0.0	57.0	23.8	33.25	1.715		
7,600.0	7,599.8	7,599.8	7,599.8	16.6	17.1	144.60	-33.4	0.0	57.7	24.0	33.69	1.713		
7,700.0	7,699.8	7,699.8	7,699.8	16.9	17.3	145.09	-33.4	0.0	58.4	24.3	34.14	1.712		
7,800.0	7,799.8	7,799.8	7,799.8	17.1	17.5	145.58	-33.4	0.0	59.2	24.6	34.58	1.711		
7,900.0	7,899.8	7,899.8	7,899.8	17.3	17.7	146.05	-33.4	0.0	59.9	24.9	35.02	1.710		
7,926.7	7,926.4	7,926.4	7,926.4	17.4	17.8	146.17	-33.4	0.0	60.1	24.9	35.14	1.709		
7,943.3	7,943.1	7,943.1	7,943.1	17.4	17.8	-123.79	-33.4	0.0	60.1	24.9	35.18	1.709		
8,000.0	7,999.8	7,999.8	7,999.8	17.5	18.0	-123.79	-33.4	0.0	60.1	24.7	35.42	1.698		
8,100.0	8,099.8	8,099.8	8,099.8	17.7	18.2	-123.79	-33.4	0.0	60.1	24.3	35.85	1.677		
8,200.0	8,199.8	8,199.8	8,199.8	17.9	18.4	-123.79	-33.4	0.0	60.1	23.8	36.28	1.657		
8,300.0	8,299.8	8,299.8	8,299.8	18.1	18.6	-123.79	-33.4	0.0	60.1	23.4	36.72	1.638		
8,400.0	8,399.8	8,399.8	8,399.8	18.4	18.8	-123.79	-33.4	0.0	60.1	23.0	37.15	1.619		
8,500.0	8,499.8	8,499.8	8,499.8	18.6	19.1	-123.79	-33.4	0.0	60.1	22.5	37.58	1.600		
8,600.0	8,599.8	8,599.8	8,599.8	18.8	19.3	-123.79	-33.4	0.0	60.1	22.1	38.01	1.582		
8,700.0	8,699.8	8,699.8	8,699.8	19.0	19.5	-123.79	-33.4	0.0	60.1	21.7	38.45	1.564		
8,800.0	8,799.8	8,799.8	8,799.8	19.2	19.7	-123.79	-33.4	0.0	60.1	21.2	38.88	1.546		
8,900.0	8,899.8	8,899.8	8,899.8	19.4	20.0	-123.79	-33.4	0.0	60.1	20.8	39.32	1.529		
9,000.0	8,999.8	8,999.8	8,999.8	19.6	20.2	-123.79	-33.4	0.0	60.1	20.4	39.75	1.513		
9,100.0	9,099.8	9,099.8	9,099.8	19.8	20.4	-123.79	-33.4	0.0	60.1	19.9	40.19	1.496 Level 3		
9,200.0	9,199.8	9,199.8	9,199.8	20.0	20.6	-123.79	-33.4	0.0	60.1	19.5	40.62	1.480 Level 3		
9,300.0	9,299.8	9,299.8	9,299.8	20.2	20.9	-123.79	-33.4	0.0	60.1	19.1	41.06	1.465 Level 3		
9,400.0	9,399.8	9,399.8	9,399.8	20.5	21.1	-123.79	-33.4	0.0	60.1	18.6	41.49	1.449 Level 3		
9,500.0	9,499.8	9,499.8	9,499.8	20.7	21.3	-123.79	-33.4	0.0	60.1	18.2	41.93	1.434 Level 3		
9,600.0	9,599.8	9,599.8	9,599.8	20.9	21.5	-123.79	-33.4	0.0	60.1	17.8	42.37	1.419 Level 3		
9,700.0	9,699.8	9,699.8	9,699.8	21.1	21.8	-123.79	-33.4	0.0	60.1	17.3	42.80	1.405 Level 3		
9,800.0	9,799.8	9,799.8	9,799.8	21.3	22.0	-123.79	-33.4	0.0	60.1	16.9	43.24	1.391 Level 3		
9,900.0	9,899.8	9,899.8	9,899.8	21.5	22.2	-123.79	-33.4	0.0	60.1	16.5	43.68	1.377 Level 3		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services

Anticollision Report



Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 13T2X - Kline Federal 5300 41-18 13T2X - Design #1													Offset Site Error: 0.0 usft	
Survey Program: 0-MWD													Offset Well Error: 2.0 usft	
Reference		Offset		Semi Major Axis			Distance							Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
10,000.0	9,999.8	9,999.8	9,999.8	21.7	22.4	-123.79	-33.4	0.0	60.1	16.0	44.11	1.363	Level 3	
10,100.0	10,099.8	10,099.8	10,099.8	21.9	22.6	-123.79	-33.4	0.0	60.1	15.6	44.55	1.350	Level 3	
10,200.0	10,199.8	10,199.8	10,199.8	22.2	22.9	-123.79	-33.4	0.0	60.1	15.1	44.99	1.336	Level 3	
10,300.0	10,299.8	10,299.8	10,299.8	22.4	23.1	-123.79	-33.4	0.0	60.1	14.7	45.43	1.324	Level 3	
10,354.8	10,354.6	10,354.6	10,354.6	22.5	23.2	-123.79	-33.4	0.0	60.1	14.5	45.67	1.317	Level 3	
10,375.0	10,374.8	10,374.8	10,374.8	22.5	23.3	126.52	-33.4	0.0	60.4	14.6	45.78	1.319	Level 3	
10,400.0	10,399.7	10,400.2	10,400.2	22.6	23.3	127.68	-33.5	0.1	61.4	15.5	45.84	1.339	Level 3	
10,425.0	10,424.5	10,426.8	10,426.8	22.7	23.4	128.94	-34.0	1.1	62.7	16.9	45.86	1.368	Level 3	
10,450.0	10,449.2	10,453.5	10,453.3	22.7	23.4	129.94	-35.3	3.4	64.2	18.4	45.84	1.401	Level 3	
10,475.0	10,473.5	10,480.3	10,479.8	22.8	23.5	130.70	-37.4	7.0	65.9	20.1	45.79	1.439	Level 3	
10,500.0	10,497.6	10,507.2	10,506.1	22.8	23.5	131.22	-40.2	12.0	67.6	22.0	45.69	1.480	Level 3	
10,525.0	10,521.2	10,534.3	10,532.2	22.9	23.6	131.51	-43.7	18.2	69.5	24.0	45.56	1.526		
10,550.0	10,544.4	10,561.3	10,557.8	23.0	23.6	131.58	-47.9	25.8	71.5	26.1	45.41	1.575		
10,575.0	10,567.1	10,588.5	10,583.0	23.0	23.7	131.45	-52.9	34.6	73.6	28.4	45.24	1.627		
10,600.0	10,589.1	10,615.7	10,607.7	23.1	23.8	131.14	-58.6	44.7	75.8	30.7	45.07	1.681		
10,625.0	10,610.6	10,643.0	10,631.7	23.2	23.8	130.66	-64.9	56.0	78.0	33.1	44.90	1.738		
10,650.0	10,631.3	10,670.3	10,654.9	23.3	23.9	130.02	-71.9	68.5	80.4	35.6	44.76	1.796		
10,675.0	10,651.3	10,697.7	10,677.3	23.4	24.0	129.25	-79.6	82.1	82.8	38.2	44.64	1.855		
10,700.0	10,670.5	10,725.0	10,698.8	23.5	24.1	128.34	-87.9	96.9	85.3	40.8	44.57	1.914		
10,725.0	10,688.8	10,752.4	10,719.3	23.6	24.2	127.33	-96.8	112.7	87.9	43.4	44.56	1.973		
10,750.0	10,706.2	10,779.8	10,738.7	23.7	24.3	126.22	-106.3	129.6	90.6	46.0	44.62	2.031		
10,775.0	10,722.6	10,807.2	10,757.0	23.9	24.4	125.03	-116.3	147.3	93.4	48.6	44.74	2.087		
10,800.0	10,738.0	10,834.6	10,774.1	24.0	24.5	123.76	-126.8	166.0	96.2	51.3	44.95	2.140		
10,825.0	10,752.4	10,861.9	10,789.9	24.2	24.7	122.44	-137.7	185.5	99.1	53.9	45.24	2.191		
10,850.0	10,765.7	10,889.3	10,804.4	24.4	24.9	121.06	-149.1	205.7	102.1	56.5	45.62	2.239		
10,875.0	10,777.8	10,916.6	10,817.5	24.6	25.0	119.64	-160.8	226.5	105.2	59.1	46.08	2.283		
10,900.0	10,788.8	10,943.8	10,829.2	24.8	25.2	118.20	-172.9	248.0	108.4	61.8	46.62	2.325		
10,925.0	10,798.6	10,971.1	10,839.5	25.0	25.5	116.73	-185.2	269.9	111.6	64.4	47.24	2.363		
10,950.0	10,807.2	10,998.2	10,848.3	25.3	25.7	115.24	-197.8	292.3	114.9	67.0	47.92	2.398		
10,975.0	10,814.5	11,025.3	10,855.7	25.6	26.0	113.75	-210.6	315.1	118.3	69.6	48.66	2.430		
11,000.0	10,820.6	11,052.4	10,861.5	25.8	26.2	112.25	-223.6	338.1	121.7	72.2	49.46	2.460		
11,025.0	10,825.4	11,079.4	10,865.8	26.1	26.5	110.76	-236.6	361.3	125.1	74.8	50.29	2.488		
11,050.0	10,828.9	11,106.3	10,868.6	26.5	26.9	109.28	-249.8	384.7	128.6	77.5	51.16	2.514		
11,075.0	10,831.1	11,133.2	10,869.9	26.8	27.2	107.81	-262.9	408.1	132.2	80.1	52.05	2.539		
11,100.0	10,832.0	11,159.3	10,870.0	27.1	27.5	106.51	-275.7	430.9	135.8	82.9	52.93	2.566		
11,104.0	10,832.0	11,163.4	10,870.0	27.2	27.6	106.36	-277.7	434.5	136.4	83.3	53.06	2.570		
11,200.0	10,832.2	11,262.9	10,870.2	28.6	29.0	104.72	-324.2	522.4	151.8	95.8	56.01	2.711		
11,300.0	10,832.4	11,366.8	10,870.4	30.3	30.7	103.16	-369.6	615.8	170.4	111.0	59.35	2.870		
11,400.0	10,832.6	11,470.8	10,870.5	32.2	32.6	101.77	-411.6	711.0	191.4	128.5	62.89	3.043		
11,500.0	10,832.7	11,575.2	10,870.7	34.1	34.7	100.54	-450.2	807.9	214.9	148.4	66.51	3.231		
11,600.0	10,832.9	11,679.9	10,870.9	36.2	36.9	99.47	-485.4	906.5	240.8	170.6	70.16	3.432		
11,680.4	10,833.1	11,764.4	10,871.1	38.0	38.7	98.70	-511.1	987.0	263.4	190.3	73.06	3.605		
11,700.0	10,833.1	11,785.0	10,871.1	38.4	39.2	98.50	-517.0	1,006.7	269.0	195.1	73.94	3.638		
11,800.0	10,833.3	11,891.7	10,871.3	40.7	41.6	97.66	-545.4	1,109.6	295.7	217.0	78.62	3.761		
11,900.0	10,833.4	12,000.6	10,871.5	43.0	44.2	97.04	-570.4	1,215.6	318.7	235.2	83.51	3.817		
12,000.0	10,833.6	12,111.3	10,871.7	45.4	46.9	96.59	-591.6	1,324.2	338.1	249.5	88.60	3.816		
12,100.0	10,833.8	12,223.5	10,871.9	47.9	49.6	96.26	-608.9	1,435.1	353.7	259.8	93.85	3.768		
12,200.0	10,834.0	12,337.0	10,872.1	50.4	52.4	96.03	-621.9	1,547.8	365.4	266.1	99.24	3.682		
12,300.0	10,834.1	12,451.3	10,872.3	53.0	55.3	95.89	-630.4	1,661.8	373.1	268.4	104.75	3.562		
12,400.0	10,834.3	12,566.2	10,872.5	55.5	58.2	95.82	-634.5	1,776.7	376.9	266.6	110.33	3.416		
12,500.0	10,834.5	12,671.8	10,872.7	58.2	60.9	95.81	-634.8	1,882.3	377.5	261.8	115.73	3.262		
12,600.0	10,834.7	12,771.8	10,872.9	60.8	63.4	95.80	-634.8	1,982.3	377.8	256.8	121.04	3.121		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 13T2X - Kline Federal 5300 41-18 13T2X - Design #1												Offset Site Error:	0.0 usft
Survey Program: 0-MWID												Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis		Highside Toolface (°)	Distance		Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)		Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)					
12,700.0	10,834.8	12,871.8	10,873.0	63.5	66.0	95.80	-634.8	2,082.3	378.1	251.7	126.40	2.991	
12,800.0	10,835.0	12,971.8	10,873.2	66.2	68.7	95.79	-634.8	2,182.3	378.4	246.6	131.80	2.871	
12,900.0	10,835.2	13,071.8	10,873.4	68.9	71.3	95.79	-634.8	2,282.3	378.7	241.5	137.24	2.760	
13,000.0	10,835.4	13,171.8	10,873.6	71.7	74.0	95.79	-634.8	2,382.3	379.0	236.3	142.70	2.656	
13,100.0	10,835.5	13,271.8	10,873.7	74.4	76.7	95.78	-634.8	2,482.3	379.3	231.2	148.20	2.560	
13,200.0	10,835.7	13,371.8	10,873.9	77.2	79.4	95.78	-634.8	2,582.3	379.7	225.9	153.72	2.470	
13,300.0	10,835.9	13,471.8	10,874.1	79.9	82.1	95.77	-634.8	2,682.3	380.0	220.7	159.26	2.386	
13,400.0	10,836.1	13,571.8	10,874.3	82.7	84.8	95.77	-634.8	2,782.3	380.3	215.4	164.82	2.307	
13,500.0	10,836.2	13,671.8	10,874.4	85.5	87.6	95.76	-634.8	2,882.3	380.6	210.2	170.40	2.233	
13,600.0	10,836.4	13,771.8	10,874.6	88.3	90.3	95.76	-634.8	2,982.3	380.9	204.9	176.00	2.164	
13,700.0	10,836.6	13,871.8	10,874.8	91.1	93.1	95.75	-634.8	3,082.3	381.2	199.6	181.61	2.099	
13,800.0	10,836.7	13,971.8	10,875.0	94.0	95.9	95.75	-634.8	3,182.3	381.5	194.3	187.24	2.037	
13,900.0	10,836.9	14,071.8	10,875.1	96.8	98.7	95.74	-634.8	3,282.3	381.8	188.9	192.88	1.979	
14,000.0	10,837.1	14,171.8	10,875.3	99.6	101.5	95.74	-634.8	3,382.3	382.1	183.6	198.53	1.925	
14,100.0	10,837.3	14,271.8	10,875.5	102.5	104.3	95.74	-634.8	3,482.3	382.4	178.2	204.19	1.873	
14,200.0	10,837.4	14,371.8	10,875.7	105.3	107.1	95.73	-634.8	3,582.3	382.7	172.9	209.86	1.824	
14,300.0	10,837.6	14,471.8	10,875.8	108.2	109.9	95.73	-634.8	3,682.3	383.0	167.5	215.54	1.777	
14,400.0	10,837.8	14,571.8	10,876.0	111.0	112.7	95.72	-634.8	3,782.3	383.3	162.1	221.23	1.733	
14,500.0	10,838.0	14,671.8	10,876.2	113.9	115.6	95.72	-634.8	3,882.3	383.6	156.7	226.93	1.691	
14,600.0	10,838.1	14,771.8	10,876.4	116.7	118.4	95.71	-634.8	3,982.3	384.0	151.3	232.63	1.650	
14,700.0	10,838.3	14,871.8	10,876.5	119.6	121.2	95.71	-634.8	4,082.3	384.3	145.9	238.34	1.612	
14,800.0	10,838.5	14,971.8	10,876.7	122.5	124.1	95.70	-634.8	4,182.3	384.6	140.5	244.06	1.576	
14,900.0	10,838.7	15,071.8	10,876.9	125.3	126.9	95.70	-634.8	4,282.3	384.9	135.1	249.78	1.541	
15,000.0	10,838.8	15,171.8	10,877.1	128.2	129.8	95.69	-634.8	4,382.3	385.2	129.7	255.51	1.508	
15,100.0	10,839.0	15,271.8	10,877.2	131.1	132.6	95.69	-634.8	4,482.3	385.5	124.3	261.24	1.476 Level 3	
15,200.0	10,839.2	15,371.8	10,877.4	133.9	135.5	95.69	-634.8	4,582.3	385.8	118.8	266.98	1.445 Level 3	
15,300.0	10,839.4	15,471.8	10,877.6	136.8	138.3	95.68	-634.8	4,682.3	386.1	113.4	272.72	1.416 Level 3	
15,400.0	10,839.5	15,571.8	10,877.8	139.7	141.2	95.68	-634.8	4,782.3	386.4	108.0	278.46	1.388 Level 3	
15,500.0	10,839.7	15,671.8	10,877.9	142.6	144.1	95.67	-634.8	4,882.3	386.7	102.5	284.21	1.361 Level 3	
15,600.0	10,839.9	15,771.8	10,878.1	145.5	146.9	95.67	-634.8	4,982.3	387.0	97.1	289.96	1.335 Level 3	
15,700.0	10,840.1	15,871.8	10,878.3	148.4	149.8	95.66	-634.8	5,082.3	387.3	91.6	295.72	1.310 Level 3	
15,800.0	10,840.2	15,971.8	10,878.5	151.3	152.7	95.66	-634.8	5,182.3	387.6	86.2	301.48	1.286 Level 3	
15,900.0	10,840.4	16,071.8	10,878.6	154.1	155.5	95.65	-634.8	5,282.3	387.9	80.7	307.24	1.263 Level 3	
16,000.0	10,840.6	16,171.8	10,878.8	157.0	158.4	95.65	-634.8	5,382.3	388.3	75.2	313.01	1.240 Level 2	
16,100.0	10,840.8	16,271.8	10,879.0	159.9	161.3	95.65	-634.8	5,482.3	388.6	69.8	318.77	1.219 Level 2	
16,200.0	10,840.9	16,371.8	10,879.1	162.8	164.2	95.64	-634.8	5,582.3	388.9	64.3	324.54	1.198 Level 2	
16,300.0	10,841.1	16,471.8	10,879.3	165.7	167.1	95.64	-634.8	5,682.3	389.2	58.9	330.32	1.178 Level 2	
16,400.0	10,841.3	16,571.8	10,879.5	168.6	169.9	95.63	-634.8	5,782.3	389.5	53.4	336.09	1.159 Level 2	
16,500.0	10,841.4	16,671.8	10,879.7	171.5	172.8	95.63	-634.8	5,882.3	389.8	47.9	341.87	1.140 Level 2	
16,600.0	10,841.6	16,771.8	10,879.8	174.4	175.7	95.62	-634.8	5,982.3	390.1	42.4	347.65	1.122 Level 2	
16,700.0	10,841.8	16,871.8	10,880.0	177.3	178.6	95.62	-634.8	6,082.3	390.4	37.0	353.43	1.105 Level 2	
16,800.0	10,842.0	16,971.8	10,880.2	180.2	181.5	95.62	-634.8	6,182.3	390.7	31.5	359.22	1.088 Level 2	
16,900.0	10,842.1	17,071.8	10,880.4	183.1	184.4	95.61	-634.8	6,282.3	391.0	26.0	365.00	1.071 Level 2	
17,000.0	10,842.3	17,171.8	10,880.5	186.0	187.3	95.61	-634.8	6,382.3	391.3	20.5	370.79	1.055 Level 2	
17,100.0	10,842.5	17,271.8	10,880.7	188.9	190.2	95.60	-634.8	6,482.3	391.6	15.1	376.58	1.040 Level 2	
17,200.0	10,842.7	17,371.8	10,880.9	191.8	193.1	95.60	-634.8	6,582.3	391.9	9.6	382.37	1.025 Level 2	
17,300.0	10,842.8	17,471.8	10,881.1	194.7	196.0	95.59	-634.8	6,682.3	392.2	4.1	388.16	1.011 Level 2	
17,400.0	10,843.0	17,571.8	10,881.2	197.6	198.9	95.59	-634.8	6,782.3	392.6	-1.4	393.95	0.996 Level 1	
17,500.0	10,843.2	17,671.8	10,881.4	200.6	201.8	95.58	-634.8	6,882.3	392.9	-6.9	399.75	0.983 Level 1	
17,600.0	10,843.4	17,771.8	10,881.6	203.5	204.7	95.58	-634.8	6,982.3	393.2	-12.4	405.55	0.969 Level 1	
17,700.0	10,843.5	17,871.8	10,881.8	206.4	207.6	95.58	-634.8	7,082.3	393.5	-17.9	411.34	0.957 Level 1	
17,800.0	10,843.7	17,971.8	10,881.9	209.3	210.5	95.57	-634.8	7,182.3	393.8	-23.4	417.14	0.944 Level 1	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 13T2X - Kline Federal 5300 41-18 13T2X - Design #1												Offset Site Error:	0.0 usft
Survey Program: 0-MWD												Offset Well Error:	2.0 usft
Reference	Offset	Semi Major Axis		Distance									
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
17,900.0	10,843.9	18,071.8	10,882.1	212.2	213.4	95.57	-634.8	7,282.3	394.1	-28.9	422.94	0.932	Level 1
18,000.0	10,844.1	18,171.8	10,882.3	215.1	216.3	95.56	-634.8	7,382.3	394.4	-34.3	428.74	0.920	Level 1
18,100.0	10,844.2	18,271.8	10,882.5	218.0	219.2	95.56	-634.8	7,482.3	394.7	-39.8	434.55	0.908	Level 1
18,200.0	10,844.4	18,371.8	10,882.6	220.9	222.1	95.55	-634.8	7,582.3	395.0	-45.3	440.35	0.897	Level 1
18,300.0	10,844.6	18,471.8	10,882.8	223.8	225.0	95.55	-634.8	7,682.3	395.3	-50.8	446.15	0.886	Level 1
18,400.0	10,844.8	18,571.8	10,883.0	226.7	227.9	95.55	-634.8	7,782.3	395.6	-56.3	451.96	0.875	Level 1
18,500.0	10,844.9	18,671.8	10,883.2	229.7	230.8	95.54	-634.8	7,882.2	395.9	-61.8	457.77	0.865	Level 1
18,600.0	10,845.1	18,771.8	10,883.3	232.6	233.7	95.54	-634.8	7,982.2	396.2	-67.3	463.57	0.855	Level 1
18,700.0	10,845.3	18,871.8	10,883.5	235.5	236.6	95.53	-634.8	8,082.2	396.5	-72.8	469.38	0.845	Level 1
18,800.0	10,845.4	18,971.8	10,883.7	238.4	239.5	95.53	-634.8	8,182.2	396.9	-78.3	475.19	0.835	Level 1
18,900.0	10,845.6	19,071.8	10,883.9	241.3	242.4	95.53	-634.8	8,282.2	397.2	-83.8	481.00	0.826	Level 1
19,000.0	10,845.8	19,171.8	10,884.0	244.2	245.3	95.52	-634.8	8,382.2	397.5	-89.3	486.81	0.816	Level 1
19,100.0	10,846.0	19,271.8	10,884.2	247.1	248.2	95.52	-634.8	8,482.2	397.8	-94.8	492.62	0.807	Level 1
19,200.0	10,846.1	19,371.8	10,884.4	250.1	251.1	95.51	-634.8	8,582.2	398.1	-100.4	498.44	0.799	Level 1
19,300.0	10,846.3	19,471.8	10,884.6	253.0	254.1	95.51	-634.8	8,682.2	398.4	-105.9	504.25	0.790	Level 1
19,400.0	10,846.5	19,571.8	10,884.7	255.9	257.0	95.50	-634.8	8,782.2	398.7	-111.4	510.06	0.782	Level 1
19,500.0	10,846.7	19,671.8	10,884.9	258.8	259.9	95.50	-634.8	8,882.2	399.0	-116.9	515.88	0.773	Level 1
19,600.0	10,846.8	19,771.8	10,885.1	261.7	262.8	95.50	-634.8	8,982.2	399.3	-122.4	521.69	0.765	Level 1
19,700.0	10,847.0	19,871.8	10,885.3	264.6	265.7	95.49	-634.8	9,082.2	399.6	-127.9	527.51	0.758	Level 1
19,800.0	10,847.2	19,971.8	10,885.4	267.6	268.6	95.49	-634.8	9,182.2	399.9	-133.4	533.32	0.750	Level 1
19,900.0	10,847.4	20,071.8	10,885.6	270.5	271.5	95.48	-634.8	9,282.2	400.2	-138.9	539.14	0.742	Level 1
20,000.0	10,847.5	20,171.8	10,885.8	273.4	274.4	95.48	-634.8	9,382.2	400.5	-144.4	544.96	0.735	Level 1
20,100.0	10,847.7	20,271.8	10,886.0	276.3	277.4	95.48	-634.8	9,482.2	400.9	-149.9	550.78	0.728	Level 1
20,200.0	10,847.9	20,371.8	10,886.1	279.2	280.3	95.47	-634.8	9,582.2	401.2	-155.4	556.60	0.721	Level 1
20,300.0	10,848.1	20,471.8	10,886.3	282.1	283.2	95.47	-634.8	9,682.2	401.5	-161.0	562.41	0.714	Level 1
20,400.0	10,848.2	20,571.8	10,886.5	285.1	286.1	95.46	-634.8	9,782.2	401.8	-166.5	568.23	0.707	Level 1
20,500.0	10,848.4	20,671.8	10,886.7	288.0	289.0	95.46	-634.8	9,882.2	402.1	-172.0	574.05	0.700	Level 1
20,600.0	10,848.6	20,771.8	10,886.8	290.9	291.9	95.45	-634.8	9,982.2	402.4	-177.5	579.88	0.694	Level 1
20,667.7	10,848.7	20,839.5	10,886.9	292.9	293.9	95.45	-634.8	10,049.9	402.6	-181.2	583.82	0.690	Level 1, ES, SF



Ryan Directional Services

Anticollision Report



Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 14BX - Kline Federal 5300 41-18 14BX - Design #1												Offset Site Error:	0.0 usft
Survey Program: 0-MWD												Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis		Distance							
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
0.0	0.0	0.0	0.0	2.0	2.0	-180.00	-65.9	0.0	65.9				
100.0	100.0	100.0	100.0	2.0	2.0	-180.00	-65.9	0.0	65.9	61.9	4.00	16.452	
200.0	200.0	200.0	200.0	2.0	2.0	-180.00	-65.9	0.0	65.9	61.8	4.05	16.273	
300.0	300.0	300.0	300.0	2.1	2.1	-180.00	-65.9	0.0	65.9	61.7	4.14	15.909	
400.0	400.0	400.0	400.0	2.1	2.1	-180.00	-65.9	0.0	65.9	61.6	4.28	15.396	
500.0	500.0	500.0	500.0	2.2	2.2	-180.00	-65.9	0.0	65.9	61.4	4.46	14.777	
600.0	600.0	600.0	600.0	2.3	2.3	-180.00	-65.9	0.0	65.9	61.2	4.67	14.094	
700.0	700.0	700.0	700.0	2.5	2.5	-180.00	-65.9	0.0	65.9	60.9	4.92	13.385	
800.0	800.0	800.0	800.0	2.6	2.6	-180.00	-65.9	0.0	65.9	60.7	5.20	12.678	
900.0	900.0	900.0	900.0	2.7	2.7	-180.00	-65.9	0.0	65.9	60.4	5.49	11.990	
1,000.0	1,000.0	1,000.0	1,000.0	2.9	2.9	-180.00	-65.9	0.0	65.9	60.1	5.81	11.336	
1,100.0	1,100.0	1,100.0	1,100.0	3.1	3.1	-180.00	-65.9	0.0	65.9	59.7	6.14	10.720	
1,200.0	1,200.0	1,200.0	1,200.0	3.2	3.2	-180.00	-65.9	0.0	65.9	59.4	6.49	10.145	
1,300.0	1,300.0	1,300.0	1,300.0	3.4	3.4	-180.00	-65.9	0.0	65.9	59.0	6.85	9.613	
1,400.0	1,400.0	1,400.0	1,400.0	3.6	3.6	-180.00	-65.9	0.0	65.9	58.6	7.22	9.121	
1,500.0	1,500.0	1,500.0	1,500.0	3.8	3.8	-180.00	-65.9	0.0	65.9	58.3	7.60	8.667	
1,600.0	1,600.0	1,600.0	1,600.0	4.0	4.0	-180.00	-65.9	0.0	65.9	57.9	7.99	8.248	
1,700.0	1,700.0	1,700.0	1,700.0	4.2	4.2	-180.00	-65.9	0.0	65.9	57.5	8.38	7.862	
1,800.0	1,800.0	1,800.0	1,800.0	4.4	4.4	-180.00	-65.9	0.0	65.9	57.1	8.78	7.506	
1,900.0	1,900.0	1,900.0	1,900.0	4.6	4.6	-180.00	-65.9	0.0	65.9	56.7	9.18	7.177	
2,000.0	2,000.0	2,000.0	2,000.0	4.8	4.8	-180.00	-65.9	0.0	65.9	56.3	9.58	6.872	
2,100.0	2,100.0	2,100.0	2,100.0	5.0	5.0	-180.00	-65.9	0.0	65.9	55.9	9.99	6.590	
2,200.0	2,200.0	2,200.0	2,200.0	5.2	5.2	-180.00	-65.9	0.0	65.9	55.5	10.41	6.328 CC	
2,216.7	2,216.7	2,216.1	2,216.1	5.2	5.2	90.06	-65.9	0.0	65.9	55.5	10.47	6.298	
2,300.0	2,300.0	2,299.4	2,299.4	5.4	5.4	90.68	-66.7	0.0	66.7	55.9	10.75	6.201	
2,400.0	2,400.0	2,399.4	2,399.4	5.6	5.5	91.41	-67.5	0.0	67.6	56.5	11.10	6.088	
2,500.0	2,500.0	2,499.4	2,499.4	5.8	5.7	92.13	-68.4	0.0	68.5	57.0	11.45	5.979	
2,600.0	2,600.0	2,599.4	2,599.4	6.0	5.9	92.82	-69.3	0.0	69.4	57.6	11.81	5.874	
2,700.0	2,700.0	2,699.4	2,699.4	6.1	6.0	93.50	-70.1	0.0	70.3	58.1	12.17	5.773	
2,800.0	2,800.0	2,799.4	2,799.4	6.3	6.2	94.15	-71.0	0.0	71.2	58.7	12.54	5.677	
2,900.0	2,900.0	2,899.4	2,899.3	6.5	6.4	94.79	-71.9	0.0	72.2	59.2	12.92	5.584	
3,000.0	3,000.0	2,999.4	2,999.3	6.7	6.6	95.42	-72.8	0.0	73.1	59.8	13.30	5.496	
3,100.0	3,100.0	3,099.4	3,099.3	6.9	6.7	96.03	-73.6	0.0	74.1	60.4	13.69	5.411	
3,200.0	3,200.0	3,199.3	3,199.3	7.1	6.9	96.62	-74.5	0.0	75.0	60.9	14.07	5.330	
3,300.0	3,300.0	3,299.3	3,299.3	7.3	7.1	97.20	-75.4	0.0	76.0	61.5	14.47	5.253	
3,400.0	3,400.0	3,399.3	3,399.3	7.6	7.3	97.76	-76.3	0.0	77.0	62.1	14.86	5.179	
3,500.0	3,500.0	3,499.3	3,499.3	7.8	7.5	98.31	-77.1	0.0	78.0	62.7	15.26	5.108	
3,600.0	3,599.9	3,599.3	3,599.3	8.0	7.7	98.84	-78.0	0.0	78.9	63.3	15.66	5.041	
3,700.0	3,699.9	3,699.3	3,699.3	8.2	7.9	99.37	-78.9	0.0	79.9	63.9	16.07	4.976	
3,800.0	3,799.9	3,799.3	3,799.2	8.4	8.1	99.88	-79.7	0.0	81.0	64.5	16.47	4.915	
3,900.0	3,899.9	3,899.3	3,899.2	8.6	8.3	100.37	-80.6	0.0	82.0	65.1	16.88	4.856	
4,000.0	3,999.9	3,999.3	3,999.2	8.8	8.5	100.86	-81.5	0.0	83.0	65.7	17.29	4.799	
4,100.0	4,099.9	4,099.3	4,099.2	9.0	8.7	101.33	-82.4	0.0	84.0	66.3	17.70	4.745	
4,200.0	4,199.9	4,199.3	4,199.2	9.2	8.9	101.79	-83.2	0.0	85.0	66.9	18.12	4.694	
4,300.0	4,299.9	4,299.3	4,299.2	9.4	9.1	102.24	-84.1	0.0	86.1	67.5	18.53	4.644	
4,400.0	4,399.9	4,399.3	4,399.2	9.7	9.3	102.68	-85.0	0.0	87.1	68.2	18.95	4.597	
4,500.0	4,499.9	4,499.3	4,499.2	9.9	9.5	103.11	-85.9	0.0	88.2	68.8	19.37	4.551	
4,600.0	4,599.9	4,599.2	4,599.2	10.1	9.7	103.53	-86.7	0.0	89.2	69.4	19.79	4.507	
4,700.0	4,699.9	4,699.2	4,699.1	10.3	9.9	103.94	-87.6	0.0	90.3	70.0	20.21	4.466	
4,800.0	4,799.9	4,799.2	4,799.1	10.5	10.1	104.33	-88.5	0.0	91.3	70.7	20.64	4.425	
4,900.0	4,899.9	4,899.2	4,899.1	10.7	10.3	104.72	-89.3	0.0	92.4	71.3	21.06	4.387	
5,000.0	4,999.9	4,999.2	4,999.1	10.9	10.6	105.11	-90.2	0.0	93.5	72.0	21.49	4.350	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 14BX - Kline Federal 5300 41-18 14BX - Design #1													Offset Site Error: 0.0 usft
Survey Program: 0-MWD													Offset Well Error: 2.0 usft
Reference		Offset		Semi Major Axis			Distance						Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	
5,100.0	5,099.9	5,099.2	5,099.1	11.2	10.8	105.48	-91.1	0.0	94.5	72.6	21.91	4.314	
5,200.0	5,199.9	5,199.2	5,199.1	11.4	11.0	105.84	-92.0	0.0	95.6	73.3	22.34	4.280	
5,300.0	5,299.9	5,299.2	5,299.1	11.6	11.2	106.20	-92.8	0.0	96.7	73.9	22.77	4.246	
5,400.0	5,399.9	5,399.2	5,399.1	11.8	11.4	106.55	-93.7	0.0	97.8	74.6	23.20	4.215	
5,500.0	5,499.9	5,499.2	5,499.0	12.0	11.6	106.89	-94.6	0.0	98.9	75.2	23.63	4.184	
5,600.0	5,599.9	5,599.2	5,599.0	12.2	11.8	107.22	-95.5	0.0	99.9	75.9	24.06	4.154	
5,700.0	5,699.9	5,699.2	5,699.0	12.5	12.0	107.55	-96.3	0.0	101.0	76.5	24.49	4.126	
5,800.0	5,799.9	5,799.2	5,799.0	12.7	12.3	107.87	-97.2	0.0	102.1	77.2	24.92	4.098	
5,900.0	5,899.9	5,899.1	5,899.0	12.9	12.5	108.18	-98.1	0.0	103.2	77.9	25.35	4.072	
6,000.0	5,999.9	5,999.1	5,999.0	13.1	12.7	108.49	-98.9	0.0	104.3	78.5	25.79	4.046	
6,100.0	6,099.9	6,099.1	6,099.0	13.3	12.9	108.79	-99.8	0.0	105.4	79.2	26.22	4.021	
6,200.0	6,199.8	6,199.1	6,199.0	13.6	13.1	109.08	-100.7	0.0	106.5	79.9	26.65	3.997	
6,300.0	6,299.8	6,299.1	6,299.0	13.8	13.3	109.37	-101.6	0.0	107.7	80.6	27.09	3.974	
6,400.0	6,399.8	6,399.1	6,398.9	14.0	13.6	109.65	-102.4	0.0	108.8	81.2	27.53	3.952	
6,500.0	6,499.8	6,499.1	6,498.9	14.2	13.8	109.92	-103.3	0.0	109.9	81.9	27.96	3.930	
6,600.0	6,599.8	6,599.1	6,598.9	14.4	14.0	110.19	-104.2	0.0	111.0	82.6	28.40	3.909	
6,700.0	6,699.8	6,699.1	6,698.9	14.7	14.2	110.46	-105.1	0.0	112.1	83.3	28.83	3.889	
6,800.0	6,799.8	6,799.1	6,798.9	14.9	14.4	110.72	-105.9	0.0	113.3	84.0	29.27	3.869	
6,900.0	6,899.8	6,899.1	6,898.9	15.1	14.6	110.97	-106.8	0.0	114.4	84.7	29.71	3.850	
7,000.0	6,999.8	6,999.1	6,998.9	15.3	14.9	111.22	-107.7	0.0	115.5	85.4	30.15	3.831	
7,100.0	7,099.8	7,099.1	7,098.9	15.5	15.1	111.46	-108.5	0.0	116.6	86.1	30.59	3.813	
7,200.0	7,199.8	7,199.0	7,198.9	15.8	15.3	111.70	-109.4	0.0	117.8	86.7	31.02	3.796	
7,300.0	7,299.8	7,299.0	7,298.8	16.0	15.5	111.94	-110.3	0.0	118.9	87.4	31.46	3.779	
7,400.0	7,399.8	7,399.0	7,398.8	16.2	15.7	112.17	-111.2	0.0	120.0	88.1	31.90	3.763	
7,500.0	7,499.8	7,499.0	7,498.8	16.4	16.0	112.40	-112.0	0.0	121.2	88.8	32.34	3.747	
7,600.0	7,599.8	7,599.0	7,598.8	16.6	16.2	112.62	-112.9	0.0	122.3	89.5	32.78	3.731	
7,700.0	7,699.8	7,699.0	7,698.8	16.9	16.4	112.84	-113.8	0.0	123.5	90.2	33.22	3.716	
7,800.0	7,799.8	7,799.0	7,798.8	17.1	16.6	113.05	-114.7	0.0	124.6	90.9	33.66	3.701	
7,900.0	7,899.8	7,899.0	7,898.8	17.3	16.8	113.26	-115.5	0.0	125.8	91.6	34.11	3.687	
7,926.7	7,926.4	7,925.7	7,925.4	17.4	16.9	113.32	-115.8	0.0	126.1	91.8	34.22	3.683	
7,943.3	7,943.1	7,943.3	7,943.1	17.4	16.9	-156.66	-115.8	0.0	126.2	91.9	34.30	3.679	
8,000.0	7,999.8	8,000.0	7,999.8	17.5	17.1	-156.66	-115.8	0.0	126.2	91.6	34.52	3.654	
8,100.0	8,099.8	8,100.0	8,099.8	17.7	17.2	-156.66	-115.8	0.0	126.2	91.2	34.92	3.613	
8,200.0	8,199.8	8,200.0	8,199.8	17.9	17.4	-156.66	-115.8	0.0	126.2	90.8	35.32	3.572	
8,300.0	8,299.8	8,300.0	8,299.8	18.1	17.6	-156.66	-115.8	0.0	126.2	90.4	35.72	3.532	
8,400.0	8,399.8	8,400.0	8,399.8	18.4	17.8	-156.66	-115.8	0.0	126.2	90.0	36.12	3.493	
8,500.0	8,499.8	8,500.0	8,499.8	18.6	18.0	-156.66	-115.8	0.0	126.2	89.6	36.52	3.454	
8,600.0	8,599.8	8,600.0	8,599.8	18.8	18.2	-156.66	-115.8	0.0	126.2	89.2	36.93	3.416	
8,700.0	8,699.8	8,700.0	8,699.8	19.0	18.4	-156.66	-115.8	0.0	126.2	88.8	37.33	3.379	
8,800.0	8,799.8	8,800.0	8,799.8	19.2	18.6	-156.66	-115.8	0.0	126.2	88.4	37.74	3.343	
8,900.0	8,899.8	8,900.0	8,899.8	19.4	18.8	-156.66	-115.8	0.0	126.2	88.0	38.14	3.307	
9,000.0	8,999.8	9,000.0	8,999.8	19.6	19.0	-156.66	-115.8	0.0	126.2	87.6	38.55	3.273	
9,100.0	9,099.8	9,100.0	9,099.8	19.8	19.2	-156.66	-115.8	0.0	126.2	87.2	38.96	3.238	
9,200.0	9,199.8	9,200.0	9,199.8	20.0	19.4	-156.66	-115.8	0.0	126.2	86.8	39.37	3.205	
9,300.0	9,299.8	9,300.0	9,299.8	20.2	19.6	-156.66	-115.8	0.0	126.2	86.4	39.78	3.172	
9,400.0	9,399.8	9,400.0	9,399.8	20.5	19.8	-156.66	-115.8	0.0	126.2	86.0	40.19	3.139	
9,500.0	9,499.8	9,500.0	9,499.8	20.7	20.0	-156.66	-115.8	0.0	126.2	85.6	40.60	3.107	
9,600.0	9,599.8	9,600.0	9,599.8	20.9	20.2	-156.66	-115.8	0.0	126.2	85.1	41.01	3.076	
9,700.0	9,699.8	9,700.0	9,699.8	21.1	20.4	-156.66	-115.8	0.0	126.2	84.7	41.43	3.045	
9,800.0	9,799.8	9,800.0	9,799.8	21.3	20.6	-156.66	-115.8	0.0	126.2	84.3	41.84	3.015	
9,900.0	9,899.8	9,900.0	9,899.8	21.5	20.8	-156.66	-115.8	0.0	126.2	83.9	42.26	2.986	
10,000.0	9,999.8	10,000.0	9,999.8	21.7	21.0	-156.66	-115.8	0.0	126.2	83.5	42.67	2.956	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services

Anticollision Report



Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 14BX - Kline Federal 5300 41-18 14BX - Design #1													Offset Site Error: 0.0 usft
Survey Program: 0-MWD													Offset Well Error: 2.0 usft
Reference		Offset		Semi Major Axis			Distance						Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	
10,100.0	10,099.8	10,100.0	10,099.8	21.9	21.2	-156.66	-115.8	0.0	126.2	83.1	43.09	2.928	
10,162.1	10,161.9	10,162.1	10,161.9	22.1	21.3	-156.66	-115.8	0.0	126.2	82.8	43.35	2.910	
10,200.0	10,199.8	10,199.8	10,199.6	22.2	21.4	-156.73	-115.9	0.1	126.2	82.7	43.50	2.900	
10,300.0	10,299.8	10,296.4	10,295.2	22.4	21.6	-162.19	-121.4	11.0	127.5	83.6	43.93	2.903	
10,354.8	10,354.6	10,346.2	10,343.1	22.5	21.7	-168.17	-127.5	23.3	130.8	86.6	44.17	2.961	
10,375.0	10,374.8	10,363.9	10,359.7	22.5	21.8	79.20	-130.2	28.7	132.7	88.5	44.24	3.000	
10,400.0	10,399.7	10,385.5	10,379.7	22.6	21.8	76.08	-133.9	36.1	135.6	91.3	44.34	3.059	
10,425.0	10,424.5	10,406.9	10,399.1	22.7	21.9	73.14	-138.0	44.1	139.0	94.6	44.42	3.129	
10,450.0	10,449.2	10,428.0	10,417.8	22.7	21.9	70.38	-142.4	52.9	142.8	98.3	44.49	3.209	
10,475.0	10,473.5	10,450.0	10,436.8	22.8	22.0	67.70	-147.3	62.7	146.9	102.3	44.53	3.298	
10,500.0	10,497.6	10,469.5	10,453.2	22.8	22.0	65.43	-152.0	72.1	151.2	106.7	44.53	3.395	
10,525.0	10,521.2	10,489.9	10,469.9	22.9	22.1	63.24	-157.3	82.6	155.8	111.3	44.51	3.500	
10,550.0	10,544.4	10,510.1	10,486.0	23.0	22.2	61.22	-162.8	93.6	160.5	116.1	44.44	3.612	
10,575.0	10,567.1	10,530.2	10,501.4	23.0	22.2	59.39	-168.6	105.1	165.4	121.0	44.32	3.731	
10,600.0	10,589.1	10,550.0	10,516.0	23.1	22.3	57.72	-174.6	117.0	170.3	126.1	44.17	3.855	
10,625.0	10,610.6	10,569.8	10,530.1	23.2	22.4	56.20	-180.8	129.4	175.2	131.2	43.98	3.983	
10,650.0	10,631.3	10,589.3	10,543.4	23.3	22.5	54.83	-187.2	142.2	180.1	136.4	43.76	4.117	
10,675.0	10,651.3	10,608.8	10,556.1	23.4	22.6	53.60	-193.8	155.4	185.1	141.5	43.51	4.253	
10,700.0	10,670.5	10,628.1	10,568.1	23.5	22.7	52.49	-200.6	168.9	189.9	146.7	43.23	4.392	
10,725.0	10,688.8	10,647.2	10,579.3	23.6	22.8	51.51	-207.6	182.8	194.7	151.7	42.94	4.533	
10,750.0	10,706.2	10,666.3	10,589.9	23.7	22.9	50.64	-214.7	197.0	199.3	156.7	42.65	4.673	
10,775.0	10,722.6	10,685.3	10,599.8	23.9	23.1	49.87	-221.9	211.4	203.8	161.5	42.35	4.813	
10,800.0	10,738.0	10,704.2	10,609.0	24.0	23.2	49.20	-229.3	226.2	208.2	166.1	42.07	4.950	
10,825.0	10,752.4	10,725.0	10,618.4	24.2	23.4	48.60	-237.7	242.8	212.4	170.6	41.81	5.082	
10,850.0	10,765.7	10,741.7	10,625.3	24.4	23.5	48.13	-244.5	256.4	216.5	174.9	41.56	5.208	
10,875.0	10,777.8	10,760.3	10,632.4	24.6	23.7	47.72	-252.2	271.8	220.4	179.0	41.37	5.327	
10,900.0	10,788.8	10,778.9	10,638.9	24.8	23.9	47.39	-260.1	287.4	224.1	182.8	41.22	5.436	
10,925.0	10,798.6	10,800.0	10,645.3	25.0	24.1	47.14	-269.0	305.3	227.6	186.4	41.15	5.530	
10,950.0	10,807.2	10,816.0	10,649.6	25.3	24.3	46.94	-276.0	319.1	230.9	189.7	41.11	5.616	
10,975.0	10,814.5	10,834.5	10,653.9	25.6	24.5	46.82	-284.0	335.1	233.9	192.8	41.16	5.684	
11,000.0	10,820.6	10,850.0	10,657.0	25.8	24.6	46.72	-290.8	348.7	236.8	195.6	41.24	5.742	
11,025.0	10,825.4	10,871.3	10,660.4	26.1	24.9	46.77	-300.3	367.5	239.5	198.0	41.49	5.772	
11,050.0	10,828.9	10,889.7	10,662.6	26.5	25.1	46.83	-308.5	383.8	241.9	200.1	41.78	5.790	
11,075.0	10,831.1	10,908.1	10,664.1	26.8	25.4	46.96	-316.7	400.2	244.1	202.0	42.16	5.791	
11,100.0	10,832.0	10,925.0	10,664.9	27.1	25.6	47.11	-324.3	415.3	246.2	203.6	42.60	5.778	
11,104.0	10,832.0	10,929.4	10,665.0	27.2	25.7	47.18	-326.2	419.2	246.4	203.7	42.71	5.770	
11,200.0	10,832.2	11,027.9	10,665.2	28.6	27.1	48.96	-369.1	507.9	255.8	210.3	45.46	5.626	
11,300.0	10,832.4	11,134.2	10,665.4	30.3	28.9	51.02	-412.0	605.2	267.7	218.8	48.83	5.482	
11,400.0	10,832.6	11,240.7	10,665.6	32.2	30.9	53.17	-451.2	704.2	281.8	229.2	52.57	5.361	
11,500.0	10,832.7	11,347.5	10,665.8	34.1	33.1	55.36	-486.9	804.8	298.2	241.7	56.57	5.272	
11,600.0	10,832.9	11,454.6	10,666.0	36.2	35.4	57.51	-518.9	907.0	317.1	256.4	60.73	5.222	
11,680.4	10,833.1	11,541.0	10,666.1	38.0	37.3	59.20	-541.8	990.3	334.1	269.9	64.11	5.210	
11,700.0	10,833.1	11,562.1	10,666.2	38.4	37.8	59.67	-547.1	1,010.7	338.3	273.2	65.13	5.195	
11,800.0	10,833.3	11,671.0	10,666.4	40.7	40.4	61.73	-571.7	1,116.9	358.4	288.1	70.34	5.095	
11,900.0	10,833.4	11,781.8	10,666.6	43.0	43.1	63.28	-592.6	1,225.7	375.5	299.9	75.60	4.966	
12,000.0	10,833.6	11,894.1	10,666.8	45.4	45.8	64.43	-609.4	1,336.7	389.2	308.3	80.90	4.811	
12,100.0	10,833.8	12,007.7	10,667.0	47.9	48.7	65.23	-622.0	1,449.5	399.6	313.4	86.21	4.635	
12,200.0	10,834.0	12,122.0	10,667.2	50.4	51.6	65.74	-630.2	1,563.6	406.3	314.8	91.52	4.440	
12,300.0	10,834.1	12,236.9	10,667.4	53.0	54.5	65.96	-633.8	1,678.4	409.5	312.7	96.78	4.231	
12,400.0	10,834.3	12,341.1	10,667.5	55.5	57.2	65.99	-634.0	1,782.6	409.9	308.2	101.72	4.030	
12,500.0	10,834.5	12,441.1	10,667.7	58.2	59.7	66.01	-634.0	1,882.6	410.2	303.6	106.61	3.848	
12,600.0	10,834.7	12,541.1	10,667.9	60.8	62.4	66.03	-634.0	1,982.6	410.5	298.9	111.55	3.680	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services

Anticollision Report



Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 14BX - Kline Federal 5300 41-18 14BX - Design #1													Offset Site Error: 0.0 usft
Survey Program: 0-MWD													Offset Well Error: 2.0 usft
Reference		Offset		Semi Major Axis		Distance							Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	
12,700.0	10,834.8	12,641.1	10,668.1	63.5	65.0	66.05	-634.0	2,082.6	410.8	294.2	116.53	3.525	
12,800.0	10,835.0	12,741.1	10,668.2	66.2	67.7	66.06	-634.0	2,182.6	411.0	289.5	121.54	3.382	
12,900.0	10,835.2	12,841.1	10,668.4	68.9	70.4	66.08	-634.0	2,282.6	411.3	284.7	126.59	3.249	
13,000.0	10,835.4	12,941.1	10,668.6	71.7	73.1	66.10	-634.0	2,382.6	411.6	279.9	131.67	3.126	
13,100.0	10,835.5	13,041.1	10,668.8	74.4	75.8	66.12	-634.0	2,482.6	411.9	275.1	136.77	3.012	
13,200.0	10,835.7	13,141.1	10,668.9	77.2	78.5	66.13	-634.0	2,582.6	412.2	270.3	141.90	2.905	
13,300.0	10,835.9	13,241.1	10,669.1	79.9	81.3	66.15	-634.0	2,682.6	412.5	265.4	147.05	2.805	
13,400.0	10,836.1	13,341.1	10,669.3	82.7	84.1	66.17	-634.0	2,782.6	412.7	260.5	152.21	2.712	
13,500.0	10,836.2	13,441.1	10,669.5	85.5	86.8	66.19	-634.0	2,882.6	413.0	255.6	157.40	2.624	
13,600.0	10,836.4	13,541.1	10,669.6	88.3	89.6	66.20	-634.0	2,982.6	413.3	250.7	162.60	2.542	
13,700.0	10,836.6	13,641.1	10,669.8	91.1	92.4	66.22	-634.0	3,082.6	413.6	245.8	167.81	2.465	
13,800.0	10,836.7	13,741.1	10,670.0	94.0	95.2	66.24	-634.0	3,182.6	413.9	240.8	173.04	2.392	
13,900.0	10,836.9	13,841.1	10,670.2	96.8	98.0	66.25	-634.0	3,282.6	414.1	235.9	178.28	2.323	
14,000.0	10,837.1	13,941.1	10,670.3	99.6	100.9	66.27	-634.0	3,382.6	414.4	230.9	183.53	2.258	
14,100.0	10,837.3	14,041.1	10,670.5	102.5	103.7	66.29	-634.0	3,482.6	414.7	225.9	188.79	2.197	
14,200.0	10,837.4	14,141.1	10,670.7	105.3	106.5	66.31	-634.0	3,582.6	415.0	220.9	194.06	2.138	
14,300.0	10,837.6	14,241.1	10,670.9	108.2	109.3	66.32	-634.0	3,682.6	415.3	215.9	199.35	2.083	
14,400.0	10,837.8	14,341.1	10,671.0	111.0	112.2	66.34	-634.0	3,782.6	415.6	210.9	204.63	2.031	
14,500.0	10,838.0	14,441.1	10,671.2	113.9	115.0	66.36	-634.0	3,882.6	415.8	205.9	209.93	1.981	
14,600.0	10,838.1	14,541.1	10,671.4	116.7	117.9	66.38	-634.0	3,982.6	416.1	200.9	215.24	1.933	
14,700.0	10,838.3	14,641.1	10,671.6	119.6	120.7	66.39	-634.0	4,082.6	416.4	195.9	220.55	1.888	
14,800.0	10,838.5	14,741.1	10,671.7	122.5	123.6	66.41	-634.0	4,182.6	416.7	190.8	225.86	1.845	
14,900.0	10,838.7	14,841.1	10,671.9	125.3	126.4	66.43	-634.0	4,282.6	417.0	185.8	231.19	1.804	
15,000.0	10,838.8	14,941.1	10,672.1	128.2	129.3	66.44	-634.0	4,382.6	417.3	180.7	236.52	1.764	
15,100.0	10,839.0	15,041.1	10,672.3	131.1	132.2	66.46	-634.0	4,482.6	417.5	175.7	241.85	1.726	
15,200.0	10,839.2	15,141.1	10,672.4	133.9	135.0	66.48	-634.0	4,582.6	417.8	170.6	247.19	1.690	
15,300.0	10,839.4	15,241.1	10,672.6	136.8	137.9	66.49	-634.0	4,682.6	418.1	165.6	252.54	1.656	
15,400.0	10,839.5	15,341.1	10,672.8	139.7	140.8	66.51	-634.0	4,782.6	418.4	160.5	257.89	1.622	
15,500.0	10,839.7	15,441.1	10,673.0	142.6	143.7	66.53	-634.0	4,882.6	418.7	155.4	263.24	1.590	
15,600.0	10,839.9	15,541.1	10,673.1	145.5	146.6	66.54	-634.0	4,982.6	419.0	150.3	268.60	1.560	
15,700.0	10,840.1	15,641.1	10,673.3	148.4	149.4	66.56	-634.0	5,082.6	419.2	145.3	273.97	1.530	
15,800.0	10,840.2	15,741.1	10,673.5	151.3	152.3	66.58	-634.0	5,182.6	419.5	140.2	279.33	1.502	
15,900.0	10,840.4	15,841.1	10,673.6	154.1	155.2	66.60	-634.0	5,282.6	419.8	135.1	284.70	1.475 Level 3	
16,000.0	10,840.6	15,941.1	10,673.8	157.0	158.1	66.61	-634.0	5,382.6	420.1	130.0	290.08	1.448 Level 3	
16,100.0	10,840.8	16,041.1	10,674.0	159.9	161.0	66.63	-634.0	5,482.6	420.4	124.9	295.46	1.423 Level 3	
16,200.0	10,840.9	16,141.1	10,674.2	162.8	163.9	66.65	-634.0	5,582.6	420.6	119.8	300.84	1.398 Level 3	
16,300.0	10,841.1	16,241.1	10,674.3	165.7	166.8	66.66	-634.0	5,682.6	420.9	114.7	306.22	1.375 Level 3	
16,400.0	10,841.3	16,341.1	10,674.5	168.6	169.6	66.68	-634.0	5,782.6	421.2	109.6	311.61	1.352 Level 3	
16,500.0	10,841.4	16,441.1	10,674.7	171.5	172.5	66.70	-634.0	5,882.6	421.5	104.5	317.00	1.330 Level 3	
16,600.0	10,841.6	16,541.1	10,674.9	174.4	175.4	66.71	-634.0	5,982.6	421.8	99.4	322.40	1.308 Level 3	
16,700.0	10,841.8	16,641.1	10,675.0	177.3	178.3	66.73	-634.0	6,082.6	422.1	94.3	327.80	1.288 Level 3	
16,800.0	10,842.0	16,741.1	10,675.2	180.2	181.2	66.75	-634.0	6,182.6	422.3	89.2	333.20	1.268 Level 3	
16,900.0	10,842.1	16,841.1	10,675.4	183.1	184.1	66.76	-634.0	6,282.6	422.6	84.0	338.60	1.248 Level 2	
17,000.0	10,842.3	16,941.1	10,675.6	186.0	187.0	66.78	-634.0	6,382.6	422.9	78.9	344.00	1.229 Level 2	
17,100.0	10,842.5	17,041.1	10,675.7	188.9	189.9	66.80	-634.0	6,482.6	423.2	73.8	349.41	1.211 Level 2	
17,200.0	10,842.7	17,141.1	10,675.9	191.8	192.8	66.81	-634.0	6,582.6	423.5	68.7	354.82	1.194 Level 2	
17,300.0	10,842.8	17,241.1	10,676.1	194.7	195.7	66.83	-634.0	6,682.6	423.8	63.5	360.23	1.176 Level 2	
17,400.0	10,843.0	17,341.1	10,676.3	197.6	198.6	66.84	-634.0	6,782.6	424.0	58.4	365.65	1.160 Level 2	
17,500.0	10,843.2	17,441.1	10,676.4	200.6	201.5	66.86	-634.0	6,882.6	424.3	53.3	371.07	1.144 Level 2	
17,600.0	10,843.4	17,541.1	10,676.6	203.5	204.4	66.88	-634.0	6,982.6	424.6	48.1	376.49	1.128 Level 2	
17,700.0	10,843.5	17,641.1	10,676.8	206.4	207.3	66.89	-634.0	7,082.6	424.9	43.0	381.91	1.113 Level 2	
17,800.0	10,843.7	17,741.1	10,677.0	209.3	210.3	66.91	-634.0	7,182.6	425.2	37.8	387.33	1.098 Level 2	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services

Anticollision Report



Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 14BX - Kline Federal 5300 41-18 14BX - Design #1													Offset Site Error: 0.0 usft
Survey Program: 0-MWD													Offset Well Error: 2.0 usft
Reference		Offset		Semi Major Axis			Distance						Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	
17,900.0	10,843.9	17,841.1	10,677.1	212.2	213.2	66.93	-634.0	7,282.6	425.5	32.7	392.76	1.083 Level 2	
18,000.0	10,844.1	17,941.1	10,677.3	215.1	216.1	66.94	-634.0	7,382.6	425.8	27.6	398.19	1.069 Level 2	
18,100.0	10,844.2	18,041.1	10,677.5	218.0	219.0	66.96	-634.0	7,482.6	426.0	22.4	403.62	1.056 Level 2	
18,200.0	10,844.4	18,141.1	10,677.7	220.9	221.9	66.98	-634.0	7,582.6	426.3	17.3	409.05	1.042 Level 2	
18,300.0	10,844.6	18,241.1	10,677.8	223.8	224.8	66.99	-634.0	7,682.6	426.6	12.1	414.49	1.029 Level 2	
18,400.0	10,844.8	18,341.1	10,678.0	226.7	227.7	67.01	-634.0	7,782.6	426.9	7.0	419.92	1.017 Level 2	
18,500.0	10,844.9	18,441.1	10,678.2	229.7	230.6	67.02	-634.0	7,882.6	427.2	1.8	425.36	1.004 Level 2	
18,600.0	10,845.1	18,541.1	10,678.4	232.6	233.5	67.04	-634.0	7,982.6	427.5	-3.3	430.80	0.992 Level 1	
18,700.0	10,845.3	18,641.1	10,678.5	235.5	236.4	67.06	-634.0	8,082.6	427.7	-8.5	436.24	0.981 Level 1	
18,800.0	10,845.4	18,741.1	10,678.7	238.4	239.3	67.07	-634.0	8,182.6	428.0	-13.7	441.69	0.969 Level 1	
18,900.0	10,845.6	18,841.1	10,678.9	241.3	242.3	67.09	-634.0	8,282.6	428.3	-18.8	447.13	0.958 Level 1	
19,000.0	10,845.8	18,941.1	10,679.1	244.2	245.2	67.11	-634.0	8,382.6	428.6	-24.0	452.58	0.947 Level 1	
19,100.0	10,846.0	19,041.1	10,679.2	247.1	248.1	67.12	-634.0	8,482.6	428.9	-29.2	458.03	0.936 Level 1	
19,200.0	10,846.1	19,141.1	10,679.4	250.1	251.0	67.14	-634.0	8,582.6	429.2	-34.3	463.48	0.926 Level 1	
19,300.0	10,846.3	19,241.1	10,679.6	253.0	253.9	67.15	-634.0	8,682.6	429.4	-39.5	468.93	0.916 Level 1	
19,400.0	10,846.5	19,341.1	10,679.8	255.9	256.8	67.17	-634.0	8,782.6	429.7	-44.7	474.39	0.906 Level 1	
19,500.0	10,846.7	19,441.1	10,679.9	258.8	259.7	67.19	-634.0	8,882.6	430.0	-49.8	479.84	0.896 Level 1	
19,600.0	10,846.8	19,541.1	10,680.1	261.7	262.7	67.20	-634.0	8,982.6	430.3	-55.0	485.30	0.887 Level 1	
19,700.0	10,847.0	19,641.1	10,680.3	264.6	265.6	67.22	-634.0	9,082.6	430.6	-60.2	490.76	0.877 Level 1	
19,800.0	10,847.2	19,741.1	10,680.5	267.6	268.5	67.23	-634.0	9,182.6	430.9	-65.4	496.22	0.868 Level 1	
19,900.0	10,847.4	19,841.1	10,680.6	270.5	271.4	67.25	-634.0	9,282.6	431.1	-70.5	501.68	0.859 Level 1	
20,000.0	10,847.5	19,941.1	10,680.8	273.4	274.3	67.27	-634.0	9,382.6	431.4	-75.7	507.14	0.851 Level 1	
20,100.0	10,847.7	20,041.1	10,681.0	276.3	277.2	67.28	-634.0	9,482.6	431.7	-80.9	512.61	0.842 Level 1	
20,200.0	10,847.9	20,141.1	10,681.2	279.2	280.2	67.30	-634.0	9,582.6	432.0	-86.1	518.08	0.834 Level 1	
20,300.0	10,848.1	20,241.1	10,681.3	282.1	283.1	67.31	-634.0	9,682.6	432.3	-91.3	523.54	0.826 Level 1	
20,400.0	10,848.2	20,341.1	10,681.5	285.1	286.0	67.33	-634.0	9,782.6	432.6	-96.4	529.01	0.818 Level 1	
20,500.0	10,848.4	20,441.1	10,681.7	288.0	288.9	67.34	-634.0	9,882.6	432.9	-101.6	534.49	0.810 Level 1	
20,600.0	10,848.6	20,541.1	10,681.9	290.9	291.8	67.36	-634.0	9,982.6	433.1	-106.8	539.96	0.802 Level 1	
20,667.7	10,848.7	20,608.5	10,682.0	292.9	293.8	67.37	-634.0	10,050.0	433.3	-110.3	543.66	0.797 Level 1, ES, SF	

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design		153N-100W-17/18 - Kline Federal 5300 41-18 14BX - Kline Federal 5300 41-18 14BX - Design #2											Offset Site Error: 0.0 usft	
Survey Program: 0-MWD													Offset Well Error: 2.0 usft	
Reference		Offset		Semi Major Axis			Distance							Warning
Measured Depth	Vertical Depth	Measured Depth	Vertical Depth	Reference	Offset	Highside Toolface	Offset Wellbore Centre		Between Centres	Between Ellipses	Minimum Separation	Separation Factor		
(usft)	(usft)	(usft)	(usft)	(usft)	(usft)	(°)	+N/-S (usft)	+E/-W (usft)	(usft)	(usft)	(usft)			
0.0	0.0	0.0	0.0	2.0	2.0	-180.00	-65.9	0.0	65.9					
100.0	100.0	100.0	100.0	2.0	2.0	-180.00	-65.9	0.0	65.9	61.9	4.00	16.452		
200.0	200.0	200.0	200.0	2.0	2.0	-180.00	-65.9	0.0	65.9	61.8	4.05	16.273		
300.0	300.0	300.0	300.0	2.1	2.1	-180.00	-65.9	0.0	65.9	61.7	4.14	15.909		
400.0	400.0	400.0	400.0	2.1	2.1	-180.00	-65.9	0.0	65.9	61.6	4.28	15.396		
500.0	500.0	500.0	500.0	2.2	2.2	-180.00	-65.9	0.0	65.9	61.4	4.46	14.777		
600.0	600.0	600.0	600.0	2.3	2.3	-180.00	-65.9	0.0	65.9	61.2	4.67	14.094		
700.0	700.0	700.0	700.0	2.5	2.5	-180.00	-65.9	0.0	65.9	60.9	4.92	13.385		
800.0	800.0	800.0	800.0	2.6	2.6	-180.00	-65.9	0.0	65.9	60.7	5.20	12.678		
900.0	900.0	900.0	900.0	2.7	2.7	-180.00	-65.9	0.0	65.9	60.4	5.49	11.990		
1,000.0	1,000.0	1,000.0	1,000.0	2.9	2.9	-180.00	-65.9	0.0	65.9	60.1	5.81	11.336		
1,100.0	1,100.0	1,100.0	1,100.0	3.1	3.1	-180.00	-65.9	0.0	65.9	59.7	6.14	10.720		
1,200.0	1,200.0	1,200.0	1,200.0	3.2	3.2	-180.00	-65.9	0.0	65.9	59.4	6.49	10.145		
1,300.0	1,300.0	1,300.0	1,300.0	3.4	3.4	-180.00	-65.9	0.0	65.9	59.0	6.85	9.613		
1,400.0	1,400.0	1,400.0	1,400.0	3.6	3.6	-180.00	-65.9	0.0	65.9	58.6	7.22	9.121		
1,500.0	1,500.0	1,500.0	1,500.0	3.8	3.8	-180.00	-65.9	0.0	65.9	58.3	7.60	8.667		
1,600.0	1,600.0	1,600.0	1,600.0	4.0	4.0	-180.00	-65.9	0.0	65.9	57.9	7.99	8.248		
1,700.0	1,700.0	1,700.0	1,700.0	4.2	4.2	-180.00	-65.9	0.0	65.9	57.5	8.38	7.862		
1,800.0	1,800.0	1,800.0	1,800.0	4.4	4.4	-180.00	-65.9	0.0	65.9	57.1	8.78	7.506		
1,900.0	1,900.0	1,900.0	1,900.0	4.6	4.6	-180.00	-65.9	0.0	65.9	56.7	9.18	7.177		
2,000.0	2,000.0	2,000.0	2,000.0	4.8	4.8	-180.00	-65.9	0.0	65.9	56.3	9.58	6.872		
2,100.0	2,100.0	2,100.0	2,100.0	5.0	5.0	-180.00	-65.9	0.0	65.9	55.9	9.99	6.590		
2,166.7	2,166.7	2,166.7	2,166.7	5.1	5.1	-180.00	-65.9	0.0	65.9	55.6	10.27	6.413		
2,200.0	2,200.0	2,200.0	2,200.0	5.2	5.2	-180.00	-65.9	0.0	65.9	55.5	10.41	6.328 CC		
2,216.7	2,216.7	2,216.1	2,216.1	5.2	5.2	90.06	-65.9	0.0	65.9	55.5	10.47	6.298		
2,300.0	2,300.0	2,299.4	2,299.4	5.4	5.4	90.68	-66.7	0.0	66.7	55.9	10.75	6.201		
2,400.0	2,400.0	2,399.4	2,399.4	5.6	5.5	91.41	-67.5	0.0	67.6	56.5	11.10	6.088		
2,500.0	2,500.0	2,499.4	2,499.4	5.8	5.7	92.13	-68.4	0.0	68.5	57.0	11.45	5.979		
2,600.0	2,600.0	2,599.4	2,599.4	6.0	5.9	92.82	-69.3	0.0	69.4	57.6	11.81	5.874		
2,700.0	2,700.0	2,699.4	2,699.4	6.1	6.0	93.50	-70.1	0.0	70.3	58.1	12.17	5.773		
2,800.0	2,800.0	2,799.4	2,799.4	6.3	6.2	94.15	-71.0	0.0	71.2	58.7	12.54	5.677		
2,900.0	2,900.0	2,899.4	2,899.3	6.5	6.4	94.79	-71.9	0.0	72.2	59.2	12.92	5.584		
3,000.0	3,000.0	2,999.4	2,999.3	6.7	6.6	95.42	-72.8	0.0	73.1	59.8	13.30	5.496		
3,100.0	3,100.0	3,099.4	3,099.3	6.9	6.7	96.03	-73.6	0.0	74.1	60.4	13.69	5.411		
3,200.0	3,200.0	3,199.3	3,199.3	7.1	6.9	96.62	-74.5	0.0	75.0	60.9	14.07	5.330		
3,300.0	3,300.0	3,299.3	3,299.3	7.3	7.1	97.20	-75.4	0.0	76.0	61.5	14.47	5.253		
3,400.0	3,400.0	3,399.3	3,399.3	7.6	7.3	97.76	-76.3	0.0	77.0	62.1	14.86	5.179		
3,500.0	3,500.0	3,499.3	3,499.3	7.8	7.5	98.31	-77.1	0.0	78.0	62.7	15.26	5.108		
3,600.0	3,599.9	3,599.3	3,599.3	8.0	7.7	98.84	-78.0	0.0	78.9	63.3	15.66	5.041		
3,700.0	3,699.9	3,699.3	3,699.3	8.2	7.9	99.37	-78.9	0.0	79.9	63.9	16.07	4.976		
3,800.0	3,799.9	3,799.3	3,799.2	8.4	8.1	99.88	-79.7	0.0	81.0	64.5	16.47	4.915		
3,900.0	3,899.9	3,899.3	3,899.2	8.6	8.3	100.37	-80.6	0.0	82.0	65.1	16.88	4.856		
4,000.0	3,999.9	3,999.3	3,999.2	8.8	8.5	100.86	-81.5	0.0	83.0	65.7	17.29	4.799		
4,100.0	4,099.9	4,099.3	4,099.2	9.0	8.7	101.33	-82.4	0.0	84.0	66.3	17.70	4.745		
4,200.0	4,199.9	4,199.3	4,199.2	9.2	8.9	101.79	-83.2	0.0	85.0	66.9	18.12	4.694		
4,300.0	4,299.9	4,299.3	4,299.2	9.4	9.1	102.24	-84.1	0.0	86.1	67.5	18.53	4.644		
4,400.0	4,399.9	4,399.3	4,399.2	9.7	9.3	102.68	-85.0	0.0	87.1	68.2	18.95	4.597		
4,500.0	4,499.9	4,499.3	4,499.2	9.9	9.5	103.11	-85.9	0.0	88.2	68.8	19.37	4.551		
4,600.0	4,599.9	4,599.2	4,599.2	10.1	9.7	103.53	-86.7	0.0	89.2	69.4	19.79	4.507		
4,700.0	4,699.9	4,699.2	4,699.1	10.3	9.9	103.94	-87.6	0.0	90.3	70.0	20.21	4.466		
4,800.0	4,799.9	4,799.2	4,799.1	10.5	10.1	104.33	-88.5	0.0	91.3	70.7	20.64	4.425		
4,900.0	4,899.9	4,899.2	4,899.1	10.7	10.3	104.72	-89.3	0.0	92.4	71.3	21.06	4.387		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services

Anticollision Report



Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 14BX - Kline Federal 5300 41-18 14BX - Design #2												Offset Site Error:	0.0 usft
Survey Program: 0-MWD												Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis		Distance							
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
5,000.0	4,999.9	4,999.2	4,999.1	10.9	10.6	105.11	-90.2	0.0	93.5	72.0	21.49	4.350	
5,100.0	5,099.9	5,099.2	5,099.1	11.2	10.8	105.48	-91.1	0.0	94.5	72.6	21.91	4.314	
5,200.0	5,199.9	5,199.2	5,199.1	11.4	11.0	105.84	-92.0	0.0	95.6	73.3	22.34	4.280	
5,300.0	5,299.9	5,299.2	5,299.1	11.6	11.2	106.20	-92.8	0.0	96.7	73.9	22.77	4.246	
5,400.0	5,399.9	5,399.2	5,399.1	11.8	11.4	106.55	-93.7	0.0	97.8	74.6	23.20	4.215	
5,500.0	5,499.9	5,499.2	5,499.0	12.0	11.6	106.89	-94.6	0.0	98.9	75.2	23.63	4.184	
5,600.0	5,599.9	5,599.2	5,599.0	12.2	11.8	107.22	-95.5	0.0	99.9	75.9	24.06	4.154	
5,700.0	5,699.9	5,699.2	5,699.0	12.5	12.0	107.55	-96.3	0.0	101.0	76.5	24.49	4.126	
5,800.0	5,799.9	5,799.2	5,799.0	12.7	12.3	107.87	-97.2	0.0	102.1	77.2	24.92	4.098	
5,900.0	5,899.9	5,899.1	5,899.0	12.9	12.5	108.18	-98.1	0.0	103.2	77.9	25.35	4.072	
6,000.0	5,999.9	5,999.1	5,999.0	13.1	12.7	108.49	-98.9	0.0	104.3	78.5	25.79	4.046	
6,100.0	6,099.9	6,099.1	6,099.0	13.3	12.9	108.79	-99.8	0.0	105.4	79.2	26.22	4.021	
6,200.0	6,199.8	6,199.1	6,199.0	13.6	13.1	109.08	-100.7	0.0	106.5	79.9	26.65	3.997	
6,300.0	6,299.8	6,299.1	6,299.0	13.8	13.3	109.37	-101.6	0.0	107.7	80.6	27.09	3.974	
6,400.0	6,399.8	6,399.1	6,398.9	14.0	13.6	109.65	-102.4	0.0	108.8	81.2	27.53	3.952	
6,500.0	6,499.8	6,499.1	6,498.9	14.2	13.8	109.92	-103.3	0.0	109.9	81.9	27.96	3.930	
6,600.0	6,599.8	6,599.1	6,598.9	14.4	14.0	110.19	-104.2	0.0	111.0	82.6	28.40	3.909	
6,700.0	6,699.8	6,699.1	6,698.9	14.7	14.2	110.46	-105.1	0.0	112.1	83.3	28.83	3.889	
6,800.0	6,799.8	6,799.1	6,798.9	14.9	14.4	110.72	-105.9	0.0	113.3	84.0	29.27	3.869	
6,900.0	6,899.8	6,899.1	6,898.9	15.1	14.6	110.97	-106.8	0.0	114.4	84.7	29.71	3.850	
7,000.0	6,999.8	6,999.1	6,998.9	15.3	14.9	111.22	-107.7	0.0	115.5	85.4	30.15	3.831	
7,100.0	7,099.8	7,099.1	7,098.9	15.5	15.1	111.46	-108.5	0.0	116.6	86.1	30.59	3.813	
7,200.0	7,199.8	7,199.0	7,198.9	15.8	15.3	111.70	-109.4	0.0	117.8	86.7	31.02	3.796	
7,300.0	7,299.8	7,299.0	7,298.8	16.0	15.5	111.94	-110.3	0.0	118.9	87.4	31.46	3.779	
7,400.0	7,399.8	7,399.0	7,398.8	16.2	15.7	112.17	-111.2	0.0	120.0	88.1	31.90	3.763	
7,500.0	7,499.8	7,499.0	7,498.8	16.4	16.0	112.40	-112.0	0.0	121.2	88.8	32.34	3.747	
7,600.0	7,599.8	7,599.0	7,598.8	16.6	16.2	112.62	-112.9	0.0	122.3	89.5	32.78	3.731	
7,700.0	7,699.8	7,699.0	7,698.8	16.9	16.4	112.84	-113.8	0.0	123.5	90.2	33.22	3.716	
7,800.0	7,799.8	7,799.0	7,798.8	17.1	16.6	113.05	-114.7	0.0	124.6	90.9	33.66	3.701	
7,900.0	7,899.8	7,899.0	7,898.8	17.3	16.8	113.26	-115.5	0.0	125.8	91.6	34.11	3.687	
7,926.7	7,926.4	7,925.7	7,925.4	17.4	16.9	113.32	-115.8	0.0	126.1	91.8	34.22	3.683	
7,943.3	7,943.1	7,943.3	7,943.1	17.4	16.9	-156.66	-115.8	0.0	126.2	91.9	34.30	3.679	
8,000.0	7,999.8	8,000.0	7,999.8	17.5	17.1	-156.66	-115.8	0.0	126.2	91.6	34.52	3.654	
8,100.0	8,099.8	8,100.0	8,099.8	17.7	17.2	-156.66	-115.8	0.0	126.2	91.2	34.92	3.613	
8,200.0	8,199.8	8,200.0	8,199.8	17.9	17.4	-156.66	-115.8	0.0	126.2	90.8	35.32	3.572	
8,300.0	8,299.8	8,300.0	8,299.8	18.1	17.6	-156.66	-115.8	0.0	126.2	90.4	35.72	3.532	
8,400.0	8,399.8	8,400.0	8,399.8	18.4	17.8	-156.66	-115.8	0.0	126.2	90.0	36.12	3.493	
8,500.0	8,499.8	8,500.0	8,499.8	18.6	18.0	-156.66	-115.8	0.0	126.2	89.6	36.52	3.454	
8,600.0	8,599.8	8,600.0	8,599.8	18.8	18.2	-156.66	-115.8	0.0	126.2	89.2	36.93	3.416	
8,700.0	8,699.8	8,700.0	8,699.8	19.0	18.4	-156.66	-115.8	0.0	126.2	88.8	37.33	3.379	
8,800.0	8,799.8	8,800.0	8,799.8	19.2	18.6	-156.66	-115.8	0.0	126.2	88.4	37.74	3.343	
8,900.0	8,899.8	8,900.0	8,899.8	19.4	18.8	-156.66	-115.8	0.0	126.2	88.0	38.14	3.307	
9,000.0	8,999.8	9,000.0	8,999.8	19.6	19.0	-156.66	-115.8	0.0	126.2	87.6	38.55	3.273	
9,100.0	9,099.8	9,100.0	9,099.8	19.8	19.2	-156.66	-115.8	0.0	126.2	87.2	38.96	3.238	
9,200.0	9,199.8	9,200.0	9,199.8	20.0	19.4	-156.66	-115.8	0.0	126.2	86.8	39.37	3.205	
9,300.0	9,299.8	9,300.0	9,299.8	20.2	19.6	-156.66	-115.8	0.0	126.2	86.4	39.78	3.172	
9,400.0	9,399.8	9,400.0	9,399.8	20.5	19.8	-156.66	-115.8	0.0	126.2	86.0	40.19	3.139	
9,500.0	9,499.8	9,500.0	9,499.8	20.7	20.0	-156.66	-115.8	0.0	126.2	85.6	40.60	3.107	
9,600.0	9,599.8	9,600.0	9,599.8	20.9	20.2	-156.66	-115.8	0.0	126.2	85.1	41.01	3.076	
9,700.0	9,699.8	9,700.0	9,699.8	21.1	20.4	-156.66	-115.8	0.0	126.2	84.7	41.43	3.045	
9,800.0	9,799.8	9,800.0	9,799.8	21.3	20.6	-156.66	-115.8	0.0	126.2	84.3	41.84	3.015	
9,900.0	9,899.8	9,900.0	9,899.8	21.5	20.8	-156.66	-115.8	0.0	126.2	83.9	42.26	2.986	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services

Anticollision Report



Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 14BX - Kline Federal 5300 41-18 14BX - Design #2													Offset Site Error: 0.0 usft
Survey Program: 0-MWD													Offset Well Error: 2.0 usft
Reference		Offset		Semi Major Axis			Distance						Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	
10,000.0	9,999.8	10,000.0	9,999.8	21.7	21.0	-156.66	-115.8	0.0	126.2	83.5	42.67	2.956	
10,100.0	10,099.8	10,100.0	10,099.8	21.9	21.2	-156.66	-115.8	0.0	126.2	83.1	43.09	2.928	
10,200.0	10,199.8	10,200.0	10,199.8	22.2	21.4	-156.66	-115.8	0.0	126.2	82.7	43.50	2.900	
10,262.1	10,261.8	10,262.1	10,261.8	22.3	21.5	-156.66	-115.8	0.0	126.2	82.4	43.76	2.883	
10,300.0	10,299.8	10,299.8	10,299.6	22.4	21.6	-156.73	-115.9	0.1	126.2	82.2	43.92	2.873	
10,354.8	10,354.6	10,353.4	10,352.9	22.5	21.7	-158.70	-117.9	4.0	126.5	82.4	44.16	2.865	
10,375.0	10,374.8	10,372.8	10,372.2	22.5	21.7	90.06	-119.2	6.8	126.8	82.6	44.26	2.866	
10,400.0	10,399.7	10,396.8	10,395.6	22.6	21.8	88.56	-121.4	11.1	127.5	83.1	44.36	2.873	
10,425.0	10,424.5	10,420.6	10,418.7	22.7	21.9	87.07	-124.1	16.4	128.3	83.9	44.47	2.886	
10,450.0	10,449.2	10,444.2	10,441.2	22.7	21.9	85.63	-127.2	22.7	129.4	84.8	44.58	2.903	
10,475.0	10,473.5	10,467.7	10,463.3	22.8	22.0	84.23	-130.9	30.0	130.7	86.0	44.69	2.924	
10,500.0	10,497.6	10,491.1	10,484.8	22.8	22.0	82.88	-135.0	38.1	132.2	87.4	44.79	2.950	
10,525.0	10,521.2	10,514.3	10,505.7	22.9	22.1	81.59	-139.5	47.1	133.8	88.9	44.90	2.981	
10,550.0	10,544.4	10,537.4	10,525.9	23.0	22.1	80.37	-144.5	57.0	135.6	90.6	44.99	3.014	
10,575.0	10,567.1	10,560.3	10,545.6	23.0	22.2	79.20	-149.8	67.7	137.6	92.5	45.09	3.052	
10,600.0	10,589.1	10,583.2	10,564.5	23.1	22.3	78.11	-155.5	79.1	139.7	94.5	45.18	3.093	
10,625.0	10,610.6	10,605.9	10,582.7	23.2	22.3	77.09	-161.7	91.3	141.9	96.7	45.26	3.136	
10,650.0	10,631.3	10,628.5	10,600.1	23.3	22.4	76.13	-168.1	104.2	144.3	98.9	45.35	3.182	
10,675.0	10,651.3	10,650.0	10,616.0	23.4	22.5	75.27	-174.6	117.1	146.7	101.3	45.43	3.230	
10,700.0	10,670.5	10,673.5	10,632.6	23.5	22.6	74.43	-182.0	131.9	149.3	103.7	45.52	3.279	
10,725.0	10,688.8	10,695.8	10,647.7	23.6	22.7	73.69	-189.4	146.6	151.9	106.2	45.61	3.329	
10,750.0	10,706.2	10,718.1	10,661.9	23.7	22.8	73.01	-197.1	161.9	154.5	108.8	45.72	3.380	
10,775.0	10,722.6	10,740.2	10,675.3	23.9	23.0	72.40	-205.0	177.7	157.2	111.4	45.83	3.430	
10,800.0	10,738.0	10,762.3	10,687.7	24.0	23.1	71.85	-213.2	194.0	160.0	114.0	45.97	3.480	
10,825.0	10,752.4	10,784.4	10,699.3	24.2	23.3	71.37	-221.6	210.8	162.7	116.6	46.13	3.528	
10,850.0	10,765.7	10,806.3	10,710.0	24.4	23.4	70.94	-230.2	227.9	165.5	119.2	46.32	3.574	
10,875.0	10,777.8	10,828.3	10,719.8	24.6	23.6	70.58	-239.0	245.5	168.3	121.8	46.54	3.617	
10,900.0	10,788.8	10,850.0	10,728.5	24.8	23.8	70.26	-247.9	263.2	171.1	124.3	46.79	3.658	
10,925.0	10,798.6	10,872.0	10,736.5	25.0	24.0	70.01	-257.1	281.6	173.9	126.8	47.08	3.694	
10,950.0	10,807.2	10,893.8	10,743.4	25.3	24.2	69.80	-266.4	300.0	176.7	129.3	47.42	3.727	
10,975.0	10,814.5	10,915.6	10,749.4	25.6	24.4	69.64	-275.8	318.7	179.5	131.7	47.79	3.755	
11,000.0	10,820.6	10,937.3	10,754.5	25.8	24.7	69.54	-285.3	337.7	182.2	134.0	48.21	3.779	
11,025.0	10,825.4	10,959.1	10,758.5	26.1	24.9	69.47	-294.9	356.8	184.9	136.2	48.68	3.799	
11,050.0	10,828.9	10,980.8	10,761.6	26.5	25.2	69.45	-304.5	376.0	187.6	138.4	49.19	3.814	
11,075.0	10,831.1	11,002.6	10,763.7	26.8	25.5	69.48	-314.3	395.4	190.2	140.5	49.74	3.824	
11,100.0	10,832.0	11,025.0	10,764.8	27.1	25.8	69.55	-324.3	415.4	192.8	142.5	50.35	3.829	
11,104.0	10,832.0	11,027.8	10,764.9	27.2	25.8	69.55	-325.5	417.9	193.2	142.8	50.43	3.831	
11,200.0	10,832.2	11,127.7	10,765.1	28.6	27.3	70.73	-369.1	507.8	205.0	151.7	53.31	3.845	
11,300.0	10,832.4	11,233.9	10,765.3	30.3	29.1	72.00	-411.9	605.0	219.7	163.0	56.67	3.876	
11,400.0	10,832.6	11,340.4	10,765.5	32.2	31.0	73.27	-451.2	704.0	236.7	176.4	60.29	3.925	
11,500.0	10,832.7	11,447.2	10,765.7	34.1	33.2	74.50	-486.8	804.6	256.1	192.0	64.07	3.997	
11,600.0	10,832.9	11,554.3	10,765.9	36.2	35.5	75.67	-518.8	906.8	277.8	209.9	67.90	4.091	
11,680.4	10,833.1	11,640.7	10,766.1	38.0	37.5	76.56	-541.8	990.1	297.0	226.0	70.97	4.185	
11,700.0	10,833.1	11,661.8	10,766.1	38.4	38.0	76.80	-547.0	1,010.5	301.8	229.9	71.90	4.197	
11,800.0	10,833.3	11,770.8	10,766.3	40.7	40.5	77.83	-571.7	1,116.7	324.2	247.4	76.78	4.222	
11,900.0	10,833.4	11,881.6	10,766.5	43.0	43.2	78.59	-592.5	1,225.5	342.9	261.1	81.84	4.190	
12,000.0	10,833.6	11,993.9	10,766.7	45.4	45.9	79.14	-609.4	1,336.5	358.0	270.9	87.07	4.111	
12,100.0	10,833.8	12,107.4	10,766.9	47.9	48.8	79.52	-622.0	1,449.3	369.2	276.8	92.43	3.994	
12,200.0	10,834.0	12,221.8	10,767.1	50.4	51.7	79.76	-630.2	1,563.4	376.5	278.6	97.89	3.846	
12,300.0	10,834.1	12,336.6	10,767.3	53.0	54.6	79.87	-633.8	1,678.2	379.9	276.5	103.42	3.673	
12,400.0	10,834.3	12,440.9	10,767.5	55.5	57.2	79.88	-634.0	1,782.5	380.4	271.7	108.72	3.499	
12,500.0	10,834.5	12,540.9	10,767.6	58.2	59.8	79.89	-634.0	1,882.5	380.7	266.7	113.97	3.340	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 14BX - Kline Federal 5300 41-18 14BX - Design #2													Offset Site Error: 0.0 usft
Survey Program: 0-MWD													Offset Well Error: 2.0 usft
Reference		Offset		Semi Major Axis		Highside Toolface (°)	Distance		Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)		Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)					
12,600.0	10,834.7	12,640.9	10,767.8	60.8	62.4	79.90	-634.0	1,982.5	381.0	261.7	119.27	3.194	
12,700.0	10,834.8	12,740.9	10,768.0	63.5	65.1	79.90	-634.0	2,082.5	381.3	256.7	124.60	3.060	
12,800.0	10,835.0	12,840.9	10,768.2	66.2	67.7	79.91	-634.0	2,182.5	381.6	251.6	129.98	2.936	
12,900.0	10,835.2	12,940.9	10,768.3	68.9	70.4	79.92	-634.0	2,282.5	381.9	246.5	135.39	2.821	
13,000.0	10,835.4	13,040.9	10,768.5	71.7	73.1	79.93	-634.0	2,382.5	382.2	241.4	140.83	2.714	
13,100.0	10,835.5	13,140.9	10,768.7	74.4	75.9	79.94	-634.0	2,482.5	382.5	236.2	146.29	2.615	
13,200.0	10,835.7	13,240.9	10,768.9	77.2	78.6	79.95	-634.0	2,582.5	382.8	231.0	151.78	2.522	
13,300.0	10,835.9	13,340.9	10,769.0	79.9	81.3	79.95	-634.0	2,682.5	383.1	225.8	157.29	2.436	
13,400.0	10,836.1	13,440.9	10,769.2	82.7	84.1	79.96	-634.0	2,782.5	383.4	220.6	162.82	2.355	
13,500.0	10,836.2	13,540.9	10,769.4	85.5	86.9	79.97	-634.0	2,882.5	383.7	215.4	168.36	2.279	
13,600.0	10,836.4	13,640.9	10,769.6	88.3	89.7	79.98	-634.0	2,982.5	384.0	210.1	173.93	2.208	
13,700.0	10,836.6	13,740.9	10,769.7	91.1	92.5	79.99	-634.0	3,082.5	384.3	204.8	179.50	2.141	
13,800.0	10,836.7	13,840.9	10,769.9	94.0	95.3	79.99	-634.0	3,182.5	384.6	199.5	185.09	2.078	
13,900.0	10,836.9	13,940.9	10,770.1	96.8	98.1	80.00	-634.0	3,282.5	384.9	194.2	190.69	2.019	
14,000.0	10,837.1	14,040.9	10,770.3	99.6	100.9	80.01	-634.0	3,382.5	385.2	188.9	196.30	1.963	
14,100.0	10,837.3	14,140.9	10,770.4	102.5	103.7	80.02	-634.0	3,482.5	385.5	183.6	201.92	1.909	
14,200.0	10,837.4	14,240.9	10,770.6	105.3	106.5	80.03	-634.0	3,582.5	385.8	178.3	207.55	1.859	
14,300.0	10,837.6	14,340.9	10,770.8	108.2	109.4	80.03	-634.0	3,682.5	386.1	173.0	213.19	1.811	
14,400.0	10,837.8	14,440.9	10,771.0	111.0	112.2	80.04	-634.0	3,782.5	386.5	167.6	218.83	1.766	
14,500.0	10,838.0	14,540.9	10,771.1	113.9	115.1	80.05	-634.0	3,882.5	386.8	162.3	224.48	1.723	
14,600.0	10,838.1	14,640.9	10,771.3	116.7	117.9	80.06	-634.0	3,982.5	387.1	156.9	230.14	1.682	
14,700.0	10,838.3	14,740.9	10,771.5	119.6	120.8	80.07	-634.0	4,082.5	387.4	151.6	235.81	1.643	
14,800.0	10,838.5	14,840.9	10,771.7	122.5	123.6	80.07	-634.0	4,182.5	387.7	146.2	241.48	1.605	
14,900.0	10,838.7	14,940.9	10,771.8	125.3	126.5	80.08	-634.0	4,282.5	388.0	140.8	247.15	1.570	
15,000.0	10,838.8	15,040.9	10,772.0	128.2	129.3	80.09	-634.0	4,382.5	388.3	135.4	252.84	1.536	
15,100.0	10,839.0	15,140.9	10,772.2	131.1	132.2	80.10	-634.0	4,482.5	388.6	130.1	258.52	1.503	
15,200.0	10,839.2	15,240.9	10,772.4	133.9	135.1	80.11	-634.0	4,582.5	388.9	124.7	264.21	1.472 Level 3	
15,300.0	10,839.4	15,340.9	10,772.5	136.8	137.9	80.11	-634.0	4,682.5	389.2	119.3	269.91	1.442 Level 3	
15,400.0	10,839.5	15,440.9	10,772.7	139.7	140.8	80.12	-634.0	4,782.5	389.5	113.9	275.61	1.413 Level 3	
15,500.0	10,839.7	15,540.9	10,772.9	142.6	143.7	80.13	-634.0	4,882.5	389.8	108.5	281.31	1.386 Level 3	
15,600.0	10,839.9	15,640.9	10,773.1	145.5	146.6	80.14	-634.0	4,982.5	390.1	103.1	287.02	1.359 Level 3	
15,700.0	10,840.1	15,740.9	10,773.2	148.4	149.5	80.14	-634.0	5,082.5	390.4	97.7	292.73	1.334 Level 3	
15,800.0	10,840.2	15,840.9	10,773.4	151.3	152.3	80.15	-634.0	5,182.5	390.7	92.3	298.44	1.309 Level 3	
15,900.0	10,840.4	15,940.9	10,773.6	154.1	155.2	80.16	-634.0	5,282.4	391.0	86.9	304.15	1.286 Level 3	
16,000.0	10,840.6	16,040.9	10,773.8	157.0	158.1	80.17	-634.0	5,382.4	391.3	81.4	309.87	1.263 Level 3	
16,100.0	10,840.8	16,140.9	10,773.9	159.9	161.0	80.18	-634.0	5,482.4	391.6	76.0	315.59	1.241 Level 2	
16,200.0	10,840.9	16,240.9	10,774.1	162.8	163.9	80.18	-634.0	5,582.4	391.9	70.6	321.32	1.220 Level 2	
16,300.0	10,841.1	16,340.9	10,774.3	165.7	166.8	80.19	-634.0	5,682.4	392.2	65.2	327.04	1.199 Level 2	
16,400.0	10,841.3	16,440.9	10,774.5	168.6	169.7	80.20	-634.0	5,782.4	392.5	59.8	332.77	1.180 Level 2	
16,500.0	10,841.4	16,540.9	10,774.6	171.5	172.6	80.21	-634.0	5,882.4	392.8	54.3	338.50	1.161 Level 2	
16,600.0	10,841.6	16,640.9	10,774.8	174.4	175.5	80.21	-634.0	5,982.4	393.1	48.9	344.23	1.142 Level 2	
16,700.0	10,841.8	16,740.9	10,775.0	177.3	178.4	80.22	-634.0	6,082.4	393.4	43.5	349.97	1.124 Level 2	
16,800.0	10,842.0	16,840.9	10,775.2	180.2	181.2	80.23	-634.0	6,182.4	393.7	38.0	355.70	1.107 Level 2	
16,900.0	10,842.1	16,940.9	10,775.3	183.1	184.1	80.24	-634.0	6,282.4	394.1	32.6	361.44	1.090 Level 2	
17,000.0	10,842.3	17,040.9	10,775.5	186.0	187.0	80.25	-634.0	6,382.4	394.4	27.2	367.18	1.074 Level 2	
17,100.0	10,842.5	17,140.9	10,775.7	188.9	189.9	80.25	-634.0	6,482.4	394.7	21.7	372.93	1.058 Level 2	
17,200.0	10,842.7	17,240.9	10,775.9	191.8	192.8	80.26	-634.0	6,582.4	395.0	16.3	378.67	1.043 Level 2	
17,300.0	10,842.8	17,340.9	10,776.0	194.7	195.7	80.27	-634.0	6,682.4	395.3	10.9	384.41	1.028 Level 2	
17,400.0	10,843.0	17,440.9	10,776.2	197.6	198.7	80.28	-634.0	6,782.4	395.6	5.4	390.16	1.014 Level 2	
17,500.0	10,843.2	17,540.9	10,776.4	200.6	201.6	80.28	-634.0	6,882.4	395.9	0.0	395.91	1.000 Level 1	
17,600.0	10,843.4	17,640.9	10,776.5	203.5	204.5	80.29	-634.0	6,982.4	396.2	-5.5	401.66	0.986 Level 1	
17,700.0	10,843.5	17,740.9	10,776.7	206.4	207.4	80.30	-634.0	7,082.4	396.5	-10.9	407.41	0.973 Level 1	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services

Anticollision Report



Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 14BX - Kline Federal 5300 41-18 14BX - Design #2													Offset Site Error: 0.0 usft
Survey Program: 0-MWD													Offset Well Error: 2.0 usft
Reference		Offset		Semi Major Axis		Highside Toolface (°)	Distance		Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)		Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)					
17,800.0	10,843.7	17,840.9	10,776.9	209.3	210.3	80.31	-634.0	7,182.4	396.8	-16.4	413.16	0.960	Level 1
17,900.0	10,843.9	17,940.9	10,777.1	212.2	213.2	80.31	-634.0	7,282.4	397.1	-21.8	418.91	0.948	Level 1
18,000.0	10,844.1	18,040.9	10,777.2	215.1	216.1	80.32	-634.0	7,382.4	397.4	-27.3	424.67	0.936	Level 1
18,100.0	10,844.2	18,140.9	10,777.4	218.0	219.0	80.33	-634.0	7,482.4	397.7	-32.7	430.42	0.924	Level 1
18,200.0	10,844.4	18,240.9	10,777.6	220.9	221.9	80.34	-634.0	7,582.4	398.0	-38.2	436.18	0.912	Level 1
18,300.0	10,844.6	18,340.9	10,777.8	223.8	224.8	80.34	-634.0	7,682.4	398.3	-43.6	441.94	0.901	Level 1
18,400.0	10,844.8	18,440.9	10,777.9	226.7	227.7	80.35	-634.0	7,782.4	398.6	-49.1	447.70	0.890	Level 1
18,500.0	10,844.9	18,540.9	10,778.1	229.7	230.6	80.36	-634.0	7,882.4	398.9	-54.5	453.46	0.880	Level 1
18,600.0	10,845.1	18,640.9	10,778.3	232.6	233.5	80.37	-634.0	7,982.4	399.2	-60.0	459.22	0.869	Level 1
18,700.0	10,845.3	18,740.9	10,778.5	235.5	236.5	80.37	-634.0	8,082.4	399.5	-65.5	464.98	0.859	Level 1
18,800.0	10,845.4	18,840.9	10,778.6	238.4	239.4	80.38	-634.0	8,182.4	399.8	-70.9	470.74	0.849	Level 1
18,900.0	10,845.6	18,940.9	10,778.8	241.3	242.3	80.39	-634.0	8,282.4	400.1	-76.4	476.51	0.840	Level 1
19,000.0	10,845.8	19,040.9	10,779.0	244.2	245.2	80.40	-634.0	8,382.4	400.4	-81.8	482.27	0.830	Level 1
19,100.0	10,846.0	19,140.9	10,779.2	247.1	248.1	80.40	-634.0	8,482.4	400.7	-87.3	488.04	0.821	Level 1
19,200.0	10,846.1	19,240.9	10,779.3	250.1	251.0	80.41	-634.0	8,582.4	401.0	-92.8	493.81	0.812	Level 1
19,300.0	10,846.3	19,340.9	10,779.5	253.0	253.9	80.42	-634.0	8,682.4	401.3	-98.2	499.57	0.803	Level 1
19,400.0	10,846.5	19,440.9	10,779.7	255.9	256.8	80.43	-634.0	8,782.4	401.7	-103.7	505.34	0.795	Level 1
19,500.0	10,846.7	19,540.9	10,779.9	258.8	259.8	80.43	-634.0	8,882.4	402.0	-109.2	511.11	0.786	Level 1
19,600.0	10,846.8	19,640.9	10,780.0	261.7	262.7	80.44	-634.0	8,982.4	402.3	-114.6	516.88	0.778	Level 1
19,700.0	10,847.0	19,740.9	10,780.2	264.6	265.6	80.45	-634.0	9,082.4	402.6	-120.1	522.65	0.770	Level 1
19,800.0	10,847.2	19,840.9	10,780.4	267.6	268.5	80.46	-634.0	9,182.4	402.9	-125.6	528.42	0.762	Level 1
19,900.0	10,847.4	19,940.9	10,780.6	270.5	271.4	80.46	-634.0	9,282.4	403.2	-131.0	534.19	0.755	Level 1
20,000.0	10,847.5	20,040.9	10,780.7	273.4	274.3	80.47	-634.0	9,382.4	403.5	-136.5	539.97	0.747	Level 1
20,100.0	10,847.7	20,140.9	10,780.9	276.3	277.3	80.48	-634.0	9,482.4	403.8	-142.0	545.74	0.740	Level 1
20,200.0	10,847.9	20,240.9	10,781.1	279.2	280.2	80.49	-634.0	9,582.4	404.1	-147.4	551.51	0.733	Level 1
20,300.0	10,848.1	20,340.9	10,781.3	282.1	283.1	80.49	-634.0	9,682.4	404.4	-152.9	557.29	0.726	Level 1
20,400.0	10,848.2	20,440.9	10,781.4	285.1	286.0	80.50	-634.0	9,782.4	404.7	-158.4	563.06	0.719	Level 1
20,500.0	10,848.4	20,540.9	10,781.6	288.0	288.9	80.51	-634.0	9,882.4	405.0	-163.8	568.84	0.712	Level 1
20,600.0	10,848.6	20,640.9	10,781.8	290.9	291.8	80.51	-634.0	9,982.4	405.3	-169.3	574.61	0.705	Level 1
20,667.7	10,848.7	20,708.4	10,781.9	292.9	293.8	80.52	-634.0	10,050.0	405.5	-173.0	578.52	0.701	Level 1, ES, SF



Ryan Directional Services

Anticollision Report



Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 9T - Kline Federal 5300 41-18 9T - Design #1													Offset Site Error: 0.0 usft
Survey Program: 0-MWD													Offset Well Error: 2.0 usft
Reference		Offset		Semi Major Axis			Distance						Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	
0.0	0.0	0.0	0.0	2.0	2.0	-0.39	99.3	-0.7	99.3				
100.0	100.0	100.0	100.0	2.0	2.0	-0.39	99.3	-0.7	99.3	95.3	4.00	24.802	
200.0	200.0	200.0	200.0	2.0	2.0	-0.39	99.3	-0.7	99.3	95.2	4.05	24.533	
300.0	300.0	300.0	300.0	2.1	2.1	-0.39	99.3	-0.7	99.3	95.2	4.14	23.984	
400.0	400.0	400.0	400.0	2.1	2.1	-0.39	99.3	-0.7	99.3	95.0	4.28	23.210	
500.0	500.0	500.0	500.0	2.2	2.2	-0.39	99.3	-0.7	99.3	94.8	4.46	22.277	
600.0	600.0	600.0	600.0	2.3	2.3	-0.39	99.3	-0.7	99.3	94.6	4.67	21.248	
700.0	700.0	700.0	700.0	2.5	2.5	-0.39	99.3	-0.7	99.3	94.4	4.92	20.179	
800.0	800.0	800.0	800.0	2.6	2.6	-0.39	99.3	-0.7	99.3	94.1	5.20	19.113	
900.0	900.0	900.0	900.0	2.7	2.7	-0.39	99.3	-0.7	99.3	93.8	5.49	18.076	
1,000.0	1,000.0	1,000.0	1,000.0	2.9	2.9	-0.39	99.3	-0.7	99.3	93.5	5.81	17.089	
1,100.0	1,100.0	1,100.0	1,100.0	3.1	3.1	-0.39	99.3	-0.7	99.3	93.2	6.14	16.161	
1,200.0	1,200.0	1,200.0	1,200.0	3.2	3.2	-0.39	99.3	-0.7	99.3	92.8	6.49	15.295	
1,300.0	1,300.0	1,300.0	1,300.0	3.4	3.4	-0.39	99.3	-0.7	99.3	92.4	6.85	14.492	
1,400.0	1,400.0	1,400.0	1,400.0	3.6	3.6	-0.39	99.3	-0.7	99.3	92.1	7.22	13.750	
1,500.0	1,500.0	1,500.0	1,500.0	3.8	3.8	-0.39	99.3	-0.7	99.3	91.7	7.60	13.065	
1,600.0	1,600.0	1,600.0	1,600.0	4.0	4.0	-0.39	99.3	-0.7	99.3	91.3	7.99	12.434	
1,700.0	1,700.0	1,700.0	1,700.0	4.2	4.2	-0.39	99.3	-0.7	99.3	90.9	8.38	11.852	
1,800.0	1,800.0	1,800.0	1,800.0	4.4	4.4	-0.39	99.3	-0.7	99.3	90.5	8.78	11.315	
1,900.0	1,900.0	1,900.0	1,900.0	4.6	4.6	-0.39	99.3	-0.7	99.3	90.1	9.18	10.819	
2,000.0	2,000.0	2,000.0	2,000.0	4.8	4.8	-0.39	99.3	-0.7	99.3	89.7	9.58	10.360	
2,100.0	2,100.0	2,100.0	2,100.0	5.0	5.0	-0.39	99.3	-0.7	99.3	89.3	9.99	9.935	
2,200.0	2,200.0	2,200.0	2,200.0	5.2	5.2	-0.39	99.3	-0.7	99.3	88.9	10.41	9.540 CC, ES	
2,216.7	2,216.7	2,215.8	2,215.8	5.2	5.2	-90.43	99.4	-0.7	99.4	88.9	10.47	9.487	
2,300.0	2,300.0	2,299.1	2,299.1	5.4	5.4	-90.84	100.1	-0.7	100.1	89.3	10.80	9.269	
2,400.0	2,400.0	2,399.1	2,399.1	5.6	5.6	-91.33	101.0	-0.7	101.0	89.8	11.19	9.022	
2,500.0	2,500.0	2,499.1	2,499.1	5.8	5.8	-91.81	101.8	-0.7	101.9	90.3	11.59	8.788	
2,600.0	2,600.0	2,599.1	2,599.1	6.0	6.0	-92.28	102.7	-0.7	102.8	90.8	12.00	8.569	
2,700.0	2,700.0	2,699.1	2,699.1	6.1	6.3	-92.74	103.6	-0.7	103.7	91.3	12.40	8.362	
2,800.0	2,800.0	2,799.1	2,799.1	6.3	6.5	-93.20	104.4	-0.7	104.6	91.8	12.81	8.167	
2,900.0	2,900.0	2,899.1	2,899.1	6.5	6.7	-93.64	105.3	-0.7	105.5	92.3	13.22	7.982	
3,000.0	3,000.0	2,999.1	2,999.0	6.7	6.9	-94.08	106.2	-0.7	106.5	92.8	13.64	7.808	
3,100.0	3,100.0	3,099.1	3,099.0	6.9	7.1	-94.51	107.1	-0.7	107.4	93.4	14.05	7.644	
3,200.0	3,200.0	3,199.1	3,199.0	7.1	7.3	-94.94	107.9	-0.7	108.3	93.9	14.47	7.488	
3,300.0	3,300.0	3,299.1	3,299.0	7.3	7.5	-95.35	108.8	-0.7	109.3	94.4	14.89	7.340	
3,400.0	3,400.0	3,399.0	3,399.0	7.6	7.8	-95.76	109.7	-0.7	110.2	94.9	15.31	7.200	
3,500.0	3,500.0	3,499.0	3,499.0	7.8	8.0	-96.17	110.6	-0.7	111.2	95.5	15.73	7.068	
3,600.0	3,599.9	3,599.0	3,599.0	8.0	8.2	-96.56	111.4	-0.7	112.2	96.0	16.16	6.942	
3,700.0	3,699.9	3,699.0	3,699.0	8.2	8.4	-96.95	112.3	-0.7	113.1	96.6	16.58	6.822	
3,800.0	3,799.9	3,799.0	3,799.0	8.4	8.6	-97.33	113.2	-0.7	114.1	97.1	17.01	6.708	
3,900.0	3,899.9	3,899.0	3,898.9	8.6	8.9	-97.71	114.0	-0.7	115.1	97.7	17.44	6.599	
4,000.0	3,999.9	3,999.0	3,998.9	8.8	9.1	-98.08	114.9	-0.7	116.1	98.2	17.87	6.496	
4,100.0	4,099.9	4,099.0	4,098.9	9.0	9.3	-98.44	115.8	-0.7	117.1	98.8	18.30	6.397	
4,200.0	4,199.9	4,199.0	4,198.9	9.2	9.5	-98.79	116.7	-0.7	118.1	99.3	18.73	6.303	
4,300.0	4,299.9	4,299.0	4,298.9	9.4	9.7	-99.15	117.5	-0.7	119.1	99.9	19.16	6.213	
4,400.0	4,399.9	4,399.0	4,398.9	9.7	10.0	-99.49	118.4	-0.7	120.1	100.5	19.60	6.127	
4,500.0	4,499.9	4,499.0	4,498.9	9.9	10.2	-99.83	119.3	-0.7	121.1	101.0	20.03	6.044	
4,600.0	4,599.9	4,599.0	4,598.9	10.1	10.4	-100.16	120.2	-0.7	122.1	101.6	20.46	5.965	
4,700.0	4,699.9	4,698.9	4,698.8	10.3	10.6	-100.49	121.0	-0.7	123.1	102.2	20.90	5.890	
4,800.0	4,799.9	4,798.9	4,798.8	10.5	10.8	-100.81	121.9	-0.7	124.1	102.8	21.33	5.817	
4,900.0	4,899.9	4,898.9	4,898.8	10.7	11.1	-101.13	122.8	-0.7	125.1	103.4	21.77	5.748	
5,000.0	4,999.9	4,998.9	4,998.8	10.9	11.3	-101.44	123.6	-0.7	126.2	104.0	22.21	5.681	

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services

Anticollision Report



Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 9T - Kline Federal 5300 41-18 9T - Design #1													Offset Site Error:	0.0 usft
Survey Program: 0-MWD													Offset Well Error:	2.0 usft
Reference		Offset		Semi Major Axis			Distance							Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor		
5,100.0	5,099.9	5,098.9	5,098.8	11.2	11.5	-101.75	124.5	-0.7	127.2	104.5	22.64	5.617		
5,200.0	5,199.9	5,198.9	5,198.8	11.4	11.7	-102.05	125.4	-0.7	128.2	105.1	23.08	5.555		
5,300.0	5,299.9	5,298.9	5,298.8	11.6	11.9	-102.35	126.3	-0.7	129.3	105.7	23.52	5.496		
5,400.0	5,399.9	5,398.9	5,398.8	11.8	12.2	-102.64	127.1	-0.7	130.3	106.3	23.96	5.439		
5,500.0	5,499.9	5,498.9	5,498.8	12.0	12.4	-102.93	128.0	-0.7	131.3	106.9	24.40	5.384		
5,600.0	5,599.9	5,598.9	5,598.7	12.2	12.6	-103.22	128.9	-0.7	132.4	107.6	24.84	5.330		
5,700.0	5,699.9	5,698.9	5,698.7	12.5	12.8	-103.49	129.8	-0.7	133.4	108.2	25.28	5.279		
5,800.0	5,799.9	5,798.9	5,798.7	12.7	13.0	-103.77	130.6	-0.7	134.5	108.8	25.72	5.230		
5,900.0	5,899.9	5,898.9	5,898.7	12.9	13.3	-104.04	131.5	-0.7	135.6	109.4	26.16	5.182		
6,000.0	5,999.9	5,998.8	5,998.7	13.1	13.5	-104.31	132.4	-0.7	136.6	110.0	26.60	5.136		
6,100.0	6,099.9	6,098.8	6,098.7	13.3	13.7	-104.57	133.2	-0.7	137.7	110.6	27.04	5.092		
6,200.0	6,199.8	6,198.8	6,198.7	13.6	13.9	-104.83	134.1	-0.7	138.7	111.3	27.48	5.049		
6,300.0	6,299.8	6,298.8	6,298.7	13.8	14.2	-105.08	135.0	-0.7	139.8	111.9	27.92	5.007		
6,400.0	6,399.8	6,398.8	6,398.7	14.0	14.4	-105.33	135.9	-0.7	140.9	112.5	28.36	4.967		
6,500.0	6,499.8	6,498.8	6,498.6	14.2	14.6	-105.58	136.7	-0.7	142.0	113.2	28.81	4.928		
6,600.0	6,599.8	6,598.8	6,598.6	14.4	14.8	-105.82	137.6	-0.7	143.0	113.8	29.25	4.890		
6,700.0	6,699.8	6,698.8	6,698.6	14.7	15.1	-106.06	138.5	-0.7	144.1	114.4	29.69	4.854		
6,800.0	6,799.8	6,798.8	6,798.6	14.9	15.3	-106.30	139.4	-0.7	145.2	115.1	30.13	4.818		
6,900.0	6,899.8	6,898.8	6,898.6	15.1	15.5	-106.53	140.2	-0.7	146.3	115.7	30.58	4.784		
7,000.0	6,999.8	6,998.8	6,998.6	15.3	15.7	-106.76	141.1	-0.7	147.4	116.3	31.02	4.751		
7,100.0	7,099.8	7,098.8	7,098.6	15.5	15.9	-106.98	142.0	-0.7	148.5	117.0	31.46	4.718		
7,200.0	7,199.8	7,198.8	7,198.6	15.8	16.2	-107.20	142.8	-0.7	149.5	117.6	31.91	4.687		
7,300.0	7,299.8	7,298.7	7,298.6	16.0	16.4	-107.42	143.7	-0.7	150.6	118.3	32.35	4.657		
7,400.0	7,399.8	7,398.7	7,398.5	16.2	16.6	-107.64	144.6	-0.7	151.7	118.9	32.79	4.627		
7,500.0	7,499.8	7,498.7	7,498.5	16.4	16.8	-107.85	145.5	-0.7	152.8	119.6	33.24	4.598		
7,600.0	7,599.8	7,598.7	7,598.5	16.6	17.1	-108.06	146.3	-0.7	153.9	120.2	33.68	4.570		
7,700.0	7,699.8	7,698.7	7,698.5	16.9	17.3	-108.27	147.2	-0.7	155.0	120.9	34.12	4.543		
7,800.0	7,799.8	7,798.7	7,798.5	17.1	17.5	-108.47	148.1	-0.7	156.1	121.6	34.57	4.517		
7,900.0	7,899.8	7,898.7	7,898.5	17.3	17.7	-108.67	149.0	-0.7	157.2	122.2	35.01	4.491		
7,926.7	7,926.4	7,925.4	7,925.1	17.4	17.8	-108.72	149.2	-0.7	157.5	122.4	35.13	4.484		
7,943.3	7,943.1	7,943.3	7,943.1	17.4	17.8	-18.74	149.3	-0.7	157.6	122.4	35.21	4.477		
8,000.0	7,999.8	8,000.0	7,999.8	17.5	17.9	-18.74	149.3	-0.7	157.6	122.2	35.43	4.449		
8,100.0	8,099.8	8,100.0	8,099.8	17.7	18.2	-18.74	149.3	-0.7	157.6	121.8	35.86	4.395		
8,200.0	8,199.8	8,200.0	8,199.8	17.9	18.4	-18.74	149.3	-0.7	157.6	121.3	36.29	4.343		
8,300.0	8,299.8	8,300.0	8,299.8	18.1	18.6	-18.74	149.3	-0.7	157.6	120.9	36.72	4.292		
8,400.0	8,399.8	8,400.0	8,399.8	18.4	18.8	-18.74	149.3	-0.7	157.6	120.5	37.16	4.242		
8,500.0	8,499.8	8,500.0	8,499.8	18.6	19.1	-18.74	149.3	-0.7	157.6	120.0	37.59	4.194		
8,600.0	8,599.8	8,600.0	8,599.8	18.8	19.3	-18.74	149.3	-0.7	157.6	119.6	38.02	4.146		
8,700.0	8,699.8	8,700.0	8,699.8	19.0	19.5	-18.74	149.3	-0.7	157.6	119.2	38.45	4.099		
8,800.0	8,799.8	8,800.0	8,799.8	19.2	19.7	-18.74	149.3	-0.7	157.6	118.7	38.89	4.053		
8,900.0	8,899.8	8,900.0	8,899.8	19.4	19.9	-18.74	149.3	-0.7	157.6	118.3	39.32	4.009		
9,000.0	8,999.8	9,000.0	8,999.8	19.6	20.2	-18.74	149.3	-0.7	157.6	117.9	39.76	3.965		
9,100.0	9,099.8	9,100.0	9,099.8	19.8	20.4	-18.74	149.3	-0.7	157.6	117.4	40.19	3.922		
9,200.0	9,199.8	9,200.0	9,199.8	20.0	20.6	-18.74	149.3	-0.7	157.6	117.0	40.63	3.880		
9,300.0	9,299.8	9,300.0	9,299.8	20.2	20.8	-18.74	149.3	-0.7	157.6	116.6	41.06	3.839		
9,400.0	9,399.8	9,400.0	9,399.8	20.5	21.1	-18.74	149.3	-0.7	157.6	116.1	41.50	3.799		
9,500.0	9,499.8	9,500.0	9,499.8	20.7	21.3	-18.74	149.3	-0.7	157.6	115.7	41.93	3.759		
9,600.0	9,599.8	9,600.0	9,599.8	20.9	21.5	-18.74	149.3	-0.7	157.6	115.3	42.37	3.720		
9,700.0	9,699.8	9,700.0	9,699.8	21.1	21.7	-18.74	149.3	-0.7	157.6	114.8	42.80	3.683		
9,800.0	9,799.8	9,800.0	9,799.8	21.3	22.0	-18.74	149.3	-0.7	157.6	114.4	43.24	3.645		
9,900.0	9,899.8	9,900.0	9,899.8	21.5	22.2	-18.74	149.3	-0.7	157.6	113.9	43.68	3.609		
10,000.0	9,999.8	10,000.0	9,999.8	21.7	22.4	-18.74	149.3	-0.7	157.6	113.5	44.11	3.573		

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Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 9T - Kline Federal 5300 41-18 9T - Design #1													Offset Site Error: 0.0 usft
Survey Program: 0-MWD													Offset Well Error: 2.0 usft
Reference		Offset		Semi Major Axis			Distance						Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	
10,100.0	10,099.8	10,100.0	10,099.8	21.9	22.6	-18.74	149.3	-0.7	157.6	113.1	44.55	3.538	
10,200.0	10,199.8	10,200.0	10,199.8	22.2	22.9	-18.74	149.3	-0.7	157.6	112.6	44.99	3.504	
10,300.0	10,299.8	10,300.0	10,299.8	22.4	23.1	-18.74	149.3	-0.7	157.6	112.2	45.43	3.470	
10,354.8	10,354.6	10,354.8	10,354.6	22.5	23.2	-18.74	149.3	-0.7	157.6	112.0	45.67	3.452	
10,375.0	10,374.8	10,373.6	10,373.4	22.5	23.2	-128.69	149.5	-0.4	158.0	112.2	45.74	3.454	
10,400.0	10,399.7	10,396.9	10,396.6	22.6	23.3	-128.49	150.2	0.9	159.4	113.6	45.80	3.480	
10,425.0	10,424.5	10,420.0	10,419.6	22.7	23.3	-128.13	151.6	3.1	161.9	116.1	45.83	3.533	
10,450.0	10,449.2	10,443.0	10,442.3	22.7	23.4	-127.62	153.5	6.3	165.5	119.6	45.83	3.610	
10,475.0	10,473.5	10,465.7	10,464.5	22.8	23.4	-126.98	156.0	10.3	170.1	124.3	45.81	3.713	
10,500.0	10,497.6	10,488.2	10,486.3	22.8	23.5	-126.21	158.9	15.1	175.8	130.0	45.78	3.839	
10,525.0	10,521.2	10,510.3	10,507.4	22.9	23.5	-125.34	162.3	20.8	182.4	136.7	45.73	3.989	
10,550.0	10,544.4	10,532.1	10,527.8	23.0	23.6	-124.35	166.2	27.1	190.1	144.4	45.69	4.161	
10,575.0	10,567.1	10,553.5	10,547.6	23.0	23.6	-123.28	170.5	34.1	198.7	153.1	45.66	4.353	
10,600.0	10,589.1	10,575.0	10,567.1	23.1	23.7	-122.11	175.2	41.9	208.3	162.6	45.64	4.564	
10,625.0	10,610.6	10,594.9	10,584.7	23.2	23.8	-120.87	180.0	49.8	218.7	173.1	45.64	4.792	
10,650.0	10,631.3	10,614.9	10,602.0	23.3	23.8	-119.54	185.3	58.3	230.0	184.3	45.67	5.035	
10,675.0	10,651.3	10,634.4	10,618.5	23.4	23.9	-118.14	190.7	67.2	242.0	196.3	45.74	5.291	
10,700.0	10,670.5	10,653.4	10,634.1	23.5	23.9	-116.66	196.3	76.5	254.8	209.0	45.85	5.557	
10,725.0	10,688.8	10,672.0	10,648.9	23.6	24.0	-115.11	202.2	86.0	268.3	222.3	46.01	5.832	
10,750.0	10,706.2	10,690.1	10,663.0	23.7	24.0	-113.47	208.1	95.7	282.5	236.3	46.22	6.113	
10,775.0	10,722.6	10,707.7	10,676.2	23.9	24.1	-111.76	214.1	105.6	297.3	250.8	46.47	6.397	
10,800.0	10,738.0	10,725.0	10,688.8	24.0	24.2	-109.98	220.3	115.8	312.6	265.9	46.78	6.684	
10,825.0	10,752.4	10,741.5	10,700.4	24.2	24.2	-108.11	226.5	125.8	328.5	281.4	47.13	6.971	
10,850.0	10,765.7	10,757.7	10,711.4	24.4	24.3	-106.17	232.7	136.0	344.9	297.3	47.52	7.257	
10,875.0	10,777.8	10,775.0	10,722.6	24.6	24.4	-104.20	239.5	147.2	361.6	313.7	47.94	7.544	
10,900.0	10,788.8	10,789.0	10,731.4	24.8	24.5	-102.05	245.2	156.6	378.8	330.4	48.38	7.829	
10,925.0	10,798.6	10,804.1	10,740.4	25.0	24.5	-99.89	251.5	166.8	396.3	347.4	48.84	8.114	
10,950.0	10,807.2	10,818.8	10,748.9	25.3	24.6	-97.67	257.8	177.1	414.1	364.8	49.30	8.400	
10,975.0	10,814.5	10,833.2	10,756.9	25.6	24.7	-95.39	264.0	187.4	432.2	382.4	49.74	8.688	
11,000.0	10,820.6	10,847.3	10,764.3	25.8	24.8	-93.06	270.2	197.6	450.5	400.3	50.17	8.980	
11,025.0	10,825.4	10,861.1	10,771.2	26.1	24.9	-90.70	276.5	207.8	469.0	418.4	50.55	9.276	
11,050.0	10,828.9	10,875.0	10,777.8	26.5	25.0	-88.34	282.8	218.2	487.6	436.7	50.90	9.579	
11,075.0	10,831.1	10,888.0	10,783.7	26.8	25.1	-85.92	288.9	228.1	506.4	455.2	51.20	9.890	
11,100.0	10,832.0	10,900.0	10,788.8	27.1	25.1	-83.43	294.5	237.4	525.3	473.8	51.44	10.211	
11,104.0	10,832.0	10,903.2	10,790.1	27.2	25.2	-83.14	296.0	239.8	528.3	476.8	51.48	10.260	
11,200.0	10,832.2	10,957.3	10,809.4	28.6	25.6	-86.90	322.3	283.0	600.0	546.6	53.38	11.240	
11,300.0	10,832.4	11,022.3	10,824.9	30.3	26.2	-89.21	355.2	336.9	671.8	616.3	55.48	12.109	
11,400.0	10,832.6	11,094.8	10,831.9	32.2	27.0	-90.02	392.8	398.4	739.7	681.8	57.83	12.791	
11,500.0	10,832.7	11,198.8	10,832.2	34.1	28.2	-90.03	445.6	487.9	801.8	741.1	60.68	13.214	
11,600.0	10,832.9	11,317.2	10,832.4	36.2	30.0	-90.02	501.7	592.2	855.9	791.8	64.10	13.352	
11,680.4	10,833.1	11,420.0	10,832.6	38.0	31.7	-90.01	547.0	684.5	892.8	825.5	67.25	13.276	
11,700.0	10,833.1	11,445.9	10,832.7	38.4	32.1	-90.01	557.8	708.0	900.9	832.9	68.07	13.235	
11,800.0	10,833.3	11,581.5	10,832.9	40.7	34.7	-90.02	611.2	832.7	940.0	867.5	72.51	12.965	
11,900.0	10,833.4	11,722.5	10,833.1	43.0	37.5	-90.02	660.4	964.7	974.7	897.3	77.41	12.591	
12,000.0	10,833.6	11,868.3	10,833.4	45.4	40.6	-90.02	704.4	1,103.7	1,004.6	921.8	82.73	12.142	
12,100.0	10,833.8	12,018.4	10,833.7	47.9	44.0	-90.02	742.3	1,249.0	1,029.5	941.0	88.44	11.640	
12,200.0	10,834.0	12,172.3	10,834.0	50.4	47.6	-90.02	773.1	1,399.7	1,049.2	954.7	94.51	11.102	
12,300.0	10,834.1	12,329.1	10,834.2	53.0	51.4	-90.02	796.1	1,554.8	1,063.6	962.8	100.87	10.545	
12,400.0	10,834.3	12,487.9	10,834.5	55.5	55.3	-90.02	810.8	1,712.9	1,072.6	965.1	107.46	9.981	
12,500.0	10,834.5	12,647.9	10,834.8	58.2	59.3	-90.02	816.7	1,872.7	1,075.9	961.7	114.24	9.418	
12,600.0	10,834.7	12,757.5	10,835.0	60.8	62.0	-90.02	816.8	1,982.4	1,075.7	955.9	119.79	8.980	
12,700.0	10,834.8	12,857.5	10,835.2	63.5	64.5	-90.02	816.8	2,082.4	1,075.4	950.3	125.13	8.594	

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12,800.0	10,835.0	12,957.5	10,835.4	66.2	67.1	-90.02	816.8	2,182.4	1,075.1	944.6	130.52	8.237		
12,900.0	10,835.2	13,057.5	10,835.5	68.9	69.7	-90.02	816.8	2,282.4	1,074.8	938.8	135.94	7.906		
13,000.0	10,835.4	13,157.5	10,835.7	71.7	72.3	-90.02	816.8	2,382.4	1,074.5	933.1	141.40	7.599		
13,100.0	10,835.5	13,257.5	10,835.9	74.4	75.0	-90.02	816.8	2,482.4	1,074.2	927.3	146.89	7.313		
13,200.0	10,835.7	13,357.5	10,836.0	77.2	77.6	-90.02	816.8	2,582.3	1,073.9	921.4	152.41	7.046		
13,300.0	10,835.9	13,457.5	10,836.2	79.9	80.3	-90.02	816.8	2,682.3	1,073.5	915.6	157.95	6.797		
13,400.0	10,836.1	13,557.5	10,836.4	82.7	83.0	-90.02	816.8	2,782.3	1,073.2	909.7	163.52	6.563		
13,500.0	10,836.2	13,657.5	10,836.6	85.5	85.7	-90.02	816.8	2,882.3	1,072.9	903.8	169.11	6.345		
13,600.0	10,836.4	13,757.5	10,836.7	88.3	88.4	-90.02	816.8	2,982.3	1,072.6	897.9	174.71	6.139		
13,700.0	10,836.6	13,857.5	10,836.9	91.1	91.2	-90.02	816.8	3,082.3	1,072.3	892.0	180.33	5.946		
13,800.0	10,836.7	13,957.5	10,837.1	94.0	93.9	-90.02	816.8	3,182.3	1,072.0	886.0	185.97	5.765		
13,900.0	10,836.9	14,057.5	10,837.3	96.8	96.7	-90.02	816.8	3,282.3	1,071.7	880.1	191.61	5.593		
14,000.0	10,837.1	14,157.5	10,837.4	99.6	99.5	-90.02	816.8	3,382.3	1,071.4	874.1	197.28	5.431		
14,100.0	10,837.3	14,257.5	10,837.6	102.5	102.2	-90.02	816.8	3,482.3	1,071.1	868.1	202.95	5.278		
14,200.0	10,837.4	14,357.5	10,837.8	105.3	105.0	-90.02	816.8	3,582.3	1,070.8	862.1	208.63	5.132		
14,300.0	10,837.6	14,457.5	10,838.0	108.2	107.8	-90.02	816.8	3,682.3	1,070.5	856.1	214.32	4.995		
14,400.0	10,837.8	14,557.5	10,838.1	111.0	110.6	-90.02	816.8	3,782.3	1,070.2	850.1	220.03	4.864		
14,500.0	10,838.0	14,657.5	10,838.3	113.9	113.5	-90.02	816.8	3,882.3	1,069.8	844.1	225.73	4.739		
14,600.0	10,838.1	14,757.5	10,838.5	116.7	116.3	-90.02	816.8	3,982.3	1,069.5	838.1	231.45	4.621		
14,700.0	10,838.3	14,857.5	10,838.7	119.6	119.1	-90.02	816.8	4,082.3	1,069.2	832.0	237.18	4.508		
14,800.0	10,838.5	14,957.5	10,838.8	122.5	121.9	-90.02	816.8	4,182.3	1,068.9	826.0	242.91	4.401		
14,900.0	10,838.7	15,057.5	10,839.0	125.3	124.7	-90.02	816.8	4,282.3	1,068.6	820.0	248.64	4.298		
15,000.0	10,838.8	15,157.5	10,839.2	128.2	127.6	-90.02	816.8	4,382.3	1,068.3	813.9	254.39	4.199		
15,100.0	10,839.0	15,257.5	10,839.4	131.1	130.4	-90.02	816.8	4,482.3	1,068.0	807.9	260.14	4.106		
15,200.0	10,839.2	15,357.5	10,839.5	133.9	133.3	-90.02	816.8	4,582.3	1,067.7	801.8	265.89	4.016		
15,300.0	10,839.4	15,457.5	10,839.7	136.8	136.1	-90.02	816.8	4,682.3	1,067.4	795.7	271.65	3.929		
15,400.0	10,839.5	15,557.5	10,839.9	139.7	139.0	-90.02	816.8	4,782.3	1,067.1	789.7	277.41	3.847		
15,500.0	10,839.7	15,657.5	10,840.1	142.6	141.8	-90.02	816.8	4,882.3	1,066.8	783.6	283.18	3.767		
15,600.0	10,839.9	15,757.5	10,840.2	145.5	144.7	-90.02	816.8	4,982.3	1,066.4	777.5	288.95	3.691		
15,700.0	10,840.1	15,857.5	10,840.4	148.4	147.5	-90.02	816.8	5,082.3	1,066.1	771.4	294.72	3.617		
15,800.0	10,840.2	15,957.5	10,840.6	151.3	150.4	-90.02	816.8	5,182.3	1,065.8	765.3	300.50	3.547		
15,900.0	10,840.4	16,057.5	10,840.8	154.1	153.3	-90.02	816.8	5,282.3	1,065.5	759.2	306.28	3.479		
16,000.0	10,840.6	16,157.5	10,840.9	157.0	156.1	-90.02	816.8	5,382.3	1,065.2	753.2	312.06	3.413		
16,100.0	10,840.8	16,257.5	10,841.1	159.9	159.0	-90.02	816.8	5,482.3	1,064.9	747.1	317.85	3.350		
16,200.0	10,840.9	16,357.5	10,841.3	162.8	161.9	-90.02	816.8	5,582.3	1,064.6	741.0	323.63	3.290		
16,300.0	10,841.1	16,457.5	10,841.5	165.7	164.8	-90.02	816.8	5,682.3	1,064.3	734.9	329.43	3.231		
16,400.0	10,841.3	16,557.5	10,841.6	168.6	167.6	-90.02	816.8	5,782.3	1,064.0	728.8	335.22	3.174		
16,500.0	10,841.4	16,657.5	10,841.8	171.5	170.5	-90.02	816.8	5,882.3	1,063.7	722.7	341.02	3.119		
16,600.0	10,841.6	16,757.5	10,842.0	174.4	173.4	-90.02	816.8	5,982.3	1,063.4	716.5	346.81	3.066		
16,700.0	10,841.8	16,857.5	10,842.2	177.3	176.3	-90.02	816.8	6,082.3	1,063.1	710.4	352.62	3.015		
16,800.0	10,842.0	16,957.5	10,842.3	180.2	179.2	-90.02	816.8	6,182.3	1,062.7	704.3	358.42	2.965		
16,900.0	10,842.1	17,057.5	10,842.5	183.1	182.1	-90.02	816.8	6,282.3	1,062.4	698.2	364.22	2.917		
17,000.0	10,842.3	17,157.5	10,842.7	186.0	184.9	-90.02	816.8	6,382.3	1,062.1	692.1	370.03	2.870		
17,100.0	10,842.5	17,257.5	10,842.9	188.9	187.8	-90.02	816.8	6,482.3	1,061.8	686.0	375.84	2.825		
17,200.0	10,842.7	17,357.5	10,843.0	191.8	190.7	-90.02	816.8	6,582.3	1,061.5	679.9	381.65	2.781		
17,300.0	10,842.8	17,457.5	10,843.2	194.7	193.6	-90.02	816.8	6,682.3	1,061.2	673.7	387.46	2.739		
17,400.0	10,843.0	17,557.5	10,843.4	197.6	196.5	-90.02	816.8	6,782.3	1,060.9	667.6	393.27	2.698		
17,500.0	10,843.2	17,657.4	10,843.6	200.6	199.4	-90.02	816.8	6,882.3	1,060.6	661.5	399.08	2.658		
17,600.0	10,843.4	17,757.4	10,843.7	203.5	202.3	-90.02	816.8	6,982.3	1,060.3	655.4	404.90	2.619		
17,700.0	10,843.5	17,857.4	10,843.9	206.4	205.2	-90.02	816.8	7,082.3	1,060.0	649.3	410.72	2.581		
17,800.0	10,843.7	17,957.4	10,844.1	209.3	208.1	-90.02	816.8	7,182.3	1,059.7	643.1	416.54	2.544		
17,900.0	10,843.9	18,057.4	10,844.3	212.2	211.0	-90.02	816.8	7,282.3	1,059.4	637.0	422.35	2.508		

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



Ryan Directional Services

Anticollision Report



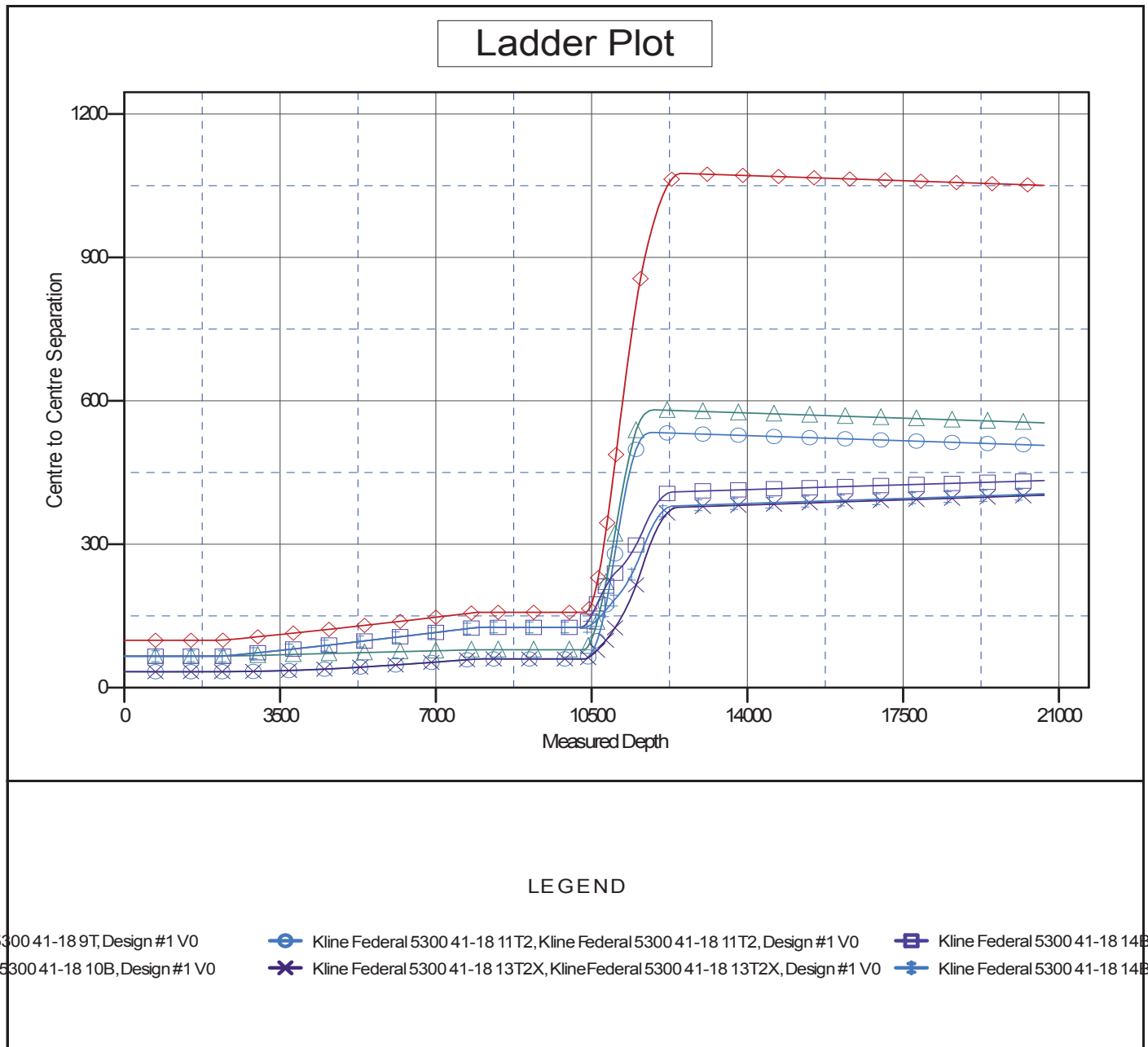
Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Offset Design 153N-100W-17/18 - Kline Federal 5300 41-18 9T - Kline Federal 5300 41-18 9T - Design #1													Offset Site Error: 0.0 usft
Survey Program: 0-MWD													Offset Well Error: 2.0 usft
Reference		Offset		Semi Major Axis			Distance						Warning
Measured Depth (usft)	Vertical Depth (usft)	Measured Depth (usft)	Vertical Depth (usft)	Reference (usft)	Offset (usft)	Highside Toolface (°)	Offset Wellbore Centre +N/-S (usft)	+E/-W (usft)	Between Centres (usft)	Between Ellipses (usft)	Minimum Separation (usft)	Separation Factor	
18,000.0	10,844.1	18,157.4	10,844.4	215.1	213.9	-90.02	816.8	7,382.3	1,059.0	630.9	428.17	2.473	
18,100.0	10,844.2	18,257.4	10,844.6	218.0	216.8	-90.02	816.8	7,482.3	1,058.7	624.7	434.00	2.439	
18,200.0	10,844.4	18,357.4	10,844.8	220.9	219.7	-90.02	816.8	7,582.3	1,058.4	618.6	439.82	2.406	
18,300.0	10,844.6	18,457.4	10,845.0	223.8	222.6	-90.02	816.8	7,682.3	1,058.1	612.5	445.64	2.374	
18,400.0	10,844.8	18,557.4	10,845.1	226.7	225.5	-90.02	816.8	7,782.3	1,057.8	606.3	451.47	2.343	
18,500.0	10,844.9	18,657.4	10,845.3	229.7	228.4	-90.02	816.8	7,882.3	1,057.5	600.2	457.29	2.313	
18,600.0	10,845.1	18,757.4	10,845.5	232.6	231.3	-90.02	816.8	7,982.3	1,057.2	594.1	463.12	2.283	
18,700.0	10,845.3	18,857.4	10,845.6	235.5	234.2	-90.02	816.8	8,082.3	1,056.9	587.9	468.95	2.254	
18,800.0	10,845.4	18,957.4	10,845.8	238.4	237.1	-90.02	816.8	8,182.3	1,056.6	581.8	474.78	2.225	
18,900.0	10,845.6	19,057.4	10,846.0	241.3	240.0	-90.02	816.8	8,282.3	1,056.3	575.7	480.60	2.198	
19,000.0	10,845.8	19,157.4	10,846.2	244.2	242.9	-90.02	816.8	8,382.3	1,056.0	569.5	486.43	2.171	
19,100.0	10,846.0	19,257.4	10,846.3	247.1	245.8	-90.02	816.8	8,482.3	1,055.6	563.4	492.26	2.144	
19,200.0	10,846.1	19,357.4	10,846.5	250.1	248.7	-90.02	816.8	8,582.3	1,055.3	557.2	498.10	2.119	
19,300.0	10,846.3	19,457.4	10,846.7	253.0	251.6	-90.02	816.8	8,682.3	1,055.0	551.1	503.93	2.094	
19,400.0	10,846.5	19,557.4	10,846.9	255.9	254.5	-90.02	816.8	8,782.3	1,054.7	545.0	509.76	2.069	
19,500.0	10,846.7	19,657.4	10,847.0	258.8	257.4	-90.02	816.8	8,882.3	1,054.4	538.8	515.59	2.045	
19,600.0	10,846.8	19,757.4	10,847.2	261.7	260.4	-90.02	816.8	8,982.3	1,054.1	532.7	521.43	2.022	
19,700.0	10,847.0	19,857.4	10,847.4	264.6	263.3	-90.02	816.8	9,082.3	1,053.8	526.5	527.26	1.999	
19,800.0	10,847.2	19,957.4	10,847.6	267.6	266.2	-90.02	816.8	9,182.3	1,053.5	520.4	533.10	1.976	
19,900.0	10,847.4	20,057.4	10,847.7	270.5	269.1	-90.02	816.8	9,282.3	1,053.2	514.2	538.93	1.954	
20,000.0	10,847.5	20,157.4	10,847.9	273.4	272.0	-90.02	816.8	9,382.3	1,052.9	508.1	544.77	1.933	
20,100.0	10,847.7	20,257.4	10,848.1	276.3	274.9	-90.02	816.8	9,482.3	1,052.6	502.0	550.61	1.912	
20,200.0	10,847.9	20,357.4	10,848.3	279.2	277.8	-90.02	816.8	9,582.3	1,052.3	495.8	556.45	1.891	
20,300.0	10,848.1	20,457.4	10,848.4	282.1	280.7	-90.02	816.8	9,682.3	1,051.9	489.7	562.28	1.871	
20,400.0	10,848.2	20,557.4	10,848.6	285.1	283.6	-90.02	816.8	9,782.3	1,051.6	483.5	568.12	1.851	
20,500.0	10,848.4	20,657.4	10,848.8	288.0	286.6	-90.02	816.8	9,882.3	1,051.3	477.4	573.96	1.832	
20,600.0	10,848.6	20,757.4	10,849.0	290.9	289.5	-90.02	816.8	9,982.3	1,051.0	471.2	579.80	1.813	
20,667.7	10,848.7	20,824.5	10,849.1	292.9	291.4	-90.02	816.8	10,049.3	1,050.8	467.1	583.73	1.800 SF	

Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Reference Depths are relative to WELL @ 2082.0usft (Original Well Ele)
Offset Depths are relative to Offset Datum
Central Meridian is 100° 30' 0.000 W

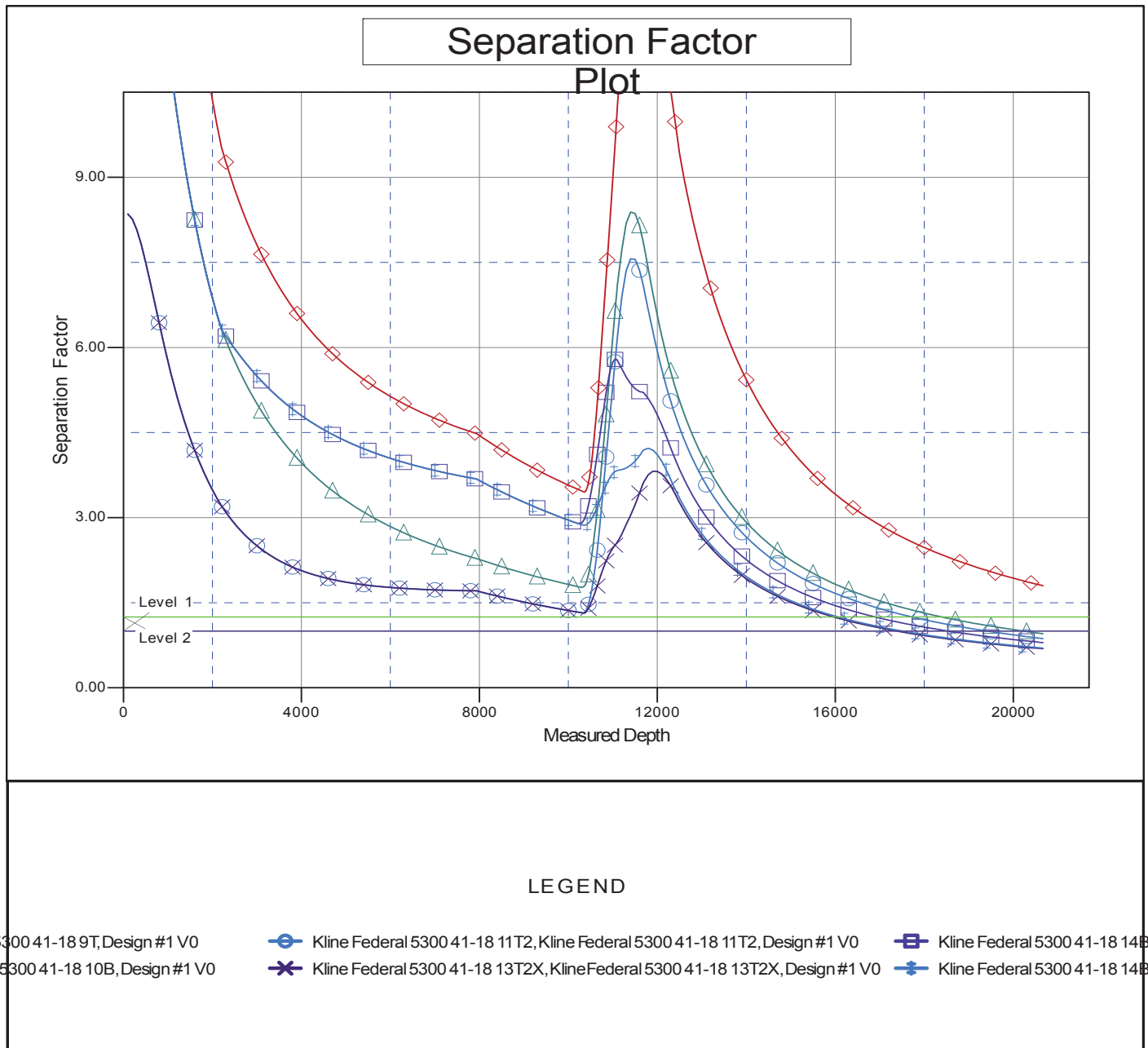
Coordinates are relative to: Kline Federal 5300 41-18 12TX
Coordinate System is US State Plane 1983, North Dakota Northern Zone
Grid Convergence at Surface is: -2.31°



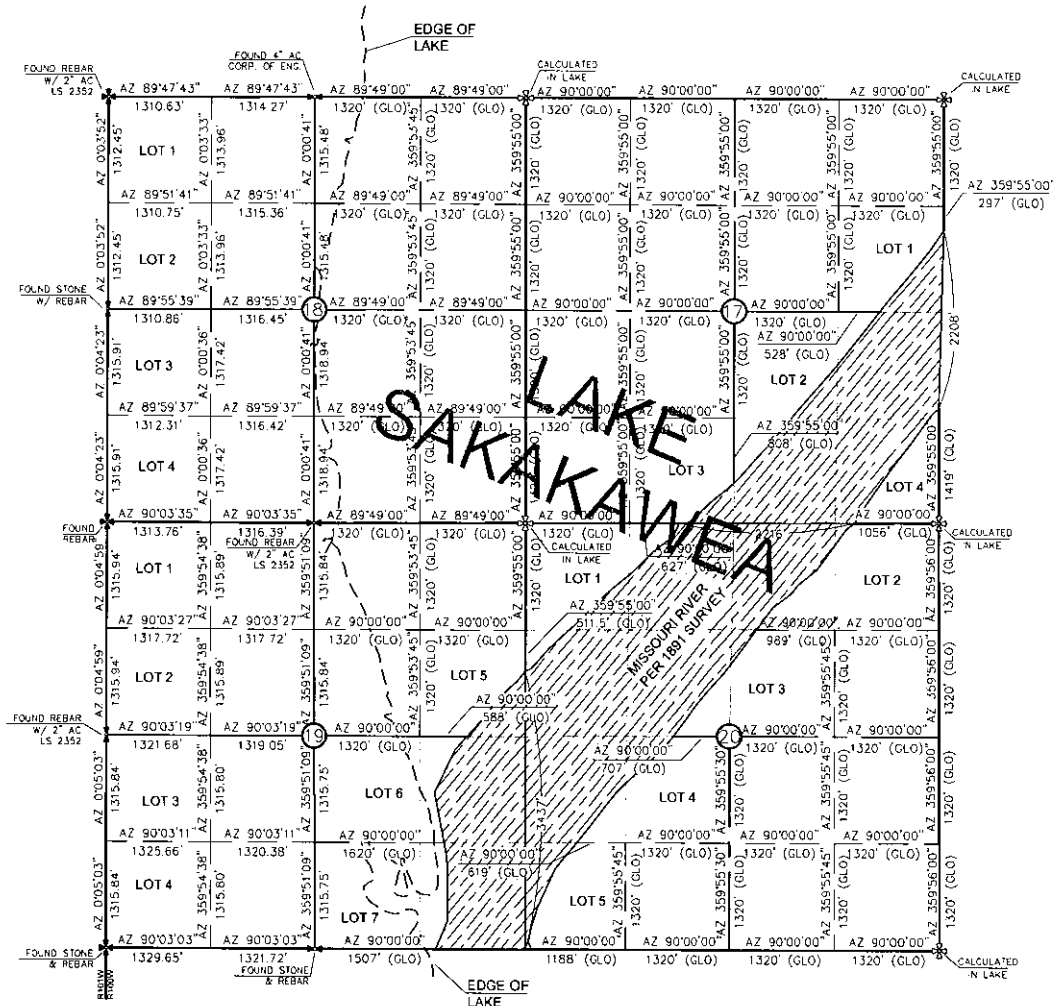
Company:	Oasis Petroleum	Local Co-ordinate Reference:	Well Kline Federal 5300 41-18 12TX
Project:	Indian Hills	TVD Reference:	WELL @ 2082.0usft (Original Well Elev)
Reference Site:	153N-100W-17/18	MD Reference:	WELL @ 2082.0usft (Original Well Elev)
Site Error:	0.0 usft	North Reference:	True
Reference Well:	Kline Federal 5300 41-18 12TX	Survey Calculation Method:	Minimum Curvature
Well Error:	2.0 usft	Output errors are at	2.00 sigma
Reference Wellbore	Kline Federal 5300 41-18 12TX	Database:	EDM 5000.1 Single User Db
Reference Design:	Design #2	Offset TVD Reference:	Offset Datum

Reference Depths are relative to WELL @ 2082.0usft (Original Well Ele
Offset Depths are relative to Offset Datum
Central Meridian is 100° 30' 0.000 W

Coordinates are relative to: Kline Federal 5300 41-18 12TX
Coordinate System is US State Plane 1983, North Dakota Northern Zone
Grid Convergence at Surface is: -2.31°

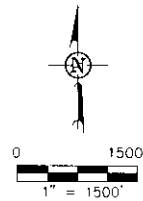
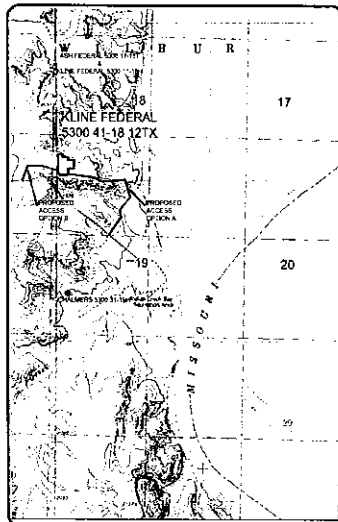


SECTION BREAKDOWN
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "KLINE FEDERAL 5300 41-18 12TX"
 434 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE
 SECTIONS 17, 18, 19, & 20, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



- MONUMENT - RECOVERED
 - MONUMENT - NOT RECOVERED

VICINITY MAP



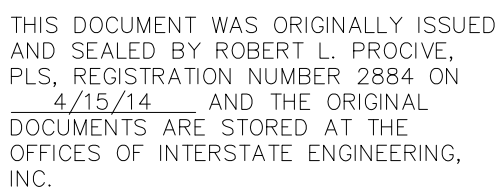
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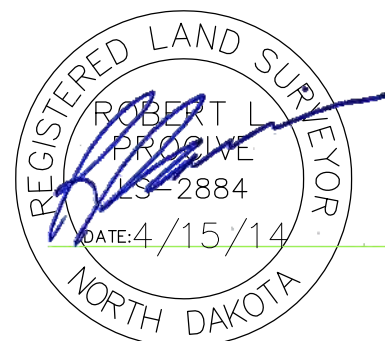
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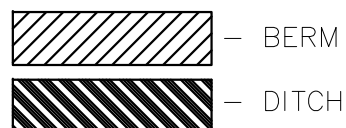
OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
"KLINE FEDERAL 5300 41-18 9T"
533 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



NOTE: Pad dimensions shown are to usable area, the v-ditch and berm areas shall be built to the outside of the pad dimensions.

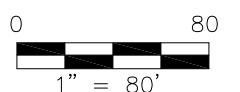
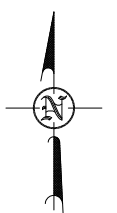


———— Proposed Contours
- - - - - Original Contours



NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

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3/8

SHEET NO.



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Interstate Engineering, Inc.
P.O. Box 648
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Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
PAD LAYOUT
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

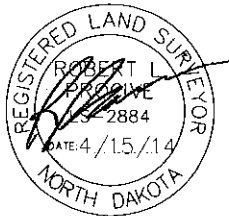
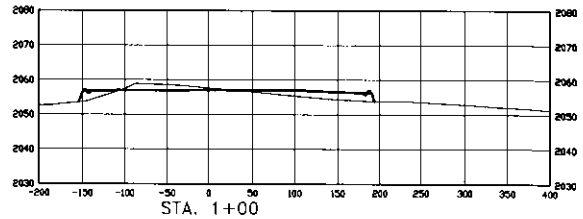
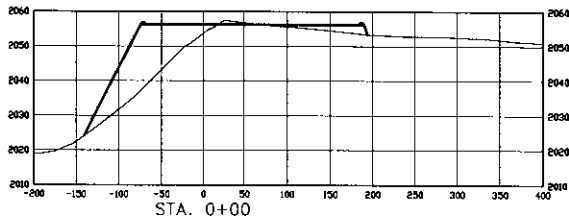
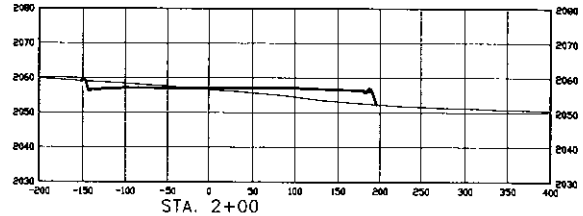
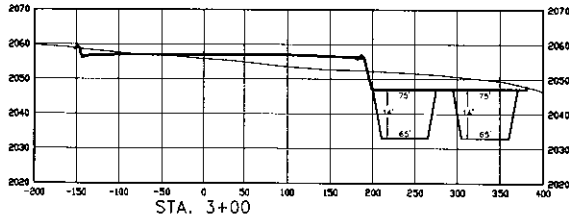
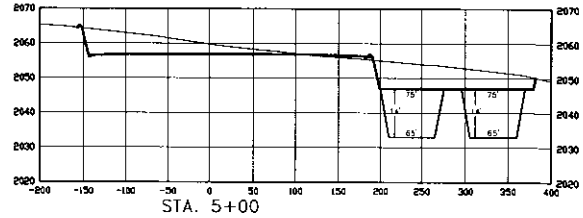
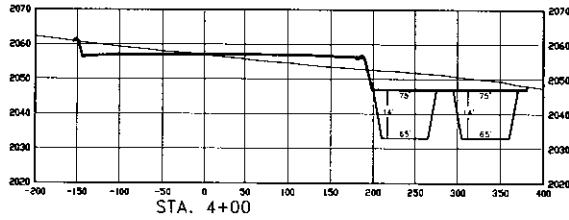
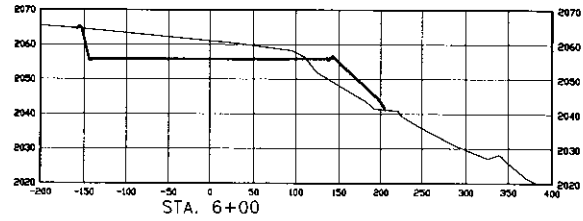
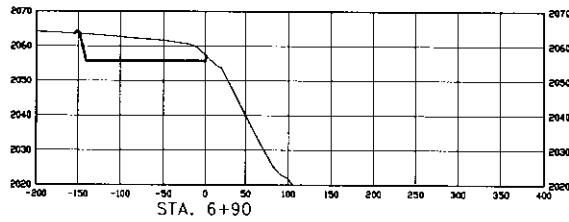
Drawn By: <u>B.H.H.</u>	Project No.: <u>S14-09-081.05</u>
Checked By: <u>D.D.K.</u>	Date: <u>APRIL 2014</u>

Revision No.	Date	By	Description

© 2014/04-09-08/05 Doss Petroleum -- Kline Federal 5300 41-18 91

G:\2014\514-09-\081.05 Oasis Petroleum - Kline Federal 5300 41-18 91

CROSS SECTIONS
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "KLINE FEDERAL 5300 41-18 12TX"
 434 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE
 SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



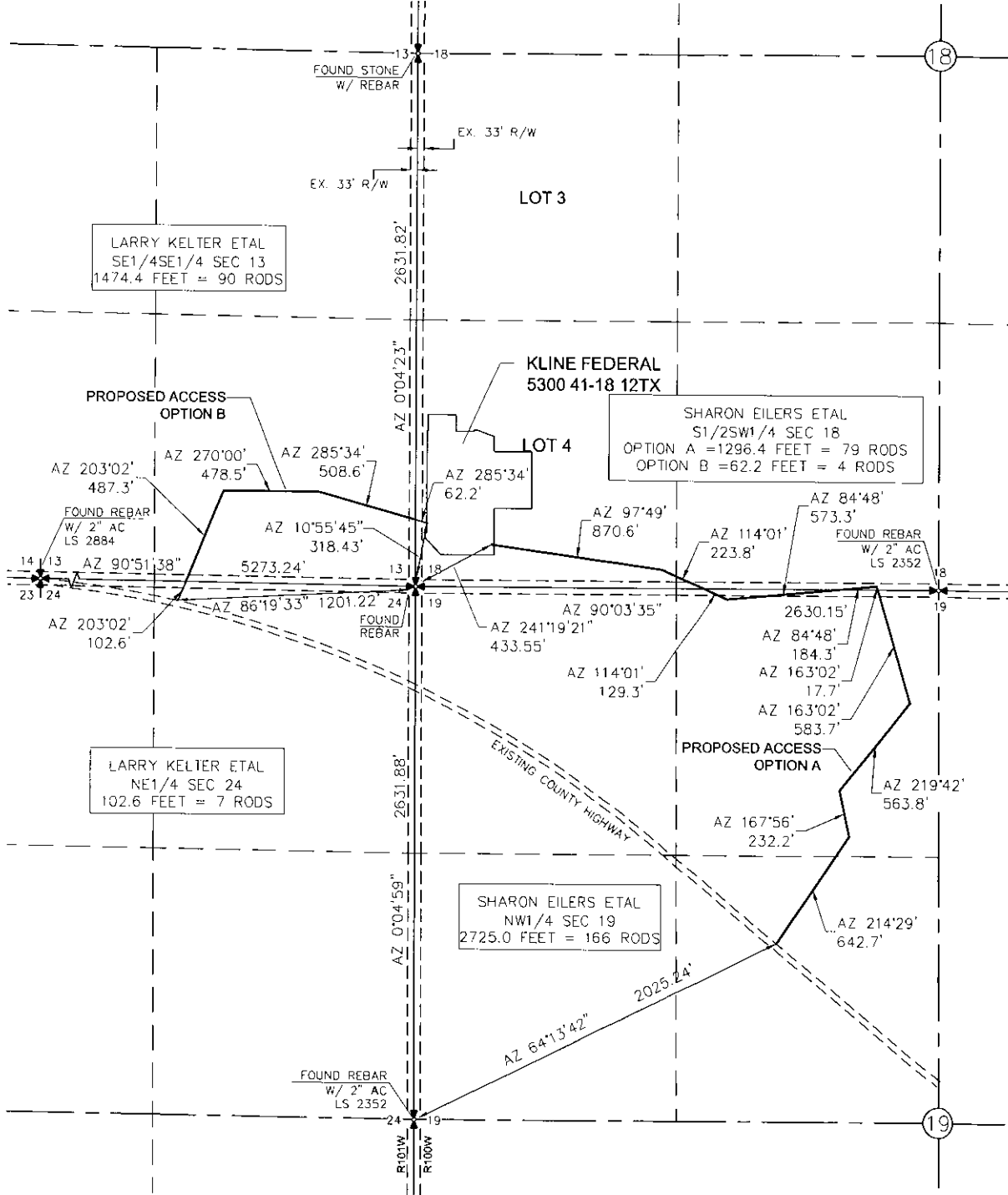
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SCALE
 HORIZ 1"=140'
 VERT 1"=35'

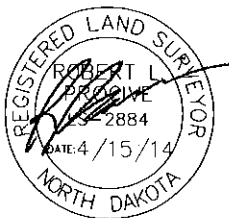
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7/8 SHEET NO.	Professional Engineering and Surveying, Inc.	Interstate Engineering, Inc. P.O. Box 649 425 East Main Street Sidney, Montana 59270 Ph (406) 433-5617 Fax (406) 433-5619 www.interstateeng.com Other offices in Minnesota, North Dakota and South Dakota	OASIS PETROLEUM NORTH AMERICA, LLC CROSS SECTIONS SECTION 18, T153N, R100W MCKENZIE COUNTY, NORTH DAKOTA Drawn By: <u>B.L.P.</u> Project No: <u>314-09-081.02</u> Checked By: <u>B.L.P.</u> Date: <u>APRIL 2014</u>	<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th>Revised No.</th> <th>Date</th> <th>By</th> <th>Description</th> </tr> </thead> <tbody> <tr><td> </td><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td><td> </td></tr> </tbody> </table>	Revised No.	Date	By	Description												
Revised No.	Date	By	Description																	

ACCESS APPROACH
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "KLINE FEDERAL 5300 41-18 12TX"
 434 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE
 SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

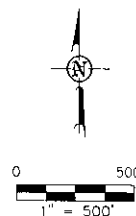


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NOTE: All utilities shown are preliminary only, a complete
 utilities location is recommended before construction.

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 425 East Main Street
 Sidney, Montana 59220
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 Fax: (406) 433-5618
 www.interstateeng.com

OASIS PETROLEUM NORTH AMERICA, LLC
 ACCESS APPROACH
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.L.P. Project No: 514-09-0611-02
 Checked By: B.L.P. Date: APRIL 2014

Revision No.	Date	By	Description

OASIS PETROLEUM NORTH AMERICA, LLC
KLINE FEDERAL 5300 41-18 12TX
434' FSL/237' FWL
QUAD LOCATION MAP
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

W 18
ASH FEDERAL 5300 11-18T
&
KLINE FEDERAL 5300 11-18H
18
KLINE FEDERAL
5300 41-18 12TX

PROPOSED
ACCESS
OPTION B

PROPOSED
ACCESS
OPTION A

Indian Creek Bay
Recreation Area

ISSOUR

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OASIS PETROLEUM NORTH AMERICA, LLO
QUAD LOCATION MAP
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: <u>B.H.H.</u>	Project No.: <u>S14-09-081.02</u>
Checked By: <u>R.L.P.</u>	Date: <u>APRIL 2014</u>

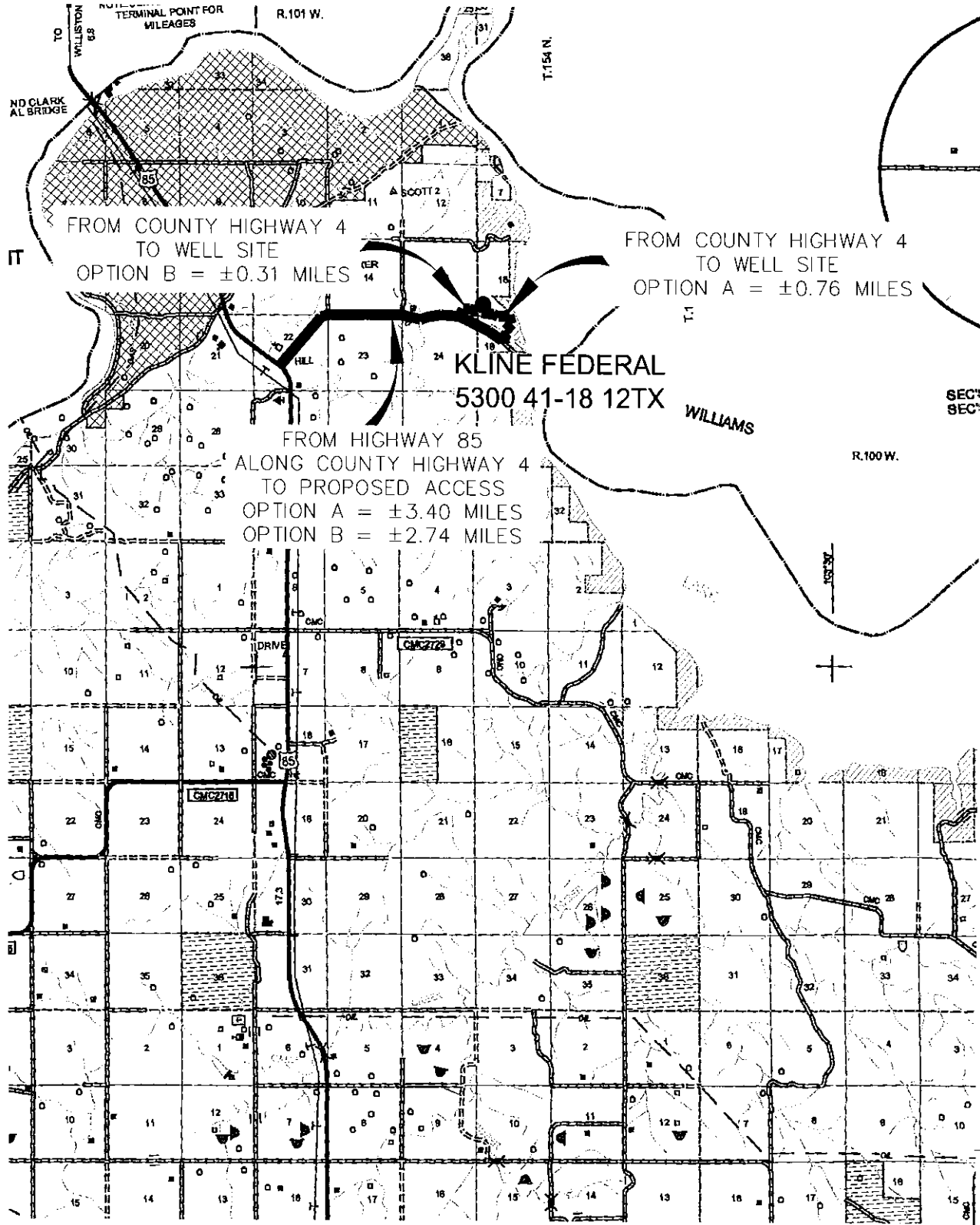
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COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

"KLINE FEDERAL 5300 41-18 12TX"

434 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE
SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



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SCALE: 1" = 2 MILE

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Fax (406) 433-5618
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Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC
COUNTY ROAD MAP
SECTION 18, T153N, R100W
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: B.H.H. Project No.: S14-09-081.02
Checked By: R.L.P. Date: APRIL 2014

Revision No.	Date	By	Description

WELL LOCATION SITE QUANTITIES
 OASIS PETROLEUM NORTH AMERICA, LLC
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002
 "KLINE FEDERAL 5300 41-18 12TX"
 434 FEET FROM SOUTH LINE AND 237 FEET FROM WEST LINE
 SECTION 18, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

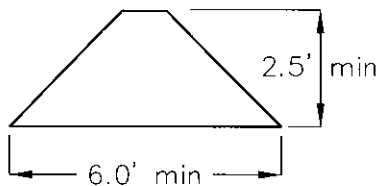
WELL SITE ELEVATION 2056.1
 WELL PAD ELEVATION 2057.0

EXCAVATION	20,682
PLUS PIT	18,900
	<u>39,582</u>
EMBANKMENT	11,538
PLUS SHRINKAGE (30%)	3,461
	<u>14,999</u>
STOCKPILE PIT	18,900
STOCKPILE TOP SOIL (6")	5,261
BERMS	1,889 LF = 612 CY
DITCHES	1,649 LF = 252 CY
DETENTION AREA	2,341 CY
STOCKPILE MATERIAL	2,403
DISTURBED AREA FROM PAD	6.52 ACRES

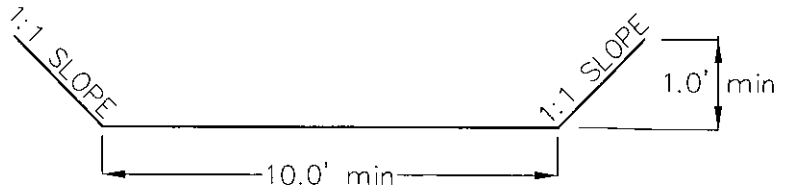
NOTE: ALL QUANTITIES ARE IN CUBIC YARDS (UNLESS NOTED)
 CUT END SLOPES AT 1:1
 FILL END SLOPES AT 1.5:1

WELL SITE LOCATION
 434' FSL
 237' FWL

BERM DETAIL



DITCH DETAIL



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OASIS PETROLEUM NORTH AMERICA, LLC
 QUANTITIES
 SECTION 18, T153N, R100W
 MCKENZIE COUNTY, NORTH DAKOTA
 Drawn By: B.H.H. Project No.: S14-09-081.02
 Checked By: R.L.P. Date: APRIL 2014

Revision No.	Date	By	Description



5/29/2014

Mineral Resources Permit Manager
North Dakota Industrial Commission
600 East Boulevard Avenue Dept. 405
Bismarck, ND 58505-0840

RE: **Kline Federal 5300 41-18 9T**
 Kline Federal 5300 41-18 10B
 Kline Federal 5300 41-18 11T2
 Kline Federal 5300 41-18 12TX
 Kline Federal 5300 41-18 13T2X
 Kline Federal 5300 41-18 14BX
 Request for a legal street address

Dear NDIC:

Oasis Petroleum has requested a physical street address for the Kline Federal 5300 41-18 9T, Kline Federal 5300 41-18 10B, Kline Federal 5300 41-18 11T2, Kline Federal 5300 41-18 12TX, Kline Federal 5300 41-18 13T2X, and Kline Federal 5300 41-18 14BX. The request was made to Aaron Chisolm, in McKenzie County. Upon receiving a legal street address, Oasis will submit the address to the NDIC on a Sundry Notice (form 4) pursuant to 43-02-03-28.

Thank you for your consideration.

Respectfully,

A handwritten signature in black ink, appearing to read "Heather McCowan". The signature is fluid and cursive, with the first name "Heather" and last name "McCowan" clearly distinguishable.

Heather McCowan
Regulatory Assistant
Oasis Petroleum North America, LLC

Surface Damage Affidavit

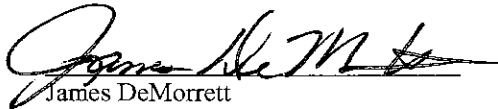
STATE OF NORTH DAKOTA)
) ss.
COUNTY OF WILLIAMS)

James DeMorrett, being duly sworn, states as follows:

The below described lands are not located in a wellhead protection/source water protection area per the website provided by the NDIC

and the adjacent landowner is fully aware that Oasis Petroleum will be drilling and operating multiple wells on spacing unit from a surface location in Section 18: Lot 4 a/k/a SW¼SW¼, Township 153 North, Range 100 West, 5th P.M., McKenzie County, North Dakota. (totaling 6.52 acres). The adjacent surface landowner is fully aware that the above referenced location will consist of the well bores and pumping units only and will not have cuttings pits, storage tanks or flare pits.

Dated this 16th day of June, 2014

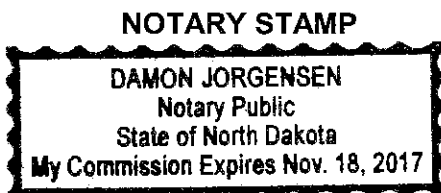

James DeMorrett

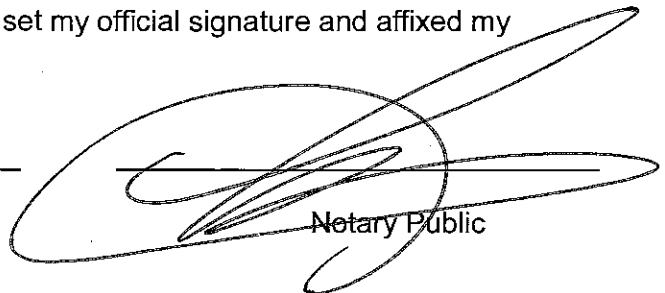
STATE OF NORTH DAKOTA }
 } ss.
COUNTY OF WILLIAMS }

BE IT REMEMBERED, That on this 16 day of June, 2014 before me, a Notary Public, in and for said County and State, personally appeared James DeMorrett, to me known to be the identical persons described in and who executed the within and foregoing instrument and acknowledged to me that he executed the same as his free and voluntary act and deed for the uses and purposes therein set forth.

IN WITNESS WHEREOF, I have hereunto set my official signature and affixed my notarial seal, the day and year first above written.

My commission expires: Nov 18, 2017




Notary Public